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# Contents

Intellectual Property Rights ..... 2
Legal Notice ..... 2
Modal verbs terminology ..... 2
Foreword ..... 16
1 Scope ..... 18
2 References ..... 18
3 Definitions and abbreviations ..... 20
3.1 Definitions ..... 20
3.2 Abbreviations ..... 20
4 Overview ..... 21
4.1 Introduction ..... 21
5 Services offered by the AMF ..... 22
5.1 Introduction ..... 22
5.2 Namf_Communication Service ..... 24
5.2.1 Service Description ..... 24
5.2.2 Service Operations ..... 24
5.2.2.1 Introduction ..... 24
5.2.2.2 UE Context Operations ..... 25
5.2.2.2.1 UEContextTransfer ..... 25
5.2.2.2.1.1 General ..... 25
5.2.2.2.1.2 Retrieve UE Context after successful UE authentication ..... 27
5.2.2.2.2 RegistrationStatusUpdate ..... 28
5.2.2.2.2.1 General ..... 28
5.2.2.2.3 CreateUEContext ..... 29
5.2.2.2.3.1 General ..... 29
5.2.2.2.3.2 Create UE Context with AMF Relocation ..... 31
5.2.2.2.4 ReleaseUEContext ..... 32
5.2.2.2.4.1 General ..... 32
5.2.2.2.5 RelocateUEContext ..... 33
5.2.2.2.5.1 General ..... 33
5.2.2.2.6 CancelRelocateUEContext ..... 34
5.2.2.2.6.1 General ..... 34
5.2.2.3 UE Specific N1N2 Message Operations ..... 35
5.2.2.3.1 N1N2MessageTransfer ..... 35
5.2.2.3.1.1 General ..... 35
5.2.2.3.1.2 Detailed behaviour of the AMF ..... 37
5.2.2.3.2 N1N2Transfer Failure Notification ..... 40
5.2.2.3.3 N1N2MessageSubscribe ..... 41
5.2.2.3.3.1 General ..... 41
5.2.2.3.4 N1N2MessageUnSubscribe ..... 42
5.2.2.3.4.1 General ..... 42
5.2.2.3.5 N1MessageNotify ..... 42
5.2.2.3.5.1 General ..... 42
5.2.2.3.5.2 Using N1MessageNotify in the Registration with AMF Re-allocation Procedure ..... 43
5.2.2.3.5.3 Using N1MessageNotify in the UE Assisted and UE Based Positioning Procedure ..... 43
5.2.2.3.5.4 Using N1MessageNotify in the UE Configuration Update for transparent UE Policy delivery ..... 44
5.2.2.3.5.5 Using N1MessageNotify in the LCS Event Report, Event Reporting in RRC INACTIVE state procedures, LCS Cancel Location and LCS Periodic-Triggered Invoke Procedures ..... 44
5.2.2.3.5.6 Using N1MessageNotify in the UE triggered policy provisioning procedure to request UE policies ..... 44
5.2.2.3.5.7 Using N1MessageNotify in the procedures applicable to a PRU ..... 45
5.2.2.3.6 N2InfoNotify ..... 45

5.2.2.3.6.1 General ..... 45
5.2.2.3.6.2 Using N2InfoNotify during Inter NG-RAN node N2 based handover procedure ..... 46
5.2.2.3.6.3 Using N2InfoNotify during Location Services procedures ..... 47
5.2.2.3.6.4 Using N2InfoNotify during AMF planned removal procedure with UDSF deployed procedure ..... 47
5.2.2.4 Non-UE N2 Message Operations ..... 48
5.2.2.4.1 NonUeN2MessageTransfer ..... 48
5.2.2.4.1.1 General ..... 48
5.2.2.4.1.2 Obtaining Non UE Associated Network Assistance Data Procedure ..... 48
5.2.2.4.1.3 Warning Request Transfer Procedure ..... 49
5.2.2.4.1.4 Configuration Transfer Procedure ..... 49
5.2.2.4.1.5 RIM Information Transfer Procedures ..... 49
5.2.2.4.1.6 Broadcast of Assistance Data by an LMF ..... 50
5.2.2.4.1.7 Management of network timing synchronization status monitoring procedures ..... 50
5.2.2.4.2 NonUeN2InfoSubscribe ..... 50
5.2.2.4.2.1 General ..... 50
5.2.2.4.3 NonUeN2InfoUnSubscribe ..... 51
5.2.2.4.3.1 General ..... 51
5.2.2.4.4 NonUeN2InfoNotify ..... 52
5.2.2.4.4.1 General ..... 52
5.2.2.4.4.2 Using NonUeN2InfoNotify during Location Services procedures ..... 52
5.2.2.4.4.3 Use of NonUeN2InfoNotify for PWS related events ..... 52
5.2.2.4.4.4 Using NonUeN2InfoNotify during network timing synchronization status monitoring procedure ..... 53
5.2.2.5 AMF Status Change Operations ..... 54
5.2.2.5.1 AMFStatusChangeSubscribe ..... 54
5.2.2.5.1.1 General ..... 54
5.2.2.5.1.2 Creation of a subscription ..... 54
5.2.2.5.1.3 Modification of a subscription ..... 54
5.2.2.5.2 AMFStatusChangeUnSubscribe ..... 55
5.2.2.5.2.1 General ..... 55
5.2.2.5.3 AMFStatusChangeNotify ..... 56
5.2.2.5.3.1 General ..... 56
5.2.2.6 EBIAssignment ..... 56
5.2.2.6.1 General ..... 56
5.3 Namf_EventExposure Service ..... 58
5.3.1 Service Description ..... 58
5.3.2 Service Operations ..... 64
5.3.2.1 Introduction ..... 64
5.3.2.2 Subscribe ..... 64
5.3.2.2.1 General ..... 64
5.3.2.2.2 Creation of a subscription ..... 65
5.3.2.2.3 Modification of a subscription ..... 67
5.3.2.2.4 Remove or add group member UE(s) for a group subscription ..... 68
5.3.2.3 Unsubscribe ..... 69
5.3.2.3.1 General ..... 69
5.3.2.4 Notify ..... 69
5.3.2.4.1 General ..... 69
5.3.2.4.2 Event Subscription Synchronization for specific UE ..... 70
5.4 Namf_MT Service ..... 71
5.4.1 Service Description ..... 71
5.4.2 Service Operations ..... 71
5.4.2.1 Introduction ..... 71
5.4.2.2 EnableUEReachability ..... 71
5.4.2.2.1 General ..... 71
5.4.2.3 ProvideDomainSelectionInfo ..... 73
5.4.2.3.1 General ..... 73
5.4.2.4 EnableGroupReachability ..... 74
5.4.2.4.1 General ..... 74
5.4.2.5 UEReachabilityInfoNotify ..... 75
5.4.2.5.1 General ..... 75
5.5 Namf_Location Service ..... 75

5.5.1 Service Description ..... 75
5.5.2 Service Operations ..... 76
5.5.2.1 Introduction ..... 76
5.5.2.2 ProvidePositioningInfo ..... 76
5.5.2.2.1 General ..... 76
5.5.2.3 EventNotify ..... 78
5.5.2.3.1 General ..... 78
5.5.2.4 ProvideLocationInfo ..... 79
5.5.2.4.1 General ..... 79
5.5.2.5 CancelLocation ..... 79
5.5.2.5.1 General ..... 79
5.6 Namf_MBSBroadcast Service ..... 80
5.6.1 Service Description ..... 80
5.6.2 Service Operations ..... 80
5.6.2.1 Introduction ..... 80
5.6.2.2 ContextCreate ..... 80
5.6.2.3 ContextUpdate ..... 82
5.6.2.4 ContextRelease ..... 83
5.6.2.5 ContextStatusNotify ..... 84
5.7 Namf_MBSCommunication Service ..... 86
5.7.1 Service Description ..... 86
5.7.2 Service Operations ..... 86
5.7.2.1 Introduction ..... 86
5.7.2.2 N2MessageTransfer ..... 86
5.7.2.3 Notify ..... 87
6 API Definitions ..... 88
6.1 Namf_Communication Service API ..... 88
6.1.1 API URI ..... 88
6.1.2 Usage of HTTP ..... 89
6.1.2.1 General ..... 89
6.1.2.2 HTTP standard headers ..... 89
6.1.2.2.1 General ..... 89
6.1.2.2.2 Content type ..... 89
6.1.2.3 HTTP custom headers ..... 89
6.1.2.3.1 General ..... 89
6.1.2.4 HTTP multipart messages ..... 89
6.1.3 Resources ..... 91
6.1.3.1 Overview ..... 91
6.1.3.2 Resource: Individual ueContext ..... 92
6.1.3.2.1 Description ..... 92
6.1.3.2.2 Resource Definition ..... 93
6.1.3.2.3 Resource Standard Methods ..... 93
6.1.3.2.3.1 PUT ..... 93
6.1.3.2.4 Resource Custom Operations ..... 95
6.1.3.2.4.1 Overview ..... 95
6.1.3.2.4.2 Operation: release (POST) ..... 95
6.1.3.2.4.2.1 Description ..... 95
6.1.3.2.4.2.2 Operation Definition ..... 95
6.1.3.2.4.3 Operation: assign-ebi (POST) ..... 96
6.1.3.2.4.3.1 Description ..... 96
6.1.3.2.4.3.2 Operation Definition ..... 96
6.1.3.2.4.4 Operation: transfer (POST) ..... 99
6.1.3.2.4.4.1 Description ..... 99
6.1.3.2.4.4.2 Operation Definition ..... 99
6.1.3.2.4.5 Operation: transfer-update (POST) ..... 100
6.1.3.2.4.5.1 Description ..... 100
6.1.3.2.4.5.2 Operation Definition ..... 101
6.1.3.2.4.6 Operation: relocate (POST) ..... 102
6.1.3.2.4.6.1 Description ..... 102
6.1.3.2.4.6.2 Operation Definition ..... 102
6.1.3.2.4.7 Operation: cancel-relocate (POST) ..... 103

6.1.3.2.4.7.1 Description ..... 103
6.1.3.2.4.7.2 Operation Definition ..... 103
6.1.3.3 Resource: N1N2 Subscriptions Collection for Individual UE Contexts ..... 104
6.1.3.3.1 Description ..... 104
6.1.3.3.2 Resource Definition ..... 104
6.1.3.3.3 Resource Standard Methods ..... 105
6.1.3.3.3.1 POST ..... 105
6.1.3.3.4 Resource Custom Operations ..... 106
6.1.3.4 Resource: N1N2 Individual Subscription ..... 106
6.1.3.4.1 Description ..... 106
6.1.3.4.2 Resource Definition ..... 106
6.1.3.4.3 Resource Standard Methods ..... 106
6.1.3.4.3.1 DELETE ..... 106
6.1.3.4.4 Resource Custom Operations ..... 107
6.1.3.5 Resource: N1N2 Messages Collection ..... 107
6.1.3.5.1 Description ..... 107
6.1.3.5.2 Resource Definition ..... 108
6.1.3.5.3 Resource Standard Methods ..... 108
6.1.3.5.3.1 POST ..... 108
6.1.3.6 Resource: subscriptions collection ..... 112
6.1.3.6.1 Description ..... 112
6.1.3.6.2 Resource Definition ..... 112
6.1.3.6.3 Resource Standard Methods ..... 113
6.1.3.6.3.1 POST ..... 113
6.1.3.7 Resource: individual subscription ..... 114
6.1.3.7.1 Description ..... 114
6.1.3.7.2 Resource Definition ..... 114
6.1.3.7.3 Resource Standard Methods ..... 114
6.1.3.7.3.1 DELETE ..... 114
6.1.3.7.3.2 PUT ..... 115
6.1.3.8 Resource: Non UE N2 Messages Collection ..... 117
6.1.3.8.1 Description ..... 117
6.1.3.8.2 Resource Definition ..... 117
6.1.3.8.3 Resource Standard Methods ..... 117
6.1.3.8.4 Resource Custom Operations ..... 117
6.1.3.8.4.1 Overview ..... 117
6.1.3.8.4.2 Operation: transfer ..... 117
6.1.3.8.4.2.1 Description ..... 117
6.1.3.8.4.2.2 Operation Definition ..... 117
6.1.3.9 Resource: Non UE N2 Messages Subscriptions Collection ..... 119
6.1.3.9.1 Description ..... 119
6.1.3.9.2 Resource Definition ..... 119
6.1.3.9.3 Resource Standard Methods ..... 119
6.1.3.9.3.1 POST ..... 119
6.1.3.9.4 Resource Custom Operations ..... 120
6.1.3.10 Resource: Non UE N2 Message Notification Individual Subscription ..... 121
6.1.3.10.1 Description ..... 121
6.1.3.10.2 Resource Definition ..... 121
6.1.3.10.3 Resource Standard Methods ..... 121
6.1.3.10.3.1 DELETE ..... 121
6.1.3.10.4 Resource Custom Operations ..... 122
6.1.4 Custom Operations without associated resources ..... 122
6.1.5 Notifications ..... 122
6.1.5.1 General ..... 122
6.1.5.2 AMF Status Change Notification ..... 123
6.1.5.2.1 Description ..... 123
6.1.5.2.2 Notification Definition ..... 123
6.1.5.2.3 Notification Standard Methods ..... 123
6.1.5.2.3.1 POST ..... 123
6.1.5.3 Non UE N2 Information Notification ..... 124
6.1.5.3.1 Description ..... 124
6.1.5.3.2 Notification Definition ..... 124

6.1.5.3.3 Notification Standard Methods ..... 125
6.1.5.3.3.1 POST ..... 125
6.1.5.4 N1 Message Notification ..... 125
6.1.5.4.1 Description ..... 125
6.1.5.4.2 Notification Definition ..... 126
6.1.5.4.3 Notification Standard Methods ..... 126
6.1.5.4.3.1 POST ..... 126
6.1.5.5 UE Specific N2 Information Notification ..... 127
6.1.5.5.1 Description ..... 127
6.1.5.5.2 Notification Definition ..... 127
6.1.5.5.3 Notification Standard Methods ..... 127
6.1.5.5.3.1 POST ..... 127
6.1.5.6 N1N2 Transfer Failure Notification ..... 128
6.1.5.6.1 Description ..... 128
6.1.5.6.2 Notification Definition ..... 128
6.1.5.6.3 Notification Standard Methods ..... 129
6.1.5.6.3.1 POST ..... 129
6.1.5.7 Void ..... 130
6.1.6 Data Model ..... 130
6.1.6.1 General ..... 130
6.1.6.2 Structured data types ..... 137
6.1.6.2.1 Introduction ..... 137
6.1.6.2.2 Type: SubscriptionData ..... 138
6.1.6.2.3 Type: AmfStatusChangeNotification ..... 138
6.1.6.2.4 Type: AmfStatusInfo ..... 138
6.1.6.2.5 Type: AssignEbiData ..... 139
6.1.6.2.6 Type: AssignedEbiData ..... 139
6.1.6.2.7 Type: AssignEbiFailed ..... 140
6.1.6.2.8 Type: UEContextRelease ..... 140
6.1.6.2.9 Type: N2InformationTransferReqData ..... 140
6.1.6.2.10 Type: NonUeN2InfoSubscriptionCreateData ..... 141
6.1.6.2.11 Type: NonUeN2InfoSubscriptionCreatedData ..... 142
6.1.6.2.12 Type: UeN1N2InfoSubscriptionCreateData ..... 142
6.1.6.2.13 Type: UeN1N2InfoSubscriptionCreatedData ..... 142
6.1.6.2.14 Type: N2InformationNotification ..... 143
6.1.6.2.15 Type: N2InfoContainer ..... 146
6.1.6.2.16 Type: N1MessageNotification ..... 147
6.1.6.2.17 Type: N1MessageContainer ..... 149
6.1.6.2.18 Type: N1N2MessageTransferReqData ..... 150
6.1.6.2.19 Type: N1N2MessageTransferRspData ..... 154
6.1.6.2.20 Type: RegistrationContextContainer ..... 155
6.1.6.2.21 Type: AreaOfValidity ..... 157
6.1.6.2.22 Void ..... 157
6.1.6.2.23 Type: UeContextTransferReqData ..... 157
6.1.6.2.24 Type: UeContextTransferRspData ..... 158
6.1.6.2.25 Type: UeContext ..... 159
6.1.6.2.26 Type: N2SmInformation ..... 170
6.1.6.2.27 Type: N2InfoContent ..... 171
6.1.6.2.28 Type: NrppaInformation ..... 171
6.1.6.2.29 Type: PwsInformation ..... 172
6.1.6.2.30 Type: N1N2MsgTxfrFailureNotification ..... 174
6.1.6.2.31 Type: N1N2MessageTransferError ..... 174
6.1.6.2.32 Type: N1N2MsgTxfrErrDetail ..... 175
6.1.6.2.33 Type: N2InformationTransferRspData ..... 175
6.1.6.2.34 Type: MmContext ..... 176
6.1.6.2.35 Type: SeafData ..... 180
6.1.6.2.36 Type: NasSecurityMode ..... 180
6.1.6.2.37 Type: PduSessionContext ..... 181
6.1.6.2.38 Type: NssaiMapping ..... 186
6.1.6.2.39 Type: UeRegStatusUpdateReqData ..... 187
6.1.6.2.40 Type: AssignEbiError ..... 187
6.1.6.2.41 Type: UeContextCreateData ..... 188

6.1.6.2.42 Type: UeContextCreatedData ..... 189
6.1.6.2.43 Type: UeContextCreateError ..... 189
6.1.6.2.44 Type: NgRanTargetId ..... 190
6.1.6.2.45 Type: N2InformationTransferError ..... 190
6.1.6.2.46 Type: PWSResponseData ..... 190
6.1.6.2.47 Type: PWSErrorData ..... 191
6.1.6.2.48 Void ..... 191
6.1.6.2.49 Type: NgKsi ..... 191
6.1.6.2.50 Type: KeyAmf ..... 191
6.1.6.2.51 Type: ExpectedUeBehavior ..... 191
6.1.6.2.52 Type: UeRegStatusUpdateRspData ..... 192
6.1.6.2.53 Type: N2RanInformation ..... 192
6.1.6.2.54 Type: N2InfoNotificationRspData ..... 192
6.1.6.2.55 Type: SmallDataRateStatusInfo ..... 192
6.1.6.2.56 Type: SmfChangeInfo ..... 193
6.1.6.2.57 Type: V2xContext ..... 193
6.1.6.2.58 Type: ImmediateMdtConf ..... 194
6.1.6.2.59 Type: V2xInformation ..... 196
6.1.6.2.60 Type: EpsNasSecurityMode ..... 196
6.1.6.2.61 Type: UeContextRelocateData ..... 197
6.1.6.2.62 Type: UeContextRelocatedData ..... 197
6.1.6.2.63 Void ..... 197
6.1.6.2.64 Type: EcRestrictionDataWb ..... 198
6.1.6.2.65 Type: ExtAmfEventSubscription ..... 198
6.1.6.2.66 Type: AmfEventSubscriptionAddInfo ..... 199
6.1.6.2.67 Type: UeContextCancelRelocateData ..... 201
6.1.6.2.68 Type: UeDifferentiationInfo ..... 201
6.1.6.2.69 Type: CeModeBInd ..... 202
6.1.6.2.70 Type: LteMInd ..... 202
6.1.6.2.71 Type: NpnAccessInfo ..... 202
6.1.6.2.72 Type: ProseContext ..... 203
6.1.6.2.73 Type: AnalyticsSubscription ..... 203
6.1.6.2.74 Type: NwdafSubscription ..... 204
6.1.6.2.75 Type: UpdpSubscriptionData ..... 204
6.1.6.2.76 Type: ProSeInformation ..... 204
6.1.6.2.77 Type: ReleaseSessionInfo ..... 204
6.1.6.2.78 Type: AreaOfInterestEventState ..... 205
6.1.6.2.79 Type: TssInformation ..... 205
6.1.6.2.80 Type: AmPolicyInfoContainer ..... 206
6.1.6.2.81 Type: RslpInformation ..... 206
6.1.6.2.82 Type: A2xContext ..... 206
6.1.6.2.83 Type: A2xInformation ..... 206
6.1.6.2.84 Type: LcsUpContext ..... 207
6.1.6.2.85 Type: DeregInactTimerInfo ..... 207
6.1.6.2.86 Type: TssRspPerNgran ..... 207
6.1.6.2.87 Type: SliceReplacementMapping ..... 207
6.1.6.2.88 Type: SliceDeregInactConfig ..... 207
6.1.6.3 Simple data types and enumerations ..... 208
6.1.6.3.1 Introduction ..... 208
6.1.6.3.2 Simple data types ..... 208
6.1.6.3.3 Enumeration: StatusChange ..... 208
6.1.6.3.4 Enumeration: N2InformationClass ..... 209
6.1.6.3.5 Enumeration: N1MessageClass ..... 209
6.1.6.3.6 Enumeration: N1N2MessageTransferCause ..... 210
6.1.6.3.7 Enumeration: UeContextTransferStatus ..... 211
6.1.6.3.8 Enumeration: N2InformationTransferResult ..... 211
6.1.6.3.9 Enumeration: CipheringAlgorithm ..... 211
6.1.6.3.10 Enumeration: IntegrityAlgorithm ..... 211
6.1.6.3.11 Enumeration: SmsSupport ..... 211
6.1.6.3.12 Enumeration: ScType ..... 212
6.1.6.3.13 Enumeration: KeyAmfType ..... 212
6.1.6.3.14 Enumeration: TransferReason ..... 212

6.1.6.3.15 Enumeration: PolicyReqTrigger ..... 212
6.1.6.3.16 Enumeration: RatSelector ..... 213
6.1.6.3.17 Enumeration: NgapIeType ..... 213
6.1.6.3.18 Enumeration: N2InfoNotifyReason ..... 213
6.1.6.3.19 Enumeration: SmfChangeIndication ..... 213
6.1.6.3.20 Enumeration: SbiBindingLevel ..... 214
6.1.6.3.21 Enumeration: EpsNasCipheringAlgorithm ..... 214
6.1.6.3.22 Enumeration: EpsNasIntegrityAlgorithm ..... 214
6.1.6.3.23 Enumeration: PeriodicCommunicationIndicator ..... 214
6.1.6.3.24 Enumeration: UuaaMmStatus ..... 214
6.1.6.3.25 Enumeration: ReleaseCause ..... 215
6.1.6.3.26 Enumeration: NgranFailureInfo ..... 215
6.1.6.3.27 Enumeration: XrDeviceWith2Rx ..... 215
6.1.6.4 Binary data ..... 215
6.1.6.4.1 Introduction ..... 215
6.1.6.4.2 N1 Message ..... 216
6.1.6.4.3 N2 Information ..... 217
6.1.6.4.3.1 Introduction ..... 217
6.1.6.4.3.2 NGAP IEs ..... 217
6.1.6.4.3.3 NGAP Messages ..... 219
6.1.6.4.4 Mobile Terminated Data ..... 220
6.1.6.4.5 GTP-C Message ..... 220
6.1.7 Error Handling ..... 221
6.1.7.1 General ..... 221
6.1.7.2 Protocol Errors ..... 221
6.1.7.3 Application Errors ..... 221
6.1.8 Feature Negotiation ..... 224
6.1.9 Security ..... 228
6.1.10 HTTP redirection ..... 228
6.2 Namf_EventExposure Service API ..... 228
6.2.1 API URI ..... 228
6.2.2 Usage of HTTP ..... 229
6.2.2.1 General ..... 229
6.2.2.2 HTTP standard headers ..... 229
6.2.2.2.1 General ..... 229
6.2.2.2.2 Content type ..... 229
6.2.2.3 HTTP custom headers ..... 230
6.2.2.3.1 General ..... 230
6.2.3 Resources ..... 230
6.2.3.1 Overview ..... 230
6.2.3.2 Resource: Subscriptions collection ..... 230
6.2.3.2.1 Description ..... 230
6.2.3.2.2 Resource Definition ..... 230
6.2.3.2.3 Resource Standard Methods ..... 231
6.2.3.2.3.1 POST ..... 231
6.2.3.2.4 Resource Custom Operations ..... 232
6.2.3.3 Resource: Individual subscription ..... 232
6.2.3.3.1 Description ..... 232
6.2.3.3.2 Resource Definition ..... 232
6.2.3.3.3 Resource Standard Methods ..... 233
6.2.3.3.3.1 PATCH ..... 233
6.2.3.3.3.2 DELETE ..... 234
6.2.3.3.4 Resource Custom Operations ..... 235
6.2.4 Custom Operations without associated resources ..... 235
6.2.5 Notifications ..... 235
6.2.5.1 Void ..... 235
6.2.5.2 AMF Event Notification ..... 235
6.2.5.2.1 Notification Definition ..... 235
6.2.5.2.3 Notification Standard Methods ..... 236
6.2.5.2.3.1 POST ..... 236
6.2.6 Data Model ..... 237
6.2.6.1 General ..... 237

6.2.6.2 Structured data types ..... 239
6.2.6.2.1 Introduction ..... 239
6.2.6.2.2 Type: AmfEventSubscription ..... 240
6.2.6.2.3 Type: AmfEvent ..... 243
6.2.6.2.4 Type: AmfEventNotification ..... 250
6.2.6.2.5 Type: AmfEventReport ..... 251
6.2.6.2.6 Type: AmfEventMode ..... 256
6.2.6.2.7 Type: AmfEventState ..... 258
6.2.6.2.8 Type: RmInfo ..... 259
6.2.6.2.9 Type: CmInfo ..... 259
6.2.6.2.10 Void ..... 259
6.2.6.2.11 Type: CommunicationFailure ..... 259
6.2.6.2.12 Type: AmfCreateEventSubscription ..... 259
6.2.6.2.13 Type: AmfCreatedEventSubscription ..... 260
6.2.6.2.14 Type: AmfUpdateEventSubscriptionItem ..... 261
6.2.6.2.15 Type: AmfUpdatedEventSubscription ..... 264
6.2.6.2.16 Type: AmfEventArea ..... 265
6.2.6.2.17 Type: LadnInfo ..... 265
6.2.6.2.18 Type: AmfUpdateEventOptionItem ..... 266
6.2.6.2.19 Type: 5GsUserStateInfo ..... 266
6.2.6.2.20 Type: TrafficDescriptor ..... 267
6.2.6.2.21 Type: UEIdExt ..... 267
6.2.6.2.22 Type: AmfEventSubsSyncInfo ..... 267
6.2.6.2.23 Type: AmfEventSubscriptionInfo ..... 268
6.2.6.2.24 Type: TargetArea ..... 268
6.2.6.2.25 Type: SnssaiTaiMapping ..... 268
6.2.6.2.26 Type: SupportedSnssai ..... 269
6.2.6.2.27 Type: UeInAreaFilter ..... 269
6.2.6.2.28 Type: IdleStatusIndication ..... 270
6.2.6.2.29 Type: UeAccessBehaviorReportItem ..... 271
6.2.6.2.30 Type: UeLocationTrendsReportItem ..... 272
6.2.6.2.31 Type: DispersionArea ..... 273
6.2.6.2.32 Type: MmTransactionLocationReportItem ..... 273
6.2.6.2.33 Type: MmTransactionSliceReportItem ..... 274
6.2.6.2.34 Type: SliceAreaRestrictionInfo ..... 274
6.2.6.3 Simple data types and enumerations ..... 274
6.2.6.3.1 Introduction ..... 274
6.2.6.3.2 Simple data types ..... 274
6.2.6.3.3 Enumeration: AmfEventType ..... 275
6.2.6.3.4 Enumeration: AmfEventTrigger ..... 277
6.2.6.3.5 Enumeration: LocationFilter ..... 278
6.2.6.3.6 Void ..... 278
6.2.6.3.7 Enumeration: UeReachability ..... 278
6.2.6.3.8 Void ..... 278
6.2.6.3.9 Enumeration: RmState ..... 278
6.2.6.3.10 Enumeration: CmState ..... 278
6.2.6.3.11 Enumeration: 5GsUserState ..... 279
6.2.6.3.12 Enumeration: LossOfConnectivityReason ..... 279
6.2.6.3.13 Enumeration: ReachabilityFilter ..... 279
6.2.6.3.14 Enumeration: UeType ..... 279
6.2.6.3.15 Enumeration: AccessStateTransitionType ..... 280
6.2.6.3.16 Enumeration: SubTerminationReason ..... 280
6.2.6.4 Binary data ..... 280
6.2.7 Error Handling ..... 280
6.2.7.1 General ..... 280
6.2.7.2 Protocol Errors ..... 280
6.2.7.3 Application Errors ..... 280
6.2.8 Feature Negotiation ..... 281
6.2.9 Security ..... 284
6.2.10 HTTP redirection ..... 284
6.3 Namf_MT Service API ..... 285
6.3.1 API URI ..... 285

6.3.2 Usage of HTTP ..... 285
6.3.2.1 General ..... 285
6.3.2.2 HTTP standard headers ..... 285
6.3.2.2.1 General ..... 285
6.3.2.2.2 Content type ..... 285
6.3.2.3 HTTP custom headers ..... 286
6.3.2.3.1 General ..... 286
6.3.3 Resources ..... 286
6.3.3.1 Overview ..... 286
6.3.3.2 Resource: ueReachInd ..... 286
6.3.3.2.1 Description ..... 286
6.3.3.2.2 Resource Definition ..... 287
6.3.3.2.3 Resource Standard Methods ..... 287
6.3.3.2.3.1 PUT ..... 287
6.3.3.2.4 Resource Custom Operations ..... 289
6.3.3.3 Resource: ueContext ..... 289
6.3.3.3.1 Description ..... 289
6.3.3.3.2 Resource Definition ..... 289
6.3.3.3.3 Resource Standard Methods ..... 289
6.3.3.3.3.1 GET ..... 289
6.3.3.3.4 Resource Custom Operations ..... 291
6.3.3.4 Resource: ueContexts collection ..... 291
6.3.3.4.1 Description ..... 291
6.3.3.4.2 Resource Definition ..... 291
6.3.3.4.3 Resource Standard Methods ..... 291
6.3.3.4.4 Resource Custom Operations ..... 291
6.3.3.4.4.1 Overview ..... 291
6.3.3.4.4.2 Operation: enable-group-reachability ..... 291
6.3.3.4.4.2.1 Description ..... 291
6.3.3.4.4.2.2 Operation Definition ..... 291
6.3.4 Custom Operations without associated resources ..... 293
6.3.5 Notifications ..... 293
6.3.5.1 General ..... 293
6.3.5.2 UE Reachability Info Notify ..... 293
6.3.5.2.1 Notification Definition ..... 293
6.3.5.2.3 Notification Standard Methods ..... 293
6.3.5.2.3.1 POST ..... 293
6.3.6 Data Model ..... 294
6.3.6.1 General ..... 294
6.3.6.2 Structured data types ..... 295
6.3.6.2.1 Introduction ..... 295
6.3.6.2.2 Type: EnableUeReachabilityReqData ..... 296
6.3.6.2.3 Type: EnableUeReachabilityRspData ..... 296
6.3.6.2.4 Type: UeContextInfo ..... 297
6.3.6.2.5 Type: ProblemDetailsEnableUeReachability ..... 297
6.3.6.2.6 Type: AdditionInfoEnableUeReachability ..... 297
6.3.6.2.7 Type: EnableGroupReachabilityReqData ..... 298
6.3.6.2.8 Type: EnableGroupReachabilityRspData ..... 298
6.3.6.2.9 Type: UeInfo ..... 298
6.3.6.2.10 Type: ReachabilityNotificationData ..... 298
6.3.6.2.11 Type: ReachableUeInfo ..... 299
6.3.6.2.12 Type: QosFlowInfo ..... 299
6.3.6.3 Simple data types and enumerations ..... 299
6.3.6.3.1 Introduction ..... 299
6.3.6.3.2 Simple data types ..... 299
6.3.6.3.3 Enumeration: UeContextInfoClass ..... 299
6.3.6.4 Binary data ..... 300
6.3.7 Error Handling ..... 300
6.3.7.1 General ..... 300
6.3.7.2 Protocol Errors ..... 300
6.3.7.3 Application Errors ..... 300
6.3.8 Feature Negotiation ..... 300

6.3.9 Security ..... 301
6.3.10 HTTP redirection ..... 302
6.4 Namf_Location Service API ..... 302
6.4.1 API URI ..... 302
6.4.2 Usage of HTTP ..... 302
6.4.2.1 General ..... 302
6.4.2.2 HTTP standard headers ..... 303
6.4.2.2.1 General ..... 303
6.4.2.2.2 Content type ..... 303
6.4.2.3 HTTP custom headers ..... 303
6.4.2.3.1 General ..... 303
6.4.3 Resources ..... 303
6.4.3.1 Overview ..... 303
6.4.3.2 Resource: Individual UE Context ..... 304
6.4.3.2.1 Description ..... 304
6.4.3.2.2 Resource Definition ..... 304
6.4.3.2.3 Resource Standard Methods ..... 304
6.4.3.2.4 Resource Custom Operations ..... 305
6.4.3.2.4.1 Overview ..... 305
6.4.3.2.4.2 Operation: provide-pos-info (POST) ..... 305
6.4.3.2.4.2.1 Description ..... 305
6.4.3.2.4.2.2 Operation Definition ..... 305
6.4.3.2.4.3 Operation: provide-loc-info (POST) ..... 307
6.4.3.2.4.3.1 Description ..... 307
6.4.3.2.4.3.2 Operation Definition ..... 307
6.4.3.2.4.4 Operation: cancel-pos-info (POST) ..... 308
6.4.3.2.4.4.1 Description ..... 308
6.4.3.2.4.4.2 Operation Definition ..... 308
6.4.4 Custom Operations without associated resources ..... 309
6.4.5 Notifications ..... 310
6.4.5.1 General ..... 310
6.4.5.2 Event Notify ..... 310
6.4.5.2.1 Description ..... 310
6.4.5.2.2 Notification Definition ..... 310
6.4.5.2.3 Notification Standard Methods ..... 310
6.4.5.2.3.1 POST ..... 310
6.4.6 Data Model ..... 311
6.4.6.1 General ..... 311
6.4.6.2 Structured data types ..... 315
6.4.6.2.1 Introduction ..... 315
6.4.6.2.2 Type: RequestPosInfo ..... 316
6.4.6.2.3 Type: ProvidePosInfo ..... 321
6.4.6.2.4 Type: NotifiedPosInfo ..... 325
6.4.6.2.5 Type: RequestLocInfo ..... 329
6.4.6.2.6 Type: ProvideLocInfo ..... 330
6.4.6.2.7 Type: CancelPosInfo ..... 330
6.4.6.2.11 Type: ProvidePosInfoExt ..... 331
6.4.6.2.12 Type: NotifiedPosInfoExt ..... 331
6.4.6.3 Simple data types and enumerations ..... 331
6.4.6.3.1 Introduction ..... 331
6.4.6.3.2 Simple data types ..... 331
6.4.6.3.3 Enumeration: LocationType ..... 332
6.4.6.3.4 Enumeration: LocationEvent ..... 332
6.4.6.3.5 Enumeration: LocationPrivacyVerResult ..... 332
6.4.6.3.6 Enumeration: LpHapType ..... 333
6.4.7 Error Handling ..... 333
6.4.7.1 General ..... 333
6.4.7.2 Protocol Errors ..... 333
6.4.7.3 Application Errors ..... 333
6.4.8 Feature Negotiation ..... 333
6.4.9 Security ..... 334
6.4.10 HTTP redirection ..... 334

6.5 Namf_MBSBroadcast Service API ..... 335
6.5.1 API URI ..... 335
6.5.2 Usage of HTTP ..... 335
6.5.2.1 General ..... 335
6.5.2.2 HTTP standard headers ..... 335
6.5.2.2.1 General ..... 335
6.5.2.2.2 Content type ..... 335
6.5.2.3 HTTP custom headers ..... 336
6.5.2.3.1 General ..... 336
6.5.2.4 HTTP multipart messages ..... 336
6.5.3 Resources ..... 337
6.5.3.1 Overview ..... 337
6.5.3.2 Resource: Broadcast MBS session contexts collection ..... 337
6.5.3.2.1 Description ..... 337
6.5.3.2.2 Resource Definition ..... 337
6.5.3.2.3 Resource Standard Methods ..... 338
6.5.3.2.3.1 POST ..... 338
6.5.3.2.4 Resource Custom Operations ..... 339
6.5.3.3 Resource: Individual broadcast MBS session context ..... 339
6.5.3.3.1 Description ..... 339
6.5.3.3.2 Resource Definition ..... 339
6.5.3.3.3 Resource Standard Methods ..... 339
6.5.3.3.3.1 DELETE ..... 339
6.5.3.3.4 Resource Custom Operations ..... 340
6.5.3.2.4.2 Operation: update (POST) ..... 340
6.5.3.2.4.2.1 Description ..... 340
6.5.3.2.4.2.2 Operation Definition ..... 340
6.5.4 Custom Operations without associated resources ..... 341
6.5.5 Notifications ..... 341
6.5.5.1 General ..... 341
6.5.5.2 Broadcast MBS Session Context Status Notification ..... 342
6.5.5.2.1 Description ..... 342
6.5.5.2.2 Target URI ..... 342
6.5.5.2.3 Notification Standard Methods ..... 342
6.5.5.2.3.1 POST ..... 342
6.5.6 Data Model ..... 343
6.5.6.1 General ..... 343
6.5.6.2 Structured data types ..... 344
6.5.6.2.1 Introduction ..... 344
6.5.6.2.2 Type: ContextCreateReqData ..... 344
6.5.6.2.3 Type: ContextCreateRspData ..... 344
6.5.6.2.4 Type: ContextStatusNotification ..... 345
6.5.6.2.5 Type: ContextUpdateReqData ..... 346
6.5.6.2.6 Type: ContextUpdateRspData ..... 346
6.5.6.2.7 Type: N2MbsSmInfo ..... 347
6.5.6.2.8 Type: OperationEvent ..... 347
6.5.6.2.9 Type: NgranFailureEvent ..... 347
6.5.6.2.10 Type: ContextStatusNotificationResponse ..... 348
6.5.6.3 Simple data types and enumerations ..... 348
6.5.6.3.1 Introduction ..... 348
6.5.6.3.2 Simple data types ..... 348
6.5.6.3.3 Enumeration: OperationStatus ..... 348
6.5.6.3.4 Enumeration: NgapIeType ..... 349
6.5.6.3.5 Enumeration: OpEventType ..... 349
6.5.6.3.6 Enumeration: NgranFailureIndication ..... 349
6.5.6.4 Binary data ..... 349
6.5.6.4.1 Introduction ..... 349
6.5.6.4.2 N2 Information ..... 350
6.5.6.4.2.1 Introduction ..... 350
6.5.6.4.2.2 NGAP IEs ..... 350
6.5.7 Error Handling ..... 350
6.5.7.1 General ..... 350

6.5.7.2 Protocol Errors ..... 350
6.5.7.3 Application Errors ..... 350
6.5.8 Feature Negotiation ..... 351
6.5.9 Security ..... 351
6.5.10 HTTP redirection ..... 351
6.6 Namf_MBSCommunication Service API ..... 351
6.6.1 API URI ..... 351
6.6.2 Usage of HTTP ..... 352
6.6.2.1 General ..... 352
6.6.2.2 HTTP standard headers ..... 352
6.6.2.2.1 General ..... 352
6.6.2.2.2 Content type ..... 352
6.6.2.3 HTTP custom headers ..... 352
6.6.2.3.1 General ..... 352
6.6.2.4 HTTP multipart messages ..... 353
6.6.3 Resources ..... 353
6.6.3.1 Overview ..... 353
6.6.3.1 Resource: N2 Message Handler (Custom Operation) ..... 354
6.6.3.1.1 Description ..... 354
6.6.3.1.2 Resource Definition ..... 354
6.6.3.1.3 Resource Standard Methods ..... 354
6.6.3.1.4 Resource Custom Operations ..... 354
6.6.3.1.4.1 Overview ..... 354
6.6.3.1.4.2 Operation: transfer ..... 354
6.6.3.1.4.2.1 Description ..... 354
6.6.3.1.4.2.2 Operation Definition ..... 354
6.6.4 Custom Operations without associated resources ..... 355
6.6.5 Notifications ..... 355
6.6.5.1 General ..... 355
6.6.5.2 Notification ..... 356
6.6.5.2.1 Description ..... 356
6.6.5.2.2 Notification Definitionn ..... 356
6.6.5.2.3 Notification Standard Methods ..... 356
6.6.5.2.3.1 POST ..... 356
6.6.6 Data Model ..... 357
6.6.6.1 General ..... 357
6.6.6.2 Structured data types ..... 358
6.6.6.2.1 Introduction ..... 358
6.6.6.2.2 Type: MbsN2MessageTransferReqData ..... 358
6.6.6.2.3 Type: MbsN2MessageTransferRspData ..... 358
6.6.6.2.4 Type: N2MbsSmInfo ..... 359
6.6.6.2.5 Type: Notification ..... 359
6.6.6.2.6 Type: RanFailure ..... 359
6.6.6.3 Simple data types and enumerations ..... 359
6.6.6.3.1 Introduction ..... 359
6.6.6.3.2 Simple data types ..... 359
6.6.6.3.3 Enumeration: MbsNgapIeType ..... 360
6.6.6.3.4 Enumeration: RanFailureIndication ..... 360
6.6.6.4 Binary data ..... 360
6.6.6.4.1 Introduction ..... 360
6.6.6.4.2 N2 Information ..... 360
6.6.6.4.2.1 Introduction ..... 360
6.6.6.4.2.2 NGAP IEs ..... 360
6.6.7 Error Handling ..... 361
6.6.7.1 General ..... 361
6.6.7.2 Protocol Errors ..... 361
6.6.7.3 Application Errors ..... 361
6.6.8 Feature Negotiation ..... 361
6.6.9 Security ..... 362
6.6.10 HTTP redirection ..... 362

**Annex A (normative): OpenAPI specification ..... 363**

A.1 General ....363
A.2 Namf_Communication API ....363
A.3 Namf_EventExposure API ....423
A.4 Namf_MT ....437
A.5 Namf_Location ....443
A.6 Namf_MBSBroadcast API ....452
A.7 Namf_MBSCommunication API ....463
**Annex B (Informative): HTTP Multipart Messages ....468**
B.1 Example of HTTP multipart message ....468
B.1.1 General ....468
B.1.2 Example HTTP multipart message with N2 Information binary data ....468
**Annex C (informative): Change history ....469**
History ....482

# Foreword

This Technical Specification has been produced by the 3rd Generation Partnership Project (3GPP).

The contents of the present document are subject to continuing work within the TSG and may change following formal TSG approval. Should the TSG modify the contents of the present document, it will be re-released by the TSG with an identifying change of release date and an increase in version number as follows:

Version x.y.z

where:

- x the first digit:
  - 1 presented to TSG for information;
  - 2 presented to TSG for approval;
  - 3 or greater indicates TSG approved document under change control.
- y the second digit is incremented for all changes of substance, i.e. technical enhancements, corrections, updates, etc.
- z the third digit is incremented when editorial only changes have been incorporated in the document.

In the present document, modal verbs have the following meanings:

**shall** indicates a mandatory requirement to do something

**shall not** indicates an interdiction (prohibition) to do something

The constructions "shall" and "shall not" are confined to the context of normative provisions, and do not appear in Technical Reports.

The constructions "must" and "must not" are not used as substitutes for "shall" and "shall not". Their use is avoided insofar as possible, and they are not used in a normative context except in a direct citation from an external, referenced, non-3GPP document, or so as to maintain continuity of style when extending or modifying the provisions of such a referenced document.

**should** indicates a recommendation to do something

**should not** indicates a recommendation not to do something

**may** indicates permission to do something

**need not** indicates permission not to do something

The construction "may not" is ambiguous and is not used in normative elements. The unambiguous constructions "might not" or "shall not" are used instead, depending upon the meaning intended.

**can** indicates that something is possible

**cannot** indicates that something is impossible

The constructions "can" and "cannot" are not substitutes for "may" and "need not".

**will** indicates that something is certain or expected to happen as a result of action taken by an agency the behaviour of which is outside the scope of the present document

**will not** indicates that something is certain or expected not to happen as a result of action taken by an agency the behaviour of which is outside the scope of the present document

**might** indicates a likelihood that something will happen as a result of action taken by some agency the behaviour of which is outside the scope of the present document

**might not** indicates a likelihood that something will not happen as a result of action taken by some agency the behaviour of which is outside the scope of the present document

In addition:

**is** (or any other verb in the indicative mood) indicates a statement of fact

**is not** (or any other negative verb in the indicative mood) indicates a statement of fact

The constructions "is" and "is not" do not indicate requirements.

# 1 Scope

The present document specifies the stage 3 protocol and data model for the Namf Service Based Interface. It provides stage 3 protocol definitions and message flows, and specifies the API for each service offered by the AMF.

The 5G System stage 2 architecture and procedures are specified in TS 23.501 [2], TS 23.502 [3] and TS 23.247 [55].

The Technical Realization of the Service Based Architecture and the Principles and Guidelines for Services Definition are specified in TS 29.500 [4] and TS 29.501 [5].

# 2 References


[1] TR 21.905: "Vocabulary for 3GPP Specifications".

[2] 3GPP TS 23.501: "System Architecture for the 5G System; Stage 2".

[3] 3GPP TS 23.502: "Procedures for the 5G System; Stage 2".

[4] 3GPP TS 29.500: "5G System; Technical Realization of Service Based Architecture; Stage 3".

[5] 3GPP TS 29.501: "5G System; Principles and Guidelines for Services Definition; Stage 3".

[6] 3GPP TS 29.571: "5G System; Common Data Types for Service Based Interfaces Stage 3".

[7] 3GPP TS 23.503: "Policy and Charging Control Framework for the 5G System; Stage 2".

[8] IETF RFC 8259: "The JavaScript Object Notation (JSON) Data Interchange Format".

[9] IETF RFC 2387: "The MIME Multipart/Related Content-type".

[10] IETF RFC 2045: "Multipurpose Internet Mail Extensions (MIME) Part One: Format of Internet Message Bodies".

[11] 3GPP TS 24.501: "Non-Access-Stratum (NAS) Protocol for 5G System (5GS); Stage 3".

[12] 3GPP TS 38.413: "NG Radio Access Network (NG-RAN); NG Application Protocol (NGAP)".

[13] 3GPP TS 36.355: "Evolved Universal Terrestrial Radio Access (E-UTRA); LTE Positioning Protocol (LPP)".

[14] IETF RFC 6902: "JavaScript Object Notation (JSON) Patch".

[15] 3GPP TS 24.007: "Mobile radio interface signalling layer 3; General Aspects".

[16] 3GPP TS 29.502: "5G System, Session Management Services; Stage 3".

[17] 3GPP TS 38.455: "NR Positioning Protocol A (NRPPa)".

[18] 3GPP TS 29.531: "Network Slice Selection Services; Stage 3".

[19] IETF RFC 9113: "Hypertext Transfer Protocol Version 2 (HTTP/2)".

[20] 3GPP TS 23.041: "Technical realization of Cell Broadcast Service (CBS)".

[21] Void.

[22] 3GPP TS 24.008: "Mobile radio interface Layer 3 specification; Core network protocols; Stage 3".

[23] OpenAPI Initiative, "OpenAPI Specification Version 3.0.0".

[24] 3GPP TS 36.413: "Evolved Universal Terrestrial Radio Access Network (E-UTRAN); S1 Application Protocol (S1AP)".

[25] 3GPP TS 29.572: "5G System, Location Management Services; Stage 3".

[26] Void.

[27] 3GPP TS 33.501: "Security architecture and procedures for 5G system".

[28] IETF RFC 6749: "The OAuth 2.0 Authorization Framework".

[29] 3GPP TS 29.510: "Network Function Repository Services; Stage 3".

[30] 3GPP TS 32.422: "Telecommunication management; Subscriber and equipment trace; Trace control and configuration management".

[31] Void.

[32] 3GPP TS 29.507: "5G System; Access and Mobility Policy Control Service; Stage 3".

[33] 3GPP TS 23.527: "5G System; Restoration Procedures".

[34] 3GPP TS 29.525: "5G System; UE Policy Control Service; Stage 3".

[35] 3GPP TS 29.503: "5G System; Unified Data Management Services; Stage 3".

[36] IETF RFC 9457: "Problem Details for HTTP APIs".

[37] 3GPP TR 21.900: "Technical Specification Group working methods".

[38] 3GPP TS 23.288: "Architecture enhancements for 5G System (5GS) to support network data analytics services".

[39] 3GPP TS 23.216: "Single Radio Voice Call Continuity (SRVCC); Stage 2".

[40] IETF RFC 6901: "JavaScript Object Notation (JSON) Pointer".

[41] 3GPP TS 29.274: "3GPP Evolved Packet System (EPS); Evolved General Packet Radio Service (GPRS) Tunnelling Protocol for Control plane (GTPv2-C); Stage 3".

[42] 3GPP TS 23.273: "5G System (5GS) Location Services (LCS); Stage 2".

[43] 3GPP TS 24.080: "Mobile radio interface layer 3 supplementary services specification; Formats and coding".

[44] 3GPP TS 23.040: "Technical realization of the Short Message Service (SMS)".

[45] 3GPP TS 24.011: "Point-to-Point (PP) Short Message Service (SMS) support on mobile radio interface".

[46] 3GPP TS 29.515: "5G System; Gateway Mobile Location Services Stage 3".

[47] 3GPP TS 23.287: "Architecture enhancements for 5G System (5GS) to support Vehicle-to-Everything (V2X) services".

[48] 3GPP TS 23.316: "Wireless and wireline convergence access support for the 5G System (5GS)".

[49] 3GPP TS 33.401: "3GPP System Architecture Evolution (SAE); Security architecture".

[50] 3GPP TS 29.010: "Information element mapping between Mobile Station - Base Station System (MS - BSS) and Base Station System - Mobile-services Switching Centre (BSS - MSC); Signalling Procedures and the Mobile Application Part (MAP)".

[51] 3GPP TS 23.304: "Proximity based Services (ProSe) in the 5G System (5GS)".

[52] 3GPP TS 29.520: "5G System; Network Data Analytics Services; Stage 3".

[53] 3GPP TS 24.587: "Vehicle-to-Everything (V2X) services in 5G System (5GS); Stage 3".

[54] 3GPP TS 24.554: " Proximity-services (ProSe) in 5G System (5GS) protocol aspects; Stage 3".

[55] 3GPP TS 23.247: "Architectural enhancements for 5G multicast-broadcast services".

[56] 3GPP TS 23.256: "Support of Uncrewed Aerial Systems (UAS) connectivity, identification and tracking; Stage 2".

[57] 3GPP TS 33.256: "Security aspects of Uncrewed Aerial Systems (UAS)".

[58] 3GPP TS 23.548: "5G System Enhancements for Edge Computing; Stage 2".

[59] 3GPP TS 23.586: "Architectural Enhancements to support Ranging based services and Sidelink Positioning".

[60] 3GPP TS 24.577: "Aircraft-to-Everything (A2X) services in 5G system (5GS); Protocol aspects; Stage 3".

[61] 3GPP TS 24.572: "User Plane Location Services (LCS) Protocols And Procedures; Stage 3".

# 3 Definitions and abbreviations

## 3.1 Definitions

For the purposes of the present document, the terms and definitions given in TR 21.905 [1] and the following apply. A term defined in the present document takes precedence over the definition of the same term, if any, in TR 21.905 [1].

**example:** text used to clarify abstract rules by applying them literally.

## 3.2 Abbreviations

For the purposes of the present document, the abbreviations given in TR 21.905 [1] and the following apply. An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, in TR 21.905 [1].

| | |
|---|---|
| 5GC | 5G Core Network |
| 5GS | 5G System |
| 5G-AN | 5G Access Network |
| 5G-GUTI | 5G Globally Unique Temporary Identifier |
| 5QI | 5G QoS Identifier |
| AMF | Access and Mobility Management Function |
| ASTI | Access Stratum TIme |
| DAPS | Dual Active Protocol Stacks |
| DCCF | Data Collection Coordination Function |
| EBI | EPS Bearer Identity |
| GAD | Universal Geographical Area Description |
| GPSI | Generic Public Subscription Identifier |
| GUAMI | Globally Unique AMF Identifier |
| JSON | JavaScript Object Notation |
| LADN | Local Area Data Network |
| LDR | Location Deferred Request |
| LIR | Location Immediate Request |
| LMF | Location Management Function |
| LPHAP | Low Power and High Accuracy Positioning |
| MA | Multi-Access |

| | |
|---|---|
| MBS | Multicast/Broadcast Service |
| MB-SMF | Multicast/Broadcast Session Management Function |
| MM | Mobility Management |
| N3IWF | Non-3GPP InterWorking Function |
| NEF | Network Exposure Function |
| NR | New Radio |
| NRF | Network Repository Function |
| NRPPa | NR Positioning Protocol A |
| NSI ID | Network Slice Instance Identifier |
| NSSAI | Network Slice Selection Assistance Information |
| NSSAA | Network Slice-Specific Authentication and Authorization |
| NWDAF | Network Data Analytics Function |
| PCF | Policy Control Function |
| PEI | Permanent Equipment Identifier |
| PRU | Positioning Reference Unit |
| RAT | Radio Access Type |
| RFSP | RAT/Frequency Selection Priority |
| SARI | Service Area Restriction Information |
| SBI | Service Based Interface |
| SM | Session Management |
| SMF | Session Management Function |
| SMSF | Short Message Service Function |
| S-NSSAI | Single Network Slice Selection Assistance Information |
| SUCI | Subscription Concealed Identifier |
| SUPI | Subscription Permanent Identifier |
| TA | Tracking Area |
| TAI | Tracking Area Identity |
| TNAP | Trusted Non-3GPP Access Point |
| TSCTSF | Time Sensitive Communication and Time Synchronization Function |
| TWAP | Trusted WLAN Access Point |
| UDM | Unified Data Management |
| UDSF | Unstructured Data Storage Function |
| UPP-CM | User Plane Positioning Connection Management |

# 4 Overview

## 4.1 Introduction

Within the 5GC, the AMF offers services to the SMF, other AMF, PCF, SMSF, LMF, GMLC, CBCF, PWS-IWF, NWDAF, DCCF, NEF, TSCTSF and MB-SMF via the Namf service based interface (see TS 23.501 [2], TS 23.502 [3], TS 23.041 [20], TS 23.288 [38] and TS 23.247 [55]).

Figure 4.1-1 provides the reference model (in service based interface representation and in reference point representation), with focus on the AMF and the scope of the present specification.

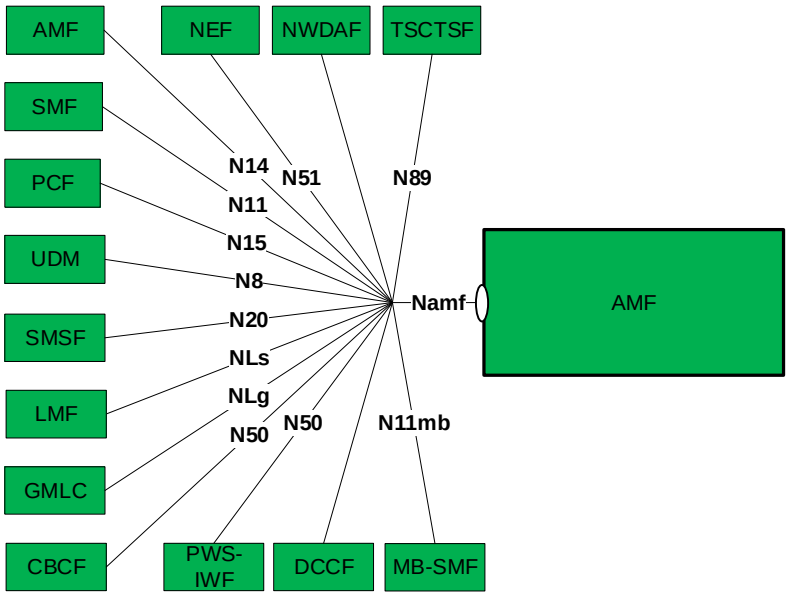



**Figure 4.1-1: Reference model – AMF**

The functionalities supported by the AMF are listed in clause 6.2.1 of TS 23.501 [2].

# 5 Services offered by the AMF

## 5.1 Introduction

The table 5.1-1 shows the AMF Services and AMF Service Operations:

**Table 5.1-1 List of AMF Services**

| Service Name | Service Operations | Operation Semantics | Example Consumer(s) |
|---|---|---|---|
| Namf_Communication | UEContextTransfer | Request/Response | Peer AMF |
| | RegistrationStatusUpdate | Request/Response | Peer AMF |
| | CreateUEContext | Request/Response | Peer AMF |
| | ReleaseUEContext | Request/Response | Peer AMF |
| | RelocateUEContext | Request/Response | Peer AMF |
| | CancelRelocateUEContext | Request/Response | Peer AMF |
| | N1MessageNotify | Subscribe/Notify | Peer AMF, LMF, PCF |
| | N2InfoNotify | | LMF, AMF |
| | N1N2MessageSubscribe | | PCF |
| | N1N2MessageUnSubscribe | | PCF |
| | N1N2MessageTransfer | Request/Response | Peer AMF, SMF, SMSF, LMF, PCF |
| | N1N2TransferFailureNotification | Subscribe/Notify | SMF, SMSF, LMF, PCF |
| | NonUeN2MessageTransfer | Request/Response | Peer AMF, LMF, CBCF, PWS-IWF, TSCTSF |
| | NonUeN2InfoSubscribe | Subscribe/Notify | CBCF, PWS-IWF, TSCTSF |
| | NonUeN2InfoUnSubscribe | | CBCF, PWS-IWF, TSCTSF |
| | NonUeN2InfoNotify | | LMF, CBCF, PWS-IWF, TSCTSF |
| | EBIAssignment | Request/Response | SMF |
| | AMFStatusChangeSubscribe | Subscribe / Notify | SMF, PCF, NEF, SMSF, UDM |
| | AMFStatusChangeUnSubscribe | Subscribe / Notify | SMF, PCF, NEF, SMSF, UDM |
| | AMFStatusChangeNotify | Subscribe / Notify | SMF, PCF, NEF, SMSF, UDM |
| Namf_EventExposure | Subscribe (see NOTE) | Subscribe/Notify | NEF, SMF, UDM, NWDAF, LMF, GMLC, DCCF, TSCTSF |
| | Unsubscribe (see NOTE) | Subscribe/Notify | NEF, SMF, UDM, NWDAF, LMF, GMLC, DCCF, TSCTSF |
| | Notify | Subscribe/Notify | NEF, SMF, UDM, NWDAF, LMF, GMLC, DCCF, TSCTSF |
| Namf_MT | EnableUEReachability | Request/Response | SMSF, SMF |
| | ProvideDomainSelectionInfo | Request/Response | UDM |
| | EnableGroupReachability | Request/Response | SMF |
| | UEReachabilityInfoNotify | Subscribe/Notify | SMF |
| Namf_Location | ProvidePositioningInfo | Request/Response | GMLC |
| | EventNotify | Subscribe / Notify | GMLC |
| | ProvideLocationInfo | Request/Response | UDM |
| | CancelLocation | Request/Response | GMLC |
| Namf_MBSBroadcast | ContextCreate | Request/Response | MB-SMF |
| | ContextUpdate | Request/Response | MB-SMF |
| | ContextRelease | Request/Response | MB-SMF |
| | ContextStatusNotify | Subscribe / Notify | MB-SMF |
| Namf_MBSCommunication | N2MessageTransfer | Request/Response | MB-SMF |
| | Notify | Subscribe/Notify | MB-SMF |
| NOTE: A subscription applies for one UE, group of UE(s) or any UE. | | | |

Table 5.1-2 summarizes the corresponding APIs defined for this specification.

**Table 5.1-2: API Descriptions**

| Service Name | Clause | Description | OpenAPI Specification File | apiName | Annex |
|---|---|---|---|---|---|
| Namf_Communication | 6.1 | AMF Communication Service | TS29518_Namf_Communication.yaml | namf-comm | A.2 |
| Namf_EventExposure | 6.2 | AMF Event Exposure Service | TS29518_Namf_EventExposure.yaml | namf-evts | A.3 |
| Namf_MT | 6.3 | AMF Mobile Terminated Service | TS29518_Namf_MT.yaml | namf-mt | A.4 |
| Namf_Location | 6.4 | AMF Location Service | TS29518_Namf_Location.yaml | namf-loc | A.5 |
| Namf_MBSBroadcast | 6.5 | AMF MBS Broadcast Service | TS29518_Namf_MBSBroadcast.yaml | namf-mbs-bc | A.6 |
| Namf_MBSCommunication | 6.6 | AMF MBS Communication Service | TS29518_Namf_MBSCommunication.yaml | namf-mbs-com | A.7 |

# 5.2 Namf_Communication Service

## 5.2.1 Service Description

This service enables an NF to communicate with the UE through N1 NAS messages or with the AN (both UE and non UE specific). The service operations defined below allow the NF to communicate with the UE and the AN. The following are the key functionalities of this NF service.

- Provide service operations for transporting N1 messages to the UE;
- Allow NFs to subscribe and unsubscribe for notifications of specific N1 messages from the UE;
- Allow NFs to subscribe and unsubscribe for notifications about specific information from AN;
- Provide service operations for initiating N2 messages towards the AN;
- Security Context Management; and
- UE information management and transfer (including its security context).

## 5.2.2 Service Operations

### 5.2.2.1 Introduction

The Namf_Communication service supports following service operations:

- UEContextTransfer
- RegistrationStatusUpdate
- N1N2MessageTransfer (UE Specific)
- N1N2TransferFailureNotification (UE Specific)
- N1N2MessageSubscribe (UE Specific)

- N1N2MessageUnsubscribe (UE Specific)
- N1MessageNotify (UE Specific)
- N2InfoNotify (UE Specific)
- NonUeN2MessageTransfer
- NonUeN2InfoSubscribe
- NonUeN2InfoUnsubscribe
- NonUeN2InfoNotify
- EBIAssignment
- CreateUEContext
- ReleaseUEContext
- RelocateUEContext
- CancelRelocateUEContext
- AMFStatusChangeSubscribe
- AMFStatusChangeUnsubscribe
- AMFStatusChangeNotify

### 5.2.2.2 UE Context Operations

#### 5.2.2.2.1 UEContextTransfer

##### 5.2.2.2.1.1 General

The UEContextTransfer service operation is used during the following procedure:

- General Registration procedure (see TS 23.502 [3], clause 4.2.2.2.2)
- Registration with Onboarding SNPN (see TS 23.502 [3], clause 4.2.2.2.4)

The UEContextTransfer service operation is invoked by a NF Service Consumer, e.g. a target AMF, towards the AMF (acting as source AMF), when the target AMF receives a Registration Request with the UE's 5G-GUTI included and the serving AMF has changed since last registration, to retrieve the UE Context, e.g. the UE's SUPI and MM Context, in the source AMF.

The NF Service Consumer (e.g. the target AMF) shall retrieve the UE Context by invoking the "transfer" custom method on the URI of an "Individual ueContext" resource identified by UE's 5G-GUTI, see clause 6.1.3.2.4. See also Figure 5.2.2.2.1.1-1.

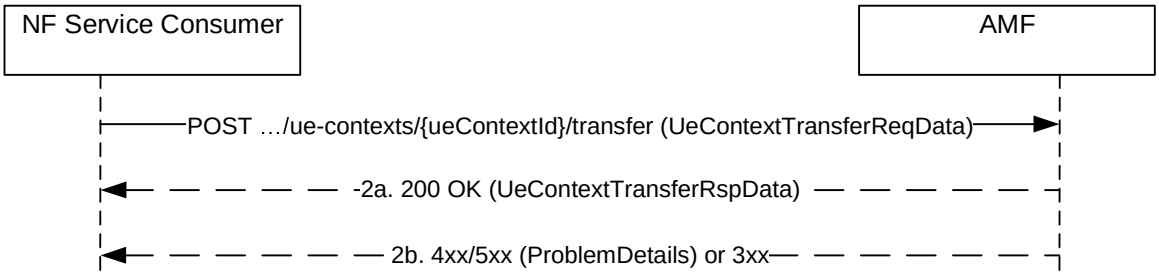



**Figure 5.2.2.2.1.1-1 UE Context Transfer**

1. The NF Service Consumer, e.g. target AMF, shall send a HTTP POST request to invoke "transfer" custom method on an "Individual ueContext" resource URI. The content of the request shall be an object of "UeContextTransferReqData" data type.

   If UE Context Transfer is triggered by UE initial registration, mobility registration, disaster roaming initial registration or disaster roaming mobility registration, the NF Service Consumer, e.g. target AMF, shall set the reason attribute to "INIT_REG" or "MOBI_REG" and include the integrity protected registration request message which triggers the UE context transfer in the content.

2a. On success:

- if the reason attribute is "INIT_REG" and integrity check is successful, the (source) AMF shall respond with the status code "200 OK". The content of the response shall be an object of "UeContextTransferRspData" data type, containing:

  case a) the representation of the requested UE Context as follows:

  - without PDU Session Contexts associated to the access type indicated in the request by the NF Service Consumer (e.g. target AMF); and
  - with PDU Session Contexts associated to the other access type, if the UE is registered for the other access type in the (source) AMF, unless the source AMF determines based on the PLMN ID or SNPN ID of the (target) AMF that there is no possibility for relocating the N2 interface for non-3GPP access to the (target) AMF;

  or

  case b) the representation of the requested UE Context only containing the "supi" attribute, if the UE is registered in a different access type in the (source) AMF and the source AMF determines based on the PLMN ID of the (target) AMF that there is no possibility for relocating the N2 interface to the (target) AMF.

- If the reason attribute is "MOBI_REG" and integrity check is successful, the (source) AMF shall respond with the status code "200 OK". The content of the response shall be an object of "UeContextTransferRspData" data type, containing:

  a) the representation of the complete UE Context including available MM and PDU Session Contexts. The source AMF shall transfer the complete UE context including both access types if the UE is registered for both 3GPP and non-3GPP accesses and if the target PLMN is the same as the source PLMN; or

  b) the representation of the requested UE Context including the available MM and PDU Session Contexts for the 3GPP access type, if the UE is registered for both 3GPP and non-3GPP accesses in the (source) AMF and the source AMF determines based on the PLMN ID or SNPN ID of the (target) AMF that there is no possibility for relocating the N2 interface for non-3GPP access to the (target) AMF.

NOTE: The source AMF can determine that it is not possible to relocate the N2 interface to the target AMF when both AMFs pertain to different PLMNs or SNPNs.

The UE context shall contain trace control and configuration parameters, if signalling based trace has been activated (see TS 32.422 [30]).

The NF Service Consumer, e.g. target AMF, starts tracing according to the received trace control and configuration parameters, if trace data is received in the UE context indicating that signalling based trace has been activated. Once the NF Service Consumer receives subscription data, trace requirements received from the UDM supersedes the trace requirements received from the AMF.

The UE context shall contain analytics subscription parameters, if the (source) AMF has created analytics subscription(s) towards NWDAF related to the UE (see clause 5.2.2.2.2 of TS 23.502 [3]) and both the source and the target AMFs support the "ASUC" feature. The NF Service Consumer, e.g. target AMF, may take over the analytics subscription(s).

The UE context shall contain event subscriptions information in the following cases:

a) Any NF Service Consumer has subscribed for UE specific event; and/or

b) Any NF Service Consumer has subscribed for UE group specific events to which the UE belongs. In this case the event subscriptions provided in the UE context shall contain the event details applicable to this specific UE in the group (e.g maxReports in options IE).

The NF Service Consumer, e.g. target AMF, shall:

- in case a) create event subscriptions for the UE specific events;
- in case b) create event subscriptions for the group Id if there are no existing event subscriptions for that group Id, subscription change notification URI (subsChangeNotifyUri) and the subscription change notification correlation Id (subsChangeNotifyCorrelationId). If there is already an existing event subscription for the group Id, and for the given subscription change notification URI (subsChangeNotifyUri) and subscription change notification correlation Id (subsChangeNotifyCorrelationId), then an event subscription shall not be created at the NF Service Consumer. The individual UE specific event details (e.g maxReports in options IE) within that group shall be taken into account.
- for both the cases, for each created event subscription, allocate a new subscription Id, if necessary (see clause 6.5.2 of TS 29.500 [4]), and if allocated, send the new subscription Id to the notification endpoint for informing the subscription Id creation, along with the notification correlation Id for the subscription Id change. If the UEContextTransfer service operation is performed towards the old AMF as part of the EPS to 5GS mobility registration procedure using N26 interface (see clause 4.11.1.3.3 of TS 23.502 [3]), the target AMF may also initiate event subscription synchronization procedure with UDM, as specified in clause 5.3.2.4.2, when both the target AMF and the UDM support the "ESSYNC" feature.

NOTE: Subscription Id can be reused if the mobility is between AMFs of same AMF Set.

If the UE context being transferred from the source AMF is the last UE context that belongs to a UE group Id related subscription, then the source AMF shall not delete the UE group Id related subscription until the expiry of that event subscription (see clause 5.3.2.2.2).

The target AMF may authorize the event subscriptions transferred from the source AMF as specified in clause 13.4.1.4 of TS 33.501 [27]. Based on local policy, the target AMF may consider that transferred subscriptions containing no or an invalid access token are not authorized. Transferred subscriptions that are not authorized by the target AMF shall not be regarded active; if the target AMF supports the STEN (Subscription Termination Event Notification) feature, and if the notification of event subscription termination was requested by the NF service consumer, the target AMF shall send a notification to the NF service consumer to report the termination of the subscription with the subscription termination cause "SUBSCRIPTION_NOT_AUTHORIZED".

The source AMF shall not transfer those PDU sessions which are not supported by the target AMF, e.g. the MA-PDU sessions shall not be transferred if the target AMF does not support ATSSS.

The UE context shall contain SNPN Onboarding indication, if the UE is registered for onboarding in an SNPN as described in clause 4.2.2.2.4 of TS 23.502 [3]. The NF Service Consumer, i.e. target AMF, may start an implementation specific timer to deregister the onboarding registered UE, i.e. if the received UE context contains SNPN Onboarding indication.

If there are ongoing Network Slice Deregistration Inactivity Timer(s) for the UE, the source AMF shall include the information (e.g. the expiry time) of the ongoing Network Slice Deregistration Inactivity Timer(s) in the UE context. The target AMF should resume the ongoing Network Slice Deregistration Inactivity Timer(s) if received for the S-NSSAI(s) that are allowed for the UE in the target AMF. The source AMF shall include the configured deregistration inactivity timer value(s) in the UE context within the "amPolicyInfoContainer" attribute, if received from PCF.

2b. On failure or redirection, one of the HTTP status code listed in Table 6.1.3.2.4.4.2-2 shall be returned. For a 4xx/5xx response, the message body shall contain a ProblemDetails structure with the "cause" attribute set to one of the application errors listed in Table 6.1.3.2.4.4.2-2.

#### 5.2.2.2.1.2 Retrieve UE Context after successful UE authentication

When a successful UE authentication has been performed after a previous integrity check failure, the NF service consumer (e.g. the target AMF) shall retrieve the UE context by invoking "transfer" service operation on the URI of the "Individual ueContext" resource identified by UE's SUPI. The same requirements in clause 5.2.2.2.1.1 shall be applied with following modifications:

1. Same as step 1 of figure 5.2.2.2.1.1-1, with following differences:
   - The {ueContextId} in the URI shall be composed using UE's SUPI, and
   - The "reason" attribute in request body shall be set to "MOBI_REG_UE_VALIDATED", and
   - The request body shall not include registration request message from UE.
2. Same as step 2a of figure 5.2.2.2.1.1-1, with following differences:
   - The (source) AMF shall skip integrity check and shall respond with the status code "200 OK "with the UE Context excluding SeafData and including available PDU Session Contexts

### 5.2.2.2.2 RegistrationStatusUpdate

#### 5.2.2.2.2.1 General

The RegistrationStatusUpdate service operation is used during the following procedure:

- General Registration procedure (see TS 23.502 [3], clause 4.2.2.2.2)
- Registration with AMF re-allocation procedure (see TS 23.502 [3], clause 4.2.2.2.3)

The RegistrationStatusUpdate service operation is invoked by a NF Service Consumer, e.g. the target AMF, towards the NF Service Producer, i.e. the source AMF, to update the status of UE registration at the target AMF, thereby indicating the result of previous UE Context transfer for a given UE (see clause 5.2.2.2.1.1).

The target AMF shall update the NF Service Producer (i.e. source AMF) with the status of the UE registration at the target AMF due to a previous UE Context transfer. The NF Service Consumer (e.g. target AMF) shall use the HTTP method POST to invoke the "transfer-update" custom operation on the URI of an "Individual ueContext" resource, see clause 6.1.3.2.4. See also Figure 5.2.2.2.2.1-1.

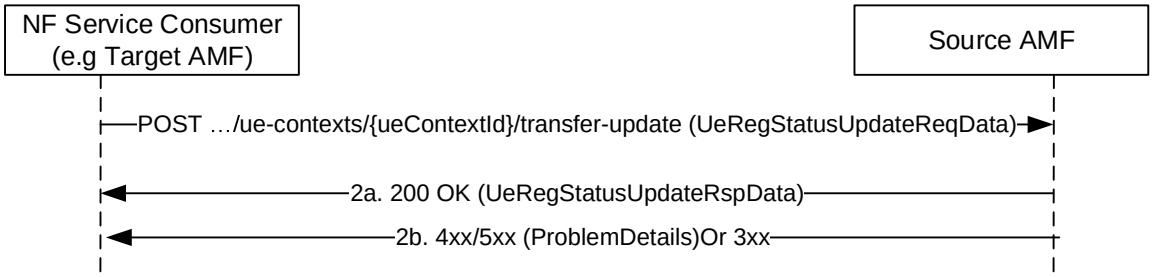



**Figure 5.2.2.2.2.1-1 Registration Status Update**

1. The NF service consumer (e.g. target AMF), shall send a POST request to invoke the "transfer-update" custom operation on the URI of an "Individual ueContext" resource, to update the source AMF with the status of the UE registration at the target AMF. The UE's 5G-GUTI is included as the UE identity.

   The request content shall include the transferStatus attribute set to "TRANSFERRED" if the UE context transfer was completed successfully (including the case where only the supi was transferred to the target AMF during the UE context transfer procedure) or to "NOT_TRANSFERRED" otherwise.

   If any network slice(s) become no longer available and there are PDU Session(s) associated with them, the target AMF shall include these PDU session(s) in the toReleaseSessionList attribute in the content. If the continuity of the PDU Session(s) cannot be supported between networks (e.g. SNPN-SNPN mobility, inter-PLMN mobility where no HR agreement exists), the target AMF shall include these PDU session(s) with release cause in the toReleaseSessionInfo attribute in the content.

   If the target AMF selects a new PCF for AM Policy and/or UE policy other than the one which was included in the UeContext by the old AMF, the target AMF shall set pcfReselectedInd to true.

NOTE: AMF selects the same PCF instance for AM policy and for UE policy, as described in clause 6.3.7.1, TS 23.501 [2].

The NF service consumer shall include the smfChangeInfoList attribute including the UE's PDU Session ID(s) for which the I-SMF or V-SMF has been changed or removed, if any, with for each such PDU session, the related smfChangeIndication attribute set to "CHANGED" or "REMOVED", if the I-SMF or V-SMF is changed or removed respectively.

If the target AMF receives analytics subscription parameters from the source AMF, and one or more analytics subscription(s) are not taken over by the target AMF, the target AMF shall include these analytics subscription(s) in the analyticsNotUsedList IE. The source AMF should unsubscribe the analytics subscriptions included in analyticsNotUsedList IE for the UE.

Once the update is received, the source AMF shall:

- remove the individual ueContext resource and release any PDU session(s) in the toReleaseSessionList attribute, if the transferStatus attribute included in the POST request body is set to "TRANSFERRED" and if the source AMF transferred the complete UE Context including all MM contexts and PDU Session Contexts. The source AMF may choose to start a timer to supervise the release of the UE context resource and may keep the individual ueContext resource until the timer expires. If the pcfReselectedInd is set to true, the source AMF shall terminate the AM Policy Association and/or the UE Policy Association that the source AMF has to the old PCF.

- keep the UE context only including the MM context and PDU session(s) associated to the non-3GPP access, if the transferStatus attribute included in the POST request body is set to " TRANSFERRED" and if the source AMF did not transfer the MM context and PDU Session Contexts for the non-3GPP access type; the AMF shall release any PDU session(s) in the toReleaseSessionList attribute. The source AMF may choose to start a timer and keep the MM context and PDU session(s) associated to the 3GPP access until the timer expires.

- keep the UE Context as if the context transfer procedure had not happened if the transferStatus attribute included in the POST request body is set to "NOT_TRANSFERRED".

2a. On Success: The source AMF shall respond with the status code "200 OK" if the request is accepted. If the smfChangeInfoList attribute was received in the request, the source AMF shall release the SM context at the I-SMF or V-SMF only, for all the PDU sessions listed in the smfChangeInfoList attribute with the smfChangeIndication attribute set to "CHANGED" or "REMOVED".

If some PDU sessions are not supported by the target AMF and thus not transferred to the target AMF, the source AMF shall release these PDU sessions after this step.

2b. On failure or redirection, one of the HTTP status code listed in Table 6.1.3.2.4.5.2-2 shall be returned. For a 4xx/5xx response, the message body shall contain a ProblemDetails structure with the "cause" attribute set to one of the application errors listed in Table 6.1.3.2.4.5.2-2, where applicable.

#### 5.2.2.2.3 CreateUEContext

##### 5.2.2.2.3.1 General

The CreateUEContext service operation is used during the following procedure:

- Inter NG-RAN node N2 based handover (see TS 23.502 [3], clause 4.9.1.3, and clause 4.23.7)

The CreateUEContext service operation is invoked by a NF Service Consumer, e.g. a source AMF, towards the AMF (acting as target AMF), when the source AMF can't serve the UE and selects the target AMF during the handover procedure, to create the UE Context in the target AMF.

The NF Service Consumer (e.g. the source AMF) shall create the UE Context by using the HTTP PUT method with the URI of the "Individual UeContext" resource (See clause 6.1.3.2.3.1). See also Figure 5.2.2.2.3.1-1.

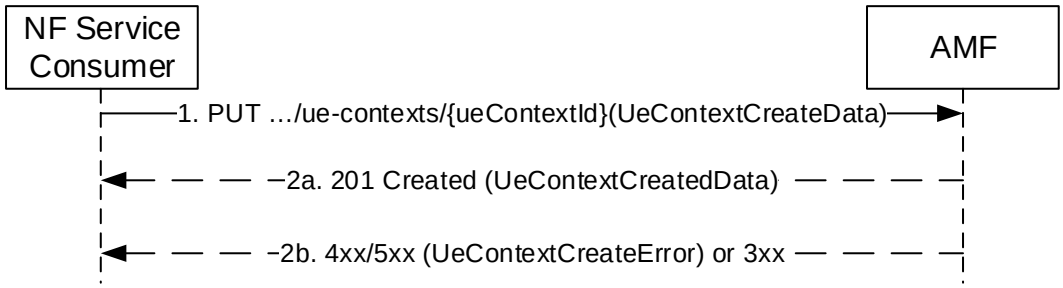



**Figure 5.2.2.2.3.1-1 Create UE Context**

1. The NF Service Consumer, e.g. source AMF, shall send a PUT request, to create the ueContext in the target AMF. The content of the PUT request shall contain a UeContextCreateData structure, including a N2 Information Notification callback URI.

   The UE context shall contain trace control and configuration parameters, if signalling based trace has been activated (see TS 32.422 [30]).

   The source AMF shall transfer the complete UE context including both access types if the UE is registered for both 3GPP and non-3GPP accesses and if the target PLMN is the same as the source PLMN.

   The source AMF shall transfer only UE context for 3GPP access if the source AMF determines there is no possibility for relocating the N2 interface for non-3GPP access to the (target) AMF, e.g. when the target AMF is in another PLMN.

   For a UE supporting 5G-SRVCC, the NF Service Consumer (i.e. AMF) shall include the Mobile Station Classmark 2, STN-SR, C-MSISDN and Supported Codec List in the request, if available, as specified in TS 23.502 [3].

   The UE context shall contain analytics subscription parameters, if the (source) AMF has created analytics subscription(s) towards NWDAF related to the UE (see clause 5.2.2.2.11 of TS 23.502 [3]) and both the source and the target AMFs support the "ASUC" feature. The NF service producer, e.g. target AMF, may take over the analytics subscription(s).

   The UE context shall contain SNPN Onboarding indication and the target AMF shall support SNPN Onboarding, if the UE is registered for onboarding in an SNPN as described in clause 4.2.2.2.4 of TS 23.502 [3].

   If there are ongoing Network Slice Deregistration Inactivity Timer(s) for the UE, the source AMF shall include the information (e.g. the expiry time) of the ongoing Network Slice Deregistration Inactivity Timer(s) in the UE context. The source AMF shall also include the configured deregistration inactivity timer value(s) in the UE context within the "amPolicyInfoContainer" attribute, if received from PCF.

2a. On success, the target AMF shall respond with the status code "201 Created" if the request is accepted, together with a HTTP Location header to provide the location of a newly created resource. The content of the PUT response shall contain the representation of the created UE Context. If the target AMF selects a new PCF for AM Policy other than the one which was included in the UeContext by the old AMF, the target AMF shall set pcfReselectedInd to true. If the pcfReselectedInd is set to true, the source AMF shall terminate the AM Policy Association to the old PCF.

   The target AMF starts tracing according to the received trace control and configuration parameters, if trace data is received in the UE context indicating that signalling based trace has been activated. Once the AMF receives subscription data, trace requirements received from the UDM supersedes the trace requirements received from the NF Service Consumer.

   If the target AMF receives analytics subscription parameters from the source AMF, and one or more analytics subscription(s) are not taken over by the target AMF, the target AMF shall include these analytics subscription(s) in the analyticsNotUsedList IE. The source AMF may unsubscribe the analytics subscriptions included in analyticsNotUsedList IE for the UE.

   The UE context shall contain event subscriptions information in the following cases:

   a) Any NF Service Consumer has subscribed for UE specific event; and/or

b) Any NF Service Consumer has subscribed for UE group specific events to which the UE belongs. In this case the event subscriptions provided in the UE context shall contain the event details applicable to this specific UE in the group (e.g maxReports in options IE).

The target AMF shall:

- in case a) create event subscriptions for the UE specific events;

- in case b) create event subscriptions for the group Id if there are no existing event subscriptions for that group Id, subscription change notification URI (subsChangeNotifyUri) and the subscription change notification correlation Id (subsChangeNotifyCorrelationId). If there is already an existing event subscription for the group Id and for the given subscription change notification URI (subsChangeNotifyUri) and subscription Id change notification correlation Id (subsChangeNotifyCorrelationId), then an event subscription shall not be created at the target AMF. The individual UE specific event details (e.g maxReports in options IE) within that group shall be taken into account.

- for both the cases, for each created event subscription, allocate a new subscription Id, if necessary (see clause 6.5.2 of TS 29.500 [4]), and if allocated send the new subscription Id to the notification endpoint for informing the subscription Id creation, along with the notification correlation Id for the subscription Id change.

NOTE: Subscription Id can be reused if the mobility is between AMFs of same AMF Set.

If the UE context being transferred from the NF service consumer (e.g. source AMF) is the last UE context that belongs to a UE group Id related subscription, then the NF service consumer (e.g. source AMF) shall not delete the UE group Id related subscription until the expiry of that event subscription (see clause 5.3.2.2.2).

The target AMF may authorize the event subscriptions transferred from the source AMF as specified in clause 13.4.1.4 of TS 33.501 [27]. Based on local policy, the target AMF may consider that transferred subscriptions containing no or an invalid access token are not authorized. Transferred subscriptions that are not authorized by the target AMF shall not be regarded active; if the target AMF supports the STEN (Subscription Termination Event Notification) feature, and if the notification of event subscription termination was requested by the NF service consumer, the target AMF shall send a notification to the NF service consumer to report the termination of the subscription with the subscription termination cause "SUBSCRIPTION_NOT_AUTHORIZED".

If the target AMF receives SNPN Onboarding indication from the source AMF, the target AMF may start an implementation specific timer to deregister the onboarding registered UE, i.e. if the received UE context contains SNPN Onboarding indication.

The source AMF shall release those PDU sessions not supported by the target AMF and thus not transferred to the target AMF.

If the source AMF includes the information of ongoing Network Slice Deregistration Inactivity Timer(s) in the UE context, the target AMF should resume the ongoing Network Slice Deregistration Inactivity Timer(s) if received for the S-NSSAI(s) that are allowed for the UE in the target AMF.

2b. On failure or redirection, one of the HTTP status code listed in Table 6.1.3.2.3.1-3 shall be returned. For a 4xx/5xx response, the message body shall contain a UeContextCreateError structure, including:

- a ProblemDetails structure with the "cause" attribute set to one of the application errors listed in Table 6.1.3.2.3.1-3. The cause in the error attribute shall be set to HANDOVER_FAILURE, if all of the PDU sessions are failed, e.g. no response from the SMF within a maximum wait timer;

- NgAPCause, if available;

- N2 information carrying the Target to Source Failure Transparent Container, if this information has been received from the target NG-RAN and if the source AMF supports the NPN feature.

#### 5.2.2.2.3.2 Create UE Context with AMF Relocation

During inter-PLMN N2 Handover, the initial AMF may relocate the UE context to a target AMF (e.g. due to slices cannot be served by initial AMF). This clause describes the procedure for this scenario.

The NF Service Consumer (e.g. the source AMF) shall create the UE Context by using the HTTP PUT method with the URI of the "Individual UeContext" resource (See clause 6.1.3.2.3.1). See also Figure 5.2.2.2.3.2-1.

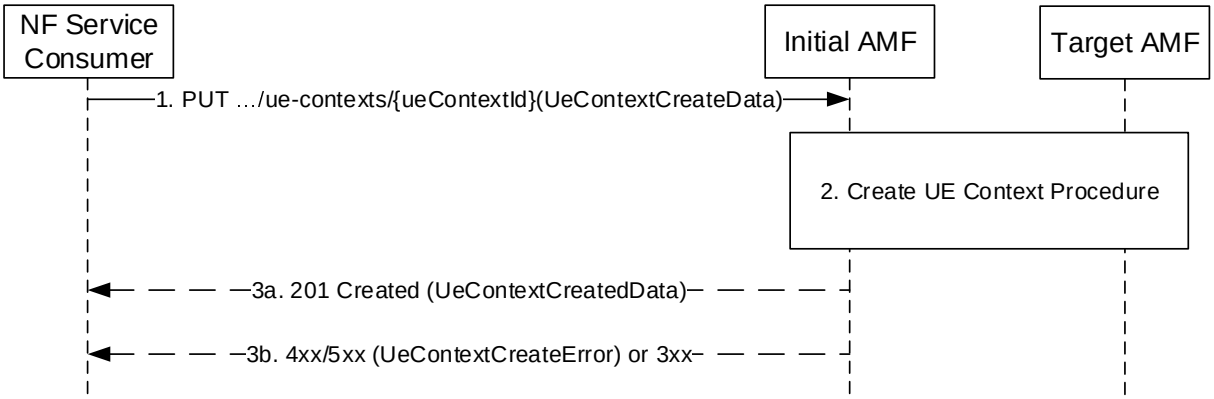



**Figure 5.2.2.2.3.2-1 Create UE Context with AMF Relocation**

Same requirement of clause 5.2.2.2.3.1 applies, with following modifications:

1. Same as step 1 of clause 5.2.2.2.3.1.
2. The initial AMF selects a target AMF and perform CreateUeContext procedure (see clause 5.2.2.2.3.1).
   - the request body shall include the information received from the source AMF in step 1, including the serving network, the supported features, etc.
   - if the received serving network (from the source AMF) is different from the PLMN of the target AMF, the resource URI in the Location header in 201 Create response shall contain the inter-PLMN API Root.

3a. Same as step 2a of clause 5.2.2.2.3.1, with following modifications:

- the request body shall contain the UE Context and other information received from the target AMF in step 2.
- the Location header shall contain the resource URI received in the "201 Created" response from target AMF in step 2.
- the initial AMF shall insert a 3gpp-Sbi-Producer-Id header indicating the target AMF.

3b. Same as step 2b of clause 5.2.2.2.3.1.

### 5.2.2.2.4 ReleaseUEContext

#### 5.2.2.2.4.1 General

The ReleaseUEContext service operation is used during the following procedure:

- Inter NG-RAN node N2 based handover, Cancel procedure (see TS 23.502 [3], clause 4.9.1.4)

The ReleaseUEContext service operation is invoked by a NF Service Consumer, e.g. a source AMF, towards the AMF (acting as target AMF), when the source AMF receives the Handover Cancel from the 5G-AN during the handover procedure, to release the UE Context in the target AMF.

The NF Service Consumer (e.g. the source AMF) shall release the UE Context by using the HTTP "release" custom operation with the URI of the "Individual UeContext" resource (See clause 6.1.3.2.4.2). See also Figure 5.2.2.2.4.1-1.

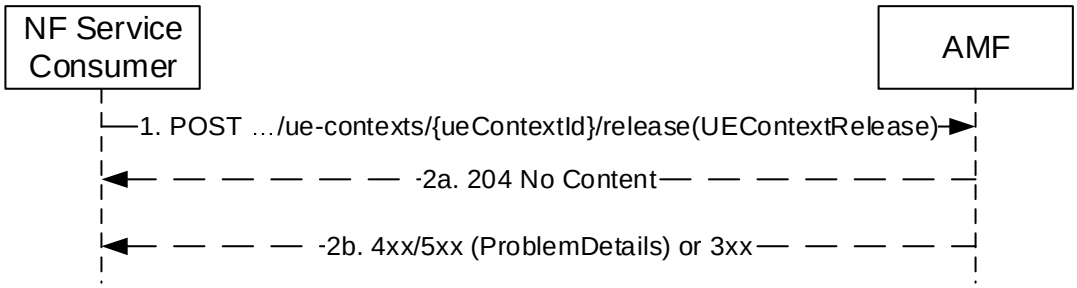



**Figure 5.2.2.2.4.1-1 Release UE Context**

1. The NF Service Consumer, e.g. source AMF, shall send a POST request, to release the ueContext in the target AMF. The content of the POST request shall contain the UEContextRelease.

2a. On success, the target AMF shall return "204 No Content" with an empty content in the POST response.

2b. On failure or redirection, one of the HTTP status code listed in Table 6.1.3.2.4.2.2-2 shall be returned. For a 4xx/5xx response, the message body shall contain a ProblemDetails structure with the "cause" attribute set to one of the application error listed in Table 6.1.3.2.4.2.2-2.

### 5.2.2.2.5 RelocateUEContext

#### 5.2.2.2.5.1 General

The RelocateUEContext service operation is used during the following procedure:

- EPS to 5GS handover using N26 interface with AMF re-allocation (see TS 23.502 [3], clause 4.11.1.2.2).

The RelocateUEContext service operation is invoked by a NF Service Consumer, e.g. an initial AMF, towards the AMF (acting as target AMF), during an EPS to 5GS handover with AMF re-allocation, to relocate the UE Context in the target AMF.

The NF Service Consumer (e.g. the initial AMF) shall relocate the UE Context in the target AMF by invoking the "relocate" custom method on the URI of an "Individual ueContext" resource (see clause 6.1.3.2.4). See also Figure 5.2.2.2.5.1-1.

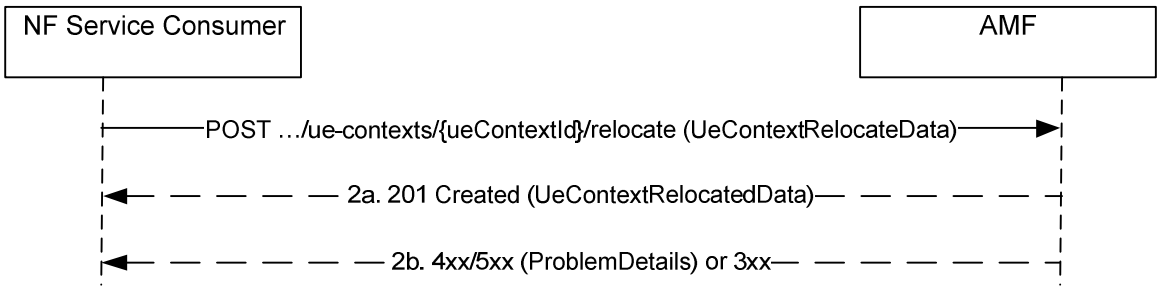



**Figure 5.2.2.2.5.1-1 Relocate UE Context**

1. The NF Service Consumer, e.g. initial AMF, shall send a POST request to relocate the UE context in the target AMF. The content of the POST request shall contain a UeContextRelocateData structure.

   The UE context shall contain trace control and configuration parameters, if signalling based trace has been activated (see TS 32.422 [30]).

   For an EPS to 5GS handover procedure, the NF Service Consumer shall carry per PDU session the S-NSSAI for serving PLMN, the MME Control Plane Address and the TEID in the request. If S-NSSAI for interworking is configured and used in initial AMF for the PDU session, the initial AMF shall also carry the configured S-NSSAI for interworking to the target AMF, as specified in clause 4.11.1.2.2 of TS 23.502 [3]. In Home Routed roaming case, the S-NSSAI for serving PLMN is derived by the initial AMF based on the S-NSSAI for home PLMN retrieved from SMF+PGW-C, as specified in TS 23.502 [3].

2a. On success, the target AMF shall respond with the status code "201 Created" if the request is accepted, together with a HTTP Location header to provide the location of the newly created resource. The content of the POST response shall contain the representation of the created UE Context.

The target AMF starts tracing according to the received trace control and configuration parameters, if trace data is received in the UE context indicating that signalling based trace has been activated. Once the AMF receives subscription data, trace requirements received from the UDM supersedes the trace requirements received from the NF Service Consumer.

2b. On failure to relocate the UE context or redirection, one of the HTTP status code listed in Table 6.1.3.2.4.6.2-2 shall be returned. For a 4xx/5xx response, the message body shall contain a ProblemDetails structure with the "cause" attribute set to one of the application errors listed in Table 6.1.3.2.4.6.2-2.

If the target RAN rejects the Handover Request, the target AMF shall send Forward Relocation Response message directly to the source MME over the N26 interface, carrying the appropriate cause value.

### 5.2.2.2.6 CancelRelocateUEContext

#### 5.2.2.2.6.1 General

The CancelRelocateUEContext service operation is used during the following procedure:

- EPS to 5GS Handover with AMF re-allocation, Handover Cancel procedure (see TS 23.502 [3], clause 4.11.1.2.3)

The CancelRelocateUEContext service operation is invoked by a NF Service Consumer (i.e. initial AMF), towards the AMF (acting as target AMF), when the initial AMF receives Forward Cancel Request from the source MME during EPS to 5GS Handover with AMF re-allocation procedure, to trigger the target AMF to release the UE Context.

The NF Service Consumer (i.e. the initial AMF) shall cancel the UE Context Relocation by using the HTTP "cancel-relocate" custom operation with the URI of the "Individual UeContext" resource (See clause 6.1.3.2.4.2). See also Figure 5.2.2.2.6.1-1.

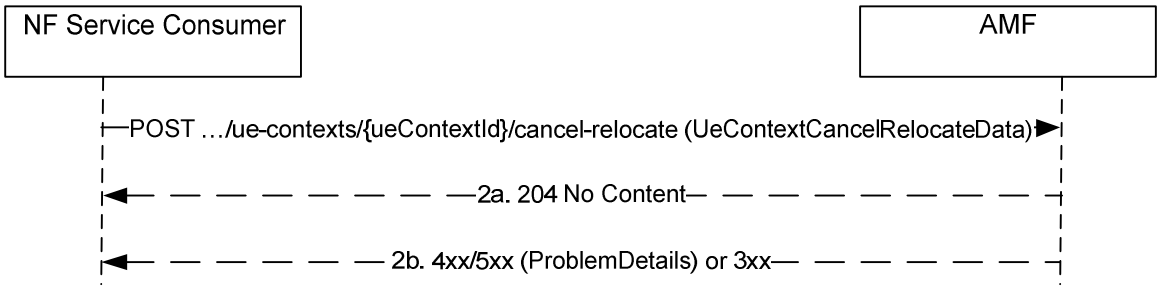



**Figure 5.2.2.2.6.1-1 Cancel Relocate UE Context**

1. The NF Service Consumer, i.e. initial AMF, shall send a POST request, to release the ueContext in the target AMF. The content of the POST request shall contain the UeContextCancelRelocateData that needs to be passed to the target AMF.

2a. On success, the target AMF shall return "204 No Content" with an empty content in the POST response.

2b. On failure or redirection, one of the HTTP status code listed in Table 6.1.3.2.4.7.2-2 shall be returned. For a 4xx/5xx response, the message body shall contain a ProblemDetails structure with the "cause" attribute set to one of the application error listed in Table 6.1.3.2.4.7.2-2.

### 5.2.2.3 UE Specific N1N2 Message Operations

#### 5.2.2.3.1 N1N2MessageTransfer

##### 5.2.2.3.1.1 General

The N1N2MessageTransfer service operation is used by a NF Service Consumer to transfer N1 and/or N2 information to the UE and/or 5G-AN through the AMF in the following procedures:

- Network triggered Service Request (see clause 4.2.3.3 of TS 23.502 [3])
- PDU Session establishment (see clause 4.3.2 of TS 23.502 [3])
- PDU Session modification (see clause 4.3.3 of TS 23.502 [3])
- PDU Session release (see clause 4.3.4 of TS 3GPP 23.502 [3])
- Session continuity, service continuity and UP path management (see clause 4.3.5 of TS 23.502 [3])
- Inter NG-RAN node N2 based handover (see clause 4.9.1.3 of TS 23.502 [3])
- SMS over NAS procedures (see clause 4.13.3 of TS 23.502 [3]
- UE assisted and UE based positioning procedure (see clause 6.11.1 of TS 23.273 [42])
- Network assisted positioning procedure (see clause 6.11.2 of TS 23.273 [42])
- LCS Event Report, Event Reporting in RRC INACTIVE state procedures, LCS Cancel Location and LCS Periodic-Triggered Invoke procedures (see clause 6.3, clause 6.7 and clause 6.20.3 of TS 23.273 [42])
- UE configuration update procedure for transparent UE policy delivery (see clause 4.2.4.3 of TS 23.502 [3])
- UPF anchored Mobile Terminated Data Transport in Control Plane CIoT 5GS Optimisation (see clause 4.24.2 of TS 23.502 [3])
- NEF Anchored Mobile Terminated Data Transport (see clause 4.25.5 of TS 23.502 [3])
- System interworking procedures with EPC (see clause 4.3 in TS 23.501 [2] and clause 4.11 in TS 23.502 [3])
- SMF triggered N3 data transfer establishment procedure (see clause 4.2.10.2 of TS 23.502 [3])
- 5G-RG requested PDU Session Establishment via W-5GAN (see clause 7.3.1 of TS 23.316 [48])
- 5G-RG or Network requested PDU Session Modification via W-5GAN (see clause 7.3.2 of TS 23.316 [48])
- 5G-RG or Network requested PDU Session Release via W-5GAN (see clause 7.3.3 of TS 23.316 [48])
- FN-RG related PDU Session Establishment via W-5GAN (see clause 7.3.4 of TS 23.316 [48])
- CN-initiated selective deactivation of UP connection of an existing PDU Session associated with W-5GAN Access (see clause 7.3.5 of TS 23.316 [48])
- FN-RG or Network Requested PDU Session Modification via W-5GAN (see clause 7.3.6 of TS 23.316 [48])
- FN-RG or Network Requested PDU Session Release via W-5GAN (see clause 7.3.7 of TS 23.316 [48])
- Non-5G capable device behind 5G-CRG and FN-CRG requested PDU Session Establishment via W-5GAN (see clause 4.10a of TS 23.316 [48])
- Non-5G capable device behind 5G-CRG and FN-CRG or Network Requested PDU Session Modification via W-5GAN (see clause 4.10a of TS 23.316 [48])
- Non-5G capable device behind 5G-CRG and FN-CRG or Network Requested PDU Session Release via W-5GAN (see clause 4.10a of TS 23.316 [48])
- Handover procedures between 3GPP access / 5GC and W-5GAN access (see clause 7.6.3 of TS 23.316 [48])

- Handover from 3GPP access / EPS to W-5GAN / 5GC (see clause 7.6.4.1 of TS 23.316 [48])
- Transfer UAV specific data via N1 SM (see clause 5.2.4.3 of TS 23.256 [56])
- MBS join and Session establishment procedure (see clause 7.2.1.3 of TS 23.247 [55])
- MBS activation procedure (see clause 7.2.5.2 of TS 23.247 [55])
- Multicast session leave requested by the network or MBS session release (see clause 7.2.2.3 of TS 23.247 [55])
- Procedures applicable to a PRU (see clause 6.17 of TS 23.273 [42])
- Procedures of SL-MT-LR involving LMF (see TS 23.273 [42], clause 6.20.3)
- Procedures of SL-MT-LR for periodic, triggered Location Events (see TS 23.273 [42], clause 6.20.4)
- 5GC-MT-LR Procedure using SL positioning (see TS 23.273 [42], clause 6.20.5)
- Procedures of user plane connection between UE and LMF for location service (see clause 6.18 of TS 23.273 [42] and TS 24.572 [61])

NOTE: Though in TS 23.502 [3] the procedure is called "UE configuration update procedure for transparent UE policy delivery", as per TS 24.501 [11] clause 5.4.5.3.1, the network initiated NAS transport procedure is used.

The NF Service Consumer shall invoke the service operation by using HTTP method POST, to request the AMF to transfer N1 and/or N2 information for a UE and/or 5G-AN, with the URI of "N1 N2 Messages Collection" resource (see clause 6.1.3.5.3.1).

The NF Service Consumer may include the following information in the HTTP Request message body:

- SUPI
- PDU Session ID or LCS Correlation ID depending on the N1/N2 message class to be transferred
- N2 SM Information (PDU Session ID, QoS profile, CN N3 Tunnel Info, S-NSSAI)
- N1 Message Container, including a N1 SM, LPP message, LCS message, SMS, UPDP message, PRU message, UPP-CM message
- N2 Information Container, including N2 SM, NRPPa message, PWS or RAN related information
- Mobile Terminated Data (i.e. CIoT user data container)
- Allocation and Retention Priority (ARP)
- Paging Policy Indication
- 5QI
- Notification URL (used for receiving Paging Failure Indication)
- Last Message Indication
- NF Instance Identifier and optionally Service Instance Identifier of the NF Service Consumer (e.g. an LMF or SMF)
- N1 SM Skipping Indication
- Area of Validity for N2 SM Information
- A MA PDU Session Accepted indication, if a MA-PDU session is established;
- Extended Buffering Support Indication, if SMF determines that Extended Buffering applies during Network triggered Service Request Procedure (see clause 4.2.3.3 of TS 23.502 [3]), UPF anchored Mobile Terminated Data Transport in Control Plane CIoT 5GS Optimisation procedure (see clause 4.24.2 of TS 23.502 [3]) or NEF Anchored Mobile Terminated Data Transport (see clause 4.25.5 of TS 23.502 [3]);

- Target Access type towards which the SMF requests to send N2 information and optionally N1 information, for a Multi-Access (MA) PDU session, or through which the LMF requests to transfer an LPP message to the UE.

During an intra-AMF handover between 3GPP and non-3GPP accesses, the SMF shall include the targetAccess IE set to the old access type in the HTTP Request message body, when releasing the N2 PDU session resources in the old access (see step 3 of Figure 4.9.2.1-1 and step 3 of Figure 4.9.2.2-1 of TS 23.502 [3]).

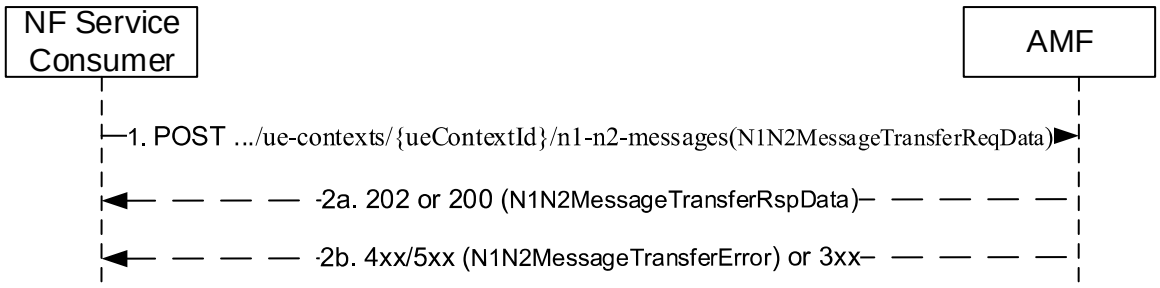



**Figure 5.2.2.3.1.1-1 N1N2MessageTransfer for UE related signalling**

1. The NF Service Consumer shall send a POST request to transfer N1 and N2 information. The NF Service Consumer may include a N1N2MessageTransfer Notification URI to AMF in the request message.

2a. On success, i.e. if the request is accepted and the AMF is able to transfer the N1/N2 message to the UE and/or the AN, the AMF shall respond with a "200 OK" status code. The AMF shall set the cause IE in the N1N2MessageTransferRspData as "N1_N2_TRANSFER_INITIATED" in this case.

The AMF shall respond with a "200 OK" status code and set the cause IE in the N1N2MessageTransferRspData to "N2_MSG_NOT_TRANSFERRED", if the N1N2MessageTransfer request included an area of validity for the N2 SM Information, the UE is in CM-CONNECTED state and outside of the area of validity.

2b. On failure or redirection, one of the HTTP status code listed in Table 6.1.3.5.3.1-2 shall be returned. For a 4xx/5xx response, the message body shall contain a N1N2MessageTransferError structure, including:

- a ProblemDetails structure with the "cause" attribute set to one of the application error listed in Table 6.1.3.5.3.1-2;

#### 5.2.2.3.1.2 Detailed behaviour of the AMF

When an NF service consumer is requesting to send N1 and/or N2 information and the UE is in CM-IDLE state for the access type for which the N1 and/or N2 information is related (called "associated access type" hereafter in this clause), the requirements specified in clause 5.2.2.3.1.1 shall apply with the following modifications:

NOTE: N1 and/or N2 Session Management information is related to the access type of the targeted PDU session for a single access PDU session, or to the Target Access received in the request for a MA PDU session; LCS related N2 (NRPPa) information is related to 3GPP access in this release of specification.

4xx and 5xx response cases shall also apply to UEs in CM-CONNECTED state, when applicable.

**2xx Response Cases:**

**Case A: When UE is CM-IDLE in 3GPP access and the associated access type is 3GPP access:**

a) Same as step 2a of Figure 5.2.2.3.1.1-1, the AMF should respond with the status code "200 OK", if "skipInd" attribute is set to "true" in the request body, with a response body that carries the cause "N1_MSG_NOT_TRANSFERRED".

b) Same as step 2a of Figure 5.2.2.3.1.1-1, the AMF shall respond with the status code "202 Accepted", if the asynchronous type communication is invoked and hence the UE is not paged, update the UE context and store N1 and/or N2 information and initiate communication with the UE and/or 5G-AN when the UE becomes reachable. In this case the AMF shall provide the URI of the resource in the AMF in the "Location" header of the response. In this release, the URI shall only be used by NF Service consumer to correlate the possible N1/N2 Message Transfer Failure Notification with the related N1/N2 Message Transfer Operation. The NF service consumer shall not send any service requests towards the URI received in the Location header.

The AMF shall also provide a response body containing the cause, "WAITING_FOR_ASYNCHRONOUS_TRANSFER" that represents the current status of the N1/N2 message transfer;

c) Same as step 2a of Figure 5.2.2.3.1.1-1, the AMF shall respond with the status code "202 Accepted", if paging is issued when the UE is in CM-IDLE and reachable for 3GPP access, with a response body that carries a cause "ATTEMPTING_TO_REACH_UE" as specified in clause 4.2.3.3 and 5.2.2.2.7 of TS 23.502 [3].

**Case B: When UE is CM-IDLE in Non-3GPP access but CM-CONNECTED in 3GPP access and the associated access type is Non-3GPP access:**

a) Same as step 2a of Figure 5.2.2.3.1.1-1, the AMF shall respond with the status code "200 OK" with cause "N1_N2_TRANSFER_INITIATED" and initiate N1 NAS SM message transfer via 3GPP access, if the NF service consumer (i.e. SMF) requests to send only N1 NAS SM message without any associated N2 SM information, and the current access type related to the PDU session is Non-3GPP access and the UE is CM-CONNECTED in 3GPP access.

b) Same as step 2a of Figure 5.2.2.3.1.1-1, the AMF shall respond with the status code "202 Accepted", if NAS Notification procedure is issued when the UE is in CM-CONNECTED in 3GPP access, with a response body that carries a cause "ATTEMPTING_TO_REACH_UE" as specified in step 4c of clause 4.2.3.3 and 5.2.2.2.7 of TS 23.502 [3].

**Case C: When UE is CM-IDLE in both Non-3GPP access and 3GPP access and the associated access type is Non-3GPP access:**

All the bullets specified in Case A are applicable.

The NF Service Consumer shall not send any further signalling for the UE if it receives a POST response body with a cause "ATTEMPTING_TO_REACH_UE" unless it has higher priority signalling. In such a case the response shall include the "Location" header containing the URI of the resource created in the AMF, e.g. ".../n1-n2-messages/{n1N2MessageId}". In this release, the URI shall only be used by NF Service consumer to correlate the possible N1/N2 Message Transfer Failure Notification with the related N1/N2 Message Transfer Operation. The NF service consumer shall not send any service requests towards the URI received in the Location header. The AMF shall:

- store the N1 and/or N2 information related to 3GPP access and, when the UE responds with a Service Request, shall initiate communication with the UE and/or 5G-AN using the stored N1 and/or N2 information;

- store the N1 NAS SM information related to Non-3GPP access if no N2 information was received and the AMF initiated paging towards the UE. Later when the UE responds with a Service Request, the AMF shall initiate communication with the UE using the stored N1 information via 3GPP access;

- inform the SMF which invoked the service operation, that the access type of the PDU Session can be changed from Non-3GPP access to 3GPP access as specified in clause 5.2.2.3.2.1 of TS 29.502 [16], when the UE responds with a "List Of Allowed PDU Sessions" and the indicated non-3GPP PDU session of the N2 (and N1 if received) information is included in the list; or

- notify the NF which invoked the service operation, as specified in clause 5.2.2.3.2, if the Notification URI is provided, when the AMF determines that the paging or NAS Notification has failed or when the UE responds with a "List Of Allowed PDU Sessions" and the indicated Non-3GPP PDU session of the N2 (and N1 if received) information is not included in the list.

**4xx Response Cases:**

- Same as step 2b of Figure 5.2.2.3.1.1-1, the AMF shall respond with status code "409 Conflict" in the following cases:

  - if the UE is in 3GPP access and there is already an ongoing paging procedure with higher or same priority, the AMF shall set the application error as "HIGHER_PRIORITY_REQUEST_ONGOING" in the "cause" attribute of the ProblemDetails structure of the POST response body. The AMF may provide a retryAfter IE to the NF Service Consumer in order for the NF Service Consumer to retry the request after the expiry of the timer. When the retryAfter IE is provided, the NF Service Consumer shall not initiate the downlink messaging until the timer expires. The AMF may also provide the ARP value of the QoS flow that has triggered the currently ongoing highest priority paging, so that the NF Service Consumer (e.g. SMF) knows that if any subsequent trigger initiating downlink messaging for a QoS flow with the same or lower priority happens.

- if there is an ongoing registration procedure (see clause 4.2.3.3 of TS 23.502 [3]) the AMF shall set the application error as "TEMPORARY_REJECT_REGISTRATION_ONGOING" in the "cause" attribute of the ProblemDetails structure in the POST response body; The AMF may provide a retryAfter IE to the NF Service Consumer in order for the NF Service Consumer to retry the request after a short period. When the retryAfter IE is provided, the NF Service Consumer should not initiate new N1/N2 Message Transfer request until the timer expires.

- if this is a request to transfer a N2 PDU Session Resource Modify Request or a N2 PDU Session Resource Release Command to a 5G-AN and if the UE is in CM-IDLE state at the AMF for the Access Network Type associated to the PDU session (see clauses 4.3.3 and 4.3.4 of TS 23.502 [3] and clause 5.3.2.1 of TS 23.527 [33]), the AMF shall set the application error "UE_IN_CM_IDLE_STATE" in the "cause" attribute of the ProblemDetails structure in the POST response body.

- if there is an ongoing Xn or N2 handover procedure (see clause 4.9.1.2.1 and 4.9.1.3.1 of TS 23.502 [3]) the AMF shall set the application error as "TEMPORARY_REJECT_HANDOVER_ONGOING" in the "cause" attribute of the ProblemDetails structure in the POST response body, if the AMF rejects the request due to the on-going handover.

- if there is an ongoing Service Request procedure (see clause 4.2.3.2 of TS 23.502 [3]), the AMF may reject a N1/N2 Message Transfer Request including a PDU Session Resource Setup Request Transfer IE due to the on-going Service Request procedure for the same PDU session, and if it does so, the AMF shall set the application error as "TEMPORARY_REJECT_SR_ONGOING" in the "cause" attribute of the ProblemDetails structure in the POST response body.

NOTE 1: In the scenario above, the AMF sends to the NG-RAN the PDU Session Resource Setup Request Transfer IE that is received in the Update SM Context Response (200 OK) for the on-going (UE triggered) Service Request.

- if the RAT Type is NB-IoT, and the UE already has 2 PDU Sessions with active user plane resources, the AMF shall set the application error as "MAX_ACTIVE_SESSIONS_EXCEEDED" in POST response body.

- if Paging Restrictions information restricts the N1N2MessageTransfer request from causing paging (see clause 4.2.3.3 of TS 23.502 [3]) the AMF shall set the application error as "REJECTION_DUE_TO_PAGING_RESTRICTION" in the "cause" attribute of the ProblemDetails structure in the POST response body.

- if the AMF determines that LCS-UPP connection already exists between the UE and one LMF, the AMF shall reject the request to transfer "UPP-CM" N1 message to the UE from another LMF (other than the target LMF during Modification of User Plane Connection between UE and LMF procedure, see clause 6.18 of TS 23.273 [42]) and set the application error as "MAX_LCS_UPP_CONN_REACHED" in POST response body.

NOTE 2: In this release, only one LCS-UPP connection is allowed for an UE. Attempts to set up additional LCS-UPP connections for the UE are not allowed.

- Same as step 2b of Figure 5.2.2.3.1.1-1, the AMF shall respond with the status code "403 Forbidden", if the UE is in a Non-Allowed Area and the service request is not for regulatory prioritized service. The AMF shall set the application error as "UE_IN_NON_ALLOWED_AREA" in POST response body.

  - The NF service consumer (i.e. the SMF) that receives this application error may supress subsequent message (e.g. N1N2MessageTransfer) to the AMF for non regulatory prioritized service. In this case, the NF service consumer (i.e. the SMF) should subscribe the Reachability-Report event for "UE Reachability Status Change" from the AMF, so as to get notified by the AMF when the UE becomes reachable again.

- Same as step 2b of Figure 5.2.2.3.1.1-1, the AMF shall respond with the status code "403 Forbidden ", if the NF service consumer (e.g. an LMF) is requesting to send N1 LPP message to the UE and the UE has indicated that it does not support LPP in N1 mode during registration procedure (see clause 5.5.1.2.2 and 5.5.1.3.2 of TS 24.501 [11]). The AMF shall set the application error to "UE_WITHOUT_N1_LPP_SUPPORT" in POST response body.

- Same as step 2b of Figure 5.2.2.3.1.1-1, the AMF shall respond with the status code "403 Forbidden", if the request body includes an nfId IE indicating an SMF instance which is different from the stored SMF instance

hosting the SM Context of the PDU session. The AMF shall set the application error to "INVALID_SM_CONTEXT" in POST response body. During procedures with SM Context relocation, e.g. UE mobility procedures with I-SMF insertion/change/removal, the AMF shall allow N1N2MessageTransfer from both SMF instances holding the old and new SM Contexts.

The NF service consumer (i.e. the SMF) that receives this application error shall remove the SM Context for the PDU session and release the PDU session resource in (H-)SMF if available. The SMF shall not send a SMContextStatusNotification to the AMF for the PDU session release.

- Same as step 2b of Figure 5.2.2.3.1.1-1, the AMF shall respond with the status code "403 Forbidden ", if the NF service consumer (e.g. an LMF) is requesting to initiate a positioning procedure towards a PRU (see clause 6.11.2 of TS 23.273 [42]), i.e. the pruInd IE with the value true was included in the request, but the UE is not a valid PRU. The AMF shall set the application error to "INVALID_PRU" in POST response body.

**5xx Response Cases:**

- Same as step 2b of Figure 5.2.2.3.1.1-1, the AMF shall respond with the status code "504 Gateway Timeout", if the UE is currently unreachable (e.g., due to the UE in MICO mode, the UE using extended idle mode DRX or the UE is only registered over Non-3GPP access and its state is CM-IDLE). The AMF shall set the application error as "UE_NOT_REACHABLE" in POST response body. If Extended Buffering Support Indication is received in the request, the AMF shall include the Estimated Maximum Waiting time in the response body when the message is rejected due to the UE in MICO mode or the UE using extended idle mode DRX. If the UE is using extended idle mode DRX, after the AMF responded with the status code "504 Gateway Timeout", when the AMF determines that the UE is reachable, then the AMF shall page the UE.

- step 2b of Figure 5.2.2.3.1.1-1, the AMF may respond with the status code "504 Gateway Timeout", if the UE is temporarily not responding (e.g., not responding to the paging). The AMF shall set the application error as "UE_NOT_RESPONDING" in POST response body. The AMF may provide a retryAfter IE to the NF Service Consumer in order for the NF Service Consumer to throttle sending further N1/N2 Message Transfer request for a short period. When the retryAfter IE is provided, the NF Service Consumer should not initiate new N1/N2 Message Transfer request until the timer expires.

### 5.2.2.3.2 N1N2Transfer Failure Notification

The AMF uses this notification to inform the NF service consumer that initiated an earlier

Namf_Communication_N1N2MessageTransfer, that the AMF failed to deliver the N1 and/or N2 message. The HTTP POST method shall be used on the notification callback URI provided by the NF service consumer as specified in clause 5.2.2.3.1.2.

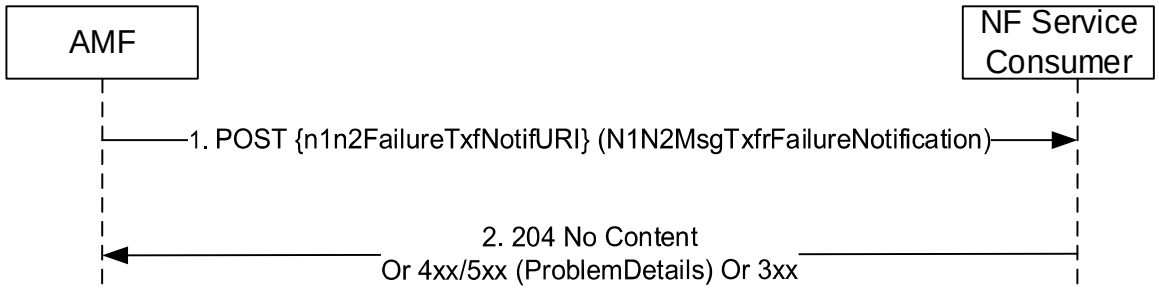



**Figure 5.2.2.3.2-1 N1N2Transfer Failure Notification for UE related signalling**

1. If the NF service consumer had provided a notification URI (see clause 5.2.2.3.1.2), the AMF shall send a POST request to the NF Service Consumer on that Notification URI when the AMF determines that:

   - the paging or NAS Notification has failed;

   - the indicated non-3GPP PDU session is not allowed to move to 3GPP access;

   - the UE has rejected the page as defined in TS 23.501 [2] clause 5.38.4;

   - the delivery of the N1 message fails, e.g. in case the UE is in RRC Inactive and NG-RAN paging was not successful or in case an Xn or N2 handover is being triggered at the NG-RAN

The AMF shall include the N1N2MessageTransfer request resource URI returned earlier in the N1N2MessageTransfer response if any (see clause 5.2.2.3.1.2), otherwise a dummy URI (see clause 6.1.6.2.30), in the POST request body. The AMF shall also include a N1/N2 message transfer cause information in the POST request body and set the value as specified in clause 6.1.5.6.3.1.

The NF Service Consumer shall delete any stored representation of the N1N2MessageTransfer request resource URI upon receiving this notification.

The AMF may also include a "retryAfter" IE in the POST request body in order for the NF consumer to throttle sending further N1/N2 Message Transfer request for a short period, e.g. to reduce unnecessary paging to an unresponding UE for a period of time to save the RAN resources.

2. The NF Service Consumer shall send a response with "204 No Content" status code.

   On failure or redirection, one of the HTTP status codes together with the response body listed Table 6.1.5.6.3.1-2 shall be returned.

### 5.2.2.3.3 N1N2MessageSubscribe

#### 5.2.2.3.3.1 General

The N1N2MessageSubscribe service operation is used by a NF Service Consumer (e.g. PCF) to subscribe to the AMF for notifying N1 messages of a specific type (e.g. UPDP) or N2 information of a specific type. For the N1 message class UPDP, a PCF shall subscribe for the N1 message notification with the AMF to receive the N1 messages from UE that are related to UE Policy.

NOTE: Step 0 of clause 4.2.4.3 of TS 23.502 [3] specifies that the PCF can split the UPDP transfer towards UE into multiple units. One UE specific callback URI is registered with the AMF by the PCF for the AMF to notify all UPDP message responses from the UE to the same callback URI. As a result, an explicit subscription per UE policy association is defined in stage 3 for this purpose.

An NF Service Consumer (e.g. PCF) may subscribe to notifications of specific N1 message type (e.g. LPP or UPDP) or N2 information type. In this case the NF Service Consumer shall subscribe by using the HTTP POST method with the URI of the "N1N2 Subscriptions Collection for Individual UE Contexts" resource (See clause 6.1.3.3). See also Figure 5.2.2.3.3.1-1.

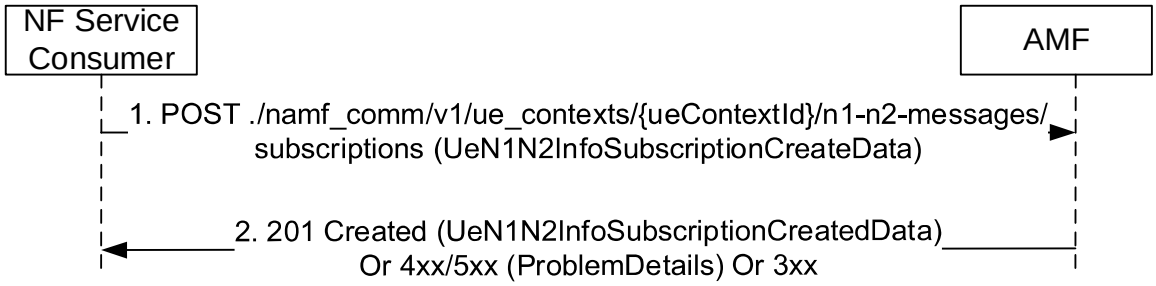



**Figure 5.2.2.3.3.1-1 N1N2 Message Subscribe**

1. The NF Service Consumer shall send a POST request to create a subscription resource in the AMF for a UE specific N1/N2 message notification. The content of the POST request shall contain:

   - N1 and/or N2 Message Type, identifying the type of N1 and/or N2 message to be notified
   - A callback URI for the notification

2. If the request is accepted, the AMF shall include a HTTP Location header to provide the location of a newly created resource (subscription) together with the status code 201 indicating the requested resource is created in the response message.

   On failure or redirection, one of the HTTP status codes together with the response body listed Table 6.1.3.3.3.1-3 shall be returned.

#### 5.2.2.3.4 N1N2MessageUnSubscribe

##### 5.2.2.3.4.1 General

The N1N2MessageUnSubscribe service operation is used by a NF Service Consumer (e.g. PCF) to unsubscribe to the AMF to stop notifying N1 messages of a specific type (e.g. LPP or UPDP).

The NF Service Consumer shall use the HTTP method DELETE with the URI of the "N1N2 Individual Subscription" resource (See clause 6.1.3.7.3.1), to request the deletion of the subscription for the N1 / N2 message towards the AMF. See also Figure 5.2.2.3.4.1-1.

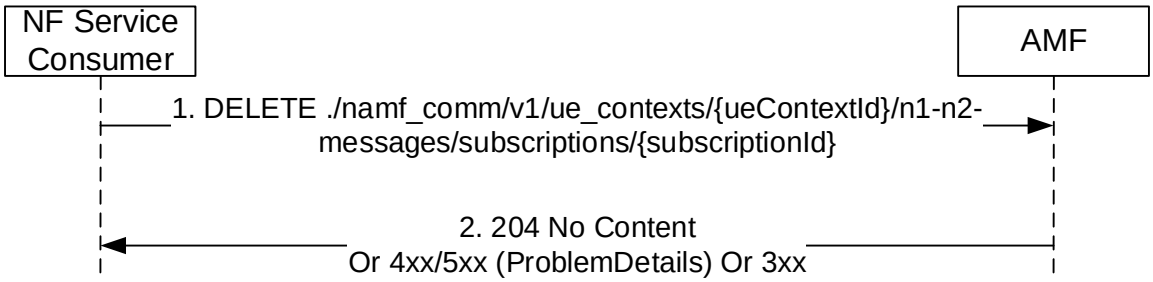



**Figure 5.2.2.3.4.1-1 N1N2 Message UnSubscribe**

1. The NF Service Consumer shall send a DELETE request to delete an existing subscription resource in the AMF.
2. If the request is accepted, the AMF shall reply with the status code 204 indicating the resource identified by subscription ID is successfully deleted, in the response message.

   On failure or redirection, one of the HTTP status codes together with the response body listed Table 6.1.3.4.3.1-3 shall be returned.

#### 5.2.2.3.5 N1MessageNotify

##### 5.2.2.3.5.1 General

The N1MessageNotify service operation is used by an AMF notifying the N1 message received from the UE to a destination CN NF, and it is used in the following procedures:

- Registration with AMF re-allocation (see clause 4.2.2.2.3 of TS 23.502 [3])
- UE assisted and UE based positioning procedure (see clause 6.11.1 of TS 23.273 [42])
- LCS Event Report, Event Reporting in RRC INACTIVE state procedures, LCS Cancel Location and LCS Periodic-Triggered Invoke procedures (see clause 6.3, clause 6.7 and clause 6.20.3 of TS 23.273 [42])
- UE configuration update procedure for transparent UE policy delivery (See clause 4.2.4.3 in TS 23.502 [3])

NOTE: Though in TS 23.502 [3] the procedure is called "UE configuration update procedure for transparent UE policy delivery", as per TS 24.501 [11] clause 5.4.5.2.1, the UE initiated NAS transport procedure is used.

- UE triggered policy provisioning procedure to request UE policies. (See clause 6.2.4 in TS 23.287 [47] and clause 6.2.4 in TS 23.304 [51])
- Procedures applicable to a PRU (see clause 6.17 of TS 23.273 [42])
- Procedures of SL-MT-LR involving LMF (see TS 23.273 [42], clause 6.20.3)
- Procedures of SL-MT-LR for periodic, triggered Location Events (see TS 23.273 [42], clause 6.20.4)
- 5GC-MT-LR Procedure using SL positioning (see TS 23.273 [42], clause 6.20.5)
- Procedures of user plane connection between UE and LMF for location service (see clause 6.18 of TS 23.273 [42] and TS 24.572 [61])

The AMF shall use HTTP POST method to the N1 Notification URI provided by the NF Service Consumer via N1N2MessageSubscribe service operation (See clause 5.2.2.3.3). See also figure 5.2.2.3.5.1-1.

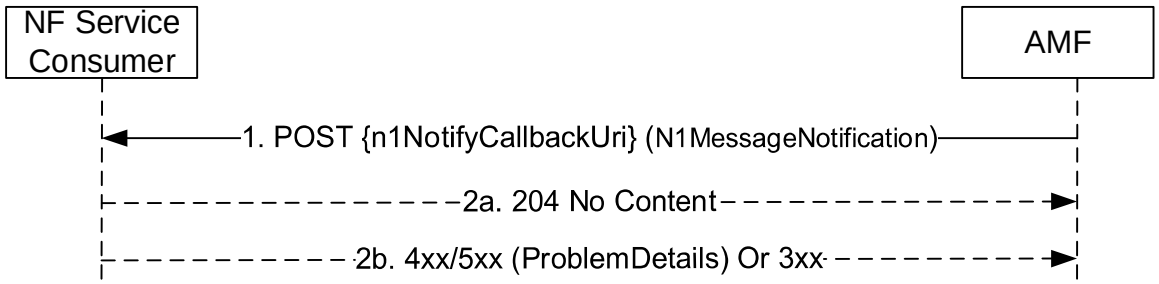



**Figure 5.2.2.3.5.1-1 N1 Message Notify**

1. The AMF shall send a HTTP POST request to the N1 Notification URI, and the content of the POST request shall contain an N1MessageNotification data structure with the subscribed N1 message.

2a. On success, "204 No Content" shall be returned and the content of the POST response shall be empty.

2b. On failure or redirection, one of the HTTP status code listed in Table 6.1.5.4.3.1-3 shall be returned. The message body shall contain a ProblemDetails object with "cause" set to one of the corresponding application errors listed in Table 6.1.5.4.3.1-3.

#### 5.2.2.3.5.2 Using N1MessageNotify in the Registration with AMF Re-allocation Procedure

In the Registration with AMF re-allocation procedure, the N1MessageNotify service operation is invoked by a NF Service Producer, i.e. an Initial AMF, towards a NF Service Consumer, e.g. the target AMF, which is selected to serve the UE, by the initial AMF.

The requirements specified in clause 5.2.2.3.5.1 shall apply with the following modifications:

1. The initial AMF discovers the NF Service Consumer (e.g. the target AMF) from the NRF, and fetch N1 Notification URI from the default notification subscription registered with "N1_MESSAGE" notification type and "5GMM" N1 message class (See Table 6.2.6.2.3-1 and Table 6.2.6.2.4-1 of TS 29.510 [29].

NOTE: The alternate AMF is expected to have registered a callback URI with the NRF.

2. Same as step 1 of Figure 5.2.2.3.5.1-1, the request content shall include the following information in the HTTP POST Request message body:

   - RAN NGAP ID and initial AMF name (the information enabling (R)AN to identify the N2 terminating point);
   - RAN identity, e.g. RAN Node Id, RAN N2 IPv4/v6 address;
   - Information from RAN, e.g. User Location, RRC Establishment Cause and UE Context Request;
   - the N1 message, which shall be the complete Registration Request message in clear text if the UE has a valid NAS security context, or as the one contained in the NAS message container IE in the Security Mode Complete message as specified in clause 4.2.2.2.3 of TS 23.502 [2];
   - the UE's SUPI and MM Context;
   - the Allowed NSSAI and if available the partially Allowed NSSAI, together with the corresponding NSI IDs (if network slicing is used and the initial AMF has obtained).

#### 5.2.2.3.5.3 Using N1MessageNotify in the UE Assisted and UE Based Positioning Procedure

In the UE assisted and UE based positioning procedure, the N1MessageNotify service operation is invoked by the AMF, towards the LMF, to notify the N1 UE positioning messages received from the UE.

The requirements specified in clause 5.2.2.3.5.1 shall apply with the following modifications:

1. If the corresponding N1 notification URI is not available, the AMF shall retrieve the NF profile of the NF Service Consumer (e.g. the LMF) from the NRF using the NF Instance Identifier received during corresponding N1N2MessageTransfer service operation (see clause 5.2.2.3.1), and further identify the corresponding service instance if Service Instance Identifier was also received, and fetch N1 Notification URI from the default notification subscription registered with "N1_MESSAGE" notification type and "LPP" N1 message class (See Table 6.2.6.2.3-1 and Table 6.2.6.2.4-1 of TS 29.510 [29]).

2. Same as step 1 of Figure 5.2.2.3.5.1-1, the request content shall include the following information:

   - the N1 Uplink Positioning Message;

   - LCS correlation identifier.

#### 5.2.2.3.5.4 Using N1MessageNotify in the UE Configuration Update for transparent UE Policy delivery

In the UE Configuration Update for transparent UE Policy delivery procedure, the N1MessageNotify service operation is invoked by the AMF, towards the PCF which subscribed to be notified with UPDP messages received from the UE.

The requirements specified in clause 5.2.2.3.5.1 shall apply with the following modifications:

1. Same as step 1 of Figure 5.2.2.3.5.1-1. The request content shall include the following information:

- the UPDP message.

#### 5.2.2.3.5.5 Using N1MessageNotify in the LCS Event Report, Event Reporting in RRC INACTIVE state procedures, LCS Cancel Location and LCS Periodic-Triggered Invoke Procedures

In the LCS Event Report, Event Reporting in RRC INACTIVE state procedures, LCS Cancel Location and LCS Periodic-Triggered Invoke procedure, the N1MessageNotify service operation is invoked by the AMF, towards the LMF, to notify the N1 UE LCS messages received from the UE.

The requirements specified in clause 5.2.2.3.5.1 shall apply with the following modifications:

1. If the corresponding N1 notification URI is not available, the AMF shall retrieve the NF profile of the NF Service Consumer (e.g. the LMF) from the NRF using the NF Instance Identifier received during corresponding N1N2MessageTransfer service operation (see clause 5.2.2.3.1), and further identify the corresponding service instance if Service Instance Identifier was also received, and fetch N1 Notification URI from the default notification subscription registered with "N1_MESSAGE" notification type and "LCS" N1 message class (See Table 6.2.6.2.3-1 and Table 6.2.6.2.4-1 of TS 29.510 [29]).

2. Same as step 1 of Figure 5.2.2.3.5.1-1, the request content shall include the following information:

   - the N1 Uplink LCS Message;

   - LCS correlation identifier;

   - indication of Control Plane CIoT 5GS Optimisation if Control Plane CIoT 5GS Optimisation is being used.

   and may include serving cell ID if it is available;

NOTE: For the EventReport message and UE initiated CancelDeferredLocation message, the AMF includes the deferred routing identifier received from UE in N1 UL NAS TRANSPORT message as LCS correlation identifier. The LCS correlation identifier can assist a serving LMF in identifying the periodic or triggered location session if the same LMF had assigned the deferred routing identifier or can indicate to the LMF that it is acting as a default LMF.

#### 5.2.2.3.5.6 Using N1MessageNotify in the UE triggered policy provisioning procedure to request UE policies

In the UE triggered policy provisioning procedure, the N1MessageNotify service operation is invoked by the AMF, towards the PCF which subscribed to be notified with UPDP messages received from the UE.

The requirements specified in clause 5.2.2.3.5.1 shall apply with the following modifications:

1. Same as step 1 of Figure 5.2.2.3.5.1-1. The request content shall include the following information:

   - the UPDP message.

#### 5.2.2.3.5.7 Using N1MessageNotify in the procedures applicable to a PRU

In the PRU Association Procedure, LMF Initiated PRU Disassociation Procedure or PRU Initiated PRU Disassociation Procedure, the N1MessageNotify service operation is invoked by the AMF, towards the LMF, to notify the N1 PRU messages received from the PRU.

The requirements specified in clause 5.2.2.3.5.1 shall apply with the following modifications:

1. If the corresponding N1 notification URI is not available, the AMF shall select the LMF and retrieve the NF profile of the LMF from the NRF (see clause 6.17 of TS 23.273 [42]), and fetch N1 Notification URI from the default notification subscription registered with "N1_MESSAGE" notification type and "LCS" N1 message class (See Table 6.2.6.2.3-1 and Table 6.2.6.2.4-1 of TS 29.510 [29]).
2. Same as step 1 of Figure 5.2.2.3.5.1-1, the request content shall include the following information if available:

   - the N1 PRU messages;
   - PRU subscription Indication;
   - the TAI and cell Id of the PRU;
   - Correlation identifier;
   - SUPI of the PRU.

### 5.2.2.3.6 N2InfoNotify

#### 5.2.2.3.6.1 General

The N2InfoNotify service operation is used during the following procedure:

- Inter NG-RAN node N2 based handover procedure (see TS 23.502 [3], clauses 4.9.1.3.3, 4.9.1.3.3a and 4.23.7.3);
- Network assisted positioning procedure (see clause 6.11.2 of TS 23.273 [42])
- AMF planned removal procedure with UDSF deployed (see clause 5.21.2.2.1 of TS 23.501 [2]), to forward uplink N2 signalling to a different AMF.

The N2InfoNotify service operation is invoked by AMF, to notify a NF Service Consumer that subscribed N2 information has been received from access network.

The AMF shall use HTTP POST method to the N2Info Notification URI provided by the NF Service Consumer via N1N2MessageSubscribe service operation (See clause 5.2.2.3.3). See also figure 5.2.2.3.6.1-1.

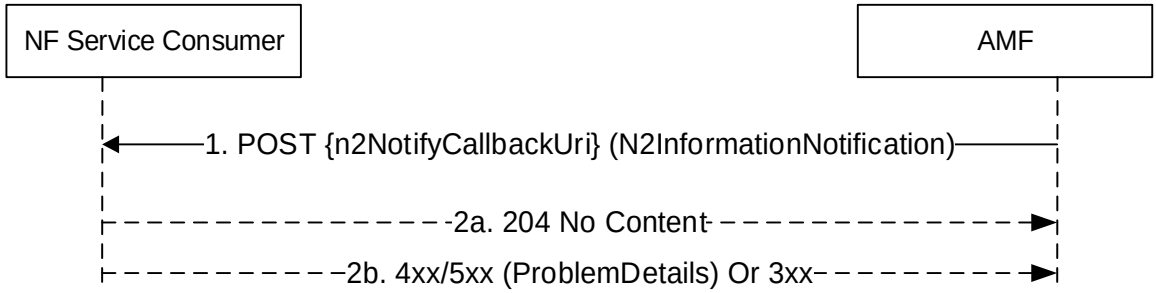



**Figure 5.2.2.3.6.1-1 N2 Information Notify**

1. The AMF shall send a HTTP POST request to the n2NotifyCallbackUri, and the content of the POST request shall contain a N2InformationNotification data structure, containing the N2 information that was subscribed by the NF Service Consumer.

2a. On success, "204 No Content" shall be returned and the content of the POST response shall be empty.

2b. On failure or redirection, one of the HTTP status code listed in Table 6.1.5.5.3.1-3 shall be returned. The message body shall contain a ProblemDetails object with "cause" set to one of the corresponding application errors listed in Table 6.1.5.5.3.1-3.

#### 5.2.2.3.6.2 Using N2InfoNotify during Inter NG-RAN node N2 based handover procedure

The N2InfoNotify service operation is invoked by a NF Service Producer, e.g. the target AMF, towards the NF Service Consumer, i.e. the source AMF, to notify that the handover procedure has been successful in the target side, for a given UE.

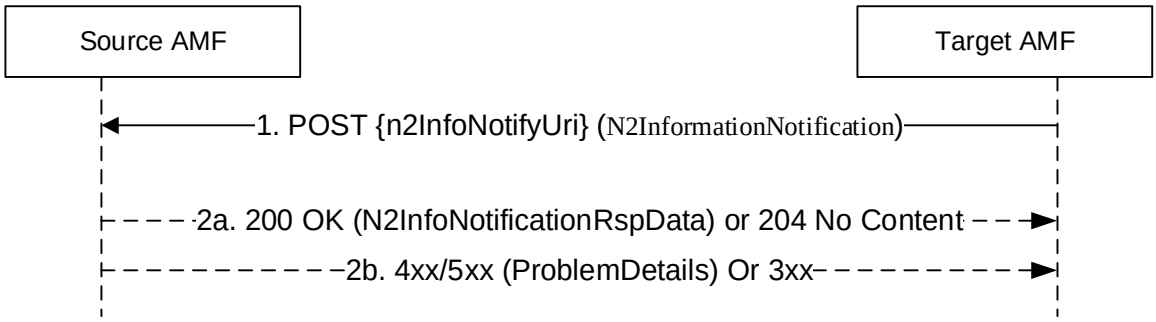



**Figure 5.2.2.3.6.2-1 N2 Information Notify during N2 Handover execution**

The requirements specified in clause 5.2.2.3.6.1 shall apply with the following modifications:

0. During an inter AMF handover procedure, the source AMF, acting as a NF Service Consumer, when invoking the CreateUEContext service operation (see clause 5.2.2.2.3), shall include a N2Info Notification URI to the target AMF in the HTTP request message.

1. Same as step 1 of Figure 5.2.2.3.6.1-1, the request content shall contain the following information:

   - notification content (see clause 6.1.5.5) without the "n2InfoContainer" attribute;
   - the "notifyReason" attribute set to "HANDOVER_COMPLETED";
   - the "smfChangeInfoList" attribute including the UE's PDU Session ID(s) for which the I-SMF or V-SMF has been changed or removed, if any, with for each such PDU session, the related "smfChangeIndication" attribute set to "CHANGED" or "REMOVED", if the I-SMF or the V-SMF is changed or removed respectively.
   - the "notifySourceNgRan" attribute set to "true" during an Inter NG-RAN node N2 based DAPS handover procedure, if the target AMF receives this indication in the Handover Notify from the target NG-RAN node (see clause 4.9.1.3.3a of TS 23.502 [3]).

   If any network slice(s) become no longer available and there are PDU Session(s) associated with them, the target AMF shall include these PDU session(s) in the toReleaseSessionList attribute in the content. If the continuity of the PDU Session(s) cannot be supported between networks (e.g. SNPN-SNPN mobility, inter-PLMN mobility where no HR agreement exists), the target AMF shall include these PDU session(s) with release cause in the toReleaseSessionInfo attribute in the content.

   The n2NotifySubscriptionId included in the notification content shall be the UE context Id.

2. Same as Step 2a of Figure 5.2.2.3.6.1-1, with the following additions/modifications:

- the source AMF shall release the PDU Session(s) listed in the toReleaseSessionList attribute and the toReleaseSessionInfo attribute in the content;

- if the smfChangeInfoList attribute was received in the request, the source AMF shall release the SM Context at the I-SMF or V-SMF only, for all the PDU sessions listed in the smfChangeInfoList attribute with the smfChangeIndication attribute set to "CHANGED" or "REMOVED";

- the source AMF shall remove the individual ueContext resource. The source AMF may choose to start a timer to supervise the release of the UE context resource and may keep the individual ueContext resource until the timer expires;

- if Secondary RAT usage data have been received from the source NG-RAN and buffered at the source AMF for one or more PDU sessions as specified in step 2a0 of clause 4.9.1.3.3 of TS 23.502 [3], the source AMF shall send a 200 OK response with the Secondary RAT usage data included in the response content for one or more PDU sessions.

- if the "notifySourceNgRan" attribute was set to "true" in the request, the source AMF shall send a HANDOVER SUCCESS to the source NG-RAN (see clause 4.9.1.3.3a of TS 23.502 [3]).

NOTE: This notification is due to an implicit subscription and hence no explicit subscription Id is created. UE context Id is included as the notification subscription Id for the NF Service Consumer (e.g. Source AMF) to co-relate the notification to an earlier initiated UE context creation during a handover procedure.

#### 5.2.2.3.6.3 Using N2InfoNotify during Location Services procedures

The N2InfoNotify service operation is invoked by a NF Service Producer, i.e. the AMF, towards the NF Service Consumer, e.g. the LMF, to notify the positioning parameters received from the 5G-AN in the NRPPa message.

The requirements specified in clause 5.2.2.3.6.1 shall apply with the following modifications:

1. If the corresponding N2 notification URI is not available, the AMF shall retrieve the NF profile of the NF Service Consumer (e.g. the LMF) from the NRF using the NF Instance Identifier received during corresponding N1N2MessageTransfer service operation (see clause 5.2.2.3.1), and further identify the corresponding service instance if Service Instance Identifier was also received, and fetch N2 Notification URI from the default subscription registered with "N2_INFORMATION" notification type and "NRPPa" N2 information class (See Table 6.2.6.2.3-1 and Table 6.2.6.2.4-1 of TS 29.510 [29]).

2. Same as step 1 of Figure 5.2.2.3.6.1-1, the request content shall contain N2 information of type NRPPa and LCS correlation identifier.

#### 5.2.2.3.6.4 Using N2InfoNotify during AMF planned removal procedure with UDSF deployed procedure

In the AMF planned removal procedure with UDSF deployed (see clause 5.21.2.2.1 of TS 23.501 [2]), the N2InfoNotify service operation is invoked by a NF Service Producer, i.e. an initial AMF, towards the NF Service Consumer, i.e. the target AMF, to forward uplink N2 signalling to the target AMF.

The requirements specified in clause 5.2.2.3.6.1 shall apply with the following modifications:

1. If the N2 notification URI is not available, the initial AMF shall discover the NF Service Consumer (i.e. the target AMF) from the NRF, and fetch the N2 Notification URI from the default notification subscription registered with "N2_INFORMATION" notification type and "RAN" N2 message class (See Table 6.2.6.2.3-1 and Table 6.2.6.2.4-1 of TS 29.510 [29].

NOTE: The target AMF is expected to have registered a callback URI with the NRF.

2. Same as step 1 of Figure 5.2.2.3.6.1-1, the request content shall contain the following information in the HTTP POST Request message body:

   - N2 information of type "RAN";

   - N2 message;

   - initial AMF name;

- RAN identity, e.g. RAN Node Id, RAN N2 IPv4/v6 address.

## 5.2.2.4 Non-UE N2 Message Operations

### 5.2.2.4.1 NonUeN2MessageTransfer

#### 5.2.2.4.1.1 General

The NonUeN2MessageTransfer service operation is used by a NF Service Consumer to transfer N2 information to the 5G-AN through the AMF in the following procedures:

- Obtaining non-UE associated network assistance data (See clause 4.13.5.6 in TS 23.502 [3]);
- Warning Request Transfer procedures (See clause 9A in TS 23.041 [20]);
- Configuration Transfer procedure (see clause 5.26 of TS 23.501 [2])
- RIM Information Transfer procedures (see clause 8.16 of TS 38.413 [12]).
- Broadcast of Assistance Data by an LMF (see clause 6.14.1 of TS 23.273 [42]).
- Management of network timing synchronization status monitoring procedures (see clause 4.15.9.5 of TS 23.502 [3]).

The NF Service Consumer shall invoke the service operation by sending POST to the URI of the "transfer" customer operation on the "Non UE N2Messages Collection" resource (See clause 6.1.3.8.4.2) on the AMF. See also figure 5.2.2.4.1.1-1.

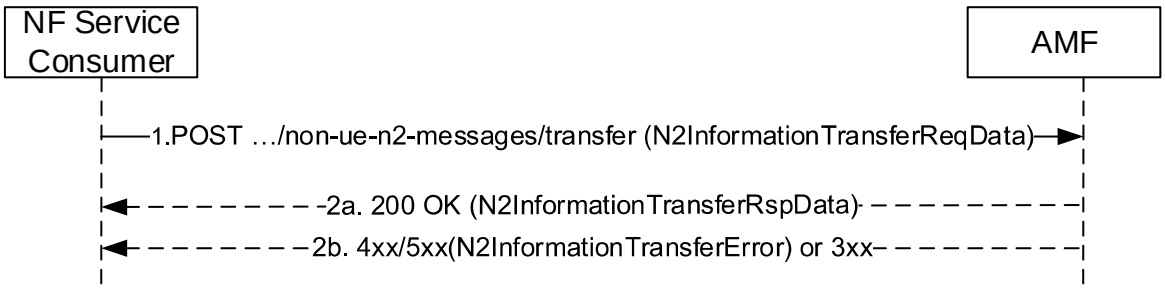



**Figure 5.2.2.4.1.1-1 Non-UE N2 Message Transfer**

1. The NF Service Consumer shall invoke the custom operation for non UE associated N2 message transfer by sending a HTTP POST request, and the request body shall carry the N2 information to be transferred.

2a. On success, AMF shall respond a "200 OK" status code with N2InformationTransferRspData data structure.

2b. On failure or redirection, one of the HTTP status code listed in Table 6.1.3.8.4.2.2-2shall be returned with the message body containing a N2InformationTransferError structure, including a ProblemDetails attribute with the "cause" attribute set to one of the application errors listed in Table 6.1.3.8.4.2.2-2.

#### 5.2.2.4.1.2 Obtaining Non UE Associated Network Assistance Data Procedure

The NonUeN2MessageTransfer service operation shall be invoked by a NF Service Consumer, e.g. LMF to transfer non UE associated N2 information of N2 information class NRPPa to NG-RAN for obtaining the network assistance data.

The requirements specified in clause 5.2.2.4.1.1 shall apply with the following modifications:

1. Same as step 1 of Figure 5.2.2.4.1.1-1, the POST request body shall carry the N2 information to be transferred together with the NG RAN node identifier(s) to which the transfer needs to be initiated. The POST request body shall also include the NF Instance Identifier of the NF Service Consumer (e.g. LMF) in "nfId" attribute.

#### 5.2.2.4.1.3 Warning Request Transfer Procedure

The NonUeN2MessageTransfer service operation shall be invoked by the NF Service Consumer, e.g. CBCF/PWS-IWF, to send non-UE specific messages of N2 information class PWS to the NG-RAN.

The requirements specified in clause 5.2.2.4.1.1 shall apply with the following modifications:

1. Same as step 1 of Figure 5.2.2.4.1.1-1, the request body shall include the N2 Message Container and:

   - the globalRanNodeList IE, or;

   - the taiList IE and the ratSelector IE, or;

   - the ratSelector IE.

   The AMF shall forward the N2 Message Container to ng-eNBs or to gNBs indicated in the globalRanNodeList IE if present. If the globalRanNodeList IE if not present, the AMF shall forward the N2 Message Container to ng-eNBs or to gNBs, subject to the value of the *ratSelector* IE, that serve Tracking Areas as listed in the *taiList* IE if present. If the *taiList* IE and the *globalRanNodeList* IE are not present, the AMF shall forward the N2 Message Container to all attached ng-eNBs or all attached gNBs, subject to the value of the *ratSelector* IE.

NOTE: The *globalRanNodeList* IE can be present when transferring WRITE-REPLACE WARNING REQUEST. When present, the *globalRanNodeList* IE only contains RAN nodes of the same type, i.e. only ng-eNBs or only gNBs.

   The request body may additionally include the *omcId* IE and/or the *sendRanResponse* IE.

2a. Same as step 2a of Figure 5.2.2.4.1.1-1, and the POST response body shall contain the mandatory elements from the Write-Replace-Warning Confirm response (see clause 9.2.17 in TS 23.041 [20]) or the mandatory elements and optionally the *unknown TAI List* IE from the Stop-Warning Confirm response (see clause 9.2.19 in TS 23.041 [20]).

   If the *sendRanResponse* IE with the value "true" was received in the request, but the corresponding N2 information subscription for PWS information from the NF service consumer is not available in the AMF, the AMF should include the *n2PwsSubMissInd* IE with the value "true" in the response.

2b. Same as step 2b of Figure 5.2.2.4.1.1-1, and the POST response body shall contain following additional information:

   - PWS specific information, if any, e.g. PWS Cause information.

#### 5.2.2.4.1.4 Configuration Transfer Procedure

The NonUeN2MessageTransfer service operation shall be invoked by the NF Service Consumer (i.e. source AMF) towards the NF Service Producer (i.e. target AMF) to transfer the RAN configuration information received from the source NG-RAN towards the target NG-RAN.

The requirements specified in clause 5.2.2.4.1.1 shall apply with the following modifications:

1. Same as step 1 of Figure 5.2.2.4.1.1-1. The POST request body shall contain the SON Configuration Transfer IE received from the source NG-RAN, the NG RAN node identifier of the destination of this configuration information, and the N2 information class "RAN".

   The target AMF shall forward the SON Configuration Transfer IE in a NGAP Downlink RAN Configuration Transfer message to the target NG-RAN.

#### 5.2.2.4.1.5 RIM Information Transfer Procedures

The NonUeN2MessageTransfer service operation shall be invoked by the NF Service Consumer (i.e. source AMF) towards the NF Service Producer (i.e. target AMF) to transfer the RIM information received from the source NG-RAN towards the target NG-RAN.

The requirements specified in clause 5.2.2.4.1.1 shall apply with the following modifications:

1. Same as step 1 of Figure 5.2.2.4.1.1-1. The POST request body shall contain the RIM Information Transfer IE received from the source NG-RAN, the NG RAN node identifier of the destination of this configuration information, and the N2 information class "RAN".

   The target AMF shall forward the RIM Information Transfer IE in a NGAP Downlink RIM Information Transfer message to the target NG-RAN.

#### 5.2.2.4.1.6 Broadcast of Assistance Data by an LMF

The NonUeN2MessageTransfer service operation shall be invoked by a NF Service Consumer, e.g. LMF to transfer non UE associated N2 information of N2 information class NRPPa to NG-RAN for sending assistance information broadcasting.

The requirements specified in clause 5.2.2.4.1.1 shall apply with the following modifications:

1. Same as step 1 of Figure 5.2.2.4.1.1-1, the POST request body shall contain NRPPa-PDU IE carrying Network Assistance Data generated by LMF to be transferred together with the target NG RAN node identifier(s) to which the transfer needs to be initiated. The POST request body shall also include the NF Instance Identifier of the NF Service Consumer (e.g. LMF) in "nfId" attribute.

#### 5.2.2.4.1.7 Management of network timing synchronization status monitoring procedures

The NonUeN2MessageTransfer service operation shall be invoked by a NF Service Consumer, e.g. TSCTSF, to transfer clock quality reporting control information to NG-RAN.

The requirements specified in clause 5.2.2.4.1.1 shall apply with the following modifications:

1. Same as step 1 of Figure 5.2.2.4.1.1-1, the POST request body shall contain a NGAP Timing Synchronization Status Request message, a list of TAIs or a list of NG-RAN IDs identifying target NG RAN node(s) towards which the NGAP message is requested to be transferred, the N2 information class set to "TSS", and the "nfId" attribute set to the NF Instance Identifier of the NF Service Consumer (e.g. TSCTSF).

2. Same as step 2a of Figure 5.2.2.4.1.1-1. The N2InformationTransferRspData data structure may contain a list of TssRspPerNgran if the AMF receives Timing Synchronization Status Failure message(s), or Timing Synchronization Status Response message(s) with the Criticality Diagnostics IE being present, from one or more NG-RAN nodes, or if the AMF is unable to reach some NG-RAN node(s).

   Upon receipt of subsequent responses from other NG-RANs after sending the 200 OK response, assuming that the TSCTSF has subscribed to the AMF for TSS information reporting (see clause 5.2.2.4.2), if additional information (e.g., Timing Synchronization Status Failure or Timing Synchronization Status Response message with the Criticality Diagnostics IE being present, as specified in TS 38.413 [12]) needs to be transferred to the TSCTSF, the AMF shall transfer such information in the TssRspPerNgran by sending one or more Namf_NonUeN2InfoNotify requests to the TSCTSF.

NOTE: The TSCTSF sends the Namf_NonUeN2InfoSubscribe request message to the AMF to subscribe for receiving TSS related information, i.e., Timing Synchronization Status Report/Failure/Response messages, before it sends the Namf_NonUeN2MessageTransfer message containing the Timing Synchronization Status Request to start the TSS reporting in the NG-RAN.

### 5.2.2.4.2 NonUeN2InfoSubscribe

#### 5.2.2.4.2.1 General

The NonUeN2InfoSubscribe service operation is used by a NF Service Consumer (e.g. CBCF, PWS-IWF, TSCTSF) to subscribe to the AMF for notifying non UE specific N2 information of a specific type (e.g. PWS Indications, TSS).

An NF Service Consumer (e.g. CBCF, PWS-IWF, TSCTSF) may subscribe to notifications of specific N2 information type (e,g PWS Indications, TSS) that are not associated with any UE. In this case, the NF Service Consumer shall subscribe by using the HTTP POST method with the URI of the "Non UE N2Messages Subscriptions Collection" resource (See clause 6.1.3.9.3.1). See also Figure 5.2.2.4.2.1-1.

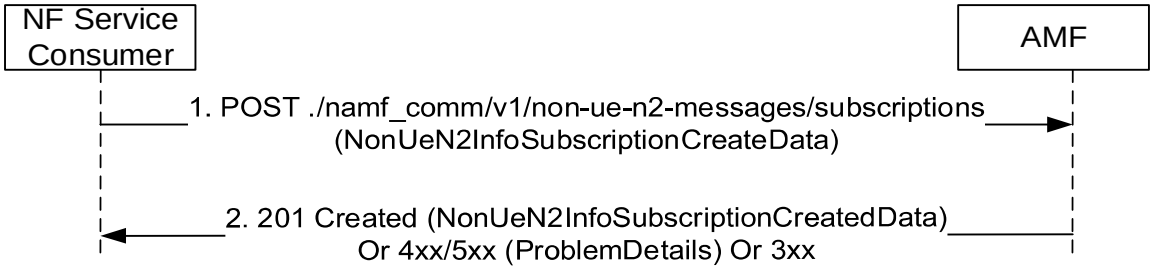



**Figure 5.2.2.4.2.1-1 N2 Information Subscription for Non UE Information**

1. The NF Service Consumer shall send a POST request to create a subscription resource in the AMF for a non UE specific N2 information notification. The content of the POST request shall contain:

   - N2 Information Type, identifying the type of N2 information to be notified
   - A callback URI for the notification

2. If the request is accepted, the AMF shall include a HTTP Location header to provide the location of a newly created resource (subscription) together with the status code 201 indicating the requested resource is created in the response message.

   On failure or redirection, one of the HTTP status codes together with the response body listed Table 6.1.3.9.3.1-3 shall be returned.

### 5.2.2.4.3 NonUeN2InfoUnSubscribe

#### 5.2.2.4.3.1 General

The NonUeN2InfoUnSubscribe service operation is used by a NF Service Consumer (e.g. CBCF, PWS-IWF, TSCTSF) to unsubscribe to the AMF to stop notifying N2 information of a specific type (e.g. PWS Indications, TSS).

The NF Service Consumer shall use the HTTP method DELETE with the URI of the "Non UE N2 Message Notification Individual Subscription" resource (See clause 6.1.3.10.3.1), to request the deletion of the subscription for non UE specific N2 information notification, towards the AMF. See also Figure 5.2.2.4.3.1-1.

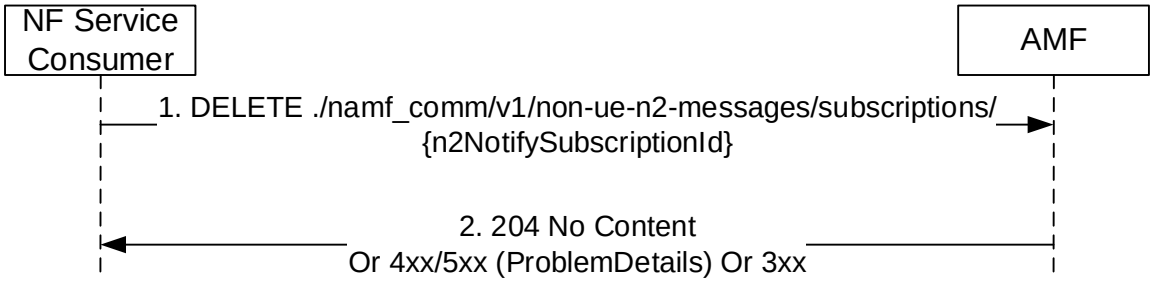



**Figure 5.2.2.4.3.1-1 NonUeN2InfoUnSubscribe for Non UE Specific Information**

1. The NF Service Consumer shall send a DELETE request to delete an existing subscription resource in the AMF.

2. If the request is accepted, the AMF shall reply with the status code 204 indicating the resource identified by subscription ID is successfully deleted, in the response message.

   On failure or redirection, one of the HTTP status codes together with the response body listed Table 6.1.3.10.3.1-3 shall be returned.

#### 5.2.2.4.4 NonUeN2InfoNotify

##### 5.2.2.4.4.1 General

The NonUeN2InfoNotify service operation is used during the following procedures:

- Obtaining non-UE associated network assistance data (See clause 4.13.5.6 in TS 23.502 [3])
- Receiving PWS related events from the NG-RAN
- Broadcast of Assistance Data by an LMF (see clause 6.14.1 of TS 23.273 [42]).
- Monitoring of Timing synchronization status information (see clause 4.15.9.5 of TS 23.502 [3]).

The NonUeN2InfoNotify service operation is invoked by the AMF to notify a NF Service Consumer that subscribed Non-UE N2 information has been received from the 5G-AN.

The AMF shall use HTTP POST method to the N2Info Notification URI provided by the NF Service Consumer via NonUeN2InfoSubscribe service operation (See clause 5.2.2.4.2). See also Figure 5.2.2.4.4.1-1.

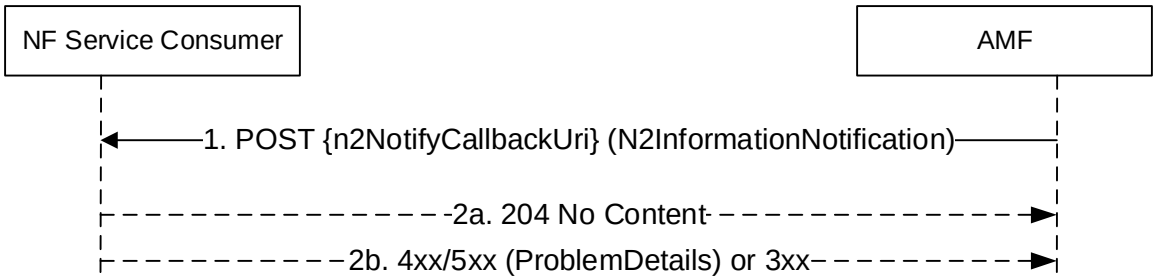



**Figure 5.2.2.4.4.1-1 Non-UE N2 Information Notify**

1. The AMF shall send a HTTP POST request to the N2Info Notification URI, and the content of the POST request shall contain a N2InformationNotification data structure, with the N2 information that was subscribed by the NF Service Consumer.

2a. On success, "204 No Content" shall be returned and the content of the POST response shall be empty.

2b. On failure or redirection, one of the HTTP status code listed in Table 6.1.5.3.3.1-3 shall be returned. The message body shall contain a ProblemDetails object with "cause" set to one of the corresponding application errors listed in Table 6.1.5.3.3.1-3.

##### 5.2.2.4.4.2 Using NonUeN2InfoNotify during Location Services procedures

The NonUeN2InfoNotify service operation is invoked by a NF Service Producer, i.e. the AMF, towards the NF Service Consumer, e.g. the LMF, to notify the assistance data received from the 5G-AN.

The requirements specified in clause 5.2.2.4.4.1 shall apply with the following modifications:

1. If the corresponding N2 notification URI is not available, the AMF shall retrieve the NF profile of the NF Service Consumer (e.g. the LMF) from the NRF using the NF Instance Identifier received during "Obtaining Non UE Associated Network Assistance Data Procedure" or "Broadcast of Assistance Data by an LMF Procedure" (see clause 5.2.2.4.1.2), and further identify the corresponding service instance if Service Instance Identifier was also received, and fetch N2 Notification URI from the default subscription registered with "N2_INFORMATION" notification type and "NRPPa" information class (See Table 6.2.6.2.3-1 and Table 6.2.6.2.4-1 of TS 29.510 [29]).

2. Same as step 1 of Figure 5.2.2.4.4.1-1, the content shall contain network assistance data.

##### 5.2.2.4.4.3 Use of NonUeN2InfoNotify for PWS related events

The NonUeN2InfoNotify service operation shall be used during the following PWS related events:

1) The AMF has received a Write-Replace-Warning-Response or a PWS-Cancel-Response from the NG-RAN over N2.

   Upon receiving the N2 Message Content the RAN Nodes return a response which may include the *Broadcast Completed Area List* IE or the *Broadcast Cancelled Area List* IE, depending on the *Message Type* IE. The AMF may aggregate the lists it receives from the RAN Nodes for the same request.

   If the *Send-Write-Replace-Warning Indication* IE was present in the Write-Replace-Warning Request message, then the AMF may forward the *Broadcast Completed Area List* IE(s) to the NF Service Consumer. If the NG-RAN node(s) have responded without the *Broadcast Completed Area List* IE then the AMF shall include the NG-RAN node ID(s) in "bcEmptyAreaList" attribute in the request body.

   If the *Send-Stop-Warning Indication* IE was present in the Stop-Warning-Request message, then the AMF may forward the *Broadcast Cancelled Area List* IE(s) to the NF Service Consumer. If the NG-RAN node(s) have responded without the *Broadcast Cancelled Area List* IE then the AMF shall include the NG-RAN node ID(s) in "bcEmptyAreaList" attribute in the request body.

   If none of the NG-RAN nodes have responded with the Broadcast Completed Area List IE (in the Write-Replace Warning Response messages) or with the Broadcast Cancelled Area List IE (in the PWS Cancel Response messages), the AMF shall send one (or more) NonUEN2InfoNotify request(s) including a Write-Replace Warning Response or a PWS Cancel Response to the CBCF/PWS-IWF, and including all the NG-RAN node(s) in the "bcEmptyAreaList" attribute.

   NOTE: TS 23.041 [20] specifies that the AMF sends a NonUEN2InfoNotify including a Write-Replace Warning Indication or a Stop Warning Indication to the CBCF/PWS-IWF. This is supported at stage 3 level by the AMF sending a NonUEN2InfoNotify including a Write-Replace Warning Response or a PWS Cancel Response to the CBCF/PWS-IWF.

2) The AMF has received a Restart Indication or a Failure Indication from a NG-RAN Node. The AMF shall forward the Restart Indication or Failure Indication to the NF Service Consumer.

   The requirements specified in clause 5.2.2.4.4.1 shall apply with the following modifications:

   - Same as step 1 of Figure 5.2.2.4.4.1-1, the request body shall include the PWS related N2 information.

#### 5.2.2.4.4.4 Using NonUeN2InfoNotify during network timing synchronization status monitoring procedure

The NonUeN2InfoNotify service operation is invoked by the AMF towards the NF Service Consumer indicated by the Routing ID IE received within the NGAP Timing Synchronization Status Report/Failure/Response messages, e.g. the TSCTSF, to notify the RAN timing synchronization status information and/or the result of the Namf_NonUeN2MessageTransfer request message which contains Timing Synchronization Status Request to start or stop the reporting of RAN timing synchronization status in the NG-RAN, as specified in clause 5.2.2.4.1.7, from the 5G-AN.

The requirements specified in clause 5.2.2.4.4.1 shall apply with the following modifications:

- Same as step 1 of Figure 5.2.2.4.4.1-1. The request body shall include:

  - the RAN Timing Synchronization Status Report message for TSS related N2 information, the Timing Synchronization Status Failure message(s), and/or the Timing Synchronization Status Response message(s) with the Criticality Diagnostics IE being present, from one or more NG-RAN nodes; and/or

  - one or more TssRspPerNgran(s) containing only the NG-RAN node ID and the ngranFailureInfo IE to report a failure related to an NG-RAN, e.g. when the AMF has detected that the NG-RAN has failed with or without restart, or the NG-RAN is not reachable, as specified in clause 8.3 of TS 23.527 [33].

  NOTE: The AMF can aggregate up to 10 N2 Information containers encapsulating Timing Synchronization Status Report, Failure and/or Response messages received from NG-RANs in an NonUeN2InfoNotify request message where each N2 Information container encapsulates a Timing Synchronization Status Report, Failure or Response message.

## 5.2.2.5 AMF Status Change Operations

### 5.2.2.5.1 AMFStatusChangeSubscribe

#### 5.2.2.5.1.1 General

This service operation is used by a NF Service Consumer to subscribe the status change of the AMF.

The AMFStatusChangeSubscribe service operation is used during the following procedure:

- AMF planned removal procedure (see TS 23.501 [2], clause 5.21.2.2)

#### 5.2.2.5.1.2 Creation of a subscription

This service operation creates a subscription so a NF Service Consumer can request to be notified when the status of the AMF is changed.

It is executed by creating a new individual resource under the collection resource "subscriptions". The operation shall be invoked by issuing a POST request on the URI of the "subscriptions collection" resource (See clause 6.1.3.6.3.1).

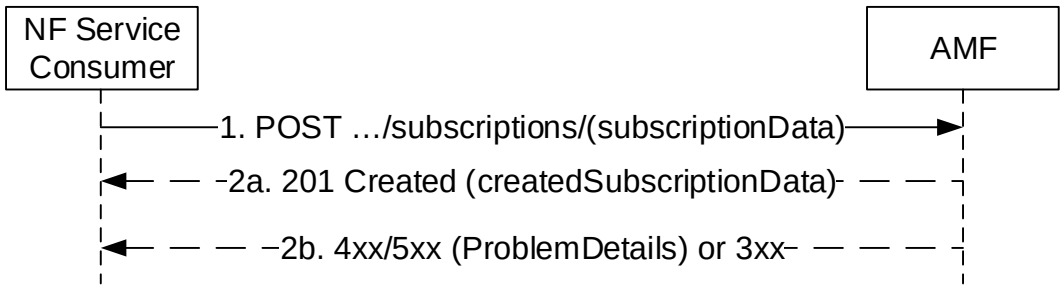



**Figure 5.2.2.5.1.1-1 NF Service Consumer Subscription to Notifications**

1. The NF Service Consumer shall send a POST request to the resource URI representing the "subscriptions" collection resource. The request body shall include the data indicating the GUAMI(s) supported by the AMF that the NF Service Consumer is interested in receiving the related status change notification. The request body also contains a callback URI, where the NF Service Consumer shall be prepared to receive the actual notification from the AMF (see AMFStatusChangeNotify operation in clause 5.2.2.5.3).

2a. On success, the AMF shall include a HTTP Location header to provide the location of a newly created resource (subscription) together with the status code 201 indicating the requested resource is created in the response message.

2b. On failure or redirection, one of the HTTP status code listed in Table 6.1.3.6.3.1-3 shall be returned. For a 4xx/5xx response, the message body containing a ProblemDetails structure with the "cause" attribute set to one of the application error listed in Table 6.1.3.6.3.1-3.

#### 5.2.2.5.1.3 Modification of a subscription

This service operation updates the subscription data of an NF Service Consumer previously subscribed in the AMF by providing the updated subscription data to the AMF. The update operation shall apply to the whole subscription data (complete replacement of the existing subscription data by a new subscription data).

The NF Service Consumer shall issue an HTTP PUT request, towards the URI of the "individual subscription" resource (See clause 6.1.3.7.3.2), as shown in Figure 5.2.2.5.1.3-1:

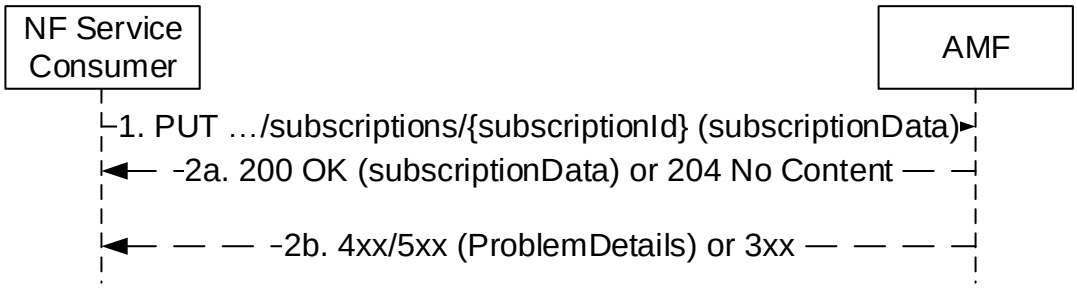



**Figure 5.2.2.5.1.3-1 Subscription Data Complete Replacement**

1. The NF Service Consumer shall send a PUT request to the resource URI representing the individual subscription. The request body shall include a representation of subscription data to replace the previous subscription data in the AMF.

2a. On success, "200 OK" shall be returned, the content of the PUT response shall contain the representation of the replaced resource. "204 No Content" may be returned, if the NF Service Producer accepts entirely the resource representation provided by the NF Service Consumer in the request.

2b. On failure or redirection, one of the HTTP status code listed in Table 6.1.3.7.3.2-3 shall be returned. For a 4xx/5xx response, the message body shall contain a ProblemDetails structure with the "cause" attribute set to one of the application error listed in Table 6.1.3.7.3.2-3.

### 5.2.2.5.2 AMFStatusChangeUnSubscribe

#### 5.2.2.5.2.1 General

This service operation removes an existing subscription to notifications.

The AMFStatusChangeUnSubscribe service operation is used during the following procedure:

- AMF planned removal procedure (see TS 23.501 [2], clause 5.21.2.2)

It is executed by deleting a given resource identified by a "subscriptionId". The operation is invoked by issuing a DELETE request on the URI of the specific " individual subscription" resource (See clause 6.1.3.7.3.1).

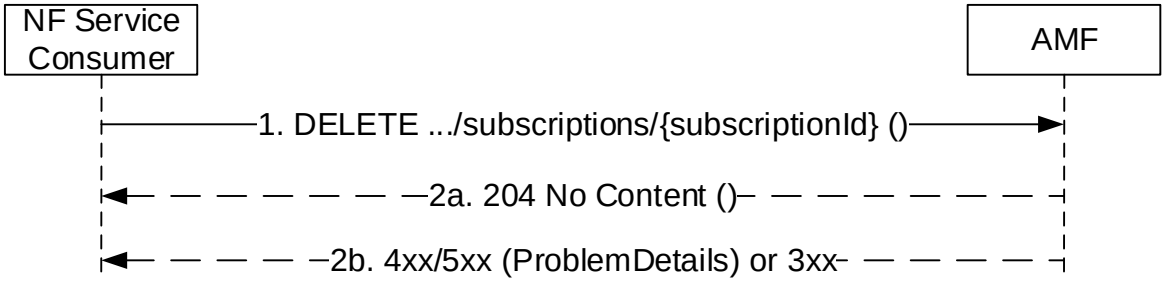



**Figure 5.2.2.5.2.1-1: NF Service Consumer Unsubscription to Notifications**

1. The NF Service Consumer shall send a DELETE request to the resource URI representing the individual subscription. The request body shall be empty.

2a. On success, "204 No Content" shall be returned. The response body shall be empty.

2b. On failure or redirection, one of the HTTP status code listed in Table 6.1.3.7.3.1-3 shall be returned. For a 4xx/5xx response, the message body shall contain a ProblemDetails structure with the "cause" attribute set to one of the application error listed in Table 6.1.3.7.3.1-3.

#### 5.2.2.5.3 AMFStatusChangeNotify

##### 5.2.2.5.3.1 General

This service operation notifies each NF Service Consumer that was previously subscribed to receiving notifications of the status change of the AMF (e.g. AMF unavailable). The notification is sent to a callback URI that each NF Service Consumer provided during the subscription (see AMFStatusChangeSubscribe operation in 5.2.2.5.1).

The AMFStatusChangeNotify service operation is used during the following procedure:

- AMF planned removal procedure (see TS 23.501 [2], clause 5.21.2.2)

The operation is invoked by issuing a POST request to each callback URI of the different NF Service Consumer.

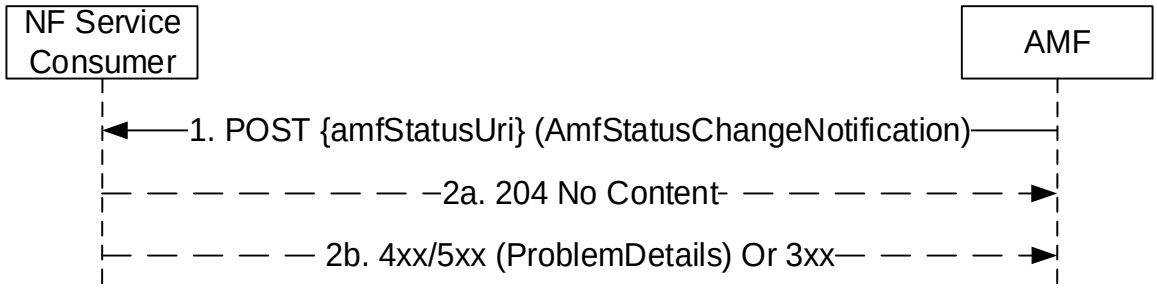



**Figure 5.2.2.5.3.1-1: AMF Notifications**

1. The AMF shall send a POST request to the callback URI. The request body shall include the GUAMI(s) and the related status change, GUAMI(s) is indicated by the NF Service Consumer during the subscription operation. For network deployment without UDSF case, the target AMF Name which is to serve the user of the indicated GUAMI(s) is also included.

2a. On success, "204 No content" shall be returned by the NF Service Consumer.

2b. On failure or redirection, one of the HTTP status code listed in Table 6.1.5.2.3.1-2 shall be returned. For a 4xx/5xx response, the message body shall contain a ProblemDetails structure with the "cause" attribute set to one of the application error listed in Table 6.1.5.2.3.1-2.

### 5.2.2.6 EBIAssignment

#### 5.2.2.6.1 General

The EBIAssignment service operation is used during the following procedures (see TS 23.502 [3], clause 4.11.1.4):

- UE requested PDU Session Establishment including Request Types "Initial Request", "Existing PDU Session", "Initial emergency request" and "Existing emergency PDU session" (Non-roaming and Roaming with Local Breakout (see TS 23.502 [3], clause 4.3.2.2.1).

- UE requested PDU Session Establishment including Request Types "Initial Request" and "Existing PDU Session" (Home-routed Roaming (see TS 23.502 [3], clause 4.3.2.2.2).

- UE or network requested PDU Session Modification (non-roaming and roaming with local breakout) (see TS 23.502 [3], clause 4.3.3.2).

- UE or network requested PDU Session Modification (home-routed roaming) (see TS 23.502 [3], clause 4.3.3.3).

- UE Triggered Service Request (see TS 23.502 [3], clause 4.2.3.2) to move PDU Session(s) from untrusted non-3GPP access to 3GPP access.

- Network requested PDU Session Modification, when the SMF needs to release the assigned EBI from a QoS flow (see TS 23.502 [3], clause 4.11.1.4.3).

The EBIAssignment service operation is invoked by a NF Service Consumer, e.g. a SMF, towards the NF Service Producer, i.e. the AMF, to request the AMF to allocate EPS bearer ID(s) towards EPS bearer(s) mapped from QoS flow(s) for an existing PDU Session for a given UE.

EBI allocation shall apply only to:

- QoS flows of Single Access PDU Session(s) via 3GPP access using SSC mode 1 and supporting EPS interworking with N26;
- Qos flows of Multi-Access PDU Session(s) using SSC mode 1 and supporting EPS interworking with N26, that are not only allowed over non-3GPP access.

EBI allocation shall not apply to:

- PDU Session(s) via 3GPP access supporting EPS interworking without N26;
- PDU Session(s) via non-3GPP access supporting EPS interworking;
- GBR QoS flow(s) that are only allowed over non-3GPP access in Multi-Access PDU Session(s) supporting EPS interworking; and
- PDU sessions using SSC mode 2 or SSC mode 3.

The EBIAssignment service operation is also invoked by an NF Service Consumer, e.g. an SMF, towards the NF Service Producer supporting the EAEA feature, i.e. the AMF, to request the AMF to update the mapping of EBI and ARP, if the ARP for a QoS flow that has already been allocated an EBI is changed during the network requested PDU Session Modification.

The NF Service Consumer (e.g. the SMF) shall perform EBIAssignment service operation by invoking "assign-ebi" custom operation on the "individual ueContext" resource (See clause 6.1.3.2.4.3). See also Figure 5.2.2.6.1-1.

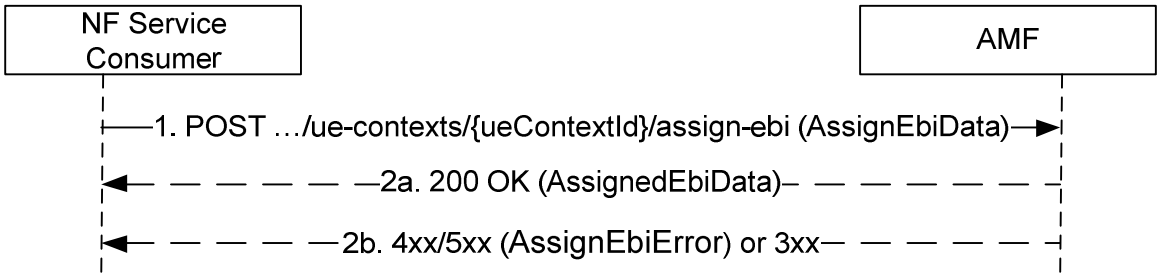



**Figure 5.2.2.6.1-1 EBI Assignment**

1. The NF Service Consumer, e.g. the SMF, shall invoke "assign-ebi" custom method on individual ueContext resource, which is identified by the UE's SUPI or PEI in the AMF. The NF Service consumer shall provide PDU Session ID and ARP list as input for the service operation. If the NF Service Consumer invokes this service operation to update the mapping of EBI and ARP for a QoS flow to which an EBI is already allocated in the AMF, the NF Service Consumer shall provide the PDU Session ID and modifiedEbiList.

2a. On success, the AMF shall assign EBI for each ARP in received ARP list, if enough EBI(s) are available. If there is not enough EBI(s) available, the AMF may revoke already assigned EBI(s) based on the ARP(s) and the S-NSSAI of the PDU session for which the request was received, EBIs information in the UE context and local policies. The AMF may only assign a subset of the requested EPS Bearer ID(s), e.g. when other PDU Sessions with higher ARP have occupied other available EPS Bearer IDs. If AMF has successfully assigned all or part of the requested EBI(s), the AMF shall respond with the status code 200 OK, together with the assigned EBI to ARP mapping(s), the list of ARPs for which the AMF failed to allocate an EBI (if any) and the list of EBI(s) released for this PDU session due to revocation based on ARP(s) and the S-NSSAI (if any).

   If the request contains "releasedEbiList", the AMF shall release the requested EBI(s). The AMF shall respond with the status code 200 OK and shall include the EBI(s) released in the "releasedEbiList" IE of the POST response body. The "releasedEbiList" in the request shall be handled before the EBI assignment in AMF.

   If the same EBI(s) are both in the "releasedEbiList"and "assignedEbiList", the NF sevice consumer considers that EBI(s) have been released and reassigned.

If the request contains "modifiedEbiList", the AMF shall store the association of the assigned EBI and ARP pair to the corresponding PDU Session ID. The AMF shall respond with the status code 200 OK and shall include the EBI(s) with ARP updated in the "modifiedEbiList" IE of the POST response body.

2b. On failure or redirection, one of the HTTP status code listed in Table 6.1.3.2.4.3.2-2 shall be returned. For a 4xx/5xx response, the message body shall contain an AssignEbiError structure, including:

- a ProblemDetails structure with the "cause" attribute set to one of the application error listed in Table 6.1.3.2.4.3.2-2;
- a failureDetails which describes the details of the failure including the list of ARPs for which the EBI assignment failed.

# 5.3 Namf_EventExposure Service

## 5.3.1 Service Description

The AMF may offer this service as a Service Producer to enable an NF to subscribe to event notifications on its own or on behalf of another NF and get notified about an event. The known Service Consumers are NEF, SMF, UDM, NWDAF, DCCF, LMF, TSCTSF and GMLC. See also clause 5.34.7 of TS 23.501 [2] and clauses 4.15.1, 4.15.3.2, 4.15.4.2 and 5.2.2.3.1 of TS 23.502 [3], clause 6.2.2 in TS 23.288 [38].

The following events are provided by Namf_EventExposure Service:

Event: Location-Report

A NF subscribes to this event to receive the Last Known Location (if the immediateFlag is set) or the Current Location (if the immediateFlag is not set) of a UE or a group of UEs or any UE, and Updated Location of any of these UEs when AMF becomes aware of a location change of any of these UEs with the granularity as requested.

This event implements the "Location Reporting" event in table 4.15.3.1-1 of TS 23.502 [3].

UE Type: One UE, Group of UEs, any UE

Report Type: One-Time Report, Continuous Report (See NOTE 1), Periodic Report (See NOTE 1 and 2)

Input: UE-ID(s), "ANY_UE", optional filters (TAI, Cell-ID, N3IWF, UE-IP, UDP-PORT, TNAP ID, TWAP ID, Global Line Id), immediateFlag

Notification: UE-ID, filtered updated location (TAI, Cell-ID for 3GPP access, most recent N3IWF node, UE local IP address and UDP source port number for non-3GPP access, TNAP ID, TWAP ID, Global Line Id).

NOTE 1: Support of Continuous Report or Periodic Report should be controlled by operator policy.

NOTE 2: For Periodic Report, UE Last Known Location is reported if the UE is in CM-IDLE state when the report is being generated.

Event: Presence-In-AOI-Report

A NF subscribe to this event to receive the current presence state of a UE or a group of UEs or any UE in a specific Area of Interest (AOI), and notification when a specified UE enters or leaves the specified area. The area could be identified by a TA list, a NG-RAN node ID list, a cell ID list, an area ID or specific interested area name like "LADN", or a specific interested area name like "S-NSSAI"(related with an S-NSSAI that is part of a Partially Allowed NSSAI or whose support is restricted to an NS-AoS).

If the AoI includes an S-NSSAI, the AMF shall report the UE presence IN or OUT of an area that support the S-NSSAI, i.e. whether the UE is IN or OUT of TAIs of the Registration Area which support the S-NSSAI (for a partially allowed S-NSSAI) or IN or OUT of the NS-AoS, or the UE presence in the reporting area is UNKNOWN.

For one-time reporting or for the first notification of Continuously reporting, the AMF shall generate the notification as following:

- when the event subscription is targeting a UE or a group of UEs, the AMF shall report the current presence status of the target UE(s);

- when the event subscription is targeting any UE, the AMF shall only report the UEs that are "IN" the Area of Interest (AOI); if no UE is currently "IN" the Area of Interest (AOI), the AMF shall generate a report only including the AnyUe indication (without any UE ID) and the subscribed AOI with the presence status set to "IN". The NF consumer should consider other UEs served by the AMF are "OUT" of the AOI or with "UNKNOWN" state.

In subsequent notifications, the AMF shall only report the UE(s) whose presence status has changed compared to the previous notification sent by the AMF.

The S-NSSAI or the combination of S-NSSAI and the NSI may be used as event filtering information, e.g. by the NWDAF to subscribe to the Total number of UE registered to a S-NSSAI or to a combination of S-NSSAI and NSI ID (see clause 6.3.2A of TS 23.288 [38]).

For an AMF supporting the AOIEF feature (AOI Event Filters, see clause 6.2.8):

- If the event subscription indicates that the AoI may be adjusted based on the UE's Registration Area, the AMF shall report that the UE is IN the AoI if the UE is inside a Registration Area which contains at least one Tracking Area that is contained within the Area of Interest (see clause 5.3.4.4 of TS 23.501 [2] and clauses 4.15.9.3.2, 4.15.9.4, 5.2.2.3.1 and Annex D.1 of TS 23.502 [3]).

- If the subscription to the Presence-In-AOI-Report event includes the "RAN timing synchronization status change event" indication and the UE indicated support for network reconnection due to RAN timing synchronization status change, the AMF shall report the UE presence in AoI based on the most recent N2 connection as described in clause 5.3.4.4 of TS 23.501 [2] and Annex D.1 of TS 23.502 [3].

- If the subscription to the Presence-In-AOI-Report event includes:

  - the notifyForSupiList IE and/or the notifyForGroupList IE, the AMF shall notify the NF service consumer about AOI events only if the event is for a UE belonging to the provided list of SUPIs or for a UE belonging to at least one Internal Group of the provided list of Internal Groups; and/or

  - the notifyForSnssaiDnnList IE, the AMF shall notify the NF service consumer about AOI events only if the event is for a UE having a PDU session established for the provided DNN(s)/S-NSSAI(s).

  The notifyForSupiList IE and the notifyForGroupList IE may only be included in a subscription request targeting Any UE.

  If a subscription to the Presence-In-AOI-Report event targeting Any UE includes the notifyForSnssaiDnnList IE in addition to the notifyForSupiList IE and/or notifyForGroupList IE, the AMF shall notify the NF service consumer about AOI events only if the event is for a UE fulfilling all the related conditions specified above.

UE Type: One UE, Group of UEs, any UE

Report Type: One-Time Report, Continuously Report

Continuous Report should be requested when subscribing to the Presence-In-AOI-Report event, unless the NF service consumer only needs to receive the current presence state.

Input: UE ID(s), "ANY_UE", Area identifier (a TA list, a NG-RAN node ID list, a cell ID list, an area Id, "LADN" or "S-NSSAI"), Adjust AoI based on RA indication, RAN timing synchronization status change indication, optional filters (Notify the NF service consumer only for UEs in the notifyForSupiList, Notify the NF service consumer only for UEs having a PDU session established with a DNN/S-NSSAI in the notifyForSnssaiDnnList IE), S-NSSAI, NSI ID, immediateFlag.

Notification: UE-ID(s), Area identifier, Presence Status (IN/OUT/UNKNOWN)

NOTE 3: If the immediateFlag is set and the AMF can determine the current UE presence state (IN/OUT/UNKNOWN) in the AoI at the time of the subscription (e.g. when the AoI comprises TAIs and the AMF can determine that the UE is IN or OUT of the AoI based on the UE Registration Area), the AMF sends an immediate report including the current UE presence in the AoI. Otherwise, if the immediateFlag is not set or if the AMF cannot determine the current UE presence state in the AoI at the time of the subscription, the AMF sends a subsequent notification with the requested information as soon as the AMF can determine the current UE presence state in the AoI.

Event: Time-Zone-Report

A NF subscribes to this event to receive the current time zone of a UE or a group of UEs, and updated time zone of the UE or any UE in the group when AMF becomes aware of a time zone change of the UE.

UE Type: One UE, Group of UEs

Report Type: One-Time Report, Continuous Report

Input: UE ID(s), immediateFlag

Notification: UE-ID, most recent time-zone

Event: Access-Type-Report

A NF subscribes to this event to receive the current access type(s) of a UE or a group of UEs or any UE, and updated access type(s) of any of the UEs when AMF becomes aware of the access type change of any of these UEs. The area could be identified by a TA list, an area ID or specific interested area name like "LADN".

UE Type: One UE, Group of UEs, any UE

Report Type: One-Time Report, Continuous Report

Input: UE ID(s), "ANY_UE", optionally filters: Area identifier (a TA list, an area Id or "LADN"), immediateFlag

Notification: UE ID, most recent access-types (3GPP, Non-3GPP)

Event: Registration-State-Report

A NF subscribes to this event to receive the current registration state of a UE or a group of UEs or any UE, and report for updated registration state of any of these UEs when AMF becomes aware of a registration state change of any of these UEs. The area could be identified by a TA list, an area ID or specific interested area name like "LADN".

UE Type: One UE, Group of UEs, any UE

Report Type: One-Time Report, Continuous Report

Input: UE ID(s), "ANY_UE", optionally filters: Area identifier (a TA list, an area Id or "LADN"), immediateFlag

Notification: UE ID, most recent registration state (REGISTERED/DEREGISTERED) with access type

Event: Connectivity-State-Report

A NF subscribes to this event to receive the current connection management state of a UE or a group of UEs, and report for updated connection management state of a UE or any UE in the group when AMF becomes aware of a connection management state change of the UE.

UE Type: One UE, Group of UEs

Report Type: One-Time Report, Continuous Report

Input: UE ID(s), immediateFlag

Notification: UE ID, most recent connection management state (IDLE/CONNECTED) with access type

Event: Reachability-Report

A NF subscribes to this event for "UE Reachability Status Change" to receive the current reachability state of a UE or a group of UEs in the AMF, and report for updated reachability state of a UE or any UE in the group when AMF becomes aware of a reachability state change of the UEs between REACHABLE, UNREACHABLE, REGULATORY_ONLY. In this case, the AMF shall report the current reachability state of the UE in a first notification (regardless of whether the immediateFlag is set or not) and then send event reports when the UE reachability state changes as described below:

- the AMF shall send a Reachability Report ("UNREACHABLE") if the Mobile Reachable Timer expires (see clause 5.4.1.1 of TS 23.501 [2]) or the UE enters CM-IDLE when it is only registered over the Non-3GPP access (see clause 5.5.3 of TS 23.501 [2]);
- the AMF shall send a Reachability Report ("REGULATORY_ONLY") if the UE becomes reachable only for regulatory prioritized service (see clause 4.15.4.2 of TS 23.502 [3]);
- the AMF shall send a Reachability Report ("REACHABLE") when the UE reachability state changes from any of the two above states to REACHABLE.

NOTE 4: The AMF does not send a Reachability Report ("UNREACHABLE") in particular when the UE enters extended DRX cycle (see clause 5.31.7.2.2.3 of TS 23.501 [2]), the UE enters power saving state (see clause 5.31.8 of TS 23.501 [2]), the UE enters CM IDLE in MICO mode (see clause 5.4.1.3 of TS 23.501 [2]), or when the UE does not respond to a paging request.

An NF subscribes to this event for "UE Reachable for DL Traffic" to receive reports of a UE or a group of UEs when the UE becomes reachable for sending downlink data. In this case, if the UE is not reachable at the time of the subscription to the event, the event is detected when the UE transitions to CM-CONNECTED mode or when the UE will become reachable for paging, as specified in table 4.15.3.1-1, clauses 4.2.5 and 4.3.3 of TS 23.502 [3]. If the UE is reachable at the time of the subscription to the event, the AMF shall send an event report immediately (regardless of whether the immediateFlag is set or not) in a notification request or in the subscription response (depending on whether the immediateFlag is set to true in the request, the subscription is done on behalf of another NF, and the support of the IERSR feature) as specified in clause 5.3.2.2.2. When reporting the "UE Reachable for DL Traffic", the AMF shall also indicate the access types through which the UE is reachable.

NOTE 5: The AMF does not send an event report for "UE Reachable for DL Traffic" immediately after an UECM Registration in UDM, if the AMF has previously been indicated that reachability event will be detected at UDM. The UDM will detect the UE reachability from the UECM Registration and send a notification to the NF consumer (unless the UDM is indicated that the UE is currently not reachable, as specified in clause 5.3.2.2.2 of TS 29.503 [35]), thus the notification report from AMF is omitted.

UE Type: One UE, Group of UEs

Report Type: One-Time Report, Continuous Report

Continuous Report should be requested when subscribing to "UE Reachability Status Change", unless the NF service consumer only needs to receive the current reachability state.

Continuous Report may be requested when subscribing to "UE Reachability for DL Traffic" with direct reporting

NOTE 6: AMFs not supporting continuous reporting can set in the event subscription response the maximum number of reports to 1.

One-Time Report shall be requested when subscribing to "UE Reachable for DL Traffic" with indirect reporting via the UDM.

Input: UE ID(s), (optional) Reachability Filter, immediateFlag

Notification: UE ID, AMF Id, most recent reachability state (i.e. the REACHABLE, UNRACHABLE or REGULATORY_ONLY state for the "UE Reachability Status Change" event, and the "REACHABLE" state for the "UE Reachable for DL Traffic" event), access type(s) through which the UE is reachable.

Event: Communication-Failure-Report

A NF subscribes to this event to receive the Communication failure report of a UE or group of UEs or any UE, when the AMF becomes aware of a RAN or NAS failure event.

This event implements the "Communication failure" event in table 4.15.3.1-1 of TS 23.502 [3], which is an unexpected termination of the communication.

UE Type: One UE, Group of UEs, any UE

Report Type: One-Time Report, Continuous Report

Input: UE ID(s), "ANY_UE", optionally filters: Area identifier (a TA list, an area Id or "LADN")

Notification: UE ID, RAN/NAS release code.

Event: UEs-In-Area-Report

A NF subscribes to this event to receive the number of UEs in a specific area. A NF may ask AMF for the UEs within the area based on Last Known Location (if the immediateFlag is set) or it may request AMF to actively look for the UEs within the area based on Current Location (if the immediateFlag is not set).

This event implements the "Number of UEs present in a geographical area" event in table 4.15.3.1-1 of TS 23.502 [3].

UE Type: any UE

Input: "ANY_UE", Area identified in a TA List or cell ID list, optionally Ue in Area filters: UE Aerial Indication, Indication of PDU session established for DNN(s) subject to aerial service, indication to omit UE IDs in the event reports, immediateFlag

Report Type: One-Time Report (See NOTE 6), Continuous Report (See NOTE 7), Periodic Report (See NOTE 7)

Notification: Number of UEs in the area, and if eNA is supported also the UE IDs

NOTE 7: For an Immediate Report, UE Last Known Location is used to count the UEs within the area.

NOTE 8: Support of Continuous Report or Periodic Report should be controlled by operator.

Event: Loss-of-Connectivity

An NF subscribes to this event to receive the event report of a UE or group of UEs when AMF detects that a target UE is no longer reachable for either signalling or user plane communication. Such condition is identified when Mobile Reachable timer expires in the AMF (see TS 23.501 [2]), when the UE detaches, when AMF deregisters from UDM for an active UE and when UE indicates Unavailability Period by including Unavailability Period Duration during Mobility Registration or Deregistration procedure. If the UE is already not reachable for either signalling or user plane communication when the event is subscribed, the AMF shall report the event immediately (regardless of whether the immediateFlag is set or not) in a notification request or in the subscription response (depending on whether the immediateFlag is set to true in the request, the subscription is done on behalf of another NF, and the support of IERSR feature) as specified in clause 5.3.2.2.2, and shall include, when applicable, the remaining value of Unavailability Period Duration to determine the foreseen Loss of Connectivity time. If the UE included Start of Unavailability Period, the event is reported when unavailability period starts, else the event is reported immediately, if subscribed.

When the AMF receives two different Unavailability Period Duration values, i.e. one reported by the UE and another one known to the AMF as specified in clause 5.4.13.3 of TS 23.501 [2], the AMF decides on the Unavailability Period as specified in clause 5.4.1.4 of TS 23.501 [2].

This event implements the "Loss of Connectivity" event in table 4.15.3.1-1 of TS 23.502 [3].

UE Type: One UE, Group of UEs.

Report Type: One-Time Report, Continuous Report

Input: UE ID(s), immediateFlag

Notification: UE ID, optionally Unavailability Period Duration.

Event: 5GS-User-State-Report

A NF subscribes to this event to receive the 5GS User State of a UE.

UE Type: One UE

Report Type: One-Time Report

Input: UE ID(s)

Notification: UE ID, 5GS User State

Event: Availability-after-DDN-failure

A NF subscribes to this event to be notified about the Availability of a UE after a DDN failure.

UE Type: One UE, Group of UEs

Report Type: One-Time Report, Continuous Report

Input: UE ID(s)

Notification: UE ID(s)

Event: Type-Allocation-Code-Report

A NF subscribes to this event to receive the TAC of a UE or a group of UEs or any UE.

UE Type: One UE, Group of UEs, any UE

Report Type: One-Time Report, Continuous Report

Input: UE ID(s),"ANY_UE", optionally filters: TAI, Area identifier (a TA list, an area Id or "LADN")

Notification: UE ID(s), TAC(s)

Event: Frequent-Mobility-Registration-Report

A NF subscribes to this event to receive the number of mobility registration during a period for a UE or a group of UEs or any UE.

UE Type: One UE, Group of UEs, any UE

Report Type: One-Time Report, Continuous Report

Input: UE ID(s), expiry time, "ANY_UE", optionally filters: Area identifier (a TA list, an area Id or "LADN")

Notification: UE ID(s), Frequent Registration

Event: Snssai-TA-Mapping-Report

A NF subscribes to this event to receive the related access type and the list of supported S-NSSAIs.

UE Type: any UE

Report Type: One-Time Report, Continuous Report

Input: Target Area: TA list or "ANY_TAI", optionally filters: S-NSSAI(s)

Notification: Access type, list of supported S-NSSAIs with an indication of restriction at the AMF

Event: UE-Access-Behavior-Trends

A NF subscribes to this event to receive the UE access behavior trends (e.g. access type change, handover, etc.) within a period for a UE or a group of UEs, as specified in clause 4.15.4.2 of TS 23.502 [3].

UE Type: One UE, Group of UEs

Report Type: Periodic Report

Input: UE ID(s), expiry time

Notification: UE ID(s), UE access behavior trends report.

Event: UE-Location-Trends

A NF subscribes to this event to receive the UE Location Trends within a period for a UE or a group of UEs, as specified in clause 4.15.4.2 of TS 23.502 [3].

UE Type: One UE, Group of UEs

Report Type: Periodic Report

Input: UE ID(s), expiry time, Dispersion Area (e.g. a TAI list).

Notification: UE ID(s), UE Location Trends report.

Event: UE-MM-Transaction-Report

A NF subscribes to this event to receive the Total Number of Mobility Management transactions during a period for a UE or a group of UEs, as specified in clause 5.2.2.3.1 of TS 23.502 [3]. The Total number of transactions is incremented when the NAS signalling transactions from Authentication, Registration, De-Registration, Service Request and UE Configuration Update procedures is completed

UE Type: One UE, Group of UEs

Report Type: Periodic Report

Input: UE ID(s), expiry time, filters: Dispersion Area (e.g. a TA list) or Slice filter (i.e. a list of S-NSSAIs)

Notification: UE ID(s), List of UE transaction numbers per location or List of UE transaction numbers per slice.

NOTE 9: The immediateFlag parameter does only apply to the events listed above including the immediateFlag within the list of input parameters. If the immediateFlag is received in a subscription for an event for which the immediateFlag does not apply, it is ignored by the AMF.

## 5.3.2 Service Operations

### 5.3.2.1 Introduction

For the Namf_EventExposure service the following service operations are defined:

- Subscribe;
- Unsubscribe;
- Notify.

### 5.3.2.2 Subscribe

#### 5.3.2.2.1 General

The Service Operation is used by a NF Service Consumer (e.g. NEF) to subscribe to an event(s) for one UE, group of UE(s) or any UE.

#### 5.3.2.2.2 Creation of a subscription

The Subscribe service operation is invoked by a NF Service Consumer, e.g. NEF, towards the AMF, when it needs to create a subscription to monitor at least one event relevant to the AMF. The NF Service Consumer may subscribe to multiple events in a subscription. A subscription may be associated with one UE, a group of UEs or any UE.

The NF Service Consumer shall request to create a new subscription by using HTTP method POST with URI of the subscriptions collection, see clause 6.2.3.2.

The NF Service Consumer shall include the following information in the HTTP message body:

- NF ID, indicates the identity of the network function instance initiating the subscription;
- Subscription Target, indicates the target(s) to be monitored, as one of the following types:
  - A specific UE, identified with a SUPI, a PEI or a GPSI;
  - A group of UEs, identified with a group identity;
  - Any UE, identified by the "anyUE" flag.
- Notification URI, indicates the address to deliver the event notifications generated by the subscription;
- Notification Correlation ID, indicates the correlation identity to be carried in the event notifications generated by the subscription;
- List of events to be subscribed;
- Event Types per event, as specified in clause 5.3.1.

The NF Service Consumer may include the following information in the HTTP message body:

- Immediate Report Flag per event, indicates an immediate report to be generated with current event status;
- Event Trigger, indicates how the events shall be reported (One-time Reporting or Continuously Reporting).
- Maximum Number of Reports, defines the maximum number of reports after which the event subscription ceases to exist;
- Expiry, defines maximum duration after which the event subscription ceases to exist;
- Sampling ratio, defines the random subset of UEs among target UEs, and AMF only report the event(s) related to the selected subset of UEs;
- partitioning criteria, that defines Criteria for partitioning UEs before applying sampling ratio;
- Periodic Report Flag per event, indicates the report to be generated periodically;
- Repetition Period, defines the period for periodic reporting;
- Variable reporting periodicity information, defines the list of conditions related to Reporting periodicity and the period per condition.
- Event Filters per applicable event, defines further options on when/how the event shall be reported;
- Reference Id per event, indicates the value of the Reference Id associated with the event to be monitored. If provided, the Reference Id shall be included in the reports triggered by the event;
- a notification flag as "notifFlag" attribute if the EneNA feature is supported; and/or
- Muting Exception Instructions, which specify instructions to apply to the subscription and the stored events when an exception occurs at the AMF while the event is muted (e.g., the buffer of stored event reports is full, or the number of stored event reports exceeds a certain number), if the ENAPH3 feature is supported (see clause 6.2.8).

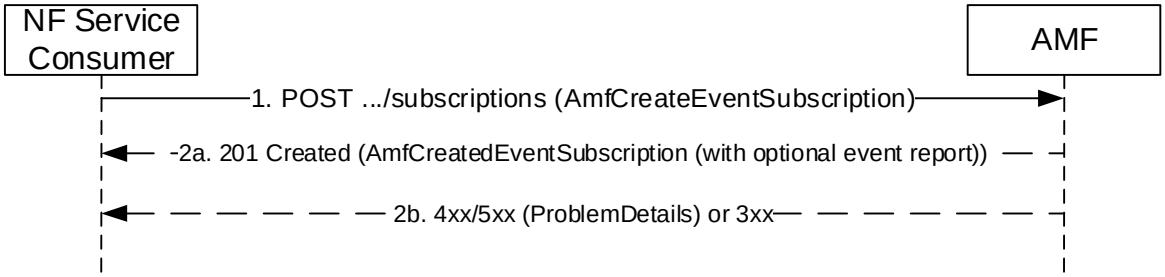



**Figure 5.3.2.2.2-1 Subscribe for Creation**

1. The NF Service Consumer shall send a POST request to create a subscription resource in the AMF. The content of the POST request shall contain a representation of the individual subscription resource to be created. The request may contain an expiry time, suggested by the NF Service Consumer as a hint, representing the time upto which the subscription is desired to be kept active and the time after which the subscribed event(s) shall stop generating report.

2a. On success, the request is accepted, the AMF shall include a HTTP Location header to provide the location of a newly created resource (subscription) together with the status code 201 indicating the requested resource is created in the response message. If the NF Service Consumer has included more than one events in the event subscription and some of the events are failed to be subscribed, the AMF shall accept the message and provide the successfully subscribed event(s) in AmfEventSubscription. If the NF Service Consumer has included the immediateFlag with value as "true" in the event subscription, the AMF shall include the current status of the events subscribed, if available (e.g. last known location information is included if the subscribed event is LOCATION_REPORT). If the events with immediateFlag set to "true" are subscribed by an NF service consumer on behalf of a third NF and the NF service consumer has not indicated supporting of IERSR feature (see 6.2.8), the notification will be sent to the third NF directly, i.e. subsChangeNotifyUri is included in the event subscription, the current status of the events subscribed shall not be included in response. The AMF shall subsequently send a notification to the third NF including the current status of the events subscribed.

   If the NF Service Consumer has set the event reporting option as ONE_TIME and if the AMF has included the current status of the events subscribed in the response, then the AMF shall not do any subsequent event notification for the events given in the AmfCreateEventSubscription parameter. If the NF Service Consumer has set the event reporting option as ONE_TIME, the subscribed event as LOCATION_REPORT and the immediateFlag is set to false or absent, the AMF shall send an event notification to notify the current location of the UE after the subscription; if the UE is in RM-REGISTERED and CM-IDLE state over 3GPP access and the UE does not respond to the paging, or if the UE is in RM-REGISTERED over non-3GPP access, the event notification shall include the last known location and the ageOfLocationInformation IE set to a value other than "0", which indicates to the NF service consumer that the AMF returned the last known location.

   If the NF Service Consumer has set the CONTINUOUS or PERIODIC event reporting option, the subscribed event as LOCATION_REPORT and the immediateFlag is set to false or absent, the AMF shall send a first event notification to notify the current location of the UE after the subscription is created and then subsequent event notifications when the user location changes or according to the requested period respectively; if at the time of the subscription creation the UE is in RM-REGISTERED and CM-IDLE state over 3GPP access and the UE does not respond to the paging, or if the UE is in RM-REGISTERED over non-3GPP access, the AMF shall send the first event notification including the last known location and the ageOfLocationInformation IE set to a value other than "0", which indicates to the NF service consumer that the AMF returned the last known location.

   The response, based on operator policy and taking into account the expiry time included in the request, may contain the expiry time, as determined by the AMF, after which the subscription becomes invalid. Once the subscription expires, if the NF Service Consumer wants to keep receiving notifications, it shall create a new subscription in the AMF. The AMF shall not provide the same expiry time for many subscriptions in order to avoid all of them expiring and recreating the subscription at the same time. If the expiry time is not included in the response, the NF Service Consumer shall consider the subscription to be valid without an expiry time.

   If the sampling ratio ("sampRatio") attribute is included in the subscription without a partitioningCriteria, the AMF shall select a random subset of UEs among target UEs according to the sampling ratio and only report the event(s) related to the selected subset of UEs. If the partitioningCriteria attribute is also included along with sampling ratio, the AMF shall apply the sampling ratio on the group of UEs determined according to the partitioning criteria.

If the AMF supports the EneNA feature and the "notifFlag" attribute is included and set to "DEACTIVATE" in the request (by e.g. the NWDAF or DCCF), the AMF shall mute the event notification and store the available events. Additionally, if the AMF also supports the ENAPH3 feature (see clause 6.2.8) and the NF service consumer also included event muting instructions in the request, the AMF should evaluate the received event muting instructions against to local actions (if configured) and, if the subscription creation request is accepted, the AMF may indicate the following information to the NF service consumer in the response:

- the maximum number of notifications that the AMF expects to be able to store for the subscription;
- an estimate of the duration for which notifications can be buffered.

If the NF service consumer is a UDM, the AMF and the UDM both support the "ESSYNC" feature and the subscription is targeting a specific UE with Reference Id(s) included in the subscription, the AMF shall locally store the information that the event subscription is subject to the Event Subscription Synchronization with UDM during EPS to 5GS mobility as specified in clause 5.3.2.4.2. During inter-AMF mobility procedures, the source AMF shall include the "eventSyncInd" IE (in AmfEventSubscriptionAddInfo data type) with the value "true" in the UE Context for the event subscriptions that are subject to Event Subscription Synchronization with UDM.

If the subscription creation request targets a group of UE or any UE, the AMF shall accept the request and create a subscription even if the AMF does not currently serve any UE of the group or any UE respectively, unless other reasons exist to reject the request.

2b. On failure or redirection, one of the HTTP status code listed in Table 6.2.3.2.3.1-3 shall be returned. For a 4xx/5xx response, the message body shall contain a ProblemDetails structure with the "cause" attribute set to one of the application error listed in Table 6.2.3.2.3.1-3.

If the subscription creation request targets a specific UE and this UE is not served by the AMF (i.e. it is not known to the AMF), the AMF shall reject the request with a 403 Forbidden response and the application error "UE_NOT_SERVED_BY_AMF", unless the request can be redirected to another AMF known to serve the UE (e.g. another AMF of the same AMF set).

If the AMF supports the EneNA and ENAPH3 features (see clause 6.2.8), the NF service consumer sets the "notifFlag" attribute to "DEACTIVATE" and event muting instructions in the request, but the AMF cannot accept the received instructions, the AMF may reject the request with a 403 Forbidden response and the application error "MUTING_EXC_INSTR_NOT_ACCEPTED".

#### 5.3.2.2.3 Modification of a subscription

The Subscribe service operation is invoked by a NF Service Consumer, e.g. NEF, towards the AMF, when it needs to modify an existing subscription previously created by itself at the AMF.

The NF Service Consumer shall modify the subscription by using HTTP method PATCH with the URI of the individual subscription resource (see clause 6.2.3.3) to be modified.

See also Figure 5.3.2.2.3-1 below.

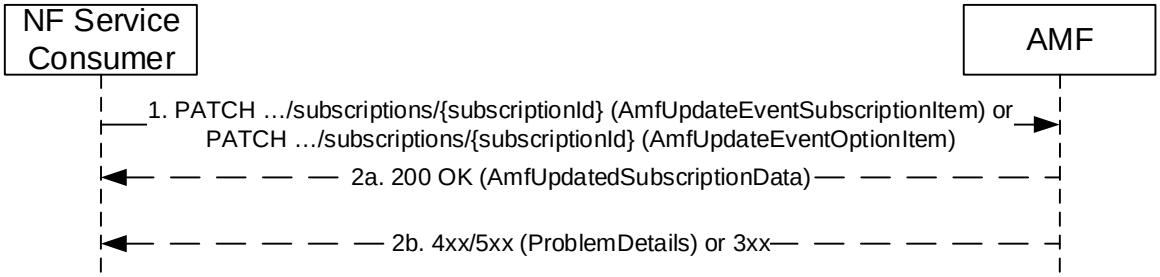



**Figure 5.3.2.2.3-1 Modification of a Subscription**

1. The NF Service Consumer shall send a PATCH request to modify a subscription resource in the AMF. The modification may be for the events subscribed or for updating the event options.

2a. On success, the request is accepted, the AMF shall return the representation of the modified subscription resource or its sub-resource together with the status code 200 OK. When the PATCH request is for modifying the expiry attribute of the options IE of the subscription, then the AMF based on operator policies and taking into

account the expiry time included in the request, shall include an expiry time, after which the subscription becomes invalid. Once the subscription expires, if the NF Service Consumer wants to keep receiving notifications, it shall create a new subscription in the AMF, as specified in clause 5.3.2.2.2. The AMF shall not provide the same expiry time for many subscriptions in order to avoid all of them expiring and recreating the subscription at the same time.

The PATCH request may be used to modify the "notifFlag" attribute of the options IE of the subscription, when both the AMF and NF Service Consumer support the EneNA feature as defined in clause 6.2.8:

- if the "notifFlag" attribute is set to "DEACTIVATE" in the request and the event notification is currently not muted, the AMF shall mute the event notification and store the available events, or
- if the "notifFlag" is set to "RETRIEVAL" in the request and the event notification is currently muted, the AMF shall send the stored events to the NF service consumer, mute the event notification again and store available events; or
- if the "notifFlag" is set to "ACTIVATE" in the request and the event notification is currently muted, the AMF shall send the stored events to the NF service consumer and stop muting the event notification.

In addition, if both the AMF and the NF service consumer (e.g. NWDAF or DCCF) also support the ENAPH3 feature (see clause 6.2.8), the PATCH request modifies the "notifFlag" attribute to "DEACTIVATE" and contains muting exception instructions, the AMF should evaluate the received event muting instructions against to local actions (if configured) and, if the subscription modification request is accepted, the AMF may indicate the following information to the NF service consumer in the response:

- the maximum number of notifications that the AMF expects to be able to store;
- an estimate of the duration for which notifications can be buffered.

If the subscription requested to be modified targets a group of UE or any UE, the AMF shall accept the request and modify the subscription even if the AMF does not currently serve any UE of the group or any UE respectively, unless other reasons exist to reject the request.

2b. On failure or redirection, one of the HTTP status codes listed in Table 6.2.3.3.3.1-3 shall be returned. For a 4xx/5xx response, the message body shall contain a ProblemDetails structure with the "cause" attribute set to one of the application errors listed in Table 6.2.3.3.3.1-3.

If the AMF cannot accept the received muting exception instructions, the AMF may reject the request with a 403 Forbidden response and the application error "MUTING_EXC_INSTR_NOT_ACCEPTED".

#### 5.3.2.2.4 Remove or add group member UE(s) for a group subscription

The Subscribe service operation is invoked by a NF Service Consumer, e.g. UDM, towards the AMF, to remove or add group member UE(s) for an existing group subscription.

The NF Service Consumer shall modify the subscription by using HTTP method PATCH with the URI of the individual subscription resource (see clause 6.2.3.3) to be modified.

See also Figure 5.3.2.2.4-1 below.

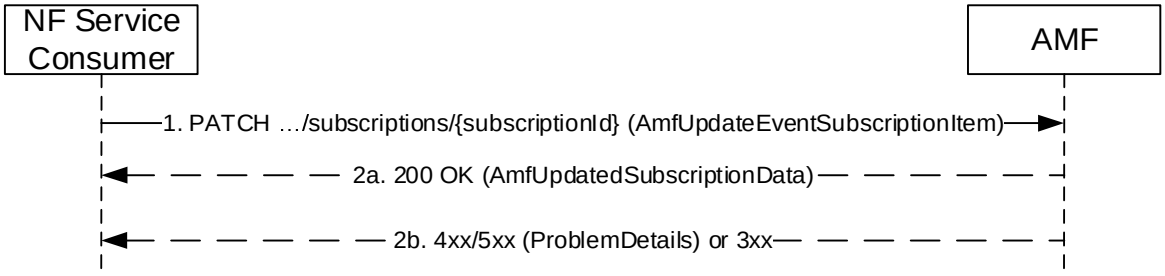



**Figure 5.3.2.2.4-1 Remove or add group member UE(s) for a group subscription**

1. The NF Service Consumer shall send a PATCH request to modify a subscription resource targeting a group of UEs in the AMF. The modification shall indicate the group member UE(s) to be excluded or added for the group subscription.

2a. On success, the request is accepted, the AMF shall return the representation of the modified subscription resource with the status code 200 OK.

The AMF shall stop monitoring events for excluded member UE(s). If Maximum number of Reports is applied, the AMF shall set the number of reports of the indicated UE(s) to Maximum Number of Reports.

2b. On failure or redirection, one of the HTTP status code listed in Table 6.2.3.3.3.1-3 shall be returned. For a 4xx/5xx response, the message body shall contain a ProblemDetails structure with the "cause" attribute set to one of the application error listed in Table 6.2.3.3.3.1-3.

### 5.3.2.3 Unsubscribe

#### 5.3.2.3.1 General

The Unsubscribe service operation is invoked by a NF Service Consumer, e.g. NEF, towards the AMF, to remove an existing subscription previously created by itself at the AMF.

The NF Service Consumer shall unsubscribe to the subscription by using HTTP method DELETE with the URI of the individual subscription resource (see clause 6.2.3.3) to be deleted.

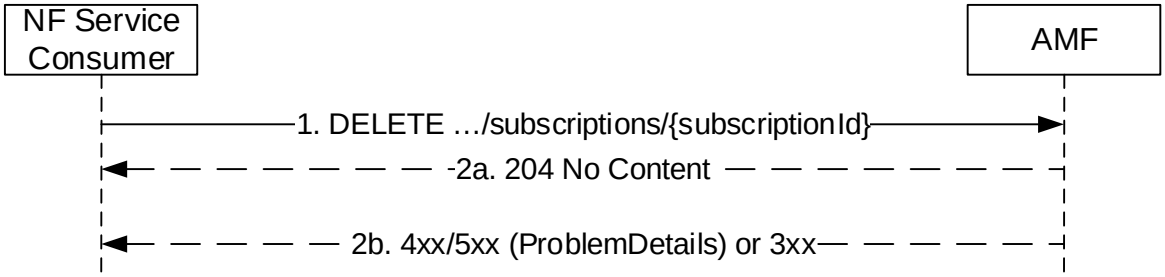



**Figure 5.3.2.3.1-1 Unsubscribe a subscription**

1. The NF Service Consumer shall send a DELETE request to delete an existing subscription resource in the AMF.

2a. On success, the request is accepted, the AMF shall reply with the status code 204 indicating the resource identified by subscription ID is successfully deleted in the response message.

2b. On failure or redirection, one of the HTTP status code listed in Table 6.2.3.3.3.2-3 shall be returned. For a 4xx/5xx response, the message body shall contain a ProblemDetails structure with the "cause" attribute set to one of the application error listed in Table 6.2.3.3.3.2-3.

### 5.3.2.4 Notify

#### 5.3.2.4.1 General

The Notify service operation is invoked by the AMF, to send a notification, towards the notification URI, when certain event included in the subscription has taken place.

The AMF shall use the HTTP method POST, using the notification URI received in the subscription creation as specified in clause 5.3.2.2.2, including e.g. the subscription ID, Event ID(s) for which event has happened, notification correlation ID provided by the NF service consumer at the time of event subscription, to send a notification. See Figure 5.3.2.4.1-1.

Additionally, the Notify service operation shall also be invoked by the AMF, when:

- there is a change of AMF during UE mobility procedures, if the subscription Id changes (i.e. Registration procedures and Handover procedures), or

- the subscription is terminated by the AMF, if event subscription termination notification is requested by the NF consumer.

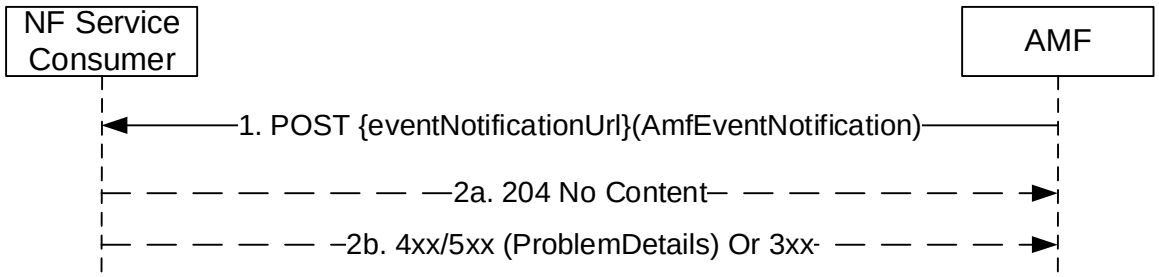



**Figure 5.3.2.4.1-1 Notify**

1. The AMF shall send a POST request to send a notification.

2a. On success, "204 No content" shall be returned by the NF Service Consumer.

2b. On failure or redirection, the appropriate HTTP status code (e.g. "403 Forbidden") indicating the error shall be returned and appropriate additional error information should be returned.

When the AMF received the following response code (and application error), the AMF should consider the subscription is no longer valid and terminate the subscription:

- "400 Bad Request" with application error "RESOURCE_CONTEXT_NOT_FOUND"
- "404 Not Found"

When AMF terminates the subscription in above scenarios, if the subscription is created by the NF consumer on behalf of another NF (e.g. the UDM subscribes to the AMF on behalf of the NEF) and notification of event subscription termination is requested by the NF consumer, the AMF supporting the 'STEN" feature shall send a notification to the NF consumer (e.g. the UDM) to report the termination of the subscription.

#### 5.3.2.4.2 Event Subscription Synchronization for specific UE

When the AMF and the UDM both support the "ESSYNC" feature, the AMF may initiate synchronization for event subscriptions with the UDM for the specific UE during EPS to 5GS mobility registration procedure (see clause 4.11.5.2 of TS 23.502 [3]), if UE specific event subscriptions from the UDM are available in UE Context.

To initiate event subscription synchronization, when sending notification for subscription change to the UDM, the AMF shall include the event subscription information in the notification request. If subscription change notification is not needed, e.g. when UE registers to the same AMF after moving from EPS, the AMF may send a notification to the subscription change notification URI. The notification request in this case only includes the event subscription information but no event report list.

The AMF shall only include active event subscriptions that are subject to Event Subscription Synchronization with UDM (determined as defined in clause 5.3.2.2.2) in the event subscription information.

For each active subscription, the following information shall include:

- URI of the subscription resource in the AMF; and
- Notification Correlation Id of the subscription; and
- list of Reference Ids, one per event in the subscription; and
- optionally, the URI of old subscription resource on the source AMF, if the subscription Id is changed during the mobility procedure.

When the UDM receives event subscription information from AMF, the UDM shall compare the active event subscriptions in AMF with the active UDM Event Exposure subscriptions using Reference Id(s) and Notification Correlation Id, and perform the following:

- if an event is to be detected by AMF but not existing in the AMF, the UDM shall subscribe the event in AMF by creating a new AMF event subscription or updating an existing AMF event subscription;
- if an event exists in AMF but does not exist in UDM, the UDM shall unsubscribe the event from AMF by removing or update an AMF event subscription.

When the AMF identified that event synchronization with UDM is required, but either the UDM or the AMF or both do not support the "ESSYNC" feature, the AMF may require the UDM to re-subscribe the stored event exposure subscriptions to the AMF, see clause 5.3.2.2.2 and 5.3.2.2.3 of TS 29.503 [35].

# 5.4 Namf_MT Service

## 5.4.1 Service Description

Namf_MT service allows a NF to request information related to capabilities to send MT signalling or data to a target UE. The following are the key functionalities of this NF service:

- enabling UE reachability by:
  - paging the UE if the UE is in CM-IDLE state and responding to the requester NF after the UE enters CM-CONNECTED state, or
  - responding to the requester NF if UE is in CM-CONNECTED state.
- providing the terminating domain selection information for IMS voice to the consumer NF.
- enabling reachability of a list of UEs by:
  - paging UEs for an MBS session if the UEs are in CM-IDLE state, and
  - responding to the requester NF, including the list of UEs that are already in CM-CONNECTED state if any, and
  - sending notification with the UE reachability information and user location information to NF consumers.

## 5.4.2 Service Operations

### 5.4.2.1 Introduction

For the Namf_MT Service the following service operations are defined:

- EnableUEReachability
- ProvideDomainSelectionInfo
- EnableGroupReachability
- UEReachabilityInfoNotify

### 5.4.2.2 EnableUEReachability

#### 5.4.2.2.1 General

The EnableUEReachability service operation is used in the following procedure:

- MT SMS over NAS in CM-IDLE state (see TS 23.502 [3], clause 4.13.3.6), or in CM-CONNECTED state (see TS 23.502 [3], clause 4.13.3.7).
- UPF anchored Mobile Terminated Data Transport in Control Plane CIoT 5GS Optimisation (see clause 4.24.2 of TS 23.502 [3]).

- Network Triggered Connection Resume in RRC Inactive with CN based MT communication handling (see clause 4.8.2.2b of TS 23.502 [3]).

The EnableUEReachability service operation shall be invoked by the NF Service Consumer (e.g. SMSF, SMF) to enable the reachability of the UE.

The NF Service Consumer shall invoke the service by using the HTTP method PUT, towards the URI of a "ueReachInd" resource as specified in clause 6.3.3.2. See also figure 5.4.2.2.1-1.

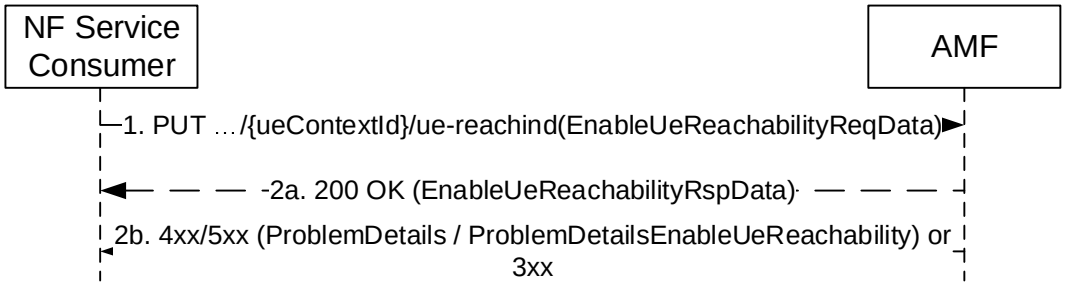



**Figure 5.4.2.2.1-1: NF Service Consumer enables the reachability of the UE**

1. The NF Service Consumer sends a PUT request to the resource representing the ueReachInd resource of the AMF. The content of the PUT request shall contain an "EnableUeReachabilityReqData" object.

   During the Network Triggered Connection Resume in RRC Inactive with CN based MT communication handling (see clause 4.8.2.2.b of TS 23.502 [3]), the SMF may include the ppi, the arp, the qfi, the 5qi and the DL data size of the received DL packets per QoS flow of the PDU session for which DL packets are received, together with the PDU session identifier, to enable NG-RAN to take this information into account when paging the UE.

   The SMF shall send a new Namf_MT_EnableUEReachability request with a higher priority or a different Paging Policy Indicator to the AMF if, while the SMF is waiting for the response for UE connection resume from the AMF as specified in clause 4.8.2.2b of TS 23.502 [3] or the SMF has received a reject response with Estimated Maximum Wait time from the AMF:

   - the SMF receives any additional Data Notification from the UPF for data packets pertaining to another QoS Flow associated with a higher priority (i.e. ARP priority level) than the priority indicated to the AMF in the previous Namf_MT_EnableUEReachability request (for the case of DL data buffering in UPF);

   - the SMF receives any additional DL data packets pertaining to another QoS Flow associated with a higher priority (i.e. ARP priority level) than the priority indicated to the AMF in the previous Namf_MT_EnableUEReachability request (for the case of DL data buffering in SMF); or

   - the SMF derives a different Paging Policy Indicator according to the additional Data Notification or the additional DL data packets.

   Based on local configuration, the SMF may send a new Namf_MT_EnableUEReachability message to AMF if:

   - the SMF receives any additional Data Notification messages for data packets pertaining to another QoS Flow associated with same or lower priority than the priority indicated to the AMF in the previous Namf_MT_EnableUEReachability (for the case of DL data buffering in UPF); or

   - the SMF receives any additional DL data packets pertaining to another QoS Flow associated with same or lower priority than the priority indicated to the AMF in the previous Namf_MT_EnableUEReachability(for the case of DL data buffering in SMF).

   Before the UE becomes reachable, the AMF may receive several Namf_MT_EnableUEReachability requests which are triggered by DL data reports for different QoS flows, the AMF shall store or update the ppi, the arp, the qfi, the 5qi and the DL data size per QoS flow if included and may provide the received ppi(s), arp(s), qfi(s), 5qi(s) and DL data size(s) for each QoS flow to the NG-RAN when paging the UE based on the local configuration.

2a. On success:

- if the UE is in CM-CONNECTED state, the AMF shall immediately respond using "200 OK" status code, with content containing an "EnableUeReachabilityRspData" object.

- if the UE is in CM-IDLE state and the NAS message is to be sent over via 3GPP access and paging is not restricted as defined in TS 23.501 [2] clause 5.38.5, the AMF shall page the UE. When UE becomes CM-CONNECTED and the UE has not rejected the page as specified in TS 23.501 [2] clause 5.38.4, "200 OK" shall be returned with content containing an "EnableUeReachabilityRspData" object.

- if the UE is in Extended DRX for RRC-INACTIVE state and with CN based MT communication handling, and the AMF determines that the UE is reachable, then the AMF shall send a N2 RAN paging request message to NG-RAN with the request for the UE's RRC connection to be resumed as specified in clause 4.8.2.2b of TS 23.502 [3]). When an N2 Notification is received by the AMF indicating that the UE is in RRC-CONNECTED state or indicating in a MT Communication Handling request that the UE is reachable for downlink data and/or signalling as specified in clause 4.8.2.2 of TS 23.502 [3], "200 OK" shall be returned with the content containing an "EnableUeReachabilityRspData" object.

2b. On failure or redirection, one of the HTTP status code listed in Table 6.3.3.2.3.1-3 shall be returned. For a 4xx/5xx response, the message body shall contain a ProblemDetails or ProblemDetailsEnableUeReachability structure with the "cause" attribute set to one of the application error listed in Table 6.3.3.2.3.1-3.

The AMF shall respond with the status code "403 Forbidden", if the UE is in a Non-Allowed Area and the service request is not for regulatory prioritized service. The AMF shall set the application error as "UE_IN_NON_ALLOWED_AREA" in POST response body.

The AMF shall respond with the status code "409 Conflict", if Paging Restriction Information restrict the EnableUEReachability request from causing paging as defined in TS 23.501 [2] clause 5.38.5 or if the UE rejects the paging as defined in TS 23.501 [2] clause 5.38.4. The AMF shall set the application error as "REJECTION_DUE_TO_PAGING_RESTRICTION" in POST response body.

The AMF shall respond with the status code "504 Gateway Timeout" and set the application error as "UE_NOT_REACHABLE" and include an Estimated Maximum Wait time in POST response body when the AMF determines the UE is unreachable (e.g. if the UE is in MICO mode or the UE has entered Extended DRX in CM-IDLE or Extended DRX for RRC-INACTIVE state) as specified in clauses 4.24.2 and 4.8.2.2b of TS 23.502 [3]), and:

- if the UE is in Extended DRX for RRC-INACTIVE state and with CN based MT communication handling,when the AMF determines that the UE is reachable, the AMF shall send a N2 RAN paging request message to NG-RAN with the request for the UE's RRC connection to be resumed as specified in clause 4.8.2.2b of TS 23.502 [3]) using the information received in the EnableUeReachabilityReqData (i.e. the ppi, the arp, the qfi, the 5qi and the DL data size of the received DL packets per QoS flow of the PDU session for which DL packets are received, together with the PDU session identifier); or

- if the UE is in Extended DRX in CM-IDLE state, and the AMF determines that the UE is reachable, the AMF shall page the UE (i.e. using CN paging).

### 5.4.2.3 ProvideDomainSelectionInfo

#### 5.4.2.3.1 General

The ProvideDomainSelectionInfo service operation shall be invoked by the NF Service Consumer (e.g. UDM) to get the UE information for terminating domain selection of IMS voice, including following information:

- Indication of supporting IMS voice over PS Session;

- Time stamp of the last radio contact with the UE;

- Current Access type and RAT type

The NF Service Consumer shall invoke the service by using the HTTP GET towards the URI of the "UeContext" resource (See clause 6.3.3.3.3.1). See also figure 5.4.2.3.1-1.

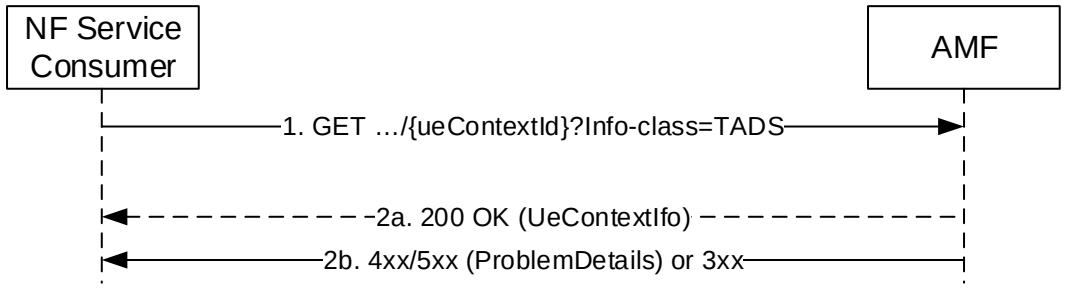



**Figure 5.4.2.3.1-1: Provide UE Information for Terminating Domain Selection**

1. The NF Service Consumer shall send a GET request to the URI of the "UeContext" resource on the AMF, with query parameter "info-class" set to value "TADS".

2a. On success, the AMF shall return "200 OK" status code with content containing an "UeContextInfo" data structure including UE information for terminating domain selection for IMS voice.

2b. On failure or redirection, one of the HTTP status code listed in Table 6.3.3.3.3.1-3 shall be returned. The message body shall contain a ProblemDetails object with "detail" set to one of the corresponding application errors listed in Table 6.3.3.3.3.1-3.

If the request cannot be completed due to a Registration procedure going-on for the target UE, the AMF shall reject the request with a 409 Conflict response and with the TEMPORARY_REJECT_REGISTRATION_ONGOING application error. The NF Service Consumer should repeat the request after a suitable delay.

If the request cannot be completed due to the target UE being in RM-DEREGISTERED state, the AMF shall reject the request with a 403 Forbidden response and with the UE_DEREGISTERED application error.

### 5.4.2.4 EnableGroupReachability

#### 5.4.2.4.1 General

The EnableGroupReachability service operation is used in the following procedure:

- MBS session activation procedure (see TS 23.247 [55], clause 7.2.5.2).

The EnableGroupReachability service operation shall be invoked by the NF Service Consumer (e.g. SMF) to enable the reachability of the list of UEs involved in the MBS Session.

The NF Service Consumer shall invoke the service by using the HTTP method POST (enable-group-reachability custom operation) as shown in figure 5.4.2.4.1-1.

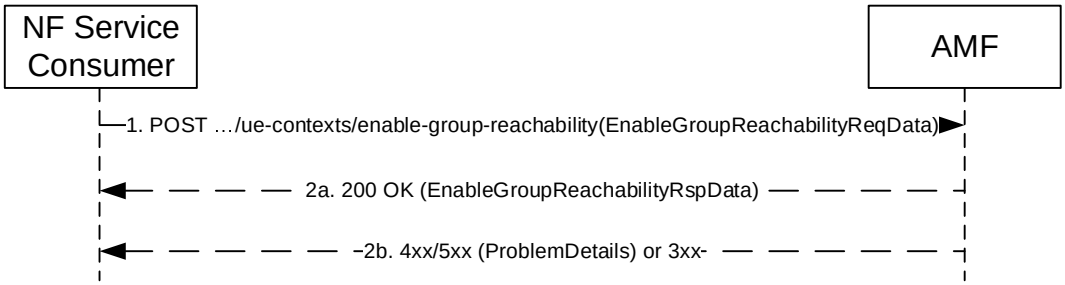



**Figure 5.4.2.4.1-1: NF Service Consumer enabling the reachability of a list of UEs**

1. The NF Service Consumer shall send a POST request to the resource representing the UeContexts resource of the AMF. The content of the POST request shall contain an "EnableGroupReachabilityReqData" object.

2a. On success:

If at least one UE in the list of UEs included in EnableGroupReachabilityReqData is in CM-CONNECTED state, the AMF shall respond using "200 OK" status code, with the content containing the list of UEs in CM-CONNECTED state in "EnableGroupReachabilityRspData" object; or

If all the UEs in the list of UEs included in EnableGroupReachabilityReqData are in CM-IDLE state, the AMF shall respond with "200 OK" status code.

The AMF shall page UEs in CM-IDLE state as specified in clause 7.2.5.2 of TS 23.247 [55].

2b. On failure or redirection, one of the HTTP status code listed in Table 6.3.3.4.4.2.2-2 shall be returned. For a 4xx/5xx response, the message body shall contain a ProblemDetails structure with the "cause" attribute set to one of the application error listed in Table 6.3.3.4.4.2.2-2.

### 5.4.2.5 UEReachabilityInfoNotify

#### 5.4.2.5.1 General

The UEReachabilityInfoNotify service operation is used in the following procedure:

- MBS session activation procedure (see TS 23.247 [55], clause 7.2.5.2).

The UEReachabilityInfoNotify service operation shall be invoked by the AMF to send a notification towards the notification URI for the UE(s) which are reachable or do not respond to paging.

The AMF shall use the HTTP method POST, using the notification URI received in the EnableGroupReachability request as specified in clause 5.4.2.4.1, to send a notification. See Figure 5.4.2.5.1-1.

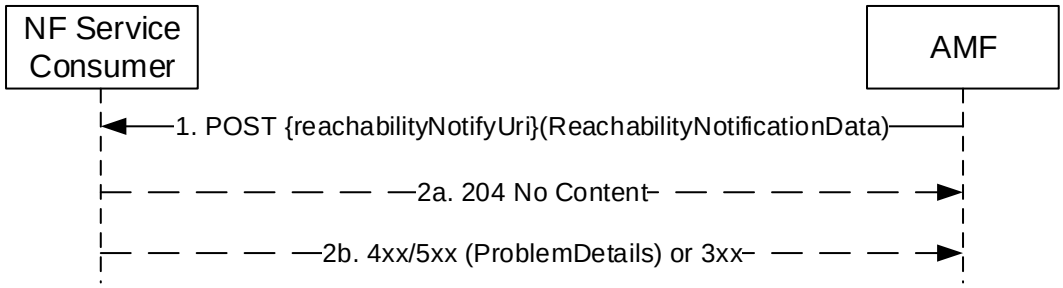



**Figure 5.4.2.5.1-1: UE Reachability Info Notify**

1. The AMF shall send a POST request to send a notification.

2a. On success, "204 No content" shall be returned by the NF Service Consumer.

2b. On failure or redirection, the appropriate HTTP status code (e.g. "403 Forbidden") indicating the error shall be returned and appropriate additional error information should be returned.

## 5.5 Namf_Location Service

### 5.5.1 Service Description

The Namf_Location service is used by NF service consumers to request the AMF for initiating positioning requests and provide the location information. It is also used to subsequently notify the location change events towards the NF service consumers. The following are the key functionalities of this NF service:

- Allow NFs to request the current geodetic and optionally local and/or civic location of a target UE.
- Allow NFs to be notified of event information related to emergency sessions.
- Allow NFs to request Network Provided Location Information (NPLI) and/or local time zone corresponding to the location of a target UE.

- Allow NFs to request the ranging and sidelink positioning location results for a group of n UEs (n≥2), the ranging and sidelink positioning location results may include absolute locations, relative locations or distance and directions related to the UEs.
- Allow NFs to enable the location reporting over user plane.

## 5.5.2 Service Operations

### 5.5.2.1 Introduction

For the Namf_Location Service the following service operations are defined:

- ProvidePositioningInfo;
- EventNotify; and
- ProvideLocationInfo.
- CancelLocation

### 5.5.2.2 ProvidePositioningInfo

#### 5.5.2.2.1 General

The ProvidePositioningInfo service operation is used in the following procedure:

- 5GC-MT-LR Procedure without UDM Query (see TS 23.273 [42], clause 6.10.2)
- 5GC-MT-LR Procedure (see TS 23.273 [42], clause 6.1)
- Initiation and Reporting of Location Events (see TS 23.273 [42], clause 6.3.1)
- Location Continuity for Handover of an Emergency session from NG-RAN (see TS 23.273 [42], clause 6.10.3)
- 5GC-MT-LR multiple location procedure without UDM Query (see TS 23.273 [42], clause 6.10.4)
- Procedures of SL-MT-LR involving LMF (see TS 23.273 [42], clause 6.20.3)
- Procedures of SL-MT-LR for periodic, triggered Location Events (see TS 23.273 [42], clause 6.20.4)
- 5GC-MT-LR Procedure using SL positioning (see TS 23.273 [42], clause 6.20.5)

The ProvidePositioningInfo service operation shall be invoked by the NF Service Consumer (e.g. GMLC) to request the current or deferred geodetic and optionally local and/or civic location of the UE. The service operation triggers the AMF to invoke the service towards the LMF.

The NF Service Consumer shall invoke the service operation by sending POST to the URI of the "provide-pos-info" custom operation on the "Individual UE Context" resource (See clause 6.4.3.2.4.2). See also figure 5.5.2.2.1-1.

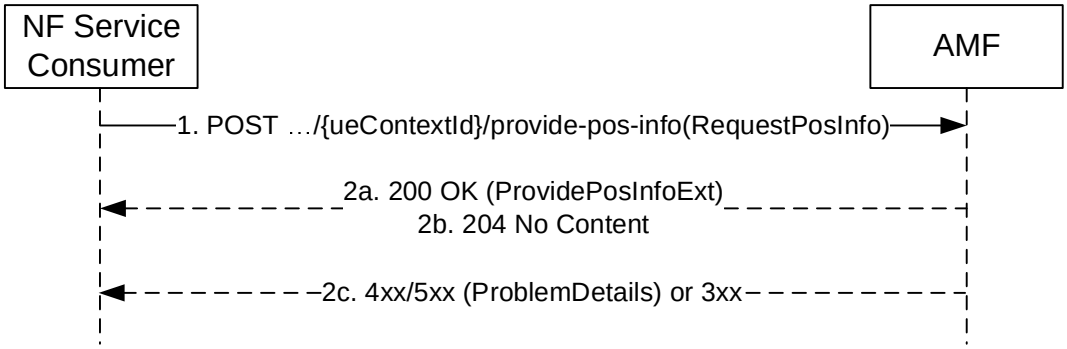



**Figure 5.5.2.2.1-1: NF Service Consumer requests the positioning information of the UE**

1. The NF Service Consumer shall send a POST request to the resource URI of "provide-pos-info" custom operation of the "Individual UE context" resource of the AMF. The content of the POST request may contain:

   - an indication of a positioning request from emergency services or commercial services client,
   - the required location QoS, and additionally the mapped location QoS applicable to EPS if multiple location QoS is required,
   - Supported GAD shapes,
   - scheduled location time,
   - reliable UE Location Request,
   - UE unaware indication,
   - the LMF ID that should be used for selecting the LMF,
   - the reporting indication,
   - the integrity requirements
   - the requested ranging_SL location results, including absolute locations, relative locations or distances and directions related to the UEs for ranging and sidelink positioning, and/or
   - the information of the related UEs, including application layer ID(s) and the related UE type for each related UE for ranging and sidelink positioning.

   If the NF service consumer wants the location change information or deferred location information to be notified (e.g. during a handover procedure or for activation or completion of deferred location), it also provides a callback URI on which the EventNotify service operation is executed (see clause 5.5.2.3).

   During 5GC-MT-LR multiple location procedure for regulatory location service, the request body may also include the indication of acceptance for intermediate response and the maximum response time, the GMLC callback address and the LIR reference number.

2a. On success, "200 OK" shall be returned. The content may contain the LCS correlation identifier, the location estimate, its age and accuracy, the information about the positioning method. If the request is invoked during a handover the response body shall also include the target AMF node identifier as specified in clause 6.10.3 of TS 23.273 [42].

   If the location determination will be sent by LMF to GMLC directly, the content shall include the directReportInd and supportedFeatures attributes.

2b. On accept, "204 No Content" shall be returned to acknowledge that AMF supports a deferred location request and a deferred location is accepted as specified in step 6 of clause 6.3.1 of TS 23.273 [42];

2c. On failure or redirection, one of the HTTP status code listed in Table 6.4.3.2.4.2.2-2 shall be returned. For a 4xx/5xx response, the message body shall contain a ProblemDetails structure with the "cause" attribute set to one of the application error listed in Table 6.4.3.2.4.2.2-2.

If the AMF received the LMF ID from the NF service consumer and the AMF does not find an LMF with the LMF ID received from GMLC, the AMF should return a 403 Forbidden response with the cause attribute set to "REQUESTED_LMF_NOT_AVAILABLE" to the NF service consumer, if no other LMF is configured as backup selection. Otherwise, the ProblemDetails structure with the "cause" attribute set to one of the application errors listed in Table 6.4.3.2.4.2.2-2 shall be applied.

### 5.5.2.3 EventNotify

#### 5.5.2.3.1 General

The EventNotify service operation is used in the following procedure:

- 5GC-NI-LR Procedure (see TS 23.273 [42], clause 6.10.1)
- Location Continuity for Handover of an Emergency session from NG-RAN (see TS 23.273 [42], clause 6.10.3)
- Completion of a deferred location for the UE available event or activation of deferred location for periodic location, area event triggered location or motion event triggered location (see TS 23.273 [42], clause 6.3.1)

The EventNotify service operation notifies the NF Service Consumer (i.e. GMLC) about the UE location related event information related to regulatory servicese.g. the initiation, handover or termination of an emergency session. The notification is delivered to:

- the callback URI registered in the NRF, if the GMLC registered to the NRF with notification endpoints for location notifications (see clauses 6.1.6.2.4 and 6.1.6.3.4 of TS 29.510 [29]); or
- GMLC URI locally provisioned in the AMF.

The EventNotify service operation notifies the NF Service Consumer (i.e. GMLC) about the UE location related event information related to deferred location request, i.e. the activation of deferred location request or the delivery of event reports. The notification is delivered to:

- the callback URI received from the GMLC during an earlier ProvidePositioningInfo service operation;

NOTE: During a handover procedure, both the source AMF and the target AMF can invoke the EventNotify service operation, based on the local configuration.

The operation is invoked by issuing a POST request to the callback URI of the NF Service Consumer (See clause 6.4.5.2.2). See also figure 5.5.2.3.1-1.

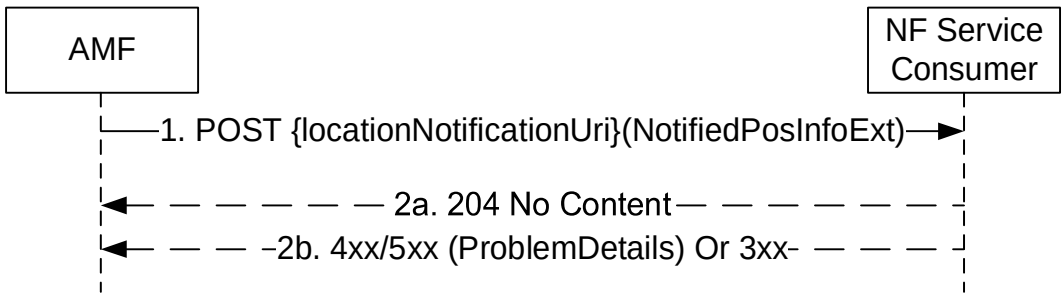



**Figure 5.5.2.3.1-1: UE Location Notification**

1. The AMF shall send a POST request to the callback URI provided by the NF service consumer determined as described above. The request body shall include the type of location related event and UE Identification (SUPI or PEI), and may include the GPSI, Geodetic Location, Local Location, Civic Location, MSC server identity, the Position methods used or a serving LMF identification for activation of periodic or triggered location.

2a. On success, "204 No content" shall be returned by the NF Service Consumer.

2b. On failure or redirection, the appropriate HTTP status code (e.g. "403 Forbidden") indicating the error shall be returned and appropriate additional error information should be returned.

### 5.5.2.4 ProvideLocationInfo

#### 5.5.2.4.1 General

The ProvideLocationInfo service operation allows an NF Service Consumer (e.g. UDM) to request the Network Provided Location Information (NPLI) of a target UE.

The NF Service Consumer shall invoke the service operation by sending POST request to the URI of the "provide-loc-info" custom operation on the "Individual UE Context" resource (see clause 6.4.3.2.4.3), as shown in figure 5.5.2.4.1-1.

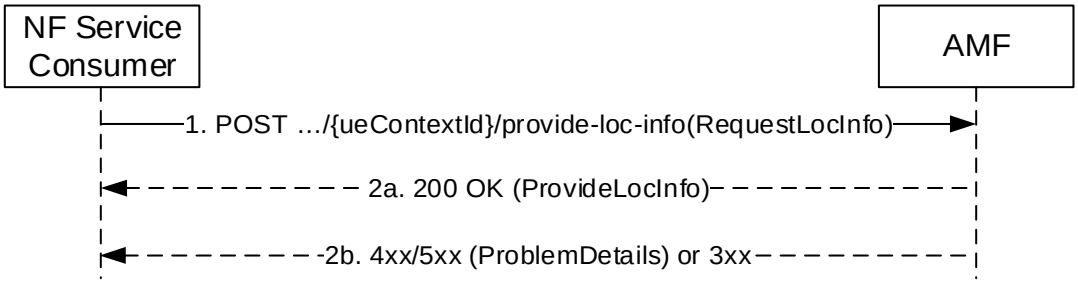



**Figure 5.5.2.4.1-1: NF Service Consumer requests the Location Information of the UE**

1. The NF Service Consumer shall send a POST request to the resource URI of "provide-loc-info" custom operation of the "Individual UE context" resource on the AMF. The content of the POST request shall contain a "requestLocInfo" data structure indicating the desired type of location information.

   If the NF Service Consumer desires the current location information of the target UE, it shall set "reqCurrentLoc" attribute to "true".

2a. On success, "200 OK" response shall be returned. The content of the response shall contain a "ProvideLocInfo" data structure including the Network Provide Location Information (NPLI) of the target UE.

   If "reqCurrentLoc" attribute is set to "true" and the UE is in RM-REGISTERED and CM-IDLE state over 3GPP access, the AMF shall initiate a paging procedure to the UE. If the paging procedure is successful, the AMF shall return the current location information and set "currentLoc" attribute to "true" in the response; if the UE does not respond to the paging, the AMF shall provide the last known location and set "currentLoc" attribute to "false" in the response.

   If "reqCurrentLoc" attribute is set to "true" and the UE is in RM-REGISTERED and CM-CONNECTED state over 3GPP access, the AMF shall follow NG-RAN Location reporting procedure, as specified in clause 4.10 of TS 23.502 [3], to trigger a single standalone report by setting "direct" event type in Location Reporting Control message. If NG-RAN reports current location of the UE, the AMF shall set "currentLoc" attribute to "true" in the response; if NG-RAN reports last known location of the UE with timestamp, the AMF shall set "currentLoc" attribute to "false" in the response.

   If the UE is in RM-REGISTERED over non-3GPP access, the AMF shall include the latest non-3GPP access location information.

2b. On failure or redirection, one of the HTTP status code listed in table 6.4.3.2.4.3.2-2 shall be returned. For a 4xx/5xx response, the message body shall contain a ProblemDetails structure with the "cause" attribute set to one of the application error listed in table 6.4.3.2.4.3.2-2.

### 5.5.2.5 CancelLocation

#### 5.5.2.5.1 General

This service operation is used in the following procedure:

- Cancellation of Reporting of Location Events by an AF or External LCS Client (see TS 23.273 [42], clause 6.3.3)

The CancelLocation service operation shall be invoked by the NF Service Consumer (e.g. GMLC) to cancel reporting periodic or events triggered location.

The NF Service Consumer shall invoke the service operation by sending a POST request to the URI of the "cancel-pos-info" custom operation on the "Individual UE Context" resource (See clause 6.4.3.2.4.4). See also figure 5.5.2.5.1-1.

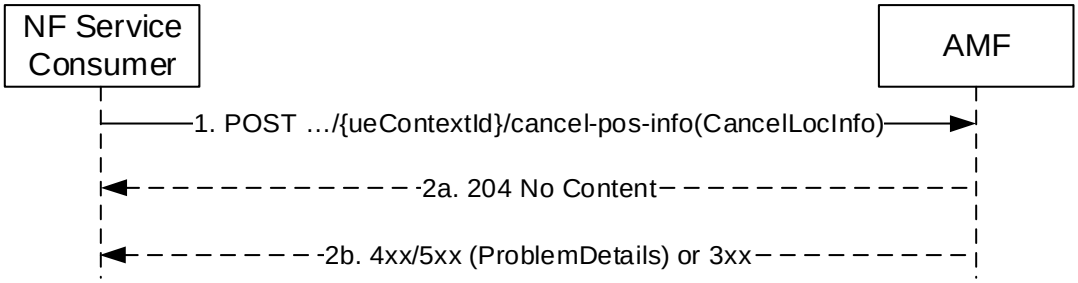



**Figure 5.5.2.5.1-1: Cancellation of reporting periodic or events triggered location of the UE**

1. The NF Service Consumer shall send a POST request to the resource URI of "cancel-pos-info" custom operation of the "Individual UE context" resource of the AMF. The content of the POST request shall contain a "CancelLocInfo" data structure indicating the desired cancellation of reporting periodic or events triggered location of the UE.

2a. On success, AMF responds with "204 No Content".

2b. On failure or redirection, one of the HTTP status code listed in Table 6.4.3.2.4.4-2 shall be returned. For a 4xx/5xx response, the message body shall contain a ProblemDetails structure with the "cause" attribute set to one of the application errors.

# 5.6 Namf_MBSBroadcast Service

## 5.6.1 Service Description

This service enables the NF Service Consumer (e.g. MB-SMF) to create, update or release a broadcast MBS session context in the AMF and to be notified about status change of the MBS broadcast context.

## 5.6.2 Service Operations

### 5.6.2.1 Introduction

The Namf_MBSBroadcast service supports following service operations:

- ContextCreate
- ContextUpdate
- ContextRelease
- ContextStatusNotify

### 5.6.2.2 ContextCreate

The ContextCreate service operation shall be used by the NF Service Consumer (e.g. MB-SMF) to request the AMF to create a broadcast MBS session context.

NOTE: For a location dependent MBS service, one single ContextCreate service operation is performed per MBS session (for a given AMF).

It is used in the following procedures:

- MBS Session Start for Broadcast (see clause 7.3.1 of TS 23.247 [55]);

- MBS Session Update for Broadcast (see clause 7.3.3 of TS 23.247 [55]);
- Support for Local Broadcast Service (see clause 7.3.4 of TS 23.247 [55]).

There shall be only one broadcast MBS session context per MBS session, or per MBS session and Area Session ID for an MBS session with Location dependent Broadcast service.

The NF Service Consumer (e.g. MB-SMF) shall create a broadcast MBS session context by using the HTTP POST method as shown in Figure 5.6.2.2-1.

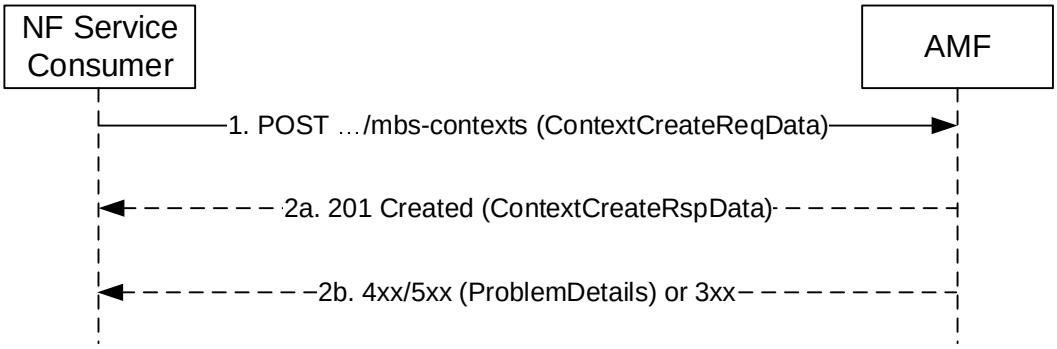



**Figure 5.6.2.2-1: Broadcast MBS session context creation**

1. The NF Service Consumer shall send a POST request targeting the Broadcast MBS session contexts collection resource of the AMF. The content of the POST request shall contain the following information:

   - MBS Session ID (i.e. TMGI, or TMGI and NID for an MBS session in an SNPN);
   - list of Area Session ID and related MBS service area, for a Location dependent broadcast MBS service;
   - MBS service area, for a Local broadcast MBS session;
   - N2 MBS Session Management container (see MBS Session Setup or Modification Request Transfer IE in TS 38.413 [12]);
   - Notification URI where to be notified about the status change of the broadcast MBS session context; and
   - SNSSAI.

   The NF Service Consumer may also include the maxResponseTime IE in the request to indicate the maximum response time to receive information about the completion of the Broadcast MBS session establishment.

   The NF Service Consumer may also include the MB-SMF instance ID and MB-SMF service instance ID to enable the AMF subscribing to the MB-SMF status notifications.

   The NF Service Consumer may also include the MBS associated session ID to enable NG-RAN to identify the multiple MBS sessions delivering the same content when AF creates multiple broadcast MBS Sessions via different Core Networks to deliver the same content.

2a. On success, "201 Created" shall be returned. The AMF should respond success when it receives the first successful response from the NG-RAN(s). The 201 Created response shall contain MBS session identifier and may contain one or more N2 MBS Session Management containers, if additional information (e.g. MBS Session Setup or Modification Response Transfer IE or MBS Session Setup or Modification Failure Transfer IE in TS 38.413 [12]) needs to be transferred to the MB-SMF. If the AMF received the NG-RAN responses from all involved NG-RAN(s), e.g. if the broadcast MBS session involves only one NG-RAN, the AMF shall include an indication of completion of the operation in all NG-RANs in the 201 Created response.

   Upon receipt of subsequent responses from other NG-RANs after sending the 201 Created response, if additional information (e.g. MBS Session Setup or Modification Response Transfer IE or MBS Session Setup or Modification Failure Transfer IE in TS 38.413 [12]) needs to be transferred to the MB-SMF, the AMF shall transfer such information by sending one or more Namf_MBSBroadcast_ContextStatusNotify requests to the MB-SMF. A Namf_MBSBroadcast_ContextStatusNotify request may include a list of N2 MBS Session

Management containers received from different NG-RANs. When the AMF receives the response from all NG-RANs, the AMF shall include an indication of the completion of the operation in the Namf_MBSBroadcast_ContextStatusNotify request.

If the AMF does not receive responses from all NG-RAN nodes before the maximum response time elapses since the reception of the Namf_MBSBroadcast_ContextCreate Request, then the AMF should send one Namf_MBSBroadcast_ContextStatusNotify request indicating the incompletion of the Broadcast MBS session establishment.

For each N2 MBS Session Management container sent towards the MB-SMF, the AMF shall insert the identifier of the NG-RAN node that generated it in the corresponding entry of the n2MbsSmInfoList attribute.

The AMF may send one or more Namf_MBSBroadcast_ContextStatusNotify request including an operationEvents attribute to report the MB-SMF about failure to reach one or more NG-RANs.

2b. On failure or redirection, one of the HTTP status code listed in Table 6.5.3.2.3.1-3 shall be returned. For a 4xx/5xx response, the message body may contain a ProblemDetails structure with the "cause" attribute set to one of the application errors listed in Table 6.5.3.2.3.1-3.

### 5.6.2.3 ContextUpdate

The ContextUpdate service operation shall be used by the NF Service Consumer (e.g. MB-SMF) to request the AMF to update a broadcast MBS session context.

It is used in the following procedures:

- MBS Session Update for Broadcast (see clause 7.3.3 of TS 23.247 [55]).
- Broadcast MBS session restoration by MB-SMF (see clause 8.3.2.3 of TS 23.527 [33].
- Selecting an alternative AMF for a Broadcast MBS Session at AMF failure (see clause 8.3.2.4 of TS 23.527 [33]).

The NF Service Consumer (e.g. MB-SMF) shall update a broadcast MBS session context by using the HTTP POST method as shown in Figure 5.6.2.3-1.

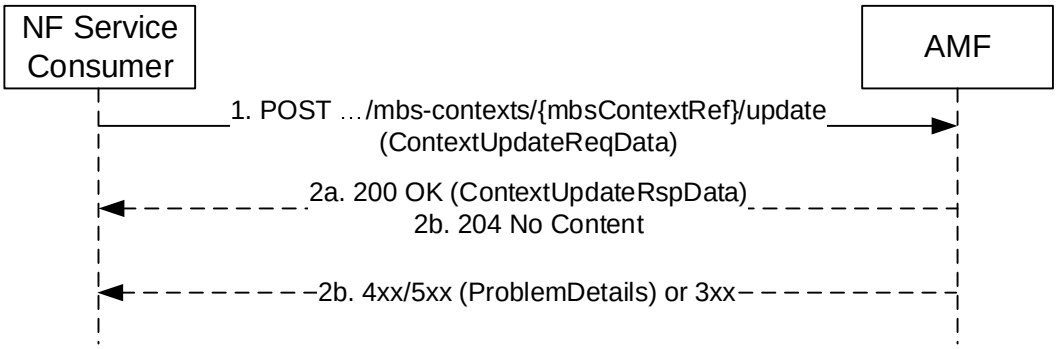



**Figure 5.6.2.3-1: Broadcast MBS session context update**

1. The NF Service Consumer shall send a POST request targeting the individual Broadcast MBS session context resource to be updated in the AMF. The content of the POST request may contain the following information:
   - N2 MBS Session Management container (see MBS Session Setup or Modification Request Transfer IE in TS 38.413 [12]);
   - Notification URI, if the NF Service Consumer wishes to modify the notification URI where to be notified about the status change of the broadcast MBS session context;
   - updated MBS service area, for a Local broadcast MBS session;

- updated list of Area Session ID and related MBS service area, for a Location dependent broadcast MBS session.

- the n2MbsInfoChangeInd IE set to "true" or "false" to indicate to the AMF whether the information within the N2 MBS Session Management container has changed or not for the MBS session

The NF Service Consumer may also include the maxResponseTime IE in the request to indicate the maximum response time to receive information about the completion of the Broadcast MBS session update.

During a broadcast MBS session restoration procedure for an NG-RAN failure with restart, the MB-SMF may include one or more ranIds attibutes to request the AMF to setup the Broadcast MBS session in a list of NG-RANs as identified by the NG-RAN ID(s), as specified in clause 8.3.2.3 of TS 23.527 [33].

During a restoration procedure upon an AMF failure without restart, for an AMF deployed in an AMF set, the MB-SMF may set the noNgapSignallingInd IE to "true" when the MB-SMF detects the original AMF has failed and then selects an alternative AMF to take over the MBS session but without a need to trigger any NGAP signalling towards NG-RANs, as specified in clause 8.3.2.4 of TS 23.527 [33].

2a. On success, "200 OK" shall be returned if additional information needs to be returned in the response. The 200 OK response may contain one or more N2 MBS Session Management containers, if such information (e.g. MBS Session Setup or Modification Response Transfer IE, MBS Session Setup or Modification Failure Transfer IE or MBS Session Release Response Transfer IE in TS 38.413 [12]) needs to be transferred to the MB-SMF. If the AMF received the NG-RAN responses from all involved NG-RAN(s), the AMF shall include an indication of completion of the operation in all NG-RANs.

2b. On success, "204 No Content" shall be returned if no additional information needs to be returned in the response.

In both 2a and 2b cases, upon receipt of subsequent responses from other NG-RANs after sending the 200 OK response or the 204 No Content response, if additional information (e.g. MBS Session Setup or Modification Response Transfer IE MBS Session Setup or Modification Failure Transfer IE or MBS Session Release Response Transfer IE in TS 38.413 [12]) needs to be transferred to the MB-SMF, the AMF shall transfer such information by sending one or more Namf_MBSBroadcast_ContextStatusNotify requests to the MB-SMF. A Namf_MBSBroadcast_ContextStatusNotify request may include a list of N2 MBS Session Management containers received from different NG-RANs. When the AMF receives the response from all NG-RANs, the AMF shall include an indication of the completion of the operation in the Namf_MBSBroadcast_ContextStatusNotify request.

If the AMF does not receive responses from all NG-RAN nodes before the maximum response time elapses since the reception of the Namf_MBSBroadcast_ContextUpdate Request, then the AMF should send one Namf_MBSBroadcast_ContextStatusNotify request indicating the incompletion of the Broadcast MBS session update.

If the n2MbsInfoChangeInd IE is present in the request and set to "false", the AMF does not need to contact NG-RAN nodes unaffected by the MBS service area change.

For each N2 MBS Session Management container sent towards the MB-SMF, the AMF shall insert the identifier of the NG-RAN node that generated it in the corresponding entry of the n2MbsSmInfoList attribute.

The AMF may send one or more Namf_MBSBroadcast_ContextStatusNotify request including an operationEvents attribute to report the MB-SMF about failure to reach one or more NG-RANs.

2c. On failure or redirection, one of the HTTP status code listed in Table 6.5.3.2.4.2.2-2 shall be returned. For a 4xx/5xx response, the message body may contain a ProblemDetails structure with the "cause" attribute set to one of the application errors listed in Table 6.5.3.2.4.2.2-2.

### 5.6.2.4 ContextRelease

The ContextRelease service operation shall be used by the NF Service Consumer (e.g. MB-SMF) to request the AMF to release a broadcast MBS session context.

It is used in the following procedures:

- MBS Session Release for Broadcast (see clause 7.3.2 of TS 23.247 [55]).

The NF Service Consumer (e.g. MB-SMF) shall release a broadcast MBS session context by using the HTTP DELETE method as shown in Figure 5.6.2.4-1.

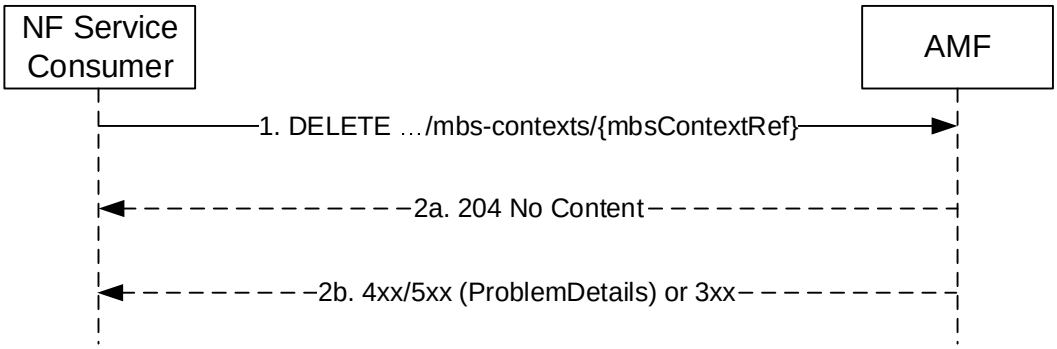



**Figure 5.6.2.4-1: Broadcast MBS session context release**

1. The NF Service Consumer shall send a DELETE request targeting the individual Broadcast MBS session context resource to be released in the AMF.

2a. On success, "204 No Content" shall be returned.

2b. On failure or redirection, one of the HTTP status code listed in Table 6.5.3.3.3.1-3 shall be returned. For a 4xx/5xx response, the message body may contain a ProblemDetails structure with the "cause" attribute set to one of the application errors listed in Table 6.5.3.3.3.1-3.

The AMF may send one or more Namf_MBSBroadcast_ContextStatusNotify request including an operationEvents attribute to report the MB-SMF about failure to reach one or more NG-RANs.

### 5.6.2.5 ContextStatusNotify

The ContextStatusNotify service operation shall be used by the AMF to notify status change of a broadcast MBS session context to the NF Service Consumer (e.g. MB-SMF).

It is used in the following procedures:

- MBS Session Start for Broadcast (see clause 7.3.1 of TS 23.247 [55]);
- MBS Session Update for Broadcast (see clause 7.3.3 of TS 23.247 [55]);
- Broadcast MBS Session Release Require (see clause 7.3.6 of TS 23.247 [55]).
- Broadcast MBS session restoration by MB-SMF (see clause 8.3.2.3 of TS 23.527 [33]).
- Selecting an alternative AMF for a Broadcast MBS Session at AMF failure (see clause 8.3.2.4 of TS 23.527 [33]).
- Transport change for resource sharing across broadcast MBS Sessions in network sharing (see clause 7.3.7 of TS 23.247 [55]).

The AMF shall notify status change of a broadcast MBS session context to the NF Service Consumer (e.g. MB-SMF) by using the HTTP POST method as shown in Figure 5.6.2.5-1.

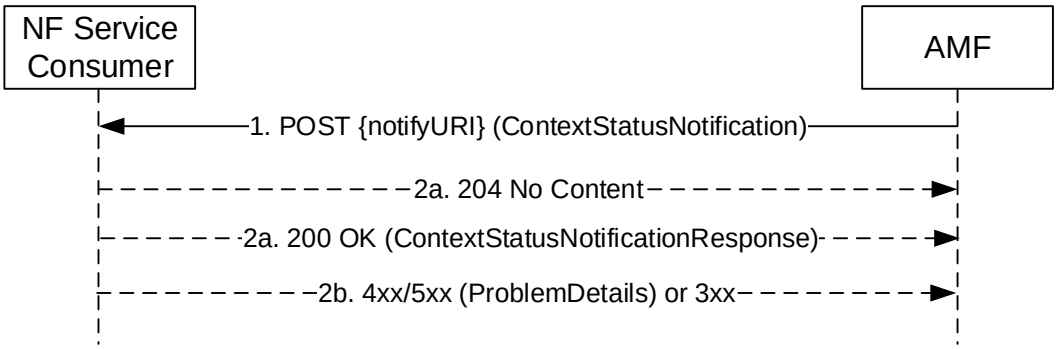



**Figure 5.6.2.5-1: Broadcast MBS session context status change notification**

1. The AMF shall send a POST request targeting the notification URI received from the NF Service Consumer. The content of the POST request shall contain the following information:

   - MBS Session ID (i.e. TMGI, or TMGI and NID for an MBS session in an SNPN);

   - Area Session ID, if this is a Location dependent broadcast MBS service;

   - one or more N2 MBS Session Management containers, if N2 MBS Session Management information (e.g. MBS Session Setup or Modification Response Transfer IE, MBS Session Setup or Modification Failure Transfer IE or MBS Session Release Response Transfer IE or Broadcast Session Transport Request Transfer IE in TS 38.413 [12]) has been received from one or more NG-RANs that needs to be transferred to the NF Service Consumer; for each N2 MBS Session Management container sent towards the MB-SMF, the AMF shall insert the identifier of the NG-RAN node that generated it in the corresponding entry of the n2MbsSmInfoList attribute.

   - the operationStatus IE indicating the completion of the Broadcast MBS session establishment or update, if the NF Service Consumer has requested to establish or update the Broadcast MBS session context and a response has been received from all NG-RANs; and

   - the operationStatus IE indicating the incompletion of the Broadcast MBS session establishment or update, if the NF Service Consumer has requested to establish or update the Broadcast MBS session context including a maximum response time and the AMF has not received responses from all NG-RANs before the maximum response time elapses.

   During a Broadcast MBS Session Release Require procedure (see clause 7.3.6 of TS 23.247 [55]), one or more NG-RANs may request the AMF to release the Broadcast MBS session. In this case, based on operator's policy, the AMF may:

   - report the Broadcast MBS Session release to the MB-SMF by including the operationEvent attribute in the MBS Context Status Notification request with the opEventType set to "NG_RAN_EVENT" together with a list of "ngranFailureEvent" for each NG-RAN that requested to release the Broadcast MBS Session. Upon receiving such a notification, per local policies, the MB-SMF may attempt to re-establish the MBS session after some operator configurable time in these NG-RANs by performing the Broadcast MBS session restoration by MB-SMF procedure specified in clause 8.3.2.3 of TS 23.527 [33]; or

   - attempt to re-establish the MBS session after some operator configurable time in these NG-RANs by performing the Broadcast MBS session restoration by AMF procedure specified in clause 8.3.2.2 of TS 23.527 [33].

   If all the NG-RANs serving the Broadcast MBS session requested the AMF to release the Broadcast MBS session, the AMF shall release the Broadcast MBS session context and send a notification with the releaseInd attribute set to true to report to the MB-SMF that the Broadcast MBS session (context) is released at the AMF and NG-RANs.

   The AMF may include an operationEvents attribute in the MBS Context Status Notification request to report the MB-SMF:

- a NG-RAN failure event, e.g. the NG-RAN failure with or without restart, as specified in clause 8.3.2.3 of TS 23.527 [33]);
- that a new AMF has taken over the control of the broadcast MBS session upon an AMF failure as specified in clause 8.3.2.4 of TS 23.527 [33]).

2a. On success, if the ContextStatusNotification does not contain a N2 MBS Session Management container with a Broadcast Session Transport Request Transfer IE, the NF Service Consumer shall return a "204 No Content" response.

If the ContextStatusNotification contains one or more N2 MBS Session Management containers encapsulating a Broadcast Session Transport Request Transfer IE, the MB-SMF shall return a "200 OK" with the ContextStatusNotificationResponse containing one or more N2 MBS Session Management containers encapsulating a Broadcast Session Transport Response Transfer IE or a Broadcast Session Transport Failure Transfer IE. For each N2 MBS Session Management container sent towards the AMF, the MB-SMF shall insert the identifier of the NG-RAN node to which the information in the container relates in the corresponding entry of the n2MbsSmInfoList attribute.

2b. On failure or redirection, one of the HTTP status code listed in Table 6.5.5.2.3.1-3 shall be returned. For a 4xx/5xx response, the message body may contain a ProblemDetails attribute with the "cause" attribute set to one of the application errors listed in Table 6.5.5.2.3.1-3.

# 5.7 Namf_MBSCommunication Service

## 5.7.1 Service Description

This service enables an NF Service Consumer (e.g. MB-SMF) to request the AMF to transfer MBS multicast related N2 message towards NG-RAN(s) serving a multicast MBS session, during a multicast MBS session activation, deactivation or update.

## 5.7.2 Service Operations

### 5.7.2.1 Introduction

The Namf_MBSCommunication service supports the following service operations:

- N2MessageTransfer
- Notify

### 5.7.2.2 N2MessageTransfer

The N2MessageTransfer service operation shall be used by the NF Service Consumer (e.g. MB-SMF) to request the AMF to transfer an MBS related N2 message to the NG-RAN nodes serving the multicast MBS session. It is used during the following procedures:

- MBS session activation procedure (see clause 7.2.5.2 of TS 23.247 [55]);
- MBS session deactivation procedure (see clause 7.2.5.3 of TS 23.247 [55]); and
- Multicast session update procedure (see clause 7.2.6 of TS 23.247 [55]).

The NF Service Consumer shall invoke the service operation by sending a POST request to the URI of the "transfer" custom operation (see clause 6.6.3.1) of the AMF. See Figure 5.7.2.2-1.

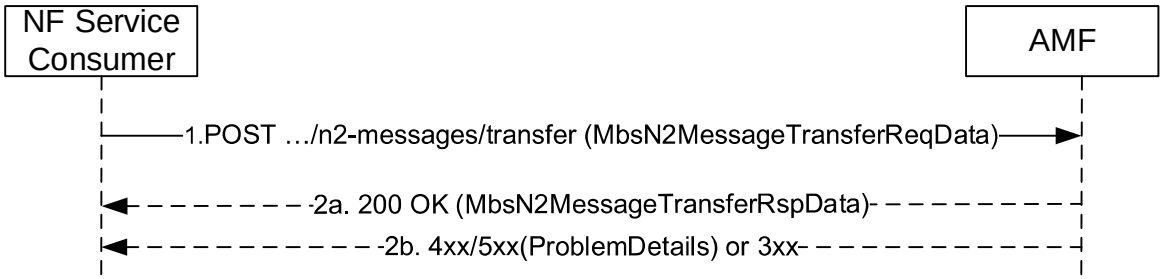



**Figure 5.7.2.2-1 N2 Message Transfer for a multicast MBS session**

1. The NF Service Consumer shall invoke the custom operation for N2 message transfer by sending a HTTP POST request and the request body shall carry the MbsN2MessageTransferReqData data structure which contains the N2 MBS Session Management information to be transferred. The MbsN2MessageTransferReqData shall contain:

   - MBS Session ID (i.e. TMGI, or TMGI and NID for an MBS session in an SNPN);

   and may also contain:

   - the Area Session ID, if this is a location dependent multicast MBS session; and/or
   - if the NF service consumer and the AMF support the RAN-ID-LIST feature (see clause 6.6.8):
     - the list of NG RAN node IDs towards which the MBS related N2 message is requested to be distributed;
     - a notification URI where to be notified about any failure of the MBS related N2 procedure for an NG RAN node in this list; and
     - an optional notification correlation ID to be sent within notifications.

   If the AMF supports the RAN-ID-LIST feature, the AMF shall distribute the MBS related N2 message to the list of NG-RAN nodes indicated by the MB-SMF, if any, otherwise to the list of NG-RAN nodes having established shared delivery that the AMF stores locally, if any.

NOTE: An AMF which does not support the the RAN-ID-LIST feature distributes the MBS related N2 message to the list of NG-RAN nodes having established shared delivery that the AMF stores locally.

2a. On success, the AMF shall respond with a "200 OK" status code with MbsN2MessageTransferRspData data structure. The AMF should respond success when it receives the first successful response from the NG-RAN(s).

   If the AMF supports the RAN-ID-LIST feature (see clause 6.6.8), and if the request included a list of NG RAN node IDs and a notification URI where to be notified about failures, the AMF shall report failure(s) of the N2 MBS related N2 procedure with an NG RAN node in this list by including the failureList IE in the "200 OK" response or in a subsequent Notify request towards the notification URI received in the request. See clause 8.4.1.2 of TS 23.527 [33].

2b. On failure or redirection, one of the HTTP status code listed in Table 6.6.3.1.4.2.2-2 shall be returned. For a 4xx/5xx response, the message body may contain a ProblemDetails attribute with the "cause" attribute set to one of the application errors listed in Table 6.6.4.2.2-2 if any.

### 5.7.2.3 Notify

The Notify service operation shall be used by the AMF to notify the NF Service Consumer about a failure of an MBS related N2 procedure with an NG RAN node (see clause 5.7.2.2).

It is used in the following procedure:

- N2 MBS session request distribution with list of NG RAN Node IDs provided by MB-SMF to AMF (see clause 8.4.1.2 of TS 23.527 [33]).

The AMF shall notify a failure of an MBS related N2 procedure with an NG RAN node to the NF Service Consumer (e.g. MB-SMF) by using the HTTP POST method as shown in Figure 5.7.2.3-1.

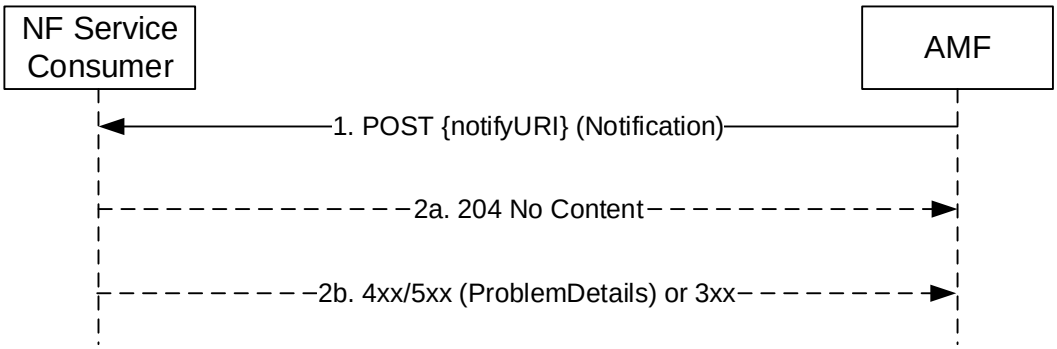



**Figure 5.7.2.3-1: Notification**

1. The AMF shall send a POST request targeting the notification URI received from the NF Service Consumer. The content of the POST request shall contain the following information:
   - MBS Session ID (i.e. TMGI, or TMGI and NID for an MBS session in an SNPN);
   - the Area Session ID, if this is a location dependent multicast MBS session; and
   - one or more failures including, for each failure, the related NG-RAN Node ID and failure cause.
   - a notification correlation ID if it is received in the N2MessageTransfer request.

2a. On success, the NF Service Consumer shall return a "204 No Content" response.

2b. On failure or redirection, one of the HTTP status code listed in Table 6.6.5.2.3.1-3 shall be returned. For a 4xx/5xx response, the message body may contain a ProblemDetails attribute with the "cause" attribute set to one of the application errors listed in Table 6.6.5.2.3.1-3.

# 6 API Definitions

## 6.1 Namf_Communication Service API

### 6.1.1 API URI

The Namf_Communication shall use the Namf_Communication API.

The API URI of the Namf_Communication API shall be:

**{apiRoot}/<apiName>/<apiVersion>/**

The request URI used in HTTP requests from the NF service consumer towards the NF service producer shall have the Resource URI structure defined in clause 4.4.1 of TS 29.501 [5], i.e.:

**{apiRoot}/<apiName>/<apiVersion>/<apiSpecificResourceUriPart>**

with the following components:

- The {apiRoot} shall be set as described in TS 29.501 [5].
- The <apiName> shall be "namf-comm".
- The <apiVersion> shall be "v1".
- The <apiSpecificResourceUriPart> shall be set as described in clause 6.1.3.

## 6.1.2 Usage of HTTP

### 6.1.2.1 General

HTTP/2, as defined in IETF RFC 9113 [19], shall be used as specified in clause 5 of TS 29.500 [4].

HTTP/2 shall be transported as specified in clause 5.3 of TS 29.500 [4].

HTTP messages and bodies for the Namf_Communication service shall comply with the OpenAPI [23] specification contained in Annex A.

### 6.1.2.2 HTTP standard headers

#### 6.1.2.2.1 General

The usage of HTTP standard headers shall be supported as specified in clause 5.2.2 of TS 29.500 [4].

#### 6.1.2.2.2 Content type

The following content types shall be supported:

- JSON, as defined in IETF RFC 8259 [8], shall be used as content type of the HTTP bodies specified in the present specification as indicated in clause 5.4 of TS 29.500 [4].
- The Problem Details JSON Object (IETF RFC 9457 [36]). The use of the Problem Details JSON object in a HTTP response body shall be signalled by the content type "application/problem+json".

Multipart messages shall also be supported (see clause 6.1.2.4) using the content type "multipart/related", comprising:

- one JSON body part with the "application/json" content type; and
- one or more binary body parts with 3gpp vendor specific content subtypes.

The 3gpp vendor specific content subtypes defined in Table 6.1.2.2.2-1 shall be supported.

**Table 6.1.2.2.2-1: 3GPP vendor specific content subtypes**

| content subtype | Description |
|---|---|
| vnd.3gpp.ngap | Binary encoded content, encoding NG Application Protocol (NGAP) IEs, as specified in clause 9.4 of TS 38.413 [12] (ASN.1 encoded). |
| vnd.3gpp.5gnas | Binary encoded content, encoding a 5GS NAS message, as specified in TS 24.501 [11]. |
| NOTE: Using 3GPP vendor content subtypes allows to describe the nature of the opaque content (e.g. NGAP or 5GS NAS information) without having to rely on metadata in the JSON content. | |

See clause 6.1.2.4 for the binary contents supported in the binary body part of multipart messages.

### 6.1.2.3 HTTP custom headers

#### 6.1.2.3.1 General

In this release of this specification, no custom headers specific to the Namf_Communication service are defined. For 3GPP specific HTTP custom headers used across all service based interfaces, see clause 5.2.3 of TS 29.500 [4].

### 6.1.2.4 HTTP multipart messages

HTTP multipart messages shall be supported, to transfer opaque N1 Information (e.g. SM, LPP) and/or N2 Information (e.g. SM, NRPPa, PWS), in the following service operations (and HTTP messages):

- N1N2MessageTransfer Request and Response (POST);

- NonUeN2MessageTransfer Request and Response (POST);
- N1MessageNotify (POST);
- N2InfoNotify (POST);
- NonUeN2InfoNotify (POST);
- UEContextTransfer (POST);
- CreateUEContext (PUT)

HTTP multipart messages shall include one JSON body part and one or more binary body parts comprising:

- N1payload, and/or N2 payload (see clause 6.1.6.4).

The JSON body part shall be the "root" body part of the multipart message. It shall be encoded as the first body part of the multipart message. The "Start" parameter does not need to be included.

The multipart message shall include a "type" parameter (see IETF RFC 2387 [9]) specifying the media type of the root body part, i.e. "application/json".

NOTE: The "root" body part (or "root" object) is the first body part the application processes when receiving a multipart/related message, see IETF RFC 2387 [9]. The default root is the first body within the multipart/related message. The "Start" parameter indicates the root body part, e.g. when this is not the first body part in the message.

For each binary body part in a HTTP multipart message, the binary body part shall include a Content-ID header (see IETF RFC 2045 [10]), and the JSON body part shall include an attribute, defined with the RefToBinaryData type, that contains the value of the Content-ID header field of the referenced binary body part.

## 6.1.3 Resources

### 6.1.3.1 Overview

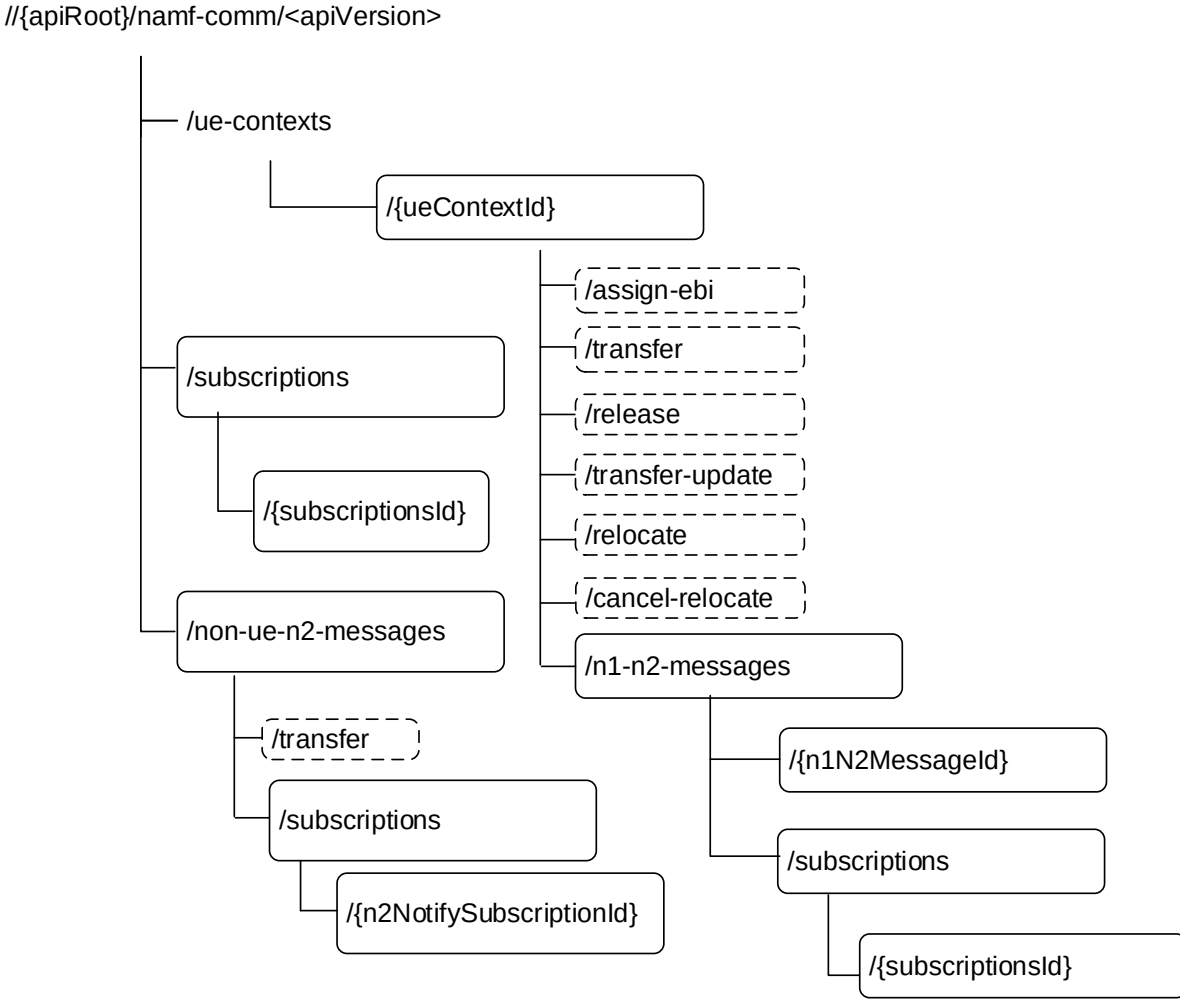



**Figure 6.1.3.1-1: Resource URI structure of the Namf_Communication API**

Table 6.1.3.1-1 provides an overview of the resources and applicable HTTP methods.

**Table 6.1.3.1-1: Resources and methods overview**

| Resource name | Resource URI | HTTP method or custom operation | Description (Mapped Service Operations) |
|---|---|---|---|
| Individual ueContext | /ue-contexts/{ueContextId} | | |
| | | PUT | CreateUEContext |
| | /ue-contexts/{ueContextId}/release | release (POST) | ReleaseUEContext |
| | /ue-contexts/{ueContextId}/assign-ebi | assign-ebi (POST) | EBIAssignment |
| | /ue-contexts/{ueContextId}/transfer | transfer (POST) | UEContextTransfer |
| | /ue-contexts/{ueContextId}/transfer-update | transfer-update (POST) | RegistrationStatusUpdate |
| | /ue-contexts/{ueContextId}/relocate | relocate (POST) | RelocateUEContext |
| | /ue-contexts/{ueContextId}/cancel-relocate | cancel-relocate (POST) | CancelRelocateUEContext |
| n1N2Message collection | /ue-contexts/{ueContextId}/n1-n2-messages | POST | N1N2MessageTransfer |
| N1N2 Subscriptions Collection for Individual UE Contexts | /ue-contexts/{ueContextId}/n1-n2-messages/subscriptions | POST | N1N2MessageSubscribe |
| N1N2 Individual Subscription | /ue-contexts/{ueContextId}/n1-n2-messages/subscriptions/{subscriptionId} | DELETE | N1N2MessageUnSubscribe |
| subscriptions collection | /subscriptions | POST | AMFStatusChangeSubscribe |
| individual subscription | /subscriptions/{subscriptionId} | PUT | AMFStatusChangeSubscribe |
| | | DELETE | AMFStatusChangeUnSubscribe |
| Non UE N2Messages collection | /non-ue-n2-messages/transfer | transfer (POST) | NonUEN2MessageTransfer |
| Non UE N2Messages Subscriptions collection | /non-ue-n2-messages/subscriptions | POST | NonUEN2InfoSubscribe |
| Non UE N2 Message Notification Individual Subscription | /non-ue-n2-messages/subscriptions/{n2NotifySubscriptionId} | DELETE | NonUEN2InfoUnsubscribe |

### 6.1.3.2 Resource: Individual ueContext

#### 6.1.3.2.1 Description

This resource represents the an individual ueContext identified by the ueContextId.

This resource is modelled as the Document resource archetype (see clause C.1 of TS 29.501 [5]).

#### 6.1.3.2.2 Resource Definition

Resource URI:{apiRoot}/namf-comm/<apiVersion>/ue-contexts/{ueContextId}

This resource shall support the resource URI variables defined in table 6.1.3.2.2-1.

**Table 6.1.3.2.2-1: Resource URI variables for this resource**

| Name | Data Type | Definition |
|---|---|---|
| apiRoot | String | See clause 6.1.1 |
| apiVersion | String | See clause 6.1.1. |
| ueContextId | String | Represents the 5G Globally Unique Temporary Identifier (See TS 23.501 [2] clause 5.9.4)<br>Pattern: "5g-guti-[0-9]{5,6}[0-9a-fA-F]{14}"<br>Or represents the Subscription Permanent Identifier (see TS 23.501 [2] clause 5.9.2)<br>pattern: see pattern of type Supi in TS 29.571 [6]<br>Or represents the Permanent Equipment Identifier (see TS 23.501 [2] clause 5.9.3)<br>pattern: "(imei-[0-9]{15}\|imeisv-[0-9]{16}\|.+)" |

When the ueContextId is composed by UE's SUPI or PEI, UE's PEI shall be used for the case:

- If the UE is emergency registered and the UE is UICCless;
- If the UE is emergency registered but SUPI is not authenticated.

For other cases, UE's SUPI shall be used.

#### 6.1.3.2.3 Resource Standard Methods

##### 6.1.3.2.3.1 PUT

This ueContextId identifies the individual ueContext resource is composed by UE's SUPI or PEI, See table 6.1.3.2.2-1.

This method shall support the URI query parameters specified in table 6.1.3.2.3.1-1.

**Table 6.1.3.2.3.1-1: URI query parameters supported by the PUT method on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| n/a | | | | |

This method shall support the request data structures specified in table 6.1.3.2.3.1-2 and the response data structures and response codes specified in table 6.1.3.2.3.1-3.

**Table 6.1.3.2.3.1-2: Data structures supported by the PUT Request Body on this resource**

| Data type | P | Cardinality | Description |
|---|---|---|---|
| UeContextCreate Data | M | 1 | Defines the UE Context to be created. |

**Table 6.1.3.2.3.1-3: Data structures supported by the PUT Response Body on this resource**

| Data type | P | Cardinality | Response codes | Description |
|---|---|---|---|---|
| UeContextCreatedData | M | 1 | 201 Created | This case represents the successful creation of a new UE Context.<br>Upon success, a response body is returned containing the newly created UE Context. |
| RedirectResponse | O | 0..1 | 307 Temporary Redirect | Temporary redirection.<br>(NOTE 2) |
| RedirectResponse | O | 0..1 | 308 Permanent Redirect | Permanent redirection.<br>(NOTE 2) |
| UeContextCreateError | O | 0..1 | 403 Forbidden | This case represents the creation of a new UE Context is not successful.<br><br>The "cause" attribute may be used to indicate one of the following application errors:<br>- HANDOVER_FAILURE |
| ProblemDetails | O | 0..1 | 403 Forbidden | This error shall only be returned by an SCP or a SEPP for errors they originate. |
| NOTE 1: The mandatory HTTP error status code for the PUT method listed in Table 5.2.7.1-1 of TS 29.500 [4] also apply, with response body containing an object of ProblemDetails data type (see clause 5.2.7 of TS 29.500 [4]).<br>NOTE 2: RedirectResponse may be inserted by an SCP or SEPP, see clause 6.10.9.1 of TS 29.500 [4]. | | | | |

**Table 6.1.3.2.3.1-4: Headers supported by the 201 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | Contains the URI of the newly created resource, according to the structure: {apiRoot}/namf-comm/<apiVersion>/ue-contexts/{ueContextId} |
| 3gpp-Sbi-Producer-Id | string | C | 0..1 | This header shall be included when the UE Context is created in a target AMF other than the initial AMF sending the response.<br><br>When included, this header shall indicate the target AMF serving the created UE Context. |

**Table 6.1.3.2.3.1-5: Headers supported by the 307 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | An alternative URI of the resource located on an alternative service instance within the same AMF or AMF (service) set.<br>For the case when a request is redirected to the same target resource via a different SCP or SEPP, see clause 6.10.9.1 in TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target NF (service) instance ID towards which the request is redirected |

**Table 6.1.3.2.3.1-6: Headers supported by the 308 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | An alternative URI of the resource located on an alternative service instance within the same AMF or AMF (service) set.<br>For the case when a request is redirected to the same target resource via a different SCP or SEPP, see clause 6.10.9.1 in TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target NF (service) instance ID towards which the request is redirected |

#### 6.1.3.2.4 Resource Custom Operations

##### 6.1.3.2.4.1 Overview

**Table 6.1.3.2.4.1-1: Custom operations**

| Operation Name | Custom operation URI | Mapped HTTP method | Description |
|---|---|---|---|
| release | /ue-contexts/{ueContextId}/release | POST | Release an existing individual ueContext resource.<br>It is used for the Release UE Context service operation. |
| assign-ebi | /ue-contexts/{ueContextId}/assign-ebi | POST | Assign EPS bearer ID(s) towards EPS bearer(s) mapped from QoS Flow(s), for a PDU session for the UE.<br>It is used for EBIAssignment service operation. |
| transfer | /ue-contexts/{ueContextId}/transfer | POST | Transfer an existing individual ueContext resource from old AMF to new AMF.<br>It is used for the UEContextTransfer service operation. |
| transfer-update | /ue-contexts/{ueContextId}/transfer-update | POST | Update the source AMF about the status of UE registration at the target AMF.<br>It is used for the RegistrationStatusUpdate service operation. |
| relocate | /ue-contexts/{ueContextId}/relocate | POST | Relocate an existing individual ueContext resource.<br>It is used for the RelocateUEContext service operation. |
| relocate | /ue-contexts/{ueContextId}/cancel-relocate | POST | Cancel a UE Context Relocation.<br>It is used for the CancelRelocateUEContext service operation. |

##### 6.1.3.2.4.2 Operation: release (POST)

###### 6.1.3.2.4.2.1 Description

This ueContextId identifies the individual ueContext resource is composed by UE's SUPI or PEI, See table 6.1.3.2.2-1.

###### 6.1.3.2.4.2.2 Operation Definition

This operation shall support the request data structures specified in table 6.1.3.2.4.2.2-1 and the response data structure and response codes specified in table 6.1.3.2.4.2.2-2.

**Table 6.1.3.2.4.2.2-1: Data structures supported by the (POST) release Request Body on this resource**

| Data type | P | Cardinality | Description |
|---|---|---|---|
| UEContextRelease | M | 1 | The information used for releasing of the UE Context |

**Table 6.1.3.2.4.2.2-2: Data structures supported by the (POST) release Response Body on this resource**

| Data type | P | Cardinality | Response codes | Description |
|---|---|---|---|---|
| n/a | | | 204 No Content | This case represents the handover is cancelled successfully. |
| RedirectResponse | O | 0..1 | 307 Temporary Redirect | Temporary redirection.<br>(NOTE 2) |
| RedirectResponse | O | 0..1 | 308 Permanent Redirect | Permanent redirection.<br>(NOTE 2) |
| ProblemDetails | O | 0..1 | 403 Forbidden | The "cause" attribute may be used to indicate one of the following application errors:<br>- UNSPECIFIED<br>- SUPI_OR_PEI_UNKNOWN<br><br>See table 6.1.7.3-1 for the description of this error. |
| ProblemDetails | O | 0..1 | 404 Not Found | The "cause" attribute may be used to indicate one of the following application errors:<br>- CONTEXT_NOT_FOUND<br><br>See table 6.1.7.3-1 for the description of this error. |
| NOTE 1: The mandatory HTTP error status code for the POST method listed in Table 5.2.7.1-1 of TS 29.500 [4] also apply, with response body containing an object of ProblemDetails data type (see clause 5.2.7 of TS 29.500 [4]).<br>NOTE 2: RedirectResponse may be inserted by an SCP or SEPP, see clause 6.10.9.1 of TS 29.500 [4]. | | | | |

**Table 6.1.3.2.4.2.2-3: Headers supported by the 307 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | An alternative URI of the resource located on an alternative service instance within the same AMF or AMF (service) set. For the case when a request is redirected to the same target resource via a different SCP or SEPP, see clause 6.10.9.1 in TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target NF (service) instance ID towards which the request is redirected |

**Table 6.1.3.2.4.2.2-4: Headers supported by the 308 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | An alternative URI of the resource located on an alternative service instance within the same AMF or AMF (service) set. For the case when a request is redirected to the same target resource via a different SCP or SEPP, see clause 6.10.9.1 in TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target NF (service) instance ID towards which the request is redirected |

### 6.1.3.2.4.3 Operation: assign-ebi (POST)

#### 6.1.3.2.4.3.1 Description

This ueContextId identifies the individual ueContext resource is composed by UE's SUPI or PEI, see Table 6.1.3.2.2-1.

#### 6.1.3.2.4.3.2 Operation Definition

This operation shall support the request data structures specified in table 6.1.3.2.4.3.2-1 and the response data structure and response codes specified in table 6.1.3.2.4.3.2-2.

**Table 6.1.3.2.4.3.2-1: Data structures supported by the (POST) assign-ebi Request Body on this resource**

| Data type | P | Cardinality | Description |
|---|---|---|---|
| AssignEbiData | M | 1 | The information required for AMF to allocate EPS bearer ID(s) or to update the mapping of EBI and ARP for a QoS flow to which an EBI is already allocated. |

**Table 6.1.3.2.4.3.2-2: Data structures supported by the (POST) assign-ebi Response Body on this resource**

| Data type | P | Cardinality | Response codes | Description |
|---|---|---|---|---|
| AssignedEbiData | M | 1 | 200 OK | Represent successful assignment of EPS bearer ID service operation, with the assigned EBIs included, or represent successful update of the mapping of EBI and ARP for a QoS flow to which an EBI is already allocated.<br>AMF may allocate only a subset of the requested EBIs, when not enough available EBI(s) can be allocated, e.g. when other PDU sessions with higher ARP have occupied too many EBIs. If the POST request body contained "releasedEbiList" the AMF shall release those EBI(s) and shall include the "releaseEbiList" IE in the POST response body.<br>(NOTE 2) |
| RedirectResponse | O | 0..1 | 307 Temporary Redirect | Temporary redirection.<br>(NOTE 3) |
| RedirectResponse | O | 0..1 | 308 Permanent Redirect | Permanent redirection.<br>(NOTE 3) |
| AssignEbiError | O | 0..1 | 403 Forbidden | This represents the case when none of the requested EBI(s) can be assigned by the AMF. The "cause" attribute of the ProblemDetails shall be set to:<br>- EBI_EXHAUSTED, if the number of EBIs allocated for the UE has already reached the maximum limit.<br>- EBI_REJECTED_LOCAL_POLICY, if the EBI allocation is rejected due to local policies at the AMF as specified in clause 4.11.1.4.1 of TS 23.502 [3].<br>- EBI_REJECTED_NO_N26, if the EBI allocation was rejected when the AMF is in a serving PLMN that does not support 5GS-EPS interworking procedures with N26 interface as specified in clause 5.17.2.3.1 of TS 23.501 [2]. |
| ProblemDetails | O | 0..1 | 403 Forbidden | This error shall only be returned by an SCP for errors it originates. |
| AssignEbiError | O | 0..1 | 409 Conflict | This represents the case when none of the requested EBI(s) can be assigned by the AMF. The "cause" attribute of the ProblemDetails shall be set to:<br>- TEMPORARY_REJECT_REGISTRATION_ONGOING, if there is an ongoing registration procedure.<br>- TEMPORARY_REJECT_HANDOVER_ONGOING, if there is an ongoing N2 handover procedure or an ongoing Xn handover procedure.<br>(NOTE 2) |

NOTE 1: The mandatory HTTP error status code for the POST method listed in Table 5.2.7.1-1 of TS 29.500 [4] also apply, with response body containing an object of ProblemDetails data type (see clause 5.2.7 of TS 29.500 [4]).
NOTE 2: When receiving EBI assignment request during Xn Handover or N2 Handover, the AMF may either reject the request with the TEMPORARY_REJECT_HANDOVER_ONGOING application error in a 409 Conflict response or proceed with assigning EBIs with a 200 OK response. In the latter case, upon receipt of the 200 OK response, the SMF shall take the assigned EBIs into account in subsequent processing.
NOTE 3: RedirectResponse may be inserted by an SCP, see clause 6.10.9.1 of TS 29.500 [4].

**Table 6.1.3.2.4.3.2-3: Headers supported by the 307 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | An alternative URI of the resource located on an alternative service instance within the same AMF or AMF (service) set. For the case when a request is redirected to the same target resource via a different SCP, see clause 6.10.9.1 in TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target NF (service) instance ID towards which the request is redirected |

**Table 6.1.3.2.4.3.2-4: Headers supported by the 308 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | An alternative URI of the resource located on an alternative service instance within the same AMF or AMF (service) set. For the case when a request is redirected to the same target resource via a different SCP, see clause 6.10.9.1 in TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target NF (service) instance ID towards which the request is redirected |

### 6.1.3.2.4.4 Operation: transfer (POST)

#### 6.1.3.2.4.4.1 Description

This ueContextId identifies the individual ueContext resource is composed by UE's 5G-GUTI or SUPI, see Table 6.1.3.2.2-1.

#### 6.1.3.2.4.4.2 Operation Definition

This operation shall support the request data structures specified in table 6.1.3.2.4.4.2-1 and the response data structure and response codes specified in table 6.1.3.2.4.4.2-2.

**Table 6.1.3.2.4.4.2-1: Data structures supported by the (POST) transfer Request Body on this resource**

| Data type | P | Cardinality | Description |
|---|---|---|---|
| UeContextTransferReqData | M | 1 | Represents to start transferring of an individual ueContext resource from old AMF to new AMF. |

**Table 6.1.3.2.4.4.2-2: Data structures supported by the (POST) transfer Response Body on this resource**

| Data type | P | Cardinality | Response codes | Description |
|---|---|---|---|---|
| UeContextTransferRspData | M | 1 | 200 OK | Indicates the transferring of the individual ueContext resource is started successfully. |
| RedirectResponse | O | 0..1 | 307 Temporary Redirect | Temporary redirection.<br>(NOTE 2) |
| RedirectResponse | O | 0..1 | 308 Permanent Redirect | Permanent redirection.<br>(NOTE 2) |
| ProblemDetails | O | 0..1 | 403 Forbidden | Indicates that AMF can understand the request but cannot fulfil the request due to errors. If the integrity check of the included complete registration message fails at the source AMF the "cause" attribute is set to:<br>- INTEGRITY_CHECK_FAIL.<br><br>See table 6.1.7.3-1 for the description of these errors. |
| ProblemDetails | O | 0..1 | 404 Not Found | If the AMF does not have the requested UE context, the AMF shall return this status code and the "cause" attribute is set to:<br>- CONTEXT_NOT_FOUND<br><br>See table 6.1.7.3-1 for the description of these errors. |
| NOTE 1: The mandatory HTTP error status code for the POST method listed in Table 5.2.7.1-1 of TS 29.500 [4] also apply, with response body containing an object of ProblemDetails data type (see clause 5.2.7 of TS 29.500 [4]).<br>NOTE 2: RedirectResponse may be inserted by an SCP or SEPP, see clause 6.10.9.1 of TS 29.500 [4]. | | | | |

**Table 6.1.3.2.4.4.2-3: Headers supported by the 307 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | An alternative URI of the resource located on an alternative service instance within the same AMF or AMF (service) set. For the case when a request is redirected to the same target resource via a different SCP or SEPP, see clause 6.10.9.1 in TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target NF (service) instance ID towards which the request is redirected |

**Table 6.1.3.2.4.4.2-4: Headers supported by the 308 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | An alternative URI of the resource located on an alternative service instance within the same AMF or AMF (service) set. For the case when a request is redirected to the same target resource via a different SCP or SEPP, see clause 6.10.9.1 in TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target NF (service) instance ID towards which the request is redirected |

### 6.1.3.2.4.5 Operation: transfer-update (POST)

#### 6.1.3.2.4.5.1 Description

This ueContextId identifies the individual ueContext resource is composed by UE's 5G-GUTI, see Table 6.1.3.2.2-1.

#### 6.1.3.2.4.5.2 Operation Definition

This operation shall support the request data structures specified in table 6.1.3.2.4.5.2-1 and the response data structure and response codes specified in table 6.1.3.2.4.5.2-2.

**Table 6.1.3.2.4.5.2-1: Data structures supported by the (POST) transfer-update Request Body on this resource**

| Data type | P | Cardinality | Description |
|---|---|---|---|
| UeRegStatusUpdateReqData | M | 1 | Represents to the update of status on the transferring of an individual ueContext resource from old AMF to new AMF. |

**Table 6.1.3.2.4.5.2-2: Data structures supported by the (POST) transfer-update Response Body on this resource**

| Data type | P | Cardinality | Response codes | Description |
|---|---|---|---|---|
| UeRegStatusUpdateRspData | M | 1 | 200 OK | Indicates the update of UE context transfer status is successful at the source AMF. |
| RedirectResponse | O | 0..1 | 307 Temporary Redirect | Temporary redirection.<br>(NOTE 2) |
| RedirectResponse | O | 0..1 | 308 Permanent Redirect | Permanent redirection.<br>(NOTE 2) |
| ProblemDetails | O | 0..1 | 403 Forbidden | Indicates that AMF can understand the request but cannot fulfil the request due to errors. |
| ProblemDetails | O | 0..1 | 404 Not Found | If the AMF does not have the requested UE context, the AMF shall return this status code and the "cause" attribute is set to:<br>- CONTEXT_NOT_FOUND |
| NOTE 1: The mandatory HTTP error status code for the POST method listed in Table 5.2.7.1-1 of TS 29.500 [4] also apply, with response body containing an object of ProblemDetails data type (see clause 5.2.7 of TS 29.500 [4]).<br>NOTE 2: RedirectResponse may be inserted by an SCP or SEPP, see clause 6.10.9.1 of TS 29.500 [4]. | | | | |

**Table 6.1.3.2.4.5.2-3: Headers supported by the 307 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | An alternative URI of the resource located on an alternative service instance within the same AMF or AMF (service) set. For the case when a request is redirected to the same target resource via a different SCP or SEPP, see clause 6.10.9.1 in TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target NF (service) instance ID towards which the request is redirected |

**Table 6.1.3.2.4.5.2-4: Headers supported by the 308 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | An alternative URI of the resource located on an alternative service instance within the same AMF or AMF (service) set. For the case when a request is redirected to the same target resource via a different SCP or SEPP, see clause 6.10.9.1 in TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target NF (service) instance ID towards which the request is redirected |

#### 6.1.3.2.4.6 Operation: relocate (POST)

##### 6.1.3.2.4.6.1 Description

The ueContextId identifying the individual ueContext resource is composed by UE's SUPI or PEI, see Table 6.1.3.2.2-1.

##### 6.1.3.2.4.6.2 Operation Definition

This operation shall support the request data structures specified in table 6.1.3.2.4.6.2-1 and the response data structure and response codes specified in table 6.1.3.2.4.5.2-2.

**Table 6.1.3.2.4.6.2-1: Data structures supported by the (POST) relocate Request Body on this resource**

| Data type | P | Cardinality | Description |
|---|---|---|---|
| UeContextRelocateData | M | 1 | Defines the UE Context to be relocated to a new AMF. |

**Table 6.1.3.2.4.6.2-2: Data structures supported by the (POST) relocate Response Body on this resource**

| Data type | P | Cardinality | Response codes | Description |
|---|---|---|---|---|
| UeContextRelocatedData | M | 1 | 201 Created | This case represents the successful relocation of UE Context to a new AMF.<br>Upon success, a response body is returned containing the newly created UE Context in new AMF. |
| RedirectResponse | O | 0..1 | 307 Temporary Redirect | Temporary redirection.<br>(NOTE 2) |
| RedirectResponse | O | 0..1 | 308 Permanent Redirect | Permanent redirection.<br>(NOTE 2) |
| ProblemDetail | O | 0..1 | 403 Forbidden | This case represents an unsuccessful relocation of UE Context to a new AMF.<br><br>The "cause" attribute may be used to indicate one of the following application errors:<br>- HANDOVER_FAILURE |
| NOTE 1: The mandatory HTTP error status code for the POST method listed in Table 5.2.7.1-1 of TS 29.500 [4] also apply, with response body containing an object of ProblemDetails data type (see clause 5.2.7 of TS 29.500 [4]).<br>NOTE 2: RedirectResponse may be inserted by an SCP, see clause 6.10.9.1 of TS 29.500 [4]. | | | | |

**Table 6.1.3.2.4.6.2-3: Headers supported by the 307 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | An alternative URI of the resource located on an alternative service instance within the same AMF or AMF (service) set.<br>For the case when a request is redirected to the same target resource via a different SCP, see clause 6.10.9.1 in TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target NF (service) instance ID towards which the request is redirected |

**Table 6.1.3.2.4.6.2-4: Headers supported by the 308 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | An alternative URI of the resource located on an alternative service instance within the same AMF or AMF (service) set. For the case when a request is redirected to the same target resource via a different SCP, see clause 6.10.9.1 in TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target NF (service) instance ID towards which the request is redirected |

### 6.1.3.2.4.7 Operation: cancel-relocate (POST)

#### 6.1.3.2.4.7.1 Description

This ueContextId identifying the individual ueContext resource is composed by UE's SUPI or PEI, See table 6.1.3.2.2-1.

#### 6.1.3.2.4.7.2 Operation Definition

This operation shall support the request data structures specified in table 6.1.3.2.4.7.2-1 and the response data structure and response codes specified in table 6.1.3.2.4.2.7-2.

**Table 6.1.3.2.4.7.2-1: Data structures supported by the (POST) release Request Body on this resource**

| Data type | P | Cardinality | Description |
|---|---|---|---|
| UEContextCancelRelocateData | M | 1 | The information used for cancellation of UE Context Relocation. |

**Table 6.1.3.2.4.2.7-2: Data structures supported by the (POST) release Response Body on this resource**

| Data type | P | Cardinality | Response codes | Description |
|---|---|---|---|---|
| n/a | | | 204 No Content | This case represents the handover is cancelled successfully. |
| RedirectResponse | O | 0..1 | 307 Temporary Redirect | Temporary redirection.<br>(NOTE 2) |
| RedirectResponse | O | 0..1 | 308 Permanent Redirect | Permanent redirection.<br>(NOTE 2) |
| ProblemDetails | O | 0..1 | 403 Forbidden | The "cause" attribute may be used to indicate one of the following application errors:<br>- UNSPECIFIED<br>- SUPI_OR_PEI_UNKNOWN<br><br>See table 6.1.7.3-1 for the description of this error. |
| ProblemDetails | O | 0..1 | 404 Not Found | The "cause" attribute may be used to indicate one of the following application errors:<br>- CONTEXT_NOT_FOUND<br><br>See table 6.1.7.3-1 for the description of this error. |
| NOTE 1: The mandatory HTTP error status code for the POST method listed in Table 5.2.7.1-1 of TS 29.500 [4] also apply, with response body containing an object of ProblemDetails data type (see clause 5.2.7 of TS 29.500 [4]).<br>NOTE 2: RedirectResponse may be inserted by an SCP, see clause 6.10.9.1 of TS 29.500 [4]. | | | | |

**Table 6.1.3.2.4.2.7-3: Headers supported by the 307 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | An alternative URI of the resource located on an alternative service instance within the same AMF or AMF (service) set.<br>v For the case when a request is redirected to the same target resource via a different SCP, see clause 6.10.9.1 in TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target NF (service) instance ID towards which the request is redirected |

**Table 6.1.3.2.4.2.7-4: Headers supported by the 308 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | An alternative URI of the resource located on an alternative service instance within the same AMF or AMF (service) set.<br>For the case when a request is redirected to the same target resource via a different SCP, see clause 6.10.9.1 in TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target NF (service) instance ID towards which the request is redirected |

## 6.1.3.3 Resource: N1N2 Subscriptions Collection for Individual UE Contexts

### 6.1.3.3.1 Description

This resource represents the collection under an individual UE context for storing the subscriptions for notifications of UE specific N1 and N2 message types. This resource is modelled as the Collection resource archetype (see clause C.2 of TS 29.501 [5]).

### 6.1.3.3.2 Resource Definition

Resource URI: **{apiRoot}/namf-comm/<apiVersion>/{ueContextId}/n1-n2-messages/subscriptions**

This resource shall support the resource URI variables defined in table 6.1.3.3.2-1.

**Table 6.1.3.3.2-1: Resource URI variables for this resource**

| Name | Data type | Definition |
|---|---|---|
| apiRoot | string | See clause 6.1.1 |
| apiVersion | string | See clause 6.1.1. |
| ueContextId | string | Represents the Subscription Permanent Identifier (see TS 23.501 [2] clause 5.9.2)<br>pattern: see pattern of type Supi in TS 29.571 [6]<br>Or represents the Permanent Equipment Identifier (see TS 23.501 [2] clause 5.9.3)<br>pattern: "(imei-[0-9]{15}\|imeisv-[0-9]{16}\|.+)" |

When the ueContextId is composed by UE's SUPI or PEI, UE's PEI shall be used for the case:

- If the UE is emergency registered and the UE is UICCless;
- If the UE is emergency registered but SUPI is not authenticated.

For other cases, UE's SUPI shall be used.

#### 6.1.3.3.3 Resource Standard Methods

##### 6.1.3.3.3.1 POST

This method creates an individual N1/N2 information subscription resource for UE related N1/N2 information. This method is used by NF Service Consumers (e.g. PCF) to subscribe for notifications about UE related N1/N2 Information.

This method shall support the request data structures specified in table 6.1.3.3.3.1-2 and the response data structures and response codes specified in table 6.1.3.3.3.1-3.

**Table 6.1.3.3.3.1-2: Data structures supported by the POST Request Body on this resource**

| Data type | P | Cardinality | Description |
|---|---|---|---|
| UeN1N2InfoSubscriptionCreateData | C | 0..1 | Representation of the subscription for N1 and/or N2 information notification. It shall contain the information regarding N1 and/or N2 information to be notified and the callback URI for the respective notifications. |

**Table 6.1.3.3.3.1-3: Data structures supported by the POST Response Body on this resource**

| Data type | P | Cardinality | Response codes | Description |
|---|---|---|---|---|
| UeN1N2InfoSubscriptionCreatedData | C | 0..1 | 201 Created | This case represents the successful creation of the subscription for N1 and/or N2 information notification.<br><br>Upon success, a response body is returned containing the representation describing the status of the request.<br><br>The Location header shall contain the location (URI) of the created subscription resource. |
| RedirectResponse | O | 0..1 | 307 Temporary Redirect | Temporary redirection.<br>(NOTE 2) |
| RedirectResponse | O | 0..1 | 308 Permanent Redirect | Permanent redirection.<br>(NOTE 2) |
| NOTE 1: The mandatory HTTP error status code for the POST method listed in Table 5.2.7.1-1 of TS 29.500 [4] also apply, with response body containing an object of ProblemDetails data type (see clause 5.2.7 of TS 29.500 [4]).<br>NOTE 2: RedirectResponse may be inserted by an SCP or SEPP, see clause 6.10.9.1 of TS 29.500 [4]. | | | | |

**Table 6.1.3.3.3.1-4: Headers supported by the 201 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | Contains the URI of the newly created resource, according to the structure: {apiRoot}/namf-comm/<apiVersion>/ue-contexts/{ueContextId}/n1-n2-messages/subscriptions/{subscriptionId} |

**Table 6.1.3.3.3.1-5: Headers supported by the 307 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | An alternative URI of the resource located on an alternative service instance within the same AMF or AMF (service) set. For the case when a request is redirected to the same target resource via a different SCP or SEPP, see clause 6.10.9.1 in TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target NF (service) instance ID towards which the request is redirected |

**Table 6.1.3.3.3.1-6: Headers supported by the 308 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | An alternative URI of the resource located on an alternative service instance within the same AMF or AMF (service) set. For the case when a request is redirected to the same target resource via a different SCP or SEPP, see clause 6.10.9.1 in TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target NF (service) instance ID towards which the request is redirected |

### 6.1.3.3.4 Resource Custom Operations

There are no custom operations supported on this resource.

## 6.1.3.4 Resource: N1N2 Individual Subscription

### 6.1.3.4.1 Description

This resource represents the individual subscription for the subscription for notifications of UE specific N1 and N2 message types.

This resource is modelled as the Document resource archetype (see clause C.1 of TS 29.501 [5]).

### 6.1.3.4.2 Resource Definition

Resource URI: **{apiRoot}/namf-comm/<apiVersion>/{ueContextId}/n1-n2-messages/subscriptions/{subscriptionId}**

This resource shall support the resource URI variables defined in table 6.1.3.4.2-1.

**Table 6.1.3.4.2-1: Resource URI variables for this resource**

| Name | Data type | Definition |
|---|---|---|
| apiRoot | string | See clause 6.1.1 |
| apiVersion | string | See clause 6.1.1. |
| ueContextId | string | Represents the Subscription Permanent Identifier (see TS 23.501 [2] clause 5.9.2)<br>pattern: see pattern of type Supi in TS 29.571 [6]<br>Or represents the Permanent Equipment Identifier (see TS 23.501 [2] clause 5.9.3)<br>pattern: "(imei-[0-9]{15}\|imeisv-[0-9]{16}\|.+)" |
| subscriptionId | string | Represents the individual subscription to the UE specific N1/N2 message notification. |

When the ueContextId is composed by UE's SUPI or PEI, UE's PEI shall be used for the case:

- If the UE is emergency registered and the UE is UICCless;
- If the UE is emergency registered but SUPI is not authenticated.

For other cases, UE's SUPI shall be used.

### 6.1.3.4.3 Resource Standard Methods

#### 6.1.3.4.3.1 DELETE

This method deletes an individual N1/N2 message notification subscription resource for an individual UE. This method is used by NF Service Consumers (e.g. PCF) to unsubscribe for notifications about UE related N1/N2 information.

This method shall support the request data structures specified in table 6.1.3.4.3.1-2 and the response data structures and response codes specified in table 6.1.3.4.3.1-3.

**Table 6.1.3.4.3.1-2: Data structures supported by the DELETE Request Body on this resource**

| Data type | P | Cardinality | Description |
|---|---|---|---|
| n/a | | | |

**Table 6.1.3.4.3.1-3: Data structures supported by the DELETE Response Body on this resource**

| Data type | P | Cardinality | Response codes | Description |
|---|---|---|---|---|
| n/a | | | 204 No Content | |
| RedirectResponse | O | 0..1 | 307 Temporary Redirect | Temporary redirection.<br>(NOTE 2) |
| RedirectResponse | O | 0..1 | 308 Permanent Redirect | Permanent redirection.<br>(NOTE 2) |
| ProblemDetails | O | 0..1 | 404 Not Found | If the resource corresponding to the SubscriptionId cannot be found, the AMF shall return this status code. The "cause" attribute is set to:<br>- SUBSCRIPTION_NOT_FOUND |
| NOTE 1: The mandatory HTTP error status code for the DELETE method listed in Table 5.2.7.1-1 of TS 29.500 [4] also apply, with response body containing an object of ProblemDetails data type (see clause 5.2.7 of TS 29.500 [4]).<br>NOTE 2: RedirectResponse may be inserted by an SCP or SEPP, see clause 6.10.9.1 of TS 29.500 [4]. | | | | |

**Table 6.1.3.4.3.1-4: Headers supported by the 307 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | An alternative URI of the resource located on an alternative service instance within the same AMF or AMF (service) set. For the case when a request is redirected to the same target resource via a different SCP or SEPP, see clause 6.10.9.1 in TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target NF (service) instance ID towards which the request is redirected |

**Table 6.1.3.4.3.1-5: Headers supported by the 308 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | An alternative URI of the resource located on an alternative service instance within the same AMF or AMF (service) set. For the case when a request is redirected to the same target resource via a different SCP or SEPP, see clause 6.10.9.1 in TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target NF (service) instance ID towards which the request is redirected |

### 6.1.3.4.4 Resource Custom Operations

There are no custom operations supported on this resource.

## 6.1.3.5 Resource: N1N2 Messages Collection

### 6.1.3.5.1 Description

This resource represents the collection on which UE related N1 messages and N2 information transfer are initiated and the N1 information for the UE is stored temporarily until the UE is reachable. This resource is modelled with the Collection resource archetype (see clause C.2 of TS 29.501 [5]).

#### 6.1.3.5.2 Resource Definition

Resource URI: {apiRoot}/namf-comm/<apiVersion>/ue-contexts/{ueContextId}/n1-n2-messages

This resource shall support the resource URI variables defined in table 6.1.3.5.2-1.

**Table 6.1.3.5.2-1: Resource URI variables for this resource**

| Name | Data type | Definition |
|---|---|---|
| apiRoot | string | See clause 6.1.1 |
| apiVersion | string | See clause 6.1.1. |
| ueContextId | string | Represents the Subscription Permanent Identifier (see TS 23.501 [2] clause 5.9.2)<br>pattern: see pattern of type Supi in TS 29.571 [6]<br>Or represents the Permanent Equipment Identifier (see TS 23.501 [2] clause 5.9.3)<br>pattern: "(imei-[0-9]{15}\|imeisv-[0-9]{16})"<br>Or represents the LCS Correlation ID (see TS 29.572 [25] clause 6.1.6.3.2) (NOTE)<br>pattern: "(cid-.{1,255})" |
| NOTE: The LCS Correlation ID shall only be applied when transferring LCS related UE-Specific N1 and/or N2 messages. | | |

When the ueContextId is composed by UE's SUPI or UE's PEI, UE's PEI shall be used for the case:

- If the UE is emergency registered and the UE is UICCless;
- If the UE is emergency registered but SUPI is not authenticated.

For other cases, UE's SUPI shall be used.

The ueContextId composed by LCS Correlation ID may be used for transferring LCS related UE-Specific N1 and/or N2 messages, when the LCS Correlation ID was provided by the AMF to the LMF in a previous location request.

#### 6.1.3.5.3 Resource Standard Methods

##### 6.1.3.5.3.1 POST

This method initiates a N1 message and/or N2 message transfer at the AMF and may create a resource to store the N1 and/or N2 message as specified in clause 5.2.2.3.1.2, e.g. if asynchronous type communication is invoked or if the UE is paged.

This method shall support the request data structures specified in table 6.1.3.5.3.1-1 and the response data structures and response codes specified in table 6.1.3.5.3.1-2.

**Table 6.1.3.5.3.1-1: Data structures supported by the POST Request Body on this resource**

| Data type | P | Cardinality | Description |
|---|---|---|---|
| N1N2MessageTransferReqData | M | 1 | This contains:<br>- N1 message, if the NF Service Consumer requests to transfer an N1 message to the UE or;<br>- N2 information, if the NF Service Consumer requests to transfer an N2 information to the 5G-AN or;<br>- both, if the NF Service Consumer requests to transfer both an N1 message to the UE and an N2 information to the 5G-AN. |

**Table 6.1.3.5.3.1-2: Data structures supported by the POST Response Body on this resource**

| Data type | P | Cardinality | Response codes | Description |
|---|---|---|---|---|
| N1N2MessageTransferRspData | M | 1 | 202 Accepted | This case represents the successful storage of the N1/N2 information at the AMF when asynchronous communication is invoked or when the AMF pages the UE. If the AMF pages the UE, it shall store the N1/N2 message information until the UE responds to paging.<br><br>The cause included in the response body shall be set to one of the following values:<br>- WAITING_FOR_ASYNCHRONOUS_TRANSFER<br>- ATTEMPTING_TO_REACH_UE<br><br>The HTTP response shall include a "Location" HTTP header that contains the resource URI of the created resource. In this release, the URI shall only be used by NF Service consumer to correlate the possible N1/N2 Message Transfer Failure Notification with the related N1/N2 Message Transfer Operation. The NF service consumer shall not send any service requests towards the URI received in the Location header. |
| N1N2MessageTransferRspData | M | 1 | 200 OK | This represents the cases where:<br>- the AMF is able to successfully transfer the N1/N2 message to the UE and/or the AN;<br><br>- the AMF skips sending and discards the N1 message when UE is in CM-IDLE and the "skipInd" is set to "true" in the request; or<br><br>- the AMF skips sending and discards the N2 message as well as the possibly included N1 message, when the UE is in CM-CONNECTED state and the UE is outside of the validity area included in the N1N2MessageTransfer Request.<br><br>The cause included in the response body shall be to one of the following values:<br>- N1_N2_TRANSFER_INITIATED<br>- N1_MSG_NOT_TRANSFERRED<br>- N2_MSG_NOT_TRANSFERRED |
| RedirectResponse | O | 0..1 | 307 Temporary Redirect | Temporary redirection.<br>When the related UE context is not fully available at the target NF Service Consumer (e.g. AMF) during a planned maintenance case (e.g. AMF planned maintenance without UDSF case), the "cause" attribute shall be set to:<br>- NF_CONSUMER_REDIRECT_ONE_TXN<br><br>See table 6.1.7.3-1 for the description of these errors<br><br>(NOTE 2) |
| RedirectResponse | O | 0..1 | 308 Permanent Redirect | Permanent redirection.<br><br>(NOTE 2) |
| ProblemDetails | O | 0..1 | 403 Forbidden | The "cause" attribute may be used to indicate one of the following application errors:<br>- UE_IN_NON_ALLOWED_AREA<br>- UE_WITHOUT_N1_LPP_SUPPORT<br>- UNSPECIFIED<br>- SM_CONTEXT_RELOCATION_REQUIRED<br>- INVALID_PRU<br><br>See table 6.1.7.3-1 for the description of these errors. |

| | | | | |
|---|---|---|---|---|
| ProblemDetails | O | 0..1 | 404 Not Found | When the related UE is not found in the NF Service Consumer (e.g. AMF), the "cause" attribute shall be set to:<br>- CONTEXT_NOT_FOUND<br><br>See table 6.1.7.3-1 for the description of these errors. |
| N1N2MessageTransferError | O | 0..1 | 409 Conflict | This represents the case where the AMF rejects the N1N2MessageTransfer request due to one of the following reasons. The cause attribute of the ProblemDetails structure shall be set to:<br>- HIGHER_PRIORITY_REQUEST_ONGOING, if there is already an ongoing paging procedure with higher or same priority;<br>- TEMPORARY_REJECT_REGISTRATION_ONGOING, if there is an ongoing registration procedure (see clause 4.2.3.3 of TS 23.502 [3]);<br>- TEMPORARY_REJECT_HANDOVER_ONGOING, if there is an ongoing Xn or N2 handover procedure (see clause 4.9.1.2.1 and 4.9.1.3.1 of TS 23.502 [3]).<br>- TEMPORARY_REJECT_SR_ONGOING, if the AMF rejects a N1/N2 Message Transfer Request including a PDU Session Resource Setup Request Transfer IE due to an on-going Service Request procedure for the same PDU session (see clause 4.2.3.2 of TS 23.502 [3]);<br>- UE_IN_CM_IDLE_STATE, if this is a request to transfer a N2 PDU Session Resource Modify Request or a N2 PDU Session Resource Release Command to a 5G-AN, and if the UE is in CM-IDLE state at the AMF for the Access Network Type associated to the PDU session.<br>- MAX_ACTIVE_SESSIONS_EXCEEDED, if the RAT type is NB-IoT, and the UE already has 2 PDU Sessions with active user plane resources.<br>- REJECTION_DUE_TO_PAGING_RESTRICTION, if Paging Restrictions information restricts the N1N2MessageTransfer request from causing paging as specified in TS 23.501 [2] clause 5.38.5<br>- MAX_LCS_UPP_CONN_REACHED, if LCS-UPP connection already exists between the UE and one LMF and the request is to transfer "UPP-CM" N1 message to the UE from another LMF (see clause 5.2.2.3.1.2).<br><br>See table 6.1.7.3-1 for the description of these errors.<br><br>The AMF may additionally provide the "retryAfter" IE in order for the NF service consumer to throttle sending further N1/N2 Message Transfer request for a short period, e.g. when there is ongoing paging procedure with higher or same priority or a ongoing registration procedure. |
| N1N2MessageTransferError | O | 0..1 | 504 Gateway Timeout | This represents the case where the UE is not reachable at the AMF and the AMF is unable to page the UE. The cause attribute of the ProblemDetails structure shall be set to:<br>- UE_NOT_REACHABLE, if the UE is not reachable for paging;<br>- UE_NOT_RESPONDING, if the UE is not responding for a previous paging.<br><br>See table 6.1.7.3-1 for the description of these errors.<br><br>The AMF may additionally provide the "retryAfter" IE in order for the NF service consumer to throttle sending further N1/N2 Message Transfer request for a short period, e.g. when UE is not responding to paging. |
| ProblemDetails | O | 0..1 | 504 Gateway Timeout | This error shall only be returned by an SCP or a SEPP for errors they originate. |

NOTE 1: The mandatory HTTP error status code for the POST method listed in Table 5.2.7.1-1 of TS 29.500 [4] also apply, with response body containing an object of ProblemDetails data type (see clause 5.2.7 of TS 29.500 [4]).
NOTE 2: RedirectResponse may be inserted by an SCP or SEPP, see clause 6.10.9.1 of TS 29.500 [4].

**Table 6.1.3.5.3.1-3: Headers supported by the 202 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | The URI of the resource located on the AMF to which the status of the N1N2 message transfer is held |

**Table 6.1.3.5.3.1-4: Headers supported by the 307 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | The URI of the resource located on an alternative service instance within the same AMF or AMF (service) set to which the request is redirected.<br>For the case when a request is redirected to the same target resource via a different SCP or SEPP, see clause 6.10.9.1 in TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target NF (service) instance ID towards which the request is redirected |

**Table 6.1.3.5.3.1-5: Headers supported by the 308 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | An alternative URI of the resource located on an alternative service instance within the same AMF or AMF (service) set.<br>For the case when a request is redirected to the same target resource via a different SCP or SEPP, see clause 6.10.9.1 in TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target NF (service) instance ID towards which the request is redirected |

## 6.1.3.6 Resource: subscriptions collection

### 6.1.3.6.1 Description

This resource represents a collection of subscriptions of NF service consumers to the status change of the AMF identified by the GUAMI(s).

This resource is modelled as the Collection resource archetype (see clause C.2 of TS 29.501 [5]).

### 6.1.3.6.2 Resource Definition

Resource URI:{apiRoot}/namf-comm/<apiVersion>/subscriptions

This resource shall support the resource URI variables defined in table 6.1.3.6.2-1.

**Table 6.1.3.6.2-1: Resource URI variables for this resource**

| Name | Data type | Definition |
|---|---|---|
| apiRoot | string | See clause 6.1.1 |
| apiVersion | string | See clause 6.1.1. |

### 6.1.3.6.3 Resource Standard Methods

#### 6.1.3.6.3.1 POST

This method creates a new subscription. This method shall support the URI query parameters specified in table 6.1.3.6.3.1-1.

**Table 6.1.3.6.3.1-1: URI query parameters supported by the POST method on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| n/a | | | | |

This method shall support the request data structures specified in table 6.1.3.6.3.1-2 and the response data structures and response codes specified in table 6.1.3.6.3.1-3.

**Table 6.1.3.6.3.1-2: Data structures supported by the POST Request Body on this resource**

| Data type | P | Cardinality | Description |
|---|---|---|---|
| SubscriptionData | M | 1 | The request body contains the input parameters for the subscription. These parameters include, e.g.:<br>- GUAMI(s)<br>- amfStatusUri |

**Table 6.1.3.6.3.1-3: Data structures supported by the POST Response Body on this resource**

| Data type | P | Cardinality | Response codes | Description |
|---|---|---|---|---|
| SubscriptionData | M | 1 | 201 Created | This case represents the successful creation of a subscription.<br><br>Upon success, the HTTP response shall include a "Location" HTTP header that contains the resource URI of the created resource. |
| RedirectResponse | O | 0..1 | 307 Temporary Redirect | Temporary redirection.<br>(NOTE 2) |
| RedirectResponse | O | 0..1 | 308 Permanent Redirect | Permanent redirection.<br>(NOTE 2) |
| ProblemDetails | O | 0..1 | 403 Forbidden | The "cause" attribute may be used to indicate one of the following application errors:<br>- UNSPECIFIED<br><br>See table 6.1.7.3-1 for the description of this error. |
| NOTE 1: The mandatory HTTP error status code for the POST method listed in Table 5.2.7.1-1 of TS 29.500 [4] also apply, with response body containing an object of ProblemDetails data type (see clause 5.2.7 of TS 29.500 [4]).<br>NOTE 2: RedirectResponse may be inserted by an SCP or SEPP, see clause 6.10.9.1 of TS 29.500 [4]. | | | | |

**Table 6.1.3.6.3.1-4: Headers supported by the 201 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | Contains the URI of the newly created resource, according to the structure: {apiRoot}/namf-comm/<apiVersion>/subscriptions/{subscriptionId} |

**Table 6.1.3.6.3.1-5: Headers supported by the 307 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | An alternative URI of the resource located on an alternative service instance within the same AMF or AMF (service) set. For the case when a request is redirected to the same target resource via a different SCP or SEPP, see clause 6.10.9.1 in TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target NF (service) instance ID towards which the request is redirected |

**Table 6.1.3.6.3.1-6: Headers supported by the 308 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | An alternative URI of the resource located on an alternative service instance within the same AMF or AMF (service) set. For the case when a request is redirected to the same target resource via a different SCP or SEPP, see clause 6.10.9.1 in TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target NF (service) instance ID towards which the request is redirected |

## 6.1.3.7 Resource: individual subscription

### 6.1.3.7.1 Description

This resource represents an individual subscription of a NF service consumer to the status change of the AMF identified by the GUAMI(s).

This resource is modelled as the Document resource archetype (see clause C.1 of TS 29.501 [5]).

### 6.1.3.7.2 Resource Definition

Resource URI: {apiRoot}/namf-comm/<apiVersion>/subscriptions/{subscriptionId}

This resource shall support the resource URI variables defined in table 6.1.3.7.2-1.

**Table 6.1.3.7.2-1: Resource URI variables for this resource**

| Name | Data type | Definition |
|---|---|---|
| apiRoot | string | See clause 6.1.1 |
| apiVersion | string | See clause 6.1.1. |
| subscriptionId | string | Represents a specific subscription |

### 6.1.3.7.3 Resource Standard Methods

#### 6.1.3.7.3.1 DELETE

This method terminates an existing subscription. This method shall support the URI query parameters specified in table 6.1.3.7.3.1-1.

**Table 6.1.3.4.3.1-1: URI query parameters supported by the DELETE method on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| n/a | | | | |

This method shall support the request data structures specified in table 6.1.3.7.3.1-2 and the response data structures and response codes specified in table 6.1.3.7.3.1-3.

**Table 6.1.3.7.3.1-2: Data structures supported by the DELETE Request Body on this resource**

| Data type | P | Cardinality | Description |
|---|---|---|---|
| n/a | | | |

**Table 6.1.3.7.3.1-3: Data structures supported by the DELETE Response Body on this resource**

| Data type | P | Cardinality | Response codes | Description |
|---|---|---|---|---|
| | | | 204 No Content | This case represents a successful deletion of the subscription. |
| RedirectResponse | O | 0..1 | 307 Temporary Redirect | Temporary redirection.<br>(NOTE 2) |
| RedirectResponse | O | 0..1 | 308 Permanent Redirect | Permanent redirection.<br>(NOTE 2) |
| ProblemDetails | O | 0..1 | 404 Not Found | If the resource corresponding to the SubscriptionId cannot be found, the AMF shall return this status code. The "cause" attribute is set to:<br>- SUBSCRIPTION_NOT_FOUND |
| NOTE 1: The mandatory HTTP error status code for the DELETE method listed in Table 5.2.7.1-1 of TS 29.500 [4] also apply, with response body containing an object of ProblemDetails data type (see clause 5.2.7 of TS 29.500 [4]).<br>NOTE 2: RedirectResponse may be inserted by an SCP or SEPP, see clause 6.10.9.1 of TS 29.500 [4]. | | | | |

**Table 6.1.3.7.3.1-4: Headers supported by the 307 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | An alternative URI of the resource located on an alternative service instance within the same AMF or AMF (service) set. For the case when a request is redirected to the same target resource via a different SCP or SEPP, see clause 6.10.9.1 in TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target NF (service) instance ID towards which the request is redirected |

**Table 6.1.3.7.3.1-5: Headers supported by the 308 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | An alternative URI of the resource located on an alternative service instance within the same AMF or AMF (service) set. For the case when a request is redirected to the same target resource via a different SCP or SEPP, see clause 6.10.9.1 in TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target NF (service) instance ID towards which the request is redirected |

### 6.1.3.7.3.2 PUT

This method replaces an existing subscription completely. This method shall support the URI query parameters specified in table 6.1.3.7.3.2-1.

**Table 6.1.3.7.3.2-1: URI query parameters supported by the PUT method on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| n/a | | | | |

This method shall support the request data structures specified in table 6.1.3.7.3.2-2 and the response data structures and response codes specified in table 6.1.3.7.3.2-3.

**Table 6.1.3.7.3.2-2: Data structures supported by the PUT Request Body on this resource**

| Data type | P | Cardinality | Description |
|---|---|---|---|
| SubscriptionData | M | 1 | The request body contains the input parameters for the subscription. These parameters include, e.g.:<br>- GUAMI(s)<br>- amfStatusUri |

**Table 6.1.3.7.3.2-3: Data structures supported by the PUT Response Body on this resource**

| Data type | P | Cardinality | Response codes | Description |
|---|---|---|---|---|
| SubscriptionData | M | 1 | 200 OK | This case represents a successful replacement of the subscription. |
| n/a | | | 204 No Content | Represents the events subscription modification provided by the NF Service Consumer is accepted entirely. |
| RedirectResponse | O | 0..1 | 307 Temporary Redirect | Temporary redirection.<br>(NOTE 2) |
| RedirectResponse | O | 0..1 | 308 Permanent Redirect | Permanent redirection.<br>(NOTE 2) |
| ProblemDetails | O | 0..1 | 403 Forbidden | This case represents the failure update of an existing subscription. |
| NOTE 1: The mandatory HTTP error status code for the PUT method listed in Table 5.2.7.1-1 of TS 29.500 [4] also apply, with response body containing an object of ProblemDetails data type (see clause 5.2.7 of TS 29.500 [4]).<br>NOTE 2: RedirectResponse may be inserted by an SCP or SEPP, see clause 6.10.9.1 of TS 29.500 [4]. | | | | |

**Table 6.1.3.7.3.2-4: Headers supported by the 307 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | An alternative URI of the resource located on an alternative service instance within the same AMF or AMF (service) set.<br>For the case when a request is redirected to the same target resource via a different SCP or SEPP, see clause 6.10.9.1 in TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target NF (service) instance ID towards which the request is redirected |

**Table 6.1.3.7.3.2-5: Headers supported by the 308 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | An alternative URI of the resource located on an alternative service instance within the same AMF or AMF (service) set.<br>For the case when a request is redirected to the same target resource via a different SCP or SEPP, see clause 6.10.9.1 in TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target NF (service) instance ID towards which the request is redirected |

### 6.1.3.8 Resource: Non UE N2 Messages Collection

#### 6.1.3.8.1 Description

This resource represents the collection on which custom operations to transfer the N2 message towards the 5G-AN are specified. This resource is modelled with the Collection resource archetype (see clause C.2 of TS 29.501 [5]).

#### 6.1.3.8.2 Resource Definition

Resource URI: **{apiRoot}/namf-comm/<apiVersion>/non-ue-n2-messages**

This resource shall support the resource URI variables defined in table 6.1.3.8.2-1.

**Table 6.1.3.8.2-1: Resource URI variables for this resource**

| Name | Data Type | Definition |
|---|---|---|
| apiRoot | String | See clause 6.1.1 |
| apiVersion | String | See clause 6.1.1. |

#### 6.1.3.8.3 Resource Standard Methods

There are no resource standard methods for the non-ue-n2-messages collection resource in this release of this specification.

#### 6.1.3.8.4 Resource Custom Operations

##### 6.1.3.8.4.1 Overview

**Table 6.1.3.8.4.1-1: Custom operations**

| Operation Name | Custom operaration URI | Mapped HTTP method | Description |
|---|---|---|---|
| transfer | {resourceUri}/transfer | POST | Transfer the N2 message to 5G-AN. |

##### 6.1.3.8.4.2 Operation: transfer

###### 6.1.3.8.4.2.1 Description

The {resourceUri}/transfer custom operation is used to initiate a non UE associated N2 information transfer to the identified 5G-AN nodes. This custom operation uses the HTTP POST method.

###### 6.1.3.8.4.2.2 Operation Definition

This operation shall support the request data structures specified in table 6.1.3.8.4.2.2-1 and the response data structure and response codes specified in table 6.1.3.8.4.2.2-2.

**Table 6.1.3.8.4.2.2-1: Data structures supported by the POST Request Body on this resource**

| Data type | P | Cardinality | Description |
|---|---|---|---|
| N2InformationTransferReqData | M | 1 | Representation of the data to be sent to the 5G-AN node(s) by the AMF. |

**Table 6.1.3.8.4.2.2-2: Data structures supported by the POST Response Body on this resource**

| Data type | P | Cardinality | Response codes | Description |
|---|---|---|---|---|
| N2InformationTransferRspData | M | 1 | 200 OK | Indicates AMF has successfully initiated the transferring of N2 Information to the AN.. |
| RedirectResponse | O | 0..1 | 307 Temporary Redirect | Temporary redirection.<br>(NOTE 2) |
| RedirectResponse | O | 0..1 | 308 Permanent Redirect | Permanent redirection.<br>(NOTE 2) |
| N2InformationTransferError | O | 0..1 | 400 Bad Request | The "cause" attribute may be set to one of the errors defined in Table 5.2.7.2-1 of TS 29.500 [4]. |
| ProblemDetails | O | 0..1 | 400 Bad Request | This error shall only be returned by an SCP or a SEPP for errors they originate. |
| N2InformationTransferError | O | 0..1 | 403 Forbidden | The "cause" attribute may be set to one of the following application errors:<br>- UNSPECIFIED<br>See table 6.1.7.3-1 for the description of these errors. |
| ProblemDetails | O | 0..1 | 403 Forbidden | This error shall only be returned by an SCP or a SEPP for errors they originate. |
| N2InformationTransferError | O | 0..1 | 404 Not Found | The "cause" attribute may be set to one of the following application errors:<br>- CONTEXT_NOT_FOUND<br>See table 6.1.7.3-1 for the description of these errors. |
| N2InformationTransferError | O | 0..1 | 500 Internal Server Error | The "cause" attribute may be set to one of the errors defined in Table 5.2.7.2-1 of TS 29.500 [4]. |
| ProblemDetails | O | 0..1 | 500 Internal Server Error | This error shall only be returned by an SCP or a SEPP for errors they originate. |
| N2InformationTransferError | O | 0..1 | 503 Service Unavailable | The "cause" attribute may be set to one of the errors defined in Table 5.2.7.2-1 of TS 29.500 [4]. |
| ProblemDetails | O | 0..1 | 503 Service Unavailable | This error shall only be returned by an SCP or a SEPP for errors they originate. |
| NOTE 1: The mandatory HTTP error status code for the POST method listed in Table 5.2.7.1-1 of TS 29.500 [4] also apply, with response body containing an object of ProblemDetails data type (see clause 5.2.7 of TS 29.500 [4]).<br>NOTE 2: RedirectResponse may be inserted by an SCP or SEPP, see clause 6.10.9.1 of TS 29.500 [4]. | | | | |

**Table 6.1.3.8.4.2.2-3: Headers supported by the 307 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | An alternative URI of the resource located on an alternative service instance within the same AMF or AMF (service) set.<br>For the case when a request is redirected to the same target resource via a different SCP or SEPP, see clause 6.10.9.1 in TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target NF (service) instance ID towards which the request is redirected |

**Table 6.1.3.8.4.2.2-4: Headers supported by the 308 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | An alternative URI of the resource located on an alternative service instance within the same AMF or AMF (service) set.<br>For the case when a request is redirected to the same target resource via a different SCP or SEPP, see clause 6.10.9.1 in TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target NF (service) instance ID towards which the request is redirected |

### 6.1.3.9 Resource: Non UE N2 Messages Subscriptions Collection

#### 6.1.3.9.1 Description

This resource represents the collection on which individual subscriptions for non UE N2 messages from the 5G-AN are stored. This resource is modelled with the Collection resource archetype (see clause C.2 of TS 29.501 [5]).

#### 6.1.3.9.2 Resource Definition

Resource URI: **{apiRoot}/namf-comm/<apiVersion>/non-ue-n2-messages/subscriptions**

This resource shall support the resource URI variables defined in table 6.1.3.9.2-1.

**Table 6.1.3.9.2-1: Resource URI variables for this resource**

| Name | Data type | Definition |
|---|---|---|
| apiRoot | string | See clause 6.1.1 |
| apiVersion | string | See clause 6.1.1. |

#### 6.1.3.9.3 Resource Standard Methods

##### 6.1.3.9.3.1 POST

This method creates an individual N2 information subscription resource for non UE related N2 information. This method is used by NF Service Consumers (e.g. LMF, CBCF/PWS-IWF) to subscribe for notifications about non UE related N2 Information from a specific 5G-AN node, or from any 5G-AN node.

This method shall support the request data structures specified in table 6.1.3.9.3.1-2 and the response data structures and response codes specified in table 6.1.3.9.3.1-3.

**Table 6.1.3.9.3.1-2: Data structures supported by the POST Request Body on this resource**

| Data type | P | Cardinality | Description |
|---|---|---|---|
| NonUeN2InfoSubscriptionCreateData | M | 1 | Representation of the subscription for N2 information notification. |

**Table 6.1.3.9.3.1-3: Data structures supported by the POST Response Body on this resource**

| Data type | P | Cardinality | Response codes | Description |
|---|---|---|---|---|
| NonUeN2InfoSubscriptionCreatedData | M | 1 | 201 Created | This case represents the successful creation of the subscription for N2 information notification.<br><br>Upon success, a response body is returned containing the representation describing the status of the request. The Location header shall carry the location (URI) of the created subscription resource. |
| RedirectResponse | O | 0..1 | 307 Temporary Redirect | Temporary redirection.<br>(NOTE 2) |
| RedirectResponse | O | 0..1 | 308 Permanent Redirect | Permanent redirection.<br>(NOTE 2) |
| ProblemDetails | O | 0..1 | 403 Forbidden | If the NF Service Consumer is not authorized to subscribe for non UE N2 message notifications, the AMF shall return this status code with the ProblemDetails |
| NOTE 1: The mandatory HTTP error status code for the POST method listed in Table 5.2.7.1-1 of TS 29.500 [4] also apply, with response body containing an object of ProblemDetails data type (see clause 5.2.7 of TS 29.500 [4]).<br>NOTE 2: RedirectResponse may be inserted by an SCP or SEPP, see clause 6.10.9.1 of TS 29.500 [4]. | | | | |

**Table 6.1.3.9.3.1-4: Headers supported by the 201 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | Contains the URI of the newly created resource, according to the structure: {apiRoot}/namf-comm/<apiVersion>/non-ue-n2-messages/subscriptions/{n2NotifySubscriptionId} |

**Table 6.1.3.9.3.1-5: Headers supported by the 307 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | An alternative URI of the resource located on an alternative service instance within the same AMF or AMF (service) set.<br>Or the same URI, if a request is redirected to the same target resource via a different SCP. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target NF (service) instance ID towards which the request is redirected |

**Table 6.1.3.9.3.1-6: Headers supported by the 308 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | An alternative URI of the resource located on an alternative service instance within the same AMF or AMF (service) set.<br>Or the same URI, if a request is redirected to the same target resource via a different SCP. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target NF (service) instance ID towards which the request is redirected |

### 6.1.3.9.4 Resource Custom Operations

There are no custom operations supported on this resource.

## 6.1.3.10 Resource: Non UE N2 Message Notification Individual Subscription

### 6.1.3.10.1 Description

This resource represents the individual subscription for the notifications of non UE specific N2 message types (e.g. NRPPa, PWS Notifications).

This resource is modelled with the Document resource archetype (see clause C.1 of TS 29.501 [5]).

### 6.1.3.10.2 Resource Definition

Resource URI: **{apiRoot}/namf-comm/<apiVersion>/non-ue-n2-messages/subscriptions/{n2NotifySubscriptionId}**

This resource shall support the resource URI variables defined in table 6.1.3.10.2-1.

**Table 6.1.3.7.2-1: Resource URI variables for this resource**

| Name | Data type | Definition |
|---|---|---|
| apiRoot | string | See clause 6.1.1 |
| apiVersion | string | See clause 6.1.1. |
| n2NotifySubscriptionId | string | Represents the individual subscription to the non UE specific N2 message notification. |

### 6.1.3.10.3 Resource Standard Methods

#### 6.1.3.10.3.1 DELETE

This method deletes an individual N2 message notification subscription resource for non UE associated N2 information. This method is used by NF Service Consumers (e.g. LMF) to unsubscribe for notifications about non UE related N2 information.

This method shall support the request data structures specified in table 6.1.3.10.3.1-2 and the response data structures and response codes specified in table 6.1.3.10.3.1-3.

**Table 6.1.3.10.3.1-2: Data structures supported by the DELETE Request Body on this resource**

| Data type | P | Cardinality | Description |
|---|---|---|---|
| n/a | | | |

**Table 6.1.3.10.3.1-3: Data structures supported by the DELETE Response Body on this resource**

| Data type | P | Cardinality | Response codes | Description |
|---|---|---|---|---|
| n/a | | | 204 No Content | |
| RedirectResponse | O | 0..1 | 307 Temporary Redirect | Temporary redirection.<br>(NOTE 2) |
| RedirectResponse | O | 0..1 | 308 Permanent Redirect | Permanent redirection.<br>(NOTE 2) |
| ProblemDetails | O | 0..1 | 404 Not Found | If the resource corresponding to the SubscriptionId cannot be found the AMF shall return this status code. The "cause" attribute is set to:<br>- SUBSCRIPTION_NOT_FOUND |
| NOTE 1: The mandatory HTTP error status code for the DELETE method listed in Table 5.2.7.1-1 of TS 29.500 [4] also apply, with response body containing an object of ProblemDetails data type (see clause 5.2.7 of TS 29.500 [4]).<br>NOTE 2: RedirectResponse may be inserted by an SCP or SEPP, see clause 6.10.9.1 of TS 29.500 [4]. | | | | |

**Table 6.1.3.10.3.1-5: Headers supported by the 307 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | An alternative URI of the resource located on an alternative service instance within the same AMF or AMF (service) set. For the case when a request is redirected to the same target resource via a different SCP or SEPP, see clause 6.10.9.1 in TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target NF (service) instance ID towards which the request is redirected |

**Table 6.1.3.10.3.1-6: Headers supported by the 308 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | An alternative URI of the resource located on an alternative service instance within the same AMF or AMF (service) set. For the case when a request is redirected to the same target resource via a different SCP or SEPP, see clause 6.10.9.1 in TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target NF (service) instance ID towards which the request is redirected |

#### 6.1.3.10.4 Resource Custom Operations

There are no custom operations supported on this resource.

### 6.1.4 Custom Operations without associated resources

There are no custom operations without associated resources supported on Namf_Communication Service.

### 6.1.5 Notifications

#### 6.1.5.1 General

The notifications provided by the Namf_Communication service are specified in this clause.

**Table 6.1.5.1-1: Callback overview**

| Notification | Resource URI | HTTP method or custom operation | Description (service operation) |
|---|---|---|---|
| AMF Status Change Notification | {amfStatusUri} | POST | |
| Non UE N2 Information Notification | {n2NotifyCallbackUri} | POST | |
| N1 Message Notification | {n1NotifyCallbackUri} | POST | |
| UE Specific N2 Information Notification | {n2NotifyCallbackUri} | POST | |
| N1N2 Transfer Failure Notification | {n1n2FailureTxfNotifURI} | POST | |

## 6.1.5.2 AMF Status Change Notification

### 6.1.5.2.1 Description

If a NF service consumer (e.g. SMF) has subscribed to AMF Status Change on Namf_Communication Service, when AMF aware of a change of its own status, AMF shall create a notification including the current state, and shall deliver the notification to the call-back URI, following Subscribe/Notify mechanism defined in TS 29.501 [5].

### 6.1.5.2.2 Notification Definition

Call-back URI: {amfStatusUri}

Call-back URI is provided by NF Service Consumer during creation of the subscription.

### 6.1.5.2.3 Notification Standard Methods

#### 6.1.5.2.3.1 POST

This method shall support the request data structures specified in table 6.1.5.2.3.1-1 and the response data structures and response codes specified in table 6.1.5.2.3.1-2.

**Table 6.1.5.2.3.1-1: Data structures supported by the POST Request Body**

| Data type | P | Cardinality | Description |
|---|---|---|---|
| AmfStatusChangeNotification | M | 1 | Representation of the AMF status change notification. |

**Table 6.1.5.2.3.1-2: Data structures supported by the POST Response Body**

| Data type | P | Cardinality | Response codes | Description |
|---|---|---|---|---|
| n/a | | | 204 No Content | This case represents a successful notification of the AMF status change. |
| RedirectResponse | O | 0..1 | 307 Temporary Redirect | Temporary redirection.<br>(NOTE 2) |
| RedirectResponse | O | 0..1 | 308 Permanent Redirect | Permanent redirection.<br>(NOTE 2) |
| ProblemDetails | O | 0..1 | 404 Not Found | When context of the notification is not found,the "cause" attribute shall be set to:<br>- CONTEXT_NOT_FOUND |
| NOTE 1: The mandatory HTTP error status code for the POST method listed in Table 5.2.7.1-1 of TS 29.500 [4] also apply, with response body containing an object of ProblemDetails data type (see clause 5.2.7 of TS 29.500 [4]).<br>NOTE 2: RedirectResponse may be inserted by an SCP or SEPP, see clause 6.10.9.1 of TS 29.500 [4]. | | | | |

**Table 6.1.5.2.3.1-3: Headers supported by the 307 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | A URI pointing to the endpoint of the NF service consumer to which the notification should be sent. For the case when a request is redirected to the same target resource via a different SCP or SEPP, see clause 6.10.9.1 in TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target NF (service) instance ID towards which the request is redirected |

**Table 6.1.5.2.3.1-4: Headers supported by the 308 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | A URI pointing to the endpoint of the NF service consumer to which the notification should be sent. For the case when a request is redirected to the same target resource via a different SCP or SEPP, see clause 6.10.9.1 in TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target NF (service) instance ID towards which the request is redirected |

## 6.1.5.3 Non UE N2 Information Notification

### 6.1.5.3.1 Description

This resource represents the callback reference provided by the NF Service Consumer (e.g. LMF, CBCF/PWS-IWF) to receive notifications about N2 information that are not related to a UE.

### 6.1.5.3.2 Notification Definition

Callback URI: {n2NotifyCallbackUri}

This notification shall support the callback URI variables defined in table 6.1.5.2.2-1.

**Table 6.1.5.3.2-1: Callback URI variables for this notification**

| Name | Definition |
|---|---|
| n2NotifyCallbackUri | Callback reference provided by the NF Service Consumer during the subscription to this notification. |

### 6.1.5.3.3 Notification Standard Methods

#### 6.1.5.3.3.1 POST

This method sends an N2 information notification to the NF Service Consumer (e.g. LMF, CBCF/PWS-IWF).

This method shall support the request data structures specified in table 6.1.5.3.3.1-2 and the response data structures and response codes specified in table 6.1.5.3.3.1-3.

**Table 6.1.5.3.3.1-2: Data structures supported by the POST Request Body**

| Data type | P | Cardinality | Description |
|---|---|---|---|
| N2InformationNotification | M | 1 | Representation of the N2 information notification. |

**Table 6.1.5.3.3.1-3: Data structures supported by the POST Response Body**

| Data type | P | Cardinality | Response codes | Description |
|---|---|---|---|---|
| n/a | | | 204 No Content | This case represents a successful notification of the N2 information to the NF service consumer. |
| RedirectResponse | O | 0..1 | 307 Temporary Redirect | Temporary redirection.<br>(NOTE 2) |
| RedirectResponse | O | 0..1 | 308 Permanent Redirect | Permanent redirection.<br>(NOTE 2) |
| NOTE 1: The mandatory HTTP error status code for the POST method listed in Table 5.2.7.1-1 of TS 29.500 [4] also apply, with response body containing an object of ProblemDetails data type (see clause 5.2.7 of TS 29.500 [4]).<br>NOTE 2: RedirectResponse may be inserted by an SCP or SEPP, see clause 6.10.9.1 of TS 29.500 [4]. | | | | |

**Table 6.1.5.3.3.1-4: Headers supported by the 307 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | A URI pointing to the endpoint of the NF service consumer to which the notification should be sent.<br>For the case when a request is redirected to the same target resource via a different SCP or SEPP, see clause 6.10.9.1 in TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target NF (service) instance ID towards which the request is redirected.<br>For the case when a request is redirected to the same target resource via a different SCP or SEPP, see clause 6.10.9.1 in TS 29.500 [4]. |

**Table 6.1.5.3.3.1-5: Headers supported by the 308 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | A URI pointing to the endpoint of the NF service consumer to which the notification should be sent |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target NF (service) instance ID towards which the request is redirected |

### 6.1.5.4 N1 Message Notification

#### 6.1.5.4.1 Description

This resource represents the callback reference provided by the NF Service Consumer (e.g. LMF) to receive notifications about N1 message from the UE (e.g. LPP messages).

#### 6.1.5.4.2 Notification Definition

Callback URI: { n1NotifyCallbackUri }

Callback URI is provided by the NF Service Consumer during the subscription to this notification. . The callback URI for N1 message notification may also be obtained from the NRF, if the NF Service Consumer has registered it in the NF Profile with the NRF.

#### 6.1.5.4.3 Notification Standard Methods

##### 6.1.5.4.3.1 POST

This method sends an N1 message notification to the NF Service Consumer (e.g. LMF).

This method shall support the request data structures specified in table 6.1.5.4.3.1-2 and the response data structures and response codes specified in table 6.1.5.4.3.1-3.

**Table 6.1.5.4.3.1-2: Data structures supported by the POST Request Body**

| Data type | P | Cardinality | Description |
|---|---|---|---|
| N1MessageNotification | M | 1 | Representation of the N1 message notification. |

**Table 6.1.5.4.3.1-3: Data structures supported by the POST Response Body**

| Data type | P | Cardinality | Response codes | Description |
|---|---|---|---|---|
| n/a | | | 204 No Content | This case represents a successful notification of the N1 message to the NF service consumer. |
| RedirectResponse | O | 0..1 | 307 Temporary Redirect | Temporary redirection.<br>(NOTE 2) |
| RedirectResponse | O | 0..1 | 308 Permanent Redirect | Permanent redirection.<br>(NOTE 2) |
| ProblemDetails | O | 0..1 | 403 Forbidden | This case represents, the NF service consumer failing to accept the processing of the notified N1 message. The detailed information shall be provided in the ProblemDetails structure. |
| NOTE 1: The mandatory HTTP error status code for the POST method listed in Table 5.2.7.1-1 of TS 29.500 [4] also apply, with response body containing an object of ProblemDetails data type (see clause 5.2.7 of TS 29.500 [4]).<br>NOTE 2: RedirectResponse may be inserted by an SCP or SEPP, see clause 6.10.9.1 of TS 29.500 [4]. | | | | |

**Table 6.1.5.4.3.1-4: Headers supported by the 307 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | A URI pointing to the endpoint of the NF service consumer to which the notification should be sent.<br>For the case when a request is redirected to the same target resource via a different SCP or SEPP, see clause 6.10.9.1 in TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target NF (service) instance ID towards which the request is redirected |

**Table 6.1.5.4.3.1-5: Headers supported by the 308 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | A URI pointing to the endpoint of the NF service consumer to which the notification should be sent.<br>For the case when a request is redirected to the same target resource via a different SCP or SEPP, see clause 6.10.9.1 in TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target NF (service) instance ID towards which the request is redirected |

## 6.1.5.5 UE Specific N2 Information Notification

### 6.1.5.5.1 Description

This resource represents the callback reference provided by the NF Service Consumer (e.g. LMF) to receive notifications about UE specific N2 information.

### 6.1.5.5.2 Notification Definition

Callback URI: {n2NotifyCallbackUri}

Callback URI is provided by the NF Service Consumer during the subscription to this notification.

### 6.1.5.5.3 Notification Standard Methods

#### 6.1.5.5.3.1 POST

This method sends an N2 information notification to the NF Service Consumer (e.g. LMF).

This method shall support the request data structures specified in table 6.1.5.5.3.1-2 and the response data structures and response codes specified in table 6.1.5.5.3.1-3.

**Table 6.1.5.5.3.1-2: Data structures supported by the POST Request Body**

| Data type | P | Cardinality | Description |
|---|---|---|---|
| N2InformationNotification | M | 1 | Representation of the N2 information notification. |

**Table 6.1.5.5.3.1-3: Data structures supported by the POST Response Body**

| Data type | P | Cardinality | Response codes | Description |
|---|---|---|---|---|
| n/a | | | 204 No Content | This case represents a successful notification of the N2 information to the NF service consumer. |
| N2InfoNotificationRspData | M | 1 | 200 OK | This case represents a successful notification of the N2 information to the NF service consumer when information needs to be returned in the response. |
| RedirectResponse | O | 0..1 | 307 Temporary Redirect | Temporary redirection.<br>(NOTE 2) |
| RedirectResponse | O | 0..1 | 308 Permanent Redirect | Permanent redirection.<br>(NOTE 2) |
| NOTE 1: The mandatory HTTP error status code for the POST method listed in Table 5.2.7.1-1 of TS 29.500 [4] also apply, with response body containing an object of ProblemDetails data type (see clause 5.2.7 of TS 29.500 [4]).<br>NOTE 2: RedirectResponse may be inserted by an SCP or SEPP, see clause 6.10.9.1 of TS 29.500 [4]. | | | | |

**Table 6.1.5.5.3.1-4: Headers supported by the 307 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | A URI pointing to the endpoint of the NF service consumer to which the notification should be sent.<br>For the case when a request is redirected to the same target resource via a different SCP or SEPP, see clause 6.10.9.1 in TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target NF (service) instance ID towards which the request is redirected |

**Table 6.1.5.5.3.1-5: Headers supported by the 308 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | A URI pointing to the endpoint of the NF service consumer to which the notification should be sent'.<br>For the case when a request is redirected to the same target resource via a different SCP or SEPP, see clause 6.10.9.1 in TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target NF (service) instance ID towards which the request is redirected |

## 6.1.5.6 N1N2 Transfer Failure Notification

### 6.1.5.6.1 Description

This resource represents the callback reference provided by the NF Service Consumer (e.g. SMF) to receive notifications about failure to deliver N1 / N2 message.

### 6.1.5.6.2 Notification Definition

Callback URI: {n1n2FailureTxfNotifURI}

Callback URI is provided by the NF Service Consumer during the UE specific N1N2MessageTransfer operation (see clause 6.1.3.5.3.1.

#### 6.1.5.6.3 Notification Standard Methods

##### 6.1.5.6.3.1 POST

This method sends an N1/N2 message transfer failure notification to the NF Service Consumer (e.g. SMF).

This method shall support the request data structures specified in table 6.1.5.6.3.1-1 and the response data structures and response codes specified in table 6.1.5.6.3.1-3.

**Table 6.1.5.6.3.1-1: Data structures supported by the POST Request Body**

| Data type | P | Cardinality | Description |
|---|---|---|---|
| N1N2MsgTxfrFailureNotification | M | 1 | Representation of the N1/N2 message transfer failure notification.<br><br>The "cause" attribute shall be set to one of following cause value s (see clause 6.1.6.3.6):<br>- UE_NOT_RESPONDING<br>- UE_NOT_REACHABLE_FOR_SESSION<br>- TEMPORARY_REJECT_REGISTRATION_ONGOING<br>- TEMPORARY_REJECT_HANDOVER_ONGOING<br>- REJECTION_DUE_TO_PAGING_RESTRICTION<br>- AN_NOT_RESPONDING<br>- FAILURE_CAUSE_UNSPECIFIED<br><br>The AMF may additionally provide the “retryAfter” IE in order for the NF service consumer to throttle sending further N1/N2 Message Transfer request for a short period, e.g. when UE is not responding to paging. |

**Table 6.1.5.6.3.1-2: Data structures supported by the POST Response Body**

| Data type | P | Cardinality | Response codes | Description |
|---|---|---|---|---|
| n/a | | | 204 No Content | This case represents a successful notification of the N1 / N2 message transfer to the NF service consumer. |
| RedirectResponse | O | 0..1 | 307 Temporary Redirect | Temporary redirection.<br>(NOTE 2) |
| RedirectResponse | O | 0..1 | 308 Permanent Redirect | Permanent redirection.<br>(NOTE 2) |
| NOTE 1: The mandatory HTTP error status code for the POST method listed in Table 5.2.7.1-1 of TS 29.500 [4] also apply, with response body containing an object of ProblemDetails data type (see clause 5.2.7 of TS 29.500 [4]).<br>NOTE 2: RedirectResponse may be inserted by an SCP or SEPP, see clause 6.10.9.1 of TS 29.500 [4]. | | | | |

**Table 6.1.5.6.3.1-3: Headers supported by the 307 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | A URI pointing to the endpoint of the NF service consumer to which the notification should be sent.<br>For the case when a request is redirected to the same target resource via a different SCP or SEPP, see clause 6.10.9.1 in TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target NF (service) instance ID towards which the request is redirected |

**Table 6.1.5.6.3.1-4: Headers supported by the 308 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | A URI pointing to the endpoint of the NF service consumer to which the notification should be sent.<br>For the case when a request is redirected to the same target resource via a different SCP or SEPP, see clause 6.10.9.1 in TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target NF (service) instance ID towards which the request is redirected |

### 6.1.5.7 Void

## 6.1.6 Data Model

### 6.1.6.1 General

This clause specifies the application data model supported by the API.

Table 6.1.6.1-1 specifies the data types defined for the Namf_Communication service based interface protocol.

**Table 6.1.6.1-1: Namf_Communication specific Data Types**

| Data type | Clause defined | Description |
|---|---|---|
| SubscriptionData | 6.1.6.2.2 | Data within an AMF Status Change Subscription request and response. |
| AmfStatusChangeNotification | 6.1.6.2.3 | Data within an AMF Status Change Notification request. |
| AmfStatusInfo | 6.1.6.2.4 | AMF Status Information |
| AssignEbiData | 6.1.6.2.5 | Data within an EBI assignment request. |
| AssignedEbiData | 6.1.6.2.6 | Data within a successful response to an EBI assignment request. |
| AssignEbiFailed | 6.1.6.2.7 | Represents failed assignment of EBI(s) |
| UEContextRelease | 6.1.6.2.8 | Data within a Release UE Context request. |
| N2InformationTransferReqData | 6.1.6.2.9 | Data within a N2 Information Transfer request containing the N2 information requested to be transferred to 5G AN. |
| NonUeN2InfoSubscriptionCreateData | 6.1.6.2.10 | Data within a create subscription requestfor non-UE specific N2 information notification. |
| NonUeN2InfoSubscriptionCreatedData | 6.1.6.2.11 | Data for the created subscription for non-UE specific N2 information notification. |
| UeN1N2InfoSubscriptionCreateData | 6.1.6.2.12 | Data within a create subscription request for UE specific N1 and/or N2 information notification. |
| UeN1N2InfoSubscriptionCreatedData | 6.1.6.2.13 | Data for the created subscription for UE specific N1 and/or N2 information notification. |
| N2InformationNotification | 6.1.6.2.14 | Data within a N2 information notification request. |
| N2InfoContainer | 6.1.6.2.15 | N2 information container. |
| N1MessageNotification | 6.1.6.2.16 | Data within a N1 message notification request. |
| N1MessageContainer | 6.1.6.2.17 | N1 Message Container. |
| N1N2MessageTransferReqData | 6.1.6.2.18 | Data within a N1/N2 message transfer request. |
| N1N2MessageTransferRspData | 6.1.6.2.19 | Data within a N1/N2 message transfer response. |
| RegistrationContextContainer | 6.1.6.2.20 | Registration Context Container used to send the UE context information, N1 message from UE, AN address etc during Registration with AMF re-allocation procedure. |
| AreaOfValidity | 6.1.6.2.21 | Area of validity information for N2 information transfer |
| UeContextTransferReqData | 6.1.6.2.23 | Data within a UE Context Transfer Request to start transferring of an individual ueContext resource from old AMF to new AMF. |
| UeContextTransferRspData | 6.1.6.2.24 | Data within a successful response to the UE Context Transfer request. |
| UeContext | 6.1.6.2.25 | Represents an individual ueContext resource |
| N2SmInformation | 6.1.6.2.26 | Represents the session management SMF related N2 information data part. |
| N2InfoContent | 6.1.6.2.27 | Represents a transparent N2 information content to be relayed by AMF. |
| NrppaInformation | 6.1.6.2.28 | Represents a NRPPa related N2 information data part. |
| PwsInformation | 6.1.6.2.29 | Represents a PWS related information data part. |
| N1N2MsgTxfrFailureNotification | 6.1.6.2.30 | Data within a N1/N2 Message Transfer Failure Notification request |
| N1N2MessageTransferError | 6.1.6.2.31 | Data within a N1/N2 Message Transfer Error response. |
| N1N2MsgTxfrErrDetail | 6.1.6.2.32 | N1/N2 Message Transfer Error Details |
| N2InformationTransferRspData | 6.1.6.2.33 | Data within a successful response to the N2 Information Transfer request to transfer N2 Information to the AN. |
| MmContext | 6.1.6.2.34 | Represents a Mobility Management Context in UE Context |
| SeafData | 6.1.6.2.35 | Represents SEAF data derived from data received from AUSF |

| | | |
|---|---|---|
| NasSecurityMode | 6.1.6.2.36 | Indicates the NAS Security Mode |
| PduSessionContext | 6.1.6.2.37 | Represents a PDU Session Context in UE Context |
| NssaiMapping | 6.1.6.2.38 | Represents the mapping between a S-NSSAI in serving PLMN to a S-NSSAI in home PLMN. |
| UeRegStatusUpdateReqData | 6.1.6.2.39 | Data within a UE registration status update request to indicate a completion of transferring at a target AMF. |
| AssignEbiError | 6.1.6.2.40 | Data within a failure response to the EBI assignment request. |
| UeContextCreateData | 6.1.6.2.41 | Data within a request to create an individual ueContext resource |
| UeContextCreatedData | 6.1.6.2.42 | Data within a successful response for creating an individual ueContext resource |
| UeContextCreateError | 6.1.6.2.43 | Data within a failure response for creating a UE context |
| NgRanTargetId | 6.1.6.2.44 | Indicates a NG RAN as target of the handover |
| N2InformationTransferError | 6.1.6.2.45 | Data within a failure response for a non-UE related N2 Information Transfer. |
| PWSResponseData | 6.1.6.2.46 | Data related PWS included in a N2 Information Transfer response. |
| PWSErrorData | 6.1.6.2.47 | Data related to PWS error included in a N2 Information Transfer failure response. |
| NgKsi | 6.1.6.2.49 | Represents the ngKSI (see TS 33.501 [27]) |
| KeyAmf | 6.1.6.2.50 | Represents the $K_{amf}$ or $K'_{amf.}$ (see TS 33.501 [27]). |
| ExpectedUeBehavior | 6.1.6.2.51 | Represents the expected UE behavior (e.g. UE moving trajectory) and its validity period. |
| UeRegStatusUpdateRspData | 6.1.6.2.52 | Data within a UE registration status update response to provide the status of UE context transfer status update at a source AMF. |
| N2RanInformation | 6.1.6.2.53 | Represents the RAN related N2 information data part. |
| N2InfoNotificationRspData | 6.1.6.2.54 | Data within a N2 information notification response. |
| SmallDataRateStatusInfo | 6.1.6.2.55 | Represents the small data rate status |
| SmfChangeInfo | 6.1.6.2.56 | SMF change information for PDU session(s) |
| V2xContext | 6.1.6.2.57 | Represents the V2X services related parameters |
| ImmediateMdtConf | 6.1.6.2.58 | Immediate MDT Configuration |
| V2xInformation | 6.1.6.2.59 | V2X related N2 information |
| EpsNasSecurityMode | 6.1.6.2.60 | Indicates the EPS NAS Security Mode |
| UeContextRelocateData | 6.1.6.2.61 | Data within a Relocate UE Context request |
| UeContextRelocatedData | 6.1.6.2.62 | Data within a Relocate UE Context |
| EcRestrictionDataWb | 6.1.6.2.64 | Enhanced Coverage Restriction Data for WB-N1 mode. |
| ExtAmfEventSubscription | 6.1.6.2.65 | AMF event subscription extended with additional information received for the subscription |
| AmfEventSubscriptionAddInfo | 6.1.6.2.66 | Additional information received for an AMF event subscription, e.g. binding indications. |
| UeContextCancelRelocateData | 6.1.6.2.67 | Data structure used for cancellation of UE Context Relocation. |
| UeDifferentiationInfo | 6.1.6.2.68 | Represents the UE Differentiation Information and its validity time. |
| CeModeBInd | 6.1.6.2.69 | CE-mode-B Support Indicator |
| LteMInd | 6.1.6.2.70 | LTE-M Indication |
| NpnAccessInfo | 6.1.6.2.71 | NPN Access Information |
| ProseContext | 6.1.6.2.72 | Represents the ProSE services related parameters |
| AnalyticsSubscription | 6.1.6.2.73 | Analytics subscriptions created in the NWDAF. |
| NwdafSubscription | 6.1.6.2.74 | Individual NWDAF subscription identified by the subscription Id. |
| UpdpSubscriptionData | 6.1.6.2.75 | UE policy delivery related N1 message notification subscription data |
| ProSeInformation | 6.1.6.2.76 | 5G ProSe related N2 information. |

| | | |
|---|---|---|
| ReleaseSessionInfo | 6.1.6.2.77 | PDU session Id(s) and the cause for triggering the release |
| AreaOfInterestEventState | 6.1.6.2.78 | Area Of Interest Event State in old AMF |
| TssInformation | 6.1.6.2.79 | TSS related N2 information |
| AmPolicyInfoContainer | 6.1.6.2.80 | AM Policy Information Container |
| RslpInformation | 6.1.6.2.81 | Ranging/SL positioning related N2 information |
| A2xContext | 6.1.6.2.82 | Represents the A2X services related parameters |
| A2xInformation | 6.1.6.2.83 | A2X related N2 information |
| LcsUpContext | 6.1.6.2.84 | Represents the LCS UP related parameters |
| DeregInactTimerInfo | 6.1.6.2.85 | Network Slice Deregistration Inactivity Timer Information |
| TssRspPerNgran | 6.1.6.2.86 | TSS Information from NG-RAN |
| SliceReplacementMapping | 6.1.6.2.87 | Represents the mapping between a replaced S-NSSAI in serving PLMN to its alternative S-NSSAI. |
| SliceDeregInactConfig | 6.1.6.2.88 | Deregistration inactivity configuration |
| EpsBearerId | 6.1.6.3.2 | EPS Bearer Identifier |
| Ppi | 6.1.6.3.2 | Paging Policy Indicator |
| NasCount | 6.1.6.3.2 | Represents a NAS COUNT |
| 5GMmCapability | 6.1.6.3.2 | Represents a 5GMM capability |
| UeSecurityCapability | 6.1.6.3.2 | Represents a UE Security Capability |
| S1UeNetworkCapability | 6.1.6.3.2 | Represents a S1 UE Network Capability |
| DrxParameter | 6.1.6.3.2 | Indicates the UE DRX Parameters |
| OmcIdentifier | 6.1.6.3.2 | Represents the OMC Identifier |
| MSClassmark2 | 6.1.6.3.2 | Indicates the MS Classmark 2 of a 5G SRVCC UE |
| SupportedCodec | 6.1.6.3.2 | Indicates the supported codec of a 5G SRVCC UE |
| StatusChange | 6.1.6.3.3 | Enumeration for AMF status |
| N2InformationClass | 6.1.6.3.4 | Enumeration for N2 Information Class |
| N1MessageClass | 6.1.6.3.5 | Enumeration for N1 Message Class |
| N1N2MessageTransferCause | 6.1.6.3.6 | Enumeration for N1N2Message Transfer Cause |
| UeContextTransferStatus | 6.1.6.3.7 | Describes the status of an individual ueContext resource in UE Context Transfer procedures |
| N2InformationTransferResult | 6.1.6.3.8 | Describes the result of N2 information transfer by AMF to the AN. |
| CipheringAlgorithm | 6.1.6.3.9 | Indicates the supported Ciphering Algorithm |
| IntegrityAlgorithm | 6.1.6.3.10 | Indicates the supported Integrity Algorithm |
| SmsSupport | 6.1.6.3.11 | Indicates the supported SMS delivery of a UE. |
| ScType | 6.1.6.3.12 | Indicates the security context type. |
| KeyAmfType | 6.1.6.3.13 | Indicates the $K_{amf}$ type. |
| TransferReason | 6.1.6.3.14 | Indicates UE Context Transfer Reason |
| PolicyReqTrigger | 6.1.6.3.15 | Policy Request Triggers |
| RatSelector | 6.1.6.3.16 | Indicates the RAT type for the transfer of N2 information |
| NgapIeType | 6.1.6.3.17 | Indicates the supported NGAP IE types |
| N2InfoNotifyReason | 6.1.6.3.18 | N2 Information Notify Reason |
| SmfChangeIndication | 6.1.6.3.19 | Indicates the I-SMF or V-SMF change or removal |
| SbiBindingLevel | 6.1.6.3.20 | SBI Binding Level |
| EpsNasCipheringAlgorithm | 6.1.6.3.21 | Indicates the supported EPS NAS Ciphering Algorithm |
| EpsNasIntegrityAlgorithm | 6.1.6.3.22 | Indicates the supported EPS NAS Integrity Algorithm |
| PeriodicCommunicationIndicator | 6.1.6.3.23 | Indicates the Periodic Communication Indicator |
| UuaaMmStatus | 6.1.6.3.24 | Indicates UUAA-MM status |
| ReleaseCause | 6.1.6.3.25 | The cause for triggering the release |
| NgranFailureInfo | 6.1.6.3.26 | Information related to a NG-RAN failure |
| XrDeviceWith2Rx | 6.1.6.3.27 | XR Device with 2Rx information |

Table 6.1.6.1-2 specifies data types re-used by the Namf_Communication service based interface protocol from other specifications and from other service APIs in current specification, including a reference to their respective specifications and when needed, a short description of their use within the Namf_Communication service based interface.

**Table 6.1.6.1-2: Namf_Communication re-used Data Types**

| Data type | Reference | Comments |
|---|---|---|
| Snssai | TS 29.571 [6] | |
| Arp | TS 29.571 [6] | |
| PduSesisonId | TS 29.571 [6] | |
| Guami | TS 29.571 [6] | Globally Unique AMF Identifier |
| AmfName | TS 29.571 [6] | The name of the AMF |
| Supi | TS 29.571 [6] | Subscription Permanent Identifier |
| Cause | TS 29.571 [6] | 5G-AN Cause |
| ProblemDetails | TS 29.571 [6] | Detailed problems in failure case |
| supportedFeatures | TS 29.571 [6] | Supported Features |
| TimeZone | TS 29.571 [6] | |
| UserLocation | TS 29.571 [6] | |
| AccessType | TS 29.571 [6] | |
| AllowedNssai | TS 29.531 [18] | |
| NfInstanceId | TS 29.571 [6] | |
| Uri | TS 29.571 [6] | |
| Ecgi | TS 29.571 [6] | EUTRA Cell Identifier |
| Ncgi | TS 29.571 [6] | NR Cell Identifier |
| Uint16 | TS 29.571 [6] | |
| 5Qi | TS 29.571 [6] | 5G QoS Identifier |
| CorrelationID | TS 29.572 [25] | LCS Correlation ID |
| Pei | TS 29.571 [6] | |
| Dnn | TS 29.571 [6] | |
| Gpsi | TS 29.571 [6] | |
| GroupId | TS 29.571 [6] | |
| PlmnId | TS 29.571 [6] | |
| RfspIndex | TS 29.571 [6] | |
| EbiArpMapping | TS 29.502 [16] | EBI - ARP mapping |
| NsiId | TS 29.531 [18] | |
| TraceData | TS 29.571 [6] | Trace control and configuration parameters |
| ConfiguredSnssai | TS 29.531 [18] | |
| NgApCause | TS 29.571 [6] | Represents the NG AP cause IE |
| Area | TS 29.571 [6] | |
| ServiceAreaRestriction | TS 29.571 [6] | |
| CoreNetworkType | TS 29.571 [6] | |
| Ambr | TS 29.571 [6] | |
| SliceMbr | TS 29.571 [6] | |
| GlobalRanNodeId | TS 29.571 [6] | |
| NfGroupId | TS 29.571 [6] | Network Function Group Id |
| DurationSec | TS 29.571 [6] | |
| StnSr | TS 29.571 [6] | Session Transfer Number for SRVCC |
| CMsisdn | TS 29.571 [6] | Correlation MSISDN |
| DateTime | TS 29.571 [6] | |
| SmallDataRateStatus | TS 29.571 [6] | |
| NfSetId | TS 29.571 [6] | NF Set ID |
| NfServiceSetId | TS 29.571 [6] | NF Service Set ID |
| LMFIdentification | TS 29.572 [25] | LMF Identification |
| PlmnAssiUeRadioCapId | TS 29.571 [6] | |
| ManAssiUeRadioCapId | TS 29.571 [6] | |
| NrV2xAuth | TS 29.571 [6] | NR V2X services authorized |
| LteV2xAuth | TS 29.571 [6] | LTE V2X services authorized |
| BitRate | TS 29.571 [6] | Bit Rate |
| Pc5QoSPara | TS 29.571 [6] | PC5 QoS parameters |
| PduSessionInfo | TS 29.571 [6] | The Slice and DNN combination of a PDU session. |
| PcfUeCallbackInfo | TS 29.571 [6] | The callback information of the PCF for the UE to allow the PCF for the PDU session to send SM Policy Association Establishment and Termination events notification. |
| CnAssistedRanPara | TS 29.502 [16] | SMF derived CN assisted RAN Parameters Tuning |
| MoExpDataCounter | TS 29.571 [6] | MO Exception Data Counter |
| CagData | TS 29.503 [35] | Closed Access Group Data |
| NssaaStatus | TS 29.571 [6] | Subscribed S-NSSAI subject to NSSAA procedure and the status |
| JobType | TS 29.571 [6] | Job Type in the trace |

| | | |
|---|---|---|
| MeasurementLteForMdt | TS 29.571 [6] | Measurements used for MDT in LTE in the trace |
| MeasurementNrForMdt | TS 29.571 [6] | Measurements used for MDT in NR in the trace |
| ReportingTrigger | TS 29.571 [6] | Reporting Triggers for MDT in the trace |
| ReportIntervalMdt | TS 29.571 [6] | Report Interval for MDT in LTE in the trace |
| ReportAmountMdt | TS 29.571 [6] | Report Amount for MDT in the trace |
| CollectionPeriodRmmLteMdt | TS 29.571 [6] | Collection period for RRM measurements LTE for MDT in the trace |
| MeasurementPeriodLteMdt | TS 29.571 [6] | Measurement period LTE for MDT in the trace in |
| AreaScope | TS 29.571 [6] | Area Scope |
| PositioningMethodMdt | TS 29.571 [6] | Positioning Method for MDT in the trace in LTE |
| ReportIntervalNrMdt | TS 29.571 [6] | Report Interval for MDT in NR in the trace |
| CollectionPeriodRmmNrMdt | TS 29.571 [6] | Collection period for RRM measurements NR for MDT in the trace |
| SensorMeasurement | TS 29.571 [6] | Sensor information for MDT in the trace |
| ScheduledCommunicationTime | TS 29.571 [6] | Scheduled Communication Time |
| StationaryIndication | TS 29.571 [6] | Stationary Indication |
| TrafficProfile | TS 29.571 [6] | Traffic Profile |
| BatteryIndication | TS 29.571 [6] | Battery Indication |
| NFType | TS 29.510 [29] | NF type |
| UeAuth | TS 29.571 [6] | UE authorisation for PC5 service |
| PartitioningCriteria | TS 29.571 [6] | Partitioning Criteria |
| RedirectResponse | TS 29.571 [6] | Response body of the redirect response message. |
| CagId | TS 29.571 [6] | CAG ID |
| NnwdafEventsSubscription | TS 29.520 [52] | Represents an Individual NWDAF Event Subscription resource |
| PresenceInfo | TS 29.571 [6] | |
| UePositioningCapabilities | TS 29.572 [25] | Indicates the positioning capabilities supported by the UE. |
| SmfSelectionData | TS 29.507 [32] | |
| EpsInterworkingInfo | TS 29.503 [35] | |
| IpAddress | TS 29.503 [35] | |
| Fqdn | TS 29.571 [6] | |
| PresenceState | TS 29.571 [6] | Presence State |
| SatelliteBackhaulCategory | TS 29.571 [6] | Satellite Backhaul Category |
| WirelineServiceAreaRestriction | TS 29.571 [6] | |
| PartiallyAllowedSnssai | TS 29.571 [6] | S-NSSAI with TAIs information |
| VarRepPeriod | TS 29.571 [6] | Variable Reporting Periodicity |
| AsTimeDistributionParam | TS 29.507 [32] | |
| PruInd | TS 29.503 [35] | PRU Indicator |
| SliceUsageControlInfo | TS 29.571 [6] | Slice Usage Control Information |
| MbsrOperationAllowed | TS 29.503 [35] | MBSR Operation Information |
| UpConnectionStatus | TS 29.572 [25] | User Plane Connection Status |
| LMFIdentification | TS 29.572 [25] | |
| TaiRange | TS 29.510 [29] | TAI range |
| PduSessionPriority | TS 29.502 [16] | PDU Session Priority |
| AmfEventSubscription | Clause 6.2.6.2.2 | Represents an individual event subscription resource on AMF |
| UeAccessBehaviorReportItem | Clause 6.2.6.2.29 | Report Item for UE Access Behavior Trends event. |
| UeLocationTrendsReportItem | Clause 6.2.6.2.30 | Report Item for UE Location Trends event. |

## 6.1.6.2 Structured data types

### 6.1.6.2.1 Introduction

Structured data types used in Namf_Communication service are specified in this clause.

#### 6.1.6.2.2 Type: SubscriptionData

**Table 6.1.6.2.2-1: Definition of type SubscriptionData**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| amfStatusUri | Uri | M | 1 | This IE shall include the callback URI to receive notification of AMF status change. |
| guamiList | array(Guami) | C | 1..N | This IE shall be absent for subscribing to status change for any GUAMI supported by the AMF, it shall be present for subscribing to specific GUAMIs supported by the AMF. |

#### 6.1.6.2.3 Type: AmfStatusChangeNotification

**Table 6.1.6.2.3-1: Definition of type AmfStatusChangeNotification**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| amfStatusInfoList | array(AmfStatusInfo) | M | 1..N | This IE shall contain the status change information about the AMF |

#### 6.1.6.2.4 Type: AmfStatusInfo

**Table 6.1.6.2.4-1: Definition of type AmfStatusInfo**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| guamiList | array(Guami) | M | 1..N | This IE shall contain the GUAMIs |
| statusChange | StatusChange | M | 1 | This IE shall contain the Status change of the related GUAMIs |
| targetAmfRemoval | AmfName | C | 0..1 | This IE shall contain the AMF Name of the target AMF in the AMF planned removal without UDSF scenario |
| targetAmfFailure | AmfName | C | 0..1 | This IE shall contain the AMF Name of the target AMF in the AMF Auto-recovery without UDSF scenario. |

#### 6.1.6.2.5 Type: AssignEbiData

**Table 6.1.6.2.5-1: Definition of type AssignEbiData**

| Attribute name | Data type | P | Cardinality | Description | Applicability |
|---|---|---|---|---|---|
| pduSessionId | PduSessionId | M | 1 | Represents the identifier of the PDU Session requesting EBI(s) to be assigned. | |
| arpList | array(Arp) | C | 1..N | This IE shall be present if the NF Service Consumer (e.g SMF) requests the AMF to assign EBI(s) for the PDU session. When present, this IE shall contain the list of ARP(s)of the QoS flow(s) for which EBI(s) are requested. | |
| releasedEbiList | array(EpsBearerId) | C | 1..N | This IE shall be present if the NF Service Consumer (e.g. SMF) needs to release the assigned EBI(s) from QoS flows (e.g. when the QoS flow is released). | |
| oldGuami | Guami | C | 0..1 | This IE shall be present during an AMF planned removal procedure when the NF Service Consumer initiates a request towards the target AMF, for a UE associated to an AMF that is unavailable (see clause 5.21.2.2 of TS 23.501 [2]). | |
| modifiedEbiList | array(EbiArpMapping) | C | 1..N | This IE shall be present if a PDU session modification procedure resulted in the change of ARP for a QoS flow to which an EBI is already allocated. | EAEA |

#### 6.1.6.2.6 Type: AssignedEbiData

**Table 6.1.6.2.6-1: Definition of type AssignedEbiData**

| Attribute name | Data type | P | Cardinality | Description | Applicability |
|---|---|---|---|---|---|
| pduSessionId | PduSessionId | M | 1 | Represents the identifier of the PDU Session requesting EBI(s) to be assigned. | |
| assignedEbiList | array(EbiArpMapping) | M | 0..N | This IE shall be present if the AMF assigned the requested EBI(s). This IE shall contain the successfully assigned EBIs. (NOTE) | |
| failedArpList | array(Arp) | C | 1..N | This IE shall be present if the AMF fails to allocate EBIs for a set of ARP(s). (NOTE) | |
| releasedEbiList | array(EpsBearerId) | C | 1..N | This IE shall be present if the NF Service Consumer requested the release of EBI(s) or if the AMF revoked an already assigned EBI towards the same PDU session. This IE shall contain the list of EBI(s) released at the AMF. | |
| modifiedEbiList | array(EpsBearerId) | C | 1..N | This IE shall be present if the NF Service Consumer requested to update the ARP for a QoS flow to which an EBI is already allocated. This IE shall contain the list of EBI(s) whose ARP has been updated at the AMF. | EAEA |
| NOTE: The same ARP value may be returned in the assignedEbiList and in the failedArpList, if the request included the same ARP value more than once in the arpList and the AMF is not able to allocate an EBI for every occurrence of this ARP value. | | | | | |

#### 6.1.6.2.7 Type: AssignEbiFailed

**Table 6.1.6.2.7-1: Definition of type AssignEbiFailed**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| pduSessionId | PduSessionId | M | 1 | Represents the identifier of the PDU Session requesting EBI(s) to be assigned. |
| failedArpList | array(Arp) | C | 1..N | This IE shall be present if the AMF fails to allocate EBIs for a set of ARPs. |

#### 6.1.6.2.8 Type: UEContextRelease

**Table 6.1.6.2.8-1: Definition of type UEContextRelease**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| supi | Supi | C | 0..1 | This IE shall be present if the UE is emergency registered and the SUPI is not authenticated. |
| unauthenticatedSupi | boolean | C | 0..1 | When present, this IE shall be set as follows:<br>- true: unauthenticated SUPI;<br>- false (default): authenticated SUPI.<br><br>This IE shall be present if the SUPI is present in the message but is not authenticated and is for an emergency registered UE. |
| ngapCause | NgApCause | M | 1 | This IE shall contain the cause value received from the source 5G-AN in the handover Cancel message received over the NGAP interface. |

#### 6.1.6.2.9 Type: N2InformationTransferReqData

**Table 6.1.6.2.9-1: Definition of type N2InformationTransferReqDataTransfer**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| taiList | array(Tai) | C | 1..N | This IE shall be included if the N2 information needs to be sent to the 5G-AN nodes that serve the list of tracking areas provided. |
| ratSelector | RatSelector | C | 0..1 | This IE shall be included to indicate if the N2 information shall be transferred to ng-eNBs or gNBs exclusively. |
| globalRanNodeList | array(GlobalRanNodeId) | C | 1..N | This IE shall be included if the N2 information needs to be sent to the list of RAN nodes provided. |
| n2Information | N2InfoContainer | M | 1 | This IE includes the information to be sent on the N2 interface to the identified 5G-AN nodes and additional information required for the processing of the message by the AMF. |
| supportedFeatures | SupportedFeatures | C | 0..1 | This IE shall be present if at least one feature defined in clause 6.1.8 is supported. |

### 6.1.6.2.10 Type: NonUeN2InfoSubscriptionCreateData

**Table 6.1.6.2.10-1: Definition of type NonUeN2InfoSubscriptionCreateData**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| globalRanNodeList | array(GlobalRanNodeId)) | C | 1..N | This IE shall be included if the subscription is for N2 information from RAN node(s) for which the N2 information notification is subscribed (i.e N3IWF identifier or gNB identifier or Ng-eNB identifier). (NOTE) |
| anTypeList | array(AccessType) | C | 1..N | This IE shall be included, if the globalRanNodeId IE is not included and if the N2 information from a specific access network needs to be subscribed. When included this IE shall contain the access type of the access network from which Non UE specific N2 information is to be notified. (NOTE) |
| n2InformationClass | N2InformationClass | M | 1 | This IE represents the class of N2 information that the NF Service Consumer requires to be notified. |
| n2NotifyCallbackUri | Uri | M | 1 | This IE represents the callback URI on which the N2 information shall be notified. |
| nfId | NfInstanceId | C | 0..1 | This IE shall be present if the subscription is for "NRPPa" N2 information class and/or "LPP" N1 information class. When present, this IE shall carry the value to be used for NGAP "Routing ID" IE, which identifies the Network Function (e.g. LMF) instance handling the NRPPa and/or LPP data.<br><br>This IE may also be present if the subscription is for "PWS" N2 information class. When present, this IE shall carry the instance identity of the network function (e.g. CBCF or PWS-IWF) creating the subscription. This IE should be included when more than one CBCF/PWS-IWF instances are deployed in the network. The AMF may use this IE to identify whether the same CBCF/PWS-IWF instance has subscribed for N2 PWS information to receive the PWS Response data from the RAN.<br><br>This IE shall be present if the subscription is for "TSS" N2 information class. When present, this IE shall identify the TSCTSF NF instance subscribing the TSS data. |
| supportedFeatures | SupportedFeatures | C | 0..1 | This IE shall be present if at least one feature defined in clause 6.1.8 is supported. |
| notifCorrelationId | string | O | 0..1 | When present, this IE shall contain the notification correlation ID of the subscription. |
| NOTE: Absence of both IEs means the subscription is for N2 information from all connected Access Network node(s) via any access type. | | | | |

### 6.1.6.2.11 Type: NonUeN2InfoSubscriptionCreatedData

**Table 6.1.6.2.11-1: Definition of type NonUeN2InfoSubscriptionCreatedData**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| n2NotifySubscriptionId | string | M | 1 | Represents the Id created by the AMF for the subscription to notify a non-UE related N2 information. |
| supportedFeatures | SupportedFeatures | C | 0..1 | This IE shall be present if at least one feature defined in clause 6.1.8 is supported. |
| n2InformationClass | N2InformationClass | O | 0..1 | This IE represents the class of N2 information that the NF Service Consumer subscribed to. |

### 6.1.6.2.12 Type: UeN1N2InfoSubscriptionCreateData

**Table 6.1.6.2.12-1: Definition of type UeN1N2InfoSubscriptionCreateData**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| n2InformationClass | N2InformationClass | C | 0..1 | This IE shall be present if the NF service consumer subscribes for a N2 information notification. This IE represents the class of N2 information that the NF Service Consumer requires to be notified. |
| n2NotifyCallbackUri | Uri | C | 0..1 | This IE shall be present if the NF service consumer subscribes for a N2 information notification. This IE represents the callback URI on which the N2 information shall be notified. |
| n1MessageClass | N1MessageClass | C | 0..1 | This IE shall be present if the NF service consumer subscribes for a N1 message notification.<br>This IE represents the class of N1 message that the NF Service Consumer requires to be notified. |
| n1NotifyCallbackUri | Uri | C | 0..1 | This IE shall be present if the NF service consumer subscribes for a N1 message notification. This IE represents the callback URI on which the N1 message shall be notified. |
| nfId | NfInstanceId | C | 0..1 | This IE shall be present if the subscription is for "NRPPa" N2 information class and/or "LPP" N1 information class. When present, this IE shall carry the value to be used for NGAP "Routing ID" IE, which identifies the Network Function (e.g. LMF) instance handling the NRPPa and/or LPP data. |
| supportedFeatures | SupportedFeatures | C | 0..1 | This IE shall be present if at least one feature defined in clause 6.1.8 is supported. |
| oldGuami | Guami | C | 0..1 | This IE shall be present during an AMF planned removal procedure when the NF Service Consumer initiates a request towards the target AMF, for a UE associated to an AMF that is unavailable (see clause 5.21.2.2 of TS 23.501 [2]). |

### 6.1.6.2.13 Type: UeN1N2InfoSubscriptionCreatedData

**Table 6.1.6.2.13-1: Definition of type UeN1N2InfoSubscriptionCreatedData**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| n1n2NotifySubscriptionId | string | M | 1 | Represents the Id created by the AMF for the subscription to notify a UE related N1/N2 information. |
| supportedFeatures | SupportedFeatures | C | 0..1 | This IE shall be present if at least one feature defined in clause 6.1.8 is supported. |

6.1.6.2.14 Type: N2InformationNotification

**Table 6.1.6.2.14-1: Definition of type N2InformationNotification**

| Attribute name | Data type | P | Cardinality | Description | Applicability |
|---|---|---|---|---|---|
| n2NotifySubscriptionId | string | M | 1 | Represents the subscription Id for which the notification is generated. The NF Service Consumer uses this to co-relate the notification against a corresponding subscription. If the notification is due to an implicit subscription via NRF, then the value shall be set as "implicit".<br>During the AMF planned removal procedure with UDSF deployed procedure, this IE shall be set to "" (empty string) and be ignored by the NF Service Consumer. | |
| n2InfoContainer | N2InfoContainer | C | 0..1 | This IE shall be present, except during Inter NG-RAN node N2 based handover procedure (see clause 5.2.2.3.6.2).<br><br>When present, this IE shall contain the N2 information related to the corresponding N2 information class. | |
| toReleaseSessionList | array(PduSessionId) | C | 1..N | This IE shall be present during N2 based handover procedure, if there are any PDU session(s) associated with Network Slice(s) which become no longer available.<br><br>When present, this IE shall include all the PDU session(s) associated with no longer available S-NSSAI(s). | |
| lcsCorrelationId | CorrelationID | C | 0..1 | This IE shall be present, if an LCS correlation identifier is received in corresponding N1/N2 Message Transfer service operation.<br><br>When present, this IE shall carry the LCS correlation identifier. | |
| notifyReason | N2InfoNotifyReason | C | 0..1 | This IE shall be present, if "n2InfoContainer" attribute is not present; this IE may be present otherwise.<br><br>When present, this IE indicates the reason for the N2 information notification. | |
| smfChangeInfoList | array(SmfChangeInfo) | C | 1..N | This IE shall be present during N2 based handover procedure, if there is I-SMF or V-SMF change or removal for the related PDU session(s).<br><br>When present, this IE shall indicate the I-SMF/V-SMF situation after successful HO complete. | DTSSA |
| ranNodeId | GlobalRanNodeId | C | 0..1 | This IE shall be present during the AMF planned removal procedure with UDSF deployed procedure.<br>When present, it shall contain the Global RAN Node ID. The IE shall contain either the gNB ID or the NG-eNB ID. | |
| initialAmfName | AmfName | C | 0..1 | This IE shall be present during the AMF planned removal procedure with UDSF deployed procedure.<br>When present, it shall contain the AMF Name of the initial AMF. | |

| | | | | | |
|---|---|---|---|---|---|
| anN2IPv4Addr | Ipv4Addr | C | 0..1 | This IE shall be present during the AMF planned removal procedure with UDSF deployed procedure, if the Access Network N2 interface is using IPv4 address. | |
| anN2IPv6Addr | Ipv6Addr | C | 0..1 | This IE shall be present during the AMF planned removal procedure with UDSF deployed procedure, if the Access Network N2 interface is using IPv6 address. | |
| guami | Guami | C | 0..1 | This IE shall be present during Location Services procedures (see clause 5.2.2.3.6.3) and it may be present otherwise.<br><br>When present, it shall contain the GUAMI serving the UE. | |
| notifySourceNgRan | boolean | C | 0..1 | This IE shall be present during an Inter NG-RAN node N2 based DAPS handover procedure , if the target AMF receives this indication in the Handover Notify from the target NG-RAN node (see clause 4.9.1.3.3a of TS 23.502 [3]).<br>When present, it shall be set as follows:<br><br>- true: Notify the Source NG-RAN about Handover Success<br><br>- false (default): Do not notify the Source NG-RAN about Handover Success | |
| notifCorrelationId | string | O | 0..1 | When present, this IE shall contain the notification correlation ID of the subscription. | |
| toReleaseSessionInfo | array(ReleaseSessionInfo) | C | 1..N | This IE shall be present during Handover procedure, if there are any PDU session(s) that cannot be supported in the target AMF for a reason other than no longer available S-NSSAI(s) as specified in step 1 of clause 5.2.2.3.6.2.<br><br>When present, this IE shall include list of the PDU session(s) and the release cause. | |

### 6.1.6.2.15 Type: N2InfoContainer

**Table 6.1.6.2.15-1: Definition of type N2InfoContainer**

| Attribute name | Data type | P | Cardinality | Description | Applicability |
|---|---|---|---|---|---|
| n2InformationClass | N2InformationClass | M | 1 | This IE represents the class of N2 information to be transferred. | |
| smInfo | N2SmInformation | C | 0..1 | This IE shall be present if session management N2 information is to be transferred. When present, it represents a session management SMF related N2 information data part. | |
| ranInfo | N2RanInformation | C | 0..1 | This IE shall be present if RAN related N2 information is to be transferred (i.e. n2InformationClass is "RAN"). When present, it shall contain the RAN related N2 information data part. | |
| nrppaInfo | NrppaInformation | C | 0..1 | This IE shall be present if location service related N2 information is to be transferred. When present, it represents a NRPPa related N2 information data part. | |
| pwsInfo | PwsInformation | C | 0..1 | This IE shall be present if PWS related N2 information is to be transferred. | |
| v2xInfo | V2xInformation | C | 0..1 | This IE shall be present if V2X related N2 information is to be transferred. | |
| proseInfo | ProSeInformation | C | 0..1 | This IE shall be present if 5G ProSe related N2 information is to be transferred. | ProSe |
| tssInfo | TssInformation | C | 0..1 | This IE shall be present if TSS related N2 information is to be transferred. | NTSSM |
| rslpInfo | RslpInformation | C | 0..1 | This IE shall be present if Ranging/SL positioning related N2 information is to be transferred. | Ranging_SL |
| a2xInfo | A2xInformation | C | 0..1 | This IE shall be present if A2X related N2 information is to be transferred. | A2X |

6.1.6.2.16 Type: N1MessageNotification

**Table 6.1.6.2.16-1: Definition of type N1MessageNotification**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| n1NotifySubscriptionId | string | C | 0..1 | Represents the subscription Id for which the notification is generated. The NF Service Consumer uses this to correlate the notification against a corresponding subscription. If the notification is due to an implicit subscription via NRF, then the value shall be set as "implicit".<br>This IE shall be present if the notification is based on a subscription to N1MessgeNotification. An exception is for the case when initial AMF forwards NAS message to target AMF during AMF re-allocation procedure. |
| n1MessageContainer | N1MessageContainer | M | 1 | Contains the N1 message class and N1 message content. |
| lcsCorrelationId | CorrelationID | O | 0..1 | If the N1 message notified is for LCS proceduresor PRU procedures, the NF Service Producer (e.g. AMF) may include an LCS correlation identifier. (NOTE) |
| registrationCtxtContainer | RegistrationContextContainer | C | 0..1 | If the N1 message notified is of type 5GMM (i.e. during Registration with AMF re-allocation procedure), the NF Service Producer (e.g. AMF) shall include this IE, if available. |
| newLmfIdentification | LMFIdentification | O | 0..1 | If a new LMF is selected by AMF, this IE may include the new selected LMF Identification. |
| guami | Guami | C | 0..1 | This IE shall be present during UE Assisted and UE Based Positioning Procedure (see clause 5.2.2.3.5.3) or the LCS Event Report, Event Reporting in RRC INACTIVE state procedures, LCS Cancel Location and LCS Periodic-Triggered Invoke Procedures (see clause 5.2.2.3.5.5) and it may be present otherwise.<br><br>When present, it shall contain the GUAMI serving the UE. |
| cIoT5GSOptimisation | boolean | C | 0..1 | This IE shall be present when the N1 message class is "LPP/LCS" and the N1 message is received from the UE with Control Plane CIoT 5GS Optimisation. When present, it shall be set as follows:<br>- true: Control Plane CIoT 5GS Optimisation was used and no signalling or data is currently pending for the UE at the AMF.<br>- false (default): Control Plane CIoT 5GS Optimisation was not used or signalling or data is currently pending for the UE at the AMF. |
| ecgi | Ecgi | O | 0..1 | When present, this IE shall indicate the identifier of the E-UTRAN cell serving the UE.<br>This IE may be present if the N1 message notified is for LCS procedures. |
| ncgi | Ncgi | O | 0..1 | When present, this IE shall indicate the identifier of the NR cell serving the UE or the PRU.<br>This IE may be present if the N1 message notified is for LCS procedures or PRU procedures. |
| tai | Tai | O | 0..1 | When present, this IE shall indicate the identifier of the tracking area serving the PRU.<br>This IE may be present if the N1 message notified is for PRU procedures. |

| | | | | |
|---|---|---|---|---|
| supi | Supi | C | 0..1 | Contains the SUPI of the UE or PRU.<br><br>This IE shall be present when the N1 message class is "LCS" and the default (any) LMF is used in the deferred 5GC-MT-LR procedure as described in clause 6.3.1 of TS 23.273 [42].<br><br>This IE shall also be present when the N1 message class is "UPP-CM".<br><br>This IE may be present if the N1 message notified is for PRU procedures.<br>(NOTE) |
| pruInd | PruInd | O | 0..1 | When present, this IE shall indicate whether the UE is allowed to serve as a PRU.<br>This IE may be present if the N1 message notified is for PRU procedures. |
| NOTE: During UPP-CM procedures, LMF shall determine the corresponding LCS-UPP connection based on the SUPI of the UE. | | | | |

### 6.1.6.2.17 Type: N1MessageContainer

**Table 6.1.6.2.17-1: Definition of type N1MessageContainer**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| n1MessageClass | N1MessageClass | M | 1 | This IE shall contain the N1 message class for the message content specified in n1MessageContent. |
| n1MessageContent | RefToBinaryData | M | 1 | This IE shall reference the N1 message binary data corresponding to the n1MessageClass. See TS 24.501 [11]. See clause 6.1.6.4.2. |
| nfId | NfInstanceId | C | 0..1 | This IE shall be present when the n1MessageClass IE is set to "LPP", "LCS" or "UPP-CM". It should be present when the n1MessageClass IE is set to "SM". It may be present otherwise.<br>When present, this IE shall carry the identifier of the Network Function (e.g. LMF or SMF) instance sending the N1 message. (NOTE) |
| serviceInstanceId | string | O | 0..1 | When present, this IE shall carry the Service Instance Identifier of the Service Instance (e.g. LMF) sending the N1 message. |
| NOTE: For a Home-routed PDU session, this IE shall carry the NF instance ID of the V-SMF; for a PDU session with I-SMF, this IE shall carry the NF instance ID of the I-SMF. | | | | |

6.1.6.2.18 Type: N1N2MessageTransferReqData

**Table 6.1.6.2.18-1: Definition of type N1N2MessageTransferReqData**

| Attribute name | Data type | P | Cardinality | Description | Applicability |
|---|---|---|---|---|---|
| n1MessageContainer | N1MessageContainer | C | 0..1 | This IE shall be included if a N1 message needs to be transferred. | |
| n2InfoContainer | N2InfoContainer | C | 0..1 | This IE shall be included if a N2 information needs to be transferred. | |
| mtData | RefToBinaryData | C | 0..1 | This IE shall be included if mobile terminated data (i.e. CIoT user data container) needs to be transferred. When present, it shall reference the mobile terminated data (see clause 6.1.6.4.4). | CIOT |
| skipInd | boolean | C | 0..1 | This IE shall be present and set to "true" if the service consumer (e.g. SMF) requires the N1 message to be sent to the UE only when UE is in CM-CONNECTED, e.g. during SMF initiated PDU session release procedure (see clause 4.3.4.2 of TS 23.502 [3]).<br><br>When present, this IE shall be set as following:<br>- true: AMF should skip sending N1 message to UE, when the UE is in CM-IDLE.<br>- false (default): the AMF shall send the N1 message to the UE. | |
| lastMsgIndication | boolean | O | 0..1 | This flag when present shall indicate that the message transferred is the last message. (See clause 4.13.3.3 and clause 4.13.3.6 of TS 23.502 [3]. | |
| pduSessionId | PduSessionId | O | 0..1 | PDU Session ID for which the N1 / N2 message is sent, if the N1 / N2 message class is SM. | |
| lcsCorrelationId | CorrelationID | O | 0..1 | LCS Correlation ID, for which the N1/N2 message is sent, if<br><br>- the N1 message class is LPP (see clause 6.11.1 of TS 23.273 [42]), LCS (see clause 6.3 of TS 23.273 [42]) and clause 6.17 of TS 23.273 [42]); and/or<br><br>- the N2 Information class is NRPPa (see clause 6.11.2 of TS 23.273 [42]).<br>(NOTE 4) | |
| ppi | Ppi | O | 0..1 | This IE when present shall indicate the Paging policy to be applied. The paging policies are configured at the AMF. | |

| | | | | | |
|---|---|---|---|---|---|
| arp | Arp | O | 0..1 | This IE when present shall indicate the Allocation and Retention Priority of the PDU session for which the N1/N2 message transfer is initiated. To support priority paging, the AMF shall use this IE to determine whether to include the Paging Priority IE in the NGAP Paging Message (see clause 5.4.3.3 of TS 23.501 [2]). The set of ARP values associated with priority paging and mapping to Paging Priority IE values are configured at the AMF. This IE shall not be present when the N1/N2 message class is not SM. | |
| 5qi | 5Qi | O | 0..1 | This IE when present shall indicate the 5QI associated with the PDU session for which the N1 / N2 message transfer is initiated. This IE shall not be present when the N1/N2 message class is not SM. | |
| n1n2FailureTxfNotifURI | Uri | O | 0..1 | If included, this IE represents the callback URI on which the AMF shall notify the N1/N2 message transfer failure. | |
| smfReallocationInd | boolean | O | 0..1 | This IE shall indicate that the SMF is requested to be reallocated (see clause 4.3.5.2 of TS 23.502 [3]). When present, this IE shall be set as follows:<br>- true: the SMF is requested to be reallocated.<br>- false (default): the SMF is not requested to be reallocated. | |
| areaOfValidity | AreaOfValidity | O | 0..1 | This IE represents the list of TAs where the provided N2 information is valid. See clause 5.2.2.2.7 and 4.2.3.3 of TS 23.502 [3]. | |
| supportedFeatures | SupportedFeatures | C | 0..1 | This IE shall be present if at least one feature defined in clause 6.1.8 is supported. | |
| oldGuami | Guami | C | 0..1 | This IE shall be present during an AMF planned removal procedure when the NF Service Consumer initiates a request towards the target AMF, for a UE associated to an AMF that is unavailable (see clause 5.21.2.2 of TS 23.501 [2]). | |
| maAcceptedInd | boolean | C | 0..1 | This IE shall be present if a request to establish a MA PDU session was accepted or if a single access PDU session was upgraded into a MA PDU session (see clauses 4.22.2 and 4.22.3 of TS 23.502 [3]).<br><br>When present, it shall be set as follows:<br>- true: MA PDU session<br>- false (default): single access PDU session | MAPDU |

| | | | | | |
|---|---|---|---|---|---|
| extBufSupport | boolean | O | 0..1 | This IE may be present with value "true" if Extended Buffering is permitted, during Network triggered Service Request Procedure (see clause 4.2.3.3 of TS 23.502 [3]), UPF anchored Mobile Terminated Data Transport in Control Plane CIoT 5GS Optimisation procedure (see clause 4.24.2 of TS 23.502 [3]) or NEF Anchored Mobile Terminated Data Transport (see clause 4.25.5 of TS 23.502 [3]).<br><br>When present, this IE shall indicate whether Extended Buffering applies or not:<br><br>- true: Extended Buffering applies<br>- false (default) Extended Buffering does not apply | |
| targetAccess | AccessType | C | 0..1 | This IE shall be included by a SMF for a MA PDU session to indicate the target access type (i.e. 3GPP access or Non-3GPP access) towards which the N2 information and optionally N1 information is requested to be sent.<br><br>This IE may be included by an LMF to indicate the access type through which an LPP message shall be transmitted to the UE.<br><br>This IE shall be included by an SMF and set to the old access type during an intra-AMF handover between 3GPP and non-3GPP accesses, when releasing the N2 PDU session resources in the old access. | MAPDU<br><br>ELCS<br><br>3GA-N3GA-HO |
| nfId | NfInstanceId | C | 0..1 | This IE should be included by the SMF when invoking N1N2MessageTransfer service operation, if the n1MessageContainer IE is not present.<br><br>When present, this IE shall carry the identifier of the NF instance invoking the service operation, i.e. the SMF instance hosting the SM Context for the PDU session. (NOTE 3) | |
| pruInd | boolean | O | 0..1 | When present, this IE shall be set to the value true to indicate that the LMF as NF consumer is requesting to initiate a positioning procedure towards a PRU, as specified in clause 6.11.2 of TS 23.273 [42]. | |

| | | | | | |
|---|---|---|---|---|---|
| pduSessionPrio | PduSessionPriority | C | 0..1 | This IE shall be included by the SMF if the priority of the PDU session is to be changed. See also clauses 5.22.2 and 5.22.3 of TS 23.501 [2].<br><br>The AMF shall use the priority of the PDU session indicated in this IE for further PDU Session related priority handling (i.e. to determine the SBI Message Priority (SMP) to set in subsequent signaling it sends related to the PDU session). | |

NOTE 1: For N1 message class "UPDP", as per TS 24.501 [11] Annex D, the messages between UE and PCF carry PTI which is used by the PCF to correlate the received N1 message in the notification with a prior transaction initiated by the PCF.
NOTE 2: During Downlink Data Notification procedure, if the SMF receives the PPI value (=DSCP(0..63)) from the UPF and wants to set the PPI value in the N1N2MessgeTransfer message, the SMF shall map the PPI value received from N4 message to correct PPI value (0..7) used in N11 message.
NOTE 3: If the n1MessageContainer IE is present, the nfId attribute in the n1MessageContainer IE should be used by the SMF and the AMF to identify the NF instance ID of the sending SMF.
NOTE 4: For UPP-CM messages, the AMF shall determine the corresponding LCS-UP context based on the SUPI of the UE (carried in {ueContextId} variable in the resource URI, see clause 6.1.3.5.2) and the LMF instance ID (carried in the nfId attribute of the N1MessageContainer object, see clause 6.1.6.2.17).

### 6.1.6.2.19 Type: N1N2MessageTransferRspData

**Table 6.1.6.2.19-1: Definition of type N1N2MessageTransferRspData**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| cause | N1N2MessageTransferCause | M | 1 | This IE shall provide the result of the N1/N2 message transfer processing at the AMF. |
| supportedFeatures | SupportedFeatures | C | 0..1 | This IE shall be present if at least one feature defined in clause 6.1.8 is supported. |

6.1.6.2.20 Type: RegistrationContextContainer

**Table 6.1.6.2.20-1: Definition of type RegistrationContextContainer**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| ueContext | UeContext | M | 1 | This IE shall contain the UE Context information. |
| localTimeZone | TimeZone | O | 0..1 | This IE contains the time zone UE is currently located. |
| anType | AccessType | M | 1 | This IE shall contain the current access type of the UE. |
| anN2ApId | integer | M | 1 | This IE shall contain the RAN UE NGAP ID over N2 interface. |
| ranNodeId | GlobalRanNodeId | M | 1 | This IE shall contain the Global RAN Node ID. The IE shall contain either the gNB ID or the NG-eNB ID. |
| initialAmfName | AmfName | M | 1 | This IE shall contain the AMF Name of the initial AMF. |
| userLocation | UserLocation | M | 1 | This IE shall contain the user location received from 5G-AN. |
| anN2IPv4Addr | Ipv4Addr | C | 0..1 | If the Access Network N2 interface is using IPv4 address, this IE shall be included. |
| anN2IPv6Addr | Ipv6Addr | C | 0..1 | If the Access Network N2 interface is using IPv6 address, this IE shall be included. |
| rrcEstCause | string | C | 0..1 | This IE shall contain the RRC Establishment Cause, if received from the 5G-AN (See TS 38.413 [12], clause 9.2.5.1).<br>It carries the value in hexadecimal representation<br>Pattern: '^[0-9a-fA-F]+$' |
| ueContextRequest | boolean | C | 0..1 | This IE shall contain the indication on whether UE context including security information needs to be setup at the NG-RAN, if received from the NG-RAN by the initial AMF (See TS 38.413 [12], clause 9.2.5.1).<br><br>When present, it shall be set as follows:<br>- true: UE context including security information needs to be setup at the NG-RAN.<br>- false (default): UE context including security information does not need to be setup at the NG-RAN. |
| initialAmfN2ApId | integer | C | 0..1 | This IE shall contain the AMF UE NGAP ID of the initial AMF over N2 interface, if available. |
| allowedNssai | AllowedNssai | O | 0..1 | This IE contains the allowed NSSAI of the UE. This IE also contains the mapped home network S-NSSAI for each allowed S-NSSAI. |
| configuredNssai | array(ConfiguredSnssai) | O | 1..N | This IE shall contain the configured S-NSSAI(s) authorized by the NSSF in the serving PLMN, if received from the NSSF. |
| rejectedNssaiInPlmn | array(Snssai) | O | 1..N | This IE shall contain the rejected NSSAI in the PLMN, if received from the NSSF. |
| rejectedNssaiInTa | array(Snssai) | O | 1..N | This IE shall contain the rejected NSSAI in the current TA, if received from the NSSF. |
| selectedPlmnId | PlmnId | O | 0..1 | This IE shall contain the selected PLMN Id for the non-3GPP access, if received from the 5G-AN (See TS 38.413 [12], clause 9.2.5.1). |
| iabNodeInd | boolean | C | 0..1 | This IE shall contain the IAB Node Indication, if received from the 5G-AN (See TS 38.413 [12], clause 9.2.5.1). This IE shall be absent when mbsrNodeInd exists and is set to true.<br><br>When present, it shall be set as follows:<br>- true: 5G-AN is an IAB Node.<br>- false (default): 5G-AN is not an IAB Node. |
| mbsrNodeInd | boolean | C | 0..1 | This IE shall contain the MBSR Node Indication, if received from the 5G-AN. This IE shall be absent when iabNodeInd exists and is set to true.<br><br>When present, it shall be set as follows:<br>- true: 5G-AN is an MBSR Node.<br>- false (default): 5G-AN is not an MBSR Node. |

| | | | | |
|---|---|---|---|---|
| ceModeBInd | CeModeBInd | O | 0..1 | This IE shall contain the CE-mode-B Support Indicator, if received from the 5G-AN (See TS 38.413 [12], clause 9.2.5.1). |
| lteMInd | LteMInd | O | 0..1 | This IE shall contain the LTE-M Indication, if received from the 5G-AN (See TS 38.413 [12], clause 9.2.5.1). |
| authenticatedInd | boolean | O | 0..1 | This IE shall contain the Authenticated Indication, if received from the 5G-AN (See TS 38.413 [12], clause 9.2.5.1).<br><br>This IE shall be set as follows:<br>- true: authenticated by the 5G-AN;<br>- false (default): unauthenticated by the 5G-AN. |
| npnAccessInfo | NpnAccessInfo | O | 0..1 | This IE shall contain the NPN Access Information, if received from the 5G-AN (See TS 38.413 [12], clause 9.2.5.1). |

### 6.1.6.2.21 Type: AreaOfValidity

**Table 6.1.6.2.21-1: Definition of type AreaOfValidity**

| Attribute name | Data type | P | Cardinality | Description | Applicability |
|---|---|---|---|---|---|
| taiList | array(Tai) | M | 0..N | An array of TAI representing the area of validity of the associated N2 information provided. | |
| taiRangeList | array(TaiRange) | O | 1..N | A list of TAI ranges representing the area of validity of the associated N2 information provided.<br><br>(NOTE) | AoV-En |
| NOTE: When the taiRangeList is present, the aggregation of taiList attribute and taiRangeList indicates the entire area of validity of the associated N2 information. | | | | | |

### 6.1.6.2.22 Void

### 6.1.6.2.23 Type: UeContextTransferReqData

**Table 6.1.6.2.23-1: Definition of type UeContextTransferReqData**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| reason | TransferReason | M | 1 | Indicate the reason for the UEContextTransfer service request |
| accessType | AccessType | M | 1 | This IE shall contain the access type of the UE. |
| plmnId | PlmnIdNid | O | 0..1 | If present, this IE shall contain the PLMN ID or SNPN ID of the NF service consumer (e.g target AMF). |
| regRequest | N1MessageContainer | O | 0..1 | If present, this IE shall refer to the registration request message which triggers the UE Context Transfer. The message class shall be "5GMM" and message content shall be reference to N1 Message Content binary data, See clause 6.1.6.4.2. |
| supportedFeatures | SupportedFeatures | C | 0..1 | This IE shall be present if at least one feature defined in clause 6.1.8 is supported. |

#### 6.1.6.2.24 Type: UeContextTransferRspData

**Table 6.1.6.2.24-1: Definition of type UeContextTransferRspData**

| Attribute name | Data type | P | Cardinality | Description | Applicability |
|---|---|---|---|---|---|
| ueContext | UeContext | M | 1 | Represents an individual ueContext resource after the modification is applied. | |
| supportedFeatures | SupportedFeatures | C | 0..1 | This IE shall be present if at least one feature defined in clause 6.1.8 is supported. | |
| ueRadioCapability | N2InfoContent | C | 0..1 | This IE shall be included to contain the "UE Radio Capability Information" if available during context transfer procedure.<br>UE Radio Capability Information does not include NB-IoT UE radio capability, see clause 5.4.4.1 of TS 23.501 [2]<br>(NOTE) | |
| ueRadioCapabilityForPaging | N2InfoContent | C | 0..1 | This IE shall be included to contain the "UE Radio Capability for Paging" if available during context transfer procedure.<br>(NOTE) | |
| ueNbiotRadioCapability | N2InfoContent | C | 0..1 | This IE shall be included to contain "NB-IoT UE radio capability Information" if available during context transfer procedure, see clause 5.4.4.1 of TS 23.501 [2] | CIOT |
| xrDeviceWith2Rx | XrDeviceWith2Rx | C | 0..1 | This IE shall be included to contain the "XR Device with 2Rx" as defined in TS 38.413 [12] if available during context transfer procedure. | |
| NOTE: The source AMF may decide to not include ueRadioCapability and ueRadioCapabilityForPaging if the target AMF supports the RACS feature and if:<br>a. the PlmnAssiUeRadioCapId is included in the MM Context for an intra-PLMN AMF mobility procedure; or<br>b. the ManAssiUeRadioCapId is included in the MM Context for an intra-PLMN or an inter-PLMN AMF mobility procedure. | | | | | |

6.1.6.2.25 Type: UeContext

**Table 6.1.6.2.25-1: Definition of type UeContext**

| Attribute name | Data type | P | Cardinality | Description | Applicability |
|---|---|---|---|---|---|
| supi | Supi | C | 0..1 | This IE shall be present if available. When present, this IE contains SUPI of the UE. | |
| supiUnauthInd | boolean | C | 0..1 | This IE shall be present if SUPI is present. When present, it shall indicate whether the SUPI is unauthenticated. | |
| gpsiList | array(Gpsi) | C | 1..N | This IE shall be present if available and if it is not case b) specified in clause 5.2.2.2.1.1 step 2a. When present, this IE shall contain the GPSI(s) of the UE. | |
| pei | Pei | C | 0..1 | This IE shall be present if available and if it is not case b) specified in clause 5.2.2.2.1.1 step 2a. When present, this IE shall contain Mobile Equipment Identity of the UE. | |
| udmGroupId | NfGroupId | O | 0..1 | When present, it shall indicate the identity of the UDM Group serving the UE. | |
| ausfGroupId | NfGroupId | O | 0..1 | When present, it shall indicate the identity of the AUSF Group serving the UE. | |
| pcfGroupId | NfGroupId | O | 0..1 | When present, it shall indicate the identity of the PCF Group serving the UE. | |
| routingIndicator | string | O | 0..1 | When present, it shall indicate the Routing Indicator of the UE. | |
| hNwPubKeyId | integer | O | 0..1 | When present, it shall indicate the Home Network Public Key Identifier of the UE. (NOTE 4). | |
| groupList | array(GroupId) | C | 1..N | This IE shall be present if the UE belongs to any subscribed internal group(s) and if it is not case b) specified in clause 5.2.2.2.1.1 step 2a. When present, this IE shall list the subscribed internal group(s) to which the UE belongs to. | |
| drxParameter | DrxParameter | C | 0..1 | This IE shall be present if available and if it is not case b) specified in clause 5.2.2.2.1.1 step 2a. When present, this IE shall contain the DRX parameter of the UE. | |
| subRfsp | RfspIndex | C | 0..1 | This IE shall be present if available and if it is not case b) specified in clause 5.2.2.2.1.1 step 2a. When present, it shall indicate the subscribed RFSP Index of the UE. | |
| pcfRfsp | RfspIndex | C | 0..1 | This IE shall be present if available and if it is not case b) specified in clause 5.2.2.2.1.1 step 2a.<br><br>When present, this IE shall indicate the PCF determined RFSP Index of the UE. | |
| usedRfsp | RfspIndex | C | 0..1 | This IE shall be present if available and if it is not case b) specified in clause 5.2.2.2.1.1 step 2a. When present, it shall indicate the used RFSP Index of the UE. | |
| subUeAmbr | Ambr | C | 0..1 | This IE shall be present if subscribed UE-AMBR has been retrieved from UDM and if it is not case b) specified in clause 5.2.2.2.1.1 step 2a.<br><br>When present, this IE shall indicate the value of subscribed UE AMBR of the UE. | |
| pcfUeAmbr | Ambr | C | 0..1 | This IE shall be present if available and if it is not case b) specified in clause 5.2.2.2.1.1 step 2a.<br><br>When present, this IE shall indicate the value of the PCF determined UE AMBR of the UE. | |

| | | | | | |
|---|---|---|---|---|---|
| subUeSliceMbrList | map(SliceMbr) | C | 1..N | Map of SliceMbr, where the S-NSSAI shall be used as the key of the map.<br><br>This IE shall be present if the list of subscribed UE-Slice-MBR(s) has been retrieved from UDM and if it is not case b) specified in clause 5.2.2.2.1.1 step 2a.<br><br>When present, this IE shall indicate the list of subscribed UE-Slice-MBR(s) per S-NSSAI for the UE. | |
| smsfId | NfInstanceId | C | 0..1 | This IE shall be present if the SMS service for UE is activated and if it is not case b) specified in clause 5.2.2.2.1.1 step 2a. When present, it indicates the identifier of the SMSF network function instance serving the UE. The NF service consumer (e.g. target AMF) may use this information to identify the SMSF NF service profile from among the SMSF NF service profiles it received from the NRF. | |
| seafData | SeafData | C | 0..1 | This IE shall be present if available and if it is not case b) specified in clause 5.2.2.2.1.1 step 2a or the case specified in clause 5.2.2.2.1.2. When present, this IE contains the security data derived from data received from AUSF of the UE. | |
| 5gMmCapability | 5GMmCapability | C | 0..1 | This IE shall be present if the UE had provided this IE during Registration Procedure and if it is not case b) specified in clause 5.2.2.2.1.1 step 2a. When present, this IE shall contain 5G MM capability of the UE. | |
| pcfId | NfInstanceId | C | 0..1 | This IE shall be present if available and if it is not case b) specified in clause 5.2.2.2.1.1 step 2a. When present, this IE indicates the identity of the PCF for AM Policy and/or UE Policy. | |
| pcfSetId | NfSetId | C | 0..1 | This IE shall be present, if available. When present, it shall contain the NF Set ID of the PCF for AM Policy and/or UE Policy. | |
| pcfAmpServiceSetId | NfServiceSetId | C | 0..1 | This shall be present, if available. When present, it shall contain the NF Service Set ID of the PCF's AM Policy service. | |
| pcfUepServiceSetId | NfServiceSetId | C | 0..1 | This shall be present, if available. When present, it shall contain the NF Service Set ID of the PCF's UE Policy service. | |
| pcfBindingLevel | SbiBindingLevel | C | 0..1 | This IE shall be present if available. When present, this IE shall contain the SBI binding level of the PCF's AM policy and UE Policy association resources. (NOTE 6) | |
| pcfAmPolicyUri | Uri | C | 0..1 | This IE shall be present if available and if it is not case b) specified in clause 5.2.2.2.1.1 step 2a. When present this IE shall contain the URI of the individual AM policy resource (see TS 29.507 [32] clause 5.3.3.2) used by the AMF. | |
| amPolicyReqTriggerList | array(PolicyReqTrigger) | C | 1..N | This IE shall be present if available and if it is not case b) specified in clause 5.2.2.2.1.1 step 2a. When present this IE shall indicate the AM policy request triggers towards the PCF. The NF Service Consumer (e.g. target AMF) shall use these triggers to request AM policy from the PCF whenever these triggers are met.<br><br>The possible AM policy control request triggers are specified in clause 6.1.2.5 of TS 23.503 [7]. | |

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|---|---|---|---|---|---|
| pcfUePolicyUri | Uri | C | 0..1 | This IE shall be present if available and if it is not case b) specified in clause 5.2.2.2.1.1 step 2a. When present this IE shall contain the URI of the individual UE policy resource (see TS 29.507 [32] clause 5.3.3.2) used by the AMF. | |
| uePolicyReqTriggerList | array(PolicyReqTrigger) | C | 1..N | This IE shall be present if available and if it is not case b) specified in clause 5.2.2.2.1.1 step 2a. When present this IE shall indicate the UE policy request triggers towards the PCF. The NF Service Consumer (e.g. target AMF) shall use these triggers to request UE policy from the PCF whenever these triggers are met.<br><br>The possible UE policy control request triggers are specified in clause 6.1.2.5 of TS 23.503 [7]. | |
| hpcfId | NfInstanceId | O | 0..1 | This IE indicates the identity of PCF for UE Policy in home PLMN, when the UE is roaming. | |
| hpcfSetId | NfSetId | O | 0..1 | When present, this IE shall contain the NF Set ID of the PCF for UE Policy in home PLMN, when the UE is roaming. | |
| restrictedRatList | array(RatType) | O | 1..N | When present, this IE shall indicate the list of RAT types that are restricted for the UE; see TS 29.571 [6] (NOTE 1) | |
| forbiddenAreaList | array(Area) | O | 1..N | When present, this IE shall indicate the list of forbidden areas of the UE. | |
| serviceAreaRestriction | ServiceAreaRestriction | O | 0..1 | When present, this IE shall indicate subscribed Service Area Restriction for the UE. | |
| restrictedCnList | array(CoreNetworkType) | O | 1..N | When present, this IE shall indicate the list of Core Network Types that are restricted for the UE. | |
| eventSubscriptionList | array(ExtAmfEventSubscription) | C | 1..N | This IE shall be present if available and if it is not case b) specified in clause 5.2.2.2.1.1 step 2a. When present, it shall indicate the event subscription(s) targeting the UE or the group the UE is part of.<br><br>If the source AMF supports binding procedures and if it received binding indications for event notifications (i.e. with "callback" scope) or for subscription change event notifications (i.e. with "subscription-events" scope) for certain subscriptions, these binding indications should also be included.<br><br>If the source AMF knows the NF type of the NF that created the subscription, this information should also be indicated.<br><br>The UE access behavior trends report and/or UE location trends report may be included, if the NF service consumer is subscribed to the UE_ACCESS_BEHAVIOR_TRENDS event and/or UE_LOCATION_TRENDS event respectively. This information is required for calculating the statistical data of trends. | |
| mmContextList | array(MmContext) | C | 1..2 | This IE shall be present if available and if it is not case b) specified in clause 5.2.2.2.1.1 step 2a. When present, this IE contains the MM Contexts of the UE. | |

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|---|---|---|---|---|---|
| sessionContextList | array(PduSessionContext) | C | 1..N | This IE shall be present if available and if it is neither case a) nor case b) specified in clause 5.2.2.2.1.1 step 2a. When present, this IE contains the PDU Session Contexts of the UE.<br>(NOTE 2) | |
| epsInterworkingInfo | EpsInterworkingInfo | C | 0..1 | This IE shall contain the associations between APN/DNN and PGW-C+SMF for EPS interworking, if available. | |
| traceData | TraceData | C | 0..1 | This IE shall be present if signalling based trace has been activated (see TS 32.422 [30]) and if it is not case b) specified in clause 5.2.2.2.1.1 step 2a. | |
| serviceGapExpiryTime | DateTime | C | 0..1 | This IE shall be present if Service Gap Control is enabled and if the AMF has started a Service Gap Timer which has not expired yet (see clause 5.31.16 of TS 23.501 [2]). The value of the IE shall indicate the expiry time (in UTC) of the active Service Gap Timer for the UE. | |
| stnSr | StnSr | O | 0..1 | This IE shall be present if available, for UE supporting 5G-SRVCC (see clause 5.2.2.2.11 of TS 23.502 [3]).<br>When present, this IE contains STN-SR of the UE. | |
| cMsisdn | CMsisdn | O | 0..1 | This IE shall be present if available, for UE supporting 5G-SRVCC (see clause 5.2.2.2.11 of TS 23.502 [3]).<br>When present, this IE contains C-MSISDN of the UE. | |
| msClassmark2 | MSClassmark2 | O | 0..1 | This IE shall be present if available, for UE supporting 5G-SRVCC (see clause 5.2.2.2.11 of TS 23.502 [3]).<br>When present, this IE contains Mobile Station Classmark 2 of the UE. | |
| supportedCodecList | array(SupportedCodec) | O | 1..N | This IE shall be present if available, for UE supporting 5G-SRVCC (see clause 5.2.2.2.11 of TS 23.502 [3]).<br>When present, this IE shall indicate the list of speech codecs supported by the UE. | |
| smallDataRateStatusInfos | array(SmallDataRateStatusInfo) | O | 1..N | List of Small Data Rate Control Statuses for released PDU Sessions, see clause 5.31.14.3 of TS 23.501 [2]. | CIOT |
| restrictedPrimaryRatList | array(RatType) | O | 1..N | When present, this IE shall indicate the list of RAT types that are restricted for use as primary RAT for the UE; see TS 29.571 [6] (NOTE 1) | |
| restrictedSecondaryRatList | array(RatType) | O | 1..N | When present, this IE shall indicate the list of RAT types that are restricted for use as secondary RAT for the UE; see TS 29.571 [6] (NOTE 1) | |
| v2xContext | V2xContext | O | 0..1 | This IE shall be present if available (see clause 6.5.4 of TS 23.287 [47]).<br>When present, this IE shall indicate the parameters related to the V2X services. | |
| lteCatMInd | boolean | C | 0..1 | This IE shall be present with value "true" if the UE is a LTE Category M UE based on indication provided by the NG-RAN or by the MME at EPS to 5GS handover, as specified in TS 23.502 [3].<br><br>When present, this IE shall be set as following:<br>- true: the UE is a Category M UE<br>- false (default): this UE is not a Category M UE. | |

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| redCapInd | boolean | C | 0..1 | This IE shall be present with value "true" if the UE is a NR RedCap UE based on indication provided by the NG-RAN, as specified in TS 23.502 [3].<br><br>When present, this IE shall be set as following:<br>- true: the UE is a NR RedCap UE<br>- false (default): this UE is not a NR RedCap UE. | |
| eRedCapInd | boolean | C | 0..1 | This IE shall be present with value "true" if the UE is a NR eRedCap UE based on indication provided by the NG-RAN, as specified in clause 5.41 of TS 23.501 [2].<br><br>When present, this IE shall be set as following:<br>- true: the UE is a NR eRedCap UE<br>- false (default): this UE is not a NR eRedCap UE. | |
| moExpDataCounter | MoExpDataCounter | C | 0..1 | This IE shall be present if a non-zero MO Exception counter has not been reported yet to SMF.<br><br>When present, this IE shall contain the MO Exception Data Counter, as specified in clause 5.31.14.3 of TS 23.501 [2]. | |
| cagData | CagData | O | 0..1 | Closed Access Group Data<br>When present, the provisioningTime attribute (from the CagData data type) shall be absent. | NPN |
| managementMdtInd | boolean | C | 0..1 | This flag shall be present with value "true" if Management Based Minimization of Drive Tests (MDT) is allowed, as specified in TS 32.422 [30].<br><br>When present, this IE shall be set as following:<br>- true: management based MDT is allowed.<br>- false (default): management based MDT is not allowed. | |
| adjacentPlmnMngtMdtInds | map(boolean) | C | 1..N | A map (list of key-value pairs where PlmnId converted to string serves as key; see TS 29.571 [7]) of Booleans, each indicating whether Management Based MDT is allowed in the corresponding PLMN:<br>- true: management based MDT is allowed in the PLMN.<br>- false: management based MDT is not allowed in the PLMN. | |
| immediateMdtConf | ImmediateMdtConf | C | 0..1 | This IE shall be sent by the source AMF to the target AMF, for signalling based MDT if the Job Type indicates Immediate MDT. See clause 4.10 of TS 32.422 [30]. | |
| ecRestrictionDataWb | EcRestrictionDataWb | C | 0..1 | This IE shall be present if the AMF determines whether Enhanced Coverage is restricted or not for the UE for WB-N1 mode. If absent, this IE indicates Enhanced Coverage is not restricted for WB-N1 mode. (NOTE 3) | |

| | | | | | |
|---|---|---|---|---|---|
| ecRestrictionDataNb | boolean | C | 0..1 | This IE shall be present if the AMF determines whether Enhanced Coverage is restricted or not for the UE for NB-N1 mode.<br><br>If present, this IE shall indicate whether Enhanced Coverage for NB-N1 mode is restricted or not.<br><br>- true: Enhanced Coverage for NB-N1 mode is restricted.<br><br>- false or absent: Enhanced Coverage for NB-N1 mode is allowed. (NOTE 3) | |
| iabOperationAllowed | boolean | O | 0..1 | This IE shall be present if the UE is allowed for IAB operation. It may be present otherwise.<br><br>When present, it shall indicate whether the UE is allowed for IAB operation, as follows:<br>- true: indicates that the UE is allowed for IAB operation.<br><br>- false: indicates that the UE is not allowed for IAB operation. | |
| mbsrOperationAllowed | MbsrOperationAllowed | O | 0..1 | Indicates whether the subscriber is allowed for MBSR operation as specified in clause 5.35A.4 of TS 23.501 [2], optionally, with corresponding location and time period. | |
| proseContext | ProseContext | O | 0..1 | This IE shall be present if available (see clause 6.7 of TS 23.304 [51]).<br>When present, this IE shall indicate the parameters related to the ProSe services. | ProSe |
| analyticsSubscriptionList | array(AnalyticsSubscription) | C | 1..N | This IE shall be present if the AMF has created analytics subscription(s) towards NWDAF related to the UE.<br><br>If present, this IE shall include the list of analytics subscriptions, as specified in clauses 5.2.2.2.2 and 5.2.2.2.11 of TS 23.502 [3]. | ASUC |
| pcfUepBindingInfo | string | C | 0..1 | This IE shall be present if Binding Indication was received for UE Policy Association resource from the PCF. When present, this IE shall contain the Binding indication of the PCF's UE Policy Association resource and shall be set to the value of the 3gpp-Sbi-Binding header defined in clause 5.2.3.2.6 of TS 29.500 [4], without the header name. | |
| pcfAmpBindingInfo | string | C | 0..1 | This IE shall be present if Binding Indication was received for AM Policy Association resource from the PCF. When present, this IE shall contain the Binding indications of the PCF's AM policy Association resource and shall be set to the value of the 3gpp-Sbi-Binding header defined in clause 5.2.3.2.6 of TS 29.500 [4], without the header name. | |
| usedServiceAreaRestriction | ServiceAreaRestriction | O | 0..1 | When present, this IE shall include the Service Area Restriction from PCF. | |

| | | | | | |
|---|---|---|---|---|---|
| praInAmPolicy | map(PresenceInfo) | O | 1..N | When present, this IE shall include the map of PRA Information for the subscribed "PRA_CHANGE" PolicyReqTrigger in the AM Policy Association.<br><br>The key of the map shall be the "praId" attribute within the PresenceInfo data type. The "presenceState" attribute within the PresenceInfo data type shall not be supplied here. | |
| praInUePolicy | map(PresenceInfo) | O | 1..N | When present, this IE shall include the map of PRA Information for the subscribed "PRA_CHANGE" PolicyReqTrigger in the UE Policy Association.<br><br>The key of the map shall be the "praId" attribute within the PresenceInfo data type. The "presenceState" attribute within the PresenceInfo data type shall not be supplied here. | |
| updpSubscriptionData | UpdpSubscriptionData | O | 0..1 | When present, this IE shall include the subscription resource in the AMF for a UE policy delivery related N1 message notification. | |
| smPolicyNotifyPduList | array(PduSessionInfo) | C | 1..N | This IE shall be present if it has been received from the PCF for the UE, i.e. the PCF for the AM Policy Association and possibly the UE Policy Association.<br><br>When present, this IE shall contain the information (Slice and DNN combination) of the PDU session(s) applicable for the notification of SM Policy Association Establishment and Termination events. (NOTE 5) | SPAE |
| pcfUeCallbackInfo | PcfUeCallbackInfo | C | 0..1 | This IE shall be present if the smPolicyNotifyPduList IE is present.<br><br>When present, this IE shall contain the callback information of the PCF for the UE to receive SM Policy Association Establishment and Termination events notification from the PCF for the SM Policy. (NOTE 5) | SPAE |
| uePositioningCap | UePositioningCapabilities | O | 0..1 | When present, this IE shall indicate the positioning capabilities supported by the UE. | |
| snpnOnboardInd | boolean | C | 0..1 | This IE shall be present if the UE is registered for onboarding in an SNPN.<br><br>When present, it shall indicate the following:<br>- true: indicates that the UE is registered for onboarding in an SNPN.<br><br>- false (default): indicates that the UE is not registered for onboarding in an SNPN. | eNPN |
| astiDistributionIndication | boolean | O | 0..1 | When present, this IE shall indicate whether the access stratum time distribution via Uu reference point should be activated or deactivated for the UE.<br>When present, this IE shall be set as following:<br>- true: ASTI distribution is activated for the UE.<br><br>- false (default): ASTI distribution is deactivated for the UE. | |

| | | | | | |
|---|---|---|---|---|---|
| tsErrorBudget | integer | O | 0..1 | When present, this IE shall indicate the Uu time synchronization error budget for the time synchronization service (as described in clause 5.27.1 in TS 23.501 [2]). It indicates the value in nano seconds. | |
| smfSelInfo | SmfSelectionData | C | 0..1 | This IE shall be present if conditions for SMF Selection information replacement are received from the PCF for AM Policy.<br><br>When present, It shall include the conditions for SMF selection information replacement, as determined by the PCF. | |
| pcfUeSliceMbrList | map(SliceMbr) | C | 1..N | This IE shall be present when UE Slice MBR(s) were received from the PCF for AM Policy.<br><br>When present, this IE shall include one or more UE-Slice-MBR(s) as determined by the PCF for allowed S-NSSAI(s). The key of the map is the S-NSSAI in the allowed NSSAI to which the UE-Slice-MBR belongs. | |
| smsfSetId | NfSetId | C | 0..1 | This IE shall be present if available.<br><br>When present, this IE shall contain the NF Set ID of the SMSF serving the UE. | |
| smsfServiceSetId | NfServiceSetId | C | 0..1 | This shall be present, if available.<br><br>When present, it shall contain the NF Service Set ID of the SMSF's service instance serving the UE. | |
| smsfBindingInfo | string | C | 0..1 | This IE shall be present if available.<br><br>When present, this IE shall contain the binding indication of the UE Context for SMS in SMSF and shall be set to the value of the 3gpp-Sbi-Binding header defined in clause 5.2.3.2.6 of TS 29.500 [4], without the header name. | |
| disasterRoamingInd | boolean | C | 0..1 | This IE shall be present if the UE is registered for disaster roaming. It may be present otherwise.<br><br>When present, this IE shall be set as follows:<br>- true: UE is registered for Disaster Roaming service;<br>- false (default): UE is not registered for Disaster Roaming service. | |
| disasterPlmn | PlmnId | C | 0..1 | This IE shall be included if the disasterRoamingInd is present and set to "true".<br><br>When present, this IE includes the PLMN of the UE which has faced disaster condition | |
| satelliteBackhaulCat | SatelliteBackhaulCategory | O | 0..1 | When present, this IE shall be set to the last value that has been reported to the PCF if the satellite backhaul category change is subscribed with the "SAT_BACKHAUL_CHANGE" PolicyReqTrigger in the UE Policy Association. | |

| | | | | | |
|---|---|---|---|---|---|
| wlServAreaRes | WirelineServiceAreaRestriction | C | 0..1 | This IE shall be present if available and if it is not case b) specified in clause 5.2.2.2.1.1 step 2a.<br><br>When present, this IE shall indicate the value of the PCF determined Wireline Service Area Restriction. | |
| asTimeDisParam | AsTimeDistributionParam | C | 0..1 | This IE shall be present if available and if it is not case b) specified in clause 5.2.2.2.1.1 step 2a.<br><br>When present, this IE shall contain the 5G acess stratum time distribution parameters received from the PCF for AM Policy. | |
| amPolicyInfoContainer | AmPolicyInfoContainer | C | 0..1 | This IE shall be present if any of the information in the container is available and if it is not case b) specified in clause 5.2.2.2.1.1 step 2a.<br><br>When present, this IE shall contain the AM Policy information parts received from the PCF for AM Policy. | |
| a2xContext | A2xContext | O | 0..1 | This IE shall be present if available (see clause 6.3.5.4 of TS 23.256 [56]).<br>When present, this IE shall indicate the parameters related to the A2X services. | A2X |
| lcsUpContext | LcsUpContext | O | 0..1 | This IE shall be present if available (see clause 4.3.7 of TS 23.273 [42]).<br>When present, this IE shall indicate the parameters related to the LCS UP services. | |
| reconnectInd | boolean | O | 0..1 | Indicates whether the UE is to reconnect to the network in the case the UE determines that it does not have the latest available clock quality information as described in clause 5.27.1.12 in TS 23.501 [2].<br><br>When present, this IE shall be set as follows:<br>- true: the UE reconnects to the network in the case the UE determines that the reference report ID has changed;<br>- false (default): the UE does not have an indication from CN to reconnect to the network in the case that the reference report ID has changed. | |

NOTE 1: If the restrictedPrimaryRatList and restrictedSecondaryRatList attributes are supported by the sender, the sender shall include the list of RAT Types that are restricted, if any, in the restrictedRatList attribute, shall include the list of RAT Types that are restricted for use as primary RAT, if any, in the restrictedPrimaryRatList attribute and shall include the list of RAT Types that are restricted for use as secondary RAT, if any, in the restrictedSsecondaryRatList attribute. If the restrictedPrimaryRatList and restrictedSecondaryRatList attributes are supported by the receiver, the receiver shall use the data in the restrictedPrimaryRatList attribute, if received, as the list of RAT Types that are restricted for use as primary RAT for the UE, and shall use the data in the restrictedSecondaryRatList attribute, if received, as the list of RAT Types that are restricted for use as secondary RAT for the UE, otherwise the receiver shall use the data in the restrictedRatList attribute, if received, as the list of RAT Types that are restricted for the UE.

NOTE 2: A particular PDU session not supported by the target AMF shall not be transferred, e.g. MA-PDU session context shall not be transferred if target AMF does not support ATSSS.

NOTE 3: After ecRestrictionDataWb and/or ecRestrictionDataNb attributes are sent from source AMF to target AMF to build the UeContext in the target AMF, the target AMF shall re-determine the EC restriction information based on the received subscription data from UDM and UE 5GMM capability because EC restriction information may change (e.g. due to that subscription data in UDM is changed but not notified the old AMF yet) and then compare the re-determined EC restriction information with the one received in the UeContext. If the target AMF finds EC restriction information has changed after comparing, the target AMF shall proceed as described in clause 5.31.12, TS 23.501 [2].

NOTE 4: If present, this attribute shall be used together with routingIndicator. This attribute is only used by the HPLMN in roaming scenarios.

NOTE 5: If the information as indicated in both IEs were received from the PCF for the UE or from the old AMF in UE Context, the AMF shall identify whether a non-roaming or local breakout PDU session is applicable for SM Policy Association events, i.e, whethe the slice and DNN combination of the PDU session is listed in the smPolicyNotifyPduList IE or not. If the PDU session is applicable for notification of SM Policy Association events , the AMF shall provide the callback information for the PCF of the UE contained in the pcfUeCallbackInfo IE to the SMF of a new PDU session via Create SM Context service operation, or to the SMF for an ongoing PDU session via Update SM Context service operation, together with the indication for notification of SM Policy Association events. See clause 4.3.2.2.1 and clause 4.3.3.2 of TS 23.502 [3].

NOTE 6: This IE is deprecated. An AMF complying with this version of specification shall use the pcfAmpBindingInfo IE to carry the Binding indication of the AM Policy Association resource and use the pcfUepBindingInfo IE to carry the binding indication of the UE Policy Association resource.

6.1.6.2.26 Type: N2SmInformation

**Table 6.1.6.2.26-1: Definition of type N2SmInformation**

| Attribute name | Data type | P | Cardinality | Description | Applicability |
|---|---|---|---|---|---|
| pduSessionId | PduSessionId | M | 1 | Indicates the PDU Session Identity | |
| n2InfoContent | N2InfoContent | C | 0..1 | This IE shall be present if a SMF related IE should be transferred. When present, the IE contains one of NGAP SMF related IEs specified in clause 9.3.4 of TS 38.413 [12]. | |
| sNssai | Snssai | C | 0..1 | This IE shall be present if network slice information to be transferred for session management. When present, the IE indicates the network slice the PDU session belongs to. (NOTE) | |
| homePlmnSnssai | Snssai | C | 0..1 | This IE shall be present during EPS to 5GS handover procedure for Home Routed PDU session.<br>When present, it shall carry the S-NSSAI for home PLMN. | ENS |
| iwkSnssai | Snssai | C | 0..1 | This IE shall be present during EPS to 5GS handover procedure with AMF relocation for Home Routed PDU session, or during EPS to 5GS handover using N26 interface with AMF relocation and with I-SMF insertion, if S-NSSAI for interworking is configured and used in the initial AMF, as specified in clause 4.11.1.2.2 and clause 4.23.12.7.1 of TS 23.502 [3].<br><br>When present, this IE shall carry the S-NSSAI for interworking configured and used in the initial AMF for the PDU session. | ENS |
| subjectToHo | boolean | C | 0..1 | This IE shall be present if n2InfoContent carries a " Handover Required Transfer" IE. When present, it Indicates whether the PDU session shall be subject to handover to the target node. | |
| NOTE: During EPS to 5GS handover procedure for Home Routed PDU session with AMF relocation, the source AMF shall set this IE to the S-NSSAI in the serving PLMN mapped from the S-NSSAI in home PLMN indicated by the homePlmnSnssai IE in the N2SmInformation data structure sent to target AMF. | | | | | |

#### 6.1.6.2.27 Type: N2InfoContent

**Table 6.1.6.2.27-1: Definition of type N2InfoContent**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| ngapMessageType | Uinteger | C | 0..1 | This IE shall be present if PWS or TSS related N2 information is to be transferred, or during the AMF planned removal procedure with UDSF deployed procedure to transfer a RAN N2 message.<br>When present, it shall indicate the NGAP Message type of the ngapData as specified in clause 6.1.6.4.3.3. Its value equals the value of the Procedure Code defined in ASN.1 in clause 9.4.7 in TS 38.413 [12]. |
| ngapIeType | NgapIeType | C | 0..1 | This IE shall be present if SM, RAN, V2X, ProSe, Ranging_SL or NRPPa related N2 information is to be transferred.<br>When present, it shall indicate the NGAP IE type of the ngapData as specified in clause 6.1.6.4.3.2. |
| ngapData | RefToBinaryData | M | 1 | This IE reference the N2 Information binary data corresponding to the N2 information class. See clause 6.1.6.4.3. |

#### 6.1.6.2.28 Type: NrppaInformation

**Table 6.1.6.2.28-1: Definition of type NrppaInformation**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| nfId | NfInstanceId | M | 1 | This IE shall carry the identifier of the Network Function (e.g. LMF) instance that is sending or receiving the NRPPa data. |
| nrppaPdu | N2InfoContent | M | 1 | This IE represents the encoded NGAP NRPPa-PDU IE, which is transparent to AMF. |
| serviceInstanceId | string | O | 0..1 | When present, this IE shall carry the Service Instance Identifier of the Service Instance (e.g. LMF) that is sending or receiving the NRPPa data. |

6.1.6.2.29 Type: PwsInformation

**Table 6.1.6.2.29-1: Definition of type PwsInformation**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| messageIdentifier | Uint16 | M | 1 | Identifies the warning message. Sender shall set this field to 0, if the pwsContainer IE carries PWS Failure Indication or PWS Restart Indication. The receiver shall ignore this IE if the pwsContainer IE carries PWS Failure Indication or PWS Restart Indication. |
| serialNumber | Uint16 | M | 1 | identifies a particular message from the source and type indicated by the Message Identifier. Sender shall set this field to 0, if the pwsContainer IE carries PWS Failure Indication or PWS Restart Indication. The receiver shall ignore this IE if the pwsContainer IE carries PWS Failure Indication or PWS Restart Indication. |
| pwsContainer | N2InfoContent | M | 1 | This IE represents the PWS N2 information data part to be relayed between CBCF and AN. |
| bcEmptyAreaList | array(GlobalRanNodeId) | C | 1..N | This IE shall be present if the NF consumer has previously requested the AMF to send the N2 response information for:<br>- WRITE-REPLACE-WARNING-REQUEST and the AMF has received WRITE-REPLACE-WARNING-RESPONSE from RAN node(s) not including the *Broadcast Completed Area List* IE, or<br>- PWS-CANCEL-REQUEST and the AMF has received PWS-CANCEL-RESPONSE from RAN node(s) not including the *Broadcast Cancelled Area List* IE.<br>When present, this IE shall list the RAN node(s) that has sent a<br>- WRITE-REPLACE-WARNING-RESPONSE not including the *Broadcast Completed Area List* IE, or<br>- PWS-CANCEL-RESPONSE not including the *Broadcast Cancelled Area List* IE. |
| sendRanResponse | boolean | O | 0..1 | This IE shall be present to request the AMF to send the N2 response information it has received from the RAN nodes to the NF Service Consumer.<br>When present, this IE shall be set as follows:<br>- true: send RAN response<br>- false (default): do not send RAN response.<br>The N2 information received from the RAN corresponds to the *Broadcast-Completed-Area-List* IE or the *Broadcast-Cancelled-Area-List* IE defined in TS 38.413 [12]. See clause 6.1.6.4.3.3. |
| omcId | OmcIdentifier | O | 0..1 | IE shall be present if the AMF is required to write the n2Information it has received from the RAN nodes into trace records on the OMC. When present, it indicates the identifier of OMC. |

| | | | | |
|---|---|---|---|---|
| nfId | NfInstanceId | O | 0..1 | When present, this IE shall carry the instance identity of the NF Service Consumer (e.g. CBCF or PWS-IWF).<br><br>This IE should be included when more than one CBCF/PWS-IWF instances are deployed in the network and the sendRanResponse IE is present with the value "true". The AMF may use this IE to identify whether the same CBCF/PWS-IWF instance has subscribed for N2 PWS information to receive the PWS Response data from the RAN. |

### 6.1.6.2.30 Type: N1N2MsgTxfrFailureNotification

**Table 6.1.6.2.30-1: Definition of type N1N2MsgTxfrFailureNotification**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| cause | N1N2MessageTransferCause | M | 1 | This IE shall provide the result of the N1/N2 message transfer at the AMF. |
| n1n2MsgDataUri | Uri | M | 1 | This IE shall contain the N1N2MessageTransfer request resource URI returned in the Location header when the N1/N2 message transfer was initiated (see clause 6.1.3.5.3.1).<br><br>This IE shall be used by the NF Service Consumer to correlate the notification with the UE or session for which the earlier N1/N2 message transfer was initiated.<br><br>If no Location header was returned when the N1/N2 message transfer was initiated, e.g. when a 200 OK response was sent for a UE in RRC inactive state, this IE shall be set to a dummy URI, i.e. an URI with no authority and an empty path (e.g. "http:"). |
| retryAfter | Uinteger | O | 0..1 | This IE may be included if the AMF requests the NF Service Consumer to throttle sending further N1/N2 Message Transfer request for a short period, e.g. when UE is not responding to paging.<br><br>When present, this IE indicates the period in number of seconds. The NF consumer should not send new N1/N2 message transfer request during the indicated period.<br><br>(NOTE) |
| NOTE: This IE should be configured with a value less than 10 minutes, i.e. 600 seconds. | | | | |

### 6.1.6.2.31 Type: N1N2MessageTransferError

**Table 6.1.6.2.31-1: Definition of type N1N2MessageTransferError**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| error | ProblemDetails | M | 1 | This IE shall provide the result of the N1/N2 message transfer processing at the AMF. |
| errInfo | N1N2MsgTxfrErrDetail | O | 0..1 | This IE may be included to provide additional information related to the error. |

### 6.1.6.2.32 Type: N1N2MsgTxfrErrDetail

**Table 6.1.6.2.32-1: Definition of type N1N2MsgTxfrErrDetail**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| retryAfter | Uinteger | O | 0..1 | This IE may be included if the AMF requests the NF Service Consumer to stop sending the N1/N2 message before timeout, and to retry the N1/N2 message transfer request that was rejected after a timeout. The value shall be in seconds.<br><br>When included, the value shall be set to an estimate of the AMF on how long it will take before the AMF considers paging procedure as completed. (NOTE) |
| highestPrioArp | Arp | O | 0..1 | This IE may be included if the "cause" attribute in the ProblemDetails is set to "HIGHER_PRIORITY_REQUEST_ONGOING". When included this IE shall contain the ARP value of the highest priority QoS flow for which currently paging is ongoing.<br>The NF Service Consumer shall not initiate an Namf_Communication_N1N2MessageTransfer operation for the same UE with an ARP value having a lower priority than this or the same priority as this, until the retryAfter timer expires. |
| maxWaitingTime | DurationSec | C | 0..1 | This IE shall be present when:<br>- extBufSupport attribute with value "true" received in the request; and<br>- the UE is not reachable due to the UE in MICO mode or the UE using extended idle mode DRX.<br><br>When present, this IE shall indicate the estimated maximum waiting time in seconds before the UE will be reachable.<br><br>If the UE is in MICO mode, the AMF determines the Estimated Maximum Wait time based on the next expected periodic registration by the UE or by implementation. If the UE is using extended idle mode DRX, the AMF determines the Estimated Maximum Wait time based on the start of the next Paging Time Window.<br><br>(see clause 4.25.5 of 3GPP 23.502 [3]) |
| NOTE: This IE should be configured with a value less than 10 minutes, i.e. 600 seconds. | | | | |

### 6.1.6.2.33 Type: N2InformationTransferRspData

**Table 6.1.6.2.33-1: Definition of type N2InformationTransferRspData**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| result | N2InformationTransferResult | M | 1 | This IE shall provide the result of the N2 information transfer processing at the AMF. |
| supportedFeatures | SupportedFeatures | C | 0..1 | This IE shall be present if at least one feature defined in clause 6.1.8 is supported. |
| pwsRspData | PWSResponseData | C | 0..1 | This IE shall be present if the n2InformationClass is "PWS" in N2InformationTransferReqData. |
| tssRspPerNgranList | array(TssRspPerNgran) | O | 1..10 | This IE may be present, if the n2InformationClass is "TSS" in N2InformationTransferReqData, to contain the Timing Synchronization Status Failure message, or Timing Synchronization Status Response message if the Criticality Diagnostics IE is present, or to report a failure related to a NG-RAN, |

6.1.6.2.34 Type: MmContext

**Table 6.1.6.2.34-1: Definition of type MmContext**

| Attribute name | Data type | P | Cardinality | Description | Applicability |
|---|---|---|---|---|---|
| accessType | AccessType | M | 1 | This IE shall contain the access type of the MM context. | |
| nasSecurityMode | NasSecurityMode | C | 0..1 | This IE shall be present if available in 3GPP access MM context. When present, this IE shall contain the used NAS security mode of the UE. | |
| epsNasSecurityMode | EpsNasSecurityMode | C | 0..1 | This IE shall be present in 3GPP access MM context if selected EPS NAS security algorithms have been previously provided to the UE, as specified in clause 6.7.2 of TS 33.501 [27].<br><br>When present, this IE shall contain the selected EPS NAS security algorithms provided to the UE. | |
| nasDownlinkCount | NasCount | C | 0..1 | This IE shall be present if available in 3GPP access MM context. When present, this IE shall contain the NAS downlink count of the UE. | |
| nasUplinkCount | NasCount | C | 0..1 | This IE shall be present if available in 3GPP access MM context. When present, this IE shall contain the NAS uplink count of the UE. | |
| ueSecurityCapability | UeSecurityCapability | C | 0..1 | This IE shall be present if available in 3GPP access MM context. When present, this IE shall contain the UE security capability | |
| s1UeNetworkCapability | S1UeNetworkCapability | C | 0..1 | This IE shall be present if available in 3GPP access MM context. When present, this IE shall contain the S1 UE network capabilities. | |
| allowedNssai | array(Snssai) | C | 1..N | This IE shall be present if the source AMF and the target AMF are in the same PLMN and if available. When present, this IE shall contain the allowed NSSAI for the access type.<br>(NOTE 2) | |
| nssaiMappingList | array(NssaiMapping) | C | 1..N | This IE shall be present if the source AMF and the target AMF are in the same PLMN and if available. When present, this IE shall contain the mapping of the allowed NSSAI and if available the partially allowed NSSAI for the UE. | |
| allowedHomeNssai | array(Snssai) | C | 1..N | This IE shall be present if the source AMF and the target AMF are in different PLMNs and if available. When present, this IE shall contain the home S-NSSAIs corresponding to the allowed NSSAI for the access type and if available the partially allowed NSSAI. | |
| partiallyAllowedNssai | array(PartiallyAllowedSnssai) | C | 1..N | This IE shall be present if the source and target AMF supports the partially allowed S-NSSAI feature and if the source AMF and target AMF pertain to the same PLMN.<br>When present, this IE shall contain the partially allowed S-NSSAI list and the TAI list where the S-NSSAI(s) are allowed.<br>See TS 23.502 [3] clause 4.2.2.2.2.<br>(NOTE 2) | PAR-NS |
| replacedSnssaiMappingList | array(SliceReplacementMapping) | C | 1..N | This IE shall be present if the source AMF and the target AMF are in the same PLMN, the Network Slice Replacement is supported and there is at least one S-NSSAI which has been replaced with an alternative S-NSSAI.<br>When present, this IE shall contain the mapping of the replaced S-NSSAI(s) to the alternative S-NSSAI(s).<br>See clause 5.15.19 of TS 23.501 [2]. | |
| nsInstanceList | array(NsiId) | C | 1..N | This IE shall be present if available. When present, it shall indicate the Network Slice Instances selected for the UE. | |
| expectedUEbehavior | ExpectedUeBehavior | C | 0..1 | This IE shall be present if available. When present it shall indicate the expected UE moving trajectory and its validity period. See TS 23.502 [3] clause 4.15.6.3. | |

| | | | | | |
|---|---|---|---|---|---|
| ueDifferentiationInfo | UeDifferentiationInfo | C | 0..1 | This IE shall be present if available. When present it shall indicate UE Differentiation Information and its validity period. | |
| plmnAssiUeRadioCapId | PlmnAssiUeRadioCapId | C | 0..1 | This IE shall be present if the source AMF supports RACS feature and if available. When present it shall be the PLMN-assigned UE Radio Capability ID.<br>(NOTE 1) | |
| manAssiUeRadioCapId | ManAssiUeRadioCapId | C | 0..1 | This IE shall be present if the source AMF supports RACS feature and if available. When present it shall be the Manufacturer-assigned UE Radio Capability ID. | |
| ucmfDicEntryId | string | C | 0..1 | This IE shall be present if the source AMF supports RACS feature and if available. When present it shall be the UCMF allocated dicEntryId received from the UCMF. | |
| n3IwfId | GlobalRanNodeId | C | 0..1 | This IE shall be present during Registration procedure with AMF changes as specified in clause 4.2.2.2 of TS 23.502 [3] and inter AMF handover procedure, if the old AMF holds UE context established via N3IWF and if the PDU sessions associated with the non-3GPP access are transferred to the new AMF (see clause 5.2.2.2.1.1 and 5.2.2.2.3.1).<br>When present, this IE shall contain the Global RAN Node ID of N3IWF. | |
| wagfId | GlobalRanNodeId | C | 0..1 | This IE shall be present during Registration procedure with AMF changes as specified in clause 4.2.2.2 of TS 23.502 [3] and inter AMF handover procedure, if the old AMF holds UE context established via W-AGF and if the PDU sessions associated with the non-3GPP access are transferred to the new AMF (see clause 5.2.2.2.1.1 and 5.2.2.2.3.1).<br>When present, this IE shall contain the Global RAN Node ID of W-AGF. | |
| tngfId | GlobalRanNodeId | C | 0..1 | This IE shall be present during Registration procedure with AMF changes as specified in clause 4.2.2.2 of TS 23.502 [3] and inter AMF handover procedure, if the old AMF holds UE context established via TNGF and if the PDU sessions associated with the non-3GPP access are transferred to the new AMF (see clause 5.2.2.2.1.1 and 5.2.2.2.3.1).<br>When present, this IE shall contain the Global RAN Node ID of TNGF. | |
| anN2ApId | integer | C | 0..1 | This IE shall be present during Registration procedure with AMF changes, as specified in clause 4.2.2.2 of TS 23.502 [3] and inter AMF handover procedure, if the old AMF holds UE context established via N3IWF/W-AGF/TNGF and the UE is in CM-CONNECTED state via N3IWF/W-AGF/TNGF and if the PDU sessions associated with the non-3GPP access are transferred to the new AMF (see clause 5.2.2.2.1.1 and 5.2.2.2.3.1).<br>When present, this IE shall contain the RAN UE NGAP ID over N2 interface. | |
| nssaaStatusList | array(NssaaStatus) | C | 1..N | This IE shall be present if available. When present, it shall contain the subscribed S-NSSAIs subject to NSSAA procedure and for which a status information is available. See TS 23.501 [2] clause 5.15.5.2.1 and TS 23.502 [3] clause 5.2.2.2.2. | |
| pendingNssaiMappingList | array(NssaiMapping) | C | 1..N | This IE shall be present if available. When present, this IE shall contain the mapping of the pending NSSAI for the UE. | |

| | | | | | |
|---|---|---|---|---|---|
| uuaaMmStatus | UuaaMmStatus | C | 0..1 | This IE shall be present if available in 3GPP access MM context.<br>When present, this IE shall indicate the status of UUAA-MM if the AMF is configured to perform the UAV authentication/authorization at 5GS registration as described in clause 5.2.2 of TS 23.256 [56]. | |
| deregInactTimerList | map(DeregInactTimerInfo) | C | 1..N | This IE shall be present if there are ongoing Network Slice Deregistration Inactivity Timer(s) in the old AMF for certain S-NSSAI(s) for the UE during an intra PLMN mobility procedure.<br><br>When present, this IE shall indicate the list of Ongoing Network Slice Deregistration Inactivity Timer Information per S-NSSAI for the UE. The S-NSSAI shall be used as the key of the map. | SliceUsageCtrl |
| voSupportMatchInd | boolean | O | 0..1 | This IE indicates whether the UE radio capabilities are compatible with the network configuration for IMS voice. The AMF may include this IE in intra PLMN handover when the network configuration considered in the decision for the Voice Support Match Indicator is homogenous for the entire serving PLMN. The target AMF uses it as an input for setting the IMS voice over PS Session Supported Indication over 3GPP access as specified in clause 5.16.3.2 of TS 23.501 [2].<br><br>When present, this IE shall be set as follows:<br>- true: the UE radio capabilities are compatible with the network configuration;<br>- false: the UE radio capabilities are not compatible with the network configuration. | |
| NOTE 1: If the AMF supports RACS and the AMF detects that the selected PLMN during a service request procedure is different from the currently registered PLMN for the UE, the AMF stores the UE Radio Capability ID of the newly selected PLMN in the UE context as described in clause 5.2.3.2 of TS 23.502 [3], and provides this UE Radio Capability ID to the target AMF during any subsequent inter-AMF mobility.<br>NOTE 2: The source AMF shall send the partially allowed S-NSSAIs in the allowedNssai IE, if the target AMF does not support the PAR-NS feature. | | | | | |

### 6.1.6.2.35 Type: SeafData

**Table 6.1.6.2.35-1: Definition of type SeafData**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| ngKsi | NgKsi | M | 1 | Indicates the KSI used for the derivation of the keyAmf sent. |
| keyAmf | KeyAmf | M | 1 | Indicates the $K_{amf}$ or $K'_{amf}$ |
| nh | string | C | 0..1 | This IE shall be present during N2 handover procedure as specified in clause 6.9.2.3.3 of TS 33.501 [27]. When present, this IE indicates the Next Hop value used for the key derivation. The value is encoded as a string of hexadecimal characters.<br>Pattern: '^[A-Fa-f0-9]+$' |
| ncc | integer | C | 0..1 | This IE shall be present during N2 handover procedure as specified in clause 6.9.2.3.3 of TS 33.501 [27]. When present, this IE indicates the NH Chaining Counter. The value is within the range 0 to 7. |
| keyAmfChangeInd | boolean | C | 0..1 | This IE shall be included, with a value "true", if the source AMF requires the target AMF to perform AS key re-keying, during N2 handover procedure as specified in clause 6.9.2.3.3 of TS 33.501 [27]. |
| keyAmfHDerivationInd | boolean | C | 0..1 | This IE shall be included, with a value "true", if the source AMF has performed horizontal $K_{AMF}$ derivation, which means a new $K_{AMF}$ has been calculated. |

### 6.1.6.2.36 Type: NasSecurityMode

**Table 6.1.6.2.36-1: Definition of type NasSecurityMode**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| integrityAlgorithm | IntegrityAlgorithm | M | 1 | Indicates the integrity protection algorithm |
| cipheringAlgorithm | CipheringAlgorithm | M | 1 | Indicates the ciphering algorithm |

6.1.6.2.37 Type: PduSessionContext

**Table 6.1.6.2.37-1: Definition of type PduSessionContext**

| Attribute name | Data type | P | Cardinality | Description | Applicability |
|---|---|---|---|---|---|
| pduSessionId | PduSessionId | M | 1 | Indicates the identifier of the PDU Session. | |
| smContextRef | Uri | M | 1 | Indicates the resource URI of the SM context, including the apiRoot (see clause 6.1.3.3.2 of TS 29.502 [16]).<br><br>When present, it shall carry the URI of SM Context of:<br>- I-SMF, for a PDU session with I-SMF; or<br>- V-SMF, for HR PDU session; or<br>- SMF, for non-roaming PDU session without I-SMF, or LBO roaming PDU session; | |
| sNssai | Snssai | M | 1 | Indicates the associated S-NSSAI for the PDU Session. It shall be the S-NSSAI in HPLMN in non-roaming, LBO roaming or HR roaming. | |
| altSnssai | Snssai | C | 0..1 | This IE shall be present if the S-NSSAI in the HPLMN (as indicated in the sNssai IE) was replaced by an alternative S-NSSAI in the HPLMN.<br><br>When present, this IE shall indicate the alternative S-NSSAI in the HPLMN.<br><br>See clause 5.15.19 of TS 23.501 [2]. | |
| additionalSnssai | Snssai | C | 0..1 | This IE shall be present in intra-VPLMN mobility of LBO roaming and HR roaming.<br>When present, this IE shall indicate the associated S-NSSAI in VPLMN for the PDU Session. | |
| altAdditionalSnssai | Snssai | C | 0..1 | This IE shall be present if S-NSSAI in the VPLMN (as indicated in the additionalSnssai IE) was replaced by an alternative S-NSSAI in the VPLMN.<br><br>When present, this IE shall indicate the alternative S-NSSAI in the VPLMN.<br><br>See clause 5.15.19 of TS 23.501 [2]. | |
| dnn | Dnn | M | 1 | This IE shall indicate the Data Network Name. The DNN shall be the full DNN (i.e. with both the Network Identifier and Operator Identifier) for a HR PDU session, and it should be the full DNN in LBO and non-roaming scenarios. If the Operator Identifier is absent, the serving core network operator shall be assumed. | |
| selectedDnn | Dnn | C | 0..1 | This IE shall be present, if another DNN other than the UE requested DNN is selected for this PDU session. When present, it shall contain the selected DNN. The DNN shall be the full DNN (i.e. with both the Network Identifier and Operator Identifier) for a HR PDU session, and it should be the full DNN in LBO and non-roaming scenarios. If the Operator Identifier is absent, the serving core network operator shall be assumed. | |
| accessType | AccessType | M | 1 | Indicates the access type of the PDU session. | |

| | | | | | |
|---|---|---|---|---|---|
| additionalAccessType | AccessType | C | 0..1 | Indicates the additional access type for a MA PDU session, if the UE registers to both 3GPP access and Non-3GPP access. | |
| allocatedEbiList | array(EbiArpMapping) | C | 1..N | This IE shall be present when at least one EBI is allocated to the PDU session.<br><br>When present, this IE shall contain the EBIs currently allocated to the PDU session. | |
| hsmfId | NfInstanceId | C | 0..1 | This IE shall be present for non-roaming and home-routed PDU sessions.<br><br>When present, it shall indicate the associated:<br><br>- home SMF for HR PDU Session, or<br>- SMF, for non-roaming PDU session, regardless of whether an I-SMF is involved or not. | |
| hsmfSetId | NfSetId | C | 0..1 | This IE shall be present, if available.<br><br>When present, this IE shall contain the NF Set ID of the home SMF or the SMF indicated by hsmfId. | |
| hsmfServiceSetId | NfServiceSetId | C | 0..1 | This IE shall be present, if available.<br><br>When present, this IE shall contain the NF Service Set ID of the selected PDUSession service instance of home SMF or the SMF indicated by hsmfId. | |
| smfBinding | SbiBindingLevel | C | 0..1 | This IE shall be present if available, for a non-roaming PDU session. When present, this IE shall contain the SBI binding level of the SMF's SM context resource. | |
| vsmfId | NfInstanceId | C | 0..1 | This IE shall be present for roaming PDU sessions. When present, it shall indicate the associated visited SMF for home-routed the PDU Session, or the SMF for the local-breakout PDU session (regardless of whether an I-SMF is involved or not). | |
| vsmfSetId | NfSetId | C | 0..1 | This IE shall be present, if available. When present, this IE shall contain the NF Set ID of the V-SMF. | |
| vsmfServiceSetId | NfServiceSetId | C | 0..1 | This IE shall be present, if available. When present, this IE shall contain the NF Service Set ID of the V-SMF's PDUSession service instance. | |
| vsmfBinding | SbiBindingLevel | C | 0..1 | This IE shall be present, if available. When present, this IE shall contain the SBI binding level of the V-SMF's SM context resource. | |
| ismfId | NfInstanceId | C | 0..1 | This IE shall be present if I-SMF is involved in the PDU session. When present, it shall indicate the associated I-SMF for the PDU Session. | DTSSA |
| ismfSetId | NfSetId | C | 0..1 | This IE shall be present, if available. When present, this IE shall contain the NF Set ID of the I-SMF. | DTSSA |

| | | | | | |
|---|---|---|---|---|---|
| ismfServiceSetId | NfServiceSetId | C | 0..1 | This IE shall be present, if available. When present, this IE shall contain the NF Service Set ID of the I-SMF's PDUSession service instance. | DTSSA |
| ismfBinding | SbiBindingLevel | C | 0..1 | This IE shall be present if available. When present, this IE shall contain the SBI binding level of the I-SMF's SM Context resource. | DTSSA |
| nsInstance | NsiId | C | 0..1 | This IE shall be present if available. When present, this IE shall Indicate Network Slice Instance for the PDU Session | |
| smfServiceInstanceId | string | O | 0..1 | When present, this IE shall contain the serviceInstanceId of the SMF PDUSession service instance serving the SM Context, i.e. of:<br>- the I-SMF, for a PDU session with I-SMF;<br>- the V-SMF, for a HR PDU session; or<br>- the SMF, for a non-roaming or an LBO roaming PDU session without I-SMF.<br><br>This IE may be used by the AMF to identify PDU session contexts affected by a failure or restart of the SMF service instance (see clause 6.2 of TS 23.527 [33]). | |
| maPduSession | boolean | C | 0..1 | This IE shall be present if available. When present, this IE shall indicate whether it is an MA PDU session.<br><br>true: indicates the PDU session is MA PDU session;<br>false (default): the PDU session is not MA PDU session. | |
| cnAssistedRanPara | CnAssistedRanPara | C | 0..1 | This IE shall be present if available.<br><br>When present, this IE shall contain the PDU Session specific parameters received from the SMF and used by the AMF to derive the Core Network assisted RAN parameters tuning. | |
| nrfManagementUri | Uri | C | 0..1 | If included, this IE shall contain the API URI of the NFManagement Service (see clause 6.1.1 of TS 29.510 [29]) of the NRF or hNRF.<br><br>It shall be present if it is returned from the NSSF or hNSSF (see clause 6.1.6.2.7 of TS 29.531 [18]). | |
| nrfDiscoveryUri | Uri | C | 0..1 | If included, this IE shall contain the API URI of the NFDiscovery Service (see clause 6.2.1 of TS 29.510 [29]) of the NRF or hNRF.<br><br>It shall be present if it is returned from the NSSF or hNSSF (see clause 6.1.6.2.7 of TS 29.531 [18]). | |

| | | | | | |
|---|---|---|---|---|---|
| nrfAccessTokenUri | Uri | C | 0..1 | If included, this IE shall contain the API URI of the Access Token Service (see clause 6.3.2 of TS 29.510 [29]) of the NRF or hNRF.<br><br>It shall be present if it is returned from the NSSF or hNSSF (see clause 6.1.6.2.7 of TS 29.531 [18]). | |
| smfBindingInfo | string | C | 0..1 | This IE shall be present if available, for a non-roaming PDU session. When present, this IE shall contain the Binding indications of the SMF's SM context resource and shall be set to the value of the 3gpp-Sbi-Binding header defined in clause 5.2.3.2.6 of TS 29.500 [4], without the header name. | |
| vsmfBindingInfo | string | C | 0..1 | This IE shall be present, if available. When present, this IE shall contain the Binding indications of the V-SMF's SM context resource and shall be set to the value of the 3gpp-Sbi-Binding header defined in clause 5.2.3.2.6 of TS 29.500 [4], without the header name. | |
| ismfBindingInfo | string | C | 0..1 | This IE shall be present if available. When present, this IE shall contain the Binding indications of the I-SMF's SM Context resource and shall be set to the value of the 3gpp-Sbi-Binding header defined in clause 5.2.3.2.6 of TS 29.500 [4], without the header name. | DTSSA |
| interPlmnApiRoot | Uri | C | 0..1 | This IE shall be present if this information is available.<br>When present, it shall contain the apiRoot of the SM context to be used in inter-PLMN signalling request targeting the SM context.<br>(NOTE) | |
| pgwFqdn | Fqdn | O | 0..1 | FQDN of the PGW in the PGW-C+SMF, to be included for interworking with EPS. | |
| pgwIpAddr | IpAddress | O | 0..1 | IP Address of the PGW in the PGW-C+SMF, to be included for interworking with EPS. | |
| plmnId | PlmnId | O | 0..1 | PLMN where the PGW-C+SMF is located. | |
| anchorSmfSupported Features | SupportedFeatures | O | 0..1 | When present, this IE shall include the features of the Nsmf_PDUSession service (see clause 6.1.8 of TS 29.502 [16]) that are supported by the H-SMF (or the SMF for a PDU sessions with I-SMF).<br>(NOTE 2) | |

| | | | | | |
|---|---|---|---|---|---|
| anchorSmfOauth2Required | boolean | O | 0..1 | This IE may be present when new AMF and old AMF belong to the same PLMN.<br><br>When present, this IE shall indicate whether the H-SMF (or the SMF for a PDU session with I-SMF) requires Oauth2-based authorization for accessing its Nsmf_PDUSession service<br><br>- true: OAuth2 based authorization is required.<br>- false: OAuth2 based authorization is not required.<br><br>The absence of this IE means that no indication is available about the usage of Oauth2 for authorization of the anchor SMF's Nsmf_PDUSession service. | |
| hrsboAllowedInd | boolean | C | 0..1 | This IE shall be present during an intra-PLMN N2 handover, if this information is available (see clause 6.7.2.6 of TS 23.548 [58]).<br><br>When present, it shall indicate whether HR-SBO is allowed for the PDU session and it shall be set as follows:<br>- true: HR-SBO is allowed<br>- false: HR-SBO is not allowed<br><br>The absence of this IE shall not be interpreted as meaning that HR-SBO is allowed or not allowed. | |
| pduSessionPrio | PduSessionPriority | C | 0..1 | This IE shall be present if available.<br><br>The AMF shall use the priority of the PDU session indicated in this IE for further PDU Session related priority handling (i.e. to determine the SBI Message Priority (SMP) to set in subsequent signaling it sends related to the PDU session). | |
| NOTE 1: During an inter-PLMN mobility, the target AMF shall replace the apiRoot of the smContextRef with the interPlmnApiRoot if available and send the resulting smContextRef in the Create SM Context request towards the target V-SMF, I-SMF or anchor SMF. See TS 29.502 [16].<br>NOTE 2: The new AMF may use this IE to know the supported features of the H-SMF (or the SMF for a PDU session with I-SMF) and take action based on the supported features, e.g. the new AMF shall release the PDU session when V-SMF needs to be changed but the H-SMF does not support V-SMF change. | | | | | |

### 6.1.6.2.38 Type: NssaiMapping

**Table 6.1.6.2.38-1: Definition of type NssaiMapping**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| mappedSnssai | Snssai | M | 1 | Indicates the mapped S-NSSAI in the serving PLMN |
| hSnssai | Snssai | M | 1 | Indicates the S-NSSAI in home PLMN |

#### 6.1.6.2.39 Type: UeRegStatusUpdateReqData

**Table 6.1.6.2.39-1: Definition of type UeRegStatusUpdateReqData**

| Attribute name | Data type | P | Cardinality | Description | Applicability |
|---|---|---|---|---|---|
| transferStatus | UeContextTransferStatus | M | 1 | This IE shall indicate if the previous UE context transfer was completed. | |
| toReleaseSessionList | array(PduSessionId) | C | 1..N | This IE shall be present during UE Context Transfer procedure, if there are any PDU session(s) associated with Network Slice(s) which become no longer available.<br><br>When present, this IE shall include all the PDU session(s) associated with no longer available S-NSSAI(s). | |
| pcfReselectedInd | boolean | C | 0..1 | This IE shall be present and set to true if the target AMF has decided to select a new PCF for AM Policy and/or UE Policy other than the one which was included in the UeContext by the old AMF.<br><br>When present, it shall be set as follows:<br>- true: the target AMF has selected new PCF(s) other than the one which was included in the UeContext by the old AMF for AM Policy and/or UE Policy. | |
| smfChangeInfoList | array(SmfChangeInfo) | C | 1..N | This IE shall be present during an inter-AMF registration procedure, if there is an I-SMF or V-SMF change or removal for the related PDU session(s).<br><br>When present, this IE shall indicate the I-SMF/V-SMF situation after the registration completion at the target AMF. | DTSSA |
| analyticsNotUsedList | array(Uri) | C | 1..N | This IE shall be present to include the list of resource URIs of the analytics subscription(s) that are not taken over in the target AMF. | |
| toReleaseSessionInfo | array(ReleaseSessionInfo) | C | 1..N | This IE shall be present during UE Context Transfer procedure, if there are any PDU session(s) that cannot be supported in the target AMF for a reason other than no longer available S-NSSAI(s) as specified in step 1 of clause 5.2.2.2.2.1.<br><br>When present, this IE shall include list of the PDU session(s) and the release cause. | |

#### 6.1.6.2.40 Type: AssignEbiError

**Table 6.1.6.2.40-1: Definition of type AssignEbiError**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| error | ProblemDetails | M | 1 | Represents the application error information. The application level error cause shall be encoded in the "cause" attribute. |
| failureDetails | AssignEbiFailed | M | 1 | Describes the details of the failure including the list of ARPs for which the EBI assignment failed. |

#### 6.1.6.2.41 Type: UeContextCreateData

**Table 6.1.6.2.41-1: Definition of type UeContextCreateData**

| Attribute name | Data type | P | Cardinality | Description | Applicability |
|---|---|---|---|---|---|
| ueContext | UeContext | M | 1 | Represents an individual ueContext resource to be created | |
| targetId | NgRanTargetId | M | 1 | Represents the identification of target RAN | |
| sourceToTargetData | N2InfoContent | M | 1 | This IE shall be included to contain the "Source to Target Transparent Container". | |
| pduSessionList | array(N2SmInformation) | M | 1..N | This IE shall be included to contain the list of N2SmInformation, where each N2SmInformation includes the "Handover Required Transfer" received from the source RAN per PDU session ID. | |
| n2NotifyUri | Uri | M | 1 | This IE shall contain a callback URI to receive the N2 Information Notification. | |
| ueRadioCapability | N2InfoContent | C | 0..1 | This IE shall be included to contain the "UE Radio Capability Information" if available.<br>(NOTE) | |
| ueRadioCapabilityForPaging | N2InfoContent | C | 0..1 | This IE shall be included to contain the "UE Radio Capability for Paging" if available.<br>(NOTE) | |
| ngapCause | NgApCause | C | 0..1 | This IE shall be present, if available. When present, it shall represent the NGAP Cause received from RAN. | |
| supportedFeatures | SupportedFeatures | C | 0..1 | This IE shall be present if at least one feature defined in clause 6.1.8 is supported. | |
| servingNetwork | PlmnIdNid | C | 0..1 | A Source AMF complying with this release of the specification shall include this IE to indicate the current Serving Network.<br>When present, this IE shall contain the serving core network operator PLMN ID and, for an SNPN, the NID that together with the PLMN ID identifies the SNPN. | |
| xrDeviceWith2Rx | XrDeviceWith2Rx | C | 0..1 | This IE shall be included to contain the "XR Device with 2Rx" as defined in TS 38.413 [12] if available during context transfer procedure. | |

NOTE: The source AMF may decide to not include ueRadioCapability and ueRadioCapabilityForPaging if the target AMF supports the RACS feature and if:
a. the PlmnAssiUeRadioCapId is included in the MM Context for an intra-PLMN AMF mobility procedure; or
b. the ManAssiUeRadioCapId is included in the MM Context for an intra-PLMN or an inter-PLMN AMF mobility procedure.

#### 6.1.6.2.42 Type: UeContextCreatedData

**Table 6.1.6.2.42-1: Definition of type UeContextCreatedData**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| ueContext | UeContext | M | 1 | Represents the newly created individual ueContext resource |
| targetToSourceData | N2InfoContent | M | 1 | This IE shall contain the "Target to Source Transparent Container". |
| pduSessionList | array(N2SmInformation) | M | 1..N | This IE shall be included to contain the list of N2SmInformation, where each N2SmInformation includes the "Handover Command Transfer" received from the SMF, per PDU session ID. |
| pcfReselectedInd | boolean | C | 0..1 | This IE shall be present and set to true if the target AMF has decided to select a new PCF for AM Policy and/or UE Policy other than the one which was included in the UeContext by the old AMF.<br><br>When present, it shall be set as follows:<br>- true: the target AMF has selected a new PCF other than the one which was included in the UeContext by the old AMF for AM Policy and/or UE Policy. |
| failedSessionList | array(N2SmInformation) | C | 1..N | This IE shall be included to contain a list of N2SmInformation, where each N2SmInformation includes the "Handover Preparation Unsuccessful Transfer" N2 SM content either received from the SMF for a PDU session failed to be handed over or generated by the target AMF for a PDU session not accepted by the target AMF (e.g. due to no response from the SMF within a maximum wait timer or due to non-available S-NSSAI in the target AMF). See NOTE. |
| supportedFeatures | SupportedFeatures | C | 0..1 | This IE shall be present if at least one feature defined in clause 6.1.8 is supported. |
| analyticsNotUsedList | array(Uri) | C | 1..N | This IE shall be present to include the list of resource URIs of the analytics subscription(s) that are not taken over in the target AMF. |
| NOTE: As an exception, the AMF generates N2 SM Information (Handover Preparation Unsuccessful Transfer IE) for a PDU session not accepted by the AMF, since this N2 SM IE needs to be included in the Handover Command sent by the source AMF to the source NG-RAN; this N2 SM IE carries a Cause value. | | | | |

#### 6.1.6.2.43 Type: UeContextCreateError

**Table 6.1.6.2.43-1: Definition of type UeContextCreateError**

| Attribute name | Data type | P | Cardinality | Description | Applicability |
|---|---|---|---|---|---|
| error | ProblemDetails | M | 1 | Represents the detailed application error information. The application level error cause shall be encoded in the "cause" attribute. | |
| ngapCause | NgApCause | C | 0..1 | This IE shall be present, if available. When present, it shall represent the NGAP Cause received from RAN. | |
| targetToSourceFailureData | N2InfoContent | C | 0..1 | This IE shall be present if a "Target to Source Failure Transparent Container" has been received from the target NG-RAN.<br>When present, this IE shall contain this container. | NPN |

#### 6.1.6.2.44 Type: NgRanTargetId

**Table 6.1.6.2.44-1: Definition of type NgRanTargetId**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| ranNodeId | GlobalRanNodeId | M | 1 | Indicates the identity of the RAN node. The IE shall contain either the gNB ID or the NG-eNB ID. |
| tai | Tai | M | 1 | Indicates the selected TAI. |

#### 6.1.6.2.45 Type: N2InformationTransferError

**Table 6.1.6.2.45-1: Definition of type N2InformationTransferError**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| error | ProblemDetails | M | 1 | More information on the error shall be provided in the "cause" attribute of the "ProblemDetails" structure. |
| pwsErrorInfo | PWSErrorData | C | 0..1 | This IE shall be present if the n2InformationClass is "PWS" in N2InformationTransferReqData. |

#### 6.1.6.2.46 Type: PWSResponseData

**Table 6.1.6.2.46-1: Definition of type PWSResponseData**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| ngapMessageType | Uinteger | M | 1 | This IE shall identify the message type of the message being sent. Its value is the numeric code of the Procedure Code defined in ASN.1 in clause 9.4.7 in TS 38.413 [12]. |
| serialNumber | Uint16 | M | 1 | This IE shall contain the Serial Number of the associated PWS response message. |
| messageIdentifier | integer | M | 1 | This IE shall contain the Message Identifier of the associated PWS response message. |
| unknownTAIList | array(Tai) | O | 1..N | This IE shall contain the Unknown Tracking Area List which may be present in the associated PWS response message. |
| n2PwsSubMissInd | boolean | C | 0..1 | This IE should be present when sendRanResponse IE with the value true is included in the request, i.e. the RAN responses are expected to be notified to the NF service consumer (i.e. the sender CBCF or PWS-IWF), and the corresponding N2 information subscription subscription is not available in the AMF.<br><br>When present, this IE shall be set as following:<br>- true: Corresponding N2 information subscription to receive the requested RAN responses is missing in the AMF. |

### 6.1.6.2.47 Type: PWSErrorData

**Table 6.1.6.2.47-1: Definition of type PWSErrorData**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| namfCause | integer | M | 1 | Represents the cause value for the error that the AMF detected.<br>Cause values:<br>0 - Message accepted<br>1 - Parameter not recognised<br>2 - Parameter value invalid<br>3 - Valid message not identified<br>4 - Tracking area not valid<br>5 - Unrecognised message<br>6 - Missing mandatory element<br>7 - AMF capacity exceeded<br>8 - AMF memory exceeded<br>9 - Warning broadcast not supported<br>10 - Warning broadcast not operational<br>11 - Message reference already used<br>12 - Unspecified error<br>13 - Transfer syntax error<br>14 - Semantic error<br>15 - Message not compatible with receiver state |

### 6.1.6.2.48 Void

### 6.1.6.2.49 Type: NgKsi

**Table 6.1.6.2.49-1: Definition of type NgKsi**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| tsc | ScType | M | 1 | Indicates whether the security context type is native or mapped. |
| ksi | integer | M | 1 | Indicates the key set identifier value. The value is within the range 0 to 6. |

### 6.1.6.2.50 Type: KeyAmf

**Table 6.1.6.2.50-1: Definition of type KeyAmf**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| keyType | KeyAmfType | M | 1 | Indicates whether the keyAmf represents $K_{amf}$ or $K'_{amf.}$ |
| keyVal | string | M | 1 | Indicates the key value. The key value is encoded as a string of hexadecimal characters.<br>Pattern: '^[A-Fa-f0-9]$' |

### 6.1.6.2.51 Type: ExpectedUeBehavior

**Table 6.1.6.2.51-1: Definition of type ExpectedUeBehavior**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| expMoveTrajectory | array(UserLocation) | M | 1..N | This IE shall contain a list of user location areas where the UE is expected to move. |
| validityTime | DateTime | M | 1 | This IE shall contain the UTC time upto which the UE moving trajectory is valid. |

### 6.1.6.2.52 Type: UeRegStatusUpdateRspData

**Table 6.1.6.2.52-1: Definition of type UeRegStatusUpdateRspData**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| regStatusTransferComplete | boolean | M | 1 | This IE shall indicate if the status update of UE context transfer is completed successfully at the source AMF or not.<br>The value shall be set to true if the context transfer is completed successfully and false if the context transfer did not complete successfully. |

### 6.1.6.2.53 Type: N2RanInformation

**Table 6.1.6.2.53-1: Definition of type N2RanInformation**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| n2InfoContent | N2InfoContent | M | 1 | This IE shall contain the N2 RAN information to transfer. |

### 6.1.6.2.54 Type: N2InfoNotificationRspData

**Table 6.1.6.2.54-1: Definition of type N2InfoNotificationRspData**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| secRatDataUsageList | array(N2SmInformation) | C | 1..N | This IE shall be present in the N2InfoNotify response sent by the source AMF to the target AMF during an Inter NG-RAN node N2 based handover procedure (see clause 5.2.2.3.6.2), if Secondary Rat Usage Data are available at the source AMF for one or more PDU sessions.<br><br>When present, this IE shall contain a list of N2SmInformation, where each N2SmInformation includes the "Secondary RAT Data Usage Report Transfer" information received from the source RAN for a given PDU session.<br><br>The sNssai IE and subjectToHo IE shall not be included in N2SmInformation. |

### 6.1.6.2.55 Type: SmallDataRateStatusInfo

**Table 6.1.6.2.55-1: Definition of type SmallDataRateStatusInfo**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| singleNssai | Snssai | M | 1 | S-NSSAI |
| dnn | Dnn | M | 1 | This IE shall indicate the Data Network Name.<br>The DNN shall be the full DNN (i.e. with both the Network Identifier and Operator Identifier) for a HR PDU session, and it should be the full DNN in LBO and non-roaming scenarios. If the Operator Identifier is absent, the serving core network operator shall be assumed. |
| smallDataRateStatus | SmallDataRateStatus | M | 1 | Small data rate status related to the S-NSSAI and Dnn. |

#### 6.1.6.2.56 Type: SmfChangeInfo

**Table 6.1.6.2.56-1: Definition of type SmfChangeInfo**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| pduSessionIdList | array(PduSessionId) | M | 1..N | PDU Session ID(s) for which the smfChangeInd applies. |
| smfChangeInd | SmfChangeIndication | M | 1 | Indicates the I-SMF or V-SMF change or removal. |

#### 6.1.6.2.57 Type: V2xContext

**Table 6.1.6.2.57-1: Definition of type V2xContext**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| nrV2xServicesAuth | NrV2xAuth | C | 0..1 | This IE shall be present if the UE is authorized to use the NR sidelink for V2X services. |
| lteV2xServicesAuth | LteV2xAuth | C | 0..1 | This IE shall be present if the UE is authorized to use the LTE sidelink for V2X services. |
| nrUeSidelinkAmbr | BitRate | C | 0..1 | This IE shall be present if the UE is authorized for NR V2X services.<br>When present, this IE contains subscription data on UE-PC5-AMBR for NR V2X services. |
| lteUeSidelinkAmbr | BitRate | C | 0..1 | This IE shall be present if the UE is authorized for LTE V2X services.<br>When present, this IE contains subscription data on UE-PC5-AMBR for LTE V2X services. |
| pc5QoSPara | Pc5QoSPara | C | 0..1 | This IE shall be present if the UE is authorized for NR V2X services.<br>When present, this IE contains policy data on the PC5 QoS parameters. |

6.1.6.2.58 Type: ImmediateMdtConf

**Table 6.1.6.2.58-1: Definition of type ImmediateMdtConf**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| jobType | JobType | M | 1 | This IE shall indicate the Job type for MDT, see TS 32.422 [30]. |
| measurementLteList | array(MeasurementLteForMdt) | C | 1..N | This IE shall be present if available.<br>When present, this IE shall contain a list of the measurements that shall be collected for LTE. |
| measurementNrList | array(MeasurementNrForMdt) | C | 1..N | This IE shall be present if available, when present, this IE shall contain a list of the measurements that shall be collected for NR. |
| reportingTriggerList | array(ReportingTrigger) | C | 1..N | This IE shall be present if available.<br>When present, this IE shall contain a list of the reporting triggers. |
| reportInterval | ReportIntervalMdt | C | 0..1 | This IE shall be present if available.<br>When present, this IE shall indicate the interval between the periodical measurements to be taken when UE is in connected in LTE. |
| reportIntervalNr | ReportIntervalNrMdt | C | 0..1 | This IE shall be present if available.<br>When present, this IE shall indicate the interval between the periodical measurements to be taken when UE is in connected in NR. |
| reportAmount | ReportAmountMdt | C | 0..1 | This IE shall be present if available.<br>When present, this IE shall indicate the number of measurement reports that shall be taken for periodical reporting while UE is in connected. |
| eventThresholdRsrp | integer | C | 0..1 | This IE shall be present if available.<br>When present, this IE shall indicate the Event Threshold for RSRP.<br>Minimum = 0. Maximum = 97. |
| eventThresholdRsrq | integer | C | 0..1 | This IE shall be present if available.<br>When present, this IE shall indicate the Event Threshold for RSRQ.<br>Minimum = 0. Maximum = 34. |
| eventThresholdRsrpNr | integer | C | 0..1 | This IE shall be present if available.<br>When present, this IE shall indicate the Event Threshold for RSRP in NR.<br>Minimum = 0. Maximum = 127. |
| eventThresholdRsrqNr | integer | C | 0..1 | This IE shall be present if available.<br>When present, this IE shall indicate the Event Threshold for RSRQ in NR.<br>Minimum = 0. Maximum = 127. |
| collectionPeriodRmmLte | CollectionPeriodRmmLteMdt | C | 0..1 | This IE shall be present if available.<br>When present, it shall contain the collection period that should be used to collect available measurement samples in case of RRM configured measurements. |
| collectionPeriodRmmNr | CollectionPeriodRmmNrMdt | C | 0..1 | This IE shall be present if available.<br>When present, it shall contain the collection period that should be used to collect available measurement samples in case of RRM configured measurements when UE is in NR. |
| measurementPeriodLte | MeasurementPeriodLteMdt | C | 0..1 | This IE shall be present if available.<br>When present, it shall contain the measurement period that should be used for the Data Volume and Scheduled IP Throughput measurements in LTE. |
| areaScope | AreaScope | O | 0..1 | When present, this IE shall contain the area in Cells or Tracking Areas where the MDT data collection shall take place, see TS 32.422 [30]. |
| positioningMethod | PositioningMethodMdt | O | 0..1 | When present, it shall indicate the positioning method that shall be used for the MDT job in LTE. |
| addPositioningMethodList | array(PositioningMethodMdt) | O | 1..N | This IE may be present if positioningMethod IE is present.<br><br>When present, it shall indicate a list of the additional positioning methods that shall be used for the MDT job |
| mdtAllowedPlmnIdList | array(PlmnId) | O | 1..16 | When present, this IE shall contain the PLMNs related to MDT. |

| | | | | |
|---|---|---|---|---|
| sensorMeasurementList | array(SensorMeasurement) | C | 1..N | This IE shall be present if available.<br>When present, this IE shall include a list the sensor measurements to be collected for UE in NR if they are available. |

### 6.1.6.2.59 Type: V2xInformation

**Table 6.1.6.2.59-1: Definition of type V2xInformation**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| n2Pc5Pol | N2InfoContent | C | 0..1 | This IE shall be present if N2 PC5 policy should be transferred. When present, the IE contains the NGAP V2X related IEs specified in clause 9.3.1.150 of TS 38.413 [12]. |

### 6.1.6.2.60 Type: EpsNasSecurityMode

**Table 6.1.6.2.60-1: Definition of type EpsNasSecurityMode**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| integrityAlgorithm | EpsNasIntegrityAlgorithm | M | 1 | Indicates the integrity protection algorithm for EPS NAS |
| cipheringAlgorithm | EpsNasCipheringAlgorithm | M | 1 | Indicates the ciphering algorithm for EPS NAS. |

### 6.1.6.2.61 Type: UeContextRelocateData

**Table 6.1.6.2.61-1: Definition of type UeContextRelocateData**

| Attribute name | Data type | P | Cardinality | Description | Applicability |
|---|---|---|---|---|---|
| ueContext | UeContext | M | 1 | Represents an individual ueContext resource to be relocated. | |
| targetId | NgRanTargetId | M | 1 | Represents the identification of target RAN | |
| sourceToTargetData | N2InfoContent | M | 1 | This IE shall be included to contain the "Source to Target Transparent Container". | |
| forwardRelocationRequest | RefToBinaryData | M | 1 | This IE shall be present, and it shall contain the reference to the binary data carrying the Forward Relocation Request message (see clause 6.1.6.4). | |
| pduSessionList | array(N2SmInformation) | C | 1..N | This IE shall contain the list of N2SmInformation, where each N2SmInformation includes a PDU Session Resource Setup Request Transfer IE (see clause 9.3.4.1 of TS 38.413 [24]) received from the SMF(s) per PDU session ID. | |
| ueRadioCapability | N2InfoContent | C | 0..1 | This IE shall be included to contain the "UE Radio Capability Information" if available. | |
| ngapCause | NgApCause | C | 0..1 | This IE shall be present, if available. When present, it shall represent the NGAP Cause mapped from the received S1-AP cause from the source E-UTRAN. Refer to TS 29.010 [50] for the mapping of cause values between S1AP and NGAP. | |
| supportedFeatures | SupportedFeatures | C | 0..1 | This IE shall be present if at least one feature defined in clause 6.1.8 is supported. | |
| xrDeviceWith2Rx | XrDeviceWith2Rx | C | 0..1 | This IE shall be included to contain the "XR Device with 2Rx" as defined in TS 38.413 [12] if available during context transfer procedure. | |

### 6.1.6.2.62 Type: UeContextRelocatedData

**Table 6.1.6.2.62-1: Definition of type UeContextRelocatedData**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| ueContext | UeContext | M | 1 | Represents an individual ueContext resource relocated to a new AMF. |

### 6.1.6.2.63 Void

#### 6.1.6.2.64 Type: EcRestrictionDataWb

**Table 6.1.6.2.64-1: Definition of type EcRestrictionData**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| ecModeARestricted | boolean | O | 0..1 | If present, indicates whether Enhanced Coverage Mode A is restricted or not.<br>true: Enhanced Coverage Mode A is restricted.<br>false or absent: Enhanced Coverage Mode A is not restricted. |
| ecModeBRestricted | boolean | M | 1 | This IE indicates whether Enhanced Coverage Mode B is restricted or not.<br>true: Enhanced Coverage Mode B is restricted.<br>false: Enhanced Coverage Mode B is not restricted. |

#### 6.1.6.2.65 Type: ExtAmfEventSubscription

**Table 6.1.6.2.65-1: Definition of type ExtAmfEventSubscription as a list of to be combined data types**

| Data type | Cardinality | Description | Applicability |
|---|---|---|---|
| AmfEventSubscription | 1 | AMF event subscription | |
| AmfEventSubscriptionAddInfo | 1 | Additional information for the AMF event subscription, e.g. Binding Indications, NF type of the NF that created the subscription. | |

6.1.6.2.66 Type: AmfEventSubscriptionAddInfo

**Table 6.1.6.2.66-1: Definition of type AmfEventSubscriptionAddInfo**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| bindingInfo | array(string) | O | 1..2 | Binding indications received for event notifications (i.e. with "callback" scope) or for subscription change event notifications (i.e. with "subscription-events" scope) for an AMF event subscription.<br><br>When present, entries of the array shall be set to the value of the 3gpp-Sbi-Binding header defined in clause 5.2.3.2.6 of TS 29.500 [4], without the header name.<br><br>Example of an array entry: "bl= nf-set; nfset=set1.udmset.5gc.mnc012.mcc345; servname=nudm-ee;scope=subscription-events" |
| subscribingNfType | NFType | C | 0..1 | This IE should be present if the information is available. When present, it shall contain the NF type of the NF that created the subscription.<br>(NOTE 1) |
| eventSyncInd | boolean | C | 0..1 | This IE should be present with value "true" when the event subscription shall be synchronized with UDM during EPS to 5GS mobility registration procedure, as specified in clause 5.3.2.4.2.<br><br>When present, this IE shall be set as following:<br>- true: the event subscription shall be synchronized with UDM.<br>- false: the event subscription shall not be synchronized with UDM. |
| nfConsumerInfo | array(string) | C | 1..N | When present, this IE shall contain the NF Service Consumer information received together with the AMF event subscription and entries of the array shall be set to the value of the 3gpp-Sbi-Consumer-Info header defined in clause 5.2.3.3.7 of TS 29.500 [4], without the header name. |
| aoiStateList | map(AreaOfInterestEventState) | O | 1..N | Map of subscribed Area of Interest (AoI) Event State in the old AMF.<br><br>For the subscribed AoI Event(s), the JSON pointer to an AmfEventArea element in the areaList IE (or a PresenceInfo element in presenceInfoList IE) of the AmfEvent data type (see clause 6.2.6.2.3) shall be the key of the map.<br><br>(NOTE 2) |
| accessToken | string | O | 0..1 | See clause 13.4.1.4 of TS 33.501 [27], and clause 5.2.2.2.1.1 and 5.2.2.2.3.1 for the use of this IE.<br><br>When present, this IE shall contain a JWS Compact Serialized representation of the JWS signed JSON object containing AccessTokenClaims (see clause 6.3.5.2.4 of TS 29.510 [29]) that was received by the source AMF for authorizing the subscription. |
| ueAccessBehaviorTrends | array(UeAccessBehaviorReportItem) | O | 1..N | When present, this IE shall contain the UE access behavior trend information collected and calculated by the AMF.<br><br>This IE may be present when there is a subscription for the event "UE_ACCESS_BEHAVIOR_TRENDS" and the event subscription is transferred from source AMF to a target AMF and at least one occurrence of the event is already available in the source AMF. |

| | | | | |
|---|---|---|---|---|
| ueLocationTrends | array(UeLocationTrendsReportItem) | O | 1..N | When present, this IE shall contain the UE location trend information collected and calculated by the AMF.<br><br>This IE may be present when there is a subscription for the event "UE_LOCATION_TRENDS" and the event subscription is transferred from source AMF to a target AMF and at least one occurrence of the event is already available in the source AMF. |
| NOTE 1: In scenarios where an "intermediate NF" (e.g. UDM) creates a subscription on behalf of a "source NF" (e.g. NEF), this IE shall contain the NF type of the "intermediate NF". The NF type of the "source NF" may be available in the AmfEventSubscription.<br>NOTE 2: The new AMF may use the information to determine whether the UE presence state in the AOI(s) has changed, thus should be reported to the NF consumer. | | | | |

### 6.1.6.2.67 Type: UeContextCancelRelocateData

**Table 6.1.6.2.67-1: Definition of type UeContextCancelRelocateData**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| supi | Supi | C | 0..1 | This IE shall be present if the UE is emergency registered and the SUPI is not authenticated. |
| relocationCancelRequest | RefToBinaryData | M | 1 | This IE shall be present, and it shall contain the reference to the binary data carrying the GTP-C Relocation Cancel Request message (see clause 6.1.6.4). |

### 6.1.6.2.68 Type: UeDifferentiationInfo

**Table 6.1.6.2.68-1: Definition of type UeDifferentiationInfo**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| periodicComInd | PeriodicCommunicationIndicator | O | 0..1 | This IE indicates whether the UE communicates periodically or not, e.g. only on demand. |
| periodicTime | DurationSec | O | 0..1 | This IE indicates the interval time of periodic communication (see TS 23.502 [3] clause 4.15.6.3). |
| scheduledComTime | ScheduledCommunicationTime | O | 0..1 | This IE indicates time and day of the week when the UE is available for communication (see TS 23.502 [3] clause 4.15.6.3). |
| stationaryInd | StationaryIndication | O | 0..1 | This IE indicates whether the UE is stationary or mobile (see TS 23.502 [3] clause 4.15.6.3). |
| trafficProfile | TrafficProfile | O | 0..1 | This IE indicates the type of data transmission: single packet transmission (UL or DL), dual packet transmission (UL with subsequent DL or DL with subsequent UL), multiple packets transmission |
| batteryInd | BatteryIndication | O | 0..1 | This IE indicates the power consumption type(s) of the UE (see TS 23.502 [3] clause 4.15.6.3). |
| validityTime | DateTime | O | 0..1 | When present, this IE identifies the UTC time when the expected UE behaviour parameters expire and shall be deleted locally if it expire (see TS 23.502 [3] clause 4.15.6.3).<br>When absent, no expiry for the expected UE behaviour parameters applies. |

### 6.1.6.2.69 Type: CeModeBInd

**Table 6.1.6.2.69-1: Definition of type CeModeBInd**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| ceModeBSupportInd | boolean | M | 1 | This IE shall contain the CE-mode-B Support Indicator (See TS 38.413 [12], clause 9.3.1.156).<br><br>This IE shall be set as follows:<br>- true: CE-mode-B is supported;<br>- false: CE-mode-B is not supported. |

### 6.1.6.2.70 Type: LteMInd

**Table 6.1.6.2.70-1: Definition of type LteMInd**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| lteCatMInd | boolean | M | 1 | This IE shall contain the LTE-M Indication (See TS 38.413 [12], clause 9.3.1.157).<br><br>This IE shall be set as follows:<br>- true: LTE-M is indicated by the UE;<br>- false: LTE-M is not indicated by the UE. |

### 6.1.6.2.71 Type: NpnAccessInfo

**Table 6.1.6.2.71-1: Definition of type NpnAccessInfo**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| cellCagInfo | array(CagId) | O | 1..N | This IE shall contain the CAG List of the CAG cell. |

### 6.1.6.2.72 Type: ProseContext

**Table 6.1.6.2.72-1: Definition of type ProseContext**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| directDiscovery | UeAuth | C | 0..1 | This IE shall be present if the UE is authorized to use the NR sidelink for the ProSe Direct Discovery service. |
| directComm | UeAuth | C | 0..1 | This IE shall be present if the UE is authorized to use the NR sidelink for the ProSe Direct Communication service. |
| l2Relay | UeAuth | C | 0..1 | This IE shall be present if the UE is authorized to act as a 5G ProSe Layer-2 UE-to-Network Relay. |
| l3Relay | UeAuth | C | 0..1 | This IE shall be present if the UE is authorized to act as a 5G ProSe Layer-3 UE-to-Network Relay. |
| l2Remote | UeAuth | C | 0..1 | This IE shall be present if the UE is authorized to act as a 5G ProSe Layer-2 Remote UE. |
| l3Remote | UeAuth | C | 0..1 | This IE shall be present if the UE is authorized to act as a 5G ProSe Layer-3 Remote UE. |
| l2UeRelay | UeAuth | C | 0..1 | This IE shall be present if the UE is authorized to act as a 5G ProSe Layer-2 UE-to-UE Relay. |
| l3UeRelay | UeAuth | C | 0..1 | This IE shall be present if the UE is authorized to act as a 5G ProSe Layer-3 UE-to-UE Relay. |
| l2End | UeAuth | C | 0..1 | This IE shall be present if the UE is authorized to act as a 5G ProSe Layer-2 End UE. |
| l3End | UeAuth | C | 0..1 | This IE shall be present if the UE is authorized to act as a 5G ProSe Layer-3 End UE. |
| multiPathComm | UeAuth | C | 0..1 | This IE shall be present if the UE is authorized to use multi-path communication via direct Uu path and via 5G ProSe Layer-2 UE-to-Network Relay as a 5G ProSe Layer-2 Remote UE. |
| nrUePc5Ambr | BitRate | C | 0..1 | This IE shall be present if the UE is authorized to use the NR sidelink for the ProSe services.<br>When present, this IE contains the ProSe NR UE-PC5-AMBR. |
| pc5QoSPara | Pc5QoSPara | C | 0..1 | This IE shall be present if the UE is authorized to use the NR sidelink for the ProSe services.<br>When present, this IE contains policy data on the PC5 QoS parameters for the ProSe services on the NR sidelink. |

### 6.1.6.2.73 Type: AnalyticsSubscription

**Table 6.2.6.2.73-1: Definition of type AnalyticsSubscription**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| nwdafId | NfInstanceId | C | 0..1 | This IE shall be present, if available.<br><br>When present, this IE shall contain the NF Instance ID of the NWDAF.<br>(NOTE). |
| nwdafSetId | NfSetId | C | 0..1 | This IE shall be present, if available.<br><br>When present, this IE shall contain the NF Set ID of the NWDAF.<br>(NOTE). |
| nwdafSubscriptionList | array(NwdafSubscription) | M | 1..N | List of NWDAF subscriptions identified by subscription Id. |
| NOTE: At least one nwdafId or nwdafSetId shall be included. | | | | |

### 6.1.6.2.74 Type: NwdafSubscription

**Table 6.2.6.2.74-1: Definition of type NwdafSubscription**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| nwdafEvtSubsServiceUri | Uri | M | 1 | It identifies the resource URI of the individual NWDAF subscription. |
| nwdafEventsSubscription | NnwdafEventsSubscription | M | 1 | The created Individual NWDAF Event Subscription resource |

### 6.1.6.2.75 Type: UpdpSubscriptionData

**Table 6.1.6.2.75-1: Definition of type UpdpSubscriptionData**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| updpNotifySubscriptionId | string | M | 1 | Represents the Id created by the AMF for the subscription to notify a UE policy delivery related N1 information. |
| updpNotifyCallbackUri | Uri | M | 1 | This IE represents the callback URI on which the UE policy delivery related N1 message shall be notified. |
| supportedFeatures | SupportedFeatures | C | 0..1 | This IE shall be present if at least one feature defined in clause 6.1.8 is supported by the NF service consumer. |
| updpCallbackBinding | string | C | 0..1 | This IE shall be present if Binding Indication was received for the PCF for the callback URI.<br><br>When present, this IE shall contain the Binding indication of callback URI and shall be set to the value of the 3gpp-Sbi-Binding header defined in clause 5.2.3.2.6 of TS 29.500 [4], without the header name. |

### 6.1.6.2.76 Type: ProSeInformation

**Table 6.1.6.2.76-1: Definition of type ProSeInformation**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| n2Pc5ProSePol | N2InfoContent | M | 0..1 | This attribute contains the N2 PC5 policy for 5G ProSe. This IE contains the NGAP ProSe related IEs specified in TS 38.413 [12]. |

### 6.1.6.2.77 Type: ReleaseSessionInfo

**Table 6.1.6.2.77-1: Definition of type ReleaseSessionInfo**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| releaseSessionList | array(PduSessionId) | M | 1..N | This IE shall include the PDU session Id(s) to be released. |
| releaseCause | ReleaseCause | M | 1 | This IE shall include the cause to release the PDU session(s). |

### 6.1.6.2.78 Type: AreaOfInterestEventState

**Table 6.1.6.2.78-1: Definition of type AreaOfInterestEventState**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| presence | PresenceState | M | 1 | This IE shall contain the UE presence state for the indicated area of interest. |
| individualPraIdList | array(string) | C | 1..N | This IE shall be present if the indicated area of interest referring to a set of Core Network predefined Presence Reporting Areas and the UE is in at least one individual PRA within the set of Core Network predefined Presence Reporting Areas.<br><br>When present, this IE shall contain the PRA Identifier of the individual PRA(s) where the UE is located. |

### 6.1.6.2.79 Type: TssInformation

**Table 6.1.6.2.79-1: Definition of type TssInformation**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| nfId | NfInstanceId | C | 0..1 | This IE shall be present when the N2InfoContainer containing the TssInformation is included in N2InformationTransferReqData. When present, this IE shall carry the NF instance identity of the NF Service Consumer (e.g. TSCTSF). |
| tssContainer | array(N2InfoContent) | C | 1..10 | This IE shall be present if the TSS N2 information to be relayed between TSCTSF and AN contains:<br><br>- one Timing Synchronization Status Request message sent from the TSCTSF to NG-RANs, or<br><br>- one or more Timing Synchronization Status Report message(s) sent from the NG-RAN(s) to the TSCTSF.<br><br>When present, it shall represent the N2 information information to be transferred between the TSCTSF and the NG-RAN node(s). |
| tssRspPerNgranList | array(TssRspPerNgran) | O | 1..10 | This IE may be present if TSS related N2 information is to be transferred from NG-RANs to the TSCTSF, i.e., to contain the Timing Synchronization Status Failure message, or Timing Synchronization Status Response message with the Criticality Diagnostics IE being present, or to report a failure related to a NG-RAN. |

#### 6.1.6.2.80 Type: AmPolicyInfoContainer

**Table 6.1.6.2.80-1: Definition of type AmPolicyInfoContainer**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| sliceUsgCtrlInfoSets | map(SliceUsage ControlInfo) | C | 1..N | This IE shall be present when available.<br><br>When present, this IE shall indicate the network slice usage control related information received from PCF of AM Policy.<br><br>The key of the map is the S-NSSAI to which the network slice usage control related information (the value part of the map) is related. |

#### 6.1.6.2.81 Type: RslpInformation

**Table 6.1.6.2.81-1: Definition of type RslpInformation**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| n2Pc5RslpPol | N2InfoContent | M | 1 | This attribute contains the N2 PC5 policy for Ranging/SL positioning. This IE contains the Ranging and Sidelink Positioning Service Information specified in TS 38.413 [12]. |

#### 6.1.6.2.82 Type: A2xContext

**Table 6.1.6.2.82-1: Definition of type A2xContext**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| nrA2xServicesAuth | NrA2xAuth | C | 0..1 | This IE shall be present if the UE is authorized to use the NR sidelink for A2X services. |
| lteA2xServicesAuth | LteA2xAuth | C | 0..1 | This IE shall be present if the UE is authorized to use the LTE sidelink for V2X services. |
| nrUeSidelinkAmbr | BitRate | C | 0..1 | This IE shall be present if the UE is authorized for NR A2X services.<br>When present, this IE contains subscription data on UE-PC5-AMBR for NR A2X services. |
| lteUeSidelinkAmbr | BitRate | C | 0..1 | This IE shall be present if the UE is authorized for LTE A2X services.<br>When present, this IE contains subscription data on UE-PC5-AMBR for LTE A2X services. |
| pc5QoSPara | Pc5QoSPara | C | 0..1 | This IE shall be present if the UE is authorized for NR A2X services.<br>When present, this IE contains policy data on the PC5 QoS parameters. |

#### 6.1.6.2.83 Type: A2xInformation

**Table 6.1.6.2.83-1: Definition of type A2xInformation**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| n2Pc5Pol | N2InfoContent | C | 0..1 | This IE shall be present if N2 PC5 policy should be transferred. When present, the IE contains the NGAP A2X related IEs specified in clause 9.3.1.150 of TS 38.413 [12]. |

#### 6.1.6.2.84 Type: LcsUpContext

**Table 6.1.6.2.84-1: Definition of type LcsUpContext**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| upConnectionStatus | UpConnectionStatus | O | 0..1 | UP Connection Status |
| servingLMFIdentification | LMFIdentification | O | 0..1 | Serving LMF ID |

#### 6.1.6.2.85 Type: DeregInactTimerInfo

**Table 6.1.6.2.85-1: Definition of type DeregInactTimerInfo**

| Attribute name | Data type | P | Cardinality | Description | Applicability |
|---|---|---|---|---|---|
| deregInactExpiryTime | DateTime | M | 1 | The value of the IE shall indicate the expiry time (in UTC) of the active Network Slice Deregistration Inactivity Timer of a Network Slice subject to Network-Controlled Slice Usage Control. | |

#### 6.1.6.2.86 Type: TssRspPerNgran

**Table 6.1.6.2.86-1: Definition of type TssRspPerNgran**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| ngranId | GlobalRanNodeId | M | 1 | Indicates the identity of the NG-RAN node. The IE shall contain the gNB ID. |
| ngranFailureInfo | NgranFailureInfo | O | 1 | This IE may be present to indicate a failure related to the NG-RAN, e.g. when the AMF detects that the NG-RAN has failed with or without restart, or that the NG-RAN is not reachable. |
| tssContainer | N2InfoContent | O | 1 | This IE may be present to contain the TSS N2 information data part to be relayed between TSCTSF and AN, e.g., Timing Synchronization Status Response/Failure messages. |

#### 6.1.6.2.87 Type: SliceReplacementMapping

**Table 6.1.6.2.87-1: Definition of type SliceReplacementMapping**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| replacedSnssai | Snssai | M | 1 | Indicates the replaced S-NSSAI. |
| altSnssai | Snssai | M | 1 | Indicates the alternative S-NSSAI |

#### 6.1.6.2.88 Type: SliceDeregInactConfig

**Table 6.1.6.2.88-1: Definition of type SliceDeregInactConfig**

| Attribute name | Data type | P | Cardinality | Description | Applicability |
|---|---|---|---|---|---|
| deregInactTimer | DurationSec | C | 0..1 | This IE shall indicate the value of the slice deregistration inactivity timer for a certain S-NSSAI, received from the PCF.<br>This IE shall be present if received from the PCF. | |

### 6.1.6.3 Simple data types and enumerations

#### 6.1.6.3.1 Introduction

This clause defines simple data types and enumerations that can be referenced from data structures defined in the previous clauses.

#### 6.1.6.3.2 Simple data types

The simple data types defined in table 6.1.6.3.2-1 shall be supported.

**Table 6.1.6.3.2-1: Simple data types**

| Type Name | Type Definition | Description |
|---|---|---|
| EpsBearerId | integer | Integer identifying an EPS bearer, within the range 0 to 15, as specified in clause 11.2.3.1.5, bits 5 to 8, of TS 24.007 [15]. |
| Ppi | integer | This represents the Paging Policy Indicator. The value is within the range 0 to 7. |
| NasCount | Uinteger | Unsigned integer identifying the NAS COUNT as specified in TS 33.501 [27] |
| 5GMmCapability | Bytes | String with format "byte" as defined in OpenAPI Specification [23], i.e. base64-encoded characters, encoding the "5GMM capability" IE as specified in clause 9.11.3.1 of TS 24.501 [11] (starting from octet 1). |
| UeSecurityCapability | Bytes | String with format "byte" as defined in OpenAPI Specification [23], i.e. base64-encoded characters, encoding the " UE security capability" IE as specified in clause 9.11.3.54 of TS 24.501 [11] (starting from octet 1). |
| S1UeNetworkCapability | Bytes | String with format "byte" as defined in OpenAPI Specification [23], i.e. base64-encoded characters, encoding the " S1 UE network capability" IE as specified in clause 9.11.3.48 of TS 24.501 [11] (starting from octet 1). |
| DrxParameter | Bytes | String with format "byte" as defined in OpenAPI Specification [23], i.e. base64-encoded characters, encoding the "5GS DRX Parameters" IE as specified in clause 9.11.3.2A of TS 24.501 [11] (starting from octet 1). |
| OmcIdentifier | string | The OMC Identifier indicates the identity of an Operation and Maintenance Centre to which Trace Records shall be sent.<br>minLength: 1<br>maxLength: 20 |
| MSClassmark2 | Bytes | String with format "byte" as defined in OpenAPI Specification [23], i.e. base64-encoded characters, encoding the Mobile Station Classmark 2 as specified in clause 9.11.3.31C of TS 24.501 [11]) (starting from octet 1). |
| SupportedCodec | Bytes | String with format "byte" as defined in OpenAPI Specification [23], i.e. base64-encoded characters, encoding the Supported Codec as specified in clause 9.11.3.51A of TS 24.501 [11]) (starting from octet 1). |

#### 6.1.6.3.3 Enumeration: StatusChange

**Table 6.1.6.3.3-1: Enumeration StatusChange**

| Enumeration value | Description |
|---|---|
| "AMF_UNAVAILABLE" | The AMF is unavailable to serve the UEs identified by the GUAMI(s). |
| "AMF_AVAILABLE" | The AMF is available to serve the UEs identified by the GUAMI(s). |

#### 6.1.6.3.4 Enumeration: N2InformationClass

**Table 6.1.6.3.4-1: Enumeration N2InformationClass**

| Enumeration value | Description | Applicability |
|---|---|---|
| "SM" | N2 SM information. | |
| "NRPPa" | N2 NRPPa information. | |
| "PWS" | N2 PWS information of PWS type. | |
| "PWS-BCAL" | N2 Broadcast Completed Area List or the Broadcast Cancelled Area List. | |
| "PWS-RF" | N2 Restart Indication or Failure Indication | |
| "RAN" | N2 RAN related information. | |
| "V2X" | N2 V2X information | |
| "PROSE" | N2 5G ProSe information. | ProSe |
| "TSS" | N2 Timing Synchronisation Status information | |
| "RANGING_SL" | N2 ranging and sidelink positioning information transport over PC5 | Ranging_SL |
| "A2X" | N2 A2X information | A2X |

NOTE: Some of the values in the enumeration deviate from the naming conventions indicated in clause 5.1.4 of TS 29.501 [5] (i.e. to use UPPER_WITH_UNDERSCORE); however, it is kept as currently defined in this specification to maintain backwards compatibility.

#### 6.1.6.3.5 Enumeration: N1MessageClass

**Table 6.1.6.3.5-1: Enumeration N1MessageClass**

| Enumeration value | Description |
|---|---|
| "5GMM" | The whole NAS message as received (for e.g. used in forwarding the Registration message to target AMF during Registration procedure with AMF redirection). |
| "SM" | N1 Session Management message |
| "LPP" | N1 LTE Positioning Protocol message |
| "SMS" | N1 SMS message as specified in TS 23.040 [44] and TS 24.011 [45] |
| "UPDP" | The N1 messages for UE Policy Delivery (See Annex D of TS 24.501 [11]. |
| "LCS" | The N1 message of Location service message type |
| "UPP-CM" | The N1 message type for UPP-CM messages |

#### 6.1.6.3.6 Enumeration: N1N2MessageTransferCause

**Table 6.1.6.3.6-1: Enumeration N1N2MessageTransferCause**

| Enumeration value | Description |
|---|---|
| "ATTEMPTING_TO_REACH_UE" | This cause represents the case where the AMF has initiated paging to reach the UE in order to deliver the N1 message. |
| "N1_N2_TRANSFER_INITIATED" | This cause represents the case where the AMF has initiated the N1/N2 message transfer towards the UE and/or the AN. |
| "WAITING_FOR_ASYNCHRONOUS_TRANSFER" | This cause represents the case where the AMF has stored the N1/N2 message due to Asynchronous Transfer. |
| "UE_NOT_RESPONDING" | This cause represents the case that the AMF has initiated paging to reach the UE but the UE is not responding to the paging, or the case of a UE in RRC Inactive state when NG-RAN paging is not successful (e.g. NG-RAN initiated a UE context release with a cause indicating the non-delivery of the N1 message). |
| "N1_MSG_NOT_TRANSFERRED" | This cause represents the case where the AMF has skipped sending N1 message to the UE, when UE is in CM-IDLE and the "skipInd" is set to "true" in the request. |
| "N2_MSG_NOT_TRANSFERRED" | This cause indicates that the AMF has skipped sending the N2 message (i.e. PDU Session Resource Setup Request) as well as the possibly included N1 message to the 5G-AN, when the UE is in CM-CONNECTED state and the UE is outside of the validity area included in the N1N2MessageTransfer Request. |
| "UE_NOT_REACHABLE_FOR_SESSION" | This cause indicates that the UE is not reachable for the non-3GPP PDU session, due to the UE being in CM-IDLE for non-3GPP access and the PDU session is not allowed to move to 3GPP access. |
| "TEMPORARY_REJECT_REGISTRATION_ONGOING" | This cause represents the case that the AMF has initiated paging to reach the UE but there is an ongoing registration procedure. |
| "TEMPORARY_REJECT_HANDOVER_ONGOING" | This cause indicates that the AMF has initiated N1 signalling towards the UE but the N1 message could not be delivered due to an ongoing Xn or N2 handover procedure. |
| "REJECTION_DUE_TO_PAGING_RESTRICTION" | This cause represents the case that the UE has rejected the page as specified in TS 23.501 [2] clause 5.38.4. |
| "AN_NOT_RESPONDING" | This cause indicates that the AMF has initiated a N2 request to transfer a N2 PDU (e.g. to setup, modify or release PDU session resources) but the AN is not responding to the N2 request. |
| "FAILURE_CAUSE_UNSPECIFIED" | This cause indicates that the N1/N2 message transfer failed due to unspecified reasons. |

#### 6.1.6.3.7 Enumeration: UeContextTransferStatus

**Table 6.1.6.3.7-1: Enumeration UeContextTransferStatus**

| Enumeration value | Description |
|---|---|
| "TRANSFERRED" | Indicates a UE Context Transfer procedure is completed successful for the individual ueContext resource |
| "NOT_TRANSFERRED" | Indicates a UE Context Transfer procedure either did not complete successfully or the Registration request from the UE is redirected to another NF Service Consumer (e.g. AMF). |

#### 6.1.6.3.8 Enumeration: N2InformationTransferResult

**Table 6.1.6.3.8-1: Enumeration N2InformationTransferResult**

| Enumeration value | Description |
|---|---|
| "N2_INFO_TRANSFER_INITIATED" | This cause code represents the case where the AMF has initiated the N2 information transfer towards the AN. |

#### 6.1.6.3.9 Enumeration: CipheringAlgorithm

**Table 6.1.6.3.9-1: Enumeration CipheringAlgorithm**

| Enumeration value | Description |
|---|---|
| "NEA0" | Null ciphering algorithm |
| "NEA1" | 128-bit SNOW 3G based algorithm |
| "NEA2" | 128-bit AES based algorithm |
| "NEA3" | 128-bit ZUC based algorithm |

#### 6.1.6.3.10 Enumeration: IntegrityAlgorithm

**Table 6.1.6.3.10-1: Enumeration IntegrityAlgorithm**

| Enumeration value | Description |
|---|---|
| "NIA0" | Null Integrity Protection algorithm |
| "NIA1" | 128-bit SNOW 3G based algorithm |
| "NIA2" | 128-bit AES based algorithm |
| "NIA3" | 128-bit ZUC based algorithm |

#### 6.1.6.3.11 Enumeration: SmsSupport

**Table 6.1.6.3.11-1: Enumeration SmsSupport**

| Enumeration value | Description |
|---|---|
| "3GPP" | Support SMS delivery over NAS in 3GPP access |
| "NON_3GPP" | Support SMS delivery via non-3GPP access |
| "BOTH" | Support SMS delivery over NAS or via non-3GPP access |
| "NONE" | Don't support SMS delivery |

### 6.1.6.3.12 Enumeration: ScType

**Table 6.1.6.3.12-1: Enumeration ScType**

| Enumeration value | Description |
|---|---|
| "NATIVE" | Native security context (for $KSI_{AMF}$) |
| "MAPPED" | Mapped security context (for $KSI_{ASME}$) |

### 6.1.6.3.13 Enumeration: KeyAmfType

**Table 6.1.6.3.13-1: Enumeration KeyAmfType**

| Enumeration value | Description |
|---|---|
| "KAMF" | The $K_{amf}$ value is sent. |
| "KPRIMEAMF" | The $K'_{amf.}$ value is sent. |

### 6.1.6.3.14 Enumeration: TransferReason

**Table 6.1.6.3.14-1: Enumeration TransferReason**

| Enumeration value | Description |
|---|---|
| "INIT_REG" | It indicates the AMF requests UE context for initial registration or disaster roaming initial registration. |
| "MOBI_REG" | It indicates the AMF requests UE context for mobility registration or disaster roaming mobility registration. |
| "MOBI_REG_UE_VALIDATED" | It indicates the AMF requests UE context for mobility registration or disaster roaming mobility registration of a validated UE. |

### 6.1.6.3.15 Enumeration: PolicyReqTrigger

**Table 6.1.6.3.15-1: Enumeration PolicyReqTrigger**

| Enumeration value | Description |
|---|---|
| "LOCATION_CHANGE" | The AM policy request shall be triggered when the UE's location (Tracking Area) changes. |
| "PRA_CHANGE" | The AM policy request shall be triggered when the UE is entering / leaving a Presence Reporting Area. |
| "ALLOWED_NSSAI_CHANGE" | The policy request shall be triggered when the allowed NSSAI of the UE has changed. |
| "NWDAF_DATA_CHANGE" | The AM policy request shall be triggered when the NWDAF instance IDs used for the UE and/or associated Analytics IDs have changed. |
| "PLMN_CHANGE" | The UE policy request shall be triggered when the serving PLMN of UE has changed. |
| "CON_STATE_CHANGE" | The UE policy request shall be triggered when the connectivity state of UE has changed. |
| "SMF_SELECT_CHANGE" | The AM policy request shall be triggered when the UE request for an unsupported DNN or the UE request for a DNN within the list of DNN candidates for replacement per S-NSSAI. |
| "ACCESS_TYPE_CHANGE" | The AM policy request shall be triggered when the access type and the RAT type combinations available in the AMF for a UE with simultaneous 3GPP and non-3GPP connectivity have changed. |
| "SAT_BACKHAUL_CHANGE" | The UE policy request shall be triggered when the UE's satellite backhaul category changes. |
| "CONF_NSSAI_CHANGE" | The UE policy request shall be triggered when the configured NSSAI has changed. |
| "LBO_INFO_CHANGE" | The UE policy request shall be triggered when the LBO information has changed, i.e. LBO roaming allowed or not allowed for the DNN(s) and S-NSSAI(s). |

#### 6.1.6.3.16 Enumeration: RatSelector

**Table 6.1.6.3.16-1: Enumeration RatSelector**

| Enumeration value | Description |
|---|---|
| "E-UTRA" | The N2 information shall be transferred to ng-eNBs only. |
| "NR" | The N2 information shall be transferred to gNBs only. |

NOTE: Some of the values in the enumeration deviate from the naming conventions indicated in clause 5.1.4 of TS 29.501 [5] (i.e. to use UPPER_WITH_UNDERSCORE); however, it is kept as currently defined in this specification to maintain backwards compatibility.

#### 6.1.6.3.17 Enumeration: NgapIeType

**Table 6.1.6.3.17-1: Enumeration NgapIeType**

| Enumeration value | Description |
|---|---|
| "PDU_RES_SETUP_REQ" | PDU Session Resource Setup Request Transfer |
| "PDU_RES_REL_CMD" | PDU Session Resource Release Command Transfer |
| "PDU_RES_MOD_REQ" | PDU Session Resource Modify Request Transfer |
| "HANDOVER_CMD" | Handover Command Transfer |
| "HANDOVER_REQUIRED" | Handover Required Transfer |
| "HANDOVER_PREP_FAIL" | Handover Preparation Unsuccessful Transfer |
| "SRC_TO_TAR_CONTAINER" | Source to Target Transparent Container |
| "TAR_TO_SRC_CONTAINER" | Target to Source Transparent Container |
| "TAR_TO_SRC_FAIL_CONTAINER" | Target to Source Failure Transparent Container |
| "RAN_STATUS_TRANS_CONTAINER" | RAN Status Transfer Transparent Container |
| "SON_CONFIG_TRANSFER" | SON Configuration Transfer |
| "NRPPA_PDU" | NRPPa-PDU |
| "UE_RADIO_CAPABILITY" | UE Radio Capability |
| "RIM_INFO_TRANSFER" | RIM Information Transfer |
| "SECONDARY_RAT_USAGE" | Secondary RAT Data Usage Report Transfer |
| "PC5_QOS_PARA" | PC5 QoS Parameters |
| "EARLY_STATUS_TRANS_CONTAINER" | Early Status Transfer Transparent Container |
| "UE_RADIO_CAPABILITY_FOR_PAGING" | UE Radio Capability for Paging |

#### 6.1.6.3.18 Enumeration: N2InfoNotifyReason

**Table 6.1.6.3.18-1: Enumeration N2InfoNotifyReason**

| Enumeration value | Description |
|---|---|
| "HANDOVER_COMPLETED" | Indicates that the N2 Information Notification is delivered when the handover procedure is completed successfully. |

#### 6.1.6.3.19 Enumeration: SmfChangeIndication

**Table 6.1.6.3.19-1: Enumeration SmfChangeIndication**

| Enumeration value | Description |
|---|---|
| "CHANGED" | I-SMF or V-SMF changed. (NOTE 1) |
| "REMOVED" | I-SMF or V-SMF is removed. (NOTE 2) |
| NOTE 1: This enumeration value shall also be used if the UE moves from HPLMN to a VPLMN, a V-SMF is inserted and the I-SMF is removed, or from VPLMN to HPLMN, an I-SMF is inserted and the V-SMF is removed. | |
| NOTE 2: This enumeration value shall also be used if the UE moves from VPLMN to HPLMN, without an I-SMF inserted in the HPLMN. | |

#### 6.1.6.3.20 Enumeration: SbiBindingLevel

**Table 6.1.6.3.20-1: Enumeration SbiBindingLevel**

| Enumeration value | Description |
|---|---|
| "NF_INSTANCE_BINDING" | Indicates binding to NF instance |
| "NF_SET_BINDING" | Indicates binding to NF Set |
| "NF_SERVICE_SET_BINDING" | Indicates binding to NF Service Set |
| "NF_SERVICE_INSTANCE_BINDING" | Indicates binding to NF Service instance |

#### 6.1.6.3.21 Enumeration: EpsNasCipheringAlgorithm

This data type enumerates the algorithms for data ciphering in EPS NAS, as specified in clause 5.1.3.2 of TS 33.401 [49].

**Table 6.1.6.3.21-1: Enumeration EpsNasCipheringAlgorithm**

| Enumeration value | Description |
|---|---|
| "EEA0" | Null ciphering algorithm |
| "EEA1" | 128-bit SNOW 3G based algorithm |
| "EEA2" | 128-bit AES based algorithm |
| "EEA3" | 128-bit ZUC based algorithm |

#### 6.1.6.3.22 Enumeration: EpsNasIntegrityAlgorithm

This data type enumerates the algorithms for data integrity protection in EPS NAS, as specified in clause 5.1.4.2 of TS 33.401 [49].

**Table 6.1.6.3.22-1: Enumeration EpsNasIntegrityAlgorithm**

| Enumeration value | Description |
|---|---|
| "EIA0" | Null Integrity Protection algorithm |
| "EIA1" | 128-bit SNOW 3G based algorithm |
| "EIA2" | 128-bit AES based algorithm |
| "EIA3" | 128-bit ZUC based algorithm |

#### 6.1.6.3.23 Enumeration: PeriodicCommunicationIndicator

This data type enumerates types of Periodic Communication Indicator.

**Table 6.1.6.3.23-1: Enumeration PeriodicCommunicationIndicator**

| Enumeration value | Description |
|---|---|
| "PIORIODICALLY" | Periodically |
| "ON_DEMAND" | On demand |

#### 6.1.6.3.24 Enumeration: UuaaMmStatus

This data type enumerates types of UUAA-MM status.

**Table 6.1.6.3.24-1: Enumeration UuaaMmStatus**

| Enumeration value | Description |
|---|---|
| "SUCCESS" | Success |
| "PENDING" | Pending |
| "FAILED" | Failed |

#### 6.1.6.3.25 Enumeration: ReleaseCause

**Table 6.1.6.3.25-1: Enumeration ReleaseCause**

| Enumeration value | Description |
|---|---|
| "SNPN_SNPN_MOBILITY" | This cause represents the case where the continuity of the PDU Session(s) cannot be supported between networks due to SNPN-SNPN mobility. |
| "NO_HR_AGREEMENT" | This cause represents the case where the continuity of the PDU Session(s) cannot be supported between networks due to inter-PLMN mobility where no HR agreement exists. |
| "UNSPECIFIED" | This cause indicates that the continuity of the PDU Session(s) cannot be supported between networks due to unspecified reasons. |

#### 6.1.6.3.26 Enumeration: NgranFailureInfo

The enumeration NgranFailureInfo indicates a NG-RAN failure event.

**Table 6.1.6.3.26-1: Enumeration NgranFailureInfo**

| Enumeration value | Description |
|---|---|
| "NG_RAN_NOT_REACHABLE" | This value indicates that the AMF has failed to reach the NG-RAN when sending a N2 information, e.g., a NGAP message. |
| "NG_RAN_FAILURE_WITH_RESTART" | This value indicates that the AMF has detected a restart of a NG-RAN. |
| "NG_RAN_FAILURE_WITHOUT_RESTART" | This value indicates that the AMF has detected a NG-RAN failure without a restart. |

#### 6.1.6.3.27 Enumeration: XrDeviceWith2Rx

**Table 6.1.6.3.27-1: Enumeration XrDeviceWith2Rx**

| Enumeration value | Description |
|---|---|
| "TRUE" | Indicates that the UE is a 2Rx XR UE. |

### 6.1.6.4 Binary data

#### 6.1.6.4.1 Introduction

This clause defines the binary data that shall be supported in a binary body part in an HTTP multipart message (see clauses 6.1.2.2.2 and 6.1.2.4).

**Table 6.1.6.4.1-1: Binary Data Types**

| Name | Clause defined | Content type |
|---|---|---|
| N1 Message | 6.1.6.4.2 | vnd.3gpp.5gnas |
| N2 Information | 6.1.6.4.3 | vnd.3gpp.ngap |
| Mobile Terminated Data | 6.1.6.4.3 | vnd.3gpp.5gnas |
| GTP-C message | 6.1.6.4.5 | vnd.3gpp.gtpc |

#### 6.1.6.4.2 N1 Message

N1 Message shall encode a 5GS NAS message of a specified type (e.g. SM, LPP) as specified in TS 24.501 [11], using the vnd.3gpp.5gnas content-type.

N1 Message may encode e.g. the following 5GS NAS messages:

- For message class SM:
  - PDU Session Modification Command (see clause 8.3.7 of TS 24.501 [11]) during network initiated PDU session modification procedure (see clause 4.3.3 of TS 23.502 [3]);
  - PDU Session Release Command (see clause 8.3.12 of TS 24.501 [11]) during network initiated PDU session release procedure (see clause 4.3.4 of TS 23.502 [3]).
  - PDU Session Establishment Accept (see clause 8.3.2 in TS 24.501 [11]) during UE-requested PDU Session Establishment (see clause 4.3.2.2 in TS 23.502 [3]).
- For message class LPP:
  - UE Positioning Request messages as specified in TS 36.355 [13] during UE assisted and UE based positioning procedure (see clause 6.11.1 of TS 23.273 [42]).
- For message class 5GMM:
  - REGISTRATION REQUEST message as specified in see clause 8.2.5 of TS 24.501 [11], during registration procedures (see clause 4.2.2.2 of TS 23.502 [3]).
- For message class UPDP:
  - MANAGE UE POLICY COMMAND / MANAGE UE POLICY COMPLETE / MANAGE UE POLICY REJECT (see Annex D.5.1 to Annex D.5.3 of TS 24.501 [11]) during network initiated UE policy management procedure (see Annex D.2.1 of TS 24.501 [11]);
  - UE STATE INDICATION (see Annex D.5.4 of TS 24.501 [11]) during UE initiated UE state indication procedure (see Annex D.2.2 of TS 24.501 [11]).
  - UE POLICY PROVISIONING REQUEST / UE POLICY PROVISIONING REJECT (see clause 7.2 of TS 24.587 [53]) during UE-requested V2X policy provisioning procedure and/or 5G ProSe policy provisioning procedure (see clause 5.3.2 of TS 24.587 [53] and/or clause 5.3.2 of TS 24.554 [54] respectively) and/or UE-requested A2X policy provisioning procedure (see clause 5.3.2 of TS 24.577 [60]).
- For message class LCS:
  - Location services messages between UE and LMF (lcs-PeriodicTriggeredInvoke/lcs-EventReport/lcs-CancelDeferredLocation) as specified in TS 24.080 [43] during deferred 5GC-MT-LR procedure procedure (see clause 6.3 of TS 23.273 [42]).
  - Location services messages between PRU and LMF during PRU association and disassociation procedures (see clause 6.17 of TS 23.273 [42]).
- For message class SMS:
  - SMS payload information as specified in TS 23.040 [44] and TS 24.011 [45], e.g. CP-DATA, CP-ACK, CP-ERROR.
- For message class UPP-CM:
  - UPP-CM messages as specified in clause 10.3 of TS 24.572 [61].

#### 6.1.6.4.3 N2 Information

##### 6.1.6.4.3.1 Introduction

N2 Information shall encode NG Application Protocol (NGAP) IEs, as specified in clause 9.4 of TS 38.413 [12] (ASN.1 encoded), using the vnd.3gpp.ngap content-type.

##### 6.1.6.4.3.2 NGAP IEs

For N2 information class SM, N2 Information may encode following NGAP SMF related IE specified in TS 38.413 [12], as summarized in Table 6.1.6.4.3.2-1.

**Table 6.1.6.4.3.2-1: N2 Information content for class SM**

| NGAP IE | Reference (TS 38.413 [12]) | Related NGAP message |
|---|---|---|
| PDU Session Resource Setup Request Transfer | 9.3.4.1 | PDU SESSION RESOURCE SETUP REQUEST |
| PDU Session Resource Release Command Transfer | 9.3.4.12 | PDU SESSION RESOURCE RELEASE COMMAND |
| PDU Session Resource Modify Request Transfer | 9.3.4.3 | PDU SESSION RESOURCE MODIFY REQUEST |
| Handover Command Transfer | 9.3.4.10 | HANDOVER COMMAND |
| Handover Required Transfer | 9.3.4.14 | HANDOVER REQUIRED |
| Handover Preparation Unsuccessful Transfer | 9.3.4.18 | HANDOVER COMMAND |
| Secondary RAT Data Usage Report Transfer | 9.3.4.23 | SECONDARY RAT DATA USAGE REPORT |

For N2 information class RAN, N2 Information may encode one of the following NGAP Transparent Container IEs specified in TS 38.413 [12], as summarized in Table 6.1.6.4.3.2-2.

**Table 6.1.6.4.3.2-2: N2 Information content for class RAN**

| NGAP IE | Reference (TS 38.413 [12]) | Related NGAP message |
|---|---|---|
| Source to Target Transparent Container | 9.3.1.20 | HANDOVER REQUIRED, HANDOVER REQUEST |
| Target to Source Transparent Container | 9.3.1.21 | HANDOVER COMMAND, HANDOVER REQUEST ACKNOWLEDGE |
| Target to Source Failure Transparent Container | 9.3.1.186 | HANDOVER FAILURE |
| UE Radio Capability | 9.3.1.74 | UE RADIO CAPABILITY INFO INDICATION. (NOTE 1). |
| UE Radio Capability for Paging | 9.3.1.68 | UE RADIO CAPABILITY INFO INDICATION. (NOTE 1). |
| SON Configuration Transfer | 9.3.3.6 | UPLINK RAN CONFIGURATION TRANSFER, DOWNLINK RAN CONFIGURATION TRANSFER |
| RAN Status Transfer Transparent Container | 9.2.3.13, 9.2.3.14 | UPLINK RAN STATUS TRANSFER, DOWNLINK RAN STATUS TRANSFER |
| Early Status Transfer Transparent Container | 9.2.3.16, 9.2.3.17 | UPLINK RAN EARLY STATUS TRANSFER DOWNLINK RAN EARLY STATUS TRANSFER |
| RIM Information Transfer | 9.3.3.28 | UPLINK RIM INFORMATION TRANSFER, DOWNLINK RIM INFORMATION TRANSFER |
| NOTE 1: The AMF receives the UE Radio Capability and UE Radio Capability for Paging within a UE CAPABILITY INFO INDICATION message and then the AMF shall store the UE Radio Capability information and UE Radio Capability for Paging, and transfer them to the target AMF during an inter AMF mobility procedure. | | |

For N2 information class NRPPa, N2 Information may encode the following NGAP NRPPA Transport related IE specified in TS 38.413 [12], as summarized in Table 6.1.6.4.3.2-3

**Table 6.1.6.4.3.2-3: N2 Information content for class NRPPa**

| NGAP IE | Reference (TS 38.413 [12]) | Related NGAP message |
|---|---|---|
| NRPPa-PDU | 9.3.3.14 | DOWNLINK UE ASSOCIATED NRPPA TRANSPORT<br>UPLINK UE ASSOCIATED NRPPA TRANSPORT<br>DOWNLINK NON UE ASSOCIATED NRPPA TRANSPORT<br>UPLINK NON UE ASSOCIATED NRPPA TRANSPORT |

For N2 information class V2X, N2 Information may encode the following V2X related IE specified in TS 38.413 [12], as summarized in Table 6.1.6.4.3.2-4

**Table 6.1.6.4.3.2-4: N2 Information content for class V2X**

| NGAP IE | Reference (TS 38.413 [12]) | Related NGAP message |
|---|---|---|
| PC5 QoS Parameters | 9.3.1.150 | INITIAL CONTEXT SETUP REQUEST<br>UE CONTEXT MODIFICATION REQUEST<br>HANDOVER REQUEST<br>PATH SWITCH REQUEST ACKNOWLEDGE |

For N2 information class PROSE, N2 Information may encode the following ProSe related IE specified in TS 38.413 [12], as summarized in Table 6.1.6.4.3.2-5

**Table 6.1.6.4.3.2-5: N2 Information content for class PROSE**

| NGAP IE | Reference (TS 38.413 [12]) | Related NGAP message |
|---|---|---|
| 5G ProSe PC5 QoS Parameters | 9.3.1.234 | INITIAL CONTEXT SETUP REQUEST<br>UE CONTEXT MODIFICATION REQUEST<br>HANDOVER REQUEST<br>PATH SWITCH REQUEST ACKNOWLEDGE |
| 5G ProSe Authorized | 9.3.1.233 | INITIAL CONTEXT SETUP REQUEST<br>UE CONTEXT MODIFICATION REQUEST<br>HANDOVER REQUEST<br>PATH SWITCH REQUEST ACKNOWLEDGE |
| 5G ProSe UE PC5 Aggregate Maximum Bit Rate | 9.3.1.148 | INITIAL CONTEXT SETUP REQUEST<br>UE CONTEXT MODIFICATION REQUEST<br>HANDOVER REQUEST<br>PATH SWITCH REQUEST ACKNOWLEDGE |

**Table 6.1.6.4.3.2-6: Void**

For N2 information class RANGING_SL, N2 Information may encode one of the following related IEs specified in TS 38.413 [12], as summarized in Table 6.1.6.4.3.2-7.

**Table 6.1.6.4.3.2-7: N2 Information content for class RANGING_SL**

| NGAP IE | Reference (TS 38.413 [12]) | Related NGAP message |
|---|---|---|
| Ranging and Sidelink Positioning Service Information | 9.3.1.269 | INITIAL CONTEXT SETUP REQUEST<br>UE CONTEXT MODIFICATION REQUEST<br>HANDOVER REQUEST<br>PATH SWITCH REQUEST ACKNOWLEDGE |

For N2 information class A2X, N2 Information may encode the following A2X related IE specified in TS 38.413 [12], as summarized in Table 6.1.6.4.3.2-8

**Table 6.1.6.4.3.2-8: N2 Information content for class A2X**

| NGAP IE | Reference (TS 38.413 [12]) | Related NGAP message |
|---|---|---|
| PC5 QoS Parameters | 9.3.1.150 | INITIAL CONTEXT SETUP REQUEST<br>UE CONTEXT MODIFICATION REQUEST<br>HANDOVER REQUEST<br>PATH SWITCH REQUEST ACKNOWLEDGE |

#### 6.1.6.4.3.3 NGAP Messages

For N2 information class PWS, N2 Information shall encode one of the NGAP Messages specified in TS 38.413 [12] indicated in Table 6.1.6.4.3.3-1.

**Table 6.1.6.4.3.3-1: N2 PWS Request Information content**

| NGAP message | Reference (TS 38.413 [12]) |
|---|---|
| WRITE-REPLACE WARNING REQUEST | 9.2.8.1 |
| PWS CANCEL REQUEST | 9.2.8.3 |

For N2 information class PWS-BCAL, N2 Information shall encode one of the NGAP Messages specified in TS 38.413 [12] indicated in Table 6.1.6.4.3.3-2.

**Table 6.1.6.4.3.3-2: N2 PWS Response Information content**

| NGAP message | Reference (TS 38.413 [12]) |
|---|---|
| WRITE-REPLACE WARNING RESPONSE | 9.2.8.2 |
| PWS CANCEL RESPONSE | 9.2.8.4 |

N2 Information received by the AMF for PWS may be processed by the AMF before re-encoding and transferring to the Service Consumer.

If a subscription exists for N2InformationClass "PWS-BCAL" and the received N2 Message Type is a WRITE-REPLACE-WARNING-RESPONSE, then the AMF may aggregate the Broadcast Completed Area Lists it has received from the NG-RAN nodes for a message identified by its Serial Number and Message Identifier (see table 6.1.6.4.3.3-1) and transfer the ASN.1 (re-)encoded Message Type, Message Identifier, Serial Number and the (aggregated) Broadcast Completed Area List IE in the N2 Info Container in the N2InfoNotify, and the "bcEmptyAreaList" attribute if the Broadcast Completed Area List IE is not present in the PWS N2 information.

If a subscription exists for N2InformationClass "PWS-BCAL" and the received N2 Message Type is a PWS-CANCEL-RESPONSE, then the AMF may aggregate the Broadcast Cancelled Area Lists IE it has received from the NG-RAN nodes for a message identified by its Serial Number and Message Identifier (see table 6.1.6.4.3.3-1) and transfer the ASN.1 (re-)encoded the Message Type, Message Identifier, Serial Number, the (aggregated) Broadcast Cancelled Area List IE in the N2 Info Container in the N2InfoNotify, and the "bcEmptyAreaList" attribute if the PWS-CANCEL-RESPONSE does not contain a Broadcast Cancelled Area List, in the PWS N2 information.

For the ASN.1 definition for encoding the WRITE-REPLACE-WARNING-RESPONSE and the PWS-CANCEL-RESPONSE, see clause 9.4 of TS 38.413 [12].

If a subscription exists for N2InformationClass "PWS-RF" and the received N2 Message Type is a PWS-RESTART-INDICATION, then the AMF may transfer the ASN.1 encoded string from the PWS-RESTART-INDICATION (see table 6.1.6.4.3.3-3) in the N2 Info Container in the N2InfoNotify.

If a subscription exists for N2InformationClass "PWS-RF" and the received N2 Message Type is a PWS-FAILURE-INDICATION (see table 6.1.6.4.3.3-3), then the AMF may transfer the ASN1 encoded string from the PWS-FAILURE-INDICATION in the N2 Info Container in the N2InfoNotify.

For N2 information class PWS-RF, N2 Information shall encode one of the NGAP Messages specified in TS 38.413 [12] indicated in Table 6.1.6.4.3.3-3.

**Table 6.1.6.4.3.3-3: N2 PWS Indication Information content**

| NGAP message | Reference (TS 38.413 [12]) |
|---|---|
| PWS RESTART INDICATION | 9.2.8.5 |
| PWS FAILURE INDICATION | 9.2.8.6 |

The Message Type shall be present and encoded as the first N2 PWS Indication IE in any NonUeN2InfoNotify for PWS messages to enable the receiver to decode the N2 PWS IEs.

For N2 information class RAN, N2 Information shall encode one of the following NGAP messages specified in TS 38.413 [12], as summarized in Table 6.1.6.4.3.3-4.

**Table 6.1.6.4.3.3-4: N2 Information content for class RAN**

| NGAP message | Reference (TS 38.413 [12]) |
|---|---|
| Any UE specific Uplink NGAP message | |

For N2 information class TSS, N2 Information shall encode the following NGAP message specified in TS 38.413 [12], as summarized in Table 6.1.6.4.3.3-5.

**Table 6.1.6.4.3.3-5: N2 Information content for class TSS**

| NGAP message | Reference (TS 38.413 [12]) |
|---|---|
| TIMING SYNCHRONISATION STATUS REQUEST | 9.2.18.1 |
| TIMING SYNCHRONISATION STATUS RESPONSE | 9.2.18.2 |
| TIMING SYNCHRONISATION STATUS FAILURE | 9.2.18.3 |
| TIMING SYNCHRONISATION STATUS REPORT | 9.2.18.4 |

### 6.1.6.4.4 Mobile Terminated Data

Mobile Terminated Data shall encode the user data to be sent by the AMF to the UE in the Payload Container specified in TS 24.501 [7], using the vnd.3gpp.5gnas content-type, as summarized in Table 6.1.6.4.4-1.

**Table 6.1.6.4.4-1: Mobile Terminated Data**

| Mobile Terminated Data | Reference (TS 24.501 [7]) | Related NAS SM message |
|---|---|---|
| Payload container contents in octets 4 to n | 9.11.3.39 (Figure 9.11.3.39.1) | DL NAS Transport |

### 6.1.6.4.5 GTP-C Message

GTP-C Message shall encode a GTP-C message of a specified type (e.g. Forward Relocation Request) as specified in TS 29.274 [41], using the vnd.3gpp.gtpc content-type. The GTP-C message carried in the HTTP multipart message shall include the UDP/IP headers.

GTP-C Message may encode e.g. the following GTP-C messages:

- Mobility Management message:
  - Forward Relocation Request (see clause 7.3.1 of TS 29.274 [41]) during EPS to 5GS handover with AMF re-allocation procedure (see clause 4.11.1.2.2 of TS 23.502 [3]);

- Relocation Cancel Request (see clause 7.3.16 of TS 29.274 [41]) during EPS to 5GS handover with AMF re-allocation procedure (see clause 4.11.1.2.3 of TS 23.502 [3]), if handover cancel is triggered.

## 6.1.7 Error Handling

### 6.1.7.1 General

HTTP error handling shall be supported as specified in clause 5.2.4 of TS 29.500 [4].

### 6.1.7.2 Protocol Errors

Protocol Error Handling shall be supported as specified in clause 5.2.7.2 of TS 29.500 [4].

### 6.1.7.3 Application Errors

The common application errors defined in the Table 5.2.7.2-1 in TS 29.500 [4] may also be used for the Namf_Communication service. The following application errors listed in Table 6.1.7.3-1 are specific for the Namf_Communication service.

**Table 6.1.7.3-1: Application errors**

| Application Error | HTTP status code | Description |
|---|---|---|
| NF_CONSUMER_REDIRECT_ONE_TXN | 307 Temporary Redirect | The request has been asked to be redirected to a specified target. |
| HANDOVER_FAILURE | 403 Forbidden | Creation of UE context or relocation in the target AMF failed during Handover procedure causing a failure of handover. |
| INTEGRITY_CHECK_FAIL | 403 Forbidden | Integrity check of the complete registration message included in the UE context transfer request failed. |
| EBI_EXHAUSTED | 403 Forbidden | Allocation of EPS Bearer ID failed due to exhaustion of EBI as the maximum number of EBIs has already been allocated to the UE. |
| EBI_REJECTED_LOCAL_POLICY | 403 Forbidden | Allocation of EPS Bearer ID failed due to local policy at the AMF as specified in clause 4.11.1.4.1 of TS 23.502 [3]. |
| EBI_REJECTED_NO_N26 | 403 Forbidden | The allocation of EPS Bearer ID was rejected when the AMF is in a serving PLMN that does not support 5GS-EPS interworking procedures with N26 interface. |
| SUPI_OR_PEI_UNKNOWN | 403 Forbidden | The SUPI or PEI included in the message is unknown. |
| UE_IN_NON_ALLOWED_AREA | 403 Forbidden | UE is currently in a non-allowed area hence the N1/N2 message transfer cannot be completed because the request is not associated with a regulatory prioritized service. |
| UNSPECIFIED | 403 Forbidden | The request is rejected due to unspecified reasons. |
| SM_CONTEXT_RELOCATION_REQUIRED | 403 Forbidden | The request is rejected because the SM Context should be relocated to another SMF, e.g. when AMF detects that an I-SMF or V-SMF insertion, change or removal is needed, as specified in clause 4.23 of TS 23.502 [3]. |
| UE_WITHOUT_N1_LPP_SUPPORT | 403 Forbidden | UE does not support LPP in N1 mode hence the N1 LPP message cannot be sent to the UE. |
| INVALID_SM_CONTEXT | 403 Forbidden | The request is rejected because the SM Context is invalid for the PDU session, i.e. active SM Context for the PDU session (with same PDU Session ID) has been created on another SMF. (NOTE) |
| INVALID_PRU | 403 Forbidden | The request is rejected because the request is to initiate a positioning procedure towards a PRU and the target UE is not a valid PRU. |
| CONTEXT_NOT_FOUND | 404 Not Found | The requested UE Context does not exist on the AMF |
| HIGHER_PRIORITY_REQUEST_ONGOING | 409 Conflict | Paging triggered N1/N2 transfer cannot be initiated since already there is a paging due to a higher priority session ongoing. |
| TEMPORARY_REJECT_REGISTRATION_ONGOING | 409 Conflict | N1/N2 message transfer towards UE / AN cannot be initiated or the EBI assignment fails due to an ongoing registration procedure. |
| TEMPORARY_REJECT_HANDOVER_ONGOING | 409 Conflict | N1/N2 message transfer towards UE / AN cannot be initiated due to an ongoing Xn or N2 handover procedure, or the EBI assignment fails due to an ongoing N2 handover procedure or an ongoing Xn handover procedure. |
| TEMPORARY_REJECT_SR_ONGOING | 409 Conflict | A N1/N2 Message Transfer Request including a PDU Session Resource Setup Request Transfer IE towards UE / AN cannot be initiated due to an ongoing Service Request procedure for the same PDU session. |
| UE_IN_CM_IDLE_STATE | 409 Conflict | N2 message transfer towards 5G-AN cannot be initiated due to the UE being in CM-IDLE state for the Access Network Type associated to the PDU session. |

| | | |
|---|---|---|
| MAX_ACTIVE_SESSIONS_EXCEEDED | 409 Conflict | If the RAT type is NB-IoT, and the UE already has 2 PDU Sessions with active user plane resources. |
| REJECTION_DUE_TO_PAGING_RESTRICTION | 409 Conflict | If Paging Restrictions information restricts the N1N2MessageTransfer request from causing paging as defined in TS 23.501 [2] clause 5.38.5. |
| MAX_LCS_UPP_CONN_REACHED | 409 Conflict | The request from a new LMF to transfer "UPP-CM" N1 message to the UE is rejected due to the maximum number of LCS-UPP connection of the UE has been reached. |
| UE_NOT_REACHABLE | 504 Gateway Timeout | The UE is not reachable for paging. |
| UE_NOT_RESPONDING | 504 Gateway Timeout | The UE is not responding for paging. |
| NOTE: More than one SM Contexts may be present in the network for the same PDU Session ID, e.g. when the UE established a new PDU session with the same PDU Session ID and the AMF failed to release the old SM Context in the old SMF. In such a scenario, if the old SMF tries to send N1 and/or N2 Message to the RAN/UE, the AMF shall respond with this application error if the AMF identified that service operation is invoked by the SMF holding the old SM Context. | | |

## 6.1.8 Feature Negotiation

The feature negotiation mechanism specified in clause 6.6 of TS 29.500 [4] shall be used to negotiate the features applicable between the AMF and the NF Service Consumer, for the Namf_Communication service, if any.

The NF Service Consumer shall indicate the features it supports for the Namf_Communication service, if any, by including the supportedFeatures attribute in content of the HTTP Request Message for following service operations:

- N1N2MessgeTransfer, as specified in clause 5.2.2.3.1;
- N1N2MessageSubscribe, as specified in clause 5.2.2.3.3;
- NonUeN2InfoSubscribe, as specified in clause 5.2.2.4.2;
- UeContextTransfer, as specified in clause 5.2.2.2.1;
- CreateUEContext, as specified in clause 5.2.2.2.3

The AMF shall determine the supported features for the service operations as specified in clause 6.6 of TS 29.500 [4] and shall indicate the supported features by including the supportedFeatures attribute in content of the HTTP response for the service operation.

The syntax of the supportedFeatures attribute is defined in clause 5.2.2 of TS 29.571 [6].

The following features are defined for the Namf_Communication service.

**Table 6.1.8-1: Features of supportedFeatures attribute used by Namf_Communication service**

| Feature Number | Feature | M/O | Description |
|---|---|---|---|
| 1 | DTSSA | O | Deployments Topologies with specific SMF Service Areas.<br><br>An AMF that supports this feature shall support the procedures specified in clause 5.34 of TS 23.501 [2] and in clause 4.23 of TS 23.502 [3]. |
| 2 | ENS | O | This feature bit indicates whether the AMF supports procedures related to Network Slicing (see TS 23.501 [2] clause 5.15.7). This includes supporting the RelocateUEContext service operation (see clause 5.2.2.2.5). |
| 3 | CIOT | O | Cellular IoT<br><br>Support of this feature implies the support of all the CIoT features specified in clause 5.31 of TS 23.501 [2], including in particular corresponding service's extensions to support:<br><br>- NB-IoT and LTE-M RAT types;<br><br>- Control Plane CIoT 5GS Optimisation;<br><br>- Rate control of user data. |
| 4 | MAPDU | O | This feature bit indicates whether the AMF supports Multi-Access PDU session procedures related to Access Traffic Steering, Switching and Splitting (see clauses 4.2.10 and 5.32 of TS 23.501 [2]). |
| 5 | NPN | O | Non-Public Network<br><br>Support of this feature implies support of NPN information and receipt of a Create UE context error response with a binary part during an Inter-AMF N2 Handover. |
| 6 | ELCS | O | This feature indicates supports of enhanced LCS, including the capability for the AMF to send an LCS message through the target access type requested by the LMF. |
| 7 | ES3XX | M | Extended Support of HTTP 307/308 redirection<br><br>An NF Service Consumer (e.g. SMF) that supports this feature shall support handling of HTTP 307/308 redirection for any service operation of the Namf_Communication service. An NF Service Consumer that does not support this feature does only support HTTP redirection as specified for 3GPP Release 15. |
| 8 | EAEA | O | EBI and ARP mapping update in EBIAssignment<br><br>Support of this feature implies support of updating the mapping of EBI and ARP in EBIAssignment for a QoS flow to which an EBI is already allocated. |
| 9 | ProSe | O | This feature indicates the support of UE policy and N2 information provisioning for 5G ProSe. |
| 10 | UAV | O | Uncrewed Aerial Vehicle<br>This feature indicates the support of UAV feature at AMF. When this feature is supported at the AMF, the AMF shall perform the UUAA-MM procedure defined in TS 23.256 [56]. |
| 11 | SPAE | O | SM Policy Association Events<br><br>This feature bit indicates whether the AMF supports the SM Policy Association establishment and termination event notification information handling, i.e. whereby the PCF for UE subscribes to SM Policy Association events to the PCF for SM Policy via the AMF and SMF, as specified in clause 4.3.2.2.1 and clause 4.3.3.2 of TS 23.502 [3]. |
| 12 | eNPN | O | Enhanced support of Non-Public Networks (eNPN)<br><br>This feature indicates the AMF supports UE registration for onboarding in an SNPN, see clause 5.30.2.10.2.6 and clause 5.30.2.10.2.7 in TS 23.501 [2]). |

| | | | |
|---|---|---|---|
| 13 | 3GA-N3GA-HO | O | Handover between 3GPP and non-3GPP accesses<br><br>An AMF and SMF that supports Handover between 3GPP and non-3GPP accesses shall support this feature, i.e. setting the targetAccess IE in N1N2MessageTransfer Request to the old access type when releasing the N2 PDU session resources in the old access (see clauses 5.2.2.3.1.1 and 6.1.6.2.18) |
| 14 | PAR-NS | O | Partially Allowed/Rejected Network Slice<br><br>This feature indicates the AMF supports the partially allowed network slice and the partially reject Network slice. See TS 23.501 [2] clause 5.15.17. |
| 15 | NTSSM | O | Network Timing Synchronization Status Monitoring<br><br>This feature bit indicates that the AMF supports transferring TSS related information in the N2InformationTransferReqData and a subscription, e.g. created by a TSCTSF, to get notification for Non-UE related N2 Information for the Network Timing Synchronization Status information. |
| 16 | RACS | O | Support of RACS feature as specified in clause 5.4.4.1a in TS 23.501 [2]).<br><br>During an Inter AMF mobility procedure, the source AMF may decide to not include ueRadioCapability and ueRadioCapabilityForPaging in the UeContextTransferRspData or UeContextCreateData if the target AMF supports RACS feature, the target AMF can retrieve the same using UE Radio Capability ID Id(s) included in the MM Context. |
| 17 | Ranging_SL | O | This feature indicates the support of UE policy and N2 information provisioning for Ranging/SL positioning as specified in TS 23.586 [59]. |
| 18 | A2X | O | This feature indicates the support of UE policy and N2 information provisioning for A2X communication as specified in TS 23.256 [56] |
| 19 | AoV-En | O | This feature indicates the support of enhanced AreaOfValidity handling.<br><br>When invoking N1N2MessageTransfer to deliver N2 information, the SMF may use the TAI range list to efficiently indicate the area scope to deliver the N2 information, if the AMF supports this feature. |
| 20 | ASUC | O | Analytics Subscriptions in UE Context<br><br>An AMF that supports Analytics Subscriptions in UE context shall support this feature, i.e.,<br>- As source AMF, includes Analytics Subscriptions in UE context to the target AMF if the target AMF also supports this feature; and<br><br>- As target AMF, informs the not reused Analytics Subscriptions (i.e. the Analytics Subscriptions that are not taken over by the target AMF) to the source AMF. and<br><br>- As source AMF, unsubscribes to all the not reused Analytics Subscriptions as informed by the target AMF.<br><br>See clauses 5.2.2.2.1.1, 5.2.2.2.2.1 and 5.2.2.2.3.1. |
| 21 | SliceUsageCtrl | O | Network Slice Usage Control<br><br>An AMF supporting this feature shall support the Network Slice Usage Control as specified in clause 5.15.15 in TS 23.501 [2]). |
| Feature number: The order number of the feature within the supportedFeatures attribute (starting with 1).<br>Feature: A short name that can be used to refer to the bit and to the feature.<br>M/O: Defines if the implementation of the feature is mandatory ("M") or optional ("O").<br>Description: A clear textual description of the feature. | | | |

### 6.1.9 Security

As indicated in TS 33.501 [27], the access to the Namf_Communication API may be authorized by means of the OAuth2 protocol (see IETF RFC 6749 [28]), using the "Client Credentials" authorization grant, where the NRF (see TS 29.510 [29]) plays the role of the authorization server.

If Oauth2 authorization is used, an NF Service Consumer, prior to consuming services offered by the Namf_Communication API, shall obtain a "token" from the authorization server, by invoking the Access Token Request service, as described in TS 29.510 [29], clause 5.4.2.2.

NOTE: When multiple NRFs are deployed in a network, the NRF used as authorization server is the same NRF that the NF Service Consumer used for discovering the Namf_Communication service.

The Namf_Communication API defines the following scopes for OAuth2 authorization as specified in TS 33.501 [27].

**Table 6.1.9-1: OAuth2 scopes defined in Namf_Communications API**

| Scope | Description |
|---|---|
| "namf-comm" | Access to the Namf_Communications API. |
| "namf-comm:ue-contexts:mobility" | Access to service operations applying to UE context resources for inter-AMF mobility, i.e., UEContextTransfer, RegistrationStatusUpdate, CreateUEContext, ReleaseUEContext, RelocateUEContext and CancelRelocateUEContext. |
| "namf-comm:ue-contexts:assign-ebi" | Access to service operations applying to UE context resources for EBI assignment, i.e., EBIAssignment. |
| "namf-comm:n1-n2-messages" | Access to service operations applying to the n1-n2-messages resource, i.e., N1N2MessageSubscribe, N1N2MessageUnSubscribe, N1N2MessageTransfer, N1MessageNotify, and N2InfoNotify. |
| "namf-comm:non-ue-n2-messages" | Access to service operations applying to the non-ue-n2-messages resource, i.e., NonUeN2MessageTransfer, NonUeN2InfoSubscribe, NonUeN2InfoUnSubscribe, and NonUeN2InfoNotify. |

### 6.1.10 HTTP redirection

An HTTP request may be redirected to a different AMF service instance, within the same AMF or a different AMF of an AMF set, e.g. when an AMF service instance is part of an AMF (service) set or when using indirect communications (see TS 29.500 [4]). See the ES3XX feature in clause 6.1.8.

An SCP that reselects a different AMF producer instance will return the NF Instance ID of the new AMF producer instance in the 3gpp-Sbi-Producer-Id header, as specified in clause 6.10.3.4 of TS 29.500 [4].

If an AMF within an AMF set redirects a service request to a different AMF of the set using an 307 Temporary Redirect or 308 Permanent Redirect status code, the identity of the new AMF towards which the service request is redirected shall be indicated in the 3gpp-Sbi-Target-Nf-Id header of the 307 Temporary Redirect or 308 Permanent Redirect response as specified in clause 6.10.9.1 of TS 29.500 [4].

## 6.2 Namf_EventExposure Service API

### 6.2.1 API URI

The Namf_EventExposure shall use the Namf_EventExposure API.

The API URI of the Namf_EventExposure API shall be:

**{apiRoot}/<apiName>/<apiVersion>/**

The request URI used in HTTP requests from the NF service consumer towards the NF service producer shall have the Resource URI structure defined in clause 4.4.1 of TS 29.501 [5], i.e.:

**{apiRoot}/<apiName>/<apiVersion>/<apiSpecificResourceUriPart>**

with the following components:

- The {apiRoot} shall be set as described in TS 29.501 [5].
- The <apiName> shall be "namf-evts".
- The <apiVersion> shall be "v1".
- The <apiSpecificResourceUriPart> shall be set as described in clause 6.2.3.

## 6.2.2 Usage of HTTP

### 6.2.2.1 General

HTTP/2, as defined in IETF RFC 9113 [19], shall be used as specified in clause 5 of TS 29.500 [4].

HTTP/2 shall be transported as specified in clause 5.3 of TS 29.500 [4].

HTTP messages and bodies for the Namf_EventExposure service shall comply with the OpenAPI [23] specification contained in Annex A.

### 6.2.2.2 HTTP standard headers

#### 6.2.2.2.1 General

The usage of HTTP standard headers shall be supported as specified in clause 5.2.2 of TS 29.500 [4].

#### 6.2.2.2.2 Content type

The following content types shall be supported:

- JSON, as defined in IETF RFC 8259 [8], shall be used as content type of the HTTP bodies specified in the present specification as indicated in clause 5.4 of TS 29.500 [4].
- The Problem Details JSON Object (IETF RFC 9457 [36]). The use of the Problem Details JSON object in a HTTP response body shall be signalled by the content type "application/problem+json".
- JSON Patch (IETF RFC 6902 [14]). The use of the JSON Patch format in a HTTP request body shall be signalled by the content type "application/json-patch+json".

### 6.2.2.3 HTTP custom headers

#### 6.2.2.3.1 General

In this release of this specification, no custom headers specific to the Namf_EventExposure service are defined. For 3GPP specific HTTP custom headers used across all service based interfaces, see clause 5.2.3 of TS 29.500 [4].

## 6.2.3 Resources

### 6.2.3.1 Overview

{apiRoot}/namf-evts/<apiVersion>

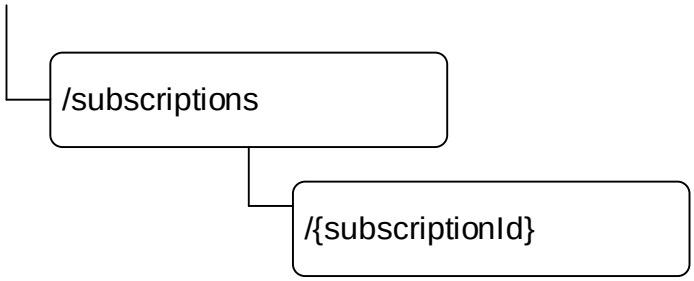



**Figure 6.2.3.1-1: Resource URI structure of the Namf_EventExposure API**

Table 6.2.3.1-1 provides an overview of the resources and applicable HTTP methods.

**Table 6.2.3.1-1: Resources and methods overview**

| Resource name | Resource URI | HTTP method or custom operation | Description |
|---|---|---|---|
| Subscriptions collection | /subscriptions | POST | Mapped to the service operation Subscribe, when to create a subscription |
| Individual subscription | /{subscriptionId} | PATCH | Mapped to the service operation Subscribe, when to modify |
| | | DELETE | Mapped to the service operation Unsubscribe |

### 6.2.3.2 Resource: Subscriptions collection

#### 6.2.3.2.1 Description

This resource represents a collection of subscriptions created by NF service consumers of Namf_EventExposure service.

This resource is modelled as the Collection resource archetype (see clause C.2 of TS 29.501 [5]).

#### 6.2.3.2.2 Resource Definition

Resource URI: **{apiRoot}/namf-evts/<apiVersion>/subscriptions**

This resource shall support the resource URI variables defined in table 6.2.3.2.2-1.

**Table 6.2.3.2.2-1: Resource URI variables for this resource**

| Name | Data type | Definition |
|---|---|---|
| apiRoot | string | See clause 6.2.1 |
| apiVersion | string | See clause 6.2.1. |

### 6.2.3.2.3 Resource Standard Methods

#### 6.2.3.2.3.1 POST

This method shall support the URI query parameters specified in table 6.2.3.2.3.1-1.

**Table 6.2.3.2.3.1-1: URI query parameters supported by the POST method on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| n/a | | | | |

This method shall support the request data structures specified in table 6.2.3.2.3.1-2 and the response data structures and response codes specified in table 6.2.3.2.3.1-3.

**Table 6.2.3.2.3.1-2: Data structures supported by the POST Request Body on this resource**

| Data type | P | Cardinality | Description |
|---|---|---|---|
| AmfCreateEventSubscription | M | 1 | Describes of an AMF Event Subscription to be created |

**Table 6.2.3.2.3.1-3: Data structures supported by the POST Response Body on this resource**

| Data type | P | Cardinality | Response codes | Description |
|---|---|---|---|---|
| AmfCreatedEventSubscription | M | 1 | 201 Created | Represents successful creation of an AMF Event Subscription |
| RedirectResponse | O | 0..1 | 307 Temporary Redirect | Temporary redirection.<br>(NOTE 2) |
| RedirectResponse | O | 0..1 | 308 Permanent Redirect | Permanent redirection.<br>(NOTE 2) |
| ProblemDetails | O | 0..1 | 403 Forbidden | Indicates the creation of subscription has failed due to application error.<br><br>The "cause" attribute may be used to indicate one of the following application errors:<br><br>- UE_NOT_SERVED_BY_AMF (for a subscription targeting a specific UE)<br>- MUTING_EXC_INSTR_NOT_ACCEPTED |
| ProblemDetails | O | 0..1 | 501 Not Implemented | Indicates the creation of subscription has failed due to application error.<br><br>The "cause" attribute may be used to indicate one of the following application errors:<br>- UNSUPPORTED_EVENT_TYPE |
| NOTE 1: The mandatory HTTP error status code for the POST method listed in Table 5.2.7.1-1 of TS 29.500 [4] also apply, with response body containing an object of ProblemDetails data type (see clause 5.2.7 of TS 29.500 [4]).<br>NOTE 2: RedirectResponse may be inserted by an SCP or SEPP, see clause 6.10.9.1 of TS 29.500 [4]. | | | | |

**Table 6.2.3.2.3.1-4: Headers supported by the 201 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | Contains the URI of the newly created resource, according to the structure: {apiRoot}/namf-evts/<apiVersion>/subscriptions/{subscriptionId} |

**Table 6.2.3.2.3.1-5: Headers supported by the 307 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | An alternative URI of the resource located on an alternative service instance within the same AMF or AMF (service) set. For the case when a request is redirected to the same target resource via a different SCP or SEPP, see clause 6.10.9.1 in TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target NF (service) instance ID towards which the request is redirected |

**Table 6.2.3.2.3.1-6: Headers supported by the 308 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | An alternative URI of the resource located on an alternative service instance within the same AMF or AMF (service) set. For the case when a request is redirected to the same target resource via a different SCP or SEPP, see clause 6.10.9.1 in TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target NF (service) instance ID towards which the request is redirected |

#### 6.2.3.2.4 Resource Custom Operations

None.

### 6.2.3.3 Resource: Individual subscription

#### 6.2.3.3.1 Description

This resource represents an individual of subscription created by NF service consumers of Namf_EventExposure service.

This resource is modelled as the Document resource archetype (see clause C.1 of TS 29.501 [5]).

#### 6.2.3.3.2 Resource Definition

Resource URI: **{apiRoot}/namf-evts/<apiVersion>/subscriptions/{subscriptionId}**

This resource shall support the resource URI variables defined in table 6.2.3.3.2-1.

**Table 6.2.3.3.2-1: Resource URI variables for this resource**

| Name | Data type | Definition |
|---|---|---|
| apiRoot | string | See clause 6.2.1 |
| apiVersion | string | See clause 6.2.1. |
| subscriptionId | string | String identifies an individual subscription to the AMF event exposure service |

#### 6.2.3.3.3 Resource Standard Methods

##### 6.2.3.3.3.1 PATCH

This method shall support the URI query parameters specified in table 6.2.3.3.3.1-1.

**Table 6.2.3.3.3.1-1: URI query parameters supported by the PATCH method on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| n/a | | | | |

This method shall support the request data structures specified in table 6.2.3.3.3.1-2 and the response data structures and response codes specified in table 6.2.3.3.3.1-3.

**Table 6.2.3.3.3.1-2: Data structures supported by the PATCH Request Body on this resource**

| Data type | P | Cardinality | Description |
|---|---|---|---|
| array(AmfUpdateEventSubscriptionItem) | M | 1..N | Document describes the modification(s) to a AMF Event Subscription |
| array(AmfUpdateEventOptionItem) | M | 1..1 | Document describing the modification to the event subscription options (e.g subscription expiry time). |

**Table 6.2.3.3.3.1-3: Data structures supported by the PATCH Response Body on this resource**

| Data type | P | Cardinality | Response codes | Description |
|---|---|---|---|---|
| AmfUpdatedEventSubscription | M | 1 | 200 OK | Represents a successful update on AMF Event Subscription |
| RedirectResponse | O | 0..1 | 307 Temporary Redirect | Temporary redirection.<br>(NOTE 2) |
| RedirectResponse | O | 0..1 | 308 Permanent Redirect | Permanent redirection.<br>(NOTE 2) |
| ProblemDetails | O | 0..1 | 403 Forbidden | Indicates the modification of subscription has failed due to application error.<br><br>The "cause" attribute may be used to indicate one of the following application errors:<br>- MUTING_EXC_INSTR_NOT_ACCEPTED |
| ProblemDetails | O | 0..1 | 404 Not Found | Indicates the modification of subscription has failed due to application error.<br><br>The "cause" attribute may be used to indicate one of the following application errors:<br>- SUBSCRIPTION_NOT_FOUND |
| ProblemDetails | O | 0..1 | 501 Not Implemented | Indicates the modification of subscription has failed due to application error.<br><br>The "cause" attribute may be used to indicate one of the following application errors:<br>- UNSUPPORTED_EVENT_TYPE |
| NOTE 1: The mandatory HTTP error status code for the PATCH method listed in Table 5.2.7.1-1 of TS 29.500 [4] also apply, with response body containing an object of ProblemDetails data type (see clause 5.2.7 of TS 29.500 [4]).<br>NOTE 2: RedirectResponse may be inserted by an SCP or SEPP, see clause 6.10.9.1 of TS 29.500 [4]. | | | | |

**Table 6.2.3.3.3.1-4: Headers supported by the 307 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | An alternative URI of the resource located on an alternative service instance within the same AMF or AMF (service) set. For the case when a request is redirected to the same target resource via a different SCP or SEPP, see clause 6.10.9.1 in TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target NF (service) instance ID towards which the request is redirected |

**Table 6.2.3.3.3.1-5: Headers supported by the 308 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | An alternative URI of the resource located on an alternative service instance within the same AMF or AMF (service) set. For the case when a request is redirected to the same target resource via a different SCP or SEPP, see clause 6.10.9.1 in TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target NF (service) instance ID towards which the request is redirected |

### 6.2.3.3.3.2 DELETE

This method shall support the URI query parameters specified in table 6.2.3.3.3.2-1.

**Table 6.2.3.3.3.2-1: URI query parameters supported by the DELETE method on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| n/a | | | | |

This method shall support the request data structures specified in table 6.2.3.3.3.2-2 and the response data structures and response codes specified in table 6.2.3.3.3.2-3.

**Table 6.2.3.3.3.2-2: Data structures supported by the DELETE Request Body on this resource**

| Data type | P | Cardinality | Description |
|---|---|---|---|
| n/a | | | |

**Table 6.2.3.3.3.2-3: Data structures supported by the DELETE Response Body on this resource**

| Data type | P | Cardinality | Response codes | Description |
|---|---|---|---|---|
| n/a | | | 204 No Content | |
| RedirectResponse | O | 0..1 | 307 Temporary Redirect | Temporary redirection.<br>(NOTE 2) |
| RedirectResponse | O | 0..1 | 308 Permanent Redirect | Permanent redirection.<br>(NOTE 2) |
| ProblemDetails | O | 0..1 | 404 Not Found | Indicates the modification of subscription has failed due to application error.<br><br>The "cause" attribute may be used to indicate one of the following application errors:<br>- SUBSCRIPTION_NOT_FOUND. |
| NOTE 1: The mandatory HTTP error status code for the DELETE method listed in Table 5.2.7.1-1 of TS 29.500 [4] also apply, with response body containing an object of ProblemDetails data type (see clause 5.2.7 of TS 29.500 [4]).<br>NOTE 2: RedirectResponse may be inserted by an SCP or SEPP, see clause 6.10.9.1 of TS 29.500 [4]. | | | | |

**Table 6.2.3.3.3.2-4: Headers supported by the 307 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | An alternative URI of the resource located on an alternative service instance within the same AMF or AMF (service) set. For the case when a request is redirected to the same target resource via a different SCP or SEPP, see clause 6.10.9.1 in TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target NF (service) instance ID towards which the request is redirected |

**Table 6.2.3.3.3.2-5: Headers supported by the 308 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | An alternative URI of the resource located on an alternative service instance within the same AMF or AMF (service) set. For the case when a request is redirected to the same target resource via a different SCP or SEPP, see clause 6.10.9.1 in TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target NF (service) instance ID towards which the request is redirected |

#### 6.2.3.3.4 Resource Custom Operations

None.

### 6.2.4 Custom Operations without associated resources

There are no custom operations without associated resources supported on Namf_EventExposure Service.

### 6.2.5 Notifications

#### 6.2.5.1 Void

This clause specifies the notifications provided by the Namf_EventExposure service.

**Table 6.2.5.1-1: Notifications overview**

| Notification | Callback URI | HTTP method or custom operation | Description (service operation) |
|---|---|---|---|
| AMF Event Notification | {eventNotifyUri} | POST | |
| AMF Event Notification | {subsChangeNotifyUri} | POST | |

#### 6.2.5.2 AMF Event Notification

If a NF service consumer has subscribed to an event(s) supported by Namf_EventExposure service, when AMF aware of a state change of the event, AMF shall create a notification including the event state report, and shall deliver the notification to the call-back URI, following Subscribe/Notify mechanism defined in TS 29.501 [5].

##### 6.2.5.2.1 Notification Definition

Call-back URI: **{callbackUri}**

Call-back URI is provided by NF Service Consumer during creation of the subscription. If the notification is to inform the change of subscription ID and if the "subsChangeNotifyUri" was provided in the AmfEventSubscription, then this callback URI shall be the "subsChangeNotifyUri" provided in the AmfEventSubscription. Otherwise, this callback URI shall be the "eventNotifyUri" provided in the AmfEventSubscription.

#### 6.2.5.2.3 Notification Standard Methods

##### 6.2.5.2.3.1 POST

This method shall support the request data structures specified in table 6.2.5.2.3.1-1 and the response data structures and response codes specified in table 6.2.5.2.3.1-2.

**Table 6.2.5.2.3.1-2: Data structures supported by the POST Request Body**

| Data type | P | Cardinality | Description |
|---|---|---|---|
| AmfEventNotification | M | 1 | Represents the notification to be delivered |

**Table 6.2.5.2.3.1-3: Data structures supported by the POST Response Body**

| Data type | P | Cardinality | Response codes | Description |
|---|---|---|---|---|
| n/a | | | 204 No Content | |
| RedirectResponse | O | 0..1 | 307 Temporary Redirect | Temporary redirection.<br>(NOTE 2) |
| RedirectResponse | O | 0..1 | 308 Permanent Redirect | Permanent redirection.<br>(NOTE 2) |
| NOTE 1: The mandatory HTTP error status code for the POST method listed in Table 5.2.7.1-1 of TS 29.500 [4] also apply, with response body containing an object of ProblemDetails data type (see clause 5.2.7 of TS 29.500 [4]).<br>NOTE 2: RedirectResponse may be inserted by an SCP or SEPP, see clause 6.10.9.1 of TS 29.500 [4]. | | | | |

**Table 6.2.5.2.3.1-4: Headers supported by the 307 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | A URI pointing to the endpoint of the NF service consumer to which the notification should be sent.<br>For the case when a request is redirected to the same target resource via a different SCP or SEPP, see clause 6.10.9.1 in TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target NF (service) instance ID towards which the request is redirected.<br>For the case when a request is redirected to the same target resource via a different SCP or SEPP, see clause 6.10.9.1 in TS 29.500 [4]. |

**Table 6.2.5.2.3.1-5: Headers supported by the 308 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | A URI pointing to the endpoint of the NF service consumer to which the notification should be sent |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target NF (service) instance ID towards which the request is redirected |

## 6.2.6 Data Model

### 6.2.6.1 General

This clause specifies the application data model supported by the API.

Table 6.2.6.1-1 specifies the data types defined for the Namf_EventExposure service based interface protocol.

**Table 6.2.6.1-1: Namf_EventExposure specific Data Types**

| Data type | Clause defined | Description |
|---|---|---|
| AmfEventSubscription | 6.2.6.2.2 | Represents an individual event subscription resource on AMF |
| AmfEvent | 6.2.6.2.3 | Describes an event to be subscribed |
| AmfEventNotification | 6.2.6.2.4 | Data within an AMF Event Notification request. |
| AmfEventReport | 6.2.6.2.5 | Represents a report triggered by a subscribed event type |
| AmfEventMode | 6.2.6.2.6 | Describes how the reports shall be generated by a subscribed event |
| AmfEventState | 6.2.6.2.7 | Represents the state of a subscribed event |
| RmInfo | 6.2.6.2.8 | Represents the registration state of a UE for an access type |
| CmInfo | 6.2.6.2.9 | Represents the connection management state of a UE for an access type |
| CommunicationFailure | 6.2.6.2.11 | Describes a communication failure detected by AMF |
| AmfCreateEventSubscription | 6.2.6.2.12 | Data within a create AMF event subscription request |
| AmfCreatedEventSubscription | 6.2.6.2.13 | Data within a create AMF event subscription response |
| AmfUpdateEventSubscriptionItem | 6.2.6.2.14 | Document describing the modification(s) to an AMF Event Subscription |
| AmfUpdatedEventSubscription | 6.2.6.2.15 | Represents a successful update on an AMF Event Subscription |
| AmfEventArea | 6.2.6.2.16 | Represents an area to be monitored by an AMF event. |
| LadnInfo | 6.2.6.2.17 | LADN Information |
| AmfUpdateEventOptionItem | 6.2.6.2.18 | Document describing the modifications to AMF event subscription options. |
| 5GsUserStateInfo | 6.2.6.2.19 | Represents the 5GS User state of the UE for an access type |
| TrafficDescriptor | 6.2.6.2.20 | Represents the Traffic Descriptor |
| UEIdExt | 6.2.6.2.21 | UE Identity |
| AmfEventSubsSyncInfo | 6.2.6.2.22 | AMF Event Subscriptions Information for synchronization |
| AmfEventSubscriptionInfo | 6.2.6.2.23 | Individual AMF Event Subscription Information |
| TargetArea | 6.2.6.2.24 | TA list or TAI range list or any TA |
| SnssaiTaiMapping | 6.2.6.2.25 | List of restricted or unrestricted S-NSSAIs per TAI(s) |
| SupportedSnssai | 6.2.6.2.26 | Supported S-NSSAIs |
| UeInAreaFilter | 6.2.6.2.27 | Describe the filter related to UEs In Area Report event. |
| IdleStatusIndication | 6.2.6.2.28 | Represents the idle status indication. |
| UeAccessBehaviorReportItem | 6.2.6.2.29 | Report Item for UE Access Behavior Trends event. |
| UeLocationTrendsReportItem | 6.2.6.2.30 | Report Item for UE Location Trends event. |
| DispersionArea | 6.2.6.2.31 | Dispersion Area |
| MmTransactionLocationReportItem | 6.2.6.2.32 | UE MM Transaction Report Item per Location |
| MmTransactionSliceReportItem | 6.2.6.2.33 | UE MM Transaction Report Item per Slice |
| SliceAreaRestrictionInfo | 6.2.6.2.34 | Information of a slice subject to area restriction |
| AmfEventType | 6.2.6.3.3 | Describes the supported event types of Namf_EventExposure Service |
| AmfEventTrigger | 6.2.6.3.4 | Describes how AMF should generate the report for the event |
| LocationFilter | 6.2.6.3.5 | Describes the supported filters of LOCATION_REPORT event type |
| UeReachability | 6.2.6.3.7 | Describes the reachability of the UE |
| RmState | 6.2.6.3.9 | Describes the registration management state of a UE |
| CmState | 6.2.6.3.10 | Describes the connection management state of a UE |
| 5GsUserState | 6.2.6.3.11 | Describes the 5GS User State of a UE |
| LossOfConnectivityReason | 6.2.6.3.12 | Describes the reason for loss of connectivity |
| ReachabilityFilter | 6.2.6.3.13 | Event filter for REACHABILITY_REPORT event type. |
| UeType | 6.2.6.3.14 | Describe UE type |
| AccessStateTransitionType | 6.2.6.3.15 | Access State Transition Type |
| SubTerminationReason | 6.2.6.3.16 | Subscription Termination Reason |

Table 6.2.6.1-2 specifies data types re-used by the Namf_EventExposure service based interface protocol from other specifications, including a reference to their respective specifications and when needed, a short description of their use within the Namf_EventExposure service based interface.

**Table 6.2.6.1-2: Namf_EventExposure re-used Data Types**

| Data type | Reference | Comments |
|---|---|---|
| Supi | TS 29.571 [6] | |
| GroupId | TS 29.571 [6] | |
| DurationSec | TS 29.571 [6] | |
| Gpsi | TS 29.571 [6] | |
| Uri | TS 29.571 [6] | |
| Pei | TS 29.571 [6] | |
| UserLocation | TS 29.571 [6] | |
| TaI | TS 29.571 [6] | |
| TimeZone | TS 29.571 [6] | |
| AccessType | TS 29.571 [6] | |
| Ecgi | TS 29.571 [6] | EUTRA Cell Identifier |
| Ncgi | TS 29.571 [6] | NR Cell Identifier |
| NfInstanceId | TS 29.571 [6] | |
| ProblemDetails | TS 29.571 [6] | Problem Details |
| SupportedFeatures | TS 29.571 [6] | Supported Features |
| DateTime | TS 29.571 [6] | |
| NgApCause | TS 29.571 [6] | |
| PresenceInfo | TS 29.571 [6] | Presence Reporting Area Information |
| PresenceState | TS 29.571 [6] | Describes the presence state of the UE to a specified area of interest |
| Dnn | TS 29.571 [6] | |
| Snssai | TS 29.571 [6] | |
| DddTrafficDescriptor | TS 29.571 [6] | Downlink Data Delivery Traffic Descriptor |
| SamplingRatio | TS 29.571 [6] | Sampling Ratio. |
| RedirectResponse | TS 29.571 [6] | Response body of the redirect response message. |
| NotificationFlag | TS 29.571 [6] | Notification flag |
| ExtSnssai | TS 29.571 [6] | |
| N3gaLocation | TS 29.571 [6] | Non-3GPP Location |
| SnssaiDnnItem | TS 29.571 [6] | Combination of S-NSSAIs and DNNs |
| ReferenceId | TS 29.503 [35] | |
| NsiId | TS 29.531 [18] | NSI ID |
| NFType | TS 29.510 [29] | NF type |
| TaiRange | TS 29.510 [29] | |
| MutingExceptionInstructions | TS 29.571 [6] | Muting exception instructions. |
| MutingNotificationsSettings | TS 29.571 [6] | Muting notifications settings. |
| Float | TS 29.571 [6] | Number with format "float" |

## 6.2.6.2 Structured data types

### 6.2.6.2.1 Introduction

Structured data types used in Namf_EventExposure service are specified in this clause.

#### 6.2.6.2.2 Type: AmfEventSubscription

**Table 6.2.6.2.2-1: Definition of type AmfEventSubscription**

| Attribute name | Data type | P | Cardinality | Description | Applicability |
|---|---|---|---|---|---|
| eventList | array(AmfEvent) | M | 1..N | Describes the events to be subscribed in subscription request or the events successfully subscribed for this subscription in subscription response. | |
| eventNotifyUri | Uri | M | 1 | Identifies the recipient of notifications sent by AMF for this subscription (NOTE 1) | |
| notifyCorrelationId | string | M | 1 | Identifies the notification correlation ID. The AMF shall include this ID in the notifications. The value of this IE shall be unique per subscription for a given NF service consumer. | |
| nfId | NfInstanceId | M | 1 | Indicates the instance identity of the network function creating the subscription. | |
| subsChangeNotifyUri | Uri | C | 0..1 | This IE shall be present if the subscription is created by an NF service consumer on behalf of another NF (e.g UDM creating event subscription at AMF for event notifications towards NEF). When present, this IE Identifies the recipient of notifications sent by AMF, for the creation of a new subscription ID, that is considered as a change of subscription ID by the NF service consumer for event subscriptions related to single UE or as the creation of a new subscription Id for event subscriptions related to UE groups (e.g during mobility procedures involving AMF change). (NOTE 3). | |
| subsChangeNotifyCorrelationId | string | C | 0..1 | This IE shall be present when an NF Service Consumer (e.g. UDM) is subscribing for events on behalf of another NF Service Consumer (e.g. NEF). When present, this IE shall contain the notification correlation ID. The AMF shall include it in the notifications for the creation of a new subcription ID that is considered as a change of subscription ID by the NF service consumer for event subscriptions related to single UE or as the creation of a new subscription Id for event subscriptions related to UE groups. The value of this IE shall be unique per subscription for a given NF service consumer that is sending this IE. (NOTE 3). | |
| supi | Supi | C | 0..1 | Subscription Permanent Identifier (NOTE 2) (NOTE 4) | |
| groupId | GroupId | C | 0..1 | Identifies a group of UEs. (NOTE 2) | |
| excludeSupiList | array(Supi) | O | 1..N | This IE may be present for a group subscription.<br><br>When present, this IE shall carry the SUPI of the group member UE(s) that are excluded from the group subscription. | DGEM |
| excludeGpsiList | array(Gpsi) | O | 1..N | This IE may be present for a group subscription.<br><br>When present, this IE shall carry the GPSI of the group member UE(s) that are excluded from the group subscription. | DGEM |
| includeSupiList | array(Supi) | O | 1..N | This IE may be present for a group subscription.<br><br>When present, this IE shall carry the SUPI of the group member UE(s) that are included for the group subscription. | DGEM |

| | | | | | |
|---|---|---|---|---|---|
| includeGpsiList | array(Gpsi) | O | 1..N | This IE may be present for a group subscription.<br><br>When present, this IE shall carry the GPSI of the group member UE(s) that are included for the group subscription. | DGEM |
| gpsi | Gpsi | C | 0..1 | Generic Public Subscription Identifier<br>(NOTE 2) (NOTE 4) | |
| pei | Pei | C | 0..1 | Permanent Equipment Identifier<br>(NOTE 2) (NOTE 4) | |
| anyUE | boolean | C | 0..1 | This IE shall be present if the event subscription is applicable to any UE. Default value "FALSE" is used, if not present. The attribute shall be set to "TRUE", when the AMF event type is "SNSSAI_TA_MAPPING_REPORT".<br><br>(NOTE 2) | |
| options | AmfEventMode | O | 0..1 | This IE may be included if the NF service consumer wants to describe how the reports of the event have to be generated. The absence of this IE, when creating an AMF event subscription or when transferring the UE context to another AMF, shall be interpreted as a "ONE_TIME" AMF event trigger. | |
| sourceNfType | NFType | C | 0..1 | This IE should be present for a subscription that is created by an "intermediate NF" (e.g. UDM) on behalf of a "source NF" (e.g. NEF). When present, it shall contain the NF type of the "source NF". | |
| termNotifyInd | boolean | O | 0..1 | When present this IE shall indicate whether the notification of event subscription termination from the AMF is requested by the NF consumer.<br><br>- true: Event subscription termination notification requested<br>- false (default): Event subscription termination notification not requested | STEN |
| NOTE 1: When an NF Service Consumer subscribes on behalf of another NF, the Notification URI identifies a resource under the authority of the other NF.<br>NOTE 2: Either information about a single UE (i.e. SUPI, GPSI, PEI) or groupId, or anyUE set to "TRUE" shall be included.<br>NOTE 3: Same values of "subsChangeNotifyUri" and "subsChangeNotifyCorrelationId" shall be provided by an NF service consumer to all the serving AMF if the subscriptions apply to a group and triggered by one subscription from another NF. This allows the NF service consumer to associate the subscription Id creation notifications received from different serving AMFs to the same group Id subscription.<br>NOTE 4: At most one of these identifier IEs may be present. | | | | | |

6.2.6.2.3 Type: AmfEvent

**Table 6.2.6.2.3-1: Definition of type AmfEvent**

| Attribute name | Data type | P | Cardinality | Description | Applicability |
|---|---|---|---|---|---|
| type | AmfEventType | M | 1 | Describes the AMF event type to be reported | |
| immediateFlag | boolean | O | 0..1 | Indicates if an immediate event report containing the currently available value / status of the event is requested. The report contains the value / status of the event currently available at the AMF at the time of the subscription (NOTE 1). If the flag is not present then immediate reporting shall not be done and the first report is sent at event detection time.<br><br>When the subscribing NF subscribes on behalf of another NF, the IERSR feature controls whether or not an immediate report is sent within the subscribe response message or within a notification request message (see clause 5.3.2.2.2). Otherwise immediate reports are always sent within the subscribe response message.<br><br>The default value is false. | |
| areaList | array(AmfEventArea) | O | 1..N | Identifies the area to be applied.<br><br>More than one instance of AmfEventArea IE shall be used only when the AmfEventArea is provided during event subscription for Presence Reporting Area subscription. | |
| locationFilterList | array(LocationFilter) | O | 1..N | Describes the filters to be applied for LOCATION_REPORT event type.<br><br>If this attribute is not present in the request, it indicates the change of the TA used by the UE should be reported. | |
| refId | ReferenceId | O | 0..1 | Indicates the Reference Id associated with the event.<br>(NOTE 3) | |
| trafficDescriptorList | array(TrafficDescriptor) | O | 1..N | Indicates the filters to be applied for AVAILABILITY_AFTER_DDN_FAILURE event type. | |

| | | | | | |
|---|---|---|---|---|---|
| reportUeReachable | boolean | C | 0..1 | This IE shall be present and set to value "true" by the source AMF to request the target AMF to notify the subscriber when UE becomes reachable, during inter-AMF mobility procedures.<br><br>When present, this IE shall be set as following:<br>- true: target AMF shall notify the subscriber when UE becomes reachable<br>- false (default): target AMF shall not notify the subscriber when UE becomes reachable, until next reporting trigger is detected, i.e. DDN failure detected (for AVAILABILITY_AFTER_DDN_FAILURE event) or UE becomes unreachable for downlink traffic (for "UE Reachable for DL Traffic" of REACHABILITY_REPORT event)<br><br>This IE only applies to following Event Types:<br>- AVAILABILITY_AFTER_DDN_FAILURE<br>- REACHABILITY_REPORT (for "UE Reachable for DL Traffic") | |
| reachabilityFilter | ReachabilityFilter | O | 0..1 | When present, this IE shall indicate the filter to be applied for the REACHABILITY_REPORT event type.<br><br>If the subscription of REACHABILITY_REPORT is for "UE Reachability Status Change", the AMF shall report the current reachability state (regardless of whether the immediateFlag is set or not) and subsequent updated reachability state of the UE, when AMF becomes aware of a UE reachability state change between REACHABLE, UNREACHABLE and REGULATORY_ONLY.<br><br>If the subscription of REACHABILITY_REPORT is for "UE Reachable for DL Traffic", the AMF shall report the "REACHABLE" state, immediately if UE is reachable at the time of the subscription to the event, otherwise when the UE transitions to CM-CONNECTED mode or when the UE will become reachable for paging, as specified in table 4.15.3.1-1, clauses 4.2.5 and 4.3.3 of TS 23.502 [3].<br><br>If this IE is absent, the subscription of REACHABILITY_REPORT is for "UE Reachability Status Change". | |

| | | | | | |
|---|---|---|---|---|---|
| udmDetectInd | boolean | O | 0..1 | The IE may be present for subscription for "UE Reachable for DL Traffic".<br><br>When present, this IE shall indicate whether the UE Reachability Event will be detected at UDM (i.e. with Nudm_UECM_Registration) or not:<br>- true: UE Reachability will be detected at UDM<br><br>- false (default) UE Reachability will not be detected at UDM | |
| maxReports | integer | O | 0..1 | This IE may be present if the trigger is set to "CONTINUOUS" or "PERIODIC". When present, this IE describes the maximum number of reports that can be generated by the subscribed event.<br><br>If the AMF event subscription is for a group of UEs, this parameter shall be applied to each individual member UE of the group.<br><br>If the event subscription is transferred from source AMF to a target AMF, this IE shall contain:<br>- the remaining number of reports for the event subscription, in the case of individual UE event subscription; or<br>- the remaining number of reports for the event subscription for this specific UE, in the case of a group event subscription. If the group subscription has not expired and all reports have been sent already for this event, the remaining number of reports shall be set to "0".<br>(NOTE 2) | |
| presenceInfoList | map(PresenceInfo) | O | 1..N | Map of PRA Information, the "praId" attribute within the PresenceInfo data type shall also be the key of the map. The "presenceState" attribute within the PresenceInfo data type shall not be supplied.<br><br>When present, the areaList shall be absent. | MPRA |
| maxResponseTime | DurationSec | C | 0..1 | This IE shall be present, when the UDM subscribes to "REACHABILITY_REPORT" event for "UE Reachable for DL Traffic" on behalf of the AF and the AF sets the Maximum Response Time in the Monitoring Configuration.<br><br>When present, this IE shall indicate the Maximum Response Time configured by the AF. | |
| targetArea | TargetArea | C | 0..1 | The IE shall be present for subscription for SNSSAI_TA_MAPPING_REPORT event type.<br><br>When present, this IE shall indicate the TAI list to be applied. | |

| | | | | | |
|---|---|---|---|---|---|
| snssaiFilter | array(ExtSnssai) | O | 1..N | The IE may be present for subscription for SNSSAI_TA_MAPPING_REPORT event type.<br><br>This IE shall be present for subscription of UE_MM_TRANSACTION_REPORT event to receive the UE Mobility Management Transaction numbers based on slices.<br><br>When present, this IE shall indicate the S-NSSAI list to be applied.<br><br>(NOTE 4) | |
| ueInAreaFilter | UeInAreaFilter | O | 0..1 | Indicates the filter to be applied for UES_IN_AREA_REPORT event type related to UAVs.<br>When present, this IE shall indicate the list of items to be applied together as filter. | UARF |
| minInterval | DurationSec | O | 0..1 | This IE may be present when the NF consumer subscribes to "REACHABILITY_REPORT" event for "UE Reachable for DL Traffic".<br><br>When present, this IE indicates the minimal interval to report the event, i.e. when an event is reported, a subsequent event report shall not be sent during the interval. | |
| nextReport | DateTime | O | 0..1 | This IE may be present when the event subscription is transferred from source AMF to a target AMF and minInterval is configured for this event.<br><br>When present, this IE shall indicate the UTC time point before when a subsequent event report shall be throttled. | |
| idleStatusInd | boolean | O | 0..1 | Idle Status Indication request.<br>May be present if type is REACHABILITY_REPORT or AVAILABILITY_AFTER_DDN_FAILURE<br>true: Idle status indication is requested<br>false (default): Idle status indication is not requested | |
| dispersionArea | DispersionArea | C | 0..1 | This IE shall be present for subscription to the UE_MM_TRANSACTION_REPORT event to receive the UE Mobility Management Transaction numbers based on location, or for subscription to the UE_LOCATION_TRENDS event.<br><br>When present, this IE indicates the target area where the related events to be reported for dispersion analytics.<br>(NOTE x) | |
| nextPeriodicReportTime | DateTime | C | 0..1 | This IE should be present when the event subscription is transferred from source AMF to a target AMF and there are periodic report(s) to be generated for the event.<br><br>When present, this IE shall indicate the timestamp (in UTC) when the next periodic report for the event to be generated and notified to the NF consumer. | |

| | | | | | |
|---|---|---|---|---|---|
| adjustAoIOnRa | boolean | O | 0..1 | This IE may be present if the (event) type IE is set to "PRESENCE_IN_AOI_REPORT".<br><br>When present, it shall be set as follows:<br><br>- true: the AMF may adjust the received AoI based on the UE's registration area.<br><br>- false (default) the AoI shall remain unchanged, i.e. it shall not be adjusted based on the UE's registration area.<br><br>See clause 5.3.4.4 of TS 23.501 [2] and clauses 4.15.9.3.2, 4.15.9.4, 5.2.2.3.1 and Annex D.1 of TS 23.502 [3]. | AOIEF |
| ranTimingSynchroStatusChange | boolean | O | 0..1 | This IE may be present if the (event) type IE is set to "PRESENCE_IN_AOI_REPORT".<br><br>When present, it shall be set as follows:<br><br>- true: this is a subscription for RAN timing synchronization status change event.<br><br>- false (default): this is not a subscription for RAN timing synchronization status change event<br><br>See clause 5.3.1 for the handling of this IE by the AMF. | AOIEF |
| notifyForSupiList | array(Supi) | O | 1..N | This IE may be present if the (event) type IE is set to "PRESENCE_IN_AOI_REPORT" and the subscription targets Any UE.<br><br>If this IE is present, the AMF shall report the AoI events only if an event concerns a UE belonging to the provided list of SUPIs, or if the notifyForGroupList IE is also present, if the event concerns a UE belonging to an Internal Group of the provided list of Internal Groups.<br><br>If neither this IE nor the notifyForGroupList IE is present, the AMF shall report the AoI events for any UE.<br><br>See clause 5.3.4.4 of TS 23.501 [2] and clause 5.2.2.3.1 of TS 23.502 [3]. | AOIEF |
| notifyForGroupList | array(GroupId) | O | 1..N | This IE may be present if the (event) type IE is set to "PRESENCE_IN_AOI_REPORT".<br><br>If this IE is present, the AMF shall report the AoI events only if an event concerns a UE belonging to an Internal Group of the provided list of Internal Groups or, if the notifyForSupiList IE is also present, if the event concerns a UE belonging to the provided list of SUPIs.<br><br>If neither this IE nor the notifyForSupiList IE is present, the AMF shall report the AoI events for any UE.<br><br>See clause 5.3.4.4 of TS 23.501 [2] and clause 5.2.2.3.1 of TS 23.502 [3]. | AOIEF |

| | | | | | |
|---|---|---|---|---|---|
| notifyForSnssaiDnnList | array(SnssaiDnnItem) | O | 1..N | This IE may be present if the (event) type IE is set to "PRESENCE_IN_AOI_REPORT".<br><br>If this IE is present, the AMF shall report the AoI events only if an event concerns a UE having a PDU session established for the provided DNN(s)/S-NSSAI(s).<br><br>If the IE is not present, the AMF shall report the AoI events for any UE, i.e. without checking DNNs/S-NSSAIs.<br><br>See clause 5.3.4.4 of TS 23.501 [2] and clause 5.2.2.3.1 of TS 23.502 [3]. | AOIEF |

NOTE 1: The requested value of the location is the last known location (i.e. age of location may be greater than zero) if the immediate Flag is set to true. An NF Service Consumer requesting to receive the current location (i.e. age of location equal to zero) shall not set the immediateFlag to true when subscribing to a location event report.

NOTE 2: When creating an AMF event subscription with multiple events, the same maximum number of reports shall apply to each event. Accordingly, maxReports in this attribute should not be present when creating an AMF event subscription; if it is present, it shall contain the same value for all events and maxReports in the AmfEventMode shall have precedence over the maxReports in this attribute. maxReports in this attribute and maxReports in the AmfEventMode have different semantics when transferring the event subscription from a source AMF to a target AMF.

NOTE 3: Each Monitoring Configuration subscribed via UDM Event Exposure service uses a Reference Id as the key. This IE shall carry the Reference Id when UDM subscribes to the AMF event for the corresponding Monitoring Configuration.

NOTE 4: For a subscription to the UE_MM_TRANSACTION_REPORT event, either the snssaiFilter IE or the dispersionArea shall be present. The AMF shall report the UE MM Transaction numbers based on slices or location according to the presence of the IE.

#### 6.2.6.2.4 Type: AmfEventNotification

**Table 6.2.6.2.4-1: Definition of type AmfEventNotification**

| Attribute name | Data type | P | Cardinality | Description | Applicability |
|---|---|---|---|---|---|
| notifyCorrelationId | string | C | 0..1 | This IE shall be included if the notification is not for informing creation of a new subscription Id.<br><br>This IE shall also be included if the notification is for informing the creation of a new subscription Id and the corresponding event subscription did not contain subsChangeNotifyCorrelationId attribute (see clause 6.2.6.2.2).<br><br>When present, this IE shall indicate the notification correlation Id provided by the NF service consumer during event subscription. This parameter can be useful if the NF service consumer uses a common call-back URI for multiple subscriptions. | |
| subsChangeNotify CorrelationId | string | C | 0..1 | This IE shall be included if the notification is for informing the creation of a new subscription Id at the AMF and the corresponding event subscription contains the subsChangeNotifyCorrelationId attribute (see clause 6.2.6.2.2).<br>When present, this IE shall be set to the value of the subsChangeNotifyCorrelationId provided during subscription (see clause 6.2.6.2.2). | |
| reportList | array(AmfEventReport) | C | 1..N | This IE shall be present if a event is reported.<br>When present, this IE represents the event reports to be delivered. | |
| eventSubsSyncInfo | AmfEventSubsSyncInfo | C | 0..1 | This IE may be present for AMF to initiate event subscription synchronization with UDM during UE mobility procedures.<br><br>When present, this IE shall contain the information for event subscription synchronization, including all active event subscriptions specificially targeting the UE. | ESSYNC |

#### 6.2.6.2.5 Type: AmfEventReport

**Table 6.2.6.2.5-1: Definition of type AmfEventReport**

| Attribute name | Data type | P | Cardinality | Description | Applicability |
|---|---|---|---|---|---|
| type | AmfEventType | M | 1 | Describes the type of the event which triggers the report | |
| state | AmfEventState | M | 1 | Describes the state of the event which triggered the report. This IE shall be set to "TRUE" when subscriptionId IE is present. | |
| timeStamp | DateTime | M | 1 | This IE shall contain the UTC time at which the event is generated. | |
| subscriptionId | Uri | C | 0..1 | This IE shall be included when the event notification is for informing the creation of a subscription Id at the AMF during mobility of a UE across AMFs.<br><br>When present, this IE shall contain the URI of the created subscription resource at the AMF; this shall contain an absolute URI set to the Resource URI specified in clause 6.2.3.3.2.<br><br>The type IE shall be set to:<br>- SUBSCRIPTION_ID_CHANGE, when the AMFcreates a subscription Id for a UE specific event subscription during mobility registration and handover procedures involving an AMF change.<br>- SUBSCRIPTION_ID_ADDITION, when the AMF creates a subscription Id for a group Id specific event subscription during mobility registration and handover procedures involving an AMF change. | |
| anyUe | boolean | C | 0..1 | This IE shall be included and shall be set to "true", if the event subscription is a bulk subscription for number of UEs and the event reported is for one of those UEs. (NOTE 3) | |
| supi | Supi | C | 0..1 | This IE shall be present if available.<br><br>When present, this IE identifies the SUPI of the UE associated with the report (NOTE 1, NOTE 3). | |
| areaList | array(AmfEventArea) | C | 1..N | This IE shall be present when the AMF event type is "PRESENCE_IN_AOI_REPORT". When present, this IE represents the specified Area(s) of Interest the UE is currently IN / OUT / UNKNOWN.<br><br>If the AMF event is subscribed towards a PRA identifier referring to a Set of Core Network predefined Presence Reporting Areas, the AMF shall report both the subscribed PRA Identifier and the additional PRA identifier of the actually individual PRA(s) where the UE is currently IN / OUT, as specified in clause 5.6.11 of TS 23.501 [2].<br>(NOTE 3) | |
| refId | ReferenceId | C | 0..1 | This IE shall be present if a Reference Id has previously been associated with the event triggering the report.<br><br>When present, this IE shall indicate the Reference Id associated with the event which triggers the report. | |
| gpsi | Gpsi | C | 0..1 | This IE shall be present if available.<br><br>When present, this IE identifies the GPSI of the UE associated with the report (NOTE 1, NOTE 3). | |

| | | | | | |
|---|---|---|---|---|---|
| pei | Pei | O | 0..1 | This IE may be included if the event reported is for a particular UE or any UE. This IE identifies the PEI of the UE associated with the report (NOTE 1, NOTE 3). | |
| location | UserLocation | O | 0..1 | Represents the location information of the UE<br><br>This IE shall convey exactly one of the following:<br>- E-UTRA user location<br>- NR user location<br>- Non-3GPP access user location.<br><br>If the additionalLocation IE is present, this IE shall contain either an E-UTRA user location or NR user location. | |
| additionalLocation | UserLocation | O | 0..1 | This IE shall be present if the "location IE" is present and the AMF reports both a 3GPP user location and a non-3GPP access user location.<br><br>When present, this IE shall convey the non-3GPP access user location. | |
| timezone | TimeZone | O | 0..1 | Describes the time zone of the UE | |
| accessTypeList | array(AccessType) | O | 1..N | Describes the access type(s) of the UE.<br><br>When reporting that the UE is reachable for DL traffic, this IE shall indicate the access type(s) through which the UE is reachable.<br><br>This attribute shall be absent if the AMF event type is "SNSSAI_TA_MAPPING_REPORT". | |
| rmInfoList | array(RmInfo) | O | 1..N | Describes the registration management state of the UE | |
| cmInfoList | array(CmInfo) | O | 1..N | Describes the connection management state of the UE | |
| reachability | UeReachability | O | 0..1 | Describes the reachability of the UE | |
| commFailure | CommunicationFailure | O | 0..1 | Describes a communication failure for the UE. | |
| numberOfUes | integer | O | 0..1 | Represents the number of UEs in the specified area | |
| 5gsUserStateList | array(5GsUserStateInfo) | O | 1..N | Represents the 5GS User State of the UE per access type | |
| typeCode | string | C | 0..1 | This IE shall be present when the AMF event type is "TYPE_ALLOCATION_CODE_REPORT". When present, this IE represents the Type Allocation code (TAC), to indicate terminal model and vendor information of the UE. Pattern: '^imeitac-[0-9]{8}$'. | ENA |
| registrationNumber | integer | C | 0..1 | This IE shall be present when the AMF event type is "FREQUENT_MOBILITY_REGISTRATION_REPORT". When present, this IE represents the number of the mobility registration procedures during a period identified by the expiry time included in the event subscription request. | ENA |

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|---|---|---|---|---|---|
| ueIdExtList | array(UEIdExt) | C | 1..N | This IE shall be present if multiple SUPIs and / or GPSIs need to be included and the subscribing NF indicated support of the ENA feature, unless the subscribing NF indicated to omit UE IDs in the event reports by including ueIdOmitInd IE with the value true.<br><br>This attribute provides additional SUPIs and / or GPSIs to the supi attribute or gpsi attribute if present. The ueIdExtList attribute may be present even if both the supi and gpsi attributes are absent, e.g., in a report of "UES_IN_AREA_REPORT" event type. (NOTE 3) | ENA |
| lossOfConnectReason | LossOfConnectivityReason | O | 0..1 | Describes the reason for loss of connectivity. This IE should be present when the AMF event type is "LOSS_OF_CONNECTIVITY". | |
| maxAvailabilityTime | DateTime | O | 0..1 | Indicates the time (in UTC) until which the UE is expected to be reachable.<br><br>This IE may be present in REACHABILITY_REPORT event report for "UE Reachable_for DL Traffic".<br><br>This information may be used by the SMS Service Center to prioritize the retransmission of pending Mobile Terminated Short Message to UEs using a power saving mechanism (eDRX, PSM etc.). | |
| snssaiTaiList | array(SnssaiTaiMapping) | C | 1..N | This IE shall be present when the AMF event type is "SNSSAI_TA_MAPPING_REPORT". When present, this IE represents the list of supported S-NSSAIs at the TAI(s). It shall also include the indication if S-NSSAI is restricted at the AMF. | |
| idleStatusIndication | IdleStatusIndication | O | 0..1 | Idle Status Indication<br>May be present when type is REACHABILITY_REPORT or AVAILABILITY_AFTER_DDN_FAILURE | |
| ueAccessBehaviorTrends | array(UeAccessBehaviorReportItem) | C | 1..N | This IE shall be present to report "UE_ACCESS_BEHAVIOR_TRENDS" event.<br><br>When present, this IE shall include the UE access behavior trends within the report period. | |
| ueLocationTrends | array(UeLocationTrendsReportItem) | C | 1..N | This IE shall be present to report "UE_LOCATION_TRENDS" event.<br><br>When present, this IE shall include the UE location trends within the report period. (NOTE 2) | |
| mmTransLocationReportList | array(MmTransactionLocationReportItem) | C | 1..N | This IE shall be present to report "UE_MM_TRANSACTION_REPORT" event based on location.<br><br>When present, this IE shall include the number of UE MM transactions per location within the report period. | |
| mmTransSliceReportList | array(MmTransactionSliceReportItem) | C | 1..N | This IE shall be present to report "UE_MM_TRANSACTION_REPORT" event based on slices.<br><br>When present, this IE shall include the number of UE MM transactions per slice within the report period. | |

| | | | | | |
|---|---|---|---|---|---|
| termReason | SubTerminationReason | O | 0..1 | This IE may be present when the event type is SUBSCRIPTION_TERMINATION.<br><br>When present, this IE shall indicate the reason for the subscription termination. | |
| unavailabilityPeriod | DurationSec | O | 0..1 | This IE shall be present when the event type is LOSS_OF_CONNECTIVITY and Unavailability Period Duration is reported by the UE or known to the AMF as specified in clause 5.4.13.3 of TS 23.501 [2].<br><br>When present, it contains the Unavailability Period Duration reported by the UE or known to the AMF as specified in clause 5.4.1.4 of TS 23.501 [2]. If the UE is already not available when the event is subscribed, it is set to the remaining value of Unavailability Period Duration. | |

NOTE 1: If the event report corresponds to an event subscription of a single UE, then the same UE identifier (i.e. SUPI and/or GPSI and/or PEI) received during subscription creation shall be included in the report. If the event report corresponds to an event subscription for group of UEs or any UE, then the SUPI and if available the GPSI shall be included in the event report. SUPI, PEI and GPSI shall not be present in report for UES_IN_AREA_REPORT event type.
NOTE 2: The items shall be listed in descending order by the value of "duration" attribute.
NOTE 3: When a subscription for "PRESENCE_IN_AOI_REPORT" event targets any UE but no UE is "IN" the AOI when the AMF generats the first notification (e.g. for one-time reporting or for the first notification for continusouly reporting), the anyUe IE shall be present with the value true and IEs indicating UE IDs (Supi, Gpsi, Pei and ueIdExtList) shall not be present; the areaList IE shall be present including the subscribed AOI with the Presence Status set to "IN", i.e. no UE is "IN" the AOI.

6.2.6.2.6 Type: AmfEventMode

**Table 6.2.6.2.6-1: Definition of type AmfEventMode**

| Attribute name | Data type | P | Cardinality | Description | Applicability |
|---|---|---|---|---|---|
| trigger | AmfEventTrigger | M | 1 | Describes how the reports are triggered. | |
| maxReports | integer | C | 0..1 | This IE shall be present if the trigger is set to "CONTINUOUS" while "expiry" attribute is not present. This IE may be present if the trigger is set to "PERIODIC". When present, this IE describes the maximum number of reports that can be generated by each subscribed event in the subscription.<br><br>If the AMF event subscription is for a list of events, this parameter shall be applied to each individual event in the list.<br><br>If the AMF event subscription is for a group of UEs, this parameter shall be applied to each individual member UE of the group.<br><br>If the event subscription is transferred from source AMF to target AMF, this IE shall contain:<br>- the remaining number of reports for the event subscription, in the case of individual UE event subscription;<br>- the maximum number of reports for each event of the AMF event subscription for each individual member of the group, in the case of a group event subscription.<br>(NOTE 1)<br>(NOTE 2) | |
| expiry | DateTime | C | 0..1 | This IE shall be included in an event subscription response, if, based on operator policy and taking into account the expiry time included in the request, the AMF needs to include an expiry time.<br><br>This IE may be included in an event subscription request.<br><br>When present, this IE shall represent the UTC time after which the subscribed event(s) shall stop generating report and the subscription becomes invalid. If the trigger value included in an event subscription response is "ONE_TIME" and if an event report is included in the subscription response then the value of the expiry included in the response shall be an immediate timestamp.<br>(NOTE 1) | |
| repPeriod | DurationSec | C | 0..1 | This IE shall be present if the trigger is set to "PERIODIC". When present, this IE describes the period time for the event reports. If the AMF event subscription is for a group of UEs, this parameter shall be applied to each individual member UE of the group.<br>(NOTE 3) | |
| sampRatio | SamplingRatio | O | 0..1 | This IE may be included in an event subscription request for a group of UEs or any UE to indicate the ratio of the random subset to target UEs. Event reports only relate to the subset.<br><br>If the AMF event subscription is for a list of AMF event, this parameter shall be applied to each individual event. | |

| | | | | | |
|---|---|---|---|---|---|
| partitioningCriteria | array(PartitioningCriteria) | O | 1..N | This IE may be included in an event subscription request for a group of UEs or any UE when sampRatio is provided.<br><br>When present, this IE shall define the criteria for determining the UEs for which the sampling ratio shall apply. | |
| notifFlag | NotificationFlag | O | 0..1 | Indicates the notification flag, which is used to mute/unmute notifications and to retrieve events stored during a period of muted notifications. | EneNA |
| mutingExcInstructions | MutingExceptionInstructions | O | 0..1 | This IE may be included in the event subscription request if the notifFlag IE is present and set to "DEACTIVATE".<br>When present, it shall indicate the instructions for the subscription and stored events when an exception (e.g. the buffer of stored event reports is full, or the number of stored event reports exceeds a certain number) occurs at AMF while the events are muted.<br>See TS 23.288 [38], clause 6.2.7.2.<br>Write-Only: true | ENAPH3 |
| mutingNotSettings | MutingNotificationsSettings | O | 0..1 | This IE may be included if the event notifications muting is activated.<br>This IE Indicates the AMF muting notification settings.<br>See TS 23.288 [38], clause 6.2.7.2.<br>Read-Only: true | ENAPH3 |
| varRepPeriodInfo | array(VarRepPeriod) | O | 1..N | This IE may be present if the trigger is set to "PERIODIC".<br>This IE Indicates the variable reporting periodicity information.<br>See TS 23.502 [3], clause 4.15.1.<br>(NOTE 3) | ENAPH3 |
| NOTE 1: If the AmfEventTrigger is set to "CONTINOUS", at least one of the "maxReports" and "expiry" attributes shall be included.<br>NOTE 2: See NOTE 2 of Table 6.2.6.2.3-1 regarding the precedence between maxReports in AmfEvent and maxReports in this attribute.<br>NOTE 3: If both repPeriod and varRepPeriodInfo attributes are present, the repPeriod shall be applied if non of the conditions trigger the variable reporting is met. | | | | | |

### 6.2.6.2.7 Type: AmfEventState

**Table 6.2.6.2.7-1: Definition of type AmfEventState**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| active | boolean | M | 1 | Represents the active state of the subscribe event. "TRUE" value indicates the event will continue generating reports; "FALSE" value indicates the event will not generate further report. |
| remainReports | integer | O | 0..1 | Represents the number of remain reports to be generated by the subscribed event. |
| remainDuration | DurationSec | O | 0..1 | Represents how long the subscribed event will continue generating reports. |

#### 6.2.6.2.8 Type: RmInfo

**Table 6.2.6.2.8-1: Definition of type RmInfo**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| rmState | RmState | M | 1 | Describes the registration management state of the UE |
| accessType | AccessType | M | 1 | Describes the access type of the UE that applies to the registration management state reported. |

#### 6.2.6.2.9 Type: CmInfo

**Table 6.2.6.2.9-1: Definition of type CmInfo**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| cmState | CmState | M | 1 | Describes the Connection management state of the UE |
| accessType | AccessType | M | 1 | Describes the access type of the UE that applies to the Connection management state reported. |

#### 6.2.6.2.10 Void

#### 6.2.6.2.11 Type: CommunicationFailure

**Table 6.2.6.2.11-1: Definition of type CommunicationFailure**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| nasReleaseCode | string | O | 0..1 | Describes the NAS release code for the communication failure. This IE shall be formatted following the regular expression pattern:<br>"^(MM\|SM)-[0-9]{1,3}$"<br><br>Examples:<br>MM-7<br>SM-27 |
| ranReleaseCode | NgApCause | O | 0..1 | Describes the RAN release code for the communication failure. If present, this IE shall contain the decimal value of the NG AP cause code values as specified in TS 38.413 [12]. |

#### 6.2.6.2.12 Type: AmfCreateEventSubscription

**Table 6.2.6.2.12-1: Definition of type AmfCreateEventSubscription**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| subscription | AmfEventSubscription | M | 1 | Represents the AMF Event Subscription resource to be created. |
| supportedFeatures | SupportedFeatures | C | 0..1 | This IE shall be present if at least one feature defined in clause 6.2.8 is supported. |
| oldGuami | Guami | C | 0..1 | This IE shall be present during an AMF planned removal procedure when the NF Service Consumer initiates a request towards the target AMF, for a UE associated to an AMF that is unavailable (see clause 5.21.2.2 of TS 23.501 [2]). |

#### 6.2.6.2.13 Type: AmfCreatedEventSubscription

**Table 6.2.6.2.13-1: Definition of type AmfCreatedEventSubscription**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| subscription | AmfEventSubscription | M | 1 | Represents the newly created AMF Event Subscription resource. |
| subscriptionId | Uri | M | 1 | Represents the URI of the newly created AMF Event Subscription resource. This shall contain an absolute URI set to the Resource URI specified in clause 6.2.3.3.2. (NOTE 2) |
| reportList | array(AmfEventReport) | O | 1..N | Represents the immediate event reports (i.e. the current value / status of the events subscribed), if available (NOTE 1). |
| supportedFeatures | SupportedFeatures | C | 0..1 | This IE shall be present if at least one feature defined in clause 6.2.8 is supported. |
| NOTE 1: If the subscription is on behalf of another NF and the NF service consumer has not indicated supporting of IERSR feature (see 6.2.8), then the reports attribute shall be absent.<br>NOTE 2: TS 23.502 [3] specifies this attribute as "Subscription Correlation ID". | | | | |

6.2.6.2.14 Type: AmfUpdateEventSubscriptionItem

**Table 6.2.6.2.14-1: Definition of type AmfUpdateEventSubscriptionItem**

| Attribute name | Data type | P | Cardinality | Description | Applicability |
|---|---|---|---|---|---|
| op | string | M | 1 | This IE indicates the patch operation as defined in IETF RFC 6902 [14] to be performed on attributes of the "Individual subscription" resource (see clause 6.2.3.3).<br>This IE shall support the following values:<br>Enum: "add"<br>Enum: "replace"<br>Enum: "remove" | |

| | | | | | |
|---|---|---|---|---|---|
| path | string | M | 1 | This IE contains a JSON pointer value (as defined in IETF RFC 6901 [40]) that references a location of attributes of the "Individual subscription" resource (see clause 6.2.3.3) on which the patch operation shall be performed.<br><br>This IE shall contain the JSON pointer to a valid index of the "/eventList" array in the AMF Event Subscription, formatted with following pattern:<br>'\/eventList\/[0-]$\|\/eventList\/[1-9][0-9]*$'<br><br>Example:<br>"/eventList/0" stands for the first member of the array;<br>"/eventList/10" stands for the 11th member of the array;<br>"/eventList/-" stands for a new (non-existent) member after the last existing array element. Only allowed with "add" operation.<br><br>To update the PRA Information, this IE shall contain the JSON pointer to a valid key of the "/presenceInfoList" object in the AMF Event Subscription, the key shall be formatted as the "praId" attribute within the PresenceInfo data type. These attributes shall only be updated when both the NF Service Consumer and the AMF support the MPRA feature.<br><br>Pattern: '^(\/eventList\/0\|\/eventList\/[1-9][0-9]*){1}(\/presenceInfoList\/0\|\/presenceInfoList\/[1-9][0-9]*)?$'<br><br>Example:<br>"/eventList/10/presenceInfoList/123" stands for the PresenceInfo with PRA ID 123 for the 11th member of the array. (NOTE 1)<br><br>To remove list of group member UE(s) from a group subscription, this IE shall contain the JSON pointer to the "/excludeSupiList" object or "/excludeGpsiList" object in the AMF Event Subscription. These attributes shall only be updated when both the NF Service Consumer and the AMF support the DGEM feature.<br><br>Pattern:<br>'^\/excludeSupiList\|\/excludeGpsiList$' (NOTE 2)<br><br>To add list of group member UE(s) into a group subscription, this IE shall contain the JSON pointer to the "/includeSupiList" object or "/includeGpsiList" object in the AMF Event Subscription. These attributes shall only be updated when both the NF Service Consumer and the AMF support the DGEM feature. | |

| | | | | | |
|---|---|---|---|---|---|
| | | | | Pattern: '^\/includeSupiList\|\/includeGpsiList$' (NOTE 3)<br><br>To modify or remove the notifyForSupiList IE, the notifyForGroupList IE or the notifyForSnssaiDnnList IE, this IE shall contain the JSON pointer to the "/notifyForSupiList", "/notifyForGroupList" or "/notifyForSnssaiDnnList" attribute in the AMF Event Subscription. When modifying these IEs, the new list of SUPIs, Internal Groups or DNN/S-NSSAIs shall replace any earlier received list of SUPIs, Internal Groups or DNNs/S-NSSAIs respectively. These attributes shall only be updated when both the NF Service Consumer and the AMF support the AOIEF feature.<br><br>Pattern: '^(\/eventList\/0\|\/eventList\/[1-9][0-9]*){1}(\/notifyForSupiList)$'<br><br>Pattern: '^(\/eventList\/0\|\/eventList\/[1-9][0-9]*){1}(\/notifyForGroupList)$'<br><br>Pattern: '^(\/eventList\/0\|\/eventList\/[1-9][0-9]*){1}(\/notifyForSnssaiDnnList)$' (NOTE 4) | |
| value | Any Type | C | 0..1 | This IE contains the new value of the attribute (as indicated in "path") be added or updated of an existing AMF event to be modified.<br>It shall be present if the patch operation is "add" or "replace" | |

NOTE 1: Update of PRA information by extending the schema of the path IE with JSON pointer to a valid key of the "/presenceInfoList" object shall only be used if the AMF supports the MPRA feature.
NOTE 2: Remove group member UE(s) by extending the schema of the path IE with JSON pointer to the "/excludeSupiList" object or "/excludeGpsiList" object shall only be used if the AMF supports the DGEM feature.
NOTE 3: Add group member UE(s) by extending the schema of the path IE with JSON pointer to the "/includeSupiList" object or "/includeGpsiList" object shall only be used if the AMF supports the DGEM feature.
NOTE 4: Modifying the list of SUPIs or DNNs/S-NSSAIs shall only be used if the AMF supports the AOIEF feature.

### 6.2.6.2.15 Type: AmfUpdatedEventSubscription

**Table 6.2.6.2.15-1: Definition of type AmfUpdatedEventSubscription**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| subscription | AmfEventSubscription | M | 1 | Represents the updated AMF Event Subscription resource. |
| reportList | array(AmfEventReport) | O | 1..N | Represents the immediate event reports (i.e. the current value / status of the events subscribed), if available (NOTE). |

NOTE: For newly added AMF event subscription(s) with the immediateFlag attribute set to true, immediate event report(s) of the corresponding AMF event subscription shall be provided if available.

6.2.6.2.16 Type: AmfEventArea

**Table 6.2.6.2.16-1: Definition of type AmfEventArea**

| Attribute name | Data type | P | Cardinality | Description | Applicability |
|---|---|---|---|---|---|
| presenceInfo | PresenceInfo | C | 0..1 | This IE shall be present if the Area of Interest subscribed is not a LADN service area or a slice service area (e.g Presence Reporting Area or a list of TAIs / cell Ids) . (See NOTE1, NOTE 2) | |
| ladnInfo | LadnInfo | C | 0..1 | This IE shall be present if the Area of Interest subscribed is a LADN service area. | |
| sliceAreaRestrictionInfo | SliceAreaRestrictionInfo | C | 0..1 | This IE shall be present if the Area of Interest subscribed is the service area supporting an S-NSSAI for a PDU session associated with a partially allowed NSSAI or with a slice restricted to NS-AoS (see clause 5.15.17 and 5.15.18 of TS 23.501 [2]). | SAR |
| sNssai | Snssai | O | 0..1 | When present, it shall contain the associated S-NSSAI of the area. (NOTE 3) | ENA |
| nsiId | NsiId | O | 0..1 | When present, this IE shall contain the associated NSI ID of the S-NSSAI. (NOTE 3) | ENA |

NOTE 1: When the AmfEventArea is provided during event subscription, then for UE specific presence reporting area subscription, the praId along with what constitutes that UE specific presence reporting area (i.e. set of Tai and/or set of ecgi and/or set of ncgi and/or set of globalRanNodeId) shall be provided.

NOTE 2: If the subscription is for a Set of Core Network Predefined Presence Reporting Areas and both the AMF and the NF service consumer support the "APRA" feature, the PRA Identifier for the Set shall be carried in the "praId" IE and the individual PRA identifier shall be carried in the "additionalPraId" IE; if the subscription is for a Set of Core Network Predefined Presence Reporting Areas and the AMF or NF service consumer does not support the "APRA" feature, the individual PRA identifier shall be carried in the "praId" IE and the "additionalPraId" IE shall not be present.

NOTE 3: If the NWDAF needs to collect input data on the number of UEs registered in a S-NSSAI or a combination of S-NSSAI and NSI ID, the NF service consumer (e.g. NWDAF) shall provide the S-NSSAI or the combination of S-NSSAI and the NSI as filtering information to the AMF (when subscribing to the Presence-In-AOI-Report event to the AMF).

6.2.6.2.17 Type: LadnInfo

**Table 6.2.6.2.17-1: Definition of type LadnInfo**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| ladn | string | M | 1 | Represents the Local Access Data Network DNN. The AMF shall identify the list of tracking areas corresponding to the LADN DNN based on local configuration. |
| presence | PresenceState | C | 0..1 | This IE shall be included when the UE presence in area of interest is reported. When present, this IE contains the status of UE presence within the Area of Interest (IN / OUT / UNKNOWN). |

#### 6.2.6.2.18 Type: AmfUpdateEventOptionItem

**Table 6.2.6.2.18-1: Definition of type AmfUpdateEventOptionItem**

| Attribute name | Data type | P | Cardinality | Description | Applicability |
|---|---|---|---|---|---|
| op | string | M | 1 | This IE indicates the patch operation as defined in IETF RFC 6902 [14] to be performed on attributes of the "Individual subscription" resource (see clause 6.2.3.3).<br>This IE shall support the following values:<br>Enum: "replace" | |
| path | string | M | 1 | This IE contains a JSON pointer value (as defined in IETF RFC 6901 [40]) that references a location of attributes of the "Individual subscription" resource (see clause 6.2.3.3) on which the patch operation shall be performed.<br><br>This IE shall contain the JSON pointer to "/options/expiry" attribute of the event subscription resource.<br><br>Pattern: "^\/options\/expiry$"<br><br>To update the notifFlag attribute, this IE shall contain the JSON pointer to "/options/notifFlag" attribute of the event subscription resource. This attribute shall only be updated when both the NF Service Consumer and the AMF support the EneNA feature.<br><br>Pattern: "^\/options\/notifFlag$"<br><br>To update the mutingExcInstructions attribute, this IE shall contain the JSON pointer to "/options/mutingExcInstructions" attribute of the event subscription resource. This attribute shall only be updated when both the NF Service Consumer and the AMF support the ENAPH3 feature.<br><br>Pattern: "^\/options\/mutingExcInstructions$" | |
| value | Any Type | M | 1 | This IE contains the new value of the attribute (as indicated in "path") to be updated. | |

#### 6.2.6.2.19 Type: 5GsUserStateInfo

**Table 6.2.6.2.19-1: Definition of type 5GsUserStateInfo**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| 5gsUserState | 5GsUserState | M | 1 | Describes the 5GS user state of the UE |
| accessType | AccessType | M | 1 | Describes the access type of the UE that applies to the 5GS user state reported. |

#### 6.2.6.2.20 Type: TrafficDescriptor

**Table 6.2.6.2.20-1: Definition of type TrafficDescriptor**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| dnn | Dnn | C | 0..1 | This IE shall be present if it is available. When present, it shall indicate the Data Network Name. |
| sNssai | Snssai | C | 0..1 | This IE shall be present if it is available. When present, it shall indicate the associated S-NSSAI for the PDU Session. |
| dddTrafficDescriptorList | array(DddTrafficDescriptor) | C | 1..N | This IE shall be present if it is available. When present, it shall indicate the Traffic Descriptor related to the traffic. |

#### 6.2.6.2.21 Type: UEIdExt

**Table 6.2.6.2.21-1: Definition of type UEIdExt**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| supi | Supi | C | 0..1 | This IE shall be present if available.<br><br>When present, this IE identifies the SUPI of the UE associated with the report. |
| gpsi | Gpsi | C | 0..1 | This IE shall be present if available.<br><br>When present, this IE identifies the GPSI of the UE associated with the report. |

#### 6.2.6.2.22 Type: AmfEventSubsSyncInfo

**Table 6.2.6.2.22-1: Definition of type AmfEventSubsSyncInfo**

| Attribute name | Data type | P | Cardinality | Description | Applicability |
|---|---|---|---|---|---|
| subscriptionList | array(AmfEventSubscriptionInfo) | M | 1..N | This IE shall contain all active subscriptions in the AMF for the target UE.<br><br>Transferred subscriptions that are not authorized by the target AMF shall not be regarded active. Based on local policy, the target AMF may consider that transferred subscriptions containing no or an invalid access token are not authorized. | |

#### 6.2.6.2.23 Type: AmfEventSubscriptionInfo

**Table 6.2.6.2.23-1: Definition of type AmfEventSubscriptionInfo**

| Attribute name | Data type | P | Cardinality | Description | Applicability |
|---|---|---|---|---|---|
| subId | Uri | M | 1 | This IE shall contain the URI of the subscription resource of events with Reference Id. | |
| notifyCorrelationId | string | M | 1 | This IE shall contain the notification correlation ID of the subscription. | |
| refIdList | array(ReferenceId) | M | 1..N | This IE shall contain the Reference Ids of the events in the subscription, one Reference Id per event. | |
| oldSubId | Uri | C | 0..1 | This IE shall be present if new event subscription Id is created in the new AMF, i.e. the event subscription has been retrieved from an old AMF in UE context during EPS to 5GS mobility.<br><br>When present, this IE shall include the URI of the subscription resouce on the source AMF. | |

#### 6.2.6.2.24 Type: TargetArea

**Table 6.2.6.2.24-1: Definition of type TargetArea**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| taList | array(Tai) | C | 1..N | When present, this IE shall contain the list of TAIs. (NOTE) |
| taiRangeList | array(TaiRange) | C | 1..N | When present, this IE shall contain range(s) of TAIs. (NOTE) |
| anyTa | boolean | C | 0..1 | This IE shall be present if the event subscription is applicable to any TA. Default value "FALSE" is used, if not present (NOTE) |
| NOTE: Either information about taList or taiRangeList or anyTa set to "TRUE" shall be included. | | | | |

#### 6.2.6.2.25 Type: SnssaiTaiMapping

**Table 6.2.6.2.25-1: Definition of type SnssaiTaiMapping**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| reportingArea | TargetArea | M | 1 | This IE shall contain the list of TAIs/TAI ranges or anyTa. The taList and taiRangeList shall be absent, and the anyTa shall be set to "TRUE", if the mapping is related to all of the TAs in the AMF. |
| accessTypeList | array(AccessType) | O | 1..N | Describes the access type(s) of the reportingArea. |
| supportedSnssaiList | array(SupportedSnssai) | C | 1..N | This IE shall be present if available.<br><br>When present, this IE represents the list of S-NSSAIs (including indication of S-NSSAIs restricted by AMF) at the reportingArea. |

#### 6.2.6.2.26 Type: SupportedSnssai

**Table 6.2.6.2.26-1: Definition of type SupportedSnssai**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| sNssai | ExtSnssai | M | 1 | This IE shall contain the supported S-NSSAI. |
| restrictionInd | boolean | O | 0..1 | If present, this IE shall contain the indication if the S-NSSAI available in sNssai IE is restricted at the AMF.<br><br>When present, this IE shall be set as follows:<br>- true: the S-NSSAI available in sNssai IE is restricted at the AMF;<br><br>- false (default): the S-NSSAI available in sNssai IE is not restricted at the AMF. |

#### 6.2.6.2.27 Type: UeInAreaFilter

**Table 6.2.6.2.27-1: Definition of type UeInAreaFilter**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| ueType | UeType | C | 0..1 | When present, this IE shall contain the list of UE types.<br><br>When this IE is received, The AMF shall report the number of UEs with the indicated UE type in the area. |
| aerialSrvDnnInd | boolean | C | 0..1 | When present, this IE shall contain an indication of DNN(s) subject to aerial service.<br>Default value "FALSE" is used, if not present.<br>This IE may be set to "TRUE" if the NF service consumer wants to retrieve the number of UEs in the area with established PDU sessions for DNN(s) subject to aerial service. |
| ueIdOmitInd | boolean | O | 0..1 | When present, this IE shall indicate whether UE ID(s) should be omitted in the event reports:<br>- true: UE ID(s) to be omitted in report<br>- false (default): UE IDs are not to be omitted in report. |
| NOTE: When the value of IE ueType is AERIAL_UE, the IE ueType and IE aerialSrvDnnInd shall be considered with "AND" for filtering. | | | | |

#### 6.2.6.2.28 Type: IdleStatusIndication

**Table 6.2.6.2.28-1: Definition of type IdleStatusIndication**

| Attribute name | Data type | P | Cardinality | Description | Applicability |
|---|---|---|---|---|---|
| timeStamp | DateTime | O | 0..1 | Point in UTC time when the UE returned to Idle | |
| activeTime | DurationSec | O | 0..1 | Active Time granted to the UE. | |
| subsregTimer | DurationSec | O | 0..1 | Indicates the currently assigned periodic registration time which may be the Subscribed periodic registration time or a value overridden/assigned by local policy. | |
| edrxCycleLength | integer | O | 0..1 | Contains the eDRX cycle length in milliseconds. | |
| suggestedNumOfDlPackets | integer | O | 0..1 | Suggested number of downlink packets to be buffered | |

#### 6.2.6.2.29 Type: UeAccessBehaviorReportItem

**Table 6.2.6.2.29-1: Definition of type UeAccessBehaviorReportItem**

| Attribute name | Data type | P | Cardinality | Description | Applicability |
|---|---|---|---|---|---|
| stateTransitionType | AccessStateTransitionType | M | 1 | Indicate the type the state transition behavior. | |
| spacing | DurationSec | M | 1 | Indicates the average and variance of the time interval separating two consecutive occurrences of the state transition as indicated by the stateTransitionType IE. | |
| spacingVar | Float | C | 0..1 | Indicates the variance of the time interval separating two consecutive occurrences of the state transition as indicated by the stateTransitionType IE.<br><br>An implementation complying with this version of the specification shall include this IE. | |
| duration | DurationSec | M | 1 | Indicate the average and variance of duration in the resulting state as indicated by stateTransitionType IE. | |
| durationVar | Float | C | 0..1 | Indicates the variance of duration in the resulting state as indicated by stateTransitionType IE.<br><br>An implementation complying with this version of the specification shall include this IE. | |
| occurrences | integer | C | 0..1 | Represents the number of occurrences of the state transition of type stateTransitionType in the current reporting interval. This information is required for calculating spacing, spacingVar, duration and durationVar after handover to a target AMF.<br><br>An implementation complying with this version of the specification shall include this IE if the event subscription is transferred from source AMF to a target AMF and at least one occurrence of the event is already available in the source AMF. | |
| timestamp | DateTime | C | 0..1 | Indicates the date and time of the last occurrences of the indicated state transition. This information is required for calculating spacing, spacingVar, duration and durationVar after handover to a target AMF.<br><br>An implementation complying with this version of the specification shall include this IE. | |

#### 6.2.6.2.30 Type: UeLocationTrendsReportItem

**Table 6.2.6.2.30-1: Definition of type UeLocationTrendsReportItem**

| Attribute name | Data type | P | Cardinality | Description | Applicability |
|---|---|---|---|---|---|
| tai | Tai | C | 0..1 | Indicates the TAI where the UE arrived.<br>(NOTE) | |
| ncgi | Ncgi | C | 0..1 | Indicates the NR cell where the UE arrived.<br>(NOTE) | |
| ecgi | Ecgi | C | 0..1 | Indicates the EUTRAN cell where the UE arrived.<br>(NOTE) | |
| n3gaLocation | N3gaLocation | C | 0..1 | Indicates the Non-3GPP location where the UE arrived.<br>(NOTE) | |
| spacing | DurationSec | M | 1 | Indicates the average and variance of the time interval separating two consecutive arrivals at the indicated location. | |
| spacingVar | Float | C | 0..1 | Indicates the variance of the time interval separating two consecutive arrivals at the indicated location.<br><br>An implementation complying with this version of the specification shall include this IE. | |
| duration | DurationSec | M | 1 | Indicate the average and variance of duration of stay in the indicated location. | |
| durationVar | Float | C | 0..1 | Indicates the variances of duration of stay in the indicated location.<br><br>An implementation complying with this version of the specification shall include this IE. | |
| occurrences | integer | C | 0..1 | Represents the number of arrivals at the indicated location in the current reporting interval. This information is required for calculating spacing, spacingVar, duration and durationVar after handover to a target AMF.<br><br>An implementation complying with this version of the specification shall include this IE if the event subscription is transferred from source AMF to a target AMF and at least one occurrence of the event is already available in the source AMF. | |
| timestamp | DateTime | M | 1 | Indicates the date and time (in UTC) of UE last arrival to the indicated location. | |
| NOTE: At least one of the "tai", "ncgi", "ecgi" and "n3gaLocation" shall be present. | | | | | |

#### 6.2.6.2.31 Type: DispersionArea

**Table 6.2.6.2.31-1: Definition of type DispersionArea**

| Attribute name | Data type | P | Cardinality | Description | Applicability |
|---|---|---|---|---|---|
| taiList | array(Tai) | C | 1..N | Indicates the TAIs where the UE information to be counted for Dispersion Analytics. (NOTE) | |
| ncgiList | array(Ncgi) | C | 1..N | Indicates the NR cells where the UE information to be counted for Dispersion Analytics. (NOTE) | |
| ecgiList | array(Ecgi) | C | 1..N | Indicates the EUTRAN cells where the UE information to be counted for Dispersion Analytics. (NOTE) | |
| n3gaInd | boolean | C | 0..1 | Indicates whether that the UE information shall be counted for Non-3GPP access or not.<br><br>When present, it should be set as following:<br>- true: The UE information shall be counted for Non-3GPP access.<br>- false (default): the UE information for Non-3GPP access shall not be counted. | |
| NOTE: One and only one of the "taiList", "ncgiList", "ecgiList" or "n3gaInd" shall be present. | | | | | |

#### 6.2.6.2.32 Type: MmTransactionLocationReportItem

**Table 6.2.6.2.32-1: Definition of type MmTransactionLocationReportItem**

| Attribute name | Data type | P | Cardinality | Description | Applicability |
|---|---|---|---|---|---|
| tai | Tai | C | 0..1 | Indicates the TAI where the UE MM transactions are counted. (NOTE) | |
| ncgi | Ncgi | C | 0..1 | Indicates the NR cell where the UE MM transations are counted. (NOTE) | |
| ecgi | Ecgi | C | 0..1 | Indicates the EUTRAN cell where the UE MM transations are counted. (NOTE) | |
| n3gaLocation | N3gaLocation | C | 0..1 | Indicates the Non-3GPP location where UE MM transations are counted. (NOTE) | |
| timestamp | DateTime | M | 1 | Indicates the timestamp (in UTC) when the UE enters the location. | |
| transactions | integer | M | 1 | Totally number of UE MM Transactions counted within the location. | |
| NOTE: At least one of the "tai", "ncgi" or "ecgi" shall be present. | | | | | |

#### 6.2.6.2.33 Type: MmTransactionSliceReportItem

**Table 6.2.6.2.33-1: Definition of type MmTransactionSliceReportItem**

| Attribute name | Data type | P | Cardinality | Description | Applicability |
|---|---|---|---|---|---|
| snssai | Snssai | M | 1 | Indicates the S-NSSAI of the slice where the UE MM Transactions are counted. | |
| timestamp | DateTime | M | 1 | Indicates the timestamp (in UTC) when the slice is assigned to the UE. (NOTE) | |
| transactions | integer | M | 1 | Totally number of UE MM Transactions counted for the indicated slice. | |
| NOTE: The timestamps for assigned slices of a UE are not passed between AMFs, i.e. when a UE moves to a new AMF the timestamps for assigned slices of the UE are set to the date and time when the mobility happened in the new AMF. | | | | | |

#### 6.2.6.2.34 Type: SliceAreaRestrictionInfo

**Table 6.2.6.2.34-1: Definition of type SliceAreaRestrictionInfo**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| sNssai | Snssai | M | 1 | Represents an S-NSSAI that is part of the partially allowed NSSAI or if the support of the S-NSSAI is restricted to a NS-AoS. This IE shall be present if the NF service consumer wants to subscribe to the AOI where the S-NSSAI is supported. See clauses 5.15.17 and 5.15.18 of TS 23.501 [2]. |
| presence | PresenceState | C | 0..1 | This IE shall be included when the UE presence in area of interest is reported. When present, this IE indicates whether the UE is inside or outside of the service area supporting an S-NSSAI for a PDU session associated with a partially allowed NSSAI or with a slice restricted to NS-AoS or whether the UE presence in the reporting area is unknown. |

### 6.2.6.3 Simple data types and enumerations

#### 6.2.6.3.1 Introduction

This clause defines simple data types and enumerations that can be referenced from data structures defined in the previous clauses.

#### 6.2.6.3.2 Simple data types

The simple data types defined in table 6.2.6.3.2-1 shall be supported.

**Table 6.2.6.3.2-1: Simple data types**

| Type Name | Type Definition | Description |
|---|---|---|
| | | |

#### 6.2.6.3.3 Enumeration: AmfEventType

**Table 6.2.6.3.3-1: Enumeration AmfEventType**

| Enumeration value | Description |
|---|---|
| "LOCATION_REPORT" | A NF subscribes to this event to receive the Last Known Location or the current Location of a UE or a group of UEs, and Updated Location of the UE or any UE in the group when AMF becomes aware of a location change of the UE. |
| "PRESENCE_IN_AOI_REPORT" | A NF subscribes to this event to receive the current present state of a UE in a specific Area of Interest (AOI), and notification when a specified UE enters or leaves the specified area. The area could be identified by a TA list, an area ID, specific interested area name like "LADN" or a specific interested area name like "S-NSSAI". |
| "TIMEZONE_REPORT" | A NF subscribes to this event to receive the current time zone of a UE or a group of UEs, and updated time zone of the UE or any UE in the group when AMF becomes aware of a time zone change of the UE. |
| "ACCESS_TYPE_REPORT" | A NF subscribes to this event to receive the current access type(s) of a UE or a group of UEs, and updated access type(s) of the UE or any UE in the group when AMF becomes aware of the access type change of the UE. |
| "REGISTRATION_STATE_REPORT" | A NF subscribes to this event to receive the current registration state of a UE or a group of UEs, and report for updated registration state of a UE or any UE in the group when AMF becomes aware of a registration state change of the UE. |
| "CONNECTIVITY_STATE_REPORT" | A NF subscribes to this event to receive the current connection management state of a UE or a group of UEs, and report for updated connection management state of a UE or any UE in the group when AMF becomes aware of a connection management state change of the UE. |
| "REACHABILITY_REPORT" | A NF subscribes to this event to receive the current reachability of a UE or a group of UEs, and report for updated reachability of a UE or any UE in the group when AMF becomes aware of a reachability change of the UE. |
| "COMMUNICATION_FAILURE_REPORT" | A NF subscribes to this event to receive the Communication failure report of a UE or group of UEs or any UE. |
| "UES_IN_AREA_REPORT" | A NF subscribes to this event to receive the number of UEs in a specific area. |
| "SUBSCRIPTION_ID_CHANGE" | This event type is used by the AMF to inform the NF service consumer that the subscription Id for the event subscription is changed (e.g. Subscription Id creation at the target AMF for individual UE level event subscriptions, during mobility registration or handover procedures involving an AMF change). This event needs no explicit subscription form an NF service consumer. |
| "SUBSCRIPTION_ID_ADDITION" | This event type is used by the AMF to inform the NF service consumer that a new subscription Id is added (e.g creation of an event subscription for a UE group level event subscription at the target AMF, during mobility registration or handover procedures involving AMF change for a UE belonging to a group Id and when such a UE is the first UE of the group registering at the target AMF). This event needs no explicit subscription form the NF service consumer. |

| | |
|---|---|
| "SUBSCRIPTION_TERMINATION" | This event type is used by the AMF to inform the NF service consumer that the subscription is terminated at the AMF, e.g. the AMF inform the UDM that the subscription is terminated because the AMF has identified the subscription is no longer valid at the NEF. |
| "LOSS_OF_CONNECTIVITY" | An NF subscribes to this event to receive the event report of a UE or group of UEs when AMF detects that a target UE is no longer reachable for either signalling or user plane communication. Such condition is identified when Mobile Reachable timer expires in the AMF (see TS 23.501 [2]), when the UE detaches and when AMF deregisters from UDM for an active UE. If the UE is already not reachable for either signalling or user plane communication when the event is subscribed, the AMF reports the event directly. |
| "5GS_USER_STATE_REPORT" | A NF subscribes to this event to receive the 5GS user state of a UE. |
| "AVAILABILITY_AFTER_DDN_FAILURE" | A NF subscribes to this event to be notified about the Availability of a UE after a DDN failure. |
| "TYPE_ALLOCATION_CODE_REPORT" | A NF subscribes to this event to receive the TAC of a UE or group of UEs. |
| "FREQUENT_MOBILITY_REGISTRATION_REPORT" | A NF subscribes to this event to receive the number of mobility registration procedures during a period of a UE or group of UEs. |
| "SNSSAI_TA_MAPPING_REPORT" | A NF subscribes to this event to receive the related access type and the list of supported S-NSSAIs (including indication of S-NSSAIs restricted by AMF) at the TAI(s). |
| "UE_ACCESS_BEHAVIOR_TRENDS" | A NF subscribes to this event to receive the UE access behavior trends (e.g. access type change, handover, etc.) during a period for a UE or a group of UEs, as specified in clause 4.15.4.2 of TS 23.502 [3]. |
| "UE_LOCATION_TRENDS" | A NF subscribes to this event to receive the UE Location Trends within a period for a UE or a group of UEs, as specified in clause 4.15.4.2 of TS 23.502 [3]. |
| "UE_MM_TRANSACTION_REPORT" | A NF subscribes to this event to receive the Total Number of Mobility Management transactions during a period for a UE or a group of UEs, as specified in clause 5.2.2.3.1 of TS 23.502 [3]. |

#### 6.2.6.3.4 Enumeration: AmfEventTrigger

**Table 6.2.6.3.4-1: Enumeration AmfEventTrigger**

| Enumeration value | Description |
|---|---|
| "ONE_TIME" | Defines that AMF should generate report for the event only once. After reporting, the subscription to this event will be terminated. |
| "CONTINUOUS" | Defines that AMF should continuously generate reports for the event, until the subscription to this event ends, due to end of report duration or up to the maximum number of reports or the event being unsubscribed explicitly |
| "PERIODIC" | Defines that AMF should periodically generate reports for the event, until the subscription to this event ends, due to end of report duration or up to the maximum number of reports or the event being unsubscribed explicitly. |

#### 6.2.6.3.5 Enumeration: LocationFilter

**Table 6.2.6.3.5-1: Enumeration LocationFilter**

| Enumeration value | Description |
|---|---|
| "TAI" | Indicates any change of the TA used by the UE should be reported |
| "CELL_ID" | Indicates any change of the Cell used by the UE should be reported |
| "RAN_NODE" | Indicates any change of the RAN node serving the UE shall be reported. |
| "N3IWF" | Indicates any change of the N3IWF node used by the UE should be reported |
| "UE_IP" | Indicates any change of the UE local IP address should be reported |
| "UDP_PORT" | Indicates any change of local UDP port used by the UE reported |
| "TNAP_ID" | Indicates any change of the TNAP ID used by the UE should be reported |
| "GLI" | Indicates any change of the Global Line Id used by the UE should be reported |
| "TWAP_ID" | Indicates any change of the TWAP ID used by the UE should be reported |

#### 6.2.6.3.6 Void

#### 6.2.6.3.7 Enumeration: UeReachability

**Table 6.2.6.3.7-1: Enumeration UeReachability**

| Enumeration value | Description |
|---|---|
| "UNREACHABLE" | Indicates the UE is not reachable, e.g. when the Mobile Reachable Timer in AMF expires. |
| "REACHABLE" | Indicates the UE is reachable for services and downlink traffic. |
| "REGULATORY_ONLY" | Indicates the UE is reachable only for Regulatory Prioritized Service as the UE is in Not Allowed Areas. |

#### 6.2.6.3.8 Void

#### 6.2.6.3.9 Enumeration: RmState

**Table 6.2.6.3.9-1: Enumeration RmState**

| Enumeration value | Description |
|---|---|
| "REGISTERED" | Indicates the UE in RM-REGISTERED state |
| "DEREGISTERED" | Indicates the UE in RM-DEREGISTERED state |

#### 6.2.6.3.10 Enumeration: CmState

**Table 6.2.6.3.10-1: Enumeration CmState**

| Enumeration value | Description |
|---|---|
| "IDLE" | Indicates the UE is in CM-IDLE state |
| "CONNECTED" | Indicates the UE is in CM-CONNECTED state |

#### 6.2.6.3.11 Enumeration: 5GsUserState

**Table 6.2.6.3.11-1: Enumeration 5GsUserState**

| Enumeration value | Description |
|---|---|
| "DEREGISTERED" | Indicates the UE in RM-DEREGISTERED state |
| "CONNECTED_NOT_REACHABLE_FOR_PAGING" | Indicates the UE is in the RM-REGISTERED state in 5GS and the UE is not reachable for paging. |
| "CONNECTED_REACHABLE_FOR_PAGING" | Indicates the UE is in the RM-REGISTERED state in 5GS and the UE is reachable for paging. |
| "NOT_PROVIDED_FROM_AMF" | Indicates that the 5GS User State cannot be retrieved from the AMF (NOTE) |
| NOTE: This value is not sent by AMF (it may be sent by UDM to HSS). | |

#### 6.2.6.3.12 Enumeration: LossOfConnectivityReason

**Table 6.2.6.3.12-1: Enumeration LossOfConnectivityReason**

| Enumeration value | Description |
|---|---|
| "DEREGISTERED" | Indicates the UE is deregistered. |
| "MAX_DETECTION_TIME_EXPIRED" | Indicates the mobile reachable timer is expired. |
| "PURGED" | Indicates the UE is purged. |
| "UNAVAILABLE_PERIOD" | Indicates the unavailability period of the UE. |

#### 6.2.6.3.13 Enumeration: ReachabilityFilter

**Table 6.2.6.3.13-1: Enumeration ReachabilityFilter**

| Enumeration value | Description |
|---|---|
| "UE_REACHABILITY_STATUS_CHANGE" | Indicates subscription for "UE Reachability Status Change". |
| "UE_REACHABLE_DL_TRAFFIC" | Indicates subscription for "UE Reachable for DL Traffic". |

#### 6.2.6.3.14 Enumeration: UeType

**Table 6.2.6.3.14-1: Enumeration UeType**

| Enumeration value | Description |
|---|---|
| "AERIAL_UE" | Indicates the UE is an Aerial UE |

#### 6.2.6.3.15 Enumeration: AccessStateTransitionType

**Table 6.2.6.3.15-1: Enumeration AccessStateTransitionType**

| Enumeration value | Description |
|---|---|
| "ACCESS_TYPE_CHANGE_3GPP" | Indicates that the UE's access type has changed to 3GPP access. |
| "ACCESS_TYPE_CHANGE_N3GPP" | Indicates that the UE's access type has changed to non-3GPP access. |
| "RM_STATE_CHANGE_DEREGISTERED" | Indicates that the UE's RM state has change to RM-DEREGISTERED. |
| "RM_STATE_CHANGE_REGISTERED" | Indicates that the UE's RM state has change to RM-REGISTERED. |
| "CM_STATE_CHANGE_IDLE" | Indicates that the UE's CM state has change to CM-IDLE. |
| "CM_STATE_CHANGE_CONNECTED" | Indicates that the UE's CM state has change to CM-CONNECTED |
| "HANDOVER" | Indicates that the UE has performed a successful handover. |
| "MOBILITY_REGISTRATION_UPDATE" | Indicates that the UE has performed a successful mobility registration update. |

#### 6.2.6.3.16 Enumeration: SubTerminationReason

**Table 6.2.6.3.16-1: Enumeration SubTerminationReason**

| Enumeration value | Description |
|---|---|
| "INVALID_SUBSCRIPTION" | Indicates that the subscription is terminated because the AMF has identified that the subscription is no longer valid on the NF hosting the notification URI. |
| "SUBSCRIPTION_NOT_AUTHORIZED" | Indicates that the subscription is terminated because the AMF identified that the subscription is no longer authorized. This may occur during an inter-AMF mobility, based on local policy of the target AMF, e.g. if the transferred subscription contains no or an invalid access token. |

### 6.2.6.4 Binary data

None.

## 6.2.7 Error Handling

### 6.2.7.1 General

HTTP error handling shall be supported as specified in clause 5.2.4 of TS 29.500 [4].

### 6.2.7.2 Protocol Errors

Protocol Error Handling shall be supported as specified in clause 5.2.7 of TS 29.500 [4].

### 6.2.7.3 Application Errors

The common application errors defined in the Table 5.2.7.2-1 in TS 29.500 [4] may also be used for the Namf_EventExposure service, and the following application errors listed in Table 6.2.7.3-1 are specific for the Namf_EventExposure service.

**Table 6.2.7.3-1: Application errors**

| Application Error | HTTP status code | Description |
|---|---|---|
| UE_NOT_SERVED_BY_AMF | 403 Forbidden | Indicates the creation of a subscription targeting a specific UE has failed due to an application error when the UE is not served by the AMF (i.e. it is not known to the AMF). |
| UNSPECIFIED | 403 Forbidden | The request is rejected due to unspecified reasons. |
| MUTING_EXC_INSTR_NOT_ACCEPTED | 403 Forbidden | Indicates the AMF does not accept the received muting exception instructions. |
| SUBSCRIPTION_NOT_FOUND | 404 Not Found | Indicates the modification of subscription has failed due to an application error when the subscription is not found in the AMF. |
| UNSUPPORTED_EVENT_TYPE | 501 Not Implemented | Indicates that the request for creation or modification of a subscription is rejected, because none of the requested events are supported by the AMF. |

## 6.2.8 Feature Negotiation

The feature negotiation mechanism specified in clause 6.6 of TS 29.500 [4] shall be used to negotiate the features applicable between the AMF and the NF Service Consumer, for the Namf_EventExposure service, if any.

The NF Service Consumer shall indicate the features it supports for the Namf_EventExposure service, if any, by including the supportedFeatures attribute in content of the HTTP Request Message for subscription resource creation.

The AMF shall determine the supported features for the service operations as specified in clause 6.6 of TS 29.500 [4] and shall indicate the supported features by including the supportedFeatures attribute in content of the HTTP response for subscription resource creation.

The syntax of the supportedFeatures attribute is defined in clause 5.2.2 of TS 29.571 [6].

The following features are defined for the Namf_EventExposure service:

**Table 6.2.8-1: Features of supportedFeatures attribute used by Namf_EventExposure service**

| Feature Number | Feature | M/O | Description |
|---|---|---|---|
| 1 | ENA | O | Enablers for Network Automation for 5G<br><br>An AMF and an NF that support this feature shall support the procedures specified in TS 23.288 [38]. |
| 2 | APRA | O | Additional Presence Reporting Area<br><br>An AMF that supports this feature shall support subscription of "PRESENCE_IN_AOI_REPORT" event with a Set of Core Network Predefined Presence Reporting Areas and generating event report including both PRA Set ID and additional PRA ID referring to an individual PRA in the Set.<br><br>An NF service consumer that supports this feature shall support receiving "PRESENCE_IN_AOI_REPORT" event with additional PRA ID referring to an individual PRA in the Set. |
| 3 | ESSYNC | O | Event Subscription Synchronization<br><br>An AMF and UDM that supports this feature shall support the event subscription synchronization procedure, as specified in clause 5.3.2.4.2. |
| 4 | ES3XX | M | Extended Support of HTTP 307/308 redirection<br><br>An NF Service Consumer (e.g. NEF) that supports this feature shall support handling of HTTP 307/308 redirection for any service operation of the Namf_EventExposure service. An NF Service Consumer that does not support this feature does only support HTTP redirection as specified for 3GPP Release 15. |
| 5 | IERSR | O | Immediate Event Report in Subscription Creation Response for Subscriptions on behalf of another NF<br><br>An NF consumer (e.g. UDM) supporting this feature shall be able to handle the immediate event reports in the Subscription Creation Response for subscriptions on behalf of another NF, as specified in clause 5.3.2.2.2. |
| 6 | EneNA | O | Enhancement of Enablers for Network Automation for 5G<br><br>An AMF and an NF that support this feature shall support the enhancement of network data analytics specified in TS 23.288 [38]. |
| 7 | DGEM | O | Dynamic Group-based Event Monitoring<br><br>An AMF supporting this feature shall allow the NF consumer to remove or add list of group member UE(s) for a group-based event monitoring subscription (see clause 5.3.2.2.4). |
| 8 | UARF | O | UEs in Area Report Filter<br><br>This feature indicates the support of enhanced filter for UEs-In-Area-Report event. When this feature is supported at the AMF, the AMF shall apply additional filters provided in ueInAreaFilter IE. |
| 9 | MPRA | O | Map type PRA information<br><br>Support of this feature implies support of map type presenceInfoList during subscription creation and support of PresenceInfo modification during subscription modification (see clauses 6.2.6.2.3 and 6.2.6.2.14). |
| 10 | STEN | O | Subscription Termination Event Notification<br><br>An AMF supporting this feature shall support sending a notification to the NF consumer to inform that the AMF event subscription is terminated if requested by NF consumer; an NF consumer supporting this feature shall support processing the Subscription Termination Event Notification from the AMF, e.g. clean-up the local context for the indicated AMF event subscription. |
| 11 | ENAPH3 | O | Enablers for Network Automation for 5G, Phase 3<br><br>An AMF supporting this feature shall support the handling of event muting exception instructions as specified in clause 6.2.7.2 of TS 23.288 [38]. |

| | | | |
|---|---|---|---|
| 12 | AOIEF | O | AOI Event Filters specified in 3GPP Rel-18, e.g. for the support of 5G Timing Resiliency and TSC & URLLC enhancements.<br><br>An AMF supporting this feature shall support subscriptions to the Presence-In-AOI-Report event including:<br>- the indication that the AoI may be adjusted based on the UE's Registration Area;<br>- the "RAN timing synchronization status change event"; and<br>- the notifyForSupiList IE, the notifyForGroupList and or the notifyForSnssaiDnnList IE and shall support notifying the NF service consumer about AOI events only if the event is for a UE belonging to the provided list of SUPIs or Internal Groups and/or for a UE having a PDU session established with the provided DNN/S-NSSAI. See clause 5.3.1. |
| 13 | SAR | O | Slice Area Restriction.<br><br>An AMF supporting this feature shall support to indicate whether the UE is inside or outside of the service area supporting an S-NSSAI for a PDU session associated with a partially allowed NSSAI or with a slice restricted to NS-AoS or whether the UE presence in the reporting area is unknown. |
| Feature number: The order number of the feature within the supportedFeatures attribute (starting with 1).<br>Feature: A short name that can be used to refer to the bit and to the feature.<br>M/O: Defines if the implementation of the feature is mandatory ("M") or optional ("O").<br>Description: A clear textual description of the feature. | | | |

## 6.2.9 Security

As indicated in TS 33.501 [27], the access to the Namf_EventExposure API may be authorized by means of the OAuth2 protocol (see IETF RFC 6749 [28]), using the "Client Credentials" authorization grant, where the NRF (see TS 29.510 [29]) plays the role of the authorization server.

If Oauth2 authorization is used, an NF Service Consumer, prior to consuming services offered by the Namf_EventExposure API, shall obtain a "token" from the authorization server, by invoking the Access Token Request service, as described in TS 29.510 [29], clause 5.4.2.2.

NOTE: When multiple NRFs are deployed in a network, the NRF used as authorization server is the same NRF that the NF Service Consumer used for discovering the Namf_EventExposure service.

The Namf_EventExposure API defines scopes for OAuth2 authorization as specified in TS 33.501 [27]; it defines a single scope consisting on the name of the service (i.e., "namf-evts"), and it does not define any additional scopes at resource or operation level.

## 6.2.10 HTTP redirection

An HTTP request may be redirected to a different AMF service instance, within the same AMF or a different AMF of an AMF set, e.g. when an AMF service instance is part of an AMF (service) set or when using indirect communications (see TS 29.500 [4]). See the ES3XX feature in clause 6.2.8.

An SCP that reselects a different AMF producer instance will return the NF Instance ID of the new AMF producer instance in the 3gpp-Sbi-Producer-Id header, as specified in clause 6.10.3.4 of TS 29.500 [4].

If an AMF within an AMF set redirects a service request to a different AMF of the set using an 307 Temporary Redirect or 308 Permanent Redirect status code, the identity of the new AMF towards which the service request is redirected shall be indicated in the 3gpp-Sbi-Target-Nf-Id header of the 307 Temporary Redirect or 308 Permanent Redirect response as specified in clause 6.10.9.1 of TS 29.500 [4].

# 6.3 Namf_MT Service API

## 6.3.1 API URI

The Namf_MT shall use the Namf_MT API.

The API URI of the Namf_MT API shall be:

**{apiRoot}/<apiName>/<apiVersion>/**

The request URI used in HTTP requests from the NF service consumer towards the NF service producer shall have the Resource URI structure defined in clause 4.4.1 of TS 29.501 [5], i.e.:

**{apiRoot}/<apiName>/<apiVersion>/<apiSpecificResourceUriPart>**

with the following components:

- The {apiRoot} shall be set as described in TS 29.501 [5].
- The <apiName> shall be "namf-mt".
- The <apiVersion> shall be "v1".
- The <apiSpecificResourceUriPart> shall be set as described in clause 6.3.3.

## 6.3.2 Usage of HTTP

### 6.3.2.1 General

HTTP/2, as defined in IETF RFC 9113 [19], shall be used as specified in clause 5 of TS 29.500 [4].

HTTP/2 shall be transported as specified in clause 5.3 of TS 29.500 [4].

HTTP messages and bodies for the Namf_MT service shall comply with the OpenAPI [23] specification contained in Annex A.

### 6.3.2.2 HTTP standard headers

#### 6.3.2.2.1 General

The usage of HTTP standard headers shall be supported as specified in clause 5.2.2 of TS 29.500 [4].

#### 6.3.2.2.2 Content type

The following content types shall be supported:

- JSON, as defined in IETF RFC 8259 [8], shall be used as content type of the HTTP bodies specified in the present specification as indicated in clause 5.4 of TS 29.500 [4].
- The Problem Details JSON Object (IETF RFC 9457 [36]). The use of the Problem Details JSON object in a HTTP response body shall be signalled by the content type "application/problem+json".

#### 6.3.2.3 HTTP custom headers

##### 6.3.2.3.1 General

In this release of this specification, no custom headers specific to the Namf_MT service are defined. For 3GPP specific HTTP custom headers used across all service based interfaces, see clause 5.2.3 of TS 29.500 [4].

### 6.3.3 Resources

#### 6.3.3.1 Overview

//{apiRoot}/namf-mt/<apiVersion>

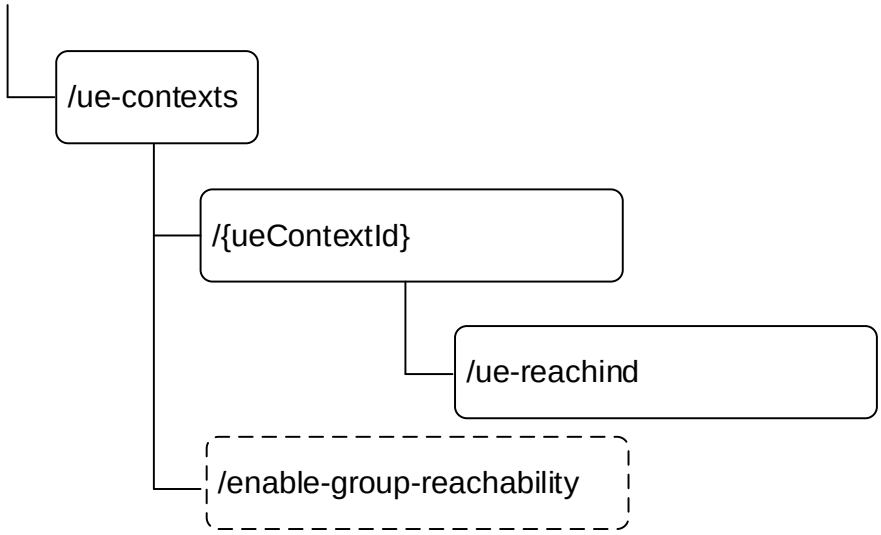



**Figure 6.3.3.1-1: Resource URI structure of the Namf_MT Service API**

Table 6.3.3.1-1 provides an overview of the resources and applicable HTTP methods.

**Table 6.3.3.1-1: Resources and methods overview**

| Resource name | Resource URI | HTTP method or custom operation | Description |
|---|---|---|---|
| ueReachInd | /ue-contexts/{ueContextId}/ue-reachind | PUT | Update the ueReachInd to UE Reachable |
| ueContext | /ue-contexts/{ueContextId} | GET | Map to following service operation:<br>- ProvideDomainSelectionInfo |
| ueContexts collection | /ue-contexts/enable-group-reachability | enable-group-reachability (POST) | Update the state of the list of UEs to CM-CONNECTED state |

#### 6.3.3.2 Resource: ueReachInd

##### 6.3.3.2.1 Description

This resource represents the ueReachInd for a SUPI.

This resource is modelled as the Document resource archetype (see clause C.1 of TS 29.501 [5]).

#### 6.3.3.2.2 Resource Definition

Resource URI: {apiRoot}/namf-mt/<apiVersion>/ue-contexts/{ueContextId}/ue-reachind

This resource shall support the resource URI variables defined in table 6.3.3.2.2-1.

**Table 6.3.3.2.2-1: Resource URI variables for this resource**

| Name | Data type | Definition |
|---|---|---|
| apiRoot | string | See clause 6.3.1 |
| apiVersion | string | See clause 6.3.1. |
| ueContextId | Supi | Represents the Subscription Permanent Identifier (see TS 23.501 [2] clause 5.9.2)<br>pattern: see pattern of type Supi in TS 29.571 [6] |

#### 6.3.3.2.3 Resource Standard Methods

##### 6.3.3.2.3.1 PUT

This method shall support the URI query parameters specified in table 6.3.3.2.3.1-1.

**Table 6.3.3.2.3.1-1: URI query parameters supported by the PUT method on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| n/a | | | | |

This method shall support the request data structures specified in table 6.3.3.2.3.1-2 and the response data structures and response codes specified in table 6.3.3.2.3.1-3.

**Table 6.3.3.2.3.1-2: Data structures supported by the PUT Request Body on this resource**

| Data type | P | Cardinality | Description |
|---|---|---|---|
| EnableUeReachabilityReqData | M | 1 | Contain the State of the UE, the value shall be set to UE Reachable. |

**Table 6.3.3.2.3.1-3: Data structures supported by the PUT Response Body on this resource**

| Data type | P | Cardinality | Response codes | Description |
|---|---|---|---|---|
| EnableUeReachabilityRspData | M | 1 | 200 OK | Indicate the ueReachInd is updated to UE Reachable. |
| RedirectResponse | O | 0..1 | 307 Temporary Redirect | Temporary redirection.<br>When the related UE context is not fully available at the target NF Service Producer (e.g. AMF) during a planned maintenance case (e.g. AMF planned maintenance without UDSF case) the "cause" attribute may be used to include the following application error:<br>- NF_CONSUMER_REDIRECT_ONE_TXN<br><br>See table 6.3.7.3-1 for the description of these errors<br><br>(NOTE 2) |
| RedirectResponse | O | 0..1 | 308 Permanent Redirect | Permanent redirection.<br>(NOTE 2) |
| ProblemDetailsEnableUeReachability | O | 0..1 | 403 Forbidden | The "cause" attribute may be used to indicate one of the following application errors:<br>- UNABLE_TO_PAGE_UE<br>- UNSPECIFIED<br>- UE_IN_NON_ALLOWED_AREA<br><br>See table 6.3.7.3-1 for the description of this error. |
| ProblemDetails | O | 0..1 | 404 Not Found | When the related UE is not found in the NF Service Producer (e.g. AMF) the "cause" attribute shall be set to:<br>- CONTEXT_NOT_FOUND<br><br>See table 6.3.7.3-1 for the description of these errors |
| ProblemDetails | O | 0..1 | 409 Conflict | The "cause" attribute may be used to indicate the following application error:<br>- REJECTION_DUE_TO_PAGING_RESTRICTION<br><br>See table 6.3.7.3-1 for the description of this error. |
| ProblemDetails | O | 0..1 | 503 Service Unavailable | The "cause" attribute may be used to indicate one of the errors defined in Table 5.2.7.2-1 of TS 29.500 [4].<br><br>The HTTP header field "Retry-After" shall not be included in this scenario. |
| ProblemDetailsEnableUeReachability | O | 0..1 | 504 Gateway Timeout | The "cause" attribute may be used to indicate one of the following application errors:<br>- UE_NOT_RESPONDING<br>- UE_NOT_REACHABLE<br><br>See table 6.3.7.3-1 for the description of this error. |
| NOTE 1: The mandatory HTTP error status code for the PUT method listed in Table 5.2.7.1-1 of TS 29.500 [4] also apply, with response body containing an object of ProblemDetails data type (see clause 5.2.7 of TS 29.500 [4]).<br>NOTE 2: RedirectResponse may be inserted by an SCP, see clause 6.10.9.1 of TS 29.500 [4]. | | | | |

**Table 6.3.3.2.3.1-4: Headers supported by the 307 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | The URI of the resource located on an alternative service instance within the same AMF or AMF (service) set to which the request is redirected.<br>For the case when a request is redirected to the same target resource via a different SCP or SEPP, see clause 6.10.9.1 in TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target NF (service) instance ID towards which the request is redirected |

**Table 6.3.3.2.3.1-5: Headers supported by the 308 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | An alternative URI of the resource located on an alternative service instance within the same AMF or AMF (service) set. For the case when a request is redirected to the same target resource via a different SCP or SEPP, see clause 6.10.9.1 in TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target NF (service) instance ID towards which the request is redirected |

### 6.3.3.2.4 Resource Custom Operations

There is no custom operation supported on this resource.

## 6.3.3.3 Resource: ueContext

### 6.3.3.3.1 Description

This resource represents the UeContext for a UE.

This resource is modelled as the Document resource archetype (see clause C.1 of TS 29.501 [5]).

### 6.3.3.3.2 Resource Definition

Resource URI: {apiRoot}/namf-mt/<apiVersion>/ue-contexts/{ueContextId}

This resource shall support the resource URI variables defined in table 6.3.3.3.2-1.

**Table 6.3.3.3.2-1: Resource URI variables for this resource**

| Name | Data type | Definition |
|---|---|---|
| apiRoot | string | See clause 6.3.1 |
| apiVersion | string | See clause 6.3.1. |
| ueContextId | Supi | Represents the Subscription Permanent Identifier (see TS 23.501 [2] clause 5.9.2)<br>pattern: see pattern of type Supi in TS 29.571 [6] |

### 6.3.3.3.3 Resource Standard Methods

#### 6.3.3.3.3.1 GET

This method shall support the URI query parameters specified in table 6.3.3.3.3.1-1.

**Table 6.3.3.3.3.1-1: URI query parameters supported by the GET method on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| info-class | UeContextInfoClass | M | 1 | Indicates the class of the UE Context information elements to be fetched. |
| supported-features | SupportedFeatures | C | 0..1 | This IE shall be present if at least one feature defined in clause 6.3.8 is supported. |
| old-guami | Guami | C | 0..1 | This IE shall be present during an AMF planned removal procedure when the NF Service Consumer initiates a request towards the target AMF, for a UE associated to an AMF that is unavailable (see clause 5.21.2.2 of TS 23.501 [2]). |

This method shall support the request data structures specified in table 6.3.3.3.3.1-2 and the response data structures and response codes specified in table 6.3.3.3.3.1-3.

**Table 6.3.3.3.3.1-2: Data structures supported by the GET Request Body on this resource**

| Data type | P | Cardinality | Description |
|---|---|---|---|
| n/a | | | |

**Table 6.3.3.3.3.1-3: Data structures supported by the GET Response Body on this resource**

| Data type | P | Cardinality | Response codes | Description |
|---|---|---|---|---|
| UeContextInfo | M | 1 | 200 OK | This represents the operation is successful and request UE Context information is returned. |
| RedirectResponse | O | 0..1 | 307 Temporary Redirect | Temporary redirection.<br>When the related UE context is not fully available at the target NF Service Consumer (e.g. AMF) during a planned maintenance case (e.g. AMF planned maintenance without UDSF case) the "cause" attribute shall be set to:<br>- NF_CONSUMER_REDIRECT_ONE_TXN<br>See table 6.3.7.3-1 for the description of these errors<br><br>(NOTE 2) |
| RedirectResponse | O | 0..1 | 308 Permanent Redirect | Permanent redirection.<br>(NOTE 2) |
| ProblemDetails | O | 0..1 | 403 Forbidden | Indicates the operation has failed due to application error.<br><br>The "cause" attribute may be used to indicate one of the following application errors:<br>- UNABLE_TO_PAGE_UE<br>- UE_DEREGISTERED<br>- UNSPECIFIED<br><br>See table 6.3.7.3-1 for the description of these errors. |
| ProblemDetails | O | 0..1 | 404 Not Found | Indicates the operation has failed due to application error.<br>The "cause" attribute may be used to indicate one of the following application errors:<br>- CONTEXT_NOT_FOUND<br><br>See table 6.3.7.3-1 for the description of these errors |
| ProblemDetails | O | 0..1 | 409 Conflict | This indicates that the request could not be completed due to a temporary conflict with the current state of the target resource.<br><br>The cause attribute of the ProblemDetails structure shall be set to:<br>- TEMPORARY_REJECT_REGISTRATION_ONGOING, if there is an on-going registration procedure. |

NOTE 1: The mandatory HTTP error status code for the GET method listed in Table 5.2.7.1-1 of TS 29.500 [4] also apply, with response body containing an object of ProblemDetails data type (see clause 5.2.7 of TS 29.500 [4]).
NOTE 2: RedirectResponse may be inserted by an SCP or SEPP, see clause 6.10.9.1 of TS 29.500 [4].

**Table 6.3.3.3.3.1-4: Headers supported by the 307 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | The URI of the resource located on the target NF Service Consumer (e.g. AMF) to which the request is redirected.<br>For the case when a request is redirected to the same target resource via a different SCP or SEPP, see clause 6.10.9.1 in TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target NF (service) instance ID towards which the request is redirected |

**Table 6.3.3.3.3.1-5: Headers supported by the 308 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | An alternative URI of the resource located on an alternative service instance within the same AMF or AMF (service) set.<br>For the case when a request is redirected to the same target resource via a different SCP or SEPP, see clause 6.10.9.1 in TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target NF (service) instance ID towards which the request is redirected |

#### 6.3.3.3.4 Resource Custom Operations

There is no custom operation supported on this resource.

### 6.3.3.4 Resource: ueContexts collection

#### 6.3.3.4.1 Description

This resource represents the collection of the individual UeContexts created in the AMF.

This resource is modelled as the Collection resource archetype (see clause C.2 of TS 29.501 [5]).

#### 6.3.3.4.2 Resource Definition

Resource URI: **{apiRoot}/namf-mt/<apiVersion>/ue-contexts**

This resource shall support the resource URI variables defined in table 6.3.3.4.2-1.

**Table 6.3.3.4.2-1: Resource URI variables for this resource**

| Name | Data type | Definition |
|---|---|---|
| apiRoot | string | See clause 6.3.1 |
| apiVersion | string | See clause 6.3.1. |

#### 6.3.3.4.3 Resource Standard Methods

There is no standard operation supported on this resource.

#### 6.3.3.4.4 Resource Custom Operations

##### 6.3.3.4.4.1 Overview

**Table 6.3.3.4.4.1-1: Custom operations**

| Operation Name | Custom operation URI | Mapped HTTP method | Description |
|---|---|---|---|
| enable-group-reachability | /ue-contexts/enable-group-reachability | POST | Enable Group Reachability service operation |

##### 6.3.3.4.4.2 Operation: enable-group-reachability

###### 6.3.3.4.4.2.1 Description

###### 6.3.3.4.4.2.2 Operation Definition

This custom operation updates the state of the list of UEs to CM-CONNECTED state.

This operation shall support the request data structures specified in table 6.3.3.4.4.2.2-1 and the response data structure and response codes specified in table 6.3.3.4.4.2.2-2.

**Table 6.3.3.4.4.2.2-1: Data structures supported by the POST Request Body on this resource**

| Data type | P | Cardinality | Description |
|---|---|---|---|
| EnableGroupReachabilityReqData | M | 1 | Contain the list of UEs involved in the MBS Session identified by the related TMGI. |

**Table 6.3.3.4.4.2.2-2: Data structures supported by the POST Response Body on this resource**

| Data type | P | Cardinality | Response codes | Description |
|---|---|---|---|---|
| EnableGroupReachabilityRspData | M | 1 | 200 OK | Successful response indicating the list of UEs in CM-CONNECTED state if any, and indicating the supported features. |
| RedirectResponse | O | 0..1 | 307 Temporary Redirect | Temporary redirection.<br>(NOTE 2) |
| RedirectResponse | O | 0..1 | 308 Permanent Redirect | Permanent redirection.<br>(NOTE 2) |
| ProblemDetails | O | 0..1 | 404 Not Found | When none of the UEs in the list of UEs are found in the AMF, the "cause" attribute shall be set to:<br>- CONTEXT_NOT_FOUND<br><br>See table 6.3.7.3-1 for the description of these errors |
| ProblemDetails | O | 0..1 | 503 Service Unavailable | The "cause" attribute may be used to indicate one of the errors defined in Table 5.2.7.2-1 of TS 29.500 [4]. |
| NOTE 1: The mandatory HTTP error status code for the POST method listed in Table 5.2.7.1-1 of TS 29.500 [4] also apply, with response body containing an object of ProblemDetails data type (see clause 5.2.7 of TS 29.500 [4]).<br>NOTE 2: RedirectResponse may be inserted by an SCP, see clause 6.10.9.1 of TS 29.500 [4]. | | | | |

**Table 6.3.3.4.4.2.2-3: Headers supported by the 307 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | The URI of the resource located on an alternative service instance within the same AMF or AMF (service) set to which the request is redirected.<br>For the case when a request is redirected to the same target resource via a different SCP, see clause 6.10.9.1 in TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target NF (service) instance ID towards which the request is redirected |

**Table 6.3.3.4.4.2.2-4: Headers supported by the 308 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | An alternative URI of the resource located on an alternative service instance within the same AMF or AMF (service) set.<br>For the case when a request is redirected to the same target resource via a different SCP, see clause 6.10.9.1 in TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target NF (service) instance ID towards which the request is redirected |

## 6.3.4 Custom Operations without associated resources

There are no custom operations without associated resources supported on Namf_MT service.

## 6.3.5 Notifications

### 6.3.5.1 General

This clause specifies the notifications provided by the Namf_MT service.

**Table 6.3.5.1-1: Notifications overview**

| Notification | Callback URI | HTTP method or custom operation | Description (service operation) |
|---|---|---|---|
| UE Reachability Info Notify | {reachabilityNotifyUri} | POST | |

### 6.3.5.2 UE Reachability Info Notify

If a NF service consumer has implicitly subscribed to the UE Reachability Info Notify supported by Namf_MT service, the AMF shall deliver the notification to the call-back URI for the UE(s) which are reachable or do not respond to paging, following Subscribe/Notify mechanism defined in TS 29.501 [5].

#### 6.3.5.2.1 Notification Definition

Call-back URI: **{callbackUri}**

This callback URI shall be the "reachabilityNotifyUri" provided in the "EnableGroupReachabilityReqData" IE.

#### 6.3.5.2.3 Notification Standard Methods

##### 6.3.5.2.3.1 POST

This method shall support the request data structures specified in table 6.3.5.2.3.1-1 and the response data structures and response codes specified in table 6.3.5.2.3.1-2.

**Table 6.3.5.2.3.1-2: Data structures supported by the POST Request Body**

| Data type | P | Cardinality | Description |
|---|---|---|---|
| ReachabilityNotification Data | M | 1 | Represents the notification to be delivered |

**Table 6.3.5.2.3.1-3: Data structures supported by the POST Response Body**

| Data type | P | Cardinality | Response codes | Description |
|---|---|---|---|---|
| n/a | | | 204 No Content | |
| RedirectResponse | O | 0..1 | 307 Temporary Redirect | Temporary redirection.<br>(NOTE 2) |
| RedirectResponse | O | 0..1 | 308 Permanent Redirect | Permanent redirection.<br>(NOTE 2) |
| NOTE 1: The mandatory HTTP error status code for the POST method listed in Table 5.2.7.1-1 of TS 29.500 [4] also apply, with response body containing an object of ProblemDetails data type (see clause 5.2.7 of TS 29.500 [4]).<br>NOTE 2: RedirectResponse may be inserted by an SCP, see clause 6.10.9.1 of TS 29.500 [4]. | | | | |

**Table 6.3.5.2.3.1-4: Headers supported by the 307 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | A URI pointing to the endpoint of the NF service consumer to which the notification should be sent.<br>For the case when a request is redirected to the same target resource via a different SCP, see clause 6.10.9.1 in TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target NF (service) instance ID towards which the request is redirected |

**Table 6.3.5.2.3.1-5: Headers supported by the 308 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | A URI pointing to the endpoint of the NF service consumer to which the notification should be sent.<br>For the case when a request is redirected to the same target resource via a different SCP, see clause 6.10.9.1 in TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target NF (service) instance ID towards which the request is redirected |

## 6.3.6 Data Model

### 6.3.6.1 General

This clause specifies the application data model supported by the API.

Table 6.3.6.3-1 specifies the data types defined for the Namf_MT service based interface protocol.

**Table 6.3.6.3-1: Namf_MT specific Data Types**

| Data type | Clause defined | Description |
|---|---|---|
| EnableUeReachabilityReqData | 6.3.6.2.2 | Data within the Enable UE Reachability Request |
| EnableUeReachabilityRspData | 6.3.6.2.3 | Data within the Enable UE Reachability Response |
| UeContextInfo | 6.3.6.2.4 | Contains the UE Context Information |
| ProblemDetailsEnableUeReachability | 6.3.6.2.5 | Enable UE Reachability Error Detail |
| AdditionInfoEnableUeReachability | 6.3.6.2.6 | Additional information to be returned in EnableUeReachability error response. |
| EnableGroupReachabilityReqData | 6.3.6.2.7 | Data within the Enable Group Reachability Request |
| EnableGroupReachabilityRspData | 6.3.6.2.8 | Data within the Enable Group Reachability Response |
| UeInfo | 6.3.6.2.9 | list of UEs requested to be made reachable for the MBS Session |
| ReachabilityNotificationData | 6.3.6.2.10 | Data within the UE Reachability Info Notify |
| ReachableUeInfo | 6.3.6.2.11 | Contains the reachable UE Information |
| QosFlowInfo | 6.3.6.2.12 | QoS Flow information |
| UeContextInfoClass | 6.3.6.3.3 | Indicates the UE Context information class |

Table 6.3.6.3-2 specifies data types re-used by the Namf_MT service based interface protocol from other specifications, including a reference to their respective specifications and when needed, a short description of their use within the Namf_MT service based interface.

**Table 6.3.6.3-2: Namf_MT re-used Data Types**

| Data type | Reference | Comments |
|---|---|---|
| ProblemDetails | TS 29.571 [6] | Common data type used in response bodies |
| supportedFeatures | TS 29.571 [6] | Supported Features |
| AccessType | TS 29.571 [6] | Access Type |
| RatType | TS 29.571 [6] | RAT Type |
| DurationSec | TS 29.571 [6] | |
| RedirectResponse | TS 29.571 [6] | Response body of the redirect response message. |
| Supi | TS 29.571 [6] | SUPI |
| Tmgi | TS 29.571 [6] | TMGI |
| PduSessionId | TS 29.571 [6] | PDU Session Id |
| Uri | TS 29.571 [6] | URI |
| UserLocation | TS 29.571 [6] | User Location Information |
| MbsServiceAreaInfo | TS 29.571 [6] | MBS Service Area Information for a Location dependent MBS session |
| Qfi | TS 29.571 [6] | QoS Flow Identifier |
| Uint16 | TS 29.571 [6] | Unsigned 16-bit integer |
| MbsServiceArea | TS 29.571 [6] | MBS Service Area Information. |
| UeReachability | 6.2.6.3.7 | Describes the reachability of the UE |
| Ppi | 6.1.6.3.2 | Paging Policy |

## 6.3.6.2 Structured data types

### 6.3.6.2.1 Introduction

Structured data types used in Namf_MT service are specified in this clause.

#### 6.3.6.2.2 Type: EnableUeReachabilityReqData

**Table 6.3.6.3.2-1: Definition of type EnableUeReachabilityReqData**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| reachability | UeReachability | M | 1 | Indicates the desired reachability of the UE |
| supportedFeatures | SupportedFeatures | C | 0..1 | This IE shall be present if at least one feature defined in clause 6.3.8 is supported. |
| oldGuami | Guami | C | 0..1 | This IE shall be present during an AMF planned removal procedure when the NF Service Consumer initiates a request towards the target AMF, for a UE associated to an AMF that is unavailable (see clause 5.21.2.2 of TS 23.501 [2]). |
| extBufSupport | boolean | C | 0..1 | This IE shall be present and set to "true", if the extended buffering is supported(see clauses 4.24.2 and clause 4.25.5 of TS 23.502 [3]),<br><br>When present, the IE shall be set as following:<br>- true: the extended buffering is supported<br>- false (default): the extended buffering is not supported |
| qosFlowInfoList | array(QosFlowInfo) | O | 1..N | This IE may be included by the SMF during the Network Triggered Connection Resume in RRC Inactive with CN based MT communication handling.<br><br>When present, it shall indicate the QoS flow information of the QoS flow(s) of the PDU session triggering the procedure, e.g., for which DL packets have been received by the UPF. |
| pduSessionId | PduSessionId | O | 0..1 | This IE may be included by the SMF during the Network Triggered Connection Resume in RRC Inactive with CN based MT communication handling.<br><br>When present, it shall indicate the PDU Session ID of the PDU session triggering the procedure, e.g., for which DL packets have been received by the UPF. |

#### 6.3.6.2.3 Type: EnableUeReachabilityRspData

**Table 6.3.6.2.3-1: Definition of type EnableUeReachabilityRspData**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| reachability | UeReachability | M | 1 | Indicates the current reachability of the UE |
| supportedFeatures | SupportedFeatures | C | 0..1 | This IE shall be present if at least one optional feature defined in clause 6.3.8 is supported. |

#### 6.3.6.2.4 Type: UeContextInfo

**Table 6.3.6.2.3-1: Definition of type UeContextInfo**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| supportVoPS | boolean | C | 0..1 | This IE shall be present when following UE Context Information class are required:<br>- "TADS"<br><br>When present, this IE shall indicate whether or not IMS voice over PS Session is supported in the registration area (s) where the UE is currently registered in 3GPP access. |
| supportVoPSn3gpp | boolean | C | 0..1 | This IE shall be present when the UE is registered in WLAN non 3GPP access and the following UE Context Information class are required:<br>- "TADS"<br><br>When present, this IE shall indicate whether or not IMS voice over PS Session Supported Indication over non-3GPP access is supported in the WLAN where the UE is currently registered. |
| lastActTime | DateTime | C | 0..1 | This IE shall be present when following UE Context Information class are required:<br>- "TADS"<br><br>When present, this IE shall indicate the UTC time stamp of the last radio contact with the UE. |
| accessType | AccessType | C | 0..1 | This IE shall be present when following UE Context Information class are required:<br>- "TADS"<br><br>When present, this IE shall indicate the current access type of the UE. |
| ratType | RatType | C | 0..1 | This IE shall be present when following UE Context Information class are required:<br>- "TADS"<br><br>When present, this IE shall indicate the current RAT type of the UE. |
| supportedFeatures | SupportedFeatures | C | 0..1 | This IE shall be present if at least one feature defined in clause 6.3.8 is supported. |

#### 6.3.6.2.5 Type: ProblemDetailsEnableUeReachability

**Table 6.3.6.2.5-1: Definition of type ProblemDetailsEnableUeReachability as a list of to be combined data**

| Data type | Cardinality | Description | Applicability |
|---|---|---|---|
| ProblemDetails | 1 | Detail information of the problem | |
| AdditionInfoEnableUeReachability | 1 | Additional information to be returned in error response. | |

#### 6.3.6.2.6 Type: AdditionInfoEnableUeReachability

**Table 6.3.6.2.6-1: Definition of type AdditionInfoEnableUeReachability**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| maxWaitingTime | DurationSec | C | 0..1 | This IE shall contain the estimated maximum wait time (see clauses 4.24.2 and 4.8.2.2b of 3GPP 23.502 [3]). |

#### 6.3.6.2.7 Type: EnableGroupReachabilityReqData

**Table 6.3.6.2.7-1: Definition of type EnableGroupReachabilityReqData**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| ueInfoList | array(UeInfo) | M | 1..N | This IE shall indicate the list of UEs requested to be made reachable for the MBS Session and may indicate the PDU Session Id of the associated PDU Session for the UE(s). |
| tmgi | Tmgi | M | 1 | This IE shall Indicate the TMGI of the MBS session. |
| reachabilityNotifyUri | Uri | O | 0..1 | The callback URI on which UEReachabilityInfoNotify is reported. |
| mbsServiceAreaInfoList | array(MbsServiceAreaInfo) | O | 1..N | List of MBS Service Area and related Area Session ID for location dependent content. |
| mbsServiceArea | MbsServiceArea | O | 0..1 | MBS Service Area for a local MBS session |
| arp | Arp | O | 0..1 | This IE when present shall indicate the most demanding Allocation and Retention Priority of all MBS QoS Flow within the MBS session.<br>The AMF may use this IE for paging differentiation (see clause 6.12 of TS 23.247 [55]). |
| 5qi | 5Qi | O | 0..1 | This IE when present shall indicate the most demanding 5QI of all MBS QoS Flow within the MBS session.<br>The AMF may use this IE for paging differentiation (see clause 6.12 of TS 23.247 [55]). |
| supportedFeatures | SupportedFeatures | C | 0..1 | This IE shall be present if at least one feature defined in clause 6.3.8 is supported. |

#### 6.3.6.2.8 Type: EnableGroupReachabilityRspData

**Table 6.3.6.2.8-1: Definition of type EnableGroupReachabilityRspData**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| ueConnectedList | array(Supi) | C | 1..N | This IE shall be present if there is at least one UE in the list of UEs received in the request that is already in CM-CONNECTED state. When present, this IE shall indicate the list of UEs in CM-CONNECTED state. |
| supportedFeatures | SupportedFeatures | C | 0..1 | This IE shall be present if at least one feature defined in clause 6.3.8 is supported. |

#### 6.3.6.2.9 Type: UeInfo

**Table 6.3.6.2.9-1: Definition of type UeInfo**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| ueList | array(Supi) | M | 1..N | This IE shall indicate the list of UEs requested to be made reachable for the MBS Session. (NOTE) |
| pduSessionId | PduSessionId | O | 0..1 | Represents the identifier of the associated PDU Session for the UEs in ueList IE. (NOTE) |
| NOTE X: When the pduSessionId is present, the ueList array shall contain only one element, i.e., a Supi. | | | | |

#### 6.3.6.2.10 Type: ReachabilityNotificationData

**Table 6.3.6.2.10-1: Definition of type ReachabilityNotificationData**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| reachableUeList | array(ReachableUeInfo) | C | 1..N | This IE shall indicate the list of reachable UEs. |
| unreachableUeList | array(Supi) | C | 1..N | This IE shall Indicate the list of unreachable UEs. |

#### 6.3.6.2.11 Type: ReachableUeInfo

**Table 6.3.6.2.11-1: Definition of type ReachableUeInfo**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| ueList | array(Supi) | M | 1..N | This IE shall indicate the list of reachable UEs. |
| userLocation | UserLocation | C | 0..1 | This IE shall Indicate the User Location of the reachable UEs. |

#### 6.3.6.2.12 Type: QosFlowInfo

**Table 6.3.6.2.12-1: Definition of type QosFlowInfo**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| qfi | Qfi | M | 1 | This IE shall indicate the QFI of the QoS flow of the PDU session triggering the procedure, e.g., for which DL packets have been received by the UPF. |
| ppi | Ppi | O | 0..1 | When present, this IE shall contain the Paging policy to be applied. |
| arp | Arp | O | 0..1 | When present, this IE shall contain the Allocation and Retention Priority of the QoS flow of the PDU session triggering the procedure, e.g., for which DL packets have been received by the UPF. |
| 5qi | 5Qi | O | 0..1 | When present, this IE shall indicate the 5QI of the QoS flow of the PDU session triggering the procedure, e.g., for which DL packets have been received by the UPF. |
| dlDataSize | Uint16 | O | 0..1 | When present, this IE shall indicate the DL data size (expressed in number of octets) that the AMF shall signal to the NG-RAN (e.g. for QoS flows that have buffered data in the CN).<br>See clause 4.8.2.2b of TS 23.502 [3]. |

### 6.3.6.3 Simple data types and enumerations

#### 6.3.6.3.1 Introduction

This clause defines simple data types and enumerations that can be referenced from data structures defined in the previous clauses.

#### 6.3.6.3.2 Simple data types

The simple data types defined in table 6.3.6.3.2-1 shall be supported.

**Table 6.3.6.3.2-1: Simple data types**

| Type Name | Type Definition | Description |
|---|---|---|
| | | |

#### 6.3.6.3.3 Enumeration: UeContextInfoClass

**Table 6.3.6.3.3-1: Enumeration UeContextInfoClass**

| Enumeration value | Description |
|---|---|
| "TADS" | Defines the UE Context Information for Terminating Domain Selection for IMS Voice over PS. |

#### 6.3.6.4 Binary data

None.

### 6.3.7 Error Handling

#### 6.3.7.1 General

HTTP error handling shall be supported as specified in clause 5.2.4 of TS 29.500 [4].

#### 6.3.7.2 Protocol Errors

Protocol Error Handling shall be supported as specified in clause 5.2.7 of TS 29.500 [4].

#### 6.3.7.3 Application Errors

The common application errors defined in the Table 5.2.7.2-1 in TS 29.500 [4] may also be used for the Namf_MT service, and the following application errors listed in Table 6.3.7.3-1 are specific for the Namf_MT service.

**Table 6.3.7.3-1: Application errors**

| Application Error | HTTP status code | Description |
|---|---|---|
| NF_CONSUMER_REDIRECT_ONE_TXN | 307 Temporary Redirect | The request has been asked to be redirected to a specified target. |
| UNABLE_TO_PAGE_UE | 403 Forbidden | AMF is unable page the UE, temporarily. |
| UE_IN_NON_ALLOWED_AREA | 403 Forbidden | UE is currently in a non-allowed area and the service request is not for a regulatory prioritized service. |
| UE_DEREGISTERED | 403 Forbidden | The user is in RM-DEREGISTERED state in the AMF. |
| UNSPECIFIED | 403 Forbidden | The request is rejected due to unspecified reasons. |
| CONTEXT_NOT_FOUND | 404 Not Found | The related UE is not found in the NF Service Consumer. |
| TEMPORARY_REJECT_REGISTRATION_ONGOING | 409 Conflict | The request fails due to an on-going registration procedure. |
| REJECTION_DUE_TO_PAGING_RESTRICTION | 409 Conflict | If Paging Restriction Information restrict the EnableUEReachability request from causing paging as defined in TS 23.501 [2] clause 5.38.5 or if the UE rejects the paging as defined in TS 23.501 [2] clause 5.38.4. |
| UE_NOT_RESPONDING | 504 Gateway Timeout | UE is not responding to the request initiated by the network, e.g. Paging. |
| UE_NOT_REACHABLE | 504 Gateway Timeout | The UE is not reachable for paging, e.g., when UE is in MICO mode or the UE has entered the UE has entered Extended DRX in CM-IDLE or Extended DRX for RRC-INACTIVE state. |

### 6.3.8 Feature Negotiation

The feature negotiation mechanism specified in clause 6.6 of TS 29.500 [4] shall be used to negotiate the features applicable between the AMF and the NF Service Consumer, for the Namf_MT service, if any.

The NF Service Consumer shall indicate the features it supports for the Namf_MT service, if any, by including the supportedFeatures attribute in content of the HTTP Request Message for following service operations:

- EnableUEReachability, as specified in clause 5.4.2.2;
- ProvideDomainSelectionInfo, as specified in clause 5.4.2.3;
- EnableGroupReachability, as specified in clause 5.4.2.4.

The AMF shall determine the supported features for the service operations as specified in clause 6.6 of TS 29.500 [4] and shall indicate the supported features by including the supportedFeatures attribute in content of the HTTP response for the service operation.

The syntax of the supportedFeatures attribute is defined in clause 5.2.2 of TS 29.571 [6].

The following features are defined for the Namf_MT service.

**Table 6.3.8-1: Features of supportedFeatures attribute used by Namf_MT service**

| Feature Number | Feature | M/O | Description |
|---|---|---|---|
| 1 | ES3XX | M | Extended Support of HTTP 307/308 redirection<br><br>An NF Service Consumer (e.g. SMSF) that supports this feature shall support handling of HTTP 307/308 redirection for any service operation of the Namf_MT service. An NF Service Consumer that does not support this feature does only support HTTP redirection as specified for 3GPP Release 15. |
| 2 | GRCAP | O | Group Reachability Capability<br><br>An AMF and SMF that supports this feature shall support the EnableGroupReachability and UEReachabilityInfoNotify service operations, as specified in clause 5.4.2.4 and clause 5.4.2.5. |
| Feature number: The order number of the feature within the supportedFeatures attribute (starting with 1).<br>Feature: A short name that can be used to refer to the bit and to the feature.<br>M/O: Defines if the implementation of the feature is mandatory ("M") or optional ("O").<br>Description: A clear textual description of the feature. | | | |

## 6.3.9 Security

As indicated in TS 33.501 [27], the access to the Namf_MT API may be authorized by means of the OAuth2 protocol (see IETF RFC 6749 [28]), using the "Client Credentials" authorization grant, where the NRF (see TS 29.510 [29]) plays the role of the authorization server.

If Oauth2 authorization is used, an NF Service Consumer, prior to consuming services offered by the Namf_MT API, shall obtain a "token" from the authorization server, by invoking the Access Token Request service, as described in TS 29.510 [29], clause 5.4.2.2.

NOTE: When multiple NRFs are deployed in a network, the NRF used as authorization server is the same NRF that the NF Service Consumer used for discovering the Namf_MT service.

The Namf_MT API defines the following scopes for OAuth2 authorization as specified in TS 33.501 [27].

**Table 6.3.9-1: OAuth2 scopes defined in Namf_MT API**

| Scope | Description |
|---|---|
| "namf-mt" | Access to the Namf_MT API. |
| "namf-mt:ue-reachind" | Access to the EnableUeReachability service operation. |
| "namf-mt:enable-group-reachability" | Access to the EnableGroupReachability service operation. |

### 6.3.10 HTTP redirection

An HTTP request may be redirected to a different AMF service instance, within the same AMF or a different AMF of an AMF set, e.g. when an AMF service instance is part of an AMF (service) set or when using indirect communications (see TS 29.500 [4]). See the ES3XX feature in clause 6.3.8.

An SCP that reselects a different AMF producer instance will return the NF Instance ID of the new AMF producer instance in the 3gpp-Sbi-Producer-Id header, as specified in clause 6.10.3.4 of TS 29.500 [4].

If an AMF within an AMF set redirects a service request to a different AMF of the set using an 307 Temporary Redirect or 308 Permanent Redirect status code, the identity of the new AMF towards which the service request is redirected shall be indicated in the 3gpp-Sbi-Target-Nf-Id header of the 307 Temporary Redirect or 308 Permanent Redirect response as specified in clause 6.10.9.1 of TS 29.500 [4].

## 6.4 Namf_Location Service API

### 6.4.1 API URI

The Namf_Location shall use the Namf_ Location API.

The API URI of the Namf_Location API shall be:

**{apiRoot}/<apiName>/<apiVersion>/**

The request URI used in HTTP requests from the NF service consumer towards the NF service producer shall have the Resource URI structure defined in clause 4.4.1 of TS 29.501 [5], i.e.:

**{apiRoot}/<apiName>/<apiVersion>/<apiSpecificResourceUriPart>**

with the following components:

- The {apiRoot} shall be set as described in TS 29.501 [5].
- The <apiName> shall be "namf-loc".
- The <apiVersion> shall be "v1".
- The <apiSpecificResourceUriPart> shall be set as described in clause 6.4.3.

### 6.4.2 Usage of HTTP

#### 6.4.2.1 General

HTTP/2, as defined in IETF RFC 9113 [19], shall be used as specified in clause 5 of TS 29.500 [4].

HTTP/2 shall be transported as specified in clause 5.3 of TS 29.500 [4].

HTTP messages and bodies for the Namf_Location service shall comply with the OpenAPI [23] specification contained in Annex A.

### 6.4.2.2 HTTP standard headers

#### 6.4.2.2.1 General

The usage of HTTP standard headers shall be supported as specified in clause 5.2.2 of TS 29.500 [4].

#### 6.4.2.2.2 Content type

The following content types shall be supported:

- JSON, as defined in IETF RFC 8259 [8], shall be used as content type of the HTTP bodies specified in the present specification as indicated in clause 5.4 of TS 29.500 [4].
- The Problem Details JSON Object (IETF RFC 9457 [36]). The use of the Problem Details JSON object in a HTTP response body shall be signalled by the content type "application/problem+json".

### 6.4.2.3 HTTP custom headers

#### 6.4.2.3.1 General

In this release of this specification, no custom headers specific to the Namf_Location service are defined. For 3GPP specific HTTP custom headers used across all service based interfaces, see clause 5.2.3 of TS 29.500 [4].

## 6.4.3 Resources

### 6.4.3.1 Overview

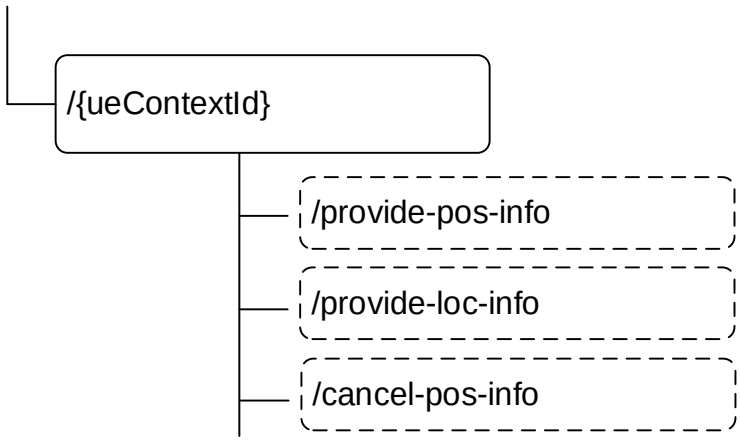



**Figure 6.4.3.1-1: Resource URI structure of the Namf_Location Service API**

Table 6.4.3.1-1 provides an overview of the resources and applicable HTTP methods.

**Table 6.4.3.1-1: Resources and methods overview**

| Resource name | Resource URI | HTTP method or custom operation | Description |
|---|---|---|---|
| Individual UE context | /provide-pos-info | provide-pos-info (POST) | ProvidePositioningInfo |
| | /provide-loc-info | provide-loc-info (POST) | ProvideLocationInfo |
| | /cancel-pos-info | cancel-pos-info (POST) | CancelLocation |

### 6.4.3.2 Resource: Individual UE Context

#### 6.4.3.2.1 Description

This resource represents an individual ueContextId.

This resource is modelled with the Document resource archetype (see clause C.1 of TS 29.501 [5]).

#### 6.4.3.2.2 Resource Definition

Resource URI:{apiRoot}/namf-loc/<apiVersion>/{ueContextId}

This resource shall support the resource URI variables defined in table 6.4.3.2.2-1.

**Table 6.4.3.2.2-1: Resource URI variables for this resource**

| Name | Data type | Definition |
|---|---|---|
| apiRoot | string | See clause 6.4.1 |
| apiVersion | string | See clause 6.4.1. |
| ueContextId | string | Represents the Subscription Permanent Identifier (see TS 23.501 [2] clause 5.9.2)<br>pattern: see pattern of type Supi in TS 29.571 [6]<br>Or represents the Permanent Equipment Identifier (see TS 23.501 [2] clause 5.9.3)<br>pattern: "(imei-[0-9]{15}\|imeisv-[0-9]{16}\|.+)"<br>Or represents the Generic Public Subscription Identifier (see TS 23.501 [2] clause 5.9.8)<br>pattern: see pattern of type Gpsi in TS 29.571 [6] |

#### 6.4.3.2.3 Resource Standard Methods

There are no standard methods supported on this resource.

#### 6.4.3.2.4 Resource Custom Operations

##### 6.4.3.2.4.1 Overview

**Table 6.4.3.2.4.1-1: Custom operations**

| Operation Name | Custom operaration URI | Mapped HTTP method | Description |
|---|---|---|---|
| provide-pos-info | /{ueContextId}/provide-pos-info | POST | Request the positioning information of the UE.<br>It is used for the ProvidePositioningInfo service operation. |
| provide-loc-info | /{ueContextId}/provide-loc-info | POST | Request the Network Provided Location Information of the UE. |
| cancel-pos-info | /{ueContextId}/cancel-pos-info | POST | Cancels periodic or triggered location for the UE. |

##### 6.4.3.2.4.2 Operation: provide-pos-info (POST)

###### 6.4.3.2.4.2.1 Description

This ueContextId identifies the individual ueContext resource is composed by UE's SUPI or PEI. If the AMF supporting the "POSGPSI" feature, the ueContextId identifies the individual ueContext resource may also be composed by UE's GPSI.

###### 6.4.3.2.4.2.2 Operation Definition

This operation shall support the request data structures specified in table 6.4.3.2.4.2.2-1 and the response data structure and response codes specified in table 6.4.3.2.4.2.2-2.

**Table 6.4.3.2.4.2.2-1: Data structures supported by (POST) the provide-pos-info operation Request Body**

| Data type | P | Cardinality | Description |
|---|---|---|---|
| RequestPosInfo | M | 1 | The information to request the positioning information of the UE. |

**Table 6.4.3.2.4.2.2-2: Data structures supported by the (POST) provide-pos-info operation Response Body**

| Data type | P | Cardinality | Response codes | Description |
|---|---|---|---|---|
| ProvidePosInfoExt | M | 1 | 200 OK | This case represents a successful query of the UE positioning information, the AMF returns the related information in the response. |
| n/a | | | 204 No Content | Shall return 204 if no information is to be returned |
| RedirectResponse | O | 0..1 | 307 Temporary Redirect | Temporary redirection.<br>(NOTE 2) |
| RedirectResponse | O | 0..1 | 308 Permanent Redirect | Permanent redirection.<br>(NOTE 2) |
| ProblemDetails | O | 0..1 | 403 Forbidden | The "cause" attribute may be used to indicate one of the following application errors:<br>- USER_UNKNOWN<br>- DETACHED_USER<br>- POSITIONING_DENIED<br>- UNSPECIFIED<br>- REQUESTED_LMF_NOT_AVAILABLE<br><br>See table 6.4.7.3-1 for the description of these errors. |
| ProblemDetailsProvidePosInfo | O | 0..1 | 409 Conflict | The request could not be completed due to a conflict with the current state of the resource.<br>The "cause" attribute may be used to indicate<br>- HO_TO_EPS<br><br>See table 6.4.7.3-1 for the description of these errors.<br><br>The response should contain the target MME name and realm, if available, and it may contain the location information available from the LMF. |
| ProblemDetails | O | 0..1 | 500 Internal Server Error | The "cause" attribute may be used to indicate one of the following application errors:<br>- POSITIONING_FAILED<br><br>See table 6.4.7.3-1 for the description of these errors. |
| ProblemDetails | O | 0..1 | 504 Gateway Timeout | The "cause" attribute may be used to indicate one of the following application errors:<br>- UNREACHABLE_USER<br>- PEER_NOT_RESPONDING<br><br>See table 6.4.7.3-1 for the description of this error. |
| NOTE 1: The mandatory HTTP error status code for the POST method listed in Table 5.2.7.1-1 of TS 29.500 [4] also apply, with response body containing an object of ProblemDetails data type (see clause 5.2.7 of TS 29.500 [4]).<br>NOTE 2: RedirectResponse may be inserted by an SCP, see clause 6.10.9.1 of TS 29.500 [4]. | | | | |

**Table 6.4.3.2.4.2.2-3: Headers supported by the 307 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | An alternative URI of the resource located on an alternative service instance within the same AMF or AMF (service) set.<br>For the case when a request is redirected to the same target resource via a different SCP, see clause 6.10.9.1 in TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target NF (service) instance ID towards which the request is redirected |

**Table 6.4.3.2.4.2.2-4: Headers supported by the 308 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | An alternative URI of the resource located on an alternative service instance within the same AMF or AMF (service) set. For the case when a request is redirected to the same target resource via a different SCP, see clause 6.10.9.1 in TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target NF (service) instance ID towards which the request is redirected |

### 6.4.3.2.4.3 Operation: provide-loc-info (POST)

#### 6.4.3.2.4.3.1 Description

This ueContextId identifies the individual ueContext resource is composed by UE's SUPI or PEI.

#### 6.4.3.2.4.3.2 Operation Definition

This operation shall support the request data structures specified in table 6.4.3.2.4.3.2-1 and the response data structure and response codes specified in table 6.4.3.2.4.3.2-2.

**Table 6.4.3.2.4.3.2-1: Data structures supported by the (POST) povideLocInfo operation Request Body**

| Data type | P | Cardinality | Description |
|---|---|---|---|
| RequestLocInfo | M | 1 | The information to request the NPLI of the UE. |

**Table 6.4.3.2.4.3.2-2: Data structures supported by the (POST) provide-loc-info operation Response Body**

| Data type | P | Cardinality | Response codes | Description |
|---|---|---|---|---|
| ProvideLocInfo | M | 1 | 200 OK | This case represents a successful query of the NPLI of the target UE, the AMF returns the related information in the response. |
| RedirectResponse | O | 0..1 | 307 Temporary Redirect | Temporary redirection.<br>(NOTE 2) |
| RedirectResponse | O | 0..1 | 308 Permanent Redirect | Permanent redirection.<br>(NOTE 2) |
| ProblemDetails | O | 0..1 | 403 Forbidden | The "cause" attribute may be used to indicate one of the following application errors:<br>- UNSPECIFIED<br><br>See table 6.4.7.3-1 for the description of these errors. |
| ProblemDetails | O | 0..1 | 404 Not Found | The "cause" attribute may be used to indicate one of the following application errors:<br>- CONTEXT NOT_FOUND<br><br>See table 6.4.7.3-1 for the description of these errors. |
| NOTE 1: The mandatory HTTP error status code for the POST method listed in Table 5.2.7.1-1 of TS 29.500 [4] also apply, with response body containing an object of ProblemDetails data type (see clause 5.2.7 of TS 29.500 [4]).<br>NOTE 2: RedirectResponse may be inserted by an SCP or SEPP, see clause 6.10.9.1 of TS 29.500 [4]. | | | | |

**Table 6.4.3.2.4.3.2-3: Headers supported by the 307 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | An alternative URI of the resource located on an alternative service instance within the same AMF or AMF (service) set. For the case when a request is redirected to the same target resource via a different SCP or SEPP, see clause 6.10.9.1 in TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target NF (service) instance ID towards which the request is redirected |

**Table 6.4.3.2.4.3.2-4: Headers supported by the 308 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | An alternative URI of the resource located on an alternative service instance within the same AMF or AMF (service) set. For the case when a request is redirected to the same target resource via a different SCP or SEPP, see clause 6.10.9.1 in TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target NF (service) instance ID towards which the request is redirected |

### 6.4.3.2.4.4 Operation: cancel-pos-info (POST)

#### 6.4.3.2.4.4.1 Description

This ueContextId identifies the individual ueContext resource and is composed by UE's SUPI or PEI.

#### 6.4.3.2.4.4.2 Operation Definition

This operation shall support the request data structures specified in table 6.4.3.2.4.4.2-1 and the response data structure and response codes specified in table 6.4.3.2.4.4.2-2.

**Table 6.4.3.2.4.4.2-1: Data structures supported by the (POST) cancel-pos-info operation Request Body**

| Data type | P | Cardinality | Description |
|---|---|---|---|
| CancelPosInfo | M | 1 | The information to identify the location session to be cancelled. |

**Table 6.4.3.2.4.4.2-2: Data structures supported by the (POST) cancel-pos-info operation Response Body**

| Data type | P | Cardinality | Response codes | Description |
|---|---|---|---|---|
| n/a | | | 204 No Content | This case represents successful cancellation of location. |
| RedirectResponse | O | 0..1 | 307 Temporary Redirect | Temporary redirection.<br>(NOTE 2) |
| RedirectResponse | O | 0..1 | 308 Permanent Redirect | Permanent redirection.<br>(NOTE 2) |
| ProblemDetails | O | 0..1 | 403 Forbidden | The "cause" attribute may be used to indicate one of the following application errors:<br>- USER_UNKNOWN<br>- LOCATION_SESSION_UNKNOWN<br>- UNSPECIFIED<br><br>See table 6.4.7.3-1 for the description of these errors. |
| ProblemDetails | O | 0..1 | 504 Gateway Timeout | The "cause" attribute may be used to indicate one of the following application errors:<br>- UNREACHABLE_USER<br>- PEER_NOT_RESPONDING<br><br>See table 6.4.7.3-1 for the description of this error. |
| NOTE 1: The mandatory HTTP error status code for the POST method listed in Table 5.2.7.1-1 of TS 29.500 [4] also apply, with response body containing an object of ProblemDetails data type (see clause 5.2.7 of TS 29.500 [4]).<br>NOTE 2: RedirectResponse may be inserted by an SCP, see clause 6.10.9.1 of TS 29.500 [4]. | | | | |

**Table 6.4.3.2.4.4.2-3: Headers supported by the 307 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | An alternative URI of the resource located on an alternative service instance within the same AMF or AMF (service) set. For the case when a request is redirected to the same target resource via a different SCP, see clause 6.10.9.1 in TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target NF (service) instance ID towards which the request is redirected |

**Table 6.4.3.2.4.4.2-4: Headers supported by the 308 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | An alternative URI of the resource located on an alternative service instance within the same AMF or AMF (service) set. For the case when a request is redirected to the same target resource via a different SCP, see clause 6.10.9.1 in TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target NF (service) instance ID towards which the request is redirected |

## 6.4.4 Custom Operations without associated resources

There are no custom operations without associated resources supported on Namf_Location service.

## 6.4.5 Notifications

### 6.4.5.1 General

This clause provides the definition of the EventNotify notification of the Namf_Location service.

**Table 6.4.5.1-1: Notifications overview**

| Notification | Callback URI | HTTP method or custom operation | Description (service operation) |
|---|---|---|---|
| Event Notify | {locationNotificationUri} | POST | |

### 6.4.5.2 Event Notify

#### 6.4.5.2.1 Description

This resource represents the callback reference of the NF Service Consumer (e.g. GMLC) to receive LCS event notify.

#### 6.4.5.2.2 Notification Definition

Callback URI: {locationNotificationUri}

See clause 5.5.2.3.1 for the description of how the AMF obtains the Callback URI of the NF Service Consumer (e.g. GMLC).

#### 6.4.5.2.3 Notification Standard Methods

##### 6.4.5.2.3.1 POST

This method sends an LCS event notify to the NF Service Consumer.

This method shall support the request data structures specified in table 6.4.5.2.3.1-1 and the response data structures and response codes specified in table 6.4.5.2.3.1-2.

**Table 6.4.5.2.3.1-1: Data structures supported by the POST Request Body**

| Data type | P | Cardinality | Description |
|---|---|---|---|
| NotifiedPosInfoExt | M | 1 | Representation of the LCS event(s) notify. |

**Table 6.4.5.2.3.1-2: Data structures supported by the POST Response Body**

| Data type | P | Cardinality | Response codes | Description |
|---|---|---|---|---|
| n/a | | | 204 No Content | This case represents a successful notification of the LCS event. |
| RedirectResponse | O | 0..1 | 307 Temporary Redirect | Temporary redirection.<br>(NOTE 2) |
| RedirectResponse | O | 0..1 | 308 Permanent Redirect | Permanent redirection.<br>(NOTE 2) |
| NOTE 1: The mandatory HTTP error status code for the POST method listed in Table 5.2.7.1-1 of TS 29.500 [4] also apply, with response body containing an object of ProblemDetails data type (see clause 5.2.7 of TS 29.500 [4]).<br>NOTE 2: RedirectResponse may be inserted by an SCP, see clause 6.10.9.1 of TS 29.500 [4]. | | | | |

**Table 6.4.5.2.3.1-3: Headers supported by the 307 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | A URI pointing to the endpoint of the NF service consumer to which the notification should be sent.<br>For the case when a request is redirected to the same target resource via a different SCP, see clause 6.10.9.1 in TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target NF (service) instance ID towards which the request is redirected |

**Table 6.4.5.2.3.1-4: Headers supported by the 308 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | A URI pointing to the endpoint of the NF service consumer to which the notification should be sent.<br>For the case when a request is redirected to the same target resource via a different SCP, see clause 6.10.9.1 in TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target NF (service) instance ID towards which the request is redirected |

## 6.4.6 Data Model

### 6.4.6.1 General

This clause specifies the application data model supported by the API.

Table 6.4.6.1-1 specifies the data types defined for the Namf_Location service based interface protocol.

**Table 6.4.6.1-1: Namf_Location specific Data Types**

| Data type | Clause defined | Description |
|---|---|---|
| RequestPosInfo | 6.4.6.2.2 | Data within Provide Positioning Information Request |
| ProvidePosInfo | 6.4.6.2.3 | Data within Provide Positioning Information Response |
| NotifiedPosInfo | 6.4.6.2.4 | Data within EventNotify notification |
| RequestLocInfo | 6.4.6.2.5 | Data within Provide Location Information Request |
| ProvideLocInfo | 6.4.6.2.6 | Data within Provide Location Information Response |
| CancelPosInfo | 6.4.6.2.7 | Data within a Cancel Location Request |
| ProblemDetailsProvidePosInfo | 6.4.6.2.8 | Detailed problems with positioning information in failure case |
| AddProvidePosInfos | 6.4.6.2.9 | Additional UE positioning information with more than one corresponding UEs |
| AddNotifiedPosInfos | 6.4.6.2.10 | Additional LCS event notify with more than one corresponding UEs. |
| ProvidePosInfoExt | 6.4.6.2.11 | Data within Provide Positioning Information Response in addition to ProvidePosInfo for the UE positioning information with more than one corresponding UEs |
| NotifiedPosInfoExt | 6.4.6.2.12 | Data within EventNotify notification in addition to NotifiedPosInfo for LCS event notify with more than one corresponding UEs. |
| LocationType | 6.4.6.3.3 | Type of location measurement requested |
| LocationEvent | 6.4.6.3.4 | Type of events initiating location procedures |
| LocationPrivacyVerResult | 6.4.6.3.5 | The result of location privacy verification by UE |
| LpHapType | 6.4.6.3.6 | Type of Low Power and/or High Accuracy Positioning |

Table 6.4.6.1-2 specifies data types re-used by the Namf_Location service based interface protocol from other specifications, including a reference to their respective specifications and when needed, a short description of their use within the Namf_Location service based interface.

**Table 6.4.6.1-2: Namf_Location re-used Data Types**

| Data type | Reference | Comments |
|---|---|---|
| Supi | TS 29.571 [6] | Subscription Permanent Identifier |
| Gpsi | TS 29.571 [6] | General Public Subscription Identifier |
| Pei | TS 29.571 [6] | Permanent Equipment Identifier |
| ExternalClientType | TS 29.572 [25] | LCS Client Type (Emergency, Lawful Interception …) |
| LocationQoS | TS 29.572 [25] | LCS QoS (accuracy, response time) |
| SupportedGADShapes | TS 29.572 [25] | LCS supported GAD shapes |
| GeographicArea | TS 29.572 [25] | Estimate of the location of the UE |
| AccuracyFulfilmentIndicator | TS 29.572 [25] | Requested accuracy was fulfilled or not |
| AgeOfLocationEstimate | TS 29.572 [25] | Age Of Location Estimate |
| PositioningMethodAndUsage | TS 29.572 [25] | Usage of each non-GANSS positioning method |
| VelocityEstimate | TS 29.572 [25] | Estimate of the velocity of the target UE |
| VelocityRequested | TS 29.572 [25] | Indication of the Velocity requirement |
| LcsPriority | TS 29.572 [25] | Priority of the LCS client |
| GnssPositioningMethodAndUsage | TS 29.572 [25] | Usage of each GANSS positioning method |
| CivicAddress | TS 29.572 [25] | Civic address |
| BarometricPressure | TS 29.572 [25] | Barometric Pressure |
| Altitude | TS 29.572 [25] | Altitude estimate of the UE |
| LocalArea | TS 29.572 [25] | Local area specified by different shape |
| MinorLocationQoS | TS 29.572 [25] | Minor Location QoS |
| Ecgi | TS 29.571 [6] | UE EUTRAN cell information |
| Ncgi | TS 29.571 [6] | UE NR cell information |
| SupportedFeatures | TS 29.571 [6] | Supported Features |
| RatType | TS 29.571 [6] | RAT type |
| TimeZone | TS 29.571 [6] | Time Zone |
| DateTime | TS 29.571 [6] | Date and Time |
| UserLocation | TS 29.571 [6] | User Location |
| LcsServiceType | TS 29.572 [25] | The LCS service type |
| LdrType | TS 29.572 [25] | The type of LDR for deferred location |
| Uri | TS 29.571 [6] | URI |
| LdrReference | TS 29.572 [25] | LDR Reference Number for deferred location |
| LirReference | TS 29.572 [25] | LIR Reference Number for immediate location |
| PeriodicEventInfo | TS 29.572 [25] | Information for periodic event reporting |
| AreaEventInfo | TS 29.572 [25] | Information for area event reporting |
| MotionEventInfo | TS 29.572 [25] | Information for motion event reporting |
| ExternalClientIdentification | TS 29.515 [46] | External LCS client identification |
| NFInstanceId | TS 29.571 [6] | Identification of an NF or AF |
| CodeWord | TS 29.515 [46] | Codeword for a 5GC-MT-LR or deferred 5GC-MT-LR |
| LMFIdentification | TS 29.572 [25] | Identification of a serving LMF for periodic or triggered location |
| TerminationCause | TS 29.572 [25] | Termination cause for a deferred location |
| UePrivacyRequirements | TS 29.515 [46] | The location related privacy requirements on UE |
| DiameterIdentity | TS 29.571 [6] | Diameter Identity |
| ProblemDetails | TS 29.571 [6] | Detailed problems in failure case |
| RedirectResponse | TS 29.571 [6] | Response body of the redirect response message. |
| E164Number | TS 29.503 [35] | The E.164 number. |
| DurationSec | TS 29.571 [6] | Duration Second |
| ReportingArea | TS 29.572 [25] | Indicates an area for event reporting |
| ReportingInd | TS 29.515 [46] | Reporting Indication |
| RangingSlResult | TS 29.572 [25] | Indicates result type for ranging and sidelink positioning |
| RelatedUe | TS 29.572 [25] | Indicates information for related UE for ranging and sidelink positioning |
| LosNlosMeasureInd | TS 29.572 [25] | LOS/NLOS measurement indication |
| IndoorOutdoorInd | TS 29.572 [25] | Indicates Indoor Outdoor Indication |
| IntegrityRequirements | TS 29.515 [46] | Integrity Requirements |
| HighAccuracyGnssMetrics | TS 29.572 [25] | High Accuracy GNSS Metrics |

| | | |
|---|---|---|
| UpLocRepInfoAf | TS 29.515 [46] | Information for the location reporting over user plane |
| MappedLocationQoSEps | TS 29.572 [25] | Mapped Location QoS for EPS |
| RangeDirection | TS 29.572 [25] | Represents the distance and direction between two points. |
| 2DRelativeLocation | TS 29.572 [25] | Represents 2D local co-ordinates with origin corresponding to another known point. |
| 3DRelativeLocation | TS 29.572 [25] | Represents 3D local co-ordinates with origin corresponding to another known point. |
| IntegrityResult | TS 29.515 [46] | Integrity Result |

### 6.4.6.2 Structured data types

#### 6.4.6.2.1 Introduction

Structured data types used in Namf_Location service are specified in this clause.

#### 6.4.6.2.2 Type: RequestPosInfo

**Table 6.4.6.2.2-1: Definition of type RequestPosInfo**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| lcsClientType | ExternalClientType | M | 1 | This IE shall contain the type of LCS client (Emergency, Lawful Interception etc.,.) issuing the location request |
| lcsLocation | LocationType | M | 1 | This IE shall contain the type of location measurement requested, such as current location,current or last known location, deferred location, etc.<br>(NOTE 2) |
| supi | Supi | C | 0..1 | If the SUPI is available, this IE shall be present. |
| gpsi | Gpsi | C | 0..1 | If the GPSI is available, this IE shall be present. |
| requestedRangingSlResult | array(RangingSlResult) | C | 1..N | This IE shall contain the type of result requested for ranging and sidelink positioning, such as absolute locations, relative locations or distances and directions related to the UEs, etc. |
| relatedUes | array(RelatedUe) | C | 1..N | This IE contains a list of the information for the related UEs for the ranging and sidelink positioning. |
| lmfId | LMFIdentification | O | 0..1 | LMF identification.<br>If present, this IE shall indicate the LMF ID that should be used to select the LMF. |
| priority | LcsPriority | O | 0..1 | If present, this IE shall contain the priority of the LCS client issuing the positioning request. |
| lcsQoS | LocationQoS | O | 0..1 | If present, this IE shall contain the quality of service requested, such as the accuracy of the positioning measurement and the response time of the positioning operation.<br><br>Multiple QoS Class (lcsQosClass sets to "MULTIPLE_QOS") shall only be used when AMF support MUTIQOS feature. |
| velocityRequested | VelocityRequested | O | 0..1 | If present, this IE shall contain an indication of whether or not the Velocity of the target UE is requested. |
| lcsSupportedGADShapes | SupportedGADShapes | O | 0..1 | If present, this IE shall contain one GAD shape supported by the LCS client. |
| additionalSuppGADShapes | array(SupportedGADShapes) | C | 1..N | Shall be absent if lcsSupportedGADShapes is absent.<br>Shall be present if the LCS client supports more than one GAD shape. |
| locationNotificationUri | Uri | O | 0..1 | The callback URI on which location change event notification is reported. |
| supportedFeatures | SupportedFeatures | C | 0..1 | This IE shall be present if at least one feature defined in clause 6.4.8 is supported. |
| oldGuami | Guami | C | 0..1 | This IE shall be present during an AMF planned removal procedure when the NF Service Consumer initiates a request towards the target AMF, for a UE associated to an AMF that is unavailable (see clause 5.21.2.2 of TS 23.501 [2]). |
| pei | Pei | C | 0..1 | This IE shall be present if supi and gpsi are not available. |
| lcsServiceType | LcsServiceType | O | 0..1 | This IE contains the LCS service type for an external client.<br>(NOTE 1) |
| ldrType | LdrType | C | 0..1 | This IE contains the type of LDR for a deferred location request. This IE shall be present when lcsLocation is set to "DEFERRED_LOCATION". |
| hgmlcCallBackURI | Uri | C | 0..1 | This IE contains the callback URI of the H-GMLC. This IE shall be present when lcsLocation is set to "DEFERRED_LOCATION".<br><br>This IE shall also be present for location service in PNI-NPN with signalling optimisation, as specified in TS 23.273 [42] clause 6.1.2. |

| | | | | |
|---|---|---|---|---|
| lirGmlcCallBackUri | Uri | C | 0..1 | This IE shall be present when the intermediateLocationInd IE is present with the value "true".<br><br>When present, this IE shall contain callback URI of the GMLC to receive the intermediate location reports. |
| ldrReference | LdrReference | C | 0..1 | This IE contains the LDR Reference Number.<br>This IE shall be present when lcsLocation is set to "DEFERRED_LOCATION".<br><br>This IE shall be present for location service in PNI-NPN with signalling optimisation, as specified in TS 23.273 [42] clause 6.1.2. |
| lirReference | LirReference | C | 0..1 | This IE shall be present when the intermediateLocationInd IE is present with the value "true".<br><br>When present, this IE shall contain the LIR Reference Number for a multiple location request |
| periodicEventInfo | PeriodicEventInfo | C | 0..1 | This IE contains information for periodic event reporting for a deferred location request. This IE shall be present when ldrType is set to "PERIODIC". |
| areaEventInfo | AreaEventInfo | C | 0..1 | This IE contains information for area event reporting for a deferred location request. This IE shall be present when ldrType is set to "ENTERING_INTO_AREA", "LEAVING_FROM_AREA" or "BEING_INSIDE_AREA". |
| motionEventInfo | MotionEventInfo | C | 0..1 | This IE contains information for motion event reporting for a deferred location request. This IE shall be present when ldrType is set to "MOTION". |
| externalClientIdentification | ExternalClientIdentification | O | 0..1 | This IE provides the external LCS client identification (e.g. the name of the LCS client).<br>(NOTE 1) |
| afID | NfInstanceId | O | 0..1 | This IE provides the identification of an AF that initiated the location request.<br>(NOTE 1) |
| codeWord | CodeWord | O | 0..1 | This IE provides a codeword for a location request which is provided by an external Client or AF and is sent to and verified by a target UE as part of privacy verification.<br>(NOTE 1) |
| uePrivacyRequirements | UePrivacyRequirements | O | 0..1 | If present, the IE provides the indication of location related notification or verification for the target UE, the indication of codeword check in UE |
| scheduledLocTime | DateTime | O | 0..1 | If present, the IE provides the scheduled UTC time that the UE needs to be located. |
| reliableLocReq | boolean | C | 0..1 | This IE shall be included with the value "true" to indicate that reliable UE location information is required, as specified in TS 33.256 [57] clause 5.3.2.<br><br>When present, this IE shall be set as following:<br>- true: the reliable UE location information is required<br>- false (default): the reliable UE location information is not required |

| | | | | |
|---|---|---|---|---|
| intermediateLocationInd | boolean | C | 0..1 | This IE shall be present with the value "true" to indicate the acceptance of intermediate location responses for the NF consumer (i.e., the GMLC), during a 5GC-MT-LR multiple location procedure for the regulatory location service (see clause 6.1.3 and clause 6.10.4 of TS 23.273 [42]).<br><br>When present, this IE shall indicate the acceptance of intermediate location response at the NF consumer:<br>- true: intermediate location response acceptable<br>- false (default): intermediate location response not acceptable |
| maxRespTime | DurationSec | C | 0..1 | This IE shall be present when the intermediateLocationInd IE is present with the value "true".<br><br>When present, this IE shall contain the maximum response time for the NF consumer to receive the FINAL location response.<br><br>The AMF may overwrite the maximum response time when passing it to the LMF, e.g., to avoid timeout of the HTTP service request. |
| ueUnawareInd | boolean | C | 0..1 | This IE shall be included and set to "true", if the UE Unaware Positioning is required, as specified in clause 6.1.1 of TS 23.273 [42].<br><br>Presence of this IE with false value shall be prohibited. |
| lpHapType | LpHapType | C | 0..1 | This IE shall be included and set to "LOW_POW_HIGH_ACCU_POS" to request low power and high accuracy positioning, as specified in clause 6.1.2 of TS 23.273 [42]. |
| evtRptAllowedAreas | array(ReportingArea) | O | 1..250 | If present, this IE shall contain the list of event report allowed areas, where UE is allowed to generate and send the event report to network during the deferred 5GC-MT-LR procedure for UE power saving purpose. |
| reportingInd | ReportingInd | O | 0..1 | This IE may be present if the evtRptAllowedAreas is present.<br><br>When present, this IE shall indicate whether the UE is allowed to generate and send the reports inside or outside the event report allowed areas:<br>- Inside reporting (default)<br>- Outside reporting<br>(see TS 23.273 [42] clause 5.14 and 6.3.1) |
| integrityRequirements | IntegrityRequirements | O | 0..1 | When present, this IE shall indicate the integrity requirements. |
| upLocRepInfoAf | UpLocRepInfoAf | C | 0..1 | This IE shall be present if the request is for the location reporting over user plane.<br>When present, the IE may include additional information for the location reporting over user plane. |
| mappedQoSEps | MappedLocationQoSEps | C | 0..1 | This IE may only be present if the Multiple QoS Class is indicated in the locationQoS IE.<br><br>When present, this IE shall indicate the mapped Location QoS applicable to EPS ("BEST_EFFORT" or "ASSURED") based on the Multiple Location QoS (see clause 6.19 of TS 23.273 [42]). |

| | | | | |
|---|---|---|---|---|
| coordinateID | integer | O | 0..1 | This IE may be present when requestedRangingSlResult indicates "ABSOLUTE_LOCATION".<br><br>When present, this IE represents a local coordinate (see clause 6.20.3 of TS 23.273 [42]). |
| | | | | |

NOTE 1: At least one of these IEs should be present when uePrivacyCallSessionUnrelatedClass indicates notification and/or verification for the target UE.
NOTE 2: If the lcsLocation IE is set to value "NOTIFICATION_VERIFICATION_ONLY", then the lcsServiceAuthInfo attribute in the uePrivacyRequirements IE, if present, shall be set to either "NOTIFICATION_ONLY" or "NOTIFICATION_AND_VERIFICATION_ONLY".

#### 6.4.6.2.3 Type: ProvidePosInfo

**Table 6.4.6.2.3-1: Definition of type ProvidePosInfo**

| Attribute name | Data type | P | Cardinality | Description | Applicability |
|---|---|---|---|---|---|
| locationEstimate | GeographicArea | O | 0..1 | If present, this IE shall contain an estimate of the location of the UE in universal coordinates and the accuracy of the estimate. | |
| localLocationEstimate | LocalArea | O | 0..1 | When present, this IE shall indicate a local area in renference system. | |
| accuracyFulfilmentIndicator | AccuracyFulfilmentIndicator | O | 0..1 | If present, this IE shall contain an indication of whether the requested accuracy (as indicated in the LcsQoS in the request message) was fulfilled or not. | |
| ageOfLocationEstimate | AgeOfLocationEstimate | O | 0..1 | If present, this IE shall contain an indication of how long ago the location estimate was obtained. | |
| timestampOfLocationEstimate | DateTime | O | 0..1 | When present, this IE shall indicate the estimated UTC time when the location estimate corresponded to the UE location (i.e. when the location estimate and the actual UE location was the same). | |
| velocityEstimate | VelocityEstimate | O | 0..1 | If present, this IE shall contain an estimate of the velocity of the target UE, composed by horizontal speed, vertical speed, and their respective uncertainty. | |
| positioningDataList | array(PositioningMethodAndUsage) | O | 0..9 | If present, this IE shall indicate the usage of each non- GANSS positioning method that was attempted to determine the location estimate, either successfully or unsuccessfully. | |
| gnssPositioningDataList | array(GnssPositioningMethodAndUsage) | O | 0..9 | If present, this IE shall indicate the usage of each GANSS positioning method that was attempted to determine the location estimate, either successfully or unsuccessfully. | |
| ecgi | Ecgi | O | 0..1 | If present, this IE shall contain the current EUTRAN cell location of the target UE as delivered by the 5G-AN. | |
| ncgi | Ncgi | O | 0..1 | If present, this IE shall contain the current NR cell location of the target UE, or the NR cell location of the relay UE if remoteUeInd IE is present with the value true, as delivered by the 5G-AN. | |
| remoteUeInd | boolean | C | 0..1 | This IE shall be present if received from LMF.<br><br>Presence of this IE with the value false shall be prohibited. | |
| targetServingNode | NfInstanceId | O | 0..1 | If present, this IE shall contain the address of the target side serving node for intra-5GS handover of an IMS Emergency Call. | |
| targetMmeName | DiameterIdentity | C | 0..1 | This IE shall be present for handover of IMS emergency call to EPS, i.e. the target node is an MME.<br><br>This IE may also be present for a handover from 5GS to EPS.<br><br>When present, this IE shall indicate the Diameter host name of the target MME. | |

| | | | | | |
|---|---|---|---|---|---|
| targetMmeRealm | DiameterIdentity | C | 0..1 | This IE shall be present for handover of IMS emergency call to EPS, i.e. the target node is an MME.<br><br>This IE may also be present for a handover from 5GS to EPS.<br><br>When present, this IE shall indicate the Diameter realm of the target MME. | |
| utranSrvccInd | boolean | C | 0..1 | This IE shall be present with value "true" for 5G-SRVCC to 3GPP UTRAN of IMS emergency call, i.e. target node is an MSC.<br><br>When present, this IE shall be set for the following value:<br>- true: IMS emergency call handover to UTRAN<br><br>- false: No IMS emergency call handover to UTRAN | |
| civicAddress | CivicAddress | O | 0..1 | If present, this IE contains a location estimate for the target UE expressed as a Civic address. | |
| barometricPressure | BarometricPressure | O | 0..1 | If present, this IE contains the barometric pressure measurement as reported by the target UE. | |
| altitude | Altitude | O | 0..1 | If present, this IE indicates the altitude of the positioning estimate. | |
| supportedFeatures | SupportedFeatures | C | 0..1 | This IE shall be present if at least one feature defined in clause 6.4.8 is supported. | |
| servingLMFIdentification | LMFIdentification | O | 0..1 | If present, this IE contains the identification of a serving LMF for periodic or triggered location | |
| locationPrivacyVerResult | LocationPrivacyVerResult | O | 0..1 | If present, this IE contains the result of location privacy verification by UE (NOTE) | |
| achievedQos | MinorLocationQoS | C | 0..1 | When present, this IE shall contain the achieved Location QoS Accuracy of the estimated location.<br><br>This IE shall be present if received from LMF. | MUTIQOS |
| directReportInd | boolean | C | 0..1 | When present, this IE shall be set for the following value:<br>- true: location determination will be sent by LMF to GMLC directly<br><br>- false (default): location determination will not be sent by LMF to GMLC directly<br><br>This IE shall be present if received from LMF. | |
| acceptedPeriodicEventInfo | PeriodicEventInfo | C | 0..1 | This IE shall be present if received from LMF.<br><br>When present, this IE shall provide the accepted periodic event reporting information. | |

| | | | | | |
|---|---|---|---|---|---|
| haGnssMetrics | HighAccuracyGnssMetrics | C | 0..1 | This IE should be included when received from LMF.<br><br>When present, this IE shall indicate the high accuracy GNSS metrics for the location estimate. | |
| indoorOutdoorInd | IndoorOutdoorInd | O | 0..1 | When present, this IE shall indicate whether the UE is indoor or outdoor. | |
| losNlosMeasureInd | LosNlosMeasureInd | O | 0..1 | When present, this IE shall indicate whether LOS measurement or NLOS measurement is used. | |
| losNlosMeasureInd | LosNlosMeasureInd | O | 0..1 | When present, this IE shall indicate whether LOS measurement or NLOS measurement is used. | |
| relatedApplicationlayerId | ApplicationlayerId | O | 0..1 | Identifies the application layer ID of the related UE for ranging and sidelink positioning, such as located UE, reference UE, etc. | Ranging_SL |
| distanceDirection | RangeDirection | O | 0..1 | When present, this IE identifies a distance and direction from a point A to a point B, comprising a distance from point A to point B, an azimuth direction from point A to point B and an elevation direction from point A to point B. | Ranging_SL |
| 2dRelativeLocation | 2DRelativeLocation | O | 0..1 | When present, this IE identifies a relative 2D location with uncertainty ellipse, characterised by a point described in 2D local co-ordinates with origin corresponding to another known point, distances r1 and r2 and an angle of orientation A. | Ranging_SL |
| 3dRelativeLocation | 3DRelativeLocation | O | 0..1 | When present, this IE identifies a relative 3D location with uncertainty ellipsoid, characterised by a point described in 3D local co-ordinates with origin corresponding to another known point, distances r1 (the "semi-major uncertainty"), r2 (the "semi-minor uncertainty") and r3 (the "vertical uncertainty") and an angle of orientation A (the "angle of the major axis"). | Ranging_SL |
| relativeVelocity | VelocityEstimate | O | 0..1 | When present, this IE identifies UE velocity relative to the UE identified with relatedApplicationlayerId. | Ranging_SL |
| integrityResult | IntegrityResult | C | 0..1 | This IE should be present when the integrity requirements are present in the request.<br><br>When present, this IE shall indicate the integrity result. | |
| NOTE: The IE may be included to indicate the result of location privacy verification by UE to (H)GMLC when a location request with notification and privacy verification only indication is sent to the serving AMF by (H)GMLC during location request procedure. | | | | | |

#### 6.4.6.2.4 Type: NotifiedPosInfo

**Table 6.4.6.2.4-1: Definition of type NotifiedPosInfo**

| Attribute name | Data type | P | Cardinality | Description | Applicability |
|---|---|---|---|---|---|
| locationEvent | LocationEvent | M | 1 | This IE shall contain the type of event that caused the location procedure to be initiated. | |
| supi | Supi | C | 0..1 | This IE shall contain the SUPI if available (see NOTE 1). | |
| gpsi | Gpsi | C | 0..1 | This IE shall contain the GPSI if available (see NOTE 1). | |
| pei | Pei | C | 0..1 | This IE shall contain the PEI if available (see NOTE 1). | |
| locationEstimate | GeographicArea | O | 0..1 | If present, this IE shall contain an estimate of the location of the UE in universal coordinates and the accuracy of the estimate. | |
| localLocationEstimate | LocalArea | O | 0..1 | When present, this IE shall indicate a local area in renference system. | |
| ageOfLocationEstimate | AgeOfLocationEstimate | O | 0..1 | If present, this IE shall contain an indication of how long ago the location estimate was obtained. | |
| timestampOfLocationEstimate | DateTime | O | 0..1 | When present, this IE shall indicate the estimated UTC time when the location estimate corresponded to the UE location (i.e. when the location estimate and the actual UE location was the same). | |
| velocityEstimate | VelocityEstimate | O | 0..1 | If present, this IE shall contain an estimate of the velocity of the target UE, composed by horizontal speed, vertical speed, and their respective uncertainty. | |
| positioningDataList | array(PositioningMethodAndUsage) | O | 0..9 | If present, this IE shall indicate the usage of each non-GANSS positioning method that was attempted to determine the location estimate, either successfully or unsuccessfully. | |
| gnssPositioningDataList | array(GnssPositioningMethodAndUsage) | O | 0..9 | If present, this IE shall indicate the usage of each GANSS positioning method that was attempted to determine the location estimate, either successfully or unsuccessfully. | |
| ecgi | Ecgi | O | 0..1 | If present, this IE shall contain the current EUTRAN cell location of the target UE as delivered by the 5G-AN. | |
| ncgi | Ncgi | O | 0..1 | If present, this IE shall contain the current NR cell location of the target UE, or the NR cell location of the relay UE if remoteUeInd IE is present with the value true, as delivered by the 5G-AN. | |
| remoteUeInd | boolean | C | 0..1 | This IE shall be present if received from LMF.<br><br>Presence of this IE with the value false shall be prohibited. | |
| servingNode | NfInstanceId | O | 0..1 | If present, this IE shall contain the address of the serving node. For intra-5GS handover of an IMS Emergency Call, this IE shall contain the address of the target side serving node. For mobility of a UE with periodic or triggered location, this IE shall contain the address of the new serving node, if available. | |
| targetMmeName | DiameterIdentity | C | 0..1 | This IE shall be present for handover of IMS emergency call to EPS, i.e. the target node is an MME.<br><br>When present, this IE shall indicate the Diameter host name of the target MME. | |

| | | | | | |
|---|---|---|---|---|---|
| targetMmeRealm | DiameterIdentity | C | 0..1 | This IE shall be present for handover of IMS emergency call to EPS, i.e. the target node is an MME.<br><br>When present, this IE shall indicate the Diameter realm of the target MME. | |
| utranSrvccInd | boolean | C | 0..1 | This IE shall be present with value "true" for 5G-SRVCC to 3GPP UTRAN of IMS emergency call, i.e. target node is an MSC.<br><br>When present, this IE shall be set for the following value:<br>true: IMS emergency call handover to UTRAN<br><br>false: No IMS emergency call handover to UTRAN | |
| civicAddress | CivicAddress | O | 0..1 | If present, this IE contains a location estimate for the target UE expressed as a Civic address. | |
| barometricPressure | BarometricPressure | O | 0..1 | If present, this IE contains the barometric pressure measurement as reported by the target UE. | |
| altitude | Altitude | O | 0..1 | If present, this IE indicates the altitude of the positioning estimate. | |
| hgmlcCallBackURI | Uri | C | 0..1 | This IE contains the callback URI of the H-GMLC<br>This IE shall be included for a locationEvent related to deferred location when the consumer NF is not the H-GMLC. | |
| ldrReference | LdrReference | C | 0..1 | This IE contains an LDR Reference.<br>This IE shall be included for a locationEvent related to deferred location. | |
| servingLMFIdentification | LMFIdentification | C | 0..1 | This IE contains the identification of a serving LMF and shall be included for a locationEvent related to deferred location with periodic or triggered location if a serving LMF is used. | |
| terminationCause | TerminationCause | C | 0..1 | This IE indicates a reason for termination and shall be included for a locationEvent related to deferred location if deferred location has been terminated. | |
| achievedQos | MinorLocationQoS | O | 0..1 | When present, this IE shall contain the achieved Location QoS Accuracy of the estimated location.<br><br>This IE shall be present if received from LMF. | MUTIQOS |
| mscServerId | E164Number | O | 0..1 | This IE may be sent from AMF to GMLC, during a 5G-SRVCC handover from NG-RAN to UTRAN procedure.<br>When present, it shall contain the international E.164 number of the MSC Server selected by the MME_SRVCC. | |
| haGnssMetrics | HighAccuracyGnssMetrics | C | 0..1 | This IE should be included when received from LMF.<br><br>When present, this IE shall indicate the high accuracy GNSS metrics for the location estimate. | |
| indoorOutdoorInd | IndoorOutdoorInd | O | 0..1 | When present, this IE shall indicate whether the UE is indoor or outdoor. | |
| relatedApplicationlayerId | ApplicationlayerId | O | 0..1 | Identifies the application layer ID of the related UE for ranging and sidelink positioning, such as located UE, reference UE, etc. | Ranging_SL |

| | | | | | |
|---|---|---|---|---|---|
| distanceDirection | RangeDirection | O | 0..1 | When present, this IE identifies a distance and direction from a point A to a point B, comprising a distance from point A to point B, an azimuth direction from point A to point B and an elevation direction from point A to point B. | Ranging_SL |
| 2dRelativeLocation | 2DRelativeLocation | O | 0..1 | When present, this IE identifies a relative 2D location with uncertainty ellipse, characterised by a point described in 2D local co-ordinates with origin corresponding to another known point, distances r1 and r2 and an angle of orientation A. | Ranging_SL |
| 3dRelativeLocation | 3DRelativeLocation | O | 0..1 | When present, this IE identifies a relative 3D location with uncertainty ellipsoid, characterised by a point described in 3D local co-ordinates with origin corresponding to another known point, distances r1 (the "semi-major uncertainty"), r2 (the "semi-minor uncertainty") and r3 (the "vertical uncertainty") and an angle of orientation A (the "angle of the major axis"). | Ranging_SL |
| relativeVelocity | VelocityEstimate | O | 0..1 | When present, this IE identifies UE velocity relative to the UE identified with relatedApplicationlayerId. | Ranging_SL |
| integrityResult | IntegrityResult | C | 0..1 | This IE should be present when the integrity requirements are present in the request.<br><br>When present, this IE shall indicate the integrity result. | |
| NOTE 1: At least one of these IEs shall be present in the message. | | | | | |

#### 6.4.6.2.5 Type: RequestLocInfo

**Table 6.4.6.2.5-1: Definition of type RequestLocInfo**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| req5gsLoc | boolean | C | 0..1 | This IE shall be present and set to "true", if 5GS location information is requested in NPLI.<br><br>When present, the IE shall be set as following:<br>- true: the location of the UE is requested<br>- false (default): the location of the UE is not requested |
| reqCurrentLoc | boolean | C | 0..1 | This IE may be present if 5GS location information is requested in NPLI.<br><br>When present, the IE shall be set as following:<br>- true: the current location of the UE is requested<br>- false (default): the current location of the UE is not requested |
| reqRatType | boolean | C | 0..1 | This IE shall be present and set to "true", if the RAT Type of the UE is requested in NPLI.<br><br>When present, the IE shall be set as following:<br>- true: the RAT type of the UE is requested<br>- false (default): the RAT type of the UE is not requested |
| reqTimeZone | boolean | C | 0..1 | This IE shall be present and set to "true, if the local timezone of the UE is requested in NPLI.<br><br>When present, the IE shall be set as following:<br>- true: the local timezone of the UE is requested<br>- false (default): the local timezone of the UE is not requested. |
| supportedFeatures | SupportedFeatures | C | 0..1 | This IE shall be present if at least one feature defined in clause 6.4.8 is supported. |
| oldGuami | Guami | C | 0..1 | This IE shall be present during an AMF planned removal procedure when the NF Service Consumer initiates a request towards the target AMF, for a UE associated to an AMF that is unavailable (see clause 5.21.2.2 of TS 23.501 [2]). |

#### 6.4.6.2.6 Type: ProvideLocInfo

**Table 6.4.6.2.6-1: Definition of type ProvideLocInfo**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| currentLoc | boolean | C | 0..1 | This IE shall be present, if the 5GS location information is requested by the NF Service consumer.<br><br>When present, this IE shall be set as following:<br>- true: the current location of the UE is returned<br><br>- false: the last known location of the UE is returned. |
| location | UserLocation | O | 0..1 | If present, this IE shall contain the location information of the UE.<br><br>This IE shall convey exactly one of the following:<br>- E-UTRA user location<br>- NR user location<br>- Non-3GPP access user location.<br><br>If the additionalLocation IE is present, this IE shall contain either an E-UTRA user location or NR user location. |
| additionalLocation | UserLocation | O | 0..1 | This IE shall be present if the "location IE" is present and the AMF reports both a 3GPP user location and a non-3GPP access user location.<br><br>When present, this IE shall convey the non-3GPP access user location. |
| geoInfo | GeographicArea | O | 0..1 | If present, this IE shall contain the geographical information of the UE (see NOTE 1). |
| locationAge | AgeOfLocationEstimate | O | 0..1 | If present, this IE shall contain the age of the location information (see NOTE 2). |
| ratType | RatType | O | 0..1 | If present, this IE shall contain the current RAT type of the UE. |
| timezone | TimeZone | O | 0..1 | If present, this IE shall contain the local time zone of the UE. |
| supportedFeatures | SupportedFeatures | C | 0..1 | This IE shall be present if at least one feature defined in clause 6.4.8 is supported. |
| NOTE 1: If geographical information is returned by the AMF, it shall be encoded in the "geoInfo" attribute and the "geographicalInformation" attribute within the "location" attribute shall not be used.<br>NOTE 2: If age of location estimate is returned by the AMF, it may be provided either in the "locationAge" attribute or in the "ageOfLocationInformation" attribute within the "location" attribute. | | | | |

#### 6.4.6.2.7 Type: CancelPosInfo

**Table 6.4.6.2.7-1: Definition of type CancelPosInfo**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| supi | Supi | M | 1 | SUPI |
| hgmlcCallBackURI | Uri | M | 1 | Callback URI of the H-GMLC |
| ldrReference | LdrReference | M | 1 | LDR Reference |
| servingLMFIdentification | LMFIdentification | C | 0..1 | Serving LMF identification. This IE shall be included if available. |
| supportedFeatures | SupportedFeatures | C | 0..1 | This IE shall be present if at least one feature defined in clause 6.4.8 is supported. |

#### 6.4.6.2.8 Type: ProblemDetailsProvidePosInfo

**Table 6.4.6.2.8-1: Definition of type ProblemDetailsProvidePosInfo as a list of to be combined data types**

| Data type | Cardinality | Description | Applicability |
|---|---|---|---|
| ProblemDetails | 1 | Detail information of the problem | |
| ProvidePosInfo | 1 | Additional information to be returned in error response. (See clause 6.19.1 of 3GPP 23.273 [42]). | |

#### 6.4.6.2.9 Type: AddProvidePosInfos

**Table 6.1.6.2.9-1: Definition of type AddProvidePosInfos**

| Attribute name | Data type | P | Cardinality | Description | Applicability |
|---|---|---|---|---|---|
| addProvidePosInfos | array(ProvidePosInfo) | O | 1..N | Contains a set of ProvidePosInfo. | |

#### 6.4.6.2.10 Type: AddNotifiedPosInfos

**Table 6.1.6.2.10-1: Definition of type AddNotifiedPosInfos**

| Attribute name | Data type | P | Cardinality | Description | Applicability |
|---|---|---|---|---|---|
| addNotifiedPosInfos | array(NotifiedPosInfo) | O | 1..N | Contains a set of NotifiedPosInfo. | |

#### 6.4.6.2.11 Type: ProvidePosInfoExt

**Table 6.4.6.2.11-1: Definition of type ProvidePosInfoExt as a list of data types to be combined**

| Data type | Cardinality | Description | Applicability |
|---|---|---|---|
| ProvidePosInfo | 1 | Positioning information | |
| AddProvidePosInfos | 1 | Additional positioning information | |

#### 6.4.6.2.12 Type: NotifiedPosInfoExt

**Table 6.4.6.2.12-1: Definition of type NotifiedPosInfoExt as a list of data types to be combined data**

| Data type | Cardinality | Description | Applicability |
|---|---|---|---|
| NotifiedPosInfo | 1 | Positioning event notify | |
| AddNotifiedPosInfos | 1 | Additional positioning event(s) notify | |

### 6.4.6.3 Simple data types and enumerations

#### 6.4.6.3.1 Introduction

This clause defines simple data types and enumerations that can be referenced from data structures defined in the previous clauses.

#### 6.4.6.3.2 Simple data types

The simple data types defined in table 6.4.6.3.2-1 shall be supported.

**Table 6.4.6.3.2-1: Simple data types**

| Type Name | Type Definition | Description |
|---|---|---|
| | | |

### 6.4.6.3.3 Enumeration: LocationType

The enumeration LocationType represents the type of location measurement requested.

**Table 6.4.6.3.3-1: Enumeration LocationType**

| Enumeration value | Description |
|---|---|
| "CURRENT_LOCATION" | This value indicates that the current location of the target UE is required. |
| "CURRENT_OR_LAST_KNOWN_LOCATION" | This value indicates that the current location or last known location of the target UE is required. |
| "NOTIFICATION_VERIFICATION_ONLY" | This value indicates that notification or verification of location by the target UE is required but a location estimate shall not be obtained. |
| "DEFERRED_LOCATION" | Deferred Location Request |

### 6.4.6.3.4 Enumeration: LocationEvent

The enumeration LocationEvent represents the type of events initiating location procedures.

**Table 6.4.6.3.4-1: Enumeration LocationEvent**

| Enumeration value | Description |
|---|---|
| "EMERGENCY_CALL_ORIGINATION" | Emergency session initiation |
| "EMERGENCY_CALL_RELEASE" | Emergency session termination |
| "EMERGENCY_CALL_HANDOVER" | Handover of an Emergency session |
| "ACTIVATION_OF_DEFERRED_LOCATION" | Confirmation of activation of periodic or triggered location in the target UE |
| "UE_MOBILITY_FOR_DEFERRED_LOCATION" | Mobility of the target UE to a different NF |
| "CANCELLATION_OF_DEFERRED_LOCATION" | Cancellation of a deferred location request |

### 6.4.6.3.5 Enumeration: LocationPrivacyVerResult

The enumeration LocationPrivacyVerResult represents the type of the result of location privacy verification or represents the type of the result of ranging and sidelink positioning privacy verification by UE.

**Table 6.4.6.3.5-1: Enumeration LocationPrivacyVerResult**

| Enumeration value | Description |
|---|---|
| "LOCATION_ALLOWED" | Location is allowed by UE |
| "LOCATION_NOT_ALLOWED" | Location is not allowed by UE |
| "RESPONSE_TIME_OUT" | UE response times out |
| "RANGING_ALLOWED" | Ranging and sidelink positioning is allowed by UE |
| "RANGING_NOT_ALLOWED" | Ranging and sidelink positioning is not allowed by UE |

#### 6.4.6.3.6 Enumeration: LpHapType

The enumeration LpHapType represents the type of the Low Power and/or High Accuracy Positioning.

**Table 6.4.6.3.6-1: Enumeration LpHapType**

| Enumeration value | Description |
|---|---|
| "LOW_POW_HIGH_ACCU_POS" | Low Power and High Accuracy Positioning |

## 6.4.7 Error Handling

### 6.4.7.1 General

HTTP error handling shall be supported as specified in clause 5.2.4 of TS 29.500 [4].

### 6.4.7.2 Protocol Errors

Protocol Error Handling shall be supported as specified in clause 5.2.7 of TS 29.500 [4].

### 6.4.7.3 Application Errors

The common application errors defined in the Table 5.2.7.2-1 in TS 29.501 [5] may also be used for the Namf_Location service, and the following application errors listed in Table 6.4.7.3-1 are specific for the Namf_Location service.

**Table 6.4.7.3-1: Application errors**

| Application Error | HTTP status code | Description |
|---|---|---|
| USER_UNKNOWN | 403 Forbidden | The user is unknown. |
| DETACHED_USER | 403 Forbidden | The user is detached (i.e. it is in RM-DEREGISTERED state) in the AMF. |
| POSITIONING_DENIED | 403 Forbidden | The positioning procedure was denied. |
| UNSPECIFIED | 403 Forbidden | The request is rejected due to unspecified reasons. |
| LOCATION_SESSION_UNKNOWN | 403 Forbidden | The location session is unknown. |
| REQUESTED_LMF_NOT_AVAILABLE | 403 Forbidden | The request is rejected due to the AMF not being able to access the requested LMF. |
| CONTEXT_NOT_FOUND | 404 Not Found | The requested UE Context does not exist in the AMF. |
| HO_TO_EPS | 409 Conflict | The request is rejected due to a handover from 5GS to EPS. |
| POSITIONING_FAILED | 500 Internal Server Error | The positioning procedure failed. |
| UNREACHABLE_USER | 504 Gateway Timeout | The user could not be reached in order to perform positioning procedure. |
| PEER_NOT_RESPONDING | 504 Gateway Timeout | No response is received from a remote peer, e.g. from the LMF. |

## 6.4.8 Feature Negotiation

The feature negotiation mechanism specified in clause 6.6 of TS 29.500 [4] shall be used to negotiate the features applicable between the AMF and the NF Service Consumer, for the Namf_Location service, if any.

The NF Service Consumer shall indicate the features it supports for the Namf_Location service, if any, by including the supportedFeatures attribute in content of the HTTP Request Message for following service operations:

- ProvidePositioningInfo, as specified in clause 5.5.2.2;
- ProvideLocationInfo, as specified in clause 5.5.2.4;
- CancelLocation, as specified in clause 5.5.2.5

The AMF shall determine the supported features for the service operations as specified in clause 6.6 of TS 29.500 [4] and shall indicate the supported features by including the supportedFeatures attribute in content of the HTTP response for the service operation.

The syntax of the supportedFeatures attribute is defined in clause 5.2.2 of TS 29.571 [6].

The following features are defined for the Namf_Location service.

**Table 6.4.8-1: Features of supportedFeatures attribute used by Namf_Location service**

| Feature Number | Feature | M/O | Description |
|---|---|---|---|
| 1 | ES3XX | M | Extended Support of HTTP 307/308 redirection<br><br>An NF Service Consumer (e.g. GMLC) that supports this feature shall support handling of HTTP 307/308 redirection for any service operation of the Namf_Location service. An NF Service Consumer that does not support this feature does only support HTTP redirection as specified for 3GPP Release 15. |
| 2 | MUTIQOS | O | Support of Multiple Location QoSes.<br><br>This feature bit indicates the support of more than one Location QoSes during consuming location service. |
| 3 | Ranging_ SL | O | Support of Ranging and Sidelink Positioning<br><br>This feature indicates the support of Ranging and Sidelink Positioning. |
| 4 | POSGPSI | M | Provided Positioning Information for GPSI<br><br>An AMF supporting this feature shall allow the NF Service Consumer (e.g. GMLC) to retrieve the UE positioning information using the GPSI of the UE. |
| Feature number: The order number of the feature within the supportedFeatures attribute (starting with 1).<br>Feature: A short name that can be used to refer to the bit and to the feature.<br>M/O: Defines if the implementation of the feature is mandatory ("M") or optional ("O").<br>Description: A clear textual description of the feature. | | | |

## 6.4.9 Security

As indicated in TS 33.501 [27], the access to the Namf_Location API may be authorized by means of the OAuth2 protocol (see IETF RFC 6749 [28]), using the "Client Credentials" authorization grant, where the NRF (see TS 29.510 [29]) plays the role of the authorization server.

If Oauth2 authorization is used, an NF Service Consumer, prior to consuming services offered by the Namf_Location API, shall obtain a "token" from the authorization server, by invoking the Access Token Request service, as described in TS 29.510 [29], clause 5.4.2.2.

NOTE: When multiple NRFs are deployed in a network, the NRF used as authorization server is the same NRF that the NF Service Consumer used for discovering the Namf_Location service.

The Namf_Location API defines scopes for OAuth2 authorization as specified in TS 33.501 [27]; it defines a single scope consisting on the name of the service (i.e., "namf-loc"), and it does not define any additional scopes at resource or operation level.

## 6.4.10 HTTP redirection

An HTTP request may be redirected to a different AMF service instance, within the same AMF or a different AMF of an AMF set, e.g. when an AMF service instance is part of an AMF (service) set or when using indirect communications (see TS 29.500 [4]). See the ES3XX feature in clause 6.4.8.

An SCP that reselects a different AMF producer instance will return the NF Instance ID of the new AMF producer instance in the 3gpp-Sbi-Producer-Id header, as specified in clause 6.10.3.4 of TS 29.500 [4].

If an AMF within an AMF set redirects a service request to a different AMF of the set using an 307 Temporary Redirect or 308 Permanent Redirect status code, the identity of the new AMF towards which the service request is redirected

shall be indicated in the 3gpp-Sbi-Target-Nf-Id header of the 307 Temporary Redirect or 308 Permanent Redirect response as specified in clause 6.10.9.1 of TS 29.500 [4].

# 6.5 Namf_MBSBroadcast Service API

## 6.5.1 API URI

The Namf_MBSBroadcast service shall use the Namf_MBSBroadcast API.

The API URI of the Namf_MBSBroadcast API shall be:

**{apiRoot}/<apiName>/<apiVersion>**

The request URI used in HTTP requests from the NF service consumer towards the NF service producer shall have the Resource URI structure defined in clause 4.4.1 of TS 29.501 [5], i.e.:

**{apiRoot}/<apiName>/<apiVersion>/<apiSpecificResourceUriPart>**

with the following components:

- The {apiRoot} shall be set as described in TS 29.501 [5].
- The <apiName> shall be "namf-mbs-bc".
- The <apiVersion> shall be "v1".
- The <apiSpecificResourceUriPart> shall be set as described in clause 6.5.3.

## 6.5.2 Usage of HTTP

### 6.5.2.1 General

HTTP/2, as defined in IETF RFC 9113 [19], shall be used as specified in clause 5 of TS 29.500 [4].

HTTP/2 shall be transported as specified in clause 5.3 of TS 29.500 [4].

HTTP messages and bodies for the Namf_MBSBroadcast service shall comply with the OpenAPI [23] specification contained in Annex A.

### 6.5.2.2 HTTP standard headers

#### 6.5.2.2.1 General

The usage of HTTP standard headers shall be supported as specified in clause 5.2.2 of TS 29.500 [4].

#### 6.5.2.2.2 Content type

The following content types shall be supported:

- JSON, as defined in IETF RFC 8259 [8], shall be used as content type of the HTTP bodies specified in the present specification as indicated in clause 5.4 of TS 29.500 [4].
- The Problem Details JSON Object (IETF RFC 9457 [36]). The use of the Problem Details JSON object in a HTTP response body shall be signalled by the content type "application/problem+json".

Multipart messages shall also be supported (see clause 6.5.2.4) using the content type "multipart/related", comprising:

- one JSON body part with the "application/json" content type; and

- one or more binary body parts with 3gpp vendor specific content subtypes.

The 3gpp vendor specific content subtypes defined in Table 6.5.2.2.2-1 shall be supported.

**Table 6.5.2.2.2-1: 3GPP vendor specific content subtypes**

| content subtype | Description |
|---|---|
| vnd.3gpp.ngap | Binary encoded content, encoding NG Application Protocol (NGAP) IEs, as specified in clause 9.3 of TS 38.413 [9] (ASN.1 encoded). |
| NOTE: Using 3GPP vendor content subtypes allows to describe the nature of the opaque content (i.e. NGAP information) without having to rely on metadata in the JSON content. | |

See clause 6.5.2.4 for the binary contents supported in the binary body part of multipart messages.

### 6.5.2.3 HTTP custom headers

#### 6.5.2.3.1 General

In this release of this specification, no custom headers specific to the Namf_MBSBroadcast service are defined. For 3GPP specific HTTP custom headers used across all service-based interfaces, see clause 5.2.3 of TS 29.500 [4].

### 6.5.2.4 HTTP multipart messages

HTTP multipart messages shall be supported, to transfer opaque N2 Information in the following service operations (and HTTP messages):

- ContextCreate Request and Response (POST);
- ContextUpdate Request and Response (POST);
- ContextStatusNotify service operation (POST).

HTTP multipart messages shall include one JSON body part and one or more binary body parts comprising:

- N2 payload (see clause 6.1.6.4).

The JSON body part shall be the "root" body part of the multipart message. It shall be encoded as the first body part of the multipart message. The "Start" parameter does not need to be included.

The multipart message shall include a "type" parameter (see IETF RFC 2387 [9]) specifying the media type of the root body part, i.e. "application/json".

NOTE: The "root" body part (or "root" object) is the first body part the application processes when receiving a multipart/related message, see IETF RFC 2387 [9]. The default root is the first body within the multipart/related message. The "Start" parameter indicates the root body part, e.g. when this is not the first body part in the message.

For each binary body part in a HTTP multipart message, the binary body part shall include a Content-ID header (see IETF RFC 2045 [10]), and the JSON body part shall include an attribute, defined with the RefToBinaryData type, that contains the value of the Content-ID header field of the referenced binary body part.

## 6.5.3 Resources

### 6.5.3.1 Overview

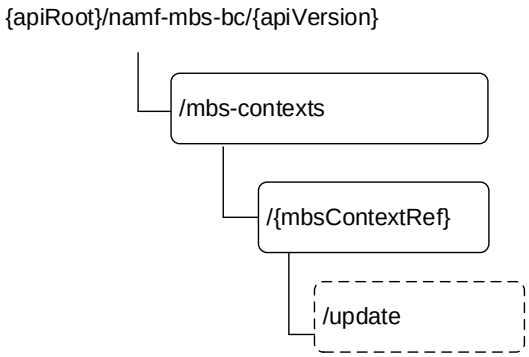



**Figure 6.5.3.1-1: Resource URI structure of the Namf_MBSBroadcast Service API**

Table 6.5.3.1-1 provides an overview of the resources and applicable HTTP methods.

**Table 6.5.3.1-1: Resources and methods overview**

| Resource name | Resource URI | HTTP method or custom operation | Description (service operation) |
|---|---|---|---|
| Broadcast MBS session contexts collection | /mbs-contexts | POST | ContextCreate |
| Individual broadcast MBS session context | /mbs-contexts/{mbsContextRef}/update | update (POST) | ContextUpdate |
| | /mbs-contexts/{mbsContextRef} | DELETE | ContextRelease |

### 6.5.3.2 Resource: Broadcast MBS session contexts collection

#### 6.5.3.2.1 Description

This resource represents a collection of Broadcast MBS session contexts created by NF service consumers of the Namf_MBSBroadcast service.

This resource is modelled as the Collection resource archetype (see clause C.2 of TS 29.501 [5]).

#### 6.5.3.2.2 Resource Definition

Resource URI: **{apiRoot}/namf-mbs-bc/<apiVersion>/mbs-contexts**

This resource shall support the resource URI variables defined in table 6.5.3.2.2-1.

**Table 6.5.3.2.2-1: Resource URI variables for this resource**

| Name | Data type | Definition |
|---|---|---|
| apiRoot | string | See clause 6.5.1 |
| apiVersion | string | See clause 6.5.1. |

#### 6.5.3.2.3 Resource Standard Methods

##### 6.5.3.2.3.1 POST

This method shall support the URI query parameters specified in table 6.5.3.2.3.1-1.

**Table 6.5.3.2.3.1-1: URI query parameters supported by the POST method on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| n/a | | | | |

This method shall support the request data structures specified in table 6.5.3.2.3.1-2 and the response data structures and response codes specified in table 6.5.3.2.3.1-3.

**Table 6.5.3.2.3.1-2: Data structures supported by the POST Request Body on this resource**

| Data type | P | Cardinality | Description |
|---|---|---|---|
| ContextCreateReqData | M | 1 | Data within ContextCreate Request |

**Table 6.5.3.2.3.1-3: Data structures supported by the POST Response Body on this resource**

| Data type | P | Cardinality | Response codes | Description |
|---|---|---|---|---|
| ContextCreateRspData | M | 1 | 201 Created | Successful creation of a broadcast MBS context |
| RedirectResponse | O | 0..1 | 307 Temporary Redirect | Temporary redirection.<br>(NOTE 2) |
| RedirectResponse | O | 0..1 | 308 Permanent Redirect | Permanent redirection.<br>(NOTE 2) |
| NOTE 1: The mandatory HTTP error status code for the POST method listed in Table 5.2.7.1-1 of TS 29.500 [4] also apply, with response body containing an object of ProblemDetails data type (see clause 5.2.7 of TS 29.500 [4]).<br>NOTE 2: RedirectResponse may be inserted by an SCP, see clause 6.10.9.1 of TS 29.500 [4]. | | | | |

**Table 6.5.3.2.3.1-4: Headers supported by the 201 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | Contains the URI of the newly created resource, according to the structure: {apiRoot}/namf-mbs-bc/<apiVersion>/mbs-contexts/{mbsContextRef} |

**Table 6.5.3.2.3.1-5: Headers supported by the 307 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | An alternative URI of the resource located on an alternative service instance within the same AMF or AMF (service) set.<br>For the case when a request is redirected to the same target resource via a different SCP, see clause 6.10.9.1 in TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target NF (service) instance ID towards which the request is redirected |

**Table 6.5.3.2.3.1-6: Headers supported by the 308 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | An alternative URI of the resource located on an alternative service instance within the same AMF or AMF (service) set. For the case when a request is redirected to the same target resource via a different SCP, see clause 6.10.9.1 in TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target NF (service) instance ID towards which the request is redirected |

### 6.5.3.2.4 Resource Custom Operations

None.

## 6.5.3.3 Resource: Individual broadcast MBS session context

### 6.5.3.3.1 Description

This resource represents an Individual broadcast MBS session context created by an NF service consumer of the Namf_MBSBroadcast service.

This resource is modelled as the Document resource archetype (see clause C.2 of TS 29.501 [5]).

### 6.5.3.3.2 Resource Definition

Resource URI: **{apiRoot}/namf-mbs-bc/<apiVersion>/mbs-contexts/{mbsContextRef}**

This resource shall support the resource URI variables defined in table 6.5.3.3.2-1.

**Table 6.5.3.3.2-1: Resource URI variables for this resource**

| Name | Data type | Definition |
|---|---|---|
| apiRoot | string | See clause 6.5.1 |
| apiVersion | string | See clause 6.5.1 |
| mbsContextRef | string | String identifying an individual broadcast MBS session context |

### 6.5.3.3.3 Resource Standard Methods

#### 6.5.3.3.3.1 DELETE

This method shall support the URI query parameters specified in table 6.5.3.3.3.1-1.

**Table 6.5.3.3.3.1-1: URI query parameters supported by the DELETE method on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| n/a | | | | |

This method shall support the request data structures specified in table 6.5.3.3.3.1-2 and the response data structures and response codes specified in table 6.5.3.3.3.1-3.

**Table 6.5.3.3.3.1-2: Data structures supported by the DELETE Request Body on this resource**

| Data type | P | Cardinality | Description |
|---|---|---|---|
| n/a | | | |

**Table 6.5.3.3.3.1-3: Data structures supported by the DELETE Response Body on this resource**

| Data type | P | Cardinality | Response codes | Description |
|---|---|---|---|---|
| n/a | | | 204 No Content | Successful deletion of a broadcast MBS context |
| RedirectResponse | O | 0..1 | 307 Temporary Redirect | Temporary redirection.<br>(NOTE 2) |
| RedirectResponse | O | 0..1 | 308 Permanent Redirect | Permanent redirection.<br>(NOTE 2) |
| NOTE 1: The mandatory HTTP error status code for the POST method listed in Table 5.2.7.1-1 of TS 29.500 [4] also apply, with response body containing an object of ProblemDetails data type (see clause 5.2.7 of TS 29.500 [4]).<br>NOTE 2: RedirectResponse may be inserted by an SCP, see clause 6.10.9.1 of TS 29.500 [4]. | | | | |

**Table 6.5.3.3.3.1-4: Headers supported by the 307 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | An alternative URI of the resource located on an alternative service instance within the same AMF or AMF (service) set.<br>For the case when a request is redirected to the same target resource via a different SCP, see clause 6.10.9.1 in TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target NF (service) instance ID towards which the request is redirected |

**Table 6.5.3.3.3.1-5: Headers supported by the 308 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | An alternative URI of the resource located on an alternative service instance within the same AMF or AMF (service) set.<br>For the case when a request is redirected to the same target resource via a different SCP, see clause 6.10.9.1 in TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target NF (service) instance ID towards which the request is redirected |

### 6.5.3.3.4 Resource Custom Operations

#### 6.5.3.2.4.1 Overview

**Table 6.5.3.2.4.1-1: Custom operations**

| Operation Name | Custom operaration URI | Mapped HTTP method | Description |
|---|---|---|---|
| update | /mbscontexts/{mbsContextRef}/update | POST | ContextUpdate service operation |

#### 6.5.3.2.4.2 Operation: update (POST)

##### 6.5.3.2.4.2.1 Description

This {mbsContextRef} identifies the individual broadcast MBS session context to be updated.

##### 6.5.3.2.4.2.2 Operation Definition

This operation shall support the request data structures specified in table 6.5.3.2.4.2.2-1 and the response data structure and response codes specified in table 6.5.3.2.4.2.2-2.

**Table 6.5.3.2.4.2.2-1: Data structures supported by (POST) the update operation Request Body**

| Data type | P | Cardinality | Description |
|---|---|---|---|
| ContextUpdateReqData | M | 1 | Data within the ContextUpdate Request |

**Table 6.5.3.2.4.2.2-2: Data structures supported by the (POST) update operation Response Body**

| Data type | P | Cardinality | Response codes | Description |
|---|---|---|---|---|
| ContextUpdateRspData | M | 1 | 200 OK | Successful update of the broadcast MBS session context, with content in the response |
| n/a | | | 204 No Content | Successful update of the broadcast MBS session context, without content in the response |
| RedirectResponse | O | 0..1 | 307 Temporary Redirect | Temporary redirection.<br>(NOTE 2) |
| RedirectResponse | O | 0..1 | 308 Permanent Redirect | Permanent redirection.<br>(NOTE 2) |
| NOTE 1: The mandatory HTTP error status code for the POST method listed in Table 5.2.7.1-1 of TS 29.500 [4] also apply, with response body containing an object of ProblemDetails data type (see clause 5.2.7 of TS 29.500 [4]).<br>NOTE 2: RedirectResponse may be inserted by an SCP, see clause 6.10.9.1 of TS 29.500 [4]. | | | | |

**Table 6.5.3.2.4.2.2-3: Headers supported by the 307 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | An alternative URI of the resource located on an alternative service instance within the same AMF or AMF (service) set.<br>For the case when a request is redirected to the same target resource via a different SCP, see clause 6.10.9.1 in TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target NF (service) instance ID towards which the request is redirected |

**Table 6.5.3.2.4.2.2-4: Headers supported by the 308 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | An alternative URI of the resource located on an alternative service instance within the same AMF or AMF (service) set.<br>For the case when a request is redirected to the same target resource via a different SCP, see clause 6.10.9.1 in TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target NF (service) instance ID towards which the request is redirected |

## 6.5.4 Custom Operations without associated resources

There are no custom operations without associated resources supported by the Namf_MBSBroadcast Service in this release of the specification.

## 6.5.5 Notifications

### 6.5.5.1 General

This clause specifies the notifications provided by the Namf_MBSBroadcast service.

**Table 6.5.5.1-1: Notifications overview**

| Notification | Callback URI | HTTP method or custom operation | Description (service operation) |
|---|---|---|---|
| Broadcast MBS Session Context Status Notification | {notifyUri} | POST | ContextStatusNotify |

## 6.5.5.2 Broadcast MBS Session Context Status Notification

### 6.5.5.2.1 Description

The Broadcast MBS session context notification is used by the AMF to report one or several status changes of a Broadcast MBS session context to a NF service consumer.

### 6.5.5.2.2 Target URI

The Callback URI **"{notifyUri}"** shall be used with the callback URI variables defined in table 6.5.5.2.2-1.

**Table 6.5.5.2.2-1: Callback URI variables**

| Name | Definition |
|---|---|
| notifyUri | String formatted as URI with the Callback URI |

### 6.5.5.2.3 Notification Standard Methods

#### 6.5.5.2.3.1 POST

This method shall support the request data structures specified in table 6.5.5.2.3.1-1 and the response data structures and response codes specified in table 6.5.5.2.3.1-2.

**Table 6.5.5.2.3.1-2: Data structures supported by the POST Request Body**

| Data type | P | Cardinality | Description |
|---|---|---|---|
| ContextStatusNotification | M | 1 | Represents the notification to be delivered |

**Table 6.5.5.2.3.1-3: Data structures supported by the POST Response Body**

| Data type | P | Cardinality | Response codes | Description |
|---|---|---|---|---|
| n/a | | | 204 No Content | Successful response without content |
| ContextStatusNotificationResponse | M | 1 | 200 OK | Successful response with content |
| RedirectResponse | O | 0..1 | 307 Temporary Redirect | Temporary redirection.<br>(NOTE 2) |
| RedirectResponse | O | 0..1 | 308 Permanent Redirect | Permanent redirection.<br>(NOTE 2) |
| NOTE 1: The mandatory HTTP error status code for the POST method listed in Table 5.2.7.1-1 of TS 29.500 [4] also apply, with response body containing an object of ProblemDetails data type (see clause 5.2.7 of TS 29.500 [4]).<br>NOTE 2: RedirectResponse may be inserted by an SCP, see clause 6.10.9.1 of TS 29.500 [4]. | | | | |

**Table 6.5.5.2.3.1-4: Headers supported by the 307 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | A URI pointing to the endpoint of the NF service consumer to which the notification should be sent |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target NF (service) instance ID towards which the request is redirected |

**Table 6.5.5.2.3.1-5: Headers supported by the 308 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | A URI pointing to the endpoint of the NF service consumer to which the notification should be sent |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target NF (service) instance ID towards which the request is redirected |

## 6.5.6 Data Model

### 6.5.6.1 General

This clause specifies the application data model supported by the API.

Table 6.5.6.1-1 specifies the data types defined for the Namf_MBSBroadcast service based interface protocol.

**Table 6.5.6.1-1: Namf_MBSBroadcast specific Data Types**

| Data type | Clause defined | Description |
|---|---|---|
| ContextCreateReqData | 6.5.6.2.2 | Data within ContextCreate Request |
| ContextCreateRspData | 6.5.6.2.3 | Data within ContextCreate Response |
| ContextStatusNotification | 6.5.6.2.4 | Data within ContextStatusNotify Request |
| ContextUpdateReqData | 6.5.6.2.5 | Data within ContextUpdate Request |
| ContextUpdateRspData | 6.5.6.2.6 | Data within ContextUpdate Response |
| N2MbsSmInfo | 6.5.6.2.7 | N2 MBS Session Management Information |
| OperationEvent | 6.5.6.2.8 | Operation Event |
| NgranFailureEvent | 6.5.6.2.9 | NG-RAN failure event |
| ContextStatusNotificationResponse | 6.5.6.2.10 | Data within ContextStatusNotify Response |
| OperationStatus | 6.5.6.3.3 | Operation Status |
| NgapIeType | 6.5.6.3.4 | NGAP Information Element Type |
| OpEventType | 6.5.6.3.5 | Operation Event Type |
| NgranFailureIndication | 6.5.6.3.6 | Indication of a NG-RAN failure event |

Table 6.5.6.1-2 specifies data types re-used by the Namf service based interface protocol from other specifications, including a reference to their respective specifications and when needed, a short description of their use within the Namf_MBSBroadcast service based interface.

**Table 6.5.6.1-2: Namf_MBSBroadcast re-used Data Types**

| Data type | Reference | Comments |
|---|---|---|
| MbsSessionId | TS 29.571 [6] | MBS Session Id |
| AreaSessionId | TS 29.571 [6] | Area Session Id |
| MbsServiceArea | TS 29.571 [6] | MBS Service Area |
| RefToBinaryData | TS 29.571 [6] | Reference to binary body part |
| Uri | TS 29.571 [6] | URI |
| DurationSec | TS 29.571 [6] | Duration in seconds |
| MbsServiceAreaInfo | TS 29.571 [6] | MBS Service Area Information for a Location dependent MBS session |
| GlobalRanNodeId | TS 29.571 [6] | Global RAN Node Identifier |
| AssociatedSessionId | TS 29.571 [6] | Associated Session ID |

## 6.5.6.2 Structured data types

### 6.5.6.2.1 Introduction

Structured data types used in Namf_MBSBroadcast service are specified in this clause.

### 6.5.6.2.2 Type: ContextCreateReqData

**Table 6.5.6.2.2-1: Definition of type ContextCreateReqData**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| mbsSessionId | MbsSessionId | M | 1 | MBS Session ID |
| mbsServiceArea | MbsServiceArea | C | 0..1 | MBS Service Area<br>This IE shall be present if this is a Local broadcast MBS session.<br>(NOTE 1) |
| mbsServiceAreaInfoList | array(MbsServiceAreaInfo) | C | 1..N | List of MBS service areas and their related Area Session IDs.<br>This IE shall be present if this is a Location dependent broadcast MBS service.<br>(NOTE 1) |
| n2MbsSmInfo | N2MbsSmInfo | M | 1 | This IE shall be present and shall contain N2 MBS Session Management related information.(see clause 6.5.6.4). |
| notifyUri | Uri | M | 1 | This IE shall contain the notification URI where to be notified about the status change of the broadcast MBS session context. |
| maxResponseTime | DurationSec | O | 0..1 | Maximum response time (in seconds) to receive information about the completion of the Broadcast MBS session establishment. |
| snssai | Snssai | M | 1 | This IE shall be included to indicate the S-NSSAI of the MBS session. (NOTE 2). |
| mbsmfId | NfInstanceId | O | 0..1 | This IE may be present to contain the NF Instance ID of the MB-SMF. |
| mbsmfServiceInstId | string | O | 0..1 | This IE may be present to contain the NF Service Instance ID within the NF Instance of the MB-SMF. |
| associatedSessionId | AssociatedSessionId | O | 0..1 | Associated Session ID to enable NG-RAN to identify the multiple MBS sessions delivering the same content when AF creates multiple broadcast MBS Sessions via different Core Networks to deliver the same content. |
| NOTE 1: Either the mbsServiceAreaInfoList IE or the mbsServiceArea IE shall be present.<br>NOTE 2: If an MB-SMF does not receive the S-NSSAI from the NEF/MBSF, the MB-SMF shall include a pre-configured default SNSSAI. | | | | |

### 6.5.6.2.3 Type: ContextCreateRspData

**Table 6.5.6.2.3-1: Definition of type ContextCreateRspData**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| mbsSessionId | MbsSessionId | M | 1 | MBS session identifier. |
| n2MbsSmInfoList | array(N2MbsSmInfo) | O | 1..10 | When present, this IE shall contain N2 MBS Session Management related information.(see clause 6.5.6.4). |
| operationStatus | OperationStatus | C | 0..1 | This IE shall be present and indicate the completion of the MBS session start operation, if the AMF received the NG-RAN responses from all involved NG-RAN(s). (NOTE) |
| NOTE: If this IE is not present, it indicates either of the following conditions (a) not all response are received from NG-RANs, while the maximum response timer is still running, or (b) the AMF has not received the maximum response timer in the request from the MB-SMF and not all response are received from NG-RANs. | | | | |

#### 6.5.6.2.4 Type: ContextStatusNotification

**Table 6.5.6.2.4-1: Definition of type ContextStatusNotification**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| mbsSessionId | MbsSessionId | M | 1 | MBS Session ID |
| areaSessionId | AreaSessionId | C | 0..1 | Area Session ID<br>This IE shall be present if this is a Location dependent broadcast MBS service. |
| n2MbsSmInfoList | array(N2MbsSmInfo) | O | 1..10 | When present, this IE shall contain N2 MBS Session Management related information.(see clause 6.5.6.4). |
| operationStatus | OperationStatus | C | 0..1 | This IE shall be present and indicate the completion of the MBS session start or update operation, if the NF Service Consumer has requested to establish or update the Broadcast MBS session context and a response has been received from all NG-RANs.<br><br>This IE shall be present and indicate the incompletion of the MBS session start or update operation, if the NF Service Consumer has requested to establish or update the Broadcast MBS session context within a maximum response time and the AMF has not received responses from all NG-RANs before the maximum response time elapses.<br>(NOTE) |
| operationEvents | array(OperationEvent) | O | 1..N | This IE may be present to report a list of operation events related to the Broadcast MBS Session, e.g., when the Broadcast MBS Session is released in one of NG-RANs as required by the NG-RAN, a failure of a NG-RAN. |
| releasedInd | boolean | C | 0..1 | This shall be present and set to "true" if all NG-RANs serving the Broadcast MBS session requested the AMF to release the Broadcast MBS session and the Broadcast MBS session (context) has been released in the AMF. |

NOTE: If this IE is not present, it indicates either of the following conditions (a) not all response are received from NG-RANs, while the maximum response timer is still running, or (b) the AMF has not received the maximum response timer in the request from the MB-SMF and not all response are received from NG-RANs.

#### 6.5.6.2.5 Type: ContextUpdateReqData

**Table 6.5.6.2.5-1: Definition of type ContextUpdateReqData**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| mbsServiceArea | MbsServiceArea | O | 0..1 | MBS Service Area<br><br>This IE may be present for a Location independent broadcast MBS session.<br>(NOTE) |
| mbsServiceAreaInfoList | array(MbsServiceAreaInfo) | O | 1..N | List of MBS service areas and their related Area Session IDs.<br>This IE may be present for a Local broadcast MBS service.<br>(NOTE) |
| n2MbsSmInfo | N2MbsSmInfo | O | 0..1 | When present, this IE shall contain N2 MBS Session Management related information.(see clause 6.5.6.4). |
| ranIdList | array(GlobalRanNodeId) | O | 1..N | This IE may be included when the MBS session update is to be performed only in a list of specific NG-RAN(s) as specified in clause 8.3.2.3 of TS 23.527 [33]). |
| noNgapSignallingInd | boolean | O | 0..1 | This IE may be present during the restoration procedure to select an alternative AMF for a Broadcast MBS Session at AMF failure as specified in clause 8.3.2.4 of TS 23.527 [33]).<br><br>When present, this IE shall be set as following:<br>- true: the AMF may consider to not trigger a NGAP signaling message to any NG-RAN. |
| notifyUri | Uri | O | 0..1 | When present, this IE shall contain the notification URI where to be notified about the status change of the broadcast MBS session context. |
| maxResponseTime | DurationSec | O | 0..1 | Maximum response time (in seconds) to receive information about the completion of the Broadcast MBS session update. |
| n2MbsInfoChangeInd | boolean | O | 0..1 | When present, this IE shall indicate whether the information within the N2 MBS Session Management container has changed or not, as follows:<br>- true: the information has changed;<br><br>- false: the information has not changed. |
| NOTE: Either the mbsServiceAreaInfoList IE or the mbsServiceArea IE may be present. | | | | |

#### 6.5.6.2.6 Type: ContextUpdateRspData

**Table 6.5.6.2.6-1: Definition of type ContextUpdateRspData**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| n2MbsSmInfoList | array(N2MbsSmInfo) | O | 1..10 | When present, this IE shall contain N2 MBS Session Management related information.(see clause 6.5.6.4). |
| operationStatus | OperationStatus | C | 0..1 | This IE shall be present and indicate the completion of the MBS session start operation, if the AMF received the NG-RAN responses from all involved NG-RAN(s). (NOTE) |
| NOTE: If this IE is not present, it indicates either of the following conditions (a) not all response are received from NG-RANs, while the maximum response timer is still running, or (b) the AMF has not received the maximum response timer in the request from the MB-SMF and not all response are received from NG-RANs. | | | | |

#### 6.5.6.2.7 Type: N2MbsSmInfo

**Table 6.5.6.2.7-1: Definition of type N2MbsSmInfo**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| ngapIeType | NgapIeType | M | 1 | This IE shall indicate the NGAP IE type of the ngapData as specified in clause 6.5.6.3.4. |
| ngapData | RefToBinaryData | M | 1 | This IE shall contain the reference to the binary data part carrying the NGAP data. |
| ranId | GlobalRanNodeId | M | 1 | This IE shall indicate the Global RAN ID of the gNB that generated the N2 MBS Session Management related information, or of the gNB towards which the N2 MBS Session Management related information is to be sent.<br><br>The IE shall be present when the N2MbsSmInfo is included in the ContextCreate Response, ContextUpdate Response, or ContextStatusNotify Request messages which are sent from the AMF to the NF service consumer (e.g., MB-SMF).<br><br>The IE shall be present when the N2MbsSmInfo is included in the ContextStatusNotify Response messages which are sent from the NF service consumer (e.g., MB-SMF) to the AMF. |

#### 6.5.6.2.8 Type: OperationEvent

**Table 6.5.6.2.8-1: Definition of type OperationEvent**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| opEventType | OpEventType | M | 1 | Indicates the event type of an operation event related to the Broadcast MBS Session. |
| amfId | NfInstanceId | C | 0..1 | This IE shall be present to contain the NF Instance ID of the AMF sending the Context Status Notify Request message if the operation event type indicates an AMF change event. |
| ngranFailureEventList | array(NgranFailureEvent) | C | 1..N | This IE shall be present if the event type is related to a NG-RAN. |

#### 6.5.6.2.9 Type: NgranFailureEvent

**Table 6.5.6.2.9-1: Definition of type NgranFailureEvent**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| ngranId | GlobalRanNodeId | M | 1 | Indicates the identity of the RAN node. The IE shall contain the gNB ID. |
| ngranFailureIndication | NgranFailureIndication | M | 1 | This IE shall contain the information related to the NG-RAN failure. |

#### 6.5.6.2.10 Type: ContextStatusNotificationResponse

**Table 6.5.6.2.10-1: Definition of type ContextStatusNotificationResponse**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| mbsSessionId | MbsSessionId | M | 1 | MBS Session ID |
| areaSessionId | AreaSessionId | C | 0..1 | Area Session ID<br>This IE shall be present if this is a Location dependent broadcast MBS service. |
| n2MbsSmInfoList | array(N2MbsSmInfo) | O | 1..10 | When present, this IE shall contain N2 MBS Session Management related information.(see clause 6.5.6.4). |

### 6.5.6.3 Simple data types and enumerations

#### 6.5.6.3.1 Introduction

This clause defines simple data types and enumerations that can be referenced from data structures defined in the previous clauses.

#### 6.5.6.3.2 Simple data types

The simple data types defined in table 6.5.6.3.2-1 shall be supported.

**Table 6.5.6.3.2-1: Simple data types**

| Type Name | Type Definition | Description |
|---|---|---|
| | | |

#### 6.5.6.3.3 Enumeration: OperationStatus

The enumeration OperationStatus represents the status of a Broadcast MBS session start or update operation.

**Table 6.5.6.3.3-1: Enumeration OperationStatus**

| Enumeration value | Description |
|---|---|
| "MBS_SESSION_START_COMPLETE" | This value indicates the completion of the Broadcast MBS session establishment, i.e. that the AMF has received a response from all NG-RANs. |
| " MBS_SESSION_START_INCOMPLETE" | This value indicates the incompletion of the Broadcast MBS session establishment because the AMF has not received responses from all NG-RANs before the maximum response time that was indicated in the request elapses. |
| "MBS_SESSION_UPDATE_COMPLETE" | This value indicates the completion of the Broadcast MBS session update, i.e. that the AMF has received a response from all NG-RANs. |
| " MBS_SESSION_UPDATE_INCOMPLETE" | This value indicates incompletion of the Broadcast MBS session update because the AMF has not received responses from all NG-RANs before the maximum response time that was indicated in the request elapses. |

#### 6.5.6.3.4 Enumeration: NgapIeType

**Table 6.5.6.3.4-1: Enumeration NgapIeType**

| Enumeration value | Description |
|---|---|
| "MBS_SES_REQ" | MBS Session Setup or Modification Request Transfer |
| "MBS_SES_RSP" | MBS Session Setup or Modification Response Transfer |
| "MBS_SES_FAIL" | MBS Session Setup or Modification Failure Transfer |
| "MBS_SES_REL_RSP" | MBS Session Release Response Transfer |
| "BC_TRA_REQ" | Broadcast Transport Request Transfer |
| "BC_TRA_RSP" | Broadcast Transport Response Transfer |
| "BC_TRA_FAIL" | Broadcast Transport Failure Transfer |

#### 6.5.6.3.5 Enumeration: OpEventType

**Table 6.5.6.3.5-1: Enumeration: OpEventType**

| Enumeration value | Description |
|---|---|
| "AMF_CHANGE" | This value indicates that the AMF has taken over of the Broadcast MBS Session. |
| "NG_RAN_EVENT" | This value indicates that an event related to a NG-RAN for the Broadcast MBS Session has taken place. |

#### 6.5.6.3.6 Enumeration: NgranFailureIndication

The enumeration NgranFailureIndication indicates a NG-RAN failure event.

**Table 6.5.6.3.6-1: Enumeration NgranFailureIndication**

| Enumeration value | Description |
|---|---|
| "NG_RAN_RESTART_OR_START" | This value indicates that the AMF has detected a (re)start of a NG-RAN. |
| "NG_RAN_FAILURE_WITHOUT_RESTART" | This value indicates that the AMF has detected a NG-RAN failure without a restart. |
| "NG_RAN_NOT_REACHABLE" | This value indicates that the AMF has failed to reach the NG-RAN when sending a NGAP MBS Session Setup/Modification/Release Request message. |
| "NG_RAN_REQUIRED_RELEASE" | This value indicates that the NG-RAN has requested to release the Broadcast MBS Session in the NG-RAN. |

### 6.5.6.4 Binary data

#### 6.5.6.4.1 Introduction

This clause defines the binary data that shall be supported in a binary body part in an HTTP multipart message (see clauses 6.5.2.2.2 and 6.5.2.4).

**Table 6.5.6.4.1-1: Binary Data Types**

| Name | Clause defined | Content type |
|---|---|---|
| N2 Information | 6.5.6.4.3 | vnd.3gpp.ngap |

### 6.5.6.4.2 N2 Information

#### 6.5.6.4.2.1 Introduction

N2 Information shall encode NG Application Protocol (NGAP) IEs, as specified in clause 9.3.A of TS 38.413 [12] (ASN.1 encoded), using the vnd.3gpp.ngap content-type.

#### 6.5.6.4.2.2 NGAP IEs

N2 Information may encode following NGAP MB-SMF related IE specified in clause 9.3.5 of TS 38.413 [12], as summarized in Table 6.5.6.4.2.2-1.

**Table 6.5.6.4.2.2-1: N2 Information content for class MBS-SM**

| NGAP IE | Reference (TS 38.413 [12]) | Related NGAP message |
|---|---|---|
| MBS Session Setup or Modification Request Transfer | 9.3.5.3 | BROADCAST SESSION SETUP REQUEST<br>BROADCAST SESSION MODIFICATION REQUEST |
| MBS Session Setup or Modification Response Transfer | 9.3.5.5 | BROADCAST SESSION SETUP RESPONSE<br>BROADCAST SESSION MODIFICATION RESPONSE |
| MBS Session Setup or Modification Failure Transfer | 9.3.5.6 | BROADCAST SESSION SETUP FAILURE<br>BROADCAST SESSION MODIFICATION FAILURE |
| MBS Session Release Response Transfer | 9.3.5.14 | BROADCAST SESSION RELEASE RESPONSE (NOTE) |
| Broadcast Transport Request Transfer | 9.3.5.15 | BROADCAST SESSION TRANSPORT REQUEST |
| Broadcast Transport Response Transfer | 9.3.5.16 | BROADCAST SESSION TRANSPORT RESPONSE |
| Broadcast Transport Failure Transfer | 9.3.5.17 | BROADCAST SESSION TRANSPORT FAILURE |
| NOTE: An MBS Session Release Response Transfer IE shall only be sent to the MB-SMF during a Broadcast MBS Session Release Require procedure (see clause 7.3.6 of TS 23.247 [55]), when unicast transport applies over N3mb, to transfer the DL F-TEID of the NG-RAN node in which the MBS session has been released and towards which the delivery of MBS data shall be stopped. | | |

## 6.5.7 Error Handling

### 6.5.7.1 General

HTTP error handling shall be supported as specified in clause 5.2.4 of TS 29.500 [4].

### 6.5.7.2 Protocol Errors

Protocol Error Handling shall be supported as specified in clause 5.2.7 of TS 29.500 [4].

### 6.5.7.3 Application Errors

The common application errors defined in the Table 5.2.7.2-1 in TS 29.500 [4] may also be used for the Namf_MBSBroadcast service, and the following application errors listed in Table 6.5.7.3-1 are specific for the Namf_MBSBroadcast service.

**Table 6.5.7.3-1: Application errors**

| Application Error | HTTP status code | Description |
|---|---|---|
| | | |

## 6.5.8 Feature Negotiation

The optional features in table 6.5.8-1 are defined for the Namf_MBSBroadcast API. They shall be negotiated using the extensibility mechanism defined in clause 6.6 of TS 29.500 [4].

**Table 6.5.8-1: Supported Features**

| Feature number | Feature Name | Description |
|---|---|---|
| | | |

## 6.5.9 Security

As indicated in TS 33.501 [8] and TS 29.500 [4], the access to the Namf_MBSBroadcast API may be authorized by means of the OAuth2 protocol (see IETF RFC 6749 [9]), based on local configuration, using the "Client Credentials" authorization grant, where the NRF (see TS 29.510 [10]) plays the role of the authorization server.

If OAuth2 is used, an NF Service Consumer, prior to consuming services offered by the Namf_MBSBroadcast API, shall obtain a "token" from the authorization server, by invoking the Access Token Request service, as described in TS 29.510 [10], clause 5.4.2.2.

NOTE: When multiple NRFs are deployed in a network, the NRF used as authorization server is the same NRF that the NF Service Consumer used for discovering the Namf_MBSBroadcast service.

The Namf_MBSBroadcast API defines a single scope "namf-mbs-bc" for the entire service, and it does not define any additional scopes at resource or operation level.

## 6.5.10 HTTP redirection

An HTTP request may be redirected to a different AMF service instance, within the same AMF or a different AMF of an AMF set, e.g. when an AMF service instance is part of an AMF (service) set or when using indirect communications (see TS 29.500 [4]).

An SCP that reselects a different AMF producer instance will return the NF Instance ID of the new AMF producer instance in the 3gpp-Sbi-Producer-Id header, as specified in clause 6.10.3.4 of TS 29.500 [4].

If an AMF within an AMF set redirects a service request to a different AMF of the set using an 307 Temporary Redirect or 308 Permanent Redirect status code, the identity of the new AMF towards which the service request is redirected shall be indicated in the 3gpp-Sbi-Target-Nf-Id header of the 307 Temporary Redirect or 308 Permanent Redirect response as specified in clause 6.10.9.1 of TS 29.500 [4].

# 6.6 Namf_MBSCommunication Service API

## 6.6.1 API URI

The Namf_MBSCommunication service shall use the Namf_MBSCommunication API.

The API URI of the Namf_ MBSCommunication API shall be:

**{apiRoot}/<apiName>/<apiVersion>**

The request URI used in HTTP requests from the NF service consumer towards the NF service producer shall have the Resource URI structure defined in clause 4.4.1 of TS 29.501 [5], i.e.:

**{apiRoot}/<apiName>/<apiVersion>/<apiSpecificResourceUriPart>**

with the following components:

- The {apiRoot} shall be set as described in TS 29.501 [5].
- The <apiName> shall be "namf-mbs-comm".
- The <apiVersion> shall be "v1".
- The <apiSpecificResourceUriPart> shall be set as described in clause 6.6.3.

## 6.6.2 Usage of HTTP

### 6.6.2.1 General

HTTP/2, as defined in IETF RFC 9113 [19], shall be used as specified in clause 5 of TS 29.500 [4].

HTTP/2 shall be transported as specified in clause 5.3 of TS 29.500 [4].

HTTP messages and bodies for the Namf_MBSCommunication service shall comply with the OpenAPI [23] specification contained in Annex A.

### 6.6.2.2 HTTP standard headers

#### 6.6.2.2.1 General

The usage of HTTP standard headers shall be supported as specified in clause 5.2.2 of TS 29.500 [4].

#### 6.6.2.2.2 Content type

The following content types shall be supported:

- JSON, as defined in IETF RFC 8259 [8], shall be used as content type of the HTTP bodies specified in the present specification as indicated in clause 5.4 of TS 29.500 [4].
- The Problem Details JSON Object (IETF RFC 9457 [36]). The use of the Problem Details JSON object in a HTTP response body shall be signalled by the content type "application/problem+json".

Multipart messages shall also be supported (see clause 6.6.2.4) using the content type "multipart/related", comprising:

- one JSON body part with the "application/json" content type; and
- one binary body part with 3gpp vendor specific content subtypes.

The 3gpp vendor specific content subtypes defined in Table 6.6.2.2.2-1 shall be supported.

**Table 6.6.2.2.2-1: 3GPP vendor specific content subtypes**

| content subtype | Description |
|---|---|
| vnd.3gpp.ngap | Binary encoded content, encoding NG Application Protocol (NGAP) IEs, as specified in clause 9.3 of TS 38.413 [9] (ASN.1 encoded). |
| NOTE: Using 3GPP vendor content subtypes allows to describe the nature of the opaque content (i.e. NGAP information) without having to rely on metadata in the JSON content. | |

See clause 6.6.2.4 for the binary contents supported in the binary body part of multipart messages.

### 6.6.2.3 HTTP custom headers

#### 6.6.2.3.1 General

In this release of this specification, no custom headers specific to the Namf_MBSCommunication service are defined. For 3GPP specific HTTP custom headers used across all service-based interfaces, see clause 5.2.3 of TS 29.500 [4].

### 6.6.2.4 HTTP multipart messages

HTTP multipart messages shall be supported, to transfer opaque N2 Information MBS, in the following service operations (and HTTP messages):

- N2MessageTransfer Request and Response (POST).

HTTP multipart messages shall include one JSON body part and one binary body part comprising:

- N2 payload (see clause 6.1.6.4).

The JSON body part shall be the "root" body part of the multipart message. It shall be encoded as the first body part of the multipart message. The "Start" parameter does not need to be included.

The multipart message shall include a "type" parameter (see IETF RFC 2387 [9]) specifying the media type of the root body part, i.e. "application/json".

NOTE: The "root" body part (or "root" object) is the first body part the application processes when receiving a multipart/related message, see IETF RFC 2387 [9]. The default root is the first body within the multipart/related message. The "Start" parameter indicates the root body part, e.g. when this is not the first body part in the message.

For each binary body part in a HTTP multipart message, the binary body part shall include a Content-ID header (see IETF RFC 2045 [10]), and the JSON body part shall include an attribute, defined with the RefToBinaryData type, that contains the value of the Content-ID header field of the referenced binary body part.

## 6.6.3 Resources

### 6.6.3.1 Overview

Figure 6.6.3.1-1 describes the resource URI structure of the Namf_MBSCommunication API.

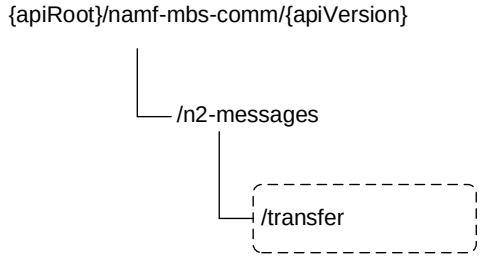



**Figure 6.6.3.1-1: Resource URI structure of the Namf_MBSCommunication API**

Table 6.6.3.1-1 provides an overview of the resources and applicable HTTP methods.

**Table 6.6.3.1-1: Resources and methods overview**

| Resource name | Resource URI | HTTP method or custom operation | Description (service operation) |
|---|---|---|---|
| N2 Message Handler (Custom Operation) | /n2-messages/transfer | transfer (POST) | N2MessageTransfer |

## 6.6.3.1 Resource: N2 Message Handler (Custom Operation)

### 6.6.3.1.1 Description

This resource represents the N2 Message Handler used to transfer a N2 message related to support a Multicast MBS session towards NG-RANs.

### 6.6.3.1.2 Resource Definition

Resource URI: **{apiRoot}/namf-comm/<apiVersion>/n2-messages**

This resource shall support the resource URI variables defined in table 6.6.3.1.2-1.

**Table 6.6.3.1.2-1: Resource URI variables for this resource**

| Name | Data Type | Definition |
|---|---|---|
| apiRoot | string | See clause 6.6.1 |
| apiVersion | string | See clause 6.6.1. |

### 6.6.3.1.3 Resource Standard Methods

There are no resource standard methods for the N2 Message Handler resource in this release of this specification

### 6.6.3.1.4 Resource Custom Operations

#### 6.6.3.1.4.1 Overview

**Table 6.6.3.1.4.1-1: Custom operations**

| Operation Name | Custom operaration URI | Mapped HTTP method | Description (service operation) |
|---|---|---|---|
| transfer | /n2-messages/transfer | POST | N2MessageTransfer |

#### 6.6.3.1.4.2 Operation: transfer

##### 6.6.3.1.4.2.1 Description

The /n2-messages/transfer custom operation is used to initiate the transfer of N2 MBS Session Management information to the NG-RAN nodes serving a multicast MBS session. This custom operation uses the HTTP POST method.

##### 6.6.3.1.4.2.2 Operation Definition

This operation shall support the request data structures specified in table 6.6.3.1.4.2-1 and the response data structure and response codes specified in table 6.6.3.1.4.2-2.

**Table 6.6.3.1.4.2.2-1: Data structures supported by the POST Request Body on this resource**

| Data type | P | Cardinality | Description |
|---|---|---|---|
| MbsN2MessageTransferReqData | M | 1 | Representation of the data related to a multicast MBS session to be sent to the NG-RAN node(s) by the AMF. |

**Table 6.6.3.1.4.2.2-2: Data structures supported by the POST Response Body on this resource**

| Data type | P | Cardinality | Response codes | Description |
|---|---|---|---|---|
| MbsN2MessageTransferRspData | M | 1 | 200 OK | Indicates that the AMF has successfully initiated the transfer of N2 Information to the AN. |
| RedirectResponse | O | 0..1 | 307 Temporary Redirect | Temporary redirection.<br>(NOTE 2) |
| RedirectResponse | O | 0..1 | 308 Permanent Redirect | Permanent redirection.<br>(NOTE 2) |
| ProblemDetails | O | 0..1 | 404 Not Found | When the MBS Session ID is not found in the NF Service Producer (i.e. AMF) the "cause" attribute shall be set to:<br>- MBS_SESSION_NOT_FOUND<br><br>See table 6.6.7.3-1 for the description of these errors |
| NOTE 1: The mandatory HTTP error status code for the POST method listed in Table 5.2.7.1-1 of TS 29.500 [4] also apply, with response body containing an object of ProblemDetails data type (see clause 5.2.7 of TS 29.500 [4]).<br>NOTE 2: RedirectResponse may be inserted by an SCP, see clause 6.10.9.1 of TS 29.500 [4]. | | | | |

**Table 6.6.3.1.4.2.2-3: Headers supported by the 307 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | An alternative URI of the resource located on an alternative service instance within the same AMF or AMF (service) set.<br>For the case when a request is redirected to the same target resource via a different SCP, see clause 6.10.9.1 in TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target NF (service) instance ID towards which the request is redirected |

**Table 6.6.3.1.4.2.2-4: Headers supported by the 308 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | An alternative URI of the resource located on an alternative service instance within the same AMF or AMF (service) set.<br>For the case when a request is redirected to the same target resource via a different SCP, see clause 6.10.9.1 in TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target NF (service) instance ID towards which the request is redirected |

## 6.6.4 Custom Operations without associated resources

## 6.6.5 Notifications

### 6.6.5.1 General

The notifications provided by the Namf_MBSCommunication service are specified in this clause.

**Table 6.6.5.1-1: Callback overview**

| Notification | Resource URI | HTTP method or custom operation | Description (service operation) |
|---|---|---|---|
| Notification | {notifyUri} | POST | Notify |

## 6.6.5.2 Notification

### 6.6.5.2.1 Description

The notification is used by the AMF to report the failure of an MBS related N2 procedure with an NG-RAN node to a NF service consumer.

### 6.6.5.2.2 Notification Definitionn

The Callback URI **"{notifyUri}"** shall be used with the callback URI variables defined in table 6.6.5.2.2-1.

**Table 6.6.5.2.2-1: Callback URI variables**

| Name | Definition |
|---|---|
| notifyUri | String formatted as URI with the Callback URI |

### 6.6.5.2.3 Notification Standard Methods

#### 6.6.5.2.3.1 POST

This method shall support the request data structures specified in table 6.6.5.2.3.1-1 and the response data structures and response codes specified in table 6.6.5.2.3.1-2.

**Table 6.6.5.2.3.1-2: Data structures supported by the POST Request Body**

| Data type | P | Cardinality | Description |
|---|---|---|---|
| Notification | M | 1 | Represents the notification to be delivered |

**Table 6.6.5.2.3.1-3: Data structures supported by the POST Response Body**

| Data type | P | Cardinality | Response codes | Description |
|---|---|---|---|---|
| n/a | | | 204 No Content | |
| RedirectResponse | O | 0..1 | 307 Temporary Redirect | Temporary redirection.<br>(NOTE 2) |
| RedirectResponse | O | 0..1 | 308 Permanent Redirect | Permanent redirection.<br>(NOTE 2) |
| NOTE 1: The mandatory HTTP error status code for the POST method listed in Table 5.2.7.1-1 of TS 29.500 [4] also apply, with response body containing an object of ProblemDetails data type (see clause 5.2.7 of TS 29.500 [4]).<br>NOTE 2: RedirectResponse may be inserted by an SCP, see clause 6.10.9.1 of TS 29.500 [4]. | | | | |

**Table 6.6.5.2.3.1-4: Headers supported by the 307 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | A URI pointing to the endpoint of the NF service consumer to which the notification should be sent.<br>For the case when a request is redirected to the same target resource via a different SCP, see clause 6.10.9.1 in TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target NF (service) instance ID towards which the request is redirected |

**Table 6.6.5.2.3.1-5: Headers supported by the 308 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | A URI pointing to the endpoint of the NF service consumer to which the notification should be sent.<br>For the case when a request is redirected to the same target resource via a different SCP, see clause 6.10.9.1 in TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target NF (service) instance ID towards which the request is redirected |

## 6.6.6 Data Model

### 6.6.6.1 General

This clause specifies the application data model supported by the API.

Table 6.6.6.1-1 specifies the data types defined for the Namf_MBSCommunication service based interface protocol.

**Table 6.6.6.1-1: Namf_MBSCommunication specific Data Types**

| Data type | Clause defined | Description |
|---|---|---|
| MbsN2MessageTransferReqData | 6.6.6.2.2 | Data within MBS N2 Message Transfer Request |
| MbsN2MessageTransferRspData | 6.6.6.2.3 | Data within MBS N2 Message Transfer Response |
| N2MbsSmInfo | 6.6.6.2.4 | N2 MBS Session Management Information |
| Notification | 6.6.6.2.5 | Data within Notify Request |
| RanFailure | 6.6.6.2.6 | Description of an NG RAN failure |
| MbsNgapIeType | 6.6.6.3.3 | NGAP Information Element Type for MBS |
| RanFailureIndication | 6.6.6.3.4 | NG RAN failure indication |

Table 6.6.6.3-2 specifies data types re-used by the Namf_MBSCommunication service based interface protocol from other specifications, including a reference to their respective specifications and when needed, a short description of their use within the Namf_MBSCommunication service based interface.

**Table 6.6.6.1-2: Namf_MBSCommunication re-used Data Types**

| Data type | Reference | Comments |
|---|---|---|
| ProblemDetails | TS 29.571 [6] | Common data type used in response bodies |
| supportedFeatures | TS 29.571 [6] | Supported Features |
| RedirectResponse | TS 29.571 [6] | Response body of the redirect response message |
| MbsSessionId | TS 29.571 [6] | MBS Session Identifier |
| NgApCause | TS 29.571 [6] | NGAP Cause |
| N2InformationTransferResult | 6.1.6.3.8 | Enumeration N2 Message Transfer Result |

## 6.6.6.2 Structured data types

### 6.6.6.2.1 Introduction

Structured data types used in Namf_MBSCommunication service are specified in this clause.

### 6.6.6.2.2 Type: MbsN2MessageTransferReqData

**Table 6.6.6.2.2-1: Definition of type MbsN2MessageTransferReqData**

| Attribute name | Data type | P | Cardinality | Description | Applicability |
|---|---|---|---|---|---|
| mbsSessionId | MbsSessionId | M | 1 | This IE shall be included to identify the MBS session to which the N2 information to be transferred is related. | |
| areaSessionId | AreaSessionId | O | 0..1 | Area Session ID<br>This IE may be present, if this is a Location dependent multicast MBS session. | |
| n2MbsSmInfo | N2MbsSmInfo | M | 1 | This IE shall contain the N2 MBS Session Information to be transferred to the NG-RAN nodes serving the MBS session and additional information required for the processing of the message by the AMF. | |
| supportedFeatures | SupportedFeatures | C | 0..1 | This IE shall be present if at least one optional feature defined in clause 6.1.8 is supported. | |
| ranNodeIdList | array(GlobalRanNodeId) | O | 1..N | When present, this IE shall contain the list of NG RAN Node IDs towards which the MBS related N2 message is requested to be distributed. | RAN-ID-LIST |
| notifyUri | Uri | O | 0..1 | When present, this IE shall contain the notification URI to be used for receiving notifications about possible failures of the MBS related N2 procedure with an NG RAN node in the ranNodeIdList. | RAN-ID-LIST |
| notifyCorrelationId | string | O | 0..1 | When present, this IE shall contain the notification correlation ID to be sent within notifications. | RAN-ID-LIST |

### 6.6.6.2.3 Type: MbsN2MessageTransferRspData

**Table 6.6.6.2.3-1: Definition of type MbsN2MessageTransferRspData**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| result | N2InformationTransferResult | M | 1 | This IE shall provide the result of the MBS N2 information transfer processing at the AMF. |
| supportedFeatures | SupportedFeatures | C | 0..1 | This IE shall be present if at least one optional feature defined in clause 6.1.8 is supported. |
| failureList | array(RanFailure) | O | 1..N | List of MBS related N2 procedure failures |

#### 6.6.6.2.4 Type: N2MbsSmInfo

**Table 6.6.6.2.4-1: Definition of type N2MbsSmInfo**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| ngapIeType | MbsNgapIeType | M | 1 | This IE shall indicate the NGAP IE type of the ngapData as specified in clause 6.6.6.4.2.2. |
| ngapData | RefToBinaryData | M | 1 | This IE shall contain the reference the binary data part carrying the NGAP data. |

#### 6.6.6.2.5 Type: Notification

**Table 6.6.6.2.5-1: Definition of type Notification**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| mbsSessionId | MbsSessionId | M | 1 | MBS Session ID |
| areaSessionId | AreaSessionId | C | 0..1 | Area Session ID<br>This IE shall be present, if present in the N2Message Transfer request. |
| failureList | array(RanFailure) | M | 1..N | List of MBS related N2 procedure failures |
| notifyCorrelationId | string | C | 0..1 | This IE shall be present if the same IE is present in the N2Message Transfer request. When present, it shall contain the same value as received in the request. |

#### 6.6.6.2.6 Type: RanFailure

**Table 6.6.6.2.6-1: Definition of type RanFailure**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| ranId | GlobalRanNodeId | M | 1 | Indicates the identity of the NG RAN node. |
| ranFailureCause | NgApCause | C | 0..1 | When present, this IE shall contain the NGAP failure cause received from the NG-RAN. (NOTE) |
| ranFailureIndication | RanFailureIndication | C | 0..1 | This IE shall be present if the AMF cannot deliver the MBS related N2 message to the NG-RAN node. (NOTE) |
| NOTE: Either the ranFailureCause IE or the ranFailureIndication IE shall be present. | | | | |

### 6.6.6.3 Simple data types and enumerations

#### 6.6.6.3.1 Introduction

This clause defines simple data types and enumerations that can be referenced from data structures defined in the previous clauses.

#### 6.6.6.3.2 Simple data types

The simple data types defined in Table 6.6.6.3.2-1 shall be supported.

#### 6.6.6.3.3 Enumeration: MbsNgapIeType

**Table 6.6.6.3.3-1: Enumeration MbsNgapIeType**

| Enumeration value | Description |
|---|---|
| "MBS_SES_ACT_REQ" | Multicast Session Activation Request Transfer |
| "MBS_SES_DEACT_REQ" | Multicast Session Deactivation Request Transfer |
| "MBS_SES_UPD_REQ" | Multicast Session Update Request Transfer |

#### 6.6.6.3.4 Enumeration: RanFailureIndication

The enumeration RanFailureIndication indicates a NG-RAN failure event.

**Table 6.5.6.3.4-1: Enumeration RanFailureIndication**

| Enumeration value | Description |
|---|---|
| "NG_RAN_FAILURE_WITHOUT_RESTART" | The NG-RAN node failed without restart. |
| "NG_RAN_NOT_REACHABLE" | The AMF cannot reach the NG RAN node when sending the MBS related N2 message. |

### 6.6.6.4 Binary data

#### 6.6.6.4.1 Introduction

This clause defines the binary data that shall be supported in a binary body part in an HTTP multipart message (see clauses 6.6.2.2.2 and 6.6.2.4).

**Table 6.6.6.4.1-1: Binary Data Types**

| Name | Clause defined | Content type |
|---|---|---|
| N2 MBS Session Management Information | 6.6.6.4.2 | vnd.3gpp.ngap |

#### 6.6.6.4.2 N2 Information

##### 6.6.6.4.2.1 Introduction

N2 Information shall encode NG Application Protocol (NGAP) IEs, as specified in clause 9.3.A of TS 38.413 [12] (ASN.1 encoded), using the vnd.3gpp.ngap content-type.

##### 6.6.6.4.2.2 NGAP IEs

N2 Information may encode following NGAP MB-SMF related IE specified in in clause 9.3.5 of TS 38.413 [12], as summarized in Table 6.6.6.4.2.2-1.

**Table 6.6.6.4.2.2-1: N2 Information content for class MBS-SM**

| NGAP IE | Reference (TS 38.413 [12]) | Related NGAP message |
|---|---|---|
| Multicast Session Activation Request Transfer | 9.3.5.11 | MULTICAST SESSION ACTIVATION REQUEST |
| Multicast Session Deactivation Request Transfer | 9.3.5.12 | MULTICAST SESSION DEACTIVATION REQUEST |
| Multicast Session Update Request Transfer | 9.3.5.13 | MULTICAST SESSION UPDATE REQUEST |

## 6.6.7 Error Handling

### 6.6.7.1 General

HTTP error handling shall be supported as specified in clause 5.2.4 of TS 29.500 [4].

### 6.6.7.2 Protocol Errors

Protocol Error Handling shall be supported as specified in clause 5.2.7 of TS 29.500 [4].

### 6.6.7.3 Application Errors

The common application errors defined in the Table 5.2.7.2-1 in TS 29.500 [4] may also be used for the Namf_MBSCommunication service, and the following application errors listed in Table 6.6.7.3-1 are specific for the Namf_MBSCommunication service.

**Table 6.6.7.3-1: Application errors**

| Application Error | HTTP status code | Description |
|---|---|---|
| MBS_SESSION_NOT_FOUND | 404 Not Found | Indicates the MBS related N2 Message transfer has failed due to the MBS Session ID being unknown to the AMF. |

## 6.6.8 Feature Negotiation

The optional features in table 6.6.8-1 are defined for the Namf_MBSCommunication API. They shall be negotiated using the extensibility mechanism defined in clause 6.6 of TS 29.500 [4].

The NF Service Consumer shall indicate the optional features it supports for the Namf_MBSCommunication service, if any, by including the supportedFeatures attribute in content of the HTTP Request Message for following service operations:

- N2MessageTransfer, as specified in clause 5.7.2.2.

The AMF shall determine the supported features for the service operations as specified in clause 6.6 of TS 29.500 [4] and shall indicate the supported features by including the supportedFeatures attribute in content of the HTTP response for the service operation.

The syntax of the supportedFeatures attribute is defined in clause 5.2.2 of TS 29.571 [6].

The following features are defined for the Namf_MBSCommunication service.

**Table 6.6.8-1: Features of supportedFeatures attribute used by Namf_MBSCommunication service**

| Feature Number | Feature | M/O | Description |
|---|---|---|---|
| 1 | RAN-ID-LIST | O | N2 MBS session request distribution with list of NG RAN Node IDs provided by NF Service Consumer to AMF<br><br>An NF Service Consumer (e.g. MB-SMF) and an AMF that support this feature shall support:<br>- Namf_MBSCommunication_N2MessageTransfer Request including the list of NG RAN node IDs towards which the MBS related N2 message is requested to be distributed; and<br>- the AMF notifying an MBS related N2 procedure failure with an NG RAN node in this list, detected by the AMF or reported by the NG-RAN node.<br>See clause 8.4.1.2 of TS 23.527 [33] and clauses 5.7.2.2 and 5.7.2.3. |
| Feature number: The order number of the feature within the supportedFeatures attribute (starting with 1).<br>Feature: A short name that can be used to refer to the bit and to the feature.<br>M/O: Defines if the implementation of the feature is mandatory ("M") or optional ("O").<br>Description: A clear textual description of the feature. | | | |

## 6.6.9 Security

As indicated in TS 33.501 [27], the access to the Namf_MBSCommunication API may be authorized by means of the OAuth2 protocol (see IETF RFC 6749 [28]), using the "Client Credentials" authorization grant, where the NRF (see TS 29.510 [29]) plays the role of the authorization server.

If Oauth2 authorization is used, an NF Service Consumer, prior to consuming services offered by the Namf_ MBSCommunication API, shall obtain a "token" from the authorization server, by invoking the Access Token Request service, as described in TS 29.510 [29], clause 5.4.2.2.

NOTE: When multiple NRFs are deployed in a network, the NRF used as authorization server is the same NRF that the NF Service Consumer used for discovering the Namf_MBSCommunication service.

The Namf_ MBSCommunication API defines scopes for OAuth2 authorization as specified in TS 33.501 [27]; it defines a single scope consisting on the name of the service (i.e., "namf-mbs-comm"), and it does not define any additional scopes at resource or operation level.

## 6.6.10 HTTP redirection

An HTTP request may be redirected to a different AMF service instance, within the same AMF or a different AMF of an AMF set, e.g. when an AMF service instance is part of an AMF (service) set or when using indirect communications (see TS 29.500 [4]).

An SCP that reselects a different AMF producer instance will return the NF Instance ID of the new AMF producer instance in the 3gpp-Sbi-Producer-Id header, as specified in clause 6.10.3.4 of TS 29.500 [4].

If an AMF within an AMF set redirects a service request to a different AMF of the set using an 307 Temporary Redirect or 308 Permanent Redirect status code, the identity of the new AMF towards which the service request is redirected shall be indicated in the 3gpp-Sbi-Target-Nf-Id header of the 307 Temporary Redirect or 308 Permanent Redirect response as specified in clause 6.10.9.1 of TS 29.500 [4].

# Annex A (normative): OpenAPI specification

## A.1 General

This Annex specifies the API definition of the service provided by AMF in this document. The APIs are defined by OpenAPI 3.0.0 specifications in YAML format, following guidelines in TS 29.501 [5].

The APIs for specified for following services:

- Namf_Communication Service
- Namf_EventExposure Service
- Namf_MT Service
- Namf_Location Service

This Annex takes precedence when being discrepant to other parts of the specification with respect to the encoding of information elements and methods within the API(s).

NOTE : The semantics and procedures, as well as conditions, e.g. for the applicability and allowed combinations of attributes or values, not expressed in the OpenAPI definitions but defined in other parts of the specification also apply.

Informative copies of the OpenAPI specification files contained in this 3GPP Technical Specification are available on a Git-based repository, that uses the GitLab software version control system (see TS 29.501 [5] clause 5.3.1 and TR 21.900 [37] clause 5B).

## A.2 Namf_Communication API

```
openapi: 3.0.0

info:
  version: 1.3.5
  title: Namf_Communication
  description: |
    AMF Communication Service.
    © 2026, 3GPP Organizational Partners (ARIB, ATIS, CCSA, ETSI, TSDSI, TTA, TTC).
    All rights reserved.

security:
  - {}
  - oAuth2ClientCredentials:
      - namf-comm

externalDocs:
  description: 3GPP TS 29.518 V18.14.0; 5G System; Access and Mobility Management Services
  url: 'https://www.3gpp.org/ftp/Specs/archive/29_series/29.518/'

servers:
  - url: '{apiRoot}/namf-comm/v1'
    variables:
      apiRoot:
        default: https://example.com
        description: apiRoot as defined in clause 4.4 of 3GPP TS 29.501

paths:
  /ue-contexts/{ueContextId}:
    put:
      summary: Namf_Communication CreateUEContext service Operation
      tags:
        - Individual ueContext (Document)
```

```
      operationId: CreateUEContext
      security:
        - {}
        - oAuth2ClientCredentials:
          - namf-comm
        - oAuth2ClientCredentials:
          - namf-comm
          - namf-comm:ue-contexts:mobility
      parameters:
        - name: ueContextId
          in: path
          description: UE Context Identifier
          required: true
          schema:
            type: string
            pattern: '^(5g-guti-[0-9]{5,6}[0-9a-fA-F]{14}|imsi-[0-9]{5,15}|nai-.+|gli-.+|gci-
.+|imei-[0-9]{15}|imeisv-[0-9]{16}|.+)$'
      requestBody:
        content:
          multipart/related:  # message with binary body part(s)
            schema:
              type: object
              properties: # Request parts
                jsonData:
                  $ref: '#/components/schemas/UeContextCreateData'
                binaryDataN2Information:
                  type: string
                  format: binary
                binaryDataN2InformationExt1:
                  type: string
                  format: binary
                binaryDataN2InformationExt2:
                  type: string
                  format: binary
                binaryDataN2InformationExt3:
                  type: string
                  format: binary
                binaryDataN2InformationExt4:
                  type: string
                  format: binary
                binaryDataN2InformationExt5:
                  type: string
                  format: binary
                binaryDataN2InformationExt6:
                  type: string
                  format: binary
                binaryDataN2InformationExt7:
                  type: string
                  format: binary
                binaryDataN2InformationExt8:
                  type: string
                  format: binary
                binaryDataN2InformationExt9:
                  type: string
                  format: binary
                binaryDataN2InformationExt10:
                  type: string
                  format: binary
                binaryDataN2InformationExt11:
                  type: string
                  format: binary
                binaryDataN2InformationExt12:
                  type: string
                  format: binary
                binaryDataN2InformationExt13:
                  type: string
                  format: binary
                binaryDataN2InformationExt14:
                  type: string
                  format: binary
                binaryDataN2InformationExt15:
                  type: string
                  format: binary
                binaryDataN2InformationExt16:
                  type: string
                  format: binary
                binaryDataN2InformationExt17:
                  type: string
```

```
        format: binary
    encoding:
      jsonData:
        contentType:  application/json
      binaryDataN2Information:
        contentType:  application/vnd.3gpp.ngap
        headers:
          Content-Id:
            schema:
              type: string
      binaryDataN2InformationExt1:
        contentType:  application/vnd.3gpp.ngap
        headers:
          Content-Id:
            schema:
              type: string
      binaryDataN2InformationExt2:
        contentType:  application/vnd.3gpp.ngap
        headers:
          Content-Id:
            schema:
              type: string
      binaryDataN2InformationExt3:
        contentType:  application/vnd.3gpp.ngap
        headers:
          Content-Id:
            schema:
              type: string
      binaryDataN2InformationExt4:
        contentType:  application/vnd.3gpp.ngap
        headers:
          Content-Id:
            schema:
              type: string
      binaryDataN2InformationExt5:
        contentType:  application/vnd.3gpp.ngap
        headers:
          Content-Id:
            schema:
              type: string
      binaryDataN2InformationExt6:
        contentType:  application/vnd.3gpp.ngap
        headers:
          Content-Id:
            schema:
              type: string
      binaryDataN2InformationExt7:
        contentType:  application/vnd.3gpp.ngap
        headers:
          Content-Id:
            schema:
              type: string
      binaryDataN2InformationExt8:
        contentType:  application/vnd.3gpp.ngap
        headers:
          Content-Id:
            schema:
              type: string
      binaryDataN2InformationExt9:
        contentType:  application/vnd.3gpp.ngap
        headers:
          Content-Id:
            schema:
              type: string
      binaryDataN2InformationExt10:
        contentType:  application/vnd.3gpp.ngap
        headers:
          Content-Id:
            schema:
              type: string
      binaryDataN2InformationExt11:
        contentType:  application/vnd.3gpp.ngap
        headers:
          Content-Id:
            schema:
              type: string
      binaryDataN2InformationExt12:
        contentType:  application/vnd.3gpp.ngap
```

```
            headers:
              Content-Id:
                schema:
                  type: string
          binaryDataN2InformationExt13:
            contentType:  application/vnd.3gpp.ngap
            headers:
              Content-Id:
                schema:
                  type: string
          binaryDataN2InformationExt14:
            contentType:  application/vnd.3gpp.ngap
            headers:
              Content-Id:
                schema:
                  type: string
          binaryDataN2InformationExt15:
            contentType:  application/vnd.3gpp.ngap
            headers:
              Content-Id:
                schema:
                  type: string
          binaryDataN2InformationExt16:
            contentType:  application/vnd.3gpp.ngap
            headers:
              Content-Id:
                schema:
                  type: string
          binaryDataN2InformationExt17:
            contentType:  application/vnd.3gpp.ngap
            headers:
              Content-Id:
                schema:
                  type: string
  required: true
callbacks:
  onN2MessageNotify:
    '{$request.body#/n2NotifyUri}':
      post:
        summary: Namf_Communication N2 Info Notify (UE Specific) service Operation
        tags:
          - N2 Info Notify
        operationId: N2InfoNotifyHandoverComplete
        requestBody:
          description: UE Specific N2 Information Notification
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/N2InformationNotification'
        responses:
          '200':
            description: N2 Information Notification Response.
            content:
              application/json:
                schema:
                  $ref: '#/components/schemas/N2InfoNotificationRspData'
              multipart/related:  # message with binary body part(s)
                schema:
                  type: object
                  properties:
                    jsonData:
                      $ref: '#/components/schemas/N2InfoNotificationRspData'
                    binaryDataN2InformationExt1:
                      type: string
                      format: binary
                    binaryDataN2InformationExt2:
                      type: string
                      format: binary
                    binaryDataN2InformationExt3:
                      type: string
                      format: binary
                    binaryDataN2InformationExt4:
                      type: string
                      format: binary
                    binaryDataN2InformationExt5:
                      type: string
                      format: binary
                    binaryDataN2InformationExt6:
```

```
        type: string
        format: binary
      binaryDataN2InformationExt7:
        type: string
        format: binary
      binaryDataN2InformationExt8:
        type: string
        format: binary
      binaryDataN2InformationExt9:
        type: string
        format: binary
      binaryDataN2InformationExt10:
        type: string
        format: binary
      binaryDataN2InformationExt11:
        type: string
        format: binary
      binaryDataN2InformationExt12:
        type: string
        format: binary
      binaryDataN2InformationExt13:
        type: string
        format: binary
      binaryDataN2InformationExt14:
        type: string
        format: binary
      binaryDataN2InformationExt15:
        type: string
        format: binary
      binaryDataN2InformationExt16:
        type: string
        format: binary
  encoding:
    jsonData:
      contentType:  application/json
    binaryDataN2InformationExt1:
      contentType:  application/vnd.3gpp.ngap
      headers:
        Content-Id:
          schema:
            type: string
    binaryDataN2InformationExt2:
      contentType:  application/vnd.3gpp.ngap
      headers:
        Content-Id:
          schema:
            type: string
    binaryDataN2InformationExt3:
      contentType:  application/vnd.3gpp.ngap
      headers:
        Content-Id:
          schema:
            type: string
    binaryDataN2InformationExt4:
      contentType:  application/vnd.3gpp.ngap
      headers:
        Content-Id:
          schema:
            type: string
    binaryDataN2InformationExt5:
      contentType:  application/vnd.3gpp.ngap
      headers:
        Content-Id:
          schema:
            type: string
    binaryDataN2InformationExt6:
      contentType:  application/vnd.3gpp.ngap
      headers:
        Content-Id:
          schema:
            type: string
    binaryDataN2InformationExt7:
      contentType:  application/vnd.3gpp.ngap
      headers:
        Content-Id:
          schema:
            type: string
    binaryDataN2InformationExt8:
```

```
              contentType:  application/vnd.3gpp.ngap
              headers:
                Content-Id:
                  schema:
                    type: string
            binaryDataN2InformationExt9:
              contentType:  application/vnd.3gpp.ngap
              headers:
                Content-Id:
                  schema:
                    type: string
            binaryDataN2InformationExt10:
              contentType:  application/vnd.3gpp.ngap
              headers:
                Content-Id:
                  schema:
                    type: string
            binaryDataN2InformationExt11:
              contentType:  application/vnd.3gpp.ngap
              headers:
                Content-Id:
                  schema:
                    type: string
            binaryDataN2InformationExt12:
              contentType:  application/vnd.3gpp.ngap
              headers:
                Content-Id:
                  schema:
                    type: string
            binaryDataN2InformationExt13:
              contentType:  application/vnd.3gpp.ngap
              headers:
                Content-Id:
                  schema:
                    type: string
            binaryDataN2InformationExt14:
              contentType:  application/vnd.3gpp.ngap
              headers:
                Content-Id:
                  schema:
                    type: string
            binaryDataN2InformationExt15:
              contentType:  application/vnd.3gpp.ngap
              headers:
                Content-Id:
                  schema:
                    type: string
            binaryDataN2InformationExt16:
              contentType:  application/vnd.3gpp.ngap
              headers:
                Content-Id:
                  schema:
                    type: string
        '204':
          description: Expected response to a successful callback processing
        '307':
          $ref: 'TS29571_CommonData.yaml#/components/responses/307'
        '308':
          $ref: 'TS29571_CommonData.yaml#/components/responses/308'
        '400':
          $ref: 'TS29571_CommonData.yaml#/components/responses/400'
        '401':
          $ref: 'TS29571_CommonData.yaml#/components/responses/401'
        '403':
          $ref: 'TS29571_CommonData.yaml#/components/responses/403'
        '404':
          $ref: 'TS29571_CommonData.yaml#/components/responses/404'
        '411':
          $ref: 'TS29571_CommonData.yaml#/components/responses/411'
        '413':
          $ref: 'TS29571_CommonData.yaml#/components/responses/413'
        '415':
          $ref: 'TS29571_CommonData.yaml#/components/responses/415'
        '429':
          $ref: 'TS29571_CommonData.yaml#/components/responses/429'
        '500':
          $ref: 'TS29571_CommonData.yaml#/components/responses/500'
        '502':
```

```
            $ref: 'TS29571_CommonData.yaml#/components/responses/502'
          '503':
            $ref: 'TS29571_CommonData.yaml#/components/responses/503'
responses:
  '201':
    description: UE context successfully created.
    headers:
      Location:
        description: >
          Contains the URI of the newly created resource, according to the structure:
          {apiRoot}/namf-comm/<apiVersion>/ue-contexts/{ueContextId}
        required: true
        schema:
          type: string
      3gpp-Sbi-Producer-Id:
        description: >
          Indicating the AMF serving the UE Context. This header shall be included when the
          UE Context is created in a target AMF other than the initial AMF sending the
          response.
        schema:
          type: string
    content:
      application/json:
        schema:
          $ref: '#/components/schemas/UeContextCreatedData'
      multipart/related:  # message with binary body part(s)
        schema:
          type: object
          properties: # Request parts
            jsonData:
              $ref: '#/components/schemas/UeContextCreatedData'
            binaryDataN2Information:
              type: string
              format: binary
            binaryDataN2InformationExt1:
              type: string
              format: binary
            binaryDataN2InformationExt2:
              type: string
              format: binary
            binaryDataN2InformationExt3:
              type: string
              format: binary
            binaryDataN2InformationExt4:
              type: string
              format: binary
            binaryDataN2InformationExt5:
              type: string
              format: binary
            binaryDataN2InformationExt6:
              type: string
              format: binary
            binaryDataN2InformationExt7:
              type: string
              format: binary
            binaryDataN2InformationExt8:
              type: string
              format: binary
            binaryDataN2InformationExt9:
              type: string
              format: binary
            binaryDataN2InformationExt10:
              type: string
              format: binary
            binaryDataN2InformationExt11:
              type: string
              format: binary
            binaryDataN2InformationExt12:
              type: string
              format: binary
            binaryDataN2InformationExt13:
              type: string
              format: binary
            binaryDataN2InformationExt14:
              type: string
              format: binary
            binaryDataN2InformationExt15:
              type: string
```

```
      format: binary
encoding:
  jsonData:
    contentType:  application/json
  binaryDataN2Information:
    contentType:  application/vnd.3gpp.ngap
    headers:
      Content-Id:
        schema:
          type: string
  binaryDataN2InformationExt1:
    contentType:  application/vnd.3gpp.ngap
    headers:
      Content-Id:
        schema:
          type: string
  binaryDataN2InformationExt2:
    contentType:  application/vnd.3gpp.ngap
    headers:
      Content-Id:
        schema:
          type: string
  binaryDataN2InformationExt3:
    contentType:  application/vnd.3gpp.ngap
    headers:
      Content-Id:
        schema:
          type: string
  binaryDataN2InformationExt4:
    contentType:  application/vnd.3gpp.ngap
    headers:
      Content-Id:
        schema:
          type: string
  binaryDataN2InformationExt5:
    contentType:  application/vnd.3gpp.ngap
    headers:
      Content-Id:
        schema:
          type: string
  binaryDataN2InformationExt6:
    contentType:  application/vnd.3gpp.ngap
    headers:
      Content-Id:
        schema:
          type: string
  binaryDataN2InformationExt7:
    contentType:  application/vnd.3gpp.ngap
    headers:
      Content-Id:
        schema:
          type: string
  binaryDataN2InformationExt8:
    contentType:  application/vnd.3gpp.ngap
    headers:
      Content-Id:
        schema:
          type: string
  binaryDataN2InformationExt9:
    contentType:  application/vnd.3gpp.ngap
    headers:
      Content-Id:
        schema:
          type: string
  binaryDataN2InformationExt10:
    contentType:  application/vnd.3gpp.ngap
    headers:
      Content-Id:
        schema:
          type: string
  binaryDataN2InformationExt11:
    contentType:  application/vnd.3gpp.ngap
    headers:
      Content-Id:
        schema:
          type: string
  binaryDataN2InformationExt12:
    contentType:  application/vnd.3gpp.ngap
```

```
          headers:
            Content-Id:
              schema:
                type: string
        binaryDataN2InformationExt13:
          contentType:  application/vnd.3gpp.ngap
          headers:
            Content-Id:
              schema:
                type: string
        binaryDataN2InformationExt14:
          contentType:  application/vnd.3gpp.ngap
          headers:
            Content-Id:
              schema:
                type: string
        binaryDataN2InformationExt15:
          contentType:  application/vnd.3gpp.ngap
          headers:
            Content-Id:
              schema:
                type: string
'307':
  $ref: 'TS29571_CommonData.yaml#/components/responses/307'
'308':
  $ref: 'TS29571_CommonData.yaml#/components/responses/308'
'400':
  description: Bad Request
  content:
    application/json:
      schema:
        $ref: '#/components/schemas/UeContextCreateError'
    application/problem+json: # error originated by an SCP or SEPP
      schema:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/ProblemDetails'
'401':
  $ref: 'TS29571_CommonData.yaml#/components/responses/401'
'403':
  description: Forbidden
  content:
    application/json:
      schema:
        $ref: '#/components/schemas/UeContextCreateError'
    application/problem+json: # error originated by an SCP or SEPP
      schema:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/ProblemDetails'
    multipart/related:  # message with binary body part(s)
      schema:
        type: object
        properties: # Response parts
          jsonData:
            $ref: '#/components/schemas/UeContextCreateError'
          binaryDataN2Information:
            type: string
            format: binary
      encoding:
        jsonData:
          contentType:  application/json
        binaryDataN2Information:
          contentType:  application/vnd.3gpp.ngap
          headers:
            Content-Id:
              schema:
                type: string
'404':
  $ref: 'TS29571_CommonData.yaml#/components/responses/404'
'411':
  $ref: 'TS29571_CommonData.yaml#/components/responses/411'
'413':
  $ref: 'TS29571_CommonData.yaml#/components/responses/413'
'415':
  $ref: 'TS29571_CommonData.yaml#/components/responses/415'
'429':
  $ref: 'TS29571_CommonData.yaml#/components/responses/429'
'500':
  description: Internal Server Error
  content:
    application/json:
```

```
              schema:
                $ref: '#/components/schemas/UeContextCreateError'
        '502':
          $ref: 'TS29571_CommonData.yaml#/components/responses/502'
        '503':
          $ref: 'TS29571_CommonData.yaml#/components/responses/503'
        default:
          description: Unexpected error
  /ue-contexts/{ueContextId}/release:
    post:
      summary: Namf_Communication ReleaseUEContext service Operation
      tags:
        - Individual ueContext (Document)
      operationId: ReleaseUEContext
      security:
        - {}
        - oAuth2ClientCredentials:
          - namf-comm
        - oAuth2ClientCredentials:
          - namf-comm
          - namf-comm:ue-contexts:mobility
      parameters:
        - name: ueContextId
          in: path
          description: UE Context Identifier
          required: true
          schema:
            type: string
            pattern: '^(5g-guti-[0-9]{5,6}[0-9a-fA-F]{14}|imsi-[0-9]{5,15}|nai-.+|gli-.+|gci-
.+|imei-[0-9]{15}|imeisv-[0-9]{16}|.+)$'
      requestBody:
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/UEContextRelease'
        required: true
      responses:
        '204':
          description: UE Context successfully released
        '307':
          $ref: 'TS29571_CommonData.yaml#/components/responses/307'
        '308':
          $ref: 'TS29571_CommonData.yaml#/components/responses/308'
        '400':
          $ref: 'TS29571_CommonData.yaml#/components/responses/400'
        '401':
          $ref: 'TS29571_CommonData.yaml#/components/responses/401'
        '403':
          $ref: 'TS29571_CommonData.yaml#/components/responses/403'
        '404':
          $ref: 'TS29571_CommonData.yaml#/components/responses/404'
        '411':
          $ref: 'TS29571_CommonData.yaml#/components/responses/411'
        '413':
          $ref: 'TS29571_CommonData.yaml#/components/responses/413'
        '415':
          $ref: 'TS29571_CommonData.yaml#/components/responses/415'
        '429':
          $ref: 'TS29571_CommonData.yaml#/components/responses/429'
        '500':
          $ref: 'TS29571_CommonData.yaml#/components/responses/500'
        '502':
          $ref: 'TS29571_CommonData.yaml#/components/responses/502'
        '503':
          $ref: 'TS29571_CommonData.yaml#/components/responses/503'
        default:
          description: Unexpected error
  /ue-contexts/{ueContextId}/assign-ebi:
    post:
      summary: Namf_Communication EBI Assignment service Operation
      tags:
        - Individual ueContext (Document)
      operationId: EBIAssignment
      security:
        - {}
        - oAuth2ClientCredentials:
          - namf-comm
        - oAuth2ClientCredentials:
```

```
        - namf-comm
        - namf-comm:ue-contexts:assign-ebi
      parameters:
        - name: ueContextId
          in: path
          description: UE Context Identifier
          required: true
          schema:
            type: string
            pattern: '^(5g-guti-[0-9]{5,6}[0-9a-fA-F]{14}|imsi-[0-9]{5,15}|nai-.+|gli-.+|gci-
.+|imei-[0-9]{15}|imeisv-[0-9]{16}|.+)$'
      requestBody:
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/AssignEbiData'
        required: true
      responses:
        '200':
          description: EBI Assignment successfully performed.
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/AssignedEbiData'
        '307':
          $ref: 'TS29571_CommonData.yaml#/components/responses/307'
        '308':
          $ref: 'TS29571_CommonData.yaml#/components/responses/308'
        '400':
          description: Bad Request
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/AssignEbiError'
            application/problem+json: # error originated by an SCP
              schema:
                $ref: 'TS29571_CommonData.yaml#/components/schemas/ProblemDetails'
        '401':
          $ref: 'TS29571_CommonData.yaml#/components/responses/401'
        '403':
          description: Forbidden
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/AssignEbiError'
            application/problem+json: # error originated by an SCP
              schema:
                $ref: 'TS29571_CommonData.yaml#/components/schemas/ProblemDetails'
        '404':
          $ref: 'TS29571_CommonData.yaml#/components/responses/404'
        '409':
          description: Conflict
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/AssignEbiError'
        '411':
          $ref: 'TS29571_CommonData.yaml#/components/responses/411'
        '413':
          $ref: 'TS29571_CommonData.yaml#/components/responses/413'
        '415':
          $ref: 'TS29571_CommonData.yaml#/components/responses/415'
        '429':
          $ref: 'TS29571_CommonData.yaml#/components/responses/429'
        '500':
          description: Internal Server Error
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/AssignEbiError'
            application/problem+json: # error originated by an SCP
              schema:
                $ref: 'TS29571_CommonData.yaml#/components/schemas/ProblemDetails'
        '502':
          $ref: 'TS29571_CommonData.yaml#/components/responses/502'
        '503':
          $ref: 'TS29571_CommonData.yaml#/components/responses/503'
        default:
```

```
          description: Unexpected error
  /ue-contexts/{ueContextId}/transfer:
    post:
      summary: Namf_Communication UEContextTransfer service Operation
      tags:
        - Individual ueContext (Document)
      operationId: UEContextTransfer
      security:
        - {}
        - oAuth2ClientCredentials:
          - namf-comm
        - oAuth2ClientCredentials:
          - namf-comm
          - namf-comm:ue-contexts:mobility
      parameters:
        - name: ueContextId
          in: path
          description: UE Context Identifier
          required: true
          schema:
            type: string
            pattern: '^(5g-guti-[0-9]{5,6}[0-9a-fA-F]{14}|imsi-[0-9]{5,15}|nai-.+|gli-.+|gci-
.+|imei-[0-9]{15}|imeisv-[0-9]{16}|.+)$'
      requestBody:
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/UeContextTransferReqData'
          multipart/related:  # message with binary body part(s)
            schema:
              type: object
              properties: # Request parts
                jsonData:
                  $ref: '#/components/schemas/UeContextTransferReqData'
                binaryDataN1Message:
                  type: string
                  format: binary
            encoding:
              jsonData:
                contentType:  application/json
              binaryDataN1Message:
                contentType:  application/vnd.3gpp.5gnas
                headers:
                  Content-Id:
                    schema:
                      type: string
        required: true
      responses:
        '200':
          description: UE context transfer successfully initiated.
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/UeContextTransferRspData'
            multipart/related:  # message with binary body part(s)
              schema:
                type: object
                properties: # Request parts
                  jsonData:
                    $ref: '#/components/schemas/UeContextTransferRspData'
                  binaryDataN2Information:
                    type: string
                    format: binary
                  binaryDataN2InformationExt1:
                    type: string
                    format: binary
                  binaryDataN2InformationExt2:
                    type: string
                    format: binary
              encoding:
                jsonData:
                  contentType:  application/json
                binaryDataN2Information:
                  contentType:  application/vnd.3gpp.ngap
                  headers:
                    Content-Id:
                      schema:
                        type: string
```

```
                binaryDataN2InformationExt1:
                  contentType:  application/vnd.3gpp.ngap
                  headers:
                    Content-Id:
                      schema:
                        type: string
                binaryDataN2InformationExt2:
                  contentType:  application/vnd.3gpp.ngap
                  headers:
                    Content-Id:
                      schema:
                        type: string
        '307':
          $ref: 'TS29571_CommonData.yaml#/components/responses/307'
        '308':
          $ref: 'TS29571_CommonData.yaml#/components/responses/308'
        '400':
          $ref: 'TS29571_CommonData.yaml#/components/responses/400'
        '401':
          $ref: 'TS29571_CommonData.yaml#/components/responses/401'
        '403':
          $ref: 'TS29571_CommonData.yaml#/components/responses/403'
        '404':
          $ref: 'TS29571_CommonData.yaml#/components/responses/404'
        '411':
          $ref: 'TS29571_CommonData.yaml#/components/responses/411'
        '413':
          $ref: 'TS29571_CommonData.yaml#/components/responses/413'
        '415':
          $ref: 'TS29571_CommonData.yaml#/components/responses/415'
        '429':
          $ref: 'TS29571_CommonData.yaml#/components/responses/429'
        '500':
          $ref: 'TS29571_CommonData.yaml#/components/responses/500'
        '502':
          $ref: 'TS29571_CommonData.yaml#/components/responses/502'
        '503':
          $ref: 'TS29571_CommonData.yaml#/components/responses/503'
        default:
          description: Unexpected error
  /ue-contexts/{ueContextId}/transfer-update:
    post:
      summary: Namf_Communication RegistrationStatusUpdate service Operation
      tags:
        - Individual ueContext (Document)
      operationId: RegistrationStatusUpdate
      security:
        - {}
        - oAuth2ClientCredentials:
          - namf-comm
        - oAuth2ClientCredentials:
          - namf-comm
          - namf-comm:ue-contexts:mobility
      parameters:
        - name: ueContextId
          in: path
          description: UE Context Identifier
          required: true
          schema:
            type: string
            pattern: '^(5g-guti-[0-9]{5,6}[0-9a-fA-F]{14}|imsi-[0-9]{5,15}|nai-.+|gli-.+|gci-
.+|imei-[0-9]{15}|imeisv-[0-9]{16}|.+)$'
      requestBody:
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/UeRegStatusUpdateReqData'
        required: true
      responses:
        '200':
          description: UE context transfer status successfully updated.
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/UeRegStatusUpdateRspData'
        '307':
          $ref: 'TS29571_CommonData.yaml#/components/responses/307'
        '308':
```

```
          $ref: 'TS29571_CommonData.yaml#/components/responses/308'
        '400':
          $ref: 'TS29571_CommonData.yaml#/components/responses/400'
        '401':
          $ref: 'TS29571_CommonData.yaml#/components/responses/401'
        '403':
          $ref: 'TS29571_CommonData.yaml#/components/responses/403'
        '404':
          $ref: 'TS29571_CommonData.yaml#/components/responses/404'
        '411':
          $ref: 'TS29571_CommonData.yaml#/components/responses/411'
        '413':
          $ref: 'TS29571_CommonData.yaml#/components/responses/413'
        '415':
          $ref: 'TS29571_CommonData.yaml#/components/responses/415'
        '429':
          $ref: 'TS29571_CommonData.yaml#/components/responses/429'
        '500':
          $ref: 'TS29571_CommonData.yaml#/components/responses/500'
        '502':
          $ref: 'TS29571_CommonData.yaml#/components/responses/502'
        '503':
          $ref: 'TS29571_CommonData.yaml#/components/responses/503'
        default:
          $ref: 'TS29571_CommonData.yaml#/components/responses/default'
  /ue-contexts/{ueContextId}/relocate:
    post:
      summary: Namf_Communication RelocateUEContext service Operation
      tags:
        - Individual ueContext (Document)
      operationId: RelocateUEContext
      security:
        - {}
        - oAuth2ClientCredentials:
          - namf-comm
        - oAuth2ClientCredentials:
          - namf-comm
          - namf-comm:ue-contexts:mobility
      parameters:
        - name: ueContextId
          in: path
          description: UE Context Identifier
          required: true
          schema:
            type: string
            pattern: '^(5g-guti-[0-9]{5,6}[0-9a-fA-F]{14}|imsi-[0-9]{5,15}|nai-.+|gli-.+|gci-
.+|imei-[0-9]{15}|imeisv-[0-9]{16}|.+)$'
      requestBody:
        content:
          multipart/related:  # message with binary body part(s)
            schema:
              type: object
              properties: # Request parts
                jsonData:
                  $ref: '#/components/schemas/UeContextRelocateData'
                binaryDataForwardRelocationRequest:
                  type: string
                  format: binary
                binaryDataN2Information:
                  type: string
                  format: binary
                binaryDataN2InformationExt1:
                  type: string
                  format: binary
                binaryDataN2InformationExt2:
                  type: string
                  format: binary
                binaryDataN2InformationExt3:
                  type: string
                  format: binary
                binaryDataN2InformationExt4:
                  type: string
                  format: binary
                binaryDataN2InformationExt5:
                  type: string
                  format: binary
                binaryDataN2InformationExt6:
                  type: string
```

```
        format: binary
      binaryDataN2InformationExt7:
        type: string
        format: binary
      binaryDataN2InformationExt8:
        type: string
        format: binary
      binaryDataN2InformationExt9:
        type: string
        format: binary
      binaryDataN2InformationExt10:
        type: string
        format: binary
      binaryDataN2InformationExt11:
        type: string
        format: binary
      binaryDataN2InformationExt12:
        type: string
        format: binary
      binaryDataN2InformationExt13:
        type: string
        format: binary
      binaryDataN2InformationExt14:
        type: string
        format: binary
      binaryDataN2InformationExt15:
        type: string
        format: binary
      binaryDataN2InformationExt16:
        type: string
        format: binary
  encoding:
    jsonData:
      contentType:  application/json
    binaryDataForwardRelocationRequest:
      contentType:  application/vnd.3gpp.gtpc
      headers:
        Content-Id:
          schema:
            type: string
    binaryDataN2Information:
      contentType:  application/vnd.3gpp.ngap
      headers:
        Content-Id:
          schema:
            type: string
    binaryDataN2InformationExt1:
      contentType:  application/vnd.3gpp.ngap
      headers:
        Content-Id:
          schema:
            type: string
    binaryDataN2InformationExt2:
      contentType:  application/vnd.3gpp.ngap
      headers:
        Content-Id:
          schema:
            type: string
    binaryDataN2InformationExt3:
      contentType:  application/vnd.3gpp.ngap
      headers:
        Content-Id:
          schema:
            type: string
    binaryDataN2InformationExt4:
      contentType:  application/vnd.3gpp.ngap
      headers:
        Content-Id:
          schema:
            type: string
    binaryDataN2InformationExt5:
      contentType:  application/vnd.3gpp.ngap
      headers:
        Content-Id:
          schema:
            type: string
    binaryDataN2InformationExt6:
      contentType:  application/vnd.3gpp.ngap
```

```
              headers:
                Content-Id:
                  schema:
                    type: string
            binaryDataN2InformationExt7:
              contentType:  application/vnd.3gpp.ngap
              headers:
                Content-Id:
                  schema:
                    type: string
            binaryDataN2InformationExt8:
              contentType:  application/vnd.3gpp.ngap
              headers:
                Content-Id:
                  schema:
                    type: string
            binaryDataN2InformationExt9:
              contentType:  application/vnd.3gpp.ngap
              headers:
                Content-Id:
                  schema:
                    type: string
            binaryDataN2InformationExt10:
              contentType:  application/vnd.3gpp.ngap
              headers:
                Content-Id:
                  schema:
                    type: string
            binaryDataN2InformationExt11:
              contentType:  application/vnd.3gpp.ngap
              headers:
                Content-Id:
                  schema:
                    type: string
            binaryDataN2InformationExt12:
              contentType:  application/vnd.3gpp.ngap
              headers:
                Content-Id:
                  schema:
                    type: string
            binaryDataN2InformationExt13:
              contentType:  application/vnd.3gpp.ngap
              headers:
                Content-Id:
                  schema:
                    type: string
            binaryDataN2InformationExt14:
              contentType:  application/vnd.3gpp.ngap
              headers:
                Content-Id:
                  schema:
                    type: string
            binaryDataN2InformationExt15:
              contentType:  application/vnd.3gpp.ngap
              headers:
                Content-Id:
                  schema:
                    type: string
            binaryDataN2InformationExt16:
              contentType:  application/vnd.3gpp.ngap
              headers:
                Content-Id:
                  schema:
                    type: string
        required: true
      responses:
        '201':
          description: UE context successfully relocated.
          headers:
            Location:
              description: 'Contains the URI of the newly created resource, according to the
structure: {apiRoot}/namf-comm/<apiVersion>/ue-contexts/{ueContextId}/relocate'
              required: true
              schema:
                type: string
          content:
            application/json:
              schema:
```

```
                $ref: '#/components/schemas/UeContextRelocatedData'
        '307':
          $ref: 'TS29571_CommonData.yaml#/components/responses/307'
        '308':
          $ref: 'TS29571_CommonData.yaml#/components/responses/308'
        '400':
          $ref: 'TS29571_CommonData.yaml#/components/responses/400'
        '401':
          $ref: 'TS29571_CommonData.yaml#/components/responses/401'
        '403':
          $ref: 'TS29571_CommonData.yaml#/components/responses/403'
        '404':
          $ref: 'TS29571_CommonData.yaml#/components/responses/404'
        '411':
          $ref: 'TS29571_CommonData.yaml#/components/responses/411'
        '413':
          $ref: 'TS29571_CommonData.yaml#/components/responses/413'
        '415':
          $ref: 'TS29571_CommonData.yaml#/components/responses/415'
        '429':
          $ref: 'TS29571_CommonData.yaml#/components/responses/429'
        '500':
          $ref: 'TS29571_CommonData.yaml#/components/responses/500'
        '502':
          $ref: 'TS29571_CommonData.yaml#/components/responses/502'
        '503':
          $ref: 'TS29571_CommonData.yaml#/components/responses/503'
        default:
          description: Unexpected error
  /ue-contexts/{ueContextId}/cancel-relocate:
    post:
      summary: Namf_Communication CancelRelocateUEContext service Operation
      tags:
        - Individual ueContext (Document)
      operationId: CancelRelocateUEContext
      security:
        - {}
        - oAuth2ClientCredentials:
          - namf-comm
        - oAuth2ClientCredentials:
          - namf-comm
          - namf-comm:ue-contexts:mobility
      parameters:
        - name: ueContextId
          in: path
          description: UE Context Identifier
          required: true
          schema:
            type: string
            pattern: '^(5g-guti-[0-9]{5,6}[0-9a-fA-F]{14}|imsi-[0-9]{5,15}|nai-.+|gli-.+|gci-
.+|imei-[0-9]{15}|imeisv-[0-9]{16}|.+)$'
      requestBody:
        content:
          multipart/related:  # message with binary body part(s)
            schema:
              type: object
              properties: # Request parts
                jsonData:
                  $ref: '#/components/schemas/UeContextCancelRelocateData'
                binaryDataGtpcMessage:
                  type: string
                  format: binary
            encoding:
              jsonData:
                contentType:  application/json
              binaryDataGtpcMessage:
                contentType:  application/vnd.3gpp.gtpc
                headers:
                  Content-Id:
                    schema:
                      type: string
        required: true
      responses:
        '204':
          description: UE Context successfully released
        '307':
          $ref: 'TS29571_CommonData.yaml#/components/responses/307'
        '308':
```

```
          $ref: 'TS29571_CommonData.yaml#/components/responses/308'
        '400':
          $ref: 'TS29571_CommonData.yaml#/components/responses/400'
        '401':
          $ref: 'TS29571_CommonData.yaml#/components/responses/401'
        '403':
          $ref: 'TS29571_CommonData.yaml#/components/responses/403'
        '404':
          $ref: 'TS29571_CommonData.yaml#/components/responses/404'
        '411':
          $ref: 'TS29571_CommonData.yaml#/components/responses/411'
        '413':
          $ref: 'TS29571_CommonData.yaml#/components/responses/413'
        '415':
          $ref: 'TS29571_CommonData.yaml#/components/responses/415'
        '429':
          $ref: 'TS29571_CommonData.yaml#/components/responses/429'
        '500':
          $ref: 'TS29571_CommonData.yaml#/components/responses/500'
        '502':
          $ref: 'TS29571_CommonData.yaml#/components/responses/502'
        '503':
          $ref: 'TS29571_CommonData.yaml#/components/responses/503'
        default:
          description: Unexpected error
  /ue-contexts/{ueContextId}/n1-n2-messages:
    post:
      summary: Namf_Communication N1N2 Message Transfer (UE Specific) service Operation
      tags:
        - n1N2Message collection (Collection)
      operationId: N1N2MessageTransfer
      security:
        - {}
        - oAuth2ClientCredentials:
          - namf-comm
        - oAuth2ClientCredentials:
          - namf-comm
          - namf-comm:n1-n2-messages
      parameters:
        - name: ueContextId
          in: path
          description: UE Context Identifier
          required: true
          schema:
            type: string
            pattern: '^(imsi-[0-9]{5,15}|nai-.+|gli-.+|gci-.+|imei-[0-9]{15}|imeisv-[0-9]{16}|cid-
.{1,255}|.+)$'
      requestBody:
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/N1N2MessageTransferReqData'
          multipart/related:  # message with binary body part(s)
            schema:
              type: object
              properties: # Request parts
                jsonData:
                  $ref: '#/components/schemas/N1N2MessageTransferReqData'
                binaryDataN1Message:
                  type: string
                  format: binary
                binaryDataN2Information:
                  type: string
                  format: binary
                binaryMtData:
                  type: string
                  format: binary
            encoding:
              jsonData:
                contentType:  application/json
              binaryDataN1Message:
                contentType:  application/vnd.3gpp.5gnas
                headers:
                  Content-Id:
                    schema:
                      type: string
              binaryDataN2Information:
                contentType:  application/vnd.3gpp.ngap
```

```
            headers:
              Content-Id:
                schema:
                  type: string
          binaryMtData:
            contentType:  application/vnd.3gpp.5gnas
            headers:
              Content-Id:
                schema:
                  type: string
    required: true
  responses:
    '202':
      description: N1N2 Message Transfer accepted.
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/N1N2MessageTransferRspData'
      headers:
        Location:
          description: >
            The URI of the resource located on the AMF In this release, the URI shall only be
            used by NF Service Consumer to correlate the possible N1/N2 Message Transfer Failure
            Notification With the related N1/N2 Message Transfer Operation. The NF service
            consumer shall not send any service requests towards the URI received in the
            Location header.
          required: true
          schema:
            type: string
    '200':
      description: N1N2 Message Transfer successfully initiated.
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/N1N2MessageTransferRspData'
    '307':
      $ref: 'TS29571_CommonData.yaml#/components/responses/307'
    '308':
      $ref: 'TS29571_CommonData.yaml#/components/responses/308'
    '400':
      $ref: 'TS29571_CommonData.yaml#/components/responses/400'
    '401':
      $ref: 'TS29571_CommonData.yaml#/components/responses/401'
    '403':
      $ref: 'TS29571_CommonData.yaml#/components/responses/403'
    '404':
      $ref: 'TS29571_CommonData.yaml#/components/responses/404'
    '409':
      description: Conflicts
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/N1N2MessageTransferError'
    '411':
      $ref: 'TS29571_CommonData.yaml#/components/responses/411'
    '413':
      $ref: 'TS29571_CommonData.yaml#/components/responses/413'
    '415':
      $ref: 'TS29571_CommonData.yaml#/components/responses/415'
    '429':
      $ref: 'TS29571_CommonData.yaml#/components/responses/429'
    '500':
      $ref: 'TS29571_CommonData.yaml#/components/responses/500'
    '502':
      $ref: 'TS29571_CommonData.yaml#/components/responses/502'
    '503':
      $ref: 'TS29571_CommonData.yaml#/components/responses/503'
    '504':
      description: Gateway Timeout
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/N1N2MessageTransferError'
        application/problem+json: # error originated by an SCP or SEPP
          schema:
            $ref: 'TS29571_CommonData.yaml#/components/schemas/ProblemDetails'
    default:
      description: Unexpected error
```

```
      callbacks:
        onN1N2TransferFailure:
          '{$request.body#/n1n2FailureTxfNotifURI}':
            post:
              summary: Namf_Communication N1N2Transfer Failure Notification service Operation
              tags:
                - N1N2 Transfer Failure Notification
              operationId: N1N2TransferFailureNotification
              requestBody:
                description: N1N2Transfer Failure Notification
                content:
                  application/json:
                    schema:
                      $ref: '#/components/schemas/N1N2MsgTxfrFailureNotification'
              responses:
                '204':
                  description: Expected response to a successful callback processing
                '307':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/307'
                '308':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/308'
                '400':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/400'
                '401':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/401'
                '403':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/403'
                '404':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/404'
                '411':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/411'
                '413':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/413'
                '415':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/415'
                '429':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/429'
                '500':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/500'
                '502':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/502'
                '503':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/503'
  /ue-contexts/{ueContextId}/n1-n2-messages/subscriptions:
    post:
      summary: Namf_Communication N1N2 Message Subscribe (UE Specific) service Operation
      tags:
        - N1N2 Subscriptions Collection for Individual UE Contexts (Collection)
      operationId: N1N2MessageSubscribe
      security:
        - {}
        - oAuth2ClientCredentials:
          - namf-comm
        - oAuth2ClientCredentials:
          - namf-comm
          - namf-comm:n1-n2-messages
      parameters:
        - name: ueContextId
          in: path
          description: UE Context Identifier
          required: true
          schema:
            type: string
            pattern: '^(imsi-[0-9]{5,15}|nai-.+|gli-.+|gci-.+|imei-[0-9]{15}|imeisv-[0-9]{16}|.+)$'
      requestBody:
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/UeN1N2InfoSubscriptionCreateData'
        required: true
      responses:
        '201':
          description: N1N2 Message Subscription successfully created.
          headers:
            Location:
              description: 'Contains the URI of the newly created resource, according to the
structure: {apiRoot}/namf-comm/<apiVersion>/ue-contexts/{ueContextId}/n1-n2-
messages/subscriptions/{subscriptionId}'
```

```
          required: true
          schema:
            type: string
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/UeN1N2InfoSubscriptionCreatedData'
    '307':
      $ref: 'TS29571_CommonData.yaml#/components/responses/307'
    '308':
      $ref: 'TS29571_CommonData.yaml#/components/responses/308'
    '400':
      $ref: 'TS29571_CommonData.yaml#/components/responses/400'
    '401':
      $ref: 'TS29571_CommonData.yaml#/components/responses/401'
    '403':
      $ref: 'TS29571_CommonData.yaml#/components/responses/403'
    '404':
      $ref: 'TS29571_CommonData.yaml#/components/responses/404'
    '411':
      $ref: 'TS29571_CommonData.yaml#/components/responses/411'
    '413':
      $ref: 'TS29571_CommonData.yaml#/components/responses/413'
    '415':
      $ref: 'TS29571_CommonData.yaml#/components/responses/415'
    '429':
      $ref: 'TS29571_CommonData.yaml#/components/responses/429'
    '500':
      $ref: 'TS29571_CommonData.yaml#/components/responses/500'
    '502':
      $ref: 'TS29571_CommonData.yaml#/components/responses/502'
    '503':
      $ref: 'TS29571_CommonData.yaml#/components/responses/503'
    default:
      description: Unexpected error
  callbacks:
    onN1N2MessageNotify:
      '{$request.body#/n1NotifyCallbackUri}':
        post:
          summary: Namf_Communication N1 Message Notify service Operation
          tags:
            - N1 Message Notify
          operationId: N1MessageNotify
          requestBody:
            description: N1 Message Notification
            content:
              multipart/related:  # message with binary body part(s)
                schema:
                  type: object
                  properties: # Request parts
                    jsonData:
                      $ref: '#/components/schemas/N1MessageNotification'
                    binaryDataN1Message:
                      type: string
                      format: binary
                encoding:
                  jsonData:
                    contentType:  application/json
                  binaryDataN1Message:
                    contentType:  application/vnd.3gpp.5gnas
                    headers:
                      Content-Id:
                        schema:
                          type: string
          responses:
            '204':
              description: Expected response to a successful callback processing
            '307':
              $ref: 'TS29571_CommonData.yaml#/components/responses/307'
            '308':
              $ref: 'TS29571_CommonData.yaml#/components/responses/308'
            '400':
              $ref: 'TS29571_CommonData.yaml#/components/responses/400'
            '401':
              $ref: 'TS29571_CommonData.yaml#/components/responses/401'
            '403':
              $ref: 'TS29571_CommonData.yaml#/components/responses/403'
            '404':
```

```
            $ref: 'TS29571_CommonData.yaml#/components/responses/404'
          '411':
            $ref: 'TS29571_CommonData.yaml#/components/responses/411'
          '413':
            $ref: 'TS29571_CommonData.yaml#/components/responses/413'
          '415':
            $ref: 'TS29571_CommonData.yaml#/components/responses/415'
          '429':
            $ref: 'TS29571_CommonData.yaml#/components/responses/429'
          '500':
            $ref: 'TS29571_CommonData.yaml#/components/responses/500'
          '502':
            $ref: 'TS29571_CommonData.yaml#/components/responses/502'
          '503':
            $ref: 'TS29571_CommonData.yaml#/components/responses/503'
      '{$request.body#/n2NotifyCallbackUri}':
        post:
          summary: Namf_Communication N2 Info Notify (UE Specific) service Operation
          tags:
            - N2 Info Notify
          operationId: N2InfoNotify
          requestBody:
            description: UE Specific N2 Information Notification
            content:
              multipart/related:  # message with binary body part(s)
                schema:
                  type: object
                  properties: # Request parts
                    jsonData:
                      $ref: '#/components/schemas/N2InformationNotification'
                    binaryDataN1Message:
                      type: string
                      format: binary
                    binaryDataN2Information:
                      type: string
                      format: binary
                encoding:
                  jsonData:
                    contentType:  application/json
                  binaryDataN1Message:
                    contentType:  application/vnd.3gpp.5gnas
                    headers:
                      Content-Id:
                        schema:
                          type: string
                  binaryDataN2Information:
                    contentType:  application/vnd.3gpp.ngap
                    headers:
                      Content-Id:
                        schema:
                          type: string
          responses:
            '204':
              description: Expected response to a successful callback processing
            '307':
              $ref: 'TS29571_CommonData.yaml#/components/responses/307'
            '308':
              $ref: 'TS29571_CommonData.yaml#/components/responses/308'
            '400':
              $ref: 'TS29571_CommonData.yaml#/components/responses/400'
            '401':
              $ref: 'TS29571_CommonData.yaml#/components/responses/401'
            '403':
              $ref: 'TS29571_CommonData.yaml#/components/responses/403'
            '404':
              $ref: 'TS29571_CommonData.yaml#/components/responses/404'
            '411':
              $ref: 'TS29571_CommonData.yaml#/components/responses/411'
            '413':
              $ref: 'TS29571_CommonData.yaml#/components/responses/413'
            '415':
              $ref: 'TS29571_CommonData.yaml#/components/responses/415'
            '429':
              $ref: 'TS29571_CommonData.yaml#/components/responses/429'
            '500':
              $ref: 'TS29571_CommonData.yaml#/components/responses/500'
            '502':
              $ref: 'TS29571_CommonData.yaml#/components/responses/502'
```

```
              '503':
                $ref: 'TS29571_CommonData.yaml#/components/responses/503'
  /ue-contexts/{ueContextId}/n1-n2-messages/subscriptions/{subscriptionId}:
    delete:
      summary: Namf_Communication N1N2 Message UnSubscribe (UE Specific) service Operation
      tags:
        - N1N2 Individual Subscription (Document)
      operationId: N1N2MessageUnSubscribe
      security:
        - {}
        - oAuth2ClientCredentials:
          - namf-comm
        - oAuth2ClientCredentials:
          - namf-comm
          - namf-comm:n1-n2-messages
      parameters:
        - name: ueContextId
          in: path
          description: UE Context Identifier
          required: true
          schema:
            type: string
            pattern: '^(imsi-[0-9]{5,15}|nai-.+|gli-.+|gci-.+|imei-[0-9]{15}|imeisv-[0-9]{16}|.+)$'
        - name: subscriptionId
          in: path
          description: Subscription Identifier
          required: true
          schema:
            type: string
      responses:
        '204':
          description: N1N2 Message Subscription successfully removed.
        '307':
          $ref: 'TS29571_CommonData.yaml#/components/responses/307'
        '308':
          $ref: 'TS29571_CommonData.yaml#/components/responses/308'
        '400':
          $ref: 'TS29571_CommonData.yaml#/components/responses/400'
        '401':
          $ref: 'TS29571_CommonData.yaml#/components/responses/401'
        '403':
          $ref: 'TS29571_CommonData.yaml#/components/responses/403'
        '404':
          $ref: 'TS29571_CommonData.yaml#/components/responses/404'
        '411':
          $ref: 'TS29571_CommonData.yaml#/components/responses/411'
        '413':
          $ref: 'TS29571_CommonData.yaml#/components/responses/413'
        '415':
          $ref: 'TS29571_CommonData.yaml#/components/responses/415'
        '429':
          $ref: 'TS29571_CommonData.yaml#/components/responses/429'
        '500':
          $ref: 'TS29571_CommonData.yaml#/components/responses/500'
        '502':
          $ref: 'TS29571_CommonData.yaml#/components/responses/502'
        '503':
          $ref: 'TS29571_CommonData.yaml#/components/responses/503'
  /non-ue-n2-messages/transfer:
    post:
      summary: Namf_Communication Non UE N2 Message Transfer service Operation
      tags:
        - Non UE N2Messages collection (Collection)
      operationId: NonUeN2MessageTransfer
      security:
        - {}
        - oAuth2ClientCredentials:
          - namf-comm
        - oAuth2ClientCredentials:
          - namf-comm
          - namf-comm:non-ue-n2-messages
      requestBody:
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/N2InformationTransferReqData'
          multipart/related:  # message with binary body part(s)
            schema:
```

```
          type: object
          properties: # Request parts
            jsonData:
              $ref: '#/components/schemas/N2InformationTransferReqData'
            binaryDataN2Information:
              type: string
              format: binary
        encoding:
          jsonData:
            contentType:  application/json
          binaryDataN2Information:
            contentType:  application/vnd.3gpp.ngap
            headers:
              Content-Id:
                schema:
                  type: string
    required: true
  responses:
    '200':
      description: Non UE N2 Message Transfer successfully initiated.
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/N2InformationTransferRspData'
        multipart/related:  # response with binary body part(s)
          schema:
            type: object
            properties:
              jsonData:
                $ref: '#/components/schemas/N2InformationTransferRspData'
              binaryDataN2Information1:
                type: string
                format: binary
              binaryDataN2Information2:
                type: string
                format: binary
              binaryDataN2Information3:
                type: string
                format: binary
              binaryDataN2Information4:
                type: string
                format: binary
              binaryDataN2Information5:
                type: string
                format: binary
              binaryDataN2Information6:
                type: string
                format: binary
              binaryDataN2Information7:
                type: string
                format: binary
              binaryDataN2Information8:
                type: string
                format: binary
              binaryDataN2Information9:
                type: string
                format: binary
              binaryDataN2Information10:
                type: string
                format: binary
          encoding:
            jsonData:
              contentType:  application/json
            binaryDataN2Information1:
              contentType:  application/vnd.3gpp.ngap
              headers:
                Content-Id:
                  schema:
                    type: string
            binaryDataN2Information2:
              contentType:  application/vnd.3gpp.ngap
              headers:
                Content-Id:
                  schema:
                    type: string
            binaryDataN2Information3:
              contentType:  application/vnd.3gpp.ngap
              headers:
```

```
            Content-Id:
              schema:
                type: string
        binaryDataN2Information4:
          contentType:  application/vnd.3gpp.ngap
          headers:
            Content-Id:
              schema:
                type: string
        binaryDataN2Information5:
          contentType:  application/vnd.3gpp.ngap
          headers:
            Content-Id:
              schema:
                type: string
        binaryDataN2Information6:
          contentType:  application/vnd.3gpp.ngap
          headers:
            Content-Id:
              schema:
                type: string
        binaryDataN2Information7:
          contentType:  application/vnd.3gpp.ngap
          headers:
            Content-Id:
              schema:
                type: string
        binaryDataN2Information8:
          contentType:  application/vnd.3gpp.ngap
          headers:
            Content-Id:
              schema:
                type: string
        binaryDataN2Information9:
          contentType:  application/vnd.3gpp.ngap
          headers:
            Content-Id:
              schema:
                type: string
        binaryDataN2Information10:
          contentType:  application/vnd.3gpp.ngap
          headers:
            Content-Id:
              schema:
                type: string
'307':
  $ref: 'TS29571_CommonData.yaml#/components/responses/307'
'308':
  $ref: 'TS29571_CommonData.yaml#/components/responses/308'
'400':
  description: Bad Request
  content:
    application/json:
      schema:
        $ref: '#/components/schemas/N2InformationTransferError'
    application/problem+json: # error originated by an SCP or SEPP
      schema:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/ProblemDetails'
'401':
  $ref: 'TS29571_CommonData.yaml#/components/responses/401'
'403':
  description: Forbidden
  content:
    application/json:
      schema:
        $ref: '#/components/schemas/N2InformationTransferError'
    application/problem+json: # error originated by an SCP or SEPP
      schema:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/ProblemDetails'
'404':
  description: Not Found
  content:
    application/json:
      schema:
        $ref: '#/components/schemas/N2InformationTransferError'
'411':
  $ref: 'TS29571_CommonData.yaml#/components/responses/411'
'413':
```

```
          $ref: 'TS29571_CommonData.yaml#/components/responses/413'
        '415':
          $ref: 'TS29571_CommonData.yaml#/components/responses/415'
        '429':
          $ref: 'TS29571_CommonData.yaml#/components/responses/429'
        '500':
          description: Internal Server Error
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/N2InformationTransferError'
            application/problem+json: # error originated by an SCP or SEPP
              schema:
                $ref: 'TS29571_CommonData.yaml#/components/schemas/ProblemDetails'
        '502':
          $ref: 'TS29571_CommonData.yaml#/components/responses/502'
        '503':
          description: Service Unavailable
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/N2InformationTransferError'
            application/problem+json: # error originated by an SCP or SEPP
              schema:
                $ref: 'TS29571_CommonData.yaml#/components/schemas/ProblemDetails'
        default:
          description: Unexpected error
  /non-ue-n2-messages/subscriptions:
    post:
      summary: Namf_Communication Non UE N2 Info Subscribe service Operation
      tags:
        - Non UE N2Messages Subscriptions collection (Collection)
      operationId: NonUeN2InfoSubscribe
      security:
        - {}
        - oAuth2ClientCredentials:
          - namf-comm
        - oAuth2ClientCredentials:
          - namf-comm
          - namf-comm:non-ue-n2-messages
      requestBody:
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/NonUeN2InfoSubscriptionCreateData'
        required: true
      responses:
        '201':
          description: Non UE N2 Info Subscription successfully created.
          headers:
            Location:
              description: 'Contains the URI of the newly created resource, according to the
structure: {apiRoot}/namf-comm/<apiVersion>/non-ue-n2-
messages/subscriptions/{n2NotifySubscriptionId}'
              required: true
              schema:
                type: string
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/NonUeN2InfoSubscriptionCreatedData'
        '307':
          $ref: 'TS29571_CommonData.yaml#/components/responses/307'
        '308':
          $ref: 'TS29571_CommonData.yaml#/components/responses/308'
        '400':
          $ref: 'TS29571_CommonData.yaml#/components/responses/400'
        '401':
          $ref: 'TS29571_CommonData.yaml#/components/responses/401'
        '403':
          $ref: 'TS29571_CommonData.yaml#/components/responses/403'
        '404':
          $ref: 'TS29571_CommonData.yaml#/components/responses/404'
        '411':
          $ref: 'TS29571_CommonData.yaml#/components/responses/411'
        '413':
          $ref: 'TS29571_CommonData.yaml#/components/responses/413'
        '415':
```

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      $ref: 'TS29571_CommonData.yaml#/components/responses/415'
    '429':
      $ref: 'TS29571_CommonData.yaml#/components/responses/429'
    '500':
      $ref: 'TS29571_CommonData.yaml#/components/responses/500'
    '502':
      $ref: 'TS29571_CommonData.yaml#/components/responses/502'
    '503':
      $ref: 'TS29571_CommonData.yaml#/components/responses/503'
    default:
      description: Unexpected error
  callbacks:
    onN2InfoNotify:
      '{$request.body#/n2NotifyCallbackUri}':
       post:
         summary: Namf_Communication Non UE N2 Info Notify service Operation
         tags:
           - Non UE N2 Info Notify
         operationId: NonUeN2InfoNotify
         requestBody:
           description: Non UE N2 Information Notification
           content:
             application/json:
               schema:
                 $ref: '#/components/schemas/N2InformationNotification'
             multipart/related:  # message with binary body part(s)
               schema:
                 type: object
                 properties: # Request parts
                   jsonData:
                     $ref: '#/components/schemas/N2InformationNotification'
                   binaryDataN2Information1:
                     type: string
                     format: binary
                   binaryDataN2Information2:
                     type: string
                     format: binary
                   binaryDataN2Information3:
                     type: string
                     format: binary
                   binaryDataN2Information4:
                     type: string
                     format: binary
                   binaryDataN2Information5:
                     type: string
                     format: binary
                   binaryDataN2Information6:
                     type: string
                     format: binary
                   binaryDataN2Information7:
                     type: string
                     format: binary
                   binaryDataN2Information8:
                     type: string
                     format: binary
                   binaryDataN2Information9:
                     type: string
                     format: binary
                   binaryDataN2Information10:
                     type: string
                     format: binary
               encoding:
                 jsonData:
                   contentType:  application/json
                 binaryDataN2Information1:
                   contentType:  application/vnd.3gpp.ngap
                   headers:
                     Content-Id:
                       schema:
                         type: string
                 binaryDataN2Information2:
                   contentType:  application/vnd.3gpp.ngap
                   headers:
                     Content-Id:
                       schema:
                         type: string
                 binaryDataN2Information3:
                   contentType:  application/vnd.3gpp.ngap
```

```
                    headers:
                      Content-Id:
                        schema:
                          type: string
                  binaryDataN2Information4:
                    contentType:  application/vnd.3gpp.ngap
                    headers:
                      Content-Id:
                        schema:
                          type: string
                  binaryDataN2Information5:
                    contentType:  application/vnd.3gpp.ngap
                    headers:
                      Content-Id:
                        schema:
                          type: string
                  binaryDataN2Information6:
                    contentType:  application/vnd.3gpp.ngap
                    headers:
                      Content-Id:
                        schema:
                          type: string
                  binaryDataN2Information7:
                    contentType:  application/vnd.3gpp.ngap
                    headers:
                      Content-Id:
                        schema:
                          type: string
                  binaryDataN2Information8:
                    contentType:  application/vnd.3gpp.ngap
                    headers:
                      Content-Id:
                        schema:
                          type: string
                  binaryDataN2Information9:
                    contentType:  application/vnd.3gpp.ngap
                    headers:
                      Content-Id:
                        schema:
                          type: string
                  binaryDataN2Information10:
                    contentType:  application/vnd.3gpp.ngap
                    headers:
                      Content-Id:
                        schema:
                          type: string
            responses:
              '204':
                description: Expected response to a successful callback processing
              '307':
                $ref: 'TS29571_CommonData.yaml#/components/responses/307'
              '308':
                $ref: 'TS29571_CommonData.yaml#/components/responses/308'
              '400':
                $ref: 'TS29571_CommonData.yaml#/components/responses/400'
              '401':
                $ref: 'TS29571_CommonData.yaml#/components/responses/401'
              '403':
                $ref: 'TS29571_CommonData.yaml#/components/responses/403'
              '404':
                $ref: 'TS29571_CommonData.yaml#/components/responses/404'
              '411':
                $ref: 'TS29571_CommonData.yaml#/components/responses/411'
              '413':
                $ref: 'TS29571_CommonData.yaml#/components/responses/413'
              '415':
                $ref: 'TS29571_CommonData.yaml#/components/responses/415'
              '429':
                $ref: 'TS29571_CommonData.yaml#/components/responses/429'
              '500':
                $ref: 'TS29571_CommonData.yaml#/components/responses/500'
              '502':
                $ref: 'TS29571_CommonData.yaml#/components/responses/502'
              '503':
                $ref: 'TS29571_CommonData.yaml#/components/responses/503'
  /non-ue-n2-messages/subscriptions/{n2NotifySubscriptionId}:
    delete:
      summary: Namf_Communication Non UE N2 Info UnSubscribe service Operation
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```
      tags:
        - Non UE N2 Message Notification Individual Subscription (Document)
      operationId: NonUeN2InfoUnSubscribe
      security:
        - {}
        - oAuth2ClientCredentials:
          - namf-comm
        - oAuth2ClientCredentials:
          - namf-comm
          - namf-comm:non-ue-n2-messages
      parameters:
        - name: n2NotifySubscriptionId
          in: path
          description: N2 info Subscription Identifier
          required: true
          schema:
            type: string
      responses:
        '204':
          description: Non UE N2 INfo Subscription successfully removed.
        '307':
          $ref: 'TS29571_CommonData.yaml#/components/responses/307'
        '308':
          $ref: 'TS29571_CommonData.yaml#/components/responses/308'
        '400':
          $ref: 'TS29571_CommonData.yaml#/components/responses/400'
        '401':
          $ref: 'TS29571_CommonData.yaml#/components/responses/401'
        '403':
          $ref: 'TS29571_CommonData.yaml#/components/responses/403'
        '404':
          $ref: 'TS29571_CommonData.yaml#/components/responses/404'
        '429':
          $ref: 'TS29571_CommonData.yaml#/components/responses/429'
        '500':
          $ref: 'TS29571_CommonData.yaml#/components/responses/500'
        '502':
          $ref: 'TS29571_CommonData.yaml#/components/responses/502'
        '503':
          $ref: 'TS29571_CommonData.yaml#/components/responses/503'
  /subscriptions:
    post:
      summary: Namf_Communication AMF Status Change Subscribe service Operation
      tags:
        - subscriptions collection (Collection)
      operationId: AMFStatusChangeSubscribe
      requestBody:
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/SubscriptionData'
        required: true
      responses:
        '201':
          description: N1N2 Message Subscription successfully created.
          headers:
            Location:
              description: 'Contains the URI of the newly created resource, according to the
structure: {apiRoot}/namf-comm/<apiVersion>/subscriptions/{subscriptionId}'
              required: true
              schema:
                type: string
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/SubscriptionData'
        '307':
          $ref: 'TS29571_CommonData.yaml#/components/responses/307'
        '308':
          $ref: 'TS29571_CommonData.yaml#/components/responses/308'
        '400':
          $ref: 'TS29571_CommonData.yaml#/components/responses/400'
        '401':
          $ref: 'TS29571_CommonData.yaml#/components/responses/401'
        '403':
          $ref: 'TS29571_CommonData.yaml#/components/responses/403'
        '404':
          $ref: 'TS29571_CommonData.yaml#/components/responses/404'
```

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        '411':
          $ref: 'TS29571_CommonData.yaml#/components/responses/411'
        '413':
          $ref: 'TS29571_CommonData.yaml#/components/responses/413'
        '415':
          $ref: 'TS29571_CommonData.yaml#/components/responses/415'
        '429':
          $ref: 'TS29571_CommonData.yaml#/components/responses/429'
        '500':
          $ref: 'TS29571_CommonData.yaml#/components/responses/500'
        '502':
          $ref: 'TS29571_CommonData.yaml#/components/responses/502'
        '503':
          $ref: 'TS29571_CommonData.yaml#/components/responses/503'
        default:
          description: Unexpected error
      callbacks:
        onAmfStatusChange:
          '{$request.body#/amfStatusUri}':
            post:
              summary: Amf Status Change Notify service Operation
              tags:
                - Amf Status Change Notify
              operationId: AmfStatusChangeNotify
              requestBody:
                description: Amf Status Change Notification
                content:
                  application/json:
                    schema:
                      $ref: '#/components/schemas/AmfStatusChangeNotification'
              responses:
                '204':
                  description: Expected response to a successful callback processing
                '307':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/307'
                '308':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/308'
                '400':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/400'
                '401':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/401'
                '403':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/403'
                '404':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/404'
                '411':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/411'
                '413':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/413'
                '415':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/415'
                '429':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/429'
                '500':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/500'
                '502':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/502'
                '503':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/503'
  /subscriptions/{subscriptionId}:
    delete:
      summary: Namf_Communication AMF Status Change UnSubscribe service Operation
      tags:
        - individual subscription (Document)
      operationId: AMFStatusChangeUnSubscribe
      parameters:
        - name: subscriptionId
          in: path
          description: AMF Status Change Subscription Identifier
          required: true
          schema:
            type: string
      responses:
        '204':
          description: N1N2 Message Subscription successfully removed.
        '307':
          $ref: 'TS29571_CommonData.yaml#/components/responses/307'
        '308':
```

```
        $ref: 'TS29571_CommonData.yaml#/components/responses/308'
      '400':
        $ref: 'TS29571_CommonData.yaml#/components/responses/400'
      '401':
        $ref: 'TS29571_CommonData.yaml#/components/responses/401'
      '403':
        $ref: 'TS29571_CommonData.yaml#/components/responses/403'
      '404':
        $ref: 'TS29571_CommonData.yaml#/components/responses/404'
      '429':
        $ref: 'TS29571_CommonData.yaml#/components/responses/429'
      '500':
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
      '502':
        $ref: 'TS29571_CommonData.yaml#/components/responses/502'
      '503':
        $ref: 'TS29571_CommonData.yaml#/components/responses/503'
      default:
        description: Unexpected error
  put:
    summary: Namf_Communication AMF Status Change Subscribe Modify service Operation
    tags:
      - individual subscription (Document)
    operationId: AMFStatusChangeSubscribeModfy
    parameters:
      - name: subscriptionId
        in: path
        description: AMF Status Change Subscription Identifier
        required: true
        schema:
          type: string
    requestBody:
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/SubscriptionData'
      required: true
    responses:
      '200':
        description: Subscription modified successfully
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/SubscriptionData'
      '204':
        description: Events subscription modification is accepted entirely
      '307':
        $ref: 'TS29571_CommonData.yaml#/components/responses/307'
      '308':
        $ref: 'TS29571_CommonData.yaml#/components/responses/308'
      '400':
        $ref: 'TS29571_CommonData.yaml#/components/responses/400'
      '401':
        $ref: 'TS29571_CommonData.yaml#/components/responses/401'
      '403':
        $ref: 'TS29571_CommonData.yaml#/components/responses/403'
      '404':
        $ref: 'TS29571_CommonData.yaml#/components/responses/404'
      '411':
        $ref: 'TS29571_CommonData.yaml#/components/responses/411'
      '413':
        $ref: 'TS29571_CommonData.yaml#/components/responses/413'
      '415':
        $ref: 'TS29571_CommonData.yaml#/components/responses/415'
      '429':
        $ref: 'TS29571_CommonData.yaml#/components/responses/429'
      '500':
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
      '502':
        $ref: 'TS29571_CommonData.yaml#/components/responses/502'
      '503':
        $ref: 'TS29571_CommonData.yaml#/components/responses/503'
      default:
        description: Unexpected error
    callbacks:
      OnAmfStatusChange:
        '{$request.body#/amfStatusUri}':
          post:
```

```
            summary: Amf Status Change Notify service Operation
            tags:
              - Amf Status Change Notify
            operationId: AmfStatusChangeNOtify
            requestBody:
              description: Amf Status Change Notification
              content:
                application/json:
                  schema:
                    $ref: '#/components/schemas/AmfStatusChangeNotification'
            responses:
              '204':
                description: Expected response to a successful callback processing
              '307':
                $ref: 'TS29571_CommonData.yaml#/components/responses/307'
              '308':
                $ref: 'TS29571_CommonData.yaml#/components/responses/308'
              '400':
                $ref: 'TS29571_CommonData.yaml#/components/responses/400'
              '401':
                $ref: 'TS29571_CommonData.yaml#/components/responses/401'
              '403':
                $ref: 'TS29571_CommonData.yaml#/components/responses/403'
              '404':
                $ref: 'TS29571_CommonData.yaml#/components/responses/404'
              '411':
                $ref: 'TS29571_CommonData.yaml#/components/responses/411'
              '413':
                $ref: 'TS29571_CommonData.yaml#/components/responses/413'
              '415':
                $ref: 'TS29571_CommonData.yaml#/components/responses/415'
              '429':
                $ref: 'TS29571_CommonData.yaml#/components/responses/429'
              '500':
                $ref: 'TS29571_CommonData.yaml#/components/responses/500'
              '502':
                $ref: 'TS29571_CommonData.yaml#/components/responses/502'
              '503':
                $ref: 'TS29571_CommonData.yaml#/components/responses/503'
components:
  securitySchemes:
    oAuth2ClientCredentials:
      type: oauth2
      flows:
        clientCredentials:
          tokenUrl: '{nrfApiRoot}/oauth2/token'
          scopes:
            namf-comm: Access to the Namf_Communication API
            namf-comm:ue-contexts:mobility: >
              Access to service operations applying to UE context resources, i.e.,
              UEContextTransfer, RegistrationStatusUpdate, CreateUEContext, ReleaseUEContext,
              RelocateUEContext, and CancelRelocateUEContext.
            namf-comm:ue-contexts:assign-ebi: >
              Access to service operations applying to UE context resources for EBI assignment,
              i.e., EBIAssignment.
            namf-comm:n1-n2-messages: >
              Access to service operations applying to n1-n2-messages resources, i.e.,
              N1N2MessageSubscribe, N1N2MessageUnSubscribe, N1N2MessageTransfer, N1MessageNotify and
              N2InfoNotify
            namf-comm:non-ue-n2-messages: >
              Access to service operations applying to the non-ue-n2-messages resources, i.e.,
              NonUeN2MessageTransfer, NonUeN2InfoSubscribe, NonUeN2InfoUnSubscribe, and
              NonUeN2InfoNotify
  schemas:
#
# STRUCTURED DATA TYPES
#
    SubscriptionData:
      description: Data within an AMF Status Change Subscription request and response
      type: object
      properties:
        amfStatusUri:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Uri'
        guamiList:
          type: array
          items:
            $ref: 'TS29571_CommonData.yaml#/components/schemas/Guami'
          minItems: 1
```

```
    required:
      - amfStatusUri
  AmfStatusChangeNotification:
    description: Data within an AMF Status Change Notification request
    type: object
    properties:
      amfStatusInfoList:
        type: array
        items:
          $ref: '#/components/schemas/AmfStatusInfo'
        minItems: 1
    required:
      - amfStatusInfoList
  AmfStatusInfo:
    description: AMF Status Information
    type: object
    properties:
      guamiList:
        type: array
        items:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Guami'
        minItems: 1
      statusChange:
        $ref: '#/components/schemas/StatusChange'
      targetAmfRemoval:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/AmfName'
      targetAmfFailure:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/AmfName'
    required:
      - guamiList
      - statusChange
  AssignEbiData:
    description: Data within an EBI assignment request
    type: object
    properties:
      pduSessionId:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/PduSessionId'
      arpList:
        type: array
        items:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Arp'
        minItems: 1
      releasedEbiList:
        type: array
        items:
          $ref: '#/components/schemas/EpsBearerId'
        minItems: 1
      oldGuami:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Guami'
      modifiedEbiList:
        type: array
        items:
          $ref: 'TS29502_Nsmf_PDUSession.yaml#/components/schemas/EbiArpMapping'
        minItems: 1
    required:
      - pduSessionId
  AssignedEbiData:
    description: Data within a successful response to an EBI assignment request
    type: object
    properties:
      pduSessionId:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/PduSessionId'
      assignedEbiList:
        type: array
        items:
          $ref: 'TS29502_Nsmf_PDUSession.yaml#/components/schemas/EbiArpMapping'
        minItems: 0
      failedArpList:
        type: array
        items:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Arp'
        minItems: 1
      releasedEbiList:
        type: array
        items:
          $ref: '#/components/schemas/EpsBearerId'
        minItems: 1
      modifiedEbiList:
```

```
          type: array
          items:
            $ref: '#/components/schemas/EpsBearerId'
          minItems: 1
      required:
        - pduSessionId
        - assignedEbiList
    AssignEbiFailed:
      description: Represents failed assignment of EBI(s)
      type: object
      properties:
        pduSessionId:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/PduSessionId'
        failedArpList:
          type: array
          items:
            $ref: 'TS29571_CommonData.yaml#/components/schemas/Arp'
          minItems: 1
      required:
        - pduSessionId
    UEContextRelease:
      description: Data within a Release UE Context request
      type: object
      properties:
        supi:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Supi'
        unauthenticatedSupi:
          type: boolean
          default: false
        ngapCause:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/NgApCause'
      required:
        - ngapCause
    N2InformationTransferReqData:
      description: Data within a N2 Information Transfer request containing the N2 information
requested to be transferred to 5G AN
      type: object
      properties:
        taiList:
          type: array
          items:
            $ref: 'TS29571_CommonData.yaml#/components/schemas/Tai'
          minItems: 1
        ratSelector:
          $ref: '#/components/schemas/RatSelector'
        globalRanNodeList:
          type: array
          items:
            $ref: 'TS29571_CommonData.yaml#/components/schemas/GlobalRanNodeId'
          minItems: 1
        n2Information:
          $ref: '#/components/schemas/N2InfoContainer'
        supportedFeatures:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/SupportedFeatures'
      required:
        - n2Information
    NonUeN2InfoSubscriptionCreateData:
      description: Data within a create subscription request for non-UE specific N2 information
notification
      type: object
      properties:
        globalRanNodeList:
          type: array
          items:
            $ref: 'TS29571_CommonData.yaml#/components/schemas/GlobalRanNodeId'
          minItems: 1
        anTypeList:
          type: array
          items:
            $ref: 'TS29571_CommonData.yaml#/components/schemas/AccessType'
          minItems: 1
        n2InformationClass:
          $ref: '#/components/schemas/N2InformationClass'
        n2NotifyCallbackUri:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Uri'
        nfId:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/NfInstanceId'
        supportedFeatures:
```

```
          $ref: 'TS29571_CommonData.yaml#/components/schemas/SupportedFeatures'
        notifCorrelationId:
          type: string
      required:
        - n2InformationClass
        - n2NotifyCallbackUri
    NonUeN2InfoSubscriptionCreatedData:
      description: Data for the created subscription for non-UE specific N2 information notification
      type: object
      properties:
        n2NotifySubscriptionId:
          type: string
        supportedFeatures:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/SupportedFeatures'
        n2InformationClass:
          $ref: '#/components/schemas/N2InformationClass'
      required:
        - n2NotifySubscriptionId
    UeN1N2InfoSubscriptionCreateData:
      description: Data within a create subscription request for UE specific N1 and/or N2
information notification
      type: object
      properties:
        n2InformationClass:
          $ref: '#/components/schemas/N2InformationClass'
        n2NotifyCallbackUri:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Uri'
        n1MessageClass:
          $ref: '#/components/schemas/N1MessageClass'
        n1NotifyCallbackUri:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Uri'
        nfId:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/NfInstanceId'
        supportedFeatures:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/SupportedFeatures'
        oldGuami:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Guami'
    UeN1N2InfoSubscriptionCreatedData:
      description: Data for the created subscription for UE specific N1 and/or N2 information
notification
      type: object
      properties:
        n1n2NotifySubscriptionId:
          type: string
        supportedFeatures:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/SupportedFeatures'
      required:
        - n1n2NotifySubscriptionId
    N2InformationNotification:
      description: Data within a N2 information notification request
      type: object
      properties:
        n2NotifySubscriptionId:
          type: string
        n2InfoContainer:
          $ref: '#/components/schemas/N2InfoContainer'
        toReleaseSessionList:
          type: array
          items:
            $ref: 'TS29571_CommonData.yaml#/components/schemas/PduSessionId'
          minItems: 1
        lcsCorrelationId:
          $ref: 'TS29572_Nlmf_Location.yaml#/components/schemas/CorrelationID'
        notifyReason:
          $ref: '#/components/schemas/N2InfoNotifyReason'
        smfChangeInfoList:
          type: array
          items:
            $ref: '#/components/schemas/SmfChangeInfo'
          minItems: 1
        ranNodeId:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/GlobalRanNodeId'
        initialAmfName:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/AmfName'
        anN2IPv4Addr:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Ipv4Addr'
        anN2IPv6Addr:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Ipv6Addr'
```

```
      guami:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Guami'
      notifySourceNgRan:
        type: boolean
        default: false
      notifCorrelationId:
        type: string
      toReleaseSessionInfo:
        type: array
        items:
          $ref: '#/components/schemas/ReleaseSessionInfo'
        minItems: 1
    required:
      - n2NotifySubscriptionId

  N2InfoContainer:
    description: N2 information container
    type: object
    properties:
      n2InformationClass:
        $ref: '#/components/schemas/N2InformationClass'
      smInfo:
        $ref: '#/components/schemas/N2SmInformation'
      ranInfo:
        $ref: '#/components/schemas/N2RanInformation'
      nrppaInfo:
        $ref: '#/components/schemas/NrppaInformation'
      pwsInfo:
        $ref: '#/components/schemas/PwsInformation'
      v2xInfo:
        $ref: '#/components/schemas/V2xInformation'
      proseInfo:
        $ref: '#/components/schemas/ProSeInformation'
      tssInfo:
        $ref: '#/components/schemas/TssInformation'
      rslpInfo:
        $ref: '#/components/schemas/RslpInformation'
      a2xInfo:
        $ref: '#/components/schemas/A2xInformation'
    required:
      - n2InformationClass
  N1MessageNotification:
    description: Data within a N1 message notification request
    type: object
    properties:
      n1NotifySubscriptionId:
        type: string
      n1MessageContainer:
        $ref: '#/components/schemas/N1MessageContainer'
      lcsCorrelationId:
        $ref: 'TS29572_Nlmf_Location.yaml#/components/schemas/CorrelationID'
      registrationCtxtContainer:
        $ref: '#/components/schemas/RegistrationContextContainer'
      newLmfIdentification:
        $ref: 'TS29572_Nlmf_Location.yaml#/components/schemas/LMFIdentification'
      guami:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Guami'
      cIoT5GSOptimisation:
        type: boolean
        default: false
      ecgi:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Ecgi'
      ncgi:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Ncgi'
      tai:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Tai'
      supi:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Supi'
      pruInd:
        $ref: 'TS29503_Nudm_SDM.yaml#/components/schemas/PruInd'
    required:
      - n1MessageContainer
  N1MessageContainer:
    description: N1 Message container
    type: object
    properties:
      n1MessageClass:
        $ref: '#/components/schemas/N1MessageClass'
```

```
        n1MessageContent:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/RefToBinaryData'
        nfId:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/NfInstanceId'
        serviceInstanceId:
          type: string
      required:
        - n1MessageClass
        - n1MessageContent
    N1N2MessageTransferReqData:
      description: Data within a N1/N2 message transfer request
      type: object
      properties:
        n1MessageContainer:
          $ref: '#/components/schemas/N1MessageContainer'
        n2InfoContainer:
          $ref: '#/components/schemas/N2InfoContainer'
        mtData:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/RefToBinaryData'
        skipInd:
          type: boolean
          default: false
        lastMsgIndication:
          type: boolean
        pduSessionId:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/PduSessionId'
        lcsCorrelationId:
          $ref: 'TS29572_Nlmf_Location.yaml#/components/schemas/CorrelationID'
        ppi:
          $ref: '#/components/schemas/Ppi'
        arp:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Arp'
        5qi:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/5Qi'
        n1n2FailureTxfNotifURI:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Uri'
        smfReallocationInd:
          type: boolean
          default: false
        areaOfValidity:
          $ref: '#/components/schemas/AreaOfValidity'
        supportedFeatures:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/SupportedFeatures'
        oldGuami:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Guami'
        maAcceptedInd:
          type: boolean
          default: false
        extBufSupport:
          type: boolean
          default: false
        targetAccess:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/AccessType'
        nfId:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/NfInstanceId'
        pruInd:
          type: boolean
          enum:
            - true
        pduSessionPrio:
          $ref: 'TS29502_Nsmf_PDUSession.yaml#/components/schemas/PduSessionPriority'



    N1N2MessageTransferRspData:
      description: Data within a N1/N2 message transfer response
      type: object
      properties:
        cause:
          $ref: '#/components/schemas/N1N2MessageTransferCause'
        supportedFeatures:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/SupportedFeatures'
      required:
        - cause
    RegistrationContextContainer:
      description: Registration Context Container used to send the UE context information, N1
message from UE, AN address etc during Registration with AMF re-allocation procedure
      type: object
```

```
    properties:
      ueContext:
        $ref: '#/components/schemas/UeContext'
      localTimeZone:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/TimeZone'
      anType:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/AccessType'
      anN2ApId:
        type: integer
      ranNodeId:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/GlobalRanNodeId'
      initialAmfName:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/AmfName'
      userLocation:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/UserLocation'
      rrcEstCause:
        type: string
        pattern: '^[0-9a-fA-F]+$'
      ueContextRequest:
        type: boolean
        default: false
      initialAmfN2ApId:
        type: integer
      anN2IPv4Addr:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Ipv4Addr'
      anN2IPv6Addr:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Ipv6Addr'
      allowedNssai:
        $ref: 'TS29531_Nnssf_NSSelection.yaml#/components/schemas/AllowedNssai'
      configuredNssai:
        type: array
        items:
          $ref: 'TS29531_Nnssf_NSSelection.yaml#/components/schemas/ConfiguredSnssai'
        minItems: 1
      rejectedNssaiInPlmn:
        type: array
        items:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Snssai'
        minItems: 1
      rejectedNssaiInTa:
        type: array
        items:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Snssai'
        minItems: 1
      selectedPlmnId:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/PlmnId'
      iabNodeInd:
        type: boolean
        default: false
      mbsrNodeInd:
        type: boolean
        default: false
      ceModeBInd:
        $ref: '#/components/schemas/CeModeBInd'
      lteMInd:
        $ref: '#/components/schemas/LteMInd'
      authenticatedInd:
        type: boolean
        default: false
      npnAccessInfo:
        $ref: '#/components/schemas/NpnAccessInfo'
    required:
      - ueContext
      - anType
      - anN2ApId
      - ranNodeId
      - initialAmfName
      - userLocation

  AreaOfValidity:
    description: Area of validity information for N2 information transfer
    type: object
    properties:
      taiList:
        type: array
        items:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Tai'
        minItems: 0
```

```
        taiRangeList:
          type: array
          items:
            $ref: 'TS29510_Nnrf_NFManagement.yaml#/components/schemas/TaiRange'
          minItems: 1
      required:
        - taiList
    UeContextTransferReqData:
      description: Data within a UE Context Transfer Request to start transferring of an individual
ueContext resource from old AMF to new AMF
      type: object
      properties:
        reason:
          $ref: '#/components/schemas/TransferReason'
        accessType:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/AccessType'
        plmnId:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/PlmnIdNid'
        regRequest:
          $ref: '#/components/schemas/N1MessageContainer'
        supportedFeatures:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/SupportedFeatures'
      required:
        - reason
        - accessType
    UeContextTransferRspData:
      description: Data within a successful response to the UE Context Transfer request
      type: object
      properties:
        ueContext:
          $ref: '#/components/schemas/UeContext'
        ueRadioCapability:
          $ref: '#/components/schemas/N2InfoContent'
        ueRadioCapabilityForPaging:
          $ref: '#/components/schemas/N2InfoContent'
        ueNbiotRadioCapability:
          $ref: '#/components/schemas/N2InfoContent'
        supportedFeatures:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/SupportedFeatures'
        xrDeviceWith2Rx:
          $ref: '#/components/schemas/XrDeviceWith2Rx'
      required:
        - ueContext

    UeContext:
      description: Represents an individual ueContext resource
      type: object
      properties:
        supi:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Supi'
        supiUnauthInd:
          type: boolean
        gpsiList:
          type: array
          items:
            $ref: 'TS29571_CommonData.yaml#/components/schemas/Gpsi'
          minItems: 1
        pei:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Pei'
        udmGroupId:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/NfGroupId'
        ausfGroupId:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/NfGroupId'
        pcfGroupId:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/NfGroupId'
        routingIndicator:
          type: string
        hNwPubKeyId:
          type: integer
        groupList:
          type: array
          items:
            $ref: 'TS29571_CommonData.yaml#/components/schemas/GroupId'
          minItems: 1
        drxParameter:
          $ref: '#/components/schemas/DrxParameter'
        subRfsp:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/RfspIndex'
```

```
pcfRfsp:
  $ref: 'TS29571_CommonData.yaml#/components/schemas/RfspIndex'
usedRfsp:
  $ref: 'TS29571_CommonData.yaml#/components/schemas/RfspIndex'
subUeAmbr:
  $ref: 'TS29571_CommonData.yaml#/components/schemas/Ambr'
pcfUeAmbr:
  $ref: 'TS29571_CommonData.yaml#/components/schemas/Ambr'
subUeSliceMbrList:
  type: object
  additionalProperties:
    $ref: 'TS29571_CommonData.yaml#/components/schemas/SliceMbr'
  minProperties: 1
  description: A map(list of key-value pairs) where Snssai serves as key.
smsfId:
  $ref: 'TS29571_CommonData.yaml#/components/schemas/NfInstanceId'
seafData:
  $ref: '#/components/schemas/SeafData'
5gMmCapability:
  $ref: '#/components/schemas/5GMmCapability'
pcfId:
  $ref: 'TS29571_CommonData.yaml#/components/schemas/NfInstanceId'
pcfSetId:
  $ref: 'TS29571_CommonData.yaml#/components/schemas/NfSetId'
pcfAmpServiceSetId:
  $ref: 'TS29571_CommonData.yaml#/components/schemas/NfServiceSetId'
pcfUepServiceSetId:
  $ref: 'TS29571_CommonData.yaml#/components/schemas/NfServiceSetId'
pcfBinding:
  $ref: '#/components/schemas/SbiBindingLevel'
pcfAmPolicyUri:
  $ref: 'TS29571_CommonData.yaml#/components/schemas/Uri'
amPolicyReqTriggerList:
  type: array
  items:
    $ref: '#/components/schemas/PolicyReqTrigger'
  minItems: 1
pcfUePolicyUri:
  $ref: 'TS29571_CommonData.yaml#/components/schemas/Uri'
uePolicyReqTriggerList:
  type: array
  items:
    $ref: '#/components/schemas/PolicyReqTrigger'
  minItems: 1
hpcfId:
  $ref: 'TS29571_CommonData.yaml#/components/schemas/NfInstanceId'
hpcfSetId:
  $ref: 'TS29571_CommonData.yaml#/components/schemas/NfSetId'
restrictedRatList:
  type: array
  items:
    $ref: 'TS29571_CommonData.yaml#/components/schemas/RatType'
  minItems: 1
forbiddenAreaList:
  type: array
  items:
    $ref: 'TS29571_CommonData.yaml#/components/schemas/Area'
  minItems: 1
serviceAreaRestriction:
  $ref: 'TS29571_CommonData.yaml#/components/schemas/ServiceAreaRestriction'
restrictedCoreNwTypeList:
  type: array
  items:
    $ref: 'TS29571_CommonData.yaml#/components/schemas/CoreNetworkType'
  minItems: 1
eventSubscriptionList:
  type: array
  items:
    $ref: '#/components/schemas/ExtAmfEventSubscription'
  minItems: 1
mmContextList:
  type: array
  items:
    $ref: '#/components/schemas/MmContext'
  minItems: 1
  maxItems: 2
sessionContextList:
  type: array
```

```
    items:
      $ref: '#/components/schemas/PduSessionContext'
    minItems: 1
  epsInterworkingInfo:
    $ref: 'TS29503_Nudm_UECM.yaml#/components/schemas/EpsInterworkingInfo'
  traceData:
    $ref: 'TS29571_CommonData.yaml#/components/schemas/TraceData'
  serviceGapExpiryTime:
    $ref: 'TS29571_CommonData.yaml#/components/schemas/DateTime'
  stnSr:
    $ref: 'TS29571_CommonData.yaml#/components/schemas/StnSr'
  cMsisdn:
    $ref: 'TS29571_CommonData.yaml#/components/schemas/CMsisdn'
  msClassmark2:
    $ref: '#/components/schemas/MSClassmark2'
  supportedCodecList:
    type: array
    items:
      $ref: '#/components/schemas/SupportedCodec'
    minItems: 1
  smallDataRateStatusInfos:
    type: array
    items:
      $ref: '#/components/schemas/SmallDataRateStatusInfo'
    minItems: 1
  restrictedPrimaryRatList:
    type: array
    items:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/RatType'
    minItems: 1
  restrictedSecondaryRatList:
    type: array
    items:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/RatType'
    minItems: 1
  v2xContext:
    $ref: '#/components/schemas/V2xContext'
  lteCatMInd:
    type: boolean
    default: false
  redCapInd:
    type: boolean
    default: false
  eRedCapInd:
    type: boolean
    default: false
  moExpDataCounter:
    $ref: 'TS29571_CommonData.yaml#/components/schemas/MoExpDataCounter'
  cagData:
    $ref: 'TS29503_Nudm_SDM.yaml#/components/schemas/CagData'
  managementMdtInd:
    type: boolean
    default: false
  adjacenPlmnMngtMdtInds:
    description: A map (list of key-value pairs where PlmnId serves as key) of Booleans
    type: object
    additionalProperties:
      type: boolean
    minProperties: 1
  immediateMdtConf:
    $ref: '#/components/schemas/ImmediateMdtConf'
  ecRestrictionDataWb:
    $ref: '#/components/schemas/EcRestrictionDataWb'
  ecRestrictionDataNb:
    type: boolean
    default: false
  iabOperationAllowed:
    type: boolean
  proseContext:
    $ref: '#/components/schemas/ProseContext'
  analyticsSubscriptionList:
    type: array
    items:
      $ref: '#/components/schemas/AnalyticsSubscription'
    minItems: 1
  pcfAmpBindingInfo:
    type: string
  pcfUepBindingInfo:
```

```
      type: string
    usedServiceAreaRestriction:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/ServiceAreaRestriction'
    praInAmPolicy:
      type: object
      additionalProperties:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/PresenceInfo'
      minProperties: 1
      description: A map(list of key-value pairs) where praId serves as key.
    praInUePolicy:
      type: object
      additionalProperties:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/PresenceInfo'
      minProperties: 1
      description: A map(list of key-value pairs) where praId serves as key.
    updpSubscriptionData:
      $ref: '#/components/schemas/UpdpSubscriptionData'
    smPolicyNotifyPduList:
      type: array
      items:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/PduSessionInfo'
      minItems: 1
    pcfUeCallbackInfo:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/PcfUeCallbackInfo'
    uePositioningCap:
      $ref: 'TS29572_Nlmf_Location.yaml#/components/schemas/UePositioningCapabilities'
    astiDistributionIndication:
      type: boolean
      default: false
    tsErrorBudget:
      type: integer
    snpnOnboardInd:
      type: boolean
      default: false
    smfSelInfo:
      $ref: 'TS29507_Npcf_AMPolicyControl.yaml#/components/schemas/SmfSelectionData'
    pcfUeSliceMbrList:
      type: object
      additionalProperties:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/SliceMbr'
      minProperties: 1
      description: A map(list of key-value pairs) where Snssai serves as key.
    smsfSetId:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/NfSetId'
    smsfServiceSetId:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/NfServiceSetId'
    smsfBindingInfo:
      type: string
    disasterRoamingInd:
      type: boolean
      default: false
    disasterPlmn:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/PlmnId'
    satelliteBackhaulCat:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/SatelliteBackhaulCategory'
    wlServAreaRes:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/WirelineServiceAreaRestriction'
    asTimeDisParam:
      $ref: 'TS29507_Npcf_AMPolicyControl.yaml#/components/schemas/AsTimeDistributionParam'
    amPolicyInfoContainer:
      $ref: '#/components/schemas/AmPolicyInfoContainer'
    a2xContext:
      $ref: '#/components/schemas/A2xContext'
    mbsrOperationAllowed:
      $ref: 'TS29503_Nudm_SDM.yaml#/components/schemas/MbsrOperationAllowed'
    lcsUpContext:
      $ref: '#/components/schemas/LcsUpContext'
    reconnectInd:
      type: boolean
      default: false


  N2SmInformation:
    description: Represents the session management SMF related N2 information data part
    type: object
    properties:
      pduSessionId:
```

```
        $ref: 'TS29571_CommonData.yaml#/components/schemas/PduSessionId'
      n2InfoContent:
        $ref: '#/components/schemas/N2InfoContent'
      sNssai:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Snssai'
      homePlmnSnssai:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Snssai'
      iwkSnssai:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Snssai'
      subjectToHo:
        type: boolean
    required:
      - pduSessionId
  N2InfoContent:
    description: Represents a transparent N2 information content to be relayed by AMF
    type: object
    properties:
      ngapMessageType:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Uinteger'
      ngapIeType:
        $ref: '#/components/schemas/NgapIeType'
      ngapData:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/RefToBinaryData'
    required:
      - ngapData
  NrppaInformation:
    description: Represents a NRPPa related N2 information data part
    type: object
    properties:
      nfId:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/NfInstanceId'
      nrppaPdu:
        $ref: '#/components/schemas/N2InfoContent'
      serviceInstanceId:
        type: string
    required:
      - nfId
      - nrppaPdu
  PwsInformation:
    description: Represents a PWS related information data part
    type: object
    properties:
      messageIdentifier:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Uint16'
      serialNumber:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Uint16'
      pwsContainer:
        $ref: '#/components/schemas/N2InfoContent'
      bcEmptyAreaList:
        type: array
        items:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/GlobalRanNodeId'
        minItems: 1
      sendRanResponse:
        type: boolean
        default: false
      omcId:
        $ref: '#/components/schemas/OmcIdentifier'
      nfId:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/NfInstanceId'
    required:
      - messageIdentifier
      - serialNumber
      - pwsContainer
  N1N2MsgTxfrFailureNotification:
    description: Data within a N1/N2 Message Transfer Failure Notification request
    type: object
    properties:
      cause:
        $ref: '#/components/schemas/N1N2MessageTransferCause'
      n1n2MsgDataUri:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Uri'
      retryAfter:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Uinteger'

    required:
      - cause
      - n1n2MsgDataUri
```

```
    N1N2MessageTransferError:
      description: Data within a N1/N2 Message Transfer Error response
      type: object
      properties:
        error:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/ProblemDetails'
        errInfo:
          $ref: '#/components/schemas/N1N2MsgTxfrErrDetail'
      required:
        - error
    N1N2MsgTxfrErrDetail:
      description: N1/N2 Message Transfer Error Details
      type: object
      properties:
        retryAfter:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Uinteger'
        highestPrioArp:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Arp'
        maxWaitingTime:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/DurationSec'
    N2InformationTransferRspData:
      description: Data within a successful response to the N2 Information Transfer request to
transfer N2 Information to the AN
      type: object
      properties:
        result:
          $ref: '#/components/schemas/N2InformationTransferResult'
        pwsRspData:
          $ref: '#/components/schemas/PWSResponseData'
        supportedFeatures:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/SupportedFeatures'
        tssRspPerNgranList:
          type: array
          items:
            $ref: '#/components/schemas/TssRspPerNgran'
          minItems: 1
          maxItems: 10
      required:
        - result
    MmContext:
      description: Represents a Mobility Management Context in UE Context
      type: object
      properties:
        accessType:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/AccessType'
        nasSecurityMode:
          $ref: '#/components/schemas/NasSecurityMode'
        epsNasSecurityMode:
          $ref: '#/components/schemas/EpsNasSecurityMode'
        nasDownlinkCount:
          $ref: '#/components/schemas/NasCount'
        nasUplinkCount:
          $ref: '#/components/schemas/NasCount'
        ueSecurityCapability:
          $ref: '#/components/schemas/UeSecurityCapability'
        s1UeNetworkCapability:
          $ref: '#/components/schemas/S1UeNetworkCapability'
        allowedNssai:
          type: array
          items:
            $ref: 'TS29571_CommonData.yaml#/components/schemas/Snssai'
          minItems: 1
        nssaiMappingList:
          type: array
          items:
            $ref: '#/components/schemas/NssaiMapping'
          minItems: 1
        allowedHomeNssai:
          type: array
          items:
            $ref: 'TS29571_CommonData.yaml#/components/schemas/Snssai'
          minItems: 1
        partiallyAllowedNssai:
          type: array
          items:
            $ref: 'TS29571_CommonData.yaml#/components/schemas/PartiallyAllowedSnssai'
          minItems: 1
        replacedSnssaiMappingList:
```

```
        type: array
        items:
          $ref: '#/components/schemas/SliceReplacementMapping'
        minItems: 1
      nsInstanceList:
        type: array
        items:
          $ref: 'TS29531_Nnssf_NSSelection.yaml#/components/schemas/NsiId'
        minItems: 1
      expectedUEbehavior:
        $ref: '#/components/schemas/ExpectedUeBehavior'
      ueDifferentiationInfo:
        $ref: '#/components/schemas/UeDifferentiationInfo'
      plmnAssiUeRadioCapId:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/PlmnAssiUeRadioCapId'
      manAssiUeRadioCapId:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/ManAssiUeRadioCapId'
      ucmfDicEntryId:
        type: string
      n3IwfId:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/GlobalRanNodeId'
      wagfId:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/GlobalRanNodeId'
      tngfId:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/GlobalRanNodeId'
      anN2ApId:
        type: integer
      nssaaStatusList:
        type: array
        items:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/NssaaStatus'
        minItems: 1
      pendingNssaiMappingList:
        type: array
        items:
          $ref: '#/components/schemas/NssaiMapping'
        minItems: 1
      uuaaMmStatus:
        $ref: '#/components/schemas/UuaaMmStatus'
      deregInactTimerList:
        description: The S-NSSAI shall be used as the key of the map.
        type: object
        additionalProperties:
          $ref: '#/components/schemas/DeregInactTimerInfo'
      voSupportMatchInd:
        type: boolean
    required:
      - accessType
  SeafData:
    description: Represents SEAF data derived from data received from AUSF
    type: object
    properties:
      ngKsi:
        $ref: '#/components/schemas/NgKsi'
      keyAmf:
        $ref: '#/components/schemas/KeyAmf'
      nh:
        type: string
        pattern: '^[A-Fa-f0-9]+$'
      ncc:
        type: integer
        minimum: 0
        maximum: 7
      keyAmfChangeInd:
        type: boolean
      keyAmfHDerivationInd:
        type: boolean
    required:
      - ngKsi
      - keyAmf
  NasSecurityMode:
    description: Indicates the NAS Security Mode
    type: object
    properties:
      integrityAlgorithm:
        $ref: '#/components/schemas/IntegrityAlgorithm'
      cipheringAlgorithm:
        $ref: '#/components/schemas/CipheringAlgorithm'
```

```
    required:
      - integrityAlgorithm
      - cipheringAlgorithm
  PduSessionContext:
    description: Represents a PDU Session Context in UE Context
    type: object
    properties:
      pduSessionId:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/PduSessionId'
      smContextRef:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Uri'
      sNssai:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Snssai'
      altSnssai:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Snssai'
      dnn:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Dnn'
      selectedDnn:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Dnn'
      accessType:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/AccessType'
      additionalAccessType:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/AccessType'
      allocatedEbiList:
        type: array
        items:
          $ref: 'TS29502_Nsmf_PDUSession.yaml#/components/schemas/EbiArpMapping'
        minItems: 1
      hsmfId:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/NfInstanceId'
      hsmfSetId:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/NfSetId'
      hsmfServiceSetId:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/NfServiceSetId'
      smfBinding:
        $ref: '#/components/schemas/SbiBindingLevel'
      vsmfId:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/NfInstanceId'
      vsmfSetId:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/NfSetId'
      vsmfServiceSetId:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/NfServiceSetId'
      vsmfBinding:
        $ref: '#/components/schemas/SbiBindingLevel'
      ismfId:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/NfInstanceId'
      ismfSetId:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/NfSetId'
      ismfServiceSetId:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/NfServiceSetId'
      ismfBinding:
        $ref: '#/components/schemas/SbiBindingLevel'
      nsInstance:
        $ref: 'TS29531_Nnssf_NSSelection.yaml#/components/schemas/NsiId'
      smfServiceInstanceId:
        type: string
      maPduSession:
        type: boolean
        default: false
      cnAssistedRanPara:
        $ref: 'TS29502_Nsmf_PDUSession.yaml#/components/schemas/CnAssistedRanPara'
      nrfManagementUri:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Uri'
      nrfDiscoveryUri:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Uri'
      nrfAccessTokenUri:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Uri'
      smfBindingInfo:
        type: string
      vsmfBindingInfo:
        type: string
      ismfBindingInfo:
        type: string
      additionalSnssai:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Snssai'
      altAdditionalSnssai:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Snssai'
      interPlmnApiRoot:
```

```
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Uri'
        pgwFqdn:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Fqdn'
        pgwIpAddr:
          $ref: 'TS29503_Nudm_SDM.yaml#/components/schemas/IpAddress'
        plmnId:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/PlmnId'
        anchorSmfSupportedFeatures:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/SupportedFeatures'
        anchorSmfOauth2Required:
          type: boolean
        hrsboAllowedInd:
          type: boolean
        pduSessionPrio:
          $ref: 'TS29502_Nsmf_PDUSession.yaml#/components/schemas/PduSessionPriority'
      required:
        - pduSessionId
        - smContextRef
        - sNssai
        - dnn
        - accessType
    NssaiMapping:
      description: Represents the mapping between a S-NSSAI in serving PLMN to a S-NSSAI in home
PLMN
      type: object
      properties:
        mappedSnssai:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Snssai'
        hSnssai:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Snssai'
      required:
        - mappedSnssai
        - hSnssai
    UeRegStatusUpdateReqData:
      description: Data within a UE registration status update request to indicate a completion of
transferring at a target AMF
      type: object
      properties:
        transferStatus:
          $ref: '#/components/schemas/UeContextTransferStatus'
        toReleaseSessionList:
          type: array
          items:
            $ref: 'TS29571_CommonData.yaml#/components/schemas/PduSessionId'
          minItems: 1
        pcfReselectedInd:
          type: boolean
        smfChangeInfoList:
          type: array
          items:
            $ref: '#/components/schemas/SmfChangeInfo'
          minItems: 1
        analyticsNotUsedList:
          type: array
          items:
            $ref: 'TS29571_CommonData.yaml#/components/schemas/Uri'
          minItems: 1
        toReleaseSessionInfo:
          type: array
          items:
            $ref: '#/components/schemas/ReleaseSessionInfo'
          minItems: 1
      required:
        - transferStatus
    UeRegStatusUpdateRspData:
      description: Data within a UE registration status update response to provide the status of UE
context transfer status update at a source AMF
      type: object
      properties:
        regStatusTransferComplete:
          type: boolean
      required:
        - regStatusTransferComplete
    AssignEbiError:
      description: Data within a failure response to the EBI assignment request
      type: object
      properties:
        error:
```

```
        $ref: 'TS29571_CommonData.yaml#/components/schemas/ProblemDetails'
      failureDetails:
        $ref: '#/components/schemas/AssignEbiFailed'
    required:
      - error
      - failureDetails

  UeContextCreateData:
    description: Data within a request to create an individual ueContext resource
    type: object
    properties:
      ueContext:
        $ref: '#/components/schemas/UeContext'
      targetId:
        $ref: '#/components/schemas/NgRanTargetId'
      sourceToTargetData:
        $ref: '#/components/schemas/N2InfoContent'
      pduSessionList:
        type: array
        items:
          $ref: '#/components/schemas/N2SmInformation'
        minItems: 1
      n2NotifyUri:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Uri'
      ueRadioCapability:
        $ref: '#/components/schemas/N2InfoContent'
      ueRadioCapabilityForPaging:
        $ref: '#/components/schemas/N2InfoContent'
      ngapCause:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/NgApCause'
      supportedFeatures:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/SupportedFeatures'
      servingNetwork:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/PlmnIdNid'
      xrDeviceWith2Rx:
        $ref: '#/components/schemas/XrDeviceWith2Rx'
    required:
      - ueContext
      - targetId
      - sourceToTargetData
      - pduSessionList

  UeContextCreatedData:
    description: Data within a successful response for creating an individual ueContext resource
    type: object
    properties:
      ueContext:
        $ref: '#/components/schemas/UeContext'
      targetToSourceData:
        $ref: '#/components/schemas/N2InfoContent'
      pduSessionList:
        type: array
        items:
          $ref: '#/components/schemas/N2SmInformation'
        minItems: 1
      failedSessionList:
        type: array
        items:
          $ref: '#/components/schemas/N2SmInformation'
        minItems: 1
      supportedFeatures:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/SupportedFeatures'
      pcfReselectedInd:
        type: boolean
      analyticsNotUsedList:
        type: array
        items:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Uri'
        minItems: 1
    required:
      - ueContext
      - targetToSourceData
      - pduSessionList
  UeContextCreateError:
    description: Data within a failure response for creating a UE context
    type: object
    properties:
      error:
```

```
        $ref: 'TS29571_CommonData.yaml#/components/schemas/ProblemDetails'
      ngapCause:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/NgApCause'
      targetToSourceFailureData:
        $ref: '#/components/schemas/N2InfoContent'
    required:
      - error
  UeContextRelocateData:
    description: Data within a Relocate UE Context request
    type: object
    properties:
      ueContext:
        $ref: '#/components/schemas/UeContext'
      targetId:
        $ref: '#/components/schemas/NgRanTargetId'
      sourceToTargetData:
        $ref: '#/components/schemas/N2InfoContent'
      forwardRelocationRequest:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/RefToBinaryData'
      pduSessionList:
        type: array
        items:
          $ref: '#/components/schemas/N2SmInformation'
        minItems: 1
      ueRadioCapability:
        $ref: '#/components/schemas/N2InfoContent'
      ngapCause:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/NgApCause'
      supportedFeatures:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/SupportedFeatures'
      xrDeviceWith2Rx:
        $ref: '#/components/schemas/XrDeviceWith2Rx'
    required:
      - ueContext
      - targetId
      - sourceToTargetData
      - forwardRelocationRequest
  UeContextRelocatedData:
    description: Data within a Relocate UE Context response
    type: object
    properties:
      ueContext:
        $ref: '#/components/schemas/UeContext'
    required:
      - ueContext

  UeContextCancelRelocateData:
    description: Data structure used for cancellation of UE Context Relocation
    type: object
    properties:
      supi:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Supi'
      relocationCancelRequest:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/RefToBinaryData'
    required:
      - relocationCancelRequest
  NgRanTargetId:
    description: Indicates a NG RAN as target of the handover
    type: object
    properties:
      ranNodeId:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/GlobalRanNodeId'
      tai:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Tai'
    required:
      - ranNodeId
      - tai
  PWSResponseData:
    description: Data related PWS included in a N2 Information Transfer response
    type: object
    properties:
      ngapMessageType:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Uinteger'
      serialNumber:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Uint16'
      messageIdentifier:
        type: integer
      unknownTaiList:
```

```
          type: array
          items:
            $ref: 'TS29571_CommonData.yaml#/components/schemas/Tai'
          minItems: 1
        n2PwsSubMissInd:
          type: boolean
          enum:
           - true
      required:
        - ngapMessageType
        - serialNumber
        - messageIdentifier
    PWSErrorData:
      description: Data related to PWS error included in a N2 Information Transfer failure response
      type: object
      properties:
        namfCause:
          type: integer
      required:
        - namfCause
    N2InformationTransferError:
      description: Data within a failure response for a non-UE related N2 Information Transfer
      type: object
      properties:
        error:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/ProblemDetails'
        pwsErrorInfo:
          $ref: '#/components/schemas/PWSErrorData'
      required:
        - error
    NgKsi:
      description: Represents the ngKSI
      type: object
      properties:
        tsc:
          $ref: '#/components/schemas/ScType'
        ksi:
          type: integer
          minimum: 0
          maximum: 6
      required:
        - tsc
        - ksi
    KeyAmf:
      description: Represents the K_amf or K'_amf
      type: object
      properties:
        keyType:
          $ref: '#/components/schemas/KeyAmfType'
        keyVal:
          type: string
      required:
        - keyType
        - keyVal
    ExpectedUeBehavior:
      description: Represents the expected UE behavior (e.g. UE moving trajectory) and its validity
period
      type: object
      properties:
        expMoveTrajectory:
          type: array
          items:
            $ref: 'TS29571_CommonData.yaml#/components/schemas/UserLocation'
          minItems: 1
        validityTime:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/DateTime'
      required:
        - expMoveTrajectory
        - validityTime
    N2RanInformation:
      description: Represents the RAN related N2 information data part
      type: object
      properties:
        n2InfoContent:
          $ref: '#/components/schemas/N2InfoContent'
      required:
        - n2InfoContent
    N2InfoNotificationRspData:
```

```
      description: Data within a N2 information notification response
      type: object
      properties:
        secRatDataUsageList:
          type: array
          items:
            $ref: '#/components/schemas/N2SmInformation'
          minItems: 1
    SmallDataRateStatusInfo:
      description: Represents the small data rate status
      type: object
      properties:
        Snssai:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Snssai'
        Dnn:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Dnn'
        SmallDataRateStatus:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/SmallDataRateStatus'
      required:
        - Snssai
        - Dnn
        - SmallDataRateStatus
    SmfChangeInfo:
      description: SMF change information for PDU session(s)
      type: object
      properties:
        pduSessionIdList:
          type: array
          items:
            $ref: 'TS29571_CommonData.yaml#/components/schemas/PduSessionId'
          minItems: 1
        smfChangeInd:
          $ref: '#/components/schemas/SmfChangeIndication'
      required:
        - pduSessionIdList
        - smfChangeInd

    V2xContext:
      description: Represents the V2X services related parameters
      type: object
      properties:
        nrV2xServicesAuth:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/NrV2xAuth'
        lteV2xServicesAuth:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/LteV2xAuth'
        nrUeSidelinkAmbr:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/BitRate'
        lteUeSidelinkAmbr:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/BitRate'
        pc5QoSPara:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Pc5QoSPara'

    V2xInformation:
      description: V2X related N2 information
      type: object
      properties:
        n2Pc5Pol:
          $ref: '#/components/schemas/N2InfoContent'

    ProSeInformation:
      description: Represents 5G ProSe related N2 information.
      type: object
      properties:
        n2Pc5ProSePol:
          $ref: '#/components/schemas/N2InfoContent'

    ImmediateMdtConf:
      description: Immediate MDT Configuration
      type: object
      properties:
        jobType:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/JobType'
        measurementLteList:
          type: array
          items:
            $ref: 'TS29571_CommonData.yaml#/components/schemas/MeasurementLteForMdt'
          minItems: 1
        measurementNrList:
```

```
        type: array
        items:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/MeasurementNrForMdt'
        minItems: 1
      reportingTriggerList:
        type: array
        items:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/ReportingTrigger'
        minItems: 1
      reportInterval:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/ReportIntervalMdt'
      reportIntervalNr:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/ReportIntervalNrMdt'
      reportAmount:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/ReportAmountMdt'
      eventThresholdRsrp:
        type: integer
        minimum: 0
        maximum: 97
      eventThresholdRsrq:
        type: integer
        minimum: 0
        maximum: 34
      eventThresholdRsrpNr:
        type: integer
        minimum: 0
        maximum: 127
      eventThresholdRsrqNr:
        type: integer
        minimum: 0
        maximum: 127
      collectionPeriodRmmLte:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/CollectionPeriodRmmLteMdt'
      collectionPeriodRmmNr:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/CollectionPeriodRmmNrMdt'
      measurementPeriodLte:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/MeasurementPeriodLteMdt'
      areaScope:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/AreaScope'
      positioningMethod:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/PositioningMethodMdt'
      addPositioningMethodList:
        type: array
        items:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/PositioningMethodMdt'
        minItems: 1
      mdtAllowedPlmnIdList:
        type: array
        items:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/PlmnId'
        minItems: 1
        maxItems: 16
      sensorMeasurementList:
        type: array
        items:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/SensorMeasurement'
        minItems: 1
    required:
      - jobType

  EpsNasSecurityMode:
    description: Indicates the EPS NAS Security Mode
    type: object
    properties:
      integrityAlgorithm:
        $ref: '#/components/schemas/EpsNasIntegrityAlgorithm'
      cipheringAlgorithm:
        $ref: '#/components/schemas/EpsNasCipheringAlgorithm'
    required:
      - integrityAlgorithm
      - cipheringAlgorithm

  EcRestrictionDataWb:
    description: Enhanced Coverage Restriction Data for WB-N1 mode
    type: object
    properties:
      ecModeARestricted:
        type: boolean
```

```
          default: false
        ecModeBRestricted:
          type: boolean
      required:
        - ecModeBRestricted

    ExtAmfEventSubscription:
      description: AMF event subscription extended with additional information received for the
subscription
      allOf:
      - $ref: 'TS29518_Namf_EventExposure.yaml#/components/schemas/AmfEventSubscription'
      - $ref: '#/components/schemas/AmfEventSubscriptionAddInfo'

    AmfEventSubscriptionAddInfo:
      description: >
        Additional information received for an AMF event subscription, e.g. binding indications,
        statistical information for UE access behavior trends report or UE location trends report.
      type: object
      properties:
        bindingInfo:
          type: array
          items:
            type: string
          minItems: 1
          maxItems: 2
        subscribingNfType:
          $ref: 'TS29510_Nnrf_NFManagement.yaml#/components/schemas/NFType'
        eventSyncInd:
          type: boolean
        nfConsumerInfo:
          type: array
          items:
            type: string
          minItems: 1
        aoiStateList:
          type: object
          description: >
            Map of subscribed Area of Interest (AoI) Event State in the old AMF. The JSON pointer to
            an AmfEventArea element in the areaList IE (or a PresenceInfo element in
            presenceInfoList IE) of the AmfEvent data type shall be the key of the map.
          additionalProperties:
            $ref: '#/components/schemas/AreaOfInterestEventState'
        accessToken:
          type: string
          description: >
            JWS Compact Serialized representation of JWS signed JSON object (AccessTokenClaims
            defined in 3GPP TS 29.510)
        ueAccessBehaviorTrends:
          type: array
          items:
            $ref: 'TS29518_Namf_EventExposure.yaml#/components/schemas/UeAccessBehaviorReportItem'
          minItems: 1
        ueLocationTrends:
          type: array
          items:
            $ref: 'TS29518_Namf_EventExposure.yaml#/components/schemas/UeLocationTrendsReportItem'
          minItems: 1


    UeDifferentiationInfo:
      description: Represents the UE Differentiation Information and its validity time
      type: object
      properties:
        periodicComInd:
          $ref: '#/components/schemas/PeriodicCommunicationIndicator'
        periodicTime:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/DurationSec'
        scheduledComTime:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/ScheduledCommunicationTime'
        stationaryInd:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/StationaryIndication'
        trafficProfile:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/TrafficProfile'
        batteryInd:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/BatteryIndication'
        validityTime:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/DateTime'
```

```
    CeModeBInd:
      description: CE-mode-B Support Indicator.
      type: object
      properties:
        ceModeBSupportInd:
          type: boolean
      required:
        - ceModeBSupportInd

    LteMInd:
      description: LTE-M Indication.
      type: object
      properties:
        lteCatMInd:
          type: boolean
      required:
        - lteCatMInd

    NpnAccessInfo:
      description: NPN Access Information.
      type: object
      properties:
        cellCagInfo:
          type: array
          items:
            $ref: 'TS29571_CommonData.yaml#/components/schemas/CagId'
          minItems: 1

    ProseContext:
      description: Represents the ProSe services related parameters.
      type: object
      properties:
        directDiscovery:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/UeAuth'
        directComm:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/UeAuth'
        l2Relay:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/UeAuth'
        l3Relay:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/UeAuth'
        l2Remote:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/UeAuth'
        l3Remote:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/UeAuth'
        l2UeRelay:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/UeAuth'
        l3UeRelay:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/UeAuth'
        l2End:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/UeAuth'
        l3End:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/UeAuth'
        multiPathComm:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/UeAuth'
        nrUePc5Ambr:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/BitRate'
        pc5QoSPara:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Pc5QoSPara'

    AnalyticsSubscription:
      description: Analytics subscriptions created in the NWDAF.
      type: object
      properties:
        nwdafId:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/NfInstanceId'
        nwdafSetId:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/NfSetId'
        nwdafSubscriptionList:
          type: array
          items:
            $ref: '#/components/schemas/NwdafSubscription'
          minItems: 1
      required:
        - nwdafSubscriptionList

    NwdafSubscription:
      description: Individual NWDAF subscription identified by the subscription Id.
      type: object
```

```
      properties:
        nwdafEvtSubsServiceUri:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Uri'
        nwdafEventsSubscription:
          $ref:
'TS29520_Nnwdaf_EventsSubscription.yaml#/components/schemas/NnwdafEventsSubscription'
      required:
        - nwdafEvtSubsServiceUri
        - nwdafEventsSubscription

    UpdpSubscriptionData:
      description: UE policy delivery related N1 message notification subscription data.
      type: object
      properties:
        updpNotifySubscriptionId:
          type: string
        updpNotifyCallbackUri:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Uri'
        supportedFeatures:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/SupportedFeatures'
        updpCallbackBinding:
          type: string
      required:
        - updpNotifySubscriptionId
        - updpNotifyCallbackUri

    ReleaseSessionInfo:
      description: PDU session Id(s) and the cause for triggering the release.
      type: object
      properties:
        releaseSessionList:
          type: array
          items:
            $ref: 'TS29571_CommonData.yaml#/components/schemas/PduSessionId'
          minItems: 1
        releaseCause:
          $ref: '#/components/schemas/ReleaseCause'
      required:
        - releaseSessionList
        - releaseCause

    AreaOfInterestEventState:
      description: Event State of AoI event in old AMF
      type: object
      required:
        - presence
      properties:
        presence:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/PresenceState'
        individualPraIdList:
          type: array
          items:
            type: string
          minItems: 1

    TssInformation:
      description: Represents a Tss related N2 information data part
      type: object
      properties:
        nfId:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/NfInstanceId'
        tssContainer:
          type: array
          items:
            $ref: '#/components/schemas/N2InfoContent'
          minItems: 1
          maxItems: 10
        tssRspPerNgranList:
          type: array
          items:
            $ref: '#/components/schemas/TssRspPerNgran'
          minItems: 1
          maxItems: 10
      required:
        - tssContainer

    RslpInformation:
      description: Represents Ranging/SL positioning related N2 information.
```

```
      type: object
      properties:
        n2Pc5RslpPol:
          $ref: '#/components/schemas/N2InfoContent'

    A2xContext:
      description: Represents the A2X services related parameters
      type: object
      properties:
        nrA2xServicesAuth:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/NrA2xAuth'
        lteA2xServicesAuth:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/LteA2xAuth'
        nrUeSidelinkAmbr:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/BitRate'
        lteUeSidelinkAmbr:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/BitRate'
        pc5QoSPara:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Pc5QoSPara'

    A2xInformation:
      description: A2X related N2 information
      type: object
      properties:
        n2Pc5Pol:
          $ref: '#/components/schemas/N2InfoContent'

    AmPolicyInfoContainer:
      description: AM Policy Information Container
      type: object
      properties:
        sliceUsgCtrlInfoSets:
          type: object
          description: A map(list of key-value pairs) where Snssai serves as key.
          additionalProperties:
            $ref: 'TS29571_CommonData.yaml#/components/schemas/SliceUsageControlInfo'
          minProperties: 1

    LcsUpContext:
      description: Represents the LCS UP related parameters
      type: object
      properties:
        upConnectionStatus:
          $ref: 'TS29572_Nlmf_Location.yaml#/components/schemas/UpConnectionStatus'
        servingLMFIdentification:
          $ref: 'TS29572_Nlmf_Location.yaml#/components/schemas/LMFIdentification'

    DeregInactTimerInfo:
      description: Network Slice Deregistration Inactivity Timer Information
      type: object
      required:
        - deregInactExpiryTime
      properties:
        deregInactExpiryTime:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/NssaaStatus'

    TssRspPerNgran:
      description: Represents a TSS related N2 information data part
      type: object
      properties:
        ngranId:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/GlobalRanNodeId'
        ngranFailureInfo:
          $ref: '#/components/schemas/NgranFailureInfo'
        tssContainer:
          $ref: '#/components/schemas/N2InfoContent'
      required:
        - ngranId

    SliceReplacementMapping:
      description: >
        Represents the mapping between a replaced S-NSSAI in serving PLMN to its alternative S-NSSAI
      type: object
      properties:
        replacedSnssai:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Snssai'
        altSnssai:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Snssai'
```

```
      required:
        - replacedSnssai
        - altSnssai


#
# SIMPLE DATA TYPES
#
    EpsBearerId:
      description: EPS Bearer Identifier
      type: integer
      minimum: 0
      maximum: 15
    Ppi:
      description: Paging Policy Indicator
      type: integer
      minimum: 0
      maximum: 7
    NasCount:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/Uinteger'
    5GMmCapability:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/Bytes'
    UeSecurityCapability:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/Bytes'
    S1UeNetworkCapability:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/Bytes'
    DrxParameter:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/Bytes'
    OmcIdentifier:
      description: Represents the OMC Identifier
      type: string
    MSClassmark2:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/Bytes'
    SupportedCodec:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/Bytes'

#
# ENUMERATIONS
#
    StatusChange:
      description: Enumeration for AMF status
      anyOf:
      - type: string
        enum:
          - AMF_UNAVAILABLE
          - AMF_AVAILABLE
      - type: string
    N2InformationClass:
      description: Enumeration for N2 Information Class
      anyOf:
      - type: string
        enum:
          - SM
          - NRPPa
          - PWS
          - PWS-BCAL
          - PWS-RF
          - RAN
          - V2X
          - PROSE
          - TSS
          - RSPP
          - A2X
      - type: string
    N1MessageClass:
      description: Enumeration for N1 Message Class
      anyOf:
      - type: string
        enum:
          - 5GMM
          - SM
          - LPP
          - SMS
          - UPDP
          - LCS
          - UPP-CM
      - type: string
    N1N2MessageTransferCause:
```

```
      description: Enumeration for N1N2Message Transfer Cause
      anyOf:
      - type: string
        enum:
          - ATTEMPTING_TO_REACH_UE
          - N1_N2_TRANSFER_INITIATED
          - WAITING_FOR_ASYNCHRONOUS_TRANSFER
          - UE_NOT_RESPONDING
          - N1_MSG_NOT_TRANSFERRED
          - N2_MSG_NOT_TRANSFERRED
          - UE_NOT_REACHABLE_FOR_SESSION
          - TEMPORARY_REJECT_REGISTRATION_ONGOING
          - TEMPORARY_REJECT_HANDOVER_ONGOING
          - REJECTION_DUE_TO_PAGING_RESTRICTION
          - AN_NOT_RESPONDING
          - FAILURE_CAUSE_UNSPECIFIED
      - type: string
    UeContextTransferStatus:
      description: Describes the status of an individual ueContext resource in UE Context Transfer
procedures
      anyOf:
      - type: string
        enum:
          - TRANSFERRED
          - NOT_TRANSFERRED
      - type: string
    N2InformationTransferResult:
      description: Describes the result of N2 information transfer by AMF to the AN
      anyOf:
      - type: string
        enum:
          - N2_INFO_TRANSFER_INITIATED
      - type: string
    CipheringAlgorithm:
      description: Indicates the supported Ciphering Algorithm
      anyOf:
      - type: string
        enum:
          - NEA0
          - NEA1
          - NEA2
          - NEA3
      - type: string
    IntegrityAlgorithm:
      description: Indicates the supported Integrity Algorithm
      anyOf:
      - type: string
        enum:
          - NIA0
          - NIA1
          - NIA2
          - NIA3
      - type: string
    SmsSupport:
      description: Indicates the supported SMS delivery of a UE
      anyOf:
      - type: string
        enum:
          - 3GPP
          - NON_3GPP
          - BOTH
          - NONE
      - type: string
    ScType:
      description: Indicates the security context type
      anyOf:
      - type: string
        enum:
          - NATIVE
          - MAPPED
      - type: string
    KeyAmfType:
      description: Indicates the K_amf type
      anyOf:
      - type: string
        enum:
          - KAMF
          - KPRIMEAMF
```

```
    - type: string
  TransferReason:
    description: Indicates UE Context Transfer Reason
    anyOf:
    - type: string
      enum:
        - INIT_REG
        - MOBI_REG
        - MOBI_REG_UE_VALIDATED
    - type: string
  PolicyReqTrigger:
    description: Policy Request Triggers
    anyOf:
    - type: string
      enum:
        - LOCATION_CHANGE
        - PRA_CHANGE
        - ALLOWED_NSSAI_CHANGE
        - NWDAF_DATA_CHANGE
        - PLMN_CHANGE
        - CON_STATE_CHANGE
        - SMF_SELECT_CHANGE
        - ACCESS_TYPE_CHANGE
        - SAT_BACKHAUL_CHANGE
        - CONF_NSSAI_CHANGE
        - LBO_INFO_CHANGE
    - type: string
  RatSelector:
    description: Indicates the RAT type for the transfer of N2 information
    anyOf:
    - type: string
      enum:
        - E-UTRA
        - NR
    - type: string
  NgapIeType:
    description: Indicates the supported NGAP IE types
    anyOf:
    - type: string
      enum:
        - PDU_RES_SETUP_REQ
        - PDU_RES_REL_CMD
        - PDU_RES_MOD_REQ
        - HANDOVER_CMD
        - HANDOVER_REQUIRED
        - HANDOVER_PREP_FAIL
        - SRC_TO_TAR_CONTAINER
        - TAR_TO_SRC_CONTAINER
        - TAR_TO_SRC_FAIL_CONTAINER
        - RAN_STATUS_TRANS_CONTAINER
        - SON_CONFIG_TRANSFER
        - NRPPA_PDU
        - UE_RADIO_CAPABILITY
        - RIM_INFO_TRANSFER
        - SECONDARY_RAT_USAGE
        - PC5_QOS_PARA
        - EARLY_STATUS_TRANS_CONTAINER
        - UE_RADIO_CAPABILITY_FOR_PAGING
    - type: string
  N2InfoNotifyReason:
    description: N2 Information Notify Reason
    anyOf:
    - type: string
      enum:
        - HANDOVER_COMPLETED
    - type: string
  SmfChangeIndication:
    description: Indicates the I-SMF or V-SMF change or removal
    anyOf:
    - type: string
      enum:
        - CHANGED
        - REMOVED
    - type: string
  SbiBindingLevel:
    description: SBI Binding Level
    anyOf:
    - type: string
```

```
      enum:
        - NF_INSTANCE_BINDING
        - NF_SET_BINDING
        - NF_SERVICE_SET_BINDING
        - NF_SERVICE_INSTANCE_BINDING
    - type: string

  EpsNasCipheringAlgorithm:
    description: Indicates the supported EPS NAS Ciphering Algorithm
    anyOf:
    - type: string
      enum:
        - EEA0
        - EEA1
        - EEA2
        - EEA3
    - type: string

  EpsNasIntegrityAlgorithm:
    description: Indicates the supported EPS NAS Integrity Algorithm
    anyOf:
    - type: string
      enum:
        - EIA0
        - EIA1
        - EIA2
        - EIA3
    - type: string

  PeriodicCommunicationIndicator:
    description: Indicates the Periodic Communication Indicator
    anyOf:
    - type: string
      enum:
        - PIORIODICALLY
        - ON_DEMAND
    - type: string

  UuaaMmStatus:
    description: Indicates the UUAA-MM status
    anyOf:
    - type: string
      enum:
        - SUCCESS
        - PENDING
        - FAILED
    - type: string

  ReleaseCause:
    description: The cause for triggering the release.
    anyOf:
    - type: string
      enum:
        - SNPN_SNPN_MOBILITY
        - NO_HR_AGREEMENT
        - UNSPECIFIED
    - type: string


  NgranFailureInfo:
    description: Indicates a NG-RAN failure event.
    anyOf:
    - type: string
      enum:
        - NG_RAN_NOT_REACHABLE
        - NG_RAN_FAILURE_WITH_RESTART
        - NG_RAN_FAILURE_WITHOUT_RESTART
    - type: string


  XrDeviceWith2Rx:
    description: Information of the XR Device with 2Rx
    anyOf:
    - type: string
      enum:
        - TRUE
    - type: string
```

# A.3 Namf_EventExposure API

```
openapi: 3.0.0

info:
  version: 1.3.1
  title: Namf_EventExposure
  description: |
    AMF Event Exposure Service.
    © 2024, 3GPP Organizational Partners (ARIB, ATIS, CCSA, ETSI, TSDSI, TTA, TTC).
    All rights reserved.

security:
  - {}
  - oAuth2ClientCredentials:
      - namf-evts

externalDocs:
  description: 3GPP TS 29.518 V18.10.0; 5G System; Access and Mobility Management Services
  url: 'https://www.3gpp.org/ftp/Specs/archive/29_series/29.518/'

servers:
  - url: '{apiRoot}/namf-evts/v1'
    variables:
      apiRoot:
        default: https://example.com
        description: apiRoot as defined in clause 4.4 of 3GPP TS 29.501

paths:
  /subscriptions:
    post:
      summary: Namf_EventExposure Subscribe service Operation
      tags:
        - Subscriptions collection (Collection)
      operationId: CreateSubscription
      requestBody:
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/AmfCreateEventSubscription'
        required: true
      responses:
        '201':
          description: Subsription Created
          headers:
            Location:
              description: 'Contains the URI of the newly created resource, according to the
structure: {apiRoot}/namf-evts/<apiVersion>/subscriptions/{subscriptionId}'
              required: true
              schema:
                type: string
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/AmfCreatedEventSubscription'
        '307':
          $ref: 'TS29571_CommonData.yaml#/components/responses/307'
        '308':
          $ref: 'TS29571_CommonData.yaml#/components/responses/308'
        '400':
          $ref: 'TS29571_CommonData.yaml#/components/responses/400'
        '401':
          $ref: 'TS29571_CommonData.yaml#/components/responses/401'
        '403':
          $ref: 'TS29571_CommonData.yaml#/components/responses/403'
        '404':
          $ref: 'TS29571_CommonData.yaml#/components/responses/404'
        '411':
          $ref: 'TS29571_CommonData.yaml#/components/responses/411'
        '413':
          $ref: 'TS29571_CommonData.yaml#/components/responses/413'
        '415':
          $ref: 'TS29571_CommonData.yaml#/components/responses/415'
        '429':
```

```
        $ref: 'TS29571_CommonData.yaml#/components/responses/429'
      '500':
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
      '501':
        $ref: 'TS29571_CommonData.yaml#/components/responses/501'
      '502':
        $ref: 'TS29571_CommonData.yaml#/components/responses/502'
      '503':
        $ref: 'TS29571_CommonData.yaml#/components/responses/503'
      default:
        description: Unexpected error
  callbacks:
    onEventReport:
      '{$request.body#/subscription/eventNotifyUri}':
        post:
          summary: Event Notificaiton Delivery
          requestBody:
            content:
              application/json:
                schema:
                  $ref: '#/components/schemas/AmfEventNotification'
            required: true
          responses:
            '204':
              description: Successful acknowledgement
            '307':
              $ref: 'TS29571_CommonData.yaml#/components/responses/307'
            '308':
              $ref: 'TS29571_CommonData.yaml#/components/responses/308'
            '400':
              $ref: 'TS29571_CommonData.yaml#/components/responses/400'
            '401':
              $ref: 'TS29571_CommonData.yaml#/components/responses/401'
            '403':
              $ref: 'TS29571_CommonData.yaml#/components/responses/403'
            '404':
              $ref: 'TS29571_CommonData.yaml#/components/responses/404'
            '411':
              $ref: 'TS29571_CommonData.yaml#/components/responses/411'
            '413':
              $ref: 'TS29571_CommonData.yaml#/components/responses/413'
            '415':
              $ref: 'TS29571_CommonData.yaml#/components/responses/415'
            '429':
              $ref: 'TS29571_CommonData.yaml#/components/responses/429'
            '500':
              $ref: 'TS29571_CommonData.yaml#/components/responses/500'
            '502':
              $ref: 'TS29571_CommonData.yaml#/components/responses/502'
            '503':
              $ref: 'TS29571_CommonData.yaml#/components/responses/503'
            default:
              description: Unexpected error
    onSubscriptionIdChangeEvtReport:
      '{$request.body#/subscription/subsChangeNotifyUri}':
        post:
          summary: Event Notificaiton Delivery For Subscription Id Change
          requestBody:
            content:
              application/json:
                schema:
                  $ref: '#/components/schemas/AmfEventNotification'
            required: true
          responses:
            '204':
              description: Successful acknowledgement
            '307':
              $ref: 'TS29571_CommonData.yaml#/components/responses/307'
            '308':
              $ref: 'TS29571_CommonData.yaml#/components/responses/308'
            '400':
              $ref: 'TS29571_CommonData.yaml#/components/responses/400'
            '401':
              $ref: 'TS29571_CommonData.yaml#/components/responses/401'
            '403':
              $ref: 'TS29571_CommonData.yaml#/components/responses/403'
            '404':
              $ref: 'TS29571_CommonData.yaml#/components/responses/404'
```

```
              '411':
                $ref: 'TS29571_CommonData.yaml#/components/responses/411'
              '413':
                $ref: 'TS29571_CommonData.yaml#/components/responses/413'
              '415':
                $ref: 'TS29571_CommonData.yaml#/components/responses/415'
              '429':
                $ref: 'TS29571_CommonData.yaml#/components/responses/429'
              '500':
                $ref: 'TS29571_CommonData.yaml#/components/responses/500'
              '502':
                $ref: 'TS29571_CommonData.yaml#/components/responses/502'
              '503':
                $ref: 'TS29571_CommonData.yaml#/components/responses/503'
              default:
                description: Unexpected error
/subscriptions/{subscriptionId}:
  patch:
    summary: Namf_EventExposure Subscribe Modify service Operation
    tags:
      - Individual subscription (Document)
    operationId: ModifySubscription
    parameters:
      - name: subscriptionId
        in: path
        required: true
        description: Unique ID of the subscription to be modified
        schema:
          type: string
    requestBody:
      content:
        application/json-patch+json:
          schema:
            oneOf:
              - type: array
                items:
                 $ref: '#/components/schemas/AmfUpdateEventSubscriptionItem'
                minItems: 1
              - type: array
                items:
                 $ref: '#/components/schemas/AmfUpdateEventOptionItem'
                minItems: 1
                maxItems: 1
      required: true
    responses:
      '200':
        description: Subsription modified successfully
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/AmfUpdatedEventSubscription'
      '307':
        $ref: 'TS29571_CommonData.yaml#/components/responses/307'
      '308':
        $ref: 'TS29571_CommonData.yaml#/components/responses/308'
      '400':
        $ref: 'TS29571_CommonData.yaml#/components/responses/400'
      '401':
        $ref: 'TS29571_CommonData.yaml#/components/responses/401'
      '403':
        $ref: 'TS29571_CommonData.yaml#/components/responses/403'
      '404':
        $ref: 'TS29571_CommonData.yaml#/components/responses/404'
      '411':
        $ref: 'TS29571_CommonData.yaml#/components/responses/411'
      '413':
        $ref: 'TS29571_CommonData.yaml#/components/responses/413'
      '415':
        $ref: 'TS29571_CommonData.yaml#/components/responses/415'
      '429':
        $ref: 'TS29571_CommonData.yaml#/components/responses/429'
      '500':
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
      '501':
        $ref: 'TS29571_CommonData.yaml#/components/responses/501'
      '502':
        $ref: 'TS29571_CommonData.yaml#/components/responses/502'
      '503':
```

```
          $ref: 'TS29571_CommonData.yaml#/components/responses/503'
        default:
          description: Unexpected error
    delete:
      summary: Namf_EventExposure Unsubscribe service Operation
      tags:
        - Individual subscription (Document)
      operationId: DeleteSubscription
      parameters:
        - name: subscriptionId
          in: path
          required: true
          description: Unique ID of the subscription to be deleted
          schema:
            type: string
      responses:
        '204':
          description: Subsription deleted successfully
        '307':
          $ref: 'TS29571_CommonData.yaml#/components/responses/307'
        '308':
          $ref: 'TS29571_CommonData.yaml#/components/responses/308'
        '400':
          $ref: 'TS29571_CommonData.yaml#/components/responses/400'
        '401':
          $ref: 'TS29571_CommonData.yaml#/components/responses/401'
        '403':
          $ref: 'TS29571_CommonData.yaml#/components/responses/403'
        '404':
          $ref: 'TS29571_CommonData.yaml#/components/responses/404'
        '411':
          $ref: 'TS29571_CommonData.yaml#/components/responses/411'
        '413':
          $ref: 'TS29571_CommonData.yaml#/components/responses/413'
        '415':
          $ref: 'TS29571_CommonData.yaml#/components/responses/415'
        '429':
          $ref: 'TS29571_CommonData.yaml#/components/responses/429'
        '500':
          $ref: 'TS29571_CommonData.yaml#/components/responses/500'
        '502':
          $ref: 'TS29571_CommonData.yaml#/components/responses/502'
        '503':
          $ref: 'TS29571_CommonData.yaml#/components/responses/503'
        default:
          description: Unexpected error
components:
  securitySchemes:
    oAuth2ClientCredentials:
      type: oauth2
      flows:
        clientCredentials:
          tokenUrl: '{nrfApiRoot}/oauth2/token'
          scopes:
            namf-evts: Access to the Namf_EventExposure API
  schemas:
    AmfEventSubscription:
      description: Represents an individual event subscription resource on AMF
      type: object
      properties:
        eventList:
          type: array
          items:
            $ref: '#/components/schemas/AmfEvent'
          minItems: 1
        eventNotifyUri:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Uri'
        notifyCorrelationId:
          type: string
        nfId:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/NfInstanceId'
        subsChangeNotifyUri:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Uri'
        subsChangeNotifyCorrelationId:
          type: string
        supi:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Supi'
        groupId:
```

```
        $ref: 'TS29571_CommonData.yaml#/components/schemas/GroupId'
      excludeSupiList:
        type: array
        items:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Supi'
        minItems: 1
      excludeGpsiList:
        type: array
        items:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Gpsi'
        minItems: 1
      includeSupiList:
        type: array
        items:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Supi'
        minItems: 1
      includeGpsiList:
        type: array
        items:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Gpsi'
        minItems: 1
      gpsi:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Gpsi'
      pei:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Pei'
      anyUE:
        type: boolean
      options:
        $ref: '#/components/schemas/AmfEventMode'
      sourceNfType:
        $ref: 'TS29510_Nnrf_NFManagement.yaml#/components/schemas/NFType'
      termNotifyInd:
        type: boolean
    required:
      - eventList
      - eventNotifyUri
      - notifyCorrelationId
      - nfId

  AmfEvent:
    description: Describes an event to be subscribed
    type: object
    properties:
      type:
        $ref: '#/components/schemas/AmfEventType'
      immediateFlag:
        type: boolean
        default: false
      areaList:
        type: array
        items:
          $ref: '#/components/schemas/AmfEventArea'
        minItems: 1
      locationFilterList:
        type: array
        items:
          $ref: '#/components/schemas/LocationFilter'
        minItems: 1
      refId:
        $ref: 'TS29503_Nudm_EE.yaml#/components/schemas/ReferenceId'
      trafficDescriptorList:
        type: array
        items:
          $ref: '#/components/schemas/TrafficDescriptor'
        minItems: 1
      reportUeReachable:
        type: boolean
        default: false
      reachabilityFilter:
        $ref: '#/components/schemas/ReachabilityFilter'
      udmDetectInd:
        type: boolean
        default: false
      maxReports:
        type: integer
      presenceInfoList:
        type: object
        additionalProperties:
```

```
          $ref: 'TS29571_CommonData.yaml#/components/schemas/PresenceInfo'
        minProperties: 1
        description: A map(list of key-value pairs) where praId serves as key.
      maxResponseTime:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/DurationSec'
      targetArea:
        $ref: '#/components/schemas/TargetArea'
      snssaiFilter:
        type: array
        items:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/ExtSnssai'
        minItems: 1
      ueInAreaFilter:
        $ref: '#/components/schemas/UeInAreaFilter'
      minInterval:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/DurationSec'
      nextReport:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/DateTime'
      idleStatusInd:
        type: boolean
        default: false
      dispersionArea:
        $ref: '#/components/schemas/DispersionArea'
      nextPeriodicReportTime:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/DateTime'
      adjustAoIOnRa:
        type: boolean
        default: false
      ranTimingSynchroStatusChange:
        type: boolean
        default: false
      notifyForSupiList:
        type: array
        items:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Supi'
        minItems: 1
      notifyForGroupList:
        type: array
        items:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/GroupId'
        minItems: 1
      notifyForSnssaiDnnList:
        type: array
        items:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/SnssaiDnnItem'
        minItems: 1
    required:
      - type
  AmfEventNotification:
    description: Data within a AMF Event Notification request
    type: object
    properties:
      notifyCorrelationId:
        type: string
      subsChangeNotifyCorrelationId:
        type: string
      reportList:
        type: array
        items:
          $ref: '#/components/schemas/AmfEventReport'
        minItems: 1
      eventSubsSyncInfo:
        $ref: '#/components/schemas/AmfEventSubsSyncInfo'
  AmfEventReport:
    description: Represents a report triggered by a subscribed event type
    type: object
    properties:
      type:
        $ref: '#/components/schemas/AmfEventType'
      state:
        $ref: '#/components/schemas/AmfEventState'
      timeStamp:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/DateTime'
      subscriptionId:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Uri'
      anyUe:
        type: boolean
      supi:
```

```
      $ref: 'TS29571_CommonData.yaml#/components/schemas/Supi'
    areaList:
      type: array
      items:
        $ref: '#/components/schemas/AmfEventArea'
      minItems: 1
    refId:
      $ref: 'TS29503_Nudm_EE.yaml#/components/schemas/ReferenceId'
    gpsi:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/Gpsi'
    pei:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/Pei'
    location:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/UserLocation'
    additionalLocation:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/UserLocation'
    timezone:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/TimeZone'
    accessTypeList:
      type: array
      items:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/AccessType'
      minItems: 1
    rmInfoList:
      type: array
      items:
        $ref: '#/components/schemas/RmInfo'
      minItems: 1
    cmInfoList:
      type: array
      items:
        $ref: '#/components/schemas/CmInfo'
      minItems: 1
    reachability:
      $ref: '#/components/schemas/UeReachability'
    commFailure:
      $ref: '#/components/schemas/CommunicationFailure'
    lossOfConnectReason:
      $ref: '#/components/schemas/LossOfConnectivityReason'
    numberOfUes:
      type: integer
    5gsUserStateList:
      type: array
      items:
        $ref: '#/components/schemas/5GsUserStateInfo'
      minItems: 1
    typeCode:
      type: string
      pattern: '^imeitac-[0-9]{8}$'
    registrationNumber:
      type: integer
    maxAvailabilityTime:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/DateTime'
    ueIdExt:
      type: array
      items:
        $ref: '#/components/schemas/UEIdExt'
      minItems: 1
    snssaiTaiList:
      type: array
      items:
        $ref: '#/components/schemas/SnssaiTaiMapping'
      minItems: 1
    idleStatusIndication:
      $ref: '#/components/schemas/IdleStatusIndication'
    ueAccessBehaviorTrends:
      type: array
      items:
        $ref: '#/components/schemas/UeAccessBehaviorReportItem'
      minItems: 1
    ueLocationTrends:
      type: array
      items:
        $ref: '#/components/schemas/UeLocationTrendsReportItem'
      minItems: 1
    mmTransLocationReportList:
      type: array
      items:
```

```
        $ref: '#/components/schemas/MmTransactionLocationReportItem'
      minItems: 1
    mmTransSliceReportList:
      type: array
      items:
        $ref: '#/components/schemas/MmTransactionSliceReportItem'
      minItems: 1
    termReason:
      $ref: '#/components/schemas/SubTerminationReason'
    unavailabilityPeriod:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/DurationSec'
  required:
    - type
    - state
    - timeStamp
AmfEventMode:
  description: Describes how the reports shall be generated by a subscribed event
  type: object
  properties:
    trigger:
      $ref: '#/components/schemas/AmfEventTrigger'
    maxReports:
      type: integer
    expiry:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/DateTime'
    repPeriod:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/DurationSec'
    sampRatio:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/SamplingRatio'
    partitioningCriteria:
      type: array
      items:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/PartitioningCriteria'
      minItems: 1
    notifFlag:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/NotificationFlag'
    mutingExcInstructions:
      writeOnly: true
      allOf:
        - $ref: 'TS29571_CommonData.yaml#/components/schemas/MutingExceptionInstructions'
    mutingNotSettings:
      readOnly: true
      allOf:
        - $ref: 'TS29571_CommonData.yaml#/components/schemas/MutingNotificationsSettings'
    varRepPeriodInfo:
      type: array
      items:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/VarRepPeriod'
      minItems: 1
  required:
    - trigger
AmfEventState:
  description: Represents the state of a subscribed event
  type: object
  properties:
    active:
      type: boolean
    remainReports:
      type: integer
    remainDuration:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/DurationSec'
  required:
    - active
RmInfo:
  description: Represents the registration state of a UE for an access type
  type: object
  properties:
    rmState:
      $ref: '#/components/schemas/RmState'
    accessType:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/AccessType'
  required:
    - rmState
    - accessType
CmInfo:
  description: Represents the connection management state of a UE for an access type
  type: object
  properties:
```

```
        cmState:
          $ref: '#/components/schemas/CmState'
        accessType:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/AccessType'
      required:
        - cmState
        - accessType
    CommunicationFailure:
      description: Describes a communication failure detected by AMF
      type: object
      properties:
        nasReleaseCode:
          type: string
        ranReleaseCode:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/NgApCause'
    AmfCreateEventSubscription:
      description: Data within a create AMF event subscription request
      type: object
      properties:
        subscription:
          $ref: '#/components/schemas/AmfEventSubscription'
        supportedFeatures:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/SupportedFeatures'
        oldGuami:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Guami'
      required:
        - subscription
    AmfCreatedEventSubscription:
      description: Data within a create AMF event subscription response
      type: object
      properties:
        subscription:
          $ref: '#/components/schemas/AmfEventSubscription'
        subscriptionId:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Uri'
        reportList:
          type: array
          items:
            $ref: '#/components/schemas/AmfEventReport'
          minItems: 1
        supportedFeatures:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/SupportedFeatures'
      required:
        - subscription
        - subscriptionId
    AmfUpdateEventSubscriptionItem:
      description: Document describing the modification(s) to an AMF Event Subscription
      type: object
      properties:
        op:
          type: string
          enum:
            - add
            - remove
            - replace
        path:
          type: string
          pattern: '^\/eventList\/-|(\/eventList\/0|\/eventList\/[1-9][0-
9]*){1}(\/presenceInfoList\/0|\/presenceInfoList\/[1-9][0-9]*
|\/notifyForSupiList|\/notifyForGroupList|\/notifyForSnssaiDnnList)?|\/excludeSupiList|\/excludeGpsi
List|\/includeSupiList|\/includeGpsiList$'
        value: {}
      required:
        - op
        - path

    AmfUpdateEventOptionItem:
      description: Document describing the modifications to AMF event subscription options
      type: object
      properties:
        op:
          type: string
          enum:
            - replace
        path:
          type: string
          pattern: '^(\/options\/expiry|\/options\/notifFlag|\/options\/mutingExcInstructions)$'
        value: {}
```

```
    required:
      - op
      - path
      - value
  AmfUpdatedEventSubscription:
    description: Represents a successful update on an AMF Event Subscription
    type: object
    properties:
      subscription:
        $ref: '#/components/schemas/AmfEventSubscription'
      reportList:
        type: array
        items:
          $ref: '#/components/schemas/AmfEventReport'
        minItems: 1
    required:
      - subscription
  AmfEventArea:
    description: Represents an area to be monitored by an AMF event
    type: object
    properties:
      presenceInfo:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/PresenceInfo'
      ladnInfo:
        $ref: '#/components/schemas/LadnInfo'
      sliceAreaRestrictionInfo:
        $ref: '#/components/schemas/SliceAreaRestrictionInfo'
      sNssai:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Snssai'
      nsiId:
        $ref: 'TS29531_Nnssf_NSSelection.yaml#/components/schemas/NsiId'
  LadnInfo:
    description: LADN Information
    type: object
    properties:
      ladn:
        type: string
      presence:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/PresenceState'
    required:
      - ladn
  5GsUserStateInfo:
    description: Represents the 5GS User state of the UE for an access type
    type: object
    properties:
      5gsUserState:
        $ref: '#/components/schemas/5GsUserState'
      accessType:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/AccessType'
    required:
      - 5gsUserState
      - accessType
  TrafficDescriptor:
    description: Represents the Traffic Descriptor
    type: object
    properties:
      dnn:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Dnn'
      sNssai:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Snssai'
      dddTrafficDescriptorList:
        type: array
        items:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/DddTrafficDescriptor'
        minItems: 1
  UEIdExt:
    description: UE Identity
    type: object
    properties:
      supi:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Supi'
      gpsi:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Gpsi'

  AmfEventSubsSyncInfo:
    description: AMF Event Subscriptions Information for synchronization
    type: object
    properties:
```

```
      subscriptionList:
        type: array
        items:
          $ref: '#/components/schemas/AmfEventSubscriptionInfo'
        minItems: 1
    required:
      - subscriptionList

  AmfEventSubscriptionInfo:
    description: Individual AMF Event Subscription Information
    type: object
    properties:
      subId:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Uri'
      notifyCorrelationId:
        type: string
      refIdList:
        type: array
        items:
          $ref: 'TS29503_Nudm_EE.yaml#/components/schemas/ReferenceId'
        minItems: 1
      oldSubId:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Uri'
    required:
      - subId
      - refIdList

  TargetArea:
    description: TA list or TAI range list or any TA
    type: object
    properties:
      taList:
        type: array
        items:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Tai'
        minItems: 1
      taiRangeList:
        type: array
        items:
          $ref: 'TS29510_Nnrf_NFManagement.yaml#/components/schemas/TaiRange'
        minItems: 1
      anyTa:
        type: boolean
        default: false

  SnssaiTaiMapping:
    description: List of restricted or unrestricted S-NSSAIs per TAI(s)
    type: object
    properties:
      reportingArea:
        $ref: '#/components/schemas/TargetArea'
      accessTypeList:
        type: array
        items:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/AccessType'
        minItems: 1
      supportedSnssaiList:
        type: array
        items:
          $ref: '#/components/schemas/SupportedSnssai'
        minItems: 1
    required:
      - reportingArea

  UeAccessBehaviorReportItem:
    description: Report Item for UE Access Behavior Trends event.
    type: object
    properties:
      stateTransitionType:
        $ref: '#/components/schemas/AccessStateTransitionType'
      spacing:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/DurationSec'
      spacingVar:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Float'
      duration:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/DurationSec'
      durationVar:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Float'
```

```
      occurrences:
        type: integer
      timestamp:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/DateTime'
    required:
      - stateTransitionType
      - spacing
      - duration

  IdleStatusIndication:
    description: Represents the idle status indication.
    type: object
    properties:
      timeStamp:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/DateTime'
      activeTime:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/DurationSec'
      subsRegTimer:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/DurationSec'
      edrxCycleLength:
        type: integer
      suggestedNumOfDlPackets:
        type: integer

  UeInAreaFilter:
    description: Additional filters for UE in Area Report event
    type: object
    properties:
      ueType:
        $ref: '#/components/schemas/UeType'
      aerialSrvDnnInd:
        type: boolean
        default: false
      ueIdOmitInd:
        type: boolean
        default: false

  SupportedSnssai:
    description: Supported S-NSSAIs
    type: object
    properties:
      sNssai:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/ExtSnssai'
      restrictionInd:
        type: boolean
        default: false
    required:
      - sNssai

  UeLocationTrendsReportItem:
    description: Report Item for UE Location Trends event.
    type: object
    properties:
      tai:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Tai'
      ncgi:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Ncgi'
      ecgi:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Ecgi'
      n3gaLocation:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/N3gaLocation'
      spacing:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/DurationSec'
      spacingVar:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Float'
      duration:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/DurationSec'
      durationVar:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Float'
      occurrences:
        type: integer
      timestamp:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/DateTime'
    required:
      - spacing
      - duration
      - timestamp
```

```
    DispersionArea:
      description: Dispersion Area
      type: object
      properties:
        taiList:
          type: array
          items:
            $ref: 'TS29571_CommonData.yaml#/components/schemas/Tai'
          minItems: 1
        ncgiList:
          type: array
          items:
            $ref: 'TS29571_CommonData.yaml#/components/schemas/Ncgi'
          minItems: 1
        ecgiList:
          type: array
          items:
            $ref: 'TS29571_CommonData.yaml#/components/schemas/Ecgi'
          minItems: 1
        n3gaInd:
          type: boolean
          default: false

    MmTransactionLocationReportItem:
      description: UE MM Transaction Report Item per Location
      type: object
      properties:
        tai:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Tai'
        ncgi:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Ncgi'
        ecgi:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Ecgi'
        n3gaLocation:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/N3gaLocation'
        timestamp:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/DateTime'
        transactions:
          type: integer
      required:
        - timestamp
        - transactions

    MmTransactionSliceReportItem:
      description: UE MM Transaction Report Item per Slice
      type: object
      properties:
        snssai:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Snssai'
        timestamp:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/DateTime'
        transactions:
          type: integer
      required:
        - timestamp
        - transactions

    SliceAreaRestrictionInfo:
      description: Slice Area Restriction Information
      type: object
      properties:
        sNssai:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Snssai'
        presence:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/PresenceState'
      required:
        - sNssai


    AmfEventType:
      description: Describes the supported event types of Namf_EventExposure Service
      anyOf:
      - type: string
        enum:
          - LOCATION_REPORT
          - PRESENCE_IN_AOI_REPORT
          - TIMEZONE_REPORT
          - ACCESS_TYPE_REPORT
```

```
        - REGISTRATION_STATE_REPORT
        - CONNECTIVITY_STATE_REPORT
        - REACHABILITY_REPORT
        - COMMUNICATION_FAILURE_REPORT
        - UES_IN_AREA_REPORT
        - SUBSCRIPTION_ID_CHANGE
        - SUBSCRIPTION_ID_ADDITION
        - SUBSCRIPTION_TERMINATION
        - LOSS_OF_CONNECTIVITY
        - 5GS_USER_STATE_REPORT
        - AVAILABILITY_AFTER_DDN_FAILURE
        - TYPE_ALLOCATION_CODE_REPORT
        - FREQUENT_MOBILITY_REGISTRATION_REPORT
        - SNSSAI_TA_MAPPING_REPORT
        - UE_LOCATION_TRENDS
        - UE_ACCESS_BEHAVIOR_TRENDS
        - UE_MM_TRANSACTION_REPORT
    - type: string
  AmfEventTrigger:
    description: Describes how AMF should generate the report for the event
    anyOf:
    - type: string
      enum:
        - ONE_TIME
        - CONTINUOUS
        - PERIODIC
    - type: string
  LocationFilter :
    description: Describes the supported filters of LOCATION_REPORT event type
    anyOf:
    - type: string
      enum:
        - TAI
        - CELL_ID
        - RAN_NODE
        - N3IWF
        - UE_IP
        - UDP_PORT
        - TNAP_ID
        - GLI
        - TWAP_ID
    - type: string
  UeReachability:
    description: Describes the reachability of the UE
    anyOf:
    - type: string
      enum:
        - UNREACHABLE
        - REACHABLE
        - REGULATORY_ONLY
    - type: string
  RmState:
    description: Describes the registration management state of a UE
    anyOf:
    - type: string
      enum:
        - REGISTERED
        - DEREGISTERED
    - type: string
  CmState:
    description: Describes the connection management state of a UE
    anyOf:
    - type: string
      enum:
        - IDLE
        - CONNECTED
    - type: string
  5GsUserState:
    description: Describes the 5GS User State of a UE
    anyOf:
    - type: string
      enum:
        - DEREGISTERED
        - CONNECTED_NOT_REACHABLE_FOR_PAGING
        - CONNECTED_REACHABLE_FOR_PAGING
        - NOT_PROVIDED_FROM_AMF
    - type: string
  LossOfConnectivityReason:
```

```
      description: Describes the reason for loss of connectivity
      anyOf:
      - type: string
        enum:
          - DEREGISTERED
          - MAX_DETECTION_TIME_EXPIRED
          - PURGED
          - UNAVAILABLE_PERIOD
      - type: string

    ReachabilityFilter:
      description: Event filter for REACHABILITY_REPORT event type
      anyOf:
      - type: string
        enum:
          - UE_REACHABILITY_STATUS_CHANGE
          - UE_REACHABLE_DL_TRAFFIC
      - type: string

    UeType:
      description: Describes the type of UEs
      anyOf:
      - type: string
        enum:
          - AERIAL_UE
      - type: string

    AccessStateTransitionType:
      description: Access State Transition Type.
      anyOf:
      - type: string
        enum:
          - ACCESS_TYPE_CHANGE_3GPP
          - ACCESS_TYPE_CHANGE_N3GPP
          - RM_STATE_CHANGE_DEREGISTERED
          - RM_STATE_CHANGE_REGISTERED
          - CM_STATE_CHANGE_IDLE
          - CM_STATE_CHANGE_CONNECTED
          - HANDOVER
          - MOBILITY_REGISTRATION_UPDATE
      - type: string

    SubTerminationReason:
      description: Subscription Termination Reason.
      anyOf:
      - type: string
        enum:
          - INVALID_SUBSCRIPTION
          - SUBSCRIPTION_NOT_AUTHORIZED
      - type: string
```

# A.4 Namf_MT

```
openapi: 3.0.0

info:
  version: 1.3.0
  title: Namf_MT
  description: |
    AMF Mobile Terminated Service.
    © 2024, 3GPP Organizational Partners (ARIB, ATIS, CCSA, ETSI, TSDSI, TTA, TTC).
    All rights reserved.

security:
  - {}
  - oAuth2ClientCredentials:
      - namf-mt

externalDocs:
  description: 3GPP TS 29.518 V18.6.0; 5G System; Access and Mobility Management Services
  url: 'https://www.3gpp.org/ftp/Specs/archive/29_series/29.518/'

servers:
  - url: '{apiRoot}/namf-mt/v1'
    variables:
      apiRoot:
```

```
        default: https://example.com
        description: apiRoot as defined in clause 4.4 of 3GPP TS 29.501

paths:
  '/ue-contexts/{ueContextId}':
    get:
      summary: Namf_MT Provide Domain Selection Info service Operation
      tags:
        - ueContext (Document)
      operationId: Provide Domain Selection Info
      parameters:
        - name: ueContextId
          in: path
          description: UE Context Identifier
          required: true
          schema:
            type: string
            pattern: '^(imsi-[0-9]{5,15}|nai-.+|gli-.+|gci-.+|.+)$'
        - name: info-class
          in: query
          description: UE Context Information Class
          schema:
            $ref: '#/components/schemas/UeContextInfoClass'
        - name: supported-features
          in: query
          description: Supported Features
          schema:
            $ref: 'TS29571_CommonData.yaml#/components/schemas/SupportedFeatures'
        - name: old-guami
          in: query
          description: Old GUAMI
          content:
            application/json:
              schema:
                $ref: 'TS29571_CommonData.yaml#/components/schemas/Guami'
      responses:
        '200':
          description: Requested UE Context Information returned
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/UeContextInfo'
        '307':
          $ref: 'TS29571_CommonData.yaml#/components/responses/307'
        '308':
          $ref: 'TS29571_CommonData.yaml#/components/responses/308'
        '400':
          $ref: 'TS29571_CommonData.yaml#/components/responses/400'
        '401':
          $ref: 'TS29571_CommonData.yaml#/components/responses/401'
        '403':
          $ref: 'TS29571_CommonData.yaml#/components/responses/403'
        '404':
          $ref: 'TS29571_CommonData.yaml#/components/responses/404'
        '409':
          $ref: 'TS29571_CommonData.yaml#/components/responses/409'
        '414':
          $ref: 'TS29571_CommonData.yaml#/components/responses/414'
        '429':
          $ref: 'TS29571_CommonData.yaml#/components/responses/429'
        '500':
          $ref: 'TS29571_CommonData.yaml#/components/responses/500'
        '502':
          $ref: 'TS29571_CommonData.yaml#/components/responses/502'
        '503':
          $ref: 'TS29571_CommonData.yaml#/components/responses/503'
        default:
          description: Unexpected error
  /ue-contexts/{ueContextId}/ue-reachind:
    put:
      summary: Namf_MT EnableUEReachability service Operation
      tags:
        - ueReachInd (Document)
      operationId: EnableUeReachability
      security:
        - {}
        - oAuth2ClientCredentials:
            - namf-mt
```

```
      - oAuth2ClientCredentials:
          - namf-mt
          - namf-mt:ue-reachind
    parameters:
      - name: ueContextId
        in: path
        description: UE Context Identifier
        required: true
        schema:
          type: string
    requestBody:
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/EnableUeReachabilityReqData'
      required: true
    responses:
      '200':
        description: UE has become reachable as desired
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/EnableUeReachabilityRspData'
      '307':
        $ref: 'TS29571_CommonData.yaml#/components/responses/307'
      '308':
        $ref: 'TS29571_CommonData.yaml#/components/responses/308'
      '400':
        $ref: 'TS29571_CommonData.yaml#/components/responses/400'
      '401':
        $ref: 'TS29571_CommonData.yaml#/components/responses/401'
      '403':
        description: Forbidden
        content:
          application/problem+json:
            schema:
              $ref: '#/components/schemas/ProblemDetailsEnableUeReachability'
      '404':
        $ref: 'TS29571_CommonData.yaml#/components/responses/404'
      '409':
        $ref: 'TS29571_CommonData.yaml#/components/responses/409'
      '411':
        $ref: 'TS29571_CommonData.yaml#/components/responses/411'
      '413':
        $ref: 'TS29571_CommonData.yaml#/components/responses/413'
      '415':
        $ref: 'TS29571_CommonData.yaml#/components/responses/415'
      '429':
        $ref: 'TS29571_CommonData.yaml#/components/responses/429'
      '500':
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
      '502':
        $ref: 'TS29571_CommonData.yaml#/components/responses/502'
      '503':
        $ref: 'TS29571_CommonData.yaml#/components/responses/503'
      '504':
        description: Gateway Timeout
        content:
          application/problem+json:
            schema:
              $ref: '#/components/schemas/ProblemDetailsEnableUeReachability'
      default:
        description: Unexpected error

  /ue-contexts/enable-group-reachability:
    post:
      summary: Namf_MT EnableGroupReachability service Operation
      tags:
        - ueContexts (collection)
      operationId: EnableGroupReachability
      security:
        - {}
        - oAuth2ClientCredentials:
            - namf-mt
        - oAuth2ClientCredentials:
            - namf-mt
            - namf-mt:enable-group-reachability
      requestBody:
```

```
    description: list of UEs requested to be made reachable for the related TMGI
    content:
      application/json:
        schema:
          $ref: '#/components/schemas/EnableGroupReachabilityReqData'
    required: true
  responses:
    '200':
      description: Successful response.
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/EnableGroupReachabilityRspData'
    '307':
      $ref: 'TS29571_CommonData.yaml#/components/responses/307'
    '308':
      $ref: 'TS29571_CommonData.yaml#/components/responses/308'
    '400':
      $ref: 'TS29571_CommonData.yaml#/components/responses/400'
    '401':
      $ref: 'TS29571_CommonData.yaml#/components/responses/401'
    '403':
      $ref: 'TS29571_CommonData.yaml#/components/responses/403'
    '404':
      $ref: 'TS29571_CommonData.yaml#/components/responses/404'
    '411':
      $ref: 'TS29571_CommonData.yaml#/components/responses/411'
    '413':
      $ref: 'TS29571_CommonData.yaml#/components/responses/413'
    '415':
      $ref: 'TS29571_CommonData.yaml#/components/responses/415'
    '429':
      $ref: 'TS29571_CommonData.yaml#/components/responses/429'
    '500':
      $ref: 'TS29571_CommonData.yaml#/components/responses/500'
    '502':
      $ref: 'TS29571_CommonData.yaml#/components/responses/502'
    '503':
      $ref: 'TS29571_CommonData.yaml#/components/responses/503'
    '504':
      $ref: 'TS29571_CommonData.yaml#/components/responses/504'
    default:
      description: Unexpected error
  callbacks:
    reachabilityNotification:
      '{$request.body#/reachabilityNotifyUri}':
        post:
          requestBody:
            required: true
            content:
              application/json:
                schema:
                  $ref: '#/components/schemas/ReachabilityNotificationData'
          responses:
            '204':
              description: UE reachability notification response
            '307':
              $ref: 'TS29571_CommonData.yaml#/components/responses/307'
            '308':
              $ref: 'TS29571_CommonData.yaml#/components/responses/308'
            '400':
              $ref: 'TS29571_CommonData.yaml#/components/responses/400'
            '401':
              $ref: 'TS29571_CommonData.yaml#/components/responses/401'
            '403':
              $ref: 'TS29571_CommonData.yaml#/components/responses/403'
            '404':
              $ref: 'TS29571_CommonData.yaml#/components/responses/404'
            '411':
              $ref: 'TS29571_CommonData.yaml#/components/responses/411'
            '413':
              $ref: 'TS29571_CommonData.yaml#/components/responses/413'
            '415':
              $ref: 'TS29571_CommonData.yaml#/components/responses/415'
            '429':
              $ref: 'TS29571_CommonData.yaml#/components/responses/429'
            '500':
              $ref: 'TS29571_CommonData.yaml#/components/responses/500'
```

```
                '502':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/502'
                '503':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/503'
                default:
                  description: Unexpected error

components:
  securitySchemes:
    oAuth2ClientCredentials:
      type: oauth2
      flows:
        clientCredentials:
          tokenUrl: '{nrfApiRoot}/oauth2/token'
          scopes:
            namf-mt: Access to the Namf_MT API
            namf-mt:ue-reachind: >
              Access to the EnableUeReachability service operation
            namf-mt:enable-group-reachability: >
              Access to the EnableGroupReachability service operation

  schemas:
    EnableUeReachabilityReqData:
      description: Data within the Enable UE Reachability Request
      type: object
      properties:
        reachability:
          $ref: 'TS29518_Namf_EventExposure.yaml#/components/schemas/UeReachability'
        supportedFeatures:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/SupportedFeatures'
        oldGuami:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Guami'
        extBufSupport:
          type: boolean
          default: false
        qosFlowInfoList:
          type: array
          items:
            $ref: '#/components/schemas/QosFlowInfo'
          minItems: 1
        pduSessionId:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/PduSessionId'
      required:
        - reachability
    EnableUeReachabilityRspData:
      description: Data within the Enable UE Reachability Response
      type: object
      properties:
        reachability:
          $ref: 'TS29518_Namf_EventExposure.yaml#/components/schemas/UeReachability'
        supportedFeatures:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/SupportedFeatures'
      required:
        - reachability
    UeContextInfo:
      description: UE Context Information
      type: object
      properties:
        supportVoPS:
          type: boolean
        supportVoPSn3gpp:
          type: boolean
        lastActTime:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/DateTime'
        accessType:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/AccessType'
        ratType:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/RatType'
        supportedFeatures:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/SupportedFeatures'
    ProblemDetailsEnableUeReachability:
      description: Enable UE Reachability Error Detail
      allOf:
      - $ref: 'TS29571_CommonData.yaml#/components/schemas/ProblemDetails'
      - $ref: '#/components/schemas/AdditionInfoEnableUeReachability'
    AdditionInfoEnableUeReachability:
      description: Additional information to be returned in EnableUeReachability error response
      type: object
```

```
    properties:
      maxWaitingTime:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/DurationSec'
  EnableGroupReachabilityReqData:
    description: Data within the Enable Group Reachability Request
    type: object
    properties:
      ueInfoList:
        type: array
        items:
          $ref: '#/components/schemas/UeInfo'
        minItems: 1
      tmgi:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Tmgi'
      reachabilityNotifyUri:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Uri'
      mbsServiceArea:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/MbsServiceArea'
      mbsServiceAreaInfoList:
        type: array
        items:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/MbsServiceAreaInfo'
        minItems: 1
      arp:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Arp'
      5qi:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/5Qi'
      supportedFeatures:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/SupportedFeatures'
    required:
      - ueInfoList
      - tmgi
  EnableGroupReachabilityRspData:
    description: Data within the Enable Group Reachability Response
    type: object
    properties:
      ueConnectedList:
        type: array
        items:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Supi'
        minItems: 1
      supportedFeatures:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/SupportedFeatures'

  UeInfo:
    description: list of UEs requested to be made reachable for the MBS Session
    type: object
    properties:
      ueList:
        type: array
        items:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Supi'
        minItems: 1
      pduSessionId:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/PduSessionId'
    required:
      - ueList

  ReachabilityNotificationData:
    description: Data within the UE Reachability Info Notify
    type: object
    properties:
      reachableUeList:
        type: array
        items:
          $ref: '#/components/schemas/ReachableUeInfo'
        minItems: 1
      unreachableUeList:
        type: array
        items:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Supi'
        minItems: 1
  ReachableUeInfo:
    description: Contains the reachable UE Information
    type: object
    properties:
      ueList:
        type: array
```

```
          items:
            $ref: 'TS29571_CommonData.yaml#/components/schemas/Supi'
          minItems: 1
        userLocation:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/UserLocation'
      required:
        - ueList

    QosFlowInfo:
      description: QOS Flow information
      type: object
      properties:
        qfi:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Qfi'
        ppi:
          $ref: 'TS29518_Namf_Communication.yaml#/components/schemas/Ppi'
        arp:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Arp'
        5qi:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/5Qi'
        dlDataSize:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Uint16'
      required:
        - qfi

    UeContextInfoClass:
      description: Indicates the UE Context information class
      anyOf:
      - type: string
        enum:
          - TADS
      - type: string
```

# A.5 Namf_Location

```
openapi: 3.0.0

info:
  version: 1.3.1
  title: Namf_Location
  description: |
    AMF Location Service.
    © 2025, 3GPP Organizational Partners (ARIB, ATIS, CCSA, ETSI, TSDSI, TTA, TTC).
    All rights reserved.

security:
  - {}
  - oAuth2ClientCredentials:
      - namf-loc

externalDocs:
  description: 3GPP TS 29.518 V18.9.0; 5G System; Access and Mobility Management Services
  url: 'https://www.3gpp.org/ftp/Specs/archive/29_series/29.518/'

servers:
  - url: '{apiRoot}/namf-loc/v1'
    variables:
      apiRoot:
        default: https://example.com
        description: apiRoot as defined in clause 4.4 of 3GPP TS 29.501

paths:
  /{ueContextId}/provide-pos-info:
    post:
      summary: Namf_Location ProvidePositioningInfo service Operation
      tags:
        - Individual UE context (Document)
      operationId: ProvidePositioningInfo
      parameters:
        - name: ueContextId
          in: path
          description: UE Context Identifier
          required: true
          schema:
            type: string
```

```
            pattern: '^(imsi-[0-9]{5,15}|nai-.+|gli-.+|gci-.+|imei-[0-9]{15}|imeisv-[0-
9]{16}|msisdn-[0-9]{5,15}|extid-.+@.+|.+)$'
      requestBody:
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/RequestPosInfo'
        required: true
      responses:
        '200':
          description: Expected response to a valid request
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/ProvidePosInfoExt'
        '204':
           description: Successful accept of location request with no information returned.
        '307':
          $ref: 'TS29571_CommonData.yaml#/components/responses/307'
        '308':
          $ref: 'TS29571_CommonData.yaml#/components/responses/308'
        '400':
          $ref: 'TS29571_CommonData.yaml#/components/responses/400'
        '401':
          $ref: 'TS29571_CommonData.yaml#/components/responses/401'
        '403':
          $ref: 'TS29571_CommonData.yaml#/components/responses/403'
        '404':
          $ref: 'TS29571_CommonData.yaml#/components/responses/404'
        '409':
          description: Conflict
          content:
            application/problem+json:
              schema:
                $ref: '#/components/schemas/ProblemDetailsProvidePosInfo'
        '411':
          $ref: 'TS29571_CommonData.yaml#/components/responses/411'
        '413':
          $ref: 'TS29571_CommonData.yaml#/components/responses/413'
        '415':
          $ref: 'TS29571_CommonData.yaml#/components/responses/415'
        '429':
          $ref: 'TS29571_CommonData.yaml#/components/responses/429'
        '500':
          $ref: 'TS29571_CommonData.yaml#/components/responses/500'
        '502':
          $ref: 'TS29571_CommonData.yaml#/components/responses/502'
        '503':
          $ref: 'TS29571_CommonData.yaml#/components/responses/503'
        '504':
          $ref: 'TS29571_CommonData.yaml#/components/responses/504'
        default:
          description: Unexpected error
      callbacks:
        onUELocationNotification:
          '{$request.body#/locationNotificationUri}':
            post:
              requestBody:
                description: UE Location Event Notification
                content:
                  application/json:
                    schema:
                      $ref: '#/components/schemas/NotifiedPosInfoExt'
              responses:
                '204':
                  description: Expected response to a successful callback processing
                '307':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/307'
                '308':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/308'
                '400':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/400'
                '401':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/401'
                '403':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/403'
                '404':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/404'
```

```
              '411':
                $ref: 'TS29571_CommonData.yaml#/components/responses/411'
              '413':
                $ref: 'TS29571_CommonData.yaml#/components/responses/413'
              '415':
                $ref: 'TS29571_CommonData.yaml#/components/responses/415'
              '429':
                $ref: 'TS29571_CommonData.yaml#/components/responses/429'
              '500':
                $ref: 'TS29571_CommonData.yaml#/components/responses/500'
              '502':
                $ref: 'TS29571_CommonData.yaml#/components/responses/502'
              '503':
                $ref: 'TS29571_CommonData.yaml#/components/responses/503'
  /{ueContextId}/provide-loc-info:
    post:
      summary: Namf_Location ProvideLocationInfo service Operation
      tags:
        - Individual UE context (Document)
      operationId: ProvideLocationInfo
      parameters:
        - name: ueContextId
          in: path
          description: UE Context Identifier
          required: true
          schema:
            type: string
            pattern: '^(imsi-[0-9]{5,15}|nai-.+|gli-.+|gci-.+|imei-[0-9]{15}|imeisv-[0-9]{16}|.+)$'
      requestBody:
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/RequestLocInfo'
        required: true
      responses:
        '200':
          description: Expected response to a valid request
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/ProvideLocInfo'
        '307':
          $ref: 'TS29571_CommonData.yaml#/components/responses/307'
        '308':
          $ref: 'TS29571_CommonData.yaml#/components/responses/308'
        '400':
          $ref: 'TS29571_CommonData.yaml#/components/responses/400'
        '401':
          $ref: 'TS29571_CommonData.yaml#/components/responses/401'
        '403':
          $ref: 'TS29571_CommonData.yaml#/components/responses/403'
        '404':
          $ref: 'TS29571_CommonData.yaml#/components/responses/404'
        '411':
          $ref: 'TS29571_CommonData.yaml#/components/responses/411'
        '413':
          $ref: 'TS29571_CommonData.yaml#/components/responses/413'
        '415':
          $ref: 'TS29571_CommonData.yaml#/components/responses/415'
        '429':
          $ref: 'TS29571_CommonData.yaml#/components/responses/429'
        '500':
          $ref: 'TS29571_CommonData.yaml#/components/responses/500'
        '502':
          $ref: 'TS29571_CommonData.yaml#/components/responses/502'
        '503':
          $ref: 'TS29571_CommonData.yaml#/components/responses/503'
        default:
          description: Unexpected error

  /{ueContextId}/cancel-pos-info:
    post:
      summary: Namf_Location CancelLocation service operation
      tags:
        - Individual UE context (Document)
      operationId: CancelLocation
      parameters:
        - name: ueContextId
```

```
          in: path
          description: UE Context Identifier
          required: true
          schema:
            type: string
            pattern: '^(imsi-[0-9]{5,15}|nai-.+|gli-.+|gci-.+|imei-[0-9]{15}|imeisv-[0-9]{16}|.+)$'
      requestBody:
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/CancelPosInfo'
        required: true
      responses:
        '204':
          description: Expected response to a successful cancellation
        '307':
          $ref: 'TS29571_CommonData.yaml#/components/responses/307'
        '308':
          $ref: 'TS29571_CommonData.yaml#/components/responses/308'
        '400':
          $ref: 'TS29571_CommonData.yaml#/components/responses/400'
        '401':
          $ref: 'TS29571_CommonData.yaml#/components/responses/401'
        '403':
          $ref: 'TS29571_CommonData.yaml#/components/responses/403'
        '404':
          $ref: 'TS29571_CommonData.yaml#/components/responses/404'
        '411':
          $ref: 'TS29571_CommonData.yaml#/components/responses/411'
        '413':
          $ref: 'TS29571_CommonData.yaml#/components/responses/413'
        '415':
          $ref: 'TS29571_CommonData.yaml#/components/responses/415'
        '429':
          $ref: 'TS29571_CommonData.yaml#/components/responses/429'
        '500':
          $ref: 'TS29571_CommonData.yaml#/components/responses/500'
        '503':
          $ref: 'TS29571_CommonData.yaml#/components/responses/503'
        '502':
          $ref: 'TS29571_CommonData.yaml#/components/responses/502'
        '504':
          $ref: 'TS29571_CommonData.yaml#/components/responses/504'
        default:
          $ref: 'TS29571_CommonData.yaml#/components/responses/default'


components:
  securitySchemes:
    oAuth2ClientCredentials:
      type: oauth2
      flows:
        clientCredentials:
          tokenUrl: '{nrfApiRoot}/oauth2/token'
          scopes:
            namf-loc: Access to the Namf_Location API
  schemas:
    RequestPosInfo:
      description: Data within Provide Positioning Information Request
      type: object
      properties:
        lcsClientType:
          $ref: 'TS29572_Nlmf_Location.yaml#/components/schemas/ExternalClientType'
        lcsLocation:
          $ref: '#/components/schemas/LocationType'
        supi:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Supi'
        gpsi:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Gpsi'
        requestedRangingSlResult:
          type: array
          items:
            $ref: 'TS29572_Nlmf_Location.yaml#/components/schemas/RangingSlResult'
          minItems: 1
        relatedUes:
          type: array
          items:
            $ref: 'TS29572_Nlmf_Location.yaml#/components/schemas/RelatedUe'
```

```
    minItems: 1
  lmfId:
    $ref: 'TS29572_Nlmf_Location.yaml#/components/schemas/LMFIdentification'
  priority:
    $ref: 'TS29572_Nlmf_Location.yaml#/components/schemas/LcsPriority'
  lcsQoS:
    $ref: 'TS29572_Nlmf_Location.yaml#/components/schemas/LocationQoS'
  velocityRequested:
    $ref: 'TS29572_Nlmf_Location.yaml#/components/schemas/VelocityRequested'
  lcsSupportedGADShapes:
    $ref: 'TS29572_Nlmf_Location.yaml#/components/schemas/SupportedGADShapes'
  additionalLcsSuppGADShapes:
    type: array
    items:
      $ref: 'TS29572_Nlmf_Location.yaml#/components/schemas/SupportedGADShapes'
    minItems: 1
  locationNotificationUri:
    $ref: 'TS29571_CommonData.yaml#/components/schemas/Uri'
  supportedFeatures:
    $ref: 'TS29571_CommonData.yaml#/components/schemas/SupportedFeatures'
  oldGuami:
    $ref: 'TS29571_CommonData.yaml#/components/schemas/Guami'
  pei:
    $ref: 'TS29571_CommonData.yaml#/components/schemas/Pei'
  lcsServiceType:
    $ref: 'TS29572_Nlmf_Location.yaml#/components/schemas/LcsServiceType'
  ldrType:
    $ref: 'TS29572_Nlmf_Location.yaml#/components/schemas/LdrType'
  hgmlcCallBackURI:
    $ref: 'TS29571_CommonData.yaml#/components/schemas/Uri'
  lirGmlcCallBackUri:
    $ref: 'TS29571_CommonData.yaml#/components/schemas/Uri'
  ldrReference:
    $ref: 'TS29572_Nlmf_Location.yaml#/components/schemas/LdrReference'
  lirReference:
    $ref: 'TS29572_Nlmf_Location.yaml#/components/schemas/LirReference'
  periodicEventInfo:
    $ref: 'TS29572_Nlmf_Location.yaml#/components/schemas/PeriodicEventInfo'
  areaEventInfo:
    $ref: 'TS29572_Nlmf_Location.yaml#/components/schemas/AreaEventInfo'
  motionEventInfo:
    $ref: 'TS29572_Nlmf_Location.yaml#/components/schemas/MotionEventInfo'
  externalClientIdentification:
    $ref: 'TS29515_Ngmlc_Location.yaml#/components/schemas/ExternalClientIdentification'
  afID:
    $ref: 'TS29571_CommonData.yaml#/components/schemas/NfInstanceId'
  codeWord:
    $ref: 'TS29515_Ngmlc_Location.yaml#/components/schemas/CodeWord'
  uePrivacyRequirements:
    $ref: 'TS29515_Ngmlc_Location.yaml#/components/schemas/UePrivacyRequirements'
  scheduledLocTime:
    $ref: 'TS29571_CommonData.yaml#/components/schemas/DateTime'
  reliableLocReq:
    type: boolean
    default: false
  intermediateLocationInd:
    type: boolean
    default: false
  maxRespTime:
    $ref: 'TS29571_CommonData.yaml#/components/schemas/DurationSec'
  ueUnawareInd:
    type: boolean
    enum:
      - true
  lpHapType:
    $ref: '#/components/schemas/LpHapType'
  evtRptAllowedAreas:
    type: array
    items:
      $ref: 'TS29572_Nlmf_Location.yaml#/components/schemas/ReportingArea'
    minItems: 1
    maxItems: 250
  reportingInd:
    allOf:
      - $ref: 'TS29515_Ngmlc_Location.yaml#/components/schemas/ReportingInd'
    default: INSIDE_REPORTING
  integrityRequirements:
    $ref: 'TS29515_Ngmlc_Location.yaml#/components/schemas/IntegrityRequirements'
```

```
      upLocRepInfoAf:
        $ref: 'TS29515_Ngmlc_Location.yaml#/components/schemas/UpLocRepInfoAf'
      mappedQoSEps:
        $ref: 'TS29572_Nlmf_Location.yaml#/components/schemas/MappedLocationQoSEps'
      coordinateID:
        type: integer
    required:
      - lcsClientType
      - lcsLocation


  ProvidePosInfoExt:
    description: Extended provided positioning information for UEs
    allOf:
      - $ref: '#/components/schemas/ProvidePosInfo'
      - $ref: '#/components/schemas/AddProvidePosInfos'

  ProvidePosInfo:
    description: Data within Provide Positioning Information Response
    type: object
    properties:
      locationEstimate:
        $ref: 'TS29572_Nlmf_Location.yaml#/components/schemas/GeographicArea'
      localLocationEstimate:
        $ref: 'TS29572_Nlmf_Location.yaml#/components/schemas/LocalArea'
      accuracyFulfilmentIndicator:
        $ref: 'TS29572_Nlmf_Location.yaml#/components/schemas/AccuracyFulfilmentIndicator'
      ageOfLocationEstimate:
        $ref: 'TS29572_Nlmf_Location.yaml#/components/schemas/AgeOfLocationEstimate'
      timestampOfLocationEstimate:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/DateTime'
      velocityEstimate:
        $ref: 'TS29572_Nlmf_Location.yaml#/components/schemas/VelocityEstimate'
      positioningDataList:
        type: array
        items:
          $ref: 'TS29572_Nlmf_Location.yaml#/components/schemas/PositioningMethodAndUsage'
        minItems: 0
        maxItems: 9
      gnssPositioningDataList:
        type: array
        items:
          $ref: 'TS29572_Nlmf_Location.yaml#/components/schemas/GnssPositioningMethodAndUsage'
        minItems: 0
        maxItems: 9
      ecgi:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Ecgi'
      ncgi:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Ncgi'
      remoteUeInd:
        type: boolean
        enum:
          - true
      targetServingNode:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/NfInstanceId'
      targetMmeName:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/DiameterIdentity'
      targetMmeRealm:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/DiameterIdentity'
      utranSrvccInd:
        type: boolean
      civicAddress:
        $ref: 'TS29572_Nlmf_Location.yaml#/components/schemas/CivicAddress'
      barometricPressure:
        $ref: 'TS29572_Nlmf_Location.yaml#/components/schemas/BarometricPressure'
      altitude:
        $ref: 'TS29572_Nlmf_Location.yaml#/components/schemas/Altitude'
      supportedFeatures:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/SupportedFeatures'
      servingLMFIdentification:
        $ref: 'TS29572_Nlmf_Location.yaml#/components/schemas/LMFIdentification'
      locationPrivacyVerResult:
        $ref: '#/components/schemas/LocationPrivacyVerResult'
      achievedQos:
        $ref: 'TS29572_Nlmf_Location.yaml#/components/schemas/MinorLocationQoS'
      directReportInd:
        type: boolean
        default: false
```

```
      acceptedPeriodicEventInfo:
        $ref: 'TS29572_Nlmf_Location.yaml#/components/schemas/PeriodicEventInfo'
      haGnssMetrics:
        $ref: 'TS29572_Nlmf_Location.yaml#/components/schemas/HighAccuracyGnssMetrics'
      indoorOutdoorInd:
        $ref: 'TS29572_Nlmf_Location.yaml#/components/schemas/IndoorOutdoorInd'
      losNlosMeasureInd:
        $ref: 'TS29572_Nlmf_Location.yaml#/components/schemas/LosNlosMeasureInd'
      relatedApplicationlayerId:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/ApplicationlayerId'
      distanceDirection:
        $ref: 'TS29572_Nlmf_Location.yaml#/components/schemas/RangeDirection'
      2dRelativeLocation:
        $ref: 'TS29572_Nlmf_Location.yaml#/components/schemas/2DRelativeLocation'
      3dRelativeLocation:
        $ref: 'TS29572_Nlmf_Location.yaml#/components/schemas/3DRelativeLocation'
      relativeVelocity:
        $ref: 'TS29572_Nlmf_Location.yaml#/components/schemas/VelocityEstimate'
      integrityResult:
        $ref: 'TS29515_Ngmlc_Location.yaml#/components/schemas/IntegrityResult'


    NotifiedPosInfoExt:
      description: Extended notified positioning information for UEs
      allOf:
        - $ref: '#/components/schemas/NotifiedPosInfo'
        - $ref: '#/components/schemas/AddNotifiedPosInfos'

    NotifiedPosInfo:
      description: Data within EventNotify notification
      type: object
      properties:
        locationEvent:
          $ref: '#/components/schemas/LocationEvent'
        supi:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Supi'
        gpsi:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Gpsi'
        pei:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Pei'
        locationEstimate:
          $ref: 'TS29572_Nlmf_Location.yaml#/components/schemas/GeographicArea'
        localLocationEstimate:
          $ref: 'TS29572_Nlmf_Location.yaml#/components/schemas/LocalArea'
        ageOfLocationEstimate:
          $ref: 'TS29572_Nlmf_Location.yaml#/components/schemas/AgeOfLocationEstimate'
        timestampOfLocationEstimate:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/DateTime'
        velocityEstimate:
          $ref: 'TS29572_Nlmf_Location.yaml#/components/schemas/VelocityEstimate'
        positioningDataList:
          type: array
          items:
            $ref: 'TS29572_Nlmf_Location.yaml#/components/schemas/PositioningMethodAndUsage'
          minItems: 0
          maxItems: 9
        gnssPositioningDataList:
          type: array
          items:
            $ref: 'TS29572_Nlmf_Location.yaml#/components/schemas/GnssPositioningMethodAndUsage'
          minItems: 0
          maxItems: 9
        ecgi:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Ecgi'
        ncgi:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Ncgi'
        remoteUeInd:
          type: boolean
          enum:
            - true
        servingNode:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/NfInstanceId'
        targetMmeName:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/DiameterIdentity'
        targetMmeRealm:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/DiameterIdentity'
        utranSrvccInd:
          type: boolean
```

```
      civicAddress:
        $ref: 'TS29572_Nlmf_Location.yaml#/components/schemas/CivicAddress'
      barometricPressure:
        $ref: 'TS29572_Nlmf_Location.yaml#/components/schemas/BarometricPressure'
      altitude:
        $ref: 'TS29572_Nlmf_Location.yaml#/components/schemas/Altitude'
      hgmlcCallBackURI:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Uri'
      ldrReference:
        $ref: 'TS29572_Nlmf_Location.yaml#/components/schemas/LdrReference'
      servingLMFIdentification:
        $ref: 'TS29572_Nlmf_Location.yaml#/components/schemas/LMFIdentification'
      terminationCause:
        $ref: 'TS29572_Nlmf_Location.yaml#/components/schemas/TerminationCause'
      achievedQos:
        $ref: 'TS29572_Nlmf_Location.yaml#/components/schemas/MinorLocationQoS'
      mscServerId:
        $ref: 'TS29503_Nudm_UECM.yaml#/components/schemas/E164Number'
      haGnssMetrics:
        $ref: 'TS29572_Nlmf_Location.yaml#/components/schemas/HighAccuracyGnssMetrics'
      indoorOutdoorInd:
        $ref: 'TS29572_Nlmf_Location.yaml#/components/schemas/IndoorOutdoorInd'
      losNlosMeasureInd:
        $ref: 'TS29572_Nlmf_Location.yaml#/components/schemas/LosNlosMeasureInd'
      relatedApplicationlayerId:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/ApplicationlayerId'
      distanceDirection:
        $ref: 'TS29572_Nlmf_Location.yaml#/components/schemas/RangeDirection'
      2dRelativeLocation:
        $ref: 'TS29572_Nlmf_Location.yaml#/components/schemas/2DRelativeLocation'
      3dRelativeLocation:
        $ref: 'TS29572_Nlmf_Location.yaml#/components/schemas/3DRelativeLocation'
      relativeVelocity:
        $ref: 'TS29572_Nlmf_Location.yaml#/components/schemas/VelocityEstimate'
      integrityResult:
        $ref: 'TS29515_Ngmlc_Location.yaml#/components/schemas/IntegrityResult'
    required:
      - locationEvent


  RequestLocInfo:
    description: Data within Provide Location Information Request
    type: object
    properties:
      req5gsLoc:
        type: boolean
        default: false
      reqCurrentLoc:
        type: boolean
        default: false
      reqRatType:
        type: boolean
        default: false
      reqTimeZone:
        type: boolean
        default: false
      supportedFeatures:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/SupportedFeatures'

  ProvideLocInfo:
    description: Data within Provide Location Information Response
    type: object
    properties:
      currentLoc:
        type: boolean
      location:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/UserLocation'
      additionalLocation:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/UserLocation'
      geoInfo:
        $ref: 'TS29572_Nlmf_Location.yaml#/components/schemas/GeographicArea'
      locationAge:
        $ref: 'TS29572_Nlmf_Location.yaml#/components/schemas/AgeOfLocationEstimate'
      ratType:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/RatType'
      timezone:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/TimeZone'
      supportedFeatures:
```

```
      $ref: 'TS29571_CommonData.yaml#/components/schemas/SupportedFeatures'
    oldGuami:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/Guami'

CancelPosInfo:
  description: Data within a Cancel Location Request
  type: object
  properties:
    supi:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/Supi'
    hgmlcCallBackURI:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/Uri'
    ldrReference:
      $ref: 'TS29572_Nlmf_Location.yaml#/components/schemas/LdrReference'
    servingLMFIdentification:
      $ref: 'TS29572_Nlmf_Location.yaml#/components/schemas/LMFIdentification'
    supportedFeatures:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/SupportedFeatures'
  required:
    - supi
    - hgmlcCallBackURI
    - ldrReference

ProblemDetailsProvidePosInfo:
  description: Handover from 5GS to EPS Error Details.
  allOf:
  - $ref: 'TS29571_CommonData.yaml#/components/schemas/ProblemDetails'
  - $ref: '#/components/schemas/ProvidePosInfo'

AddNotifiedPosInfos:
  type: object
  properties:
    addProvidePosInfos:
      type: array
      items:
        $ref: '#/components/schemas/NotifiedPosInfo'
      minItems: 1

AddProvidePosInfos:
  type: object
  properties:
    addProvidePosInfos:
      type: array
      items:
        $ref: '#/components/schemas/ProvidePosInfo'
      minItems: 1


LocationType:
  description: Type of location measurement requested
  anyOf:
  - type: string
    enum:
      - CURRENT_LOCATION
      - CURRENT_OR_LAST_KNOWN_LOCATION
      - NOTIFICATION_VERIFICATION_ONLY
      - DEFERRED_LOCATION
  - type: string

LocationEvent:
  description: Type of events initiating location procedures
  anyOf:
  - type: string
    enum:
      - EMERGENCY_CALL_ORIGINATION
      - EMERGENCY_CALL_RELEASE
      - EMERGENCY_CALL_HANDOVER
      - ACTIVATION_OF_DEFERRED_LOCATION
      - UE_MOBILITY_FOR_DEFERRED_LOCATION
      - CANCELLATION_OF_DEFERRED_LOCATION
  - type: string

LocationPrivacyVerResult:
  description: >
    The result of location privacy verification or ranging and sidelink positioning privacy
    verification by UE
  anyOf:
  - type: string
```

```
        enum:
          - LOCATION_ALLOWED
          - LOCATION_NOT_ALLOWED
          - RESPONSE_TIME_OUT
          - RANGING_ALLOWED
          - RANGING_NOT_ALLOWED
      - type: string


    LpHapType:
      description: Type of Low Power and/or High Accuracy Positioning
      anyOf:
      - type: string
        enum:
          - LOW_POW_HIGH_ACCU_POS
      - type: string
```

# A.6 Namf_MBSBroadcast API

```
openapi: 3.0.0

info:
  version: 1.1.0
  title: Namf_MBSBroadcast
  description: |
    AMF MBSBroadcast Service.
    © 2024, 3GPP Organizational Partners (ARIB, ATIS, CCSA, ETSI, TSDSI, TTA, TTC).
    All rights reserved.

externalDocs:
  description: 3GPP TS 29.518 V18.6.0; 5G System; Access and Mobility Management Services
  url: 'https://www.3gpp.org/ftp/Specs/archive/29_series/29.518/'

servers:
  - url: '{apiRoot}/namf-mbs-bc/v1'
    variables:
      apiRoot:
        default: https://example.com
        description: apiRoot as defined in clause 4.4 of 3GPP TS 29.501

security:
  - {}
  - oAuth2ClientCredentials:
      - namf-mbs-bc

paths:
  /mbs-contexts:
    post:
      summary: Namf_MBSBroadcast ContextCreate service Operation
      tags:
        - Broadcast MBS session contexts collection (Collection)
      operationId: ContextCreate
      requestBody:
        content:
          multipart/related:  # message with binary body part(s)
            schema:
              type: object
              properties: # Request parts
                jsonData:
                  $ref: '#/components/schemas/ContextCreateReqData'
                binaryDataN2Information:
                  type: string
                  format: binary
            encoding:
              jsonData:
                contentType:  application/json
              binaryDataN2Information:
                contentType:  application/vnd.3gpp.ngap
                headers:
                  Content-Id:
                    schema:
                      type: string
        required: true
      callbacks:
        contextStatusNotification:
          '{$request.body#/notifyUri}':
```

```
post:
  requestBody:  # notification request without binary body part
    required: true
    content:
      application/json:
        schema:
          $ref: '#/components/schemas/ContextStatusNotification'
      multipart/related:  # notification request with binary body part(s)
        schema:
          type: object
          properties: # Request parts
            jsonData:
              $ref: '#/components/schemas/ContextStatusNotification'
            binaryDataN2Information1:
              type: string
              format: binary
            binaryDataN2Information2:
              type: string
              format: binary
            binaryDataN2Information3:
              type: string
              format: binary
            binaryDataN2Information4:
              type: string
              format: binary
            binaryDataN2Information5:
              type: string
              format: binary
            binaryDataN2Information6:
              type: string
              format: binary
            binaryDataN2Information7:
              type: string
              format: binary
            binaryDataN2Information8:
              type: string
              format: binary
            binaryDataN2Information9:
              type: string
              format: binary
            binaryDataN2Information10:
              type: string
              format: binary
        encoding:
          jsonData:
            contentType:  application/json
          binaryDataN2Information1:
            contentType:  application/vnd.3gpp.ngap
            headers:
              Content-Id:
                schema:
                  type: string
          binaryDataN2Information2:
            contentType:  application/vnd.3gpp.ngap
            headers:
              Content-Id:
                schema:
                  type: string
          binaryDataN2Information3:
            contentType:  application/vnd.3gpp.ngap
            headers:
              Content-Id:
                schema:
                  type: string
          binaryDataN2Information4:
            contentType:  application/vnd.3gpp.ngap
            headers:
              Content-Id:
                schema:
                  type: string
          binaryDataN2Information5:
            contentType:  application/vnd.3gpp.ngap
            headers:
              Content-Id:
                schema:
                  type: string
          binaryDataN2Information6:
            contentType:  application/vnd.3gpp.ngap
```

```
              headers:
                Content-Id:
                  schema:
                    type: string
            binaryDataN2Information7:
              contentType:  application/vnd.3gpp.ngap
              headers:
                Content-Id:
                  schema:
                    type: string
            binaryDataN2Information8:
              contentType:  application/vnd.3gpp.ngap
              headers:
                Content-Id:
                  schema:
                    type: string
            binaryDataN2Information9:
              contentType:  application/vnd.3gpp.ngap
              headers:
                Content-Id:
                  schema:
                    type: string
            binaryDataN2Information10:
              contentType:  application/vnd.3gpp.ngap
              headers:
                Content-Id:
                  schema:
                    type: string
responses:
  '200':
    description: successful notification response with content
    content:
      multipart/related:  # notification response with binary body part(s)
        schema:
          type: object
          properties:
            jsonData:
              $ref: '#/components/schemas/ContextStatusNotificationResponse'
            binaryDataN2Information1:
              type: string
              format: binary
            binaryDataN2Information2:
              type: string
              format: binary
            binaryDataN2Information3:
              type: string
              format: binary
            binaryDataN2Information4:
              type: string
              format: binary
            binaryDataN2Information5:
              type: string
              format: binary
            binaryDataN2Information6:
              type: string
              format: binary
            binaryDataN2Information7:
              type: string
              format: binary
            binaryDataN2Information8:
              type: string
              format: binary
            binaryDataN2Information9:
              type: string
              format: binary
            binaryDataN2Information10:
              type: string
              format: binary
          encoding:
            jsonData:
              contentType:  application/json
            binaryDataN2Information1:
              contentType:  application/vnd.3gpp.ngap
              headers:
                Content-Id:
                  schema:
                    type: string
            binaryDataN2Information2:
```

```
              contentType:  application/vnd.3gpp.ngap
              headers:
                Content-Id:
                  schema:
                    type: string
            binaryDataN2Information3:
              contentType:  application/vnd.3gpp.ngap
              headers:
                Content-Id:
                  schema:
                    type: string
            binaryDataN2Information4:
              contentType:  application/vnd.3gpp.ngap
              headers:
                Content-Id:
                  schema:
                    type: string
            binaryDataN2Information5:
              contentType:  application/vnd.3gpp.ngap
              headers:
                Content-Id:
                  schema:
                    type: string
            binaryDataN2Information6:
              contentType:  application/vnd.3gpp.ngap
              headers:
                Content-Id:
                  schema:
                    type: string
            binaryDataN2Information7:
              contentType:  application/vnd.3gpp.ngap
              headers:
                Content-Id:
                  schema:
                    type: string
            binaryDataN2Information8:
              contentType:  application/vnd.3gpp.ngap
              headers:
                Content-Id:
                  schema:
                    type: string
            binaryDataN2Information9:
              contentType:  application/vnd.3gpp.ngap
              headers:
                Content-Id:
                  schema:
                    type: string
            binaryDataN2Information10:
              contentType:  application/vnd.3gpp.ngap
              headers:
                Content-Id:
                  schema:
                    type: string
        '204':
          description: successful notification
        '307':
          $ref: 'TS29571_CommonData.yaml#/components/responses/307'
        '308':
          $ref: 'TS29571_CommonData.yaml#/components/responses/308'
        '400':
          $ref: 'TS29571_CommonData.yaml#/components/responses/400'
        '401':
           $ref: 'TS29571_CommonData.yaml#/components/responses/401'
        '403':
          $ref: 'TS29571_CommonData.yaml#/components/responses/403'
        '404':
          $ref: 'TS29571_CommonData.yaml#/components/responses/404'
        '411':
          $ref: 'TS29571_CommonData.yaml#/components/responses/411'
        '413':
          $ref: 'TS29571_CommonData.yaml#/components/responses/413'
        '415':
          $ref: 'TS29571_CommonData.yaml#/components/responses/415'
        '429':
          $ref: 'TS29571_CommonData.yaml#/components/responses/429'
        '500':
          $ref: 'TS29571_CommonData.yaml#/components/responses/500'
        '502':
```

```
                $ref: 'TS29571_CommonData.yaml#/components/responses/502'
              '503':
                $ref: 'TS29571_CommonData.yaml#/components/responses/503'
      responses:
        '201':
          description: MBS Broadcast context created successfully
          headers:
            Location:
              description: 'Contains the URI of the newly created resource, according to the
structure: {apiRoot}/namf-mbs-bc/<apiVersion>/mbs-contexts/{mbsContextRef}'
              required: true
              schema:
                type: string
          content:
            application/json:  # message without binary body part(s)
              schema:
                $ref: '#/components/schemas/ContextCreateRspData'
            multipart/related:   # message with binary body part(s)
              schema:
                type: object
                properties: # Response parts
                  jsonData:
                    $ref: '#/components/schemas/ContextCreateRspData'
                  binaryDataN2Information1:
                    type: string
                    format: binary
                  binaryDataN2Information2:
                    type: string
                    format: binary
                  binaryDataN2Information3:
                    type: string
                    format: binary
                  binaryDataN2Information4:
                    type: string
                    format: binary
                  binaryDataN2Information5:
                    type: string
                    format: binary
                  binaryDataN2Information6:
                    type: string
                    format: binary
                  binaryDataN2Information7:
                    type: string
                    format: binary
                  binaryDataN2Information8:
                    type: string
                    format: binary
                  binaryDataN2Information9:
                    type: string
                    format: binary
                  binaryDataN2Information10:
                    type: string
                    format: binary
              encoding:
                jsonData:
                  contentType:  application/json
                binaryDataN2Information1:
                  contentType:  application/vnd.3gpp.ngap
                  headers:
                    Content-Id:
                      schema:
                        type: string
                binaryDataN2Information2:
                  contentType:  application/vnd.3gpp.ngap
                  headers:
                    Content-Id:
                      schema:
                        type: string
                binaryDataN2Information3:
                  contentType:  application/vnd.3gpp.ngap
                  headers:
                    Content-Id:
                      schema:
                        type: string
                binaryDataN2Information4:
                  contentType:  application/vnd.3gpp.ngap
                  headers:
                    Content-Id:
```

```
                    schema:
                      type: string
                binaryDataN2Information5:
                  contentType:  application/vnd.3gpp.ngap
                  headers:
                    Content-Id:
                      schema:
                        type: string
                binaryDataN2Information6:
                  contentType:  application/vnd.3gpp.ngap
                  headers:
                    Content-Id:
                      schema:
                        type: string
                binaryDataN2Information7:
                  contentType:  application/vnd.3gpp.ngap
                  headers:
                    Content-Id:
                      schema:
                        type: string
                binaryDataN2Information8:
                  contentType:  application/vnd.3gpp.ngap
                  headers:
                    Content-Id:
                      schema:
                        type: string
                binaryDataN2Information9:
                  contentType:  application/vnd.3gpp.ngap
                  headers:
                    Content-Id:
                      schema:
                        type: string
                binaryDataN2Information10:
                  contentType:  application/vnd.3gpp.ngap
                  headers:
                    Content-Id:
                      schema:
                        type: string
        '307':
          $ref: 'TS29571_CommonData.yaml#/components/responses/307'
        '308':
          $ref: 'TS29571_CommonData.yaml#/components/responses/308'
        '400':
          $ref: 'TS29571_CommonData.yaml#/components/responses/400'
        '401':
          $ref: 'TS29571_CommonData.yaml#/components/responses/401'
        '403':
          $ref: 'TS29571_CommonData.yaml#/components/responses/403'
        '404':
          $ref: 'TS29571_CommonData.yaml#/components/responses/404'
        '411':
          $ref: 'TS29571_CommonData.yaml#/components/responses/411'
        '413':
          $ref: 'TS29571_CommonData.yaml#/components/responses/413'
        '415':
          $ref: 'TS29571_CommonData.yaml#/components/responses/415'
        '429':
          $ref: 'TS29571_CommonData.yaml#/components/responses/429'
        '500':
          $ref: 'TS29571_CommonData.yaml#/components/responses/500'
        '502':
          $ref: 'TS29571_CommonData.yaml#/components/responses/502'
        '503':
          $ref: 'TS29571_CommonData.yaml#/components/responses/503'
        default:
          $ref: 'TS29571_CommonData.yaml#/components/responses/default'

  /mbs-contexts/{mbsContextRef}:
    delete:
      summary: Namf_MBSBroadcast ContextDelete service Operation
      tags:
        - Individual broadcast MBS session context (Document)
      operationId: ContextDelete
      parameters:
        - name: mbsContextRef
          in: path
          required: true
          description: Unique ID of the broadcast MSB session context to be deleted
```

```
        schema:
          type: string
    responses:
      '204':
        description: successful deletion
      '307':
        $ref: 'TS29571_CommonData.yaml#/components/responses/307'
      '308':
        $ref: 'TS29571_CommonData.yaml#/components/responses/308'
      '400':
        $ref: 'TS29571_CommonData.yaml#/components/responses/400'
      '401':
        $ref: 'TS29571_CommonData.yaml#/components/responses/401'
      '403':
        $ref: 'TS29571_CommonData.yaml#/components/responses/403'
      '404':
        $ref: 'TS29571_CommonData.yaml#/components/responses/404'
      '429':
        $ref: 'TS29571_CommonData.yaml#/components/responses/429'
      '500':
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
      '502':
        $ref: 'TS29571_CommonData.yaml#/components/responses/502'
      '503':
        $ref: 'TS29571_CommonData.yaml#/components/responses/503'
      default:
        $ref: 'TS29571_CommonData.yaml#/components/responses/default'

/mbs-contexts/{mbsContextRef}/update:
  post:
    summary: Namf_MBSBroadcast ContextUpdate service Operation
    tags:
      - Individual broadcast MBS session context (Document)
    operationId: ContextUpdate
    parameters:
      - name: mbsContextRef
        in: path
        description: Unique ID of the broadcast MSB session context to be updated
        required: true
        schema:
          type: string
    requestBody:
      content:
        multipart/related:  # message with binary body part(s)
          schema:
            type: object
            properties: # Request parts
              jsonData:
                $ref: '#/components/schemas/ContextUpdateReqData'
              binaryDataN2Information:
                type: string
                format: binary
          encoding:
            jsonData:
              contentType:  application/json
            binaryDataN2Information:
              contentType:  application/vnd.3gpp.ngap
              headers:
                Content-Id:
                  schema:
                    type: string
      required: true
    responses:
      '200':
        description: MBS Broadcast context updated successfully
        content:
          application/json:  # message without binary body part(s)
            schema:
              $ref: '#/components/schemas/ContextUpdateRspData'
          multipart/related:   # message with binary body part(s)
            schema:
              type: object
              properties: # Response parts
                jsonData:
                  $ref: '#/components/schemas/ContextUpdateRspData'
                binaryDataN2Information1:
                  type: string
                  format: binary
```

```
      binaryDataN2Information2:
        type: string
        format: binary
      binaryDataN2Information3:
        type: string
        format: binary
      binaryDataN2Information4:
        type: string
        format: binary
      binaryDataN2Information5:
        type: string
        format: binary
      binaryDataN2Information6:
        type: string
        format: binary
      binaryDataN2Information7:
        type: string
        format: binary
      binaryDataN2Information8:
        type: string
        format: binary
      binaryDataN2Information9:
        type: string
        format: binary
      binaryDataN2Information10:
        type: string
        format: binary
  encoding:
    jsonData:
      contentType:  application/json
    binaryDataN2Information1:
      contentType:  application/vnd.3gpp.ngap
      headers:
        Content-Id:
          schema:
            type: string
    binaryDataN2Information2:
      contentType:  application/vnd.3gpp.ngap
      headers:
        Content-Id:
          schema:
            type: string
    binaryDataN2Information3:
      contentType:  application/vnd.3gpp.ngap
      headers:
        Content-Id:
          schema:
            type: string
    binaryDataN2Information4:
      contentType:  application/vnd.3gpp.ngap
      headers:
        Content-Id:
          schema:
            type: string
    binaryDataN2Information5:
      contentType:  application/vnd.3gpp.ngap
      headers:
        Content-Id:
          schema:
            type: string
    binaryDataN2Information6:
      contentType:  application/vnd.3gpp.ngap
      headers:
        Content-Id:
          schema:
            type: string
    binaryDataN2Information7:
      contentType:  application/vnd.3gpp.ngap
      headers:
        Content-Id:
          schema:
            type: string
    binaryDataN2Information8:
      contentType:  application/vnd.3gpp.ngap
      headers:
        Content-Id:
          schema:
            type: string
```

```
                binaryDataN2Information9:
                  contentType:  application/vnd.3gpp.ngap
                  headers:
                    Content-Id:
                      schema:
                        type: string
                binaryDataN2Information10:
                  contentType:  application/vnd.3gpp.ngap
                  headers:
                    Content-Id:
                      schema:
                        type: string
        '204':
          description: MBS Broadcast context updated successfully. No Content.
        '307':
          $ref: 'TS29571_CommonData.yaml#/components/responses/307'
        '308':
          $ref: 'TS29571_CommonData.yaml#/components/responses/308'
        '400':
          $ref: 'TS29571_CommonData.yaml#/components/responses/400'
        '401':
          $ref: 'TS29571_CommonData.yaml#/components/responses/401'
        '403':
          $ref: 'TS29571_CommonData.yaml#/components/responses/403'
        '404':
          $ref: 'TS29571_CommonData.yaml#/components/responses/404'
        '411':
          $ref: 'TS29571_CommonData.yaml#/components/responses/411'
        '413':
          $ref: 'TS29571_CommonData.yaml#/components/responses/413'
        '415':
          $ref: 'TS29571_CommonData.yaml#/components/responses/415'
        '429':
          $ref: 'TS29571_CommonData.yaml#/components/responses/429'
        '500':
          $ref: 'TS29571_CommonData.yaml#/components/responses/500'
        '502':
          $ref: 'TS29571_CommonData.yaml#/components/responses/502'
        '503':
          $ref: 'TS29571_CommonData.yaml#/components/responses/503'
        default:
          $ref: 'TS29571_CommonData.yaml#/components/responses/default'

components:
  securitySchemes:
    oAuth2ClientCredentials:
      type: oauth2
      flows:
        clientCredentials:
          tokenUrl: '{nrfApiRoot}/oauth2/token'
          scopes:
            namf-mbs-bc: Access to the Namf_MBSBroadcast API

  schemas:

#
# STRUCTURED DATA TYPES
#
    ContextCreateReqData:
      description: Data within ContextCreate Request
      type: object
      properties:
        mbsSessionId:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/MbsSessionId'
        mbsServiceAreaInfoList:
          type: array
          items:
            $ref: 'TS29571_CommonData.yaml#/components/schemas/MbsServiceAreaInfo'
          minItems: 1
        mbsServiceArea:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/MbsServiceArea'
        n2MbsSmInfo:
          $ref: '#/components/schemas/N2MbsSmInfo'
        notifyUri:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Uri'
        maxResponseTime:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/DurationSec'
        snssai:
```

```
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Snssai'
      mbsmfId:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/NfInstanceId'
      mbsmfServiceInstId:
        type: string
      associatedSessionId:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/AssociatedSessionId'
    required:
      - mbsSessionId
      - n2MbsSmInfo
      - notifyUri
      - snssai
    oneOf:
      - required: [ mbsServiceArea ]
      - required: [ mbsServiceAreaInfoList ]

  ContextCreateRspData:
    description: Data within ContextCreate Response
    type: object
    properties:
      mbsSessionId:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/MbsSessionId'
      n2MbsSmInfoList:
        type: array
        items:
          $ref: '#/components/schemas/N2MbsSmInfo'
        minItems: 1
        maxItems: 10
      operationStatus:
        $ref: '#/components/schemas/OperationStatus'
    required:
      - mbsSessionId


  ContextUpdateReqData:
    description: Data within ContextUpdate Request
    type: object
    properties:
      mbsServiceArea:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/MbsServiceArea'
      mbsServiceAreaInfoList:
        type: array
        items:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/MbsServiceAreaInfo'
        minItems: 1
      n2MbsSmInfo:
        $ref: '#/components/schemas/N2MbsSmInfo'
      ranIdList:
        type: array
        items:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/GlobalRanNodeId'
        minItems: 1
      noNgapSignallingInd:
        type: boolean
        enum:
          - true
      notifyUri:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Uri'
      maxResponseTime:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/DurationSec'
      n2MbsInfoChangeInd:
        type: boolean
    not:
      required: [ mbsServiceArea, mbsServiceAreaInfoList ]

  ContextStatusNotification:
    description: Data within ContextStatusNotify Request
    type: object
    properties:
      mbsSessionId:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/MbsSessionId'
      areaSessionId:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/AreaSessionId'
      n2MbsSmInfoList:
        type: array
        items:
          $ref: '#/components/schemas/N2MbsSmInfo'
        minItems: 1
```

```
      maxItems: 10
    operationEvents:
      type: array
      items:
        $ref: '#/components/schemas/OperationEvent'
      minItems: 1
    operationStatus:
      $ref: '#/components/schemas/OperationStatus'
    releasedInd:
      type: boolean
      enum:
       - true
  required:
    - mbsSessionId

ContextStatusNotificationResponse:
  description: Data within ContextStatusNotify Response
  type: object
  properties:
    mbsSessionId:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/MbsSessionId'
    areaSessionId:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/AreaSessionId'
    n2MbsSmInfoList:
      type: array
      items:
        $ref: '#/components/schemas/N2MbsSmInfo'
      minItems: 1
      maxItems: 10
  required:
    - mbsSessionId


ContextUpdateRspData:
  description: Data within ContextUpdate Response
  type: object
  properties:
    n2MbsSmInfoList:
      type: array
      items:
        $ref: '#/components/schemas/N2MbsSmInfo'
      minItems: 1
      maxItems: 10
    operationStatus:
      $ref: '#/components/schemas/OperationStatus'

N2MbsSmInfo:
  description: N2 MBS Session Management information
  type: object
  properties:
    ngapIeType:
      $ref: '#/components/schemas/NgapIeType'
    ngapData:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/RefToBinaryData'
    ranId:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/GlobalRanNodeId'
  required:
    - ngapIeType
    - ngapData

OperationEvent:
  description: Operation Event for a Broadcast MBS Session.
  type: object
  properties:
    opEventType:
      $ref: '#/components/schemas/OpEventType'
    amfId:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/NfInstanceId'
    ngranFailureEventList:
      type: array
      items:
        $ref: '#/components/schemas/NgranFailureEvent'
      minItems: 1
  required:
    - opEventType


NgranFailureEvent:
```

```
      description: NG-RAN failure event for a NG-RAN
      type: object
      properties:
        ngranId:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/GlobalRanNodeId'
        ngranFailureIndication:
          $ref: '#/components/schemas/NgranFailureIndication'
      required:
        - ngranId
        - ngranFailureIndication


#
# SIMPLE DATA TYPES
#


#
# ENUMERATIONS
#

    OperationStatus:
      description: Status of a Broadcast MBS session start or update operation.
      anyOf:
      - type: string
        enum:
          - MBS_SESSION_START_COMPLETE
          - MBS_SESSION_START_INCOMPLETE
          - MBS_SESSION_UPDATE_COMPLETE
          - MBS_SESSION_UPDATE_INCOMPLETE
      - type: string

    NgapIeType:
      description: NGAP Information Element Type
      anyOf:
      - type: string
        enum:
          - MBS_SES_REQ
          - MBS_SES_RSP
          - MBS_SES_FAIL
          - MBS_SES_REL_RSP
          - BC_TRA_REQ
          - BC_TRA_RSP
          - BC_TRA_FAIL
      - type: string

    OpEventType:
      description: Operation Event Type.
      anyOf:
      - type: string
        enum:
          - AMF_CHANGE
          - NG_RAN_EVENT
      - type: string


    NgranFailureIndication:
      description: Indicates a NG-RAN failure event.
      anyOf:
      - type: string
        enum:
          - NG_RAN_RESTART_OR_START
          - NG_RAN_FAILURE_WITHOUT_RESTART
          - NG_RAN_NOT_REACHABLE
          - NG_RAN_REQUIRED_RELEASE
      - type: string
```

# A.7 Namf_MBSCommunication API

```
openapi: 3.0.0

info:
  version: 1.1.0
  title: Namf_MBSCommunication
  description: |
```

```
    AMF Communication Service for MBS.
    © 2024, 3GPP Organizational Partners (ARIB, ATIS, CCSA, ETSI, TSDSI, TTA, TTC).
    All rights reserved.

externalDocs:
  description: 3GPP TS 29.518 V18.6.0; 5G System; Access and Mobility Management Services
  url: 'https://www.3gpp.org/ftp/Specs/archive/29_series/29.518/'

servers:
  - url: '{apiRoot}/namf-mbs-comm/v1'
    variables:
      apiRoot:
        default: https://example.com
        description: apiRoot as defined in clause 4.4 of 3GPP TS 29.501

security:
  - {}
  - oAuth2ClientCredentials:
    - namf-mbs-comm

paths:
  /n2-messages/transfer:
    post:
      summary: Namf_MBSCommunication N2 Message Transfer service Operation
      tags:
        - N2Messages Handler (custom operation)
      operationId: N2MessageTransfer
      requestBody:
        content:
          multipart/related:  # message with binary body part(s)
            schema:
              type: object
              properties: # Request parts
                jsonData:
                  $ref: '#/components/schemas/MbsN2MessageTransferReqData'
                binaryDataN2Information:
                  type: string
                  format: binary
            encoding:
              jsonData:
                contentType:  application/json
              binaryDataN2Information:
                contentType:  application/vnd.3gpp.ngap
                headers:
                  Content-Id:
                    schema:
                      type: string
        required: true
      callbacks:
        notification:
          '{$request.body#/notifyUri}':
            post:
              requestBody:
                required: true
                content:
                  application/json:
                    schema:
                      $ref: '#/components/schemas/Notification'
              responses:
                '204':
                  description: successful notification
                '307':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/307'
                '308':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/308'
                '400':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/400'
                '403':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/403'
                '404':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/404'
                '411':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/411'
                '413':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/413'
                '415':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/415'
                '429':
```

```
                $ref: 'TS29571_CommonData.yaml#/components/responses/429'
              '500':
                $ref: 'TS29571_CommonData.yaml#/components/responses/500'
              '502':
                $ref: 'TS29571_CommonData.yaml#/components/responses/502'
              '503':
                $ref: 'TS29571_CommonData.yaml#/components/responses/503'
      responses:
        '200':
          description: MBS N2 Message Transfer successfully initiated
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/MbsN2MessageTransferRspData'
        '307':
          $ref: 'TS29571_CommonData.yaml#/components/responses/307'
        '308':
          $ref: 'TS29571_CommonData.yaml#/components/responses/308'
        '400':
          $ref: 'TS29571_CommonData.yaml#/components/responses/400'
        '401':
          $ref: 'TS29571_CommonData.yaml#/components/responses/401'
        '403':
          $ref: 'TS29571_CommonData.yaml#/components/responses/403'
        '404':
          description: Not Found
          content:
            application/problem+json:
              schema:
                $ref: 'TS29571_CommonData.yaml#/components/schemas/ProblemDetails'
        '411':
          $ref: 'TS29571_CommonData.yaml#/components/responses/411'
        '413':
          $ref: 'TS29571_CommonData.yaml#/components/responses/413'
        '415':
          $ref: 'TS29571_CommonData.yaml#/components/responses/415'
        '429':
          $ref: 'TS29571_CommonData.yaml#/components/responses/429'
        '500':
          $ref: 'TS29571_CommonData.yaml#/components/responses/500'
        '502':
          $ref: 'TS29571_CommonData.yaml#/components/responses/502'
        '503':
          $ref: 'TS29571_CommonData.yaml#/components/responses/503'
        default:
          description: Unexpected error


components:
  securitySchemes:
    oAuth2ClientCredentials:
      type: oauth2
      flows:
        clientCredentials:
          tokenUrl: '{nrfApiRoot}/oauth2/token'
          scopes:
            namf-mbs-comm: Access to the Namf_MBSCommunication API

  schemas:


#
# STRUCTURED DATA TYPES
#


    MbsN2MessageTransferReqData:
      description: Data within MBS N2 Message Transfer Request
      type: object
      properties:
        mbsSessionId:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/MbsSessionId'
        areaSessionId:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/AreaSessionId'
        n2MbsSmInfo:
          $ref: '#/components/schemas/N2MbsSmInfo'
        supportedFeatures:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/SupportedFeatures'
```

```
      ranNodeIdList:
        type: array
        items:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/GlobalRanNodeId'
        minItems: 1
      notifyUri:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Uri'
      notifyCorrelationId:
        type: string
    required:
      - mbsSessionId
      - n2MbsSmInfo


  MbsN2MessageTransferRspData:
    description: Data within MBS N2 Message Transfer Response
    type: object
    properties:
      result:
        $ref: 'TS29518_Namf_Communication.yaml#/components/schemas/N2InformationTransferResult'
      supportedFeatures:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/SupportedFeatures'
      failureList:
        type: array
        items:
          $ref: '#/components/schemas/RanFailure'
        minItems: 1
    required:
      - result



  N2MbsSmInfo:
    description: N2 MBS Session Management information
    type: object
    properties:
      ngapIeType:
        $ref: '#/components/schemas/MbsNgapIeType'
      ngapData:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/RefToBinaryData'
    required:
      - ngapIeType
      - ngapData

  Notification:
    description: Data within Notify Request
    type: object
    properties:
      mbsSessionId:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/MbsSessionId'
      areaSessionId:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/AreaSessionId'
      failureList:
        type: array
        items:
          $ref: '#/components/schemas/RanFailure'
        minItems: 1
      notifyCorrelationId:
        type: string
    required:
      - mbsSessionId
      - failureList

  RanFailure:
    description: Description of an MBS related N2 procedure failure
    type: object
    properties:
      ranId:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/GlobalRanNodeId'
      ranFailureCause:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/NgApCause'
      ranFailureIndication:
        $ref: '#/components/schemas/RanFailureIndication'
    required:
      - ranId
    oneOf:
```

```
        - required: [ ranFailureCause ]
        - required: [ ranFailureIndication ]

#
# SIMPLE DATA TYPES
#


#
# ENUMERATIONS
#

    MbsNgapIeType:
      description: NGAP Information Element Type
      anyOf:
      - type: string
        enum:
          - MBS_SES_ACT_REQ
          - MBS_SES_DEACT_REQ
          - MBS_SES_UPD_REQ
      - type: string

    RanFailureIndication:
      description: Indicates a NG-RAN failure event
      anyOf:
      - type: string
        enum:
          - NG_RAN_FAILURE_WITHOUT_RESTART
          - NG_RAN_NOT_REACHABLE
      - type: string
```

# Annex B (Informative): HTTP Multipart Messages

## B.1 Example of HTTP multipart message

### B.1.1 General

This clause provides a (partial) example of HTTP multipart message. The example does not aim to be a complete representation of the HTTP message, e.g. additional information or headers can be included.

This Annex is informative and the normative descriptions in this specification prevail over the description in this Annex if there is any difference.

### B.1.2 Example HTTP multipart message with N2 Information binary data

```
POST /example.com/namf-comm/v1/ue-contexts/{ueContextId}/n1-n2-messages HTTP/2
Content-Type: multipart/related; boundary=----Boundary
Content-Length: xyz

------Boundary
Content-Type: application/json

{
    "n2InfoContainer": {
      "n2InformationClass": "SM",
      "smInfo": {
        "pduSessionId": 5,
        "n2InfoContent": {
          "ngapIeType": "PDU_RES_SETUP_REQ",
          "ngapData": {
            "contentId": "n2msg"
          }
        }
      }
    },
    "pduSessionId": 5
}
------Boundary
Content-Type: application/vnd.3gpp.ngap
Content-Id: n2msg

{ … N2 Information binary data …}
------Boundary
```

# Annex C (informative): Change history

| Change history | | | | | | | |
|---|---|---|---|---|---|---|---|
| **Date** | **Meeting** | **TDoc** | **CR** | **Rev** | **Cat** | **Subject/Comment** | **New version** |
| 2017-10 | CT4#80 | C4-175297 | | | | TS Skeleton | 0.1.0 |
| 2017-10 | CT4#80 | C4-175397 | | | | Implementation of pCRs agreed at CT4#80. | 0.2.0 |
| 2017-12 | CT4#81 | C4-176441 | | | | Implementation of pCRs agreed at CT4#81, including C4-176285, C4-176290, C4-176291, C4-176292, C4-176293, C4-176375, C4-176376, C4-176378, C4-176379, C4-176380 and C4-176404. | 0.3.0 |
| 2018-01 | CT4#82 | C4-181393 | | | | Implementation of pCRs agreed at CT4#82, including C4-181090, C4-181091, C4-181258, C4-181259, C4-181260, C4-181269, C4-181270, C4-181311, C4-181312, C4-181313, C4-181314, C4-181352, C4-181353 and C4-181354 | 0.4.0 |
| 2018-03 | CT4#83 | C4-182437 | | | | Implementation of pCRs agreed at CT4#83, including C4-182287, C4-182288, C4-182290, C4-182292, C4-182293, C4-182350, C4-182353, C4-182355, C4-182358, C4-182367, C4-182385, C4-182403, C4-182414, C4-182415 | 0.5.0 |
| 2018-03 | CT#79 | CP-180033 | | | | Presented for information | 1.0.0 |
| 2018-04 | CT4#84 | C4-183518 | | | | Implementation of pCRs agreed at CT4#84, including C4-183048, C4-183054, C4-183055, C4-183064, C4-183073, C4-183074, C4-183161, C4-183166, C4-183171, C4-183345, C4-183347, C4-183351, C4-183354, C4-183356, C4-183357, C4-183359, C4-183360, C4-183361, C4-183362, C4-183406, C4-183407, C4-183408, C4-183409, C4-183410, C4-183411, C4-183412, C4-183413, C4-183414, C4-183415, C4-183417, C4-183434, C4-183435, C4-183436, C4-183437, C4-183439, C4-183445, C4-183460, C4-183461, C4-183462, C4-183463, C4-183464, C4-183493, C4-183494, C4-183495, C4-183502 | 1.1.0 |
| 2018-05 | CT4#85 | C4-184629 | | | | Implementation of pCRs agreed at CT4#85, including:<br>C4-184390, C4-184391, C4-184562, C4-184393, C4-184561, C4-184395, C4-194052, C4-184396, C4-184399, C4-184404, C4-184405, C4-184407, C4-184102, C4-184408, C4-184104, C4-184410, C4-184412, C4-184413, C4-184569, C4-184563, C4-184124, C4-184418, C4-184565, C4-184127, C4-184566, C4-184129, C4-184421, C4-184131, C4-184426, C4-184427, C4-184428, C4-184429, C4-184430, C4-184431, C4-184432, C4-184433, C4-184434, C4-184435, C4-184436, C4-184437, C4-184151, C4-184481, C4-184154, C4-184515, C4-184516, C4-184568, C4-184485, C4-184486, C4-184487, C4-184488 | 1.2.0 |
| 2018-06 | CT#80 | CP-181107 | | | | Presented for approval | 2.0.0 |
| 2018-06 | CT#80 | | | | | Approved in CT#80 | 15.0.0 |
| 2018-09 | CT#81 | CP-182062 | 0001 | 2 | F | RAT Selector for PWS | 15.1.0 |
| 2018-09 | CT#81 | CP-182062 | 0002 | 3 | F | AM Policy Triggers in MM Context | 15.1.0 |
| 2018-09 | CT#81 | CP-182062 | 0003 | 1 | F | Update UE context and MM context as per latest stage 2 agreements | 15.1.0 |
| 2018-09 | CT#81 | CP-182062 | 0004 | 1 | F | Corrections to EBI Assignment | 15.1.0 |
| 2018-09 | CT#81 | CP-182062 | 0005 | 1 | F | Clarify Max number of reports and Max duration of reporting in alignment with stage 2 | 15.1.0 |
| 2018-09 | CT#81 | CP-182062 | 0006 | | F | N1/N2 Message Transfer Temporary Reject | 15.1.0 |
| 2018-09 | CT#81 | CP-182062 | 0008 | | F | Remove AN Type from N1/N2 Message Transfer Request | 15.1.0 |
| 2018-09 | CT#81 | CP-182165 | 0009 | 2 | F | Update SeafData as per agreements in SA3 | 15.1.0 |
| 2018-09 | CT#81 | CP-182062 | 0010 | 1 | F | Include TimeStamp in AMF Event Notification | 15.1.0 |
| 2018-09 | CT#81 | CP-182062 | 0011 | | F | Provide Domain Selection Info | 15.1.0 |
| 2018-09 | CT#81 | CP-182062 | 0012 | 1 | F | RAN UE NGAP ID in RegistrationContextContainer | 15.1.0 |
| 2018-09 | CT#81 | CP-182062 | 0013 | 1 | F | NG-RAN TargetID in RegistrationContextContainer | 15.1.0 |
| 2018-09 | CT#81 | CP-182062 | 0014 | 3 | F | BackUp AMF Info | 15.1.0 |
| 2018-09 | CT#81 | CP-182062 | 0015 | | F | Description of N1N2TransferFailureNotification Operation | 15.1.0 |
| 2018-09 | CT#81 | CP-182062 | 0016 | 1 | F | Add Quotes for Runtime Expression | 15.1.0 |
| 2018-09 | CT#81 | CP-182062 | 0017 | | F | Callback URI for N2InfoNotify during N2 based handover | 15.1.0 |
| 2018-09 | CT#81 | CP-182062 | 0018 | 1 | F | Resolve Editor's Note on regular expression pattern | 15.1.0 |
| 2018-09 | CT#81 | CP-182095 | 0019 | 4 | F | Location Service ProvideLocationInfo | 15.1.0 |
| 2018-09 | CT#81 | CP-182062 | 0020 | 2 | F | Location Service ProvidePositioningInfo | 15.1.0 |
| 2018-09 | CT#81 | CP-182062 | 0021 | 2 | F | N1N2MessageTransfer Rejection due to SAR | 15.1.0 |
| 2018-09 | CT#81 | CP-182062 | 0022 | 3 | F | N2 Content Type Definition | 15.1.0 |
| 2018-09 | CT#81 | CP-182062 | 0023 | | F | Selected TAI in NgRanTargetId | 15.1.0 |
| 2018-09 | CT#81 | CP-182062 | 0024 | 2 | F | Skip Indicator | 15.1.0 |
| 2018-09 | CT#81 | CP-182062 | 0025 | 1 | F | UEContextTransfer Integrity Check Failure | 15.1.0 |
| 2018-09 | CT#81 | CP-182068 | 0026 | 1 | B | Add support for 5G Trace | 15.1.0 |
| 2018-09 | CT#81 | CP-182094 | 0027 | 3 | F | NgApCause Definition | 15.1.0 |
| 2018-09 | CT#81 | CP-182062 | 0028 | 1 | F | N1N2 Transfer Failure Notification | 15.1.0 |
| 2018-09 | CT#81 | CP-182062 | 0029 | | F | N2 Container Data Type During Handover | 15.1.0 |
| 2018-09 | CT#81 | CP-182175 | 0031 | 1 | F | Correction to RegistrationCompleteNotify | 15.1.0 |
| 2018-09 | CT#81 | CP-182062 | 0032 | 3 | F | N1N2MessageTransfer and Notify for PCF | 15.1.0 |
| 2018-09 | CT#81 | CP-182166 | 0033 | 3 | F | Regular expression pattern for UeContextId parameter in OpenAPI | 15.1.0 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2018-09 | CT#81 | CP-182062 | 0036 | 2 | F | Presence Reporting Area | 15.1.0 |
| 2018-09 | CT#81 | CP-182062 | 0037 | 1 | F | Notification Correlation Id for subscription correlation Id change | 15.1.0 |
| 2018-09 | CT#81 | CP-182062 | 0038 | 1 | F | Default Subscription for Notification to LMF | 15.1.0 |
| 2018-09 | CT#81 | CP-182062 | 0039 | 1 | F | LCS Correlation Identifier in N2Notify | 15.1.0 |
| 2018-09 | CT#81 | CP-182062 | 0040 | 1 | F | Mobility Restriction | 15.1.0 |
| 2018-09 | CT#81 | CP-182062 | 0041 | | F | Not Allowed Slice | 15.1.0 |
| 2018-09 | CT#81 | CP-182062 | 0042 | 1 | F | UE-AMBR | 15.1.0 |
| 2018-09 | CT#81 | CP-182062 | 0044 | 1 | F | Array Attributes | 15.1.0 |
| 2018-09 | CT#81 | CP-182062 | 0045 | 2 | F | Default Response Codes | 15.1.0 |
| 2018-09 | CT#81 | CP-182062 | 0046 | | F | AMF service operations | 15.1.0 |
| 2018-09 | CT#81 | CP-182048 | 0047 | 2 | F | Passing NSSF information in N1MessageNotification | 15.1.0 |
| 2018-09 | CT#81 | CP-182062 | 0049 | 3 | F | Clarification on location information in immediate report | 15.1.0 |
| 2018-09 | CT#81 | CP-182062 | 0050 | 1 | F | Resource Figures | 15.1.0 |
| 2018-09 | CT#81 | CP-182062 | 0051 | | F | Correct reference for Event Report Information | 15.1.0 |
| 2018-09 | CT#81 | CP-182062 | 0052 | | F | Consistent use of "Correlation Id" | 15.1.0 |
| 2018-09 | CT#81 | CP-182062 | 0053 | 1 | F | API version number update | 15.1.0 |
| 2018-09 | CT#81 | CP-182062 | 0054 | 1 | F | Custom Operation Name Correction for EBI Assignment | 15.1.0 |
| 2018-09 | CT#81 | CP-192096 | 0055 | | F | Correction of CorrelationId Reference in OpenAPI | 15.1.0 |
| 2018-12 | CT#82 | CP-183020 | 56 | 1 | F | Editorial Corrections | 15.2.0 |
| 2018-12 | CT#82 | CP-183020 | 57 | | F | Usage for EnableUEReachability Service Operation | 15.2.0 |
| 2018-12 | CT#82 | CP-183020 | 58 | 1 | F | Update to SeafData | 15.2.0 |
| 2018-12 | CT#82 | CP-183232 | 60 | 4 | F | Transfer UE Radio Capability between AMFs | 15.2.0 |
| 2018-12 | CT#82 | CP-183020 | 61 | 2 | F | Notification of the change of the PCF | 15.2.0 |
| 2018-12 | CT#82 | CP-183020 | 62 | 1 | F | Information in N1MessageNotify | 15.2.0 |
| 2018-12 | CT#82 | CP-183020 | 63 | | F | Event Exposure | 15.2.0 |
| 2018-12 | CT#82 | CP-183020 | 64 | | F | Correct the references | 15.2.0 |
| 2018-12 | CT#82 | CP-183020 | 65 | 5 | F | Subscription lifetime | 15.2.0 |
| 2018-12 | CT#82 | CP-183020 | 67 | | F | Corrections to TADS Query API | 15.2.0 |
| 2018-12 | CT#82 | CP-183020 | 69 | 5 | F | Transfer of Group Id Suscriptions | 15.2.0 |
| 2018-12 | CT#82 | CP-183020 | 70 | 1 | F | Attributes corrections for RegistrationContextContainer and MmContext | 15.2.0 |
| 2018-12 | CT#82 | CP-183020 | 71 | 1 | F | Correction on tables | 15.2.0 |
| 2018-12 | CT#82 | CP-183020 | 72 | | F | Mandatory Status Code Correction | 15.2.0 |
| 2018-12 | CT#82 | CP-183020 | 74 | 1 | F | N2InfoNotify correction for Handover Confirm | 15.2.0 |
| 2018-12 | CT#82 | CP-183020 | 75 | 1 | F | Naming convention of provideLocInfo and providePosInfo | 15.2.0 |
| 2018-12 | CT#82 | CP-183020 | 76 | 2 | F | OpenAPI specification alignments | 15.2.0 |
| 2018-12 | CT#82 | CP-183020 | 77 | 1 | F | Remove Duplicated Common Application Errors | 15.2.0 |
| 2018-12 | CT#82 | CP-183020 | 78 | | F | Required routingId | 15.2.0 |
| 2018-12 | CT#82 | CP-183020 | 79 | 1 | F | Resource URIs Alignment | 15.2.0 |
| 2018-12 | CT#82 | CP-183020 | 80 | | F | Seaf data type correction | 15.2.0 |
| 2018-12 | CT#82 | CP-183020 | 81 | | F | UeContextId Pattern Complement | 15.2.0 |
| 2018-12 | CT#82 | CP-183020 | 82 | | F | Use RefToBinaryData from common data types | 15.2.0 |
| 2018-12 | CT#82 | CP-183020 | 83 | 3 | F | Range Definition in OpenAPI | 15.2.0 |
| 2018-12 | CT#82 | CP-183020 | 84 | | F | sessionId in N1N2MessageTransferReqData | 15.2.0 |
| 2018-12 | CT#82 | CP-183020 | 85 | 1 | F | New rejection cause for UE in CM-IDLE state | 15.2.0 |
| 2018-12 | CT#82 | CP-183151 | 86 | 8 | F | Notifying Subscription ID Change | 15.2.0 |
| 2018-12 | CT#82 | CP-183020 | 87 | 1 | F | SMF Reallocation requested Indication | 15.2.0 |
| 2018-12 | CT#82 | CP-183020 | 88 | 1 | F | Paging Policy Indicator | 15.2.0 |
| 2018-12 | CT#82 | CP-183020 | 89 | 1 | F | EPS bearer identity | 15.2.0 |
| 2018-12 | CT#82 | CP-183020 | 90 | 1 | F | 29518 CR cardinality | 15.2.0 |
| 2018-12 | CT#82 | CP-183020 | 92 | 1 | F | Editorial Correction to PduSessionContext | 15.2.0 |
| 2018-12 | CT#82 | CP-183020 | 93 | 1 | F | Global RAN Node ID in RegistrationContextContainer | 15.2.0 |
| 2018-12 | CT#82 | CP-183154 | 97 | 2 | F | Update of Subscription Lifetime | 15.2.0 |
| 2018-12 | CT#82 | CP-183020 | 98 | 1 | F | EBI Allocation Rejection Cause | 15.2.0 |
| 2018-12 | CT#82 | CP-183020 | 100 | 2 | F | UE Context Transfer during initial registration via another access type | 15.2.0 |
| 2018-12 | CT#82 | CP-183020 | 101 | 1 | F | RAN Status Transfer Transparent Container in N2 based handover | 15.2.0 |
| 2018-12 | CT#82 | CP-183020 | 103 | 1 | F | NgapIeType for X2 and N2 based handover | 15.2.0 |
| 2018-12 | CT#82 | CP-183020 | 104 | | F | Update of N1N2 Message Operations | 15.2.0 |
| 2018-12 | CT#82 | CP-183020 | 105 | 1 | F | Clarify the handling of EBI assignment | 15.2.0 |
| 2018-12 | CT#82 | CP-183020 | 106 | | F | Align Usage of Tags | 15.2.0 |
| 2018-12 | CT#82 | CP-183020 | 107 | 1 | F | Altitude in Provide Positioning Information | 15.2.0 |
| 2018-12 | CT#82 | CP-183020 | 108 | | F | AmfStatusChangeSubcribe Modify in Resource Table | 15.2.0 |
| 2018-12 | CT#82 | CP-183020 | 109 | 1 | F | API Root | 15.2.0 |
| 2018-12 | CT#82 | CP-183020 | 110 | 1 | F | Case Convention | 15.2.0 |
| 2018-12 | CT#82 | CP-183020 | 111 | 1 | F | Clarification of ProvideLocInfo when CM-CONNECTED | 15.2.0 |
| 2018-12 | CT#82 | CP-183020 | 118 | 1 | F | N1 N2 Message for Positioning | 15.2.0 |
| 2018-12 | CT#82 | CP-183020 | 119 | 3 | F | N3GPP DDN handling when UE CM-IDLE on N3GPP | 15.2.0 |
| 2018-12 | CT#82 | CP-183020 | 121 | 1 | F | Alignment on TADS Query | 15.2.0 |
| 2018-12 | CT#82 | CP-183020 | 122 | 1 | F | Configuration Transfer procedure over N14 | 15.2.0 |
| 2018-12 | CT#82 | CP-183020 | 123 | | F | N1N2MessageTransfer Request message | 15.2.0 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2018-12 | CT#82 | CP-183020 | 124 | 2 | F | UDM group Id | 15.2.0 |
| 2018-12 | CT#82 | CP-183020 | 125 | | F | Warning Request Transfer Procedure | 15.2.0 |
| 2018-12 | CT#82 | CP-183020 | 126 | 1 | F | Location Header | 15.2.0 |
| 2018-12 | CT#82 | CP-183020 | 127 | | F | Remove duplicate references | 15.2.0 |
| 2018-12 | CT#82 | CP-183020 | 128 | 1 | F | 429 Response Codes | 15.2.0 |
| 2018-12 | CT#82 | CP-183020 | 129 | | F | API Version | 15.2.0 |
| 2018-12 | CT#82 | CP-183020 | 130 | 1 | F | Oauth2 correction | 15.2.0 |
| 2018-12 | CT#82 | CP-183191 | 131 | | F | Editorial Correction to AMF Event Type Enumeration | 15.2.0 |
| 2018-12 | CT#82 | CP-183229 | 132 | | F | Correction to OpenAPI definition of UeContextTransferRspData | 15.2.0 |
| 2019-03 | CT#83 | CP-190025 | 133 | 1 | F | OpenAPI correction for HTTP method of EnableUEReachability | 15.3.0 |
| 2019-03 | CT#83 | CP-190025 | 134 | | F | PDU sessions not accepted by target AMF in N2 based handover | 15.3.0 |
| 2019-03 | CT#83 | CP-190025 | 135 | 1 | F | Sending Secondary RAT usage over N14 during N2 handover with AMF change | 15.3.0 |
| 2019-03 | CT#83 | CP-190025 | 136 | | F | SM Context URI in UE context | 15.3.0 |
| 2019-03 | CT#83 | CP-190025 | 137 | 2 | F | UE policy delivery and control | 15.3.0 |
| 2019-03 | CT#83 | CP-190025 | 138 | | F | Correct Event Exposure Service Description | 15.3.0 |
| 2019-03 | CT#83 | CP-190025 | 139 | 2 | F | Simplify N1N2MessageTransfer when UE is in CM-IDLE | 15.3.0 |
| 2019-03 | CT#83 | CP-190025 | 140 | 2 | F | Update EBIAssignment Service Operation to Align with Stage 2 | 15.3.0 |
| 2019-03 | CT#83 | CP-190025 | 141 | 1 | F | Corrections to the HTTP methods and URI | 15.3.0 |
| 2019-03 | CT#83 | CP-190025 | 143 | 1 | F | Correction to Reponse Code for Positioning Failed | 15.3.0 |
| 2019-03 | CT#83 | CP-190025 | 144 | 1 | F | Essential Clairfication on Event Subscription Creation | 15.3.0 |
| 2019-03 | CT#83 | CP-190025 | 145 | 1 | F | OpenAPI Syntax Correction | 15.3.0 |
| 2019-03 | CT#83 | CP-190025 | 146 | 1 | F | Reference Id | 15.3.0 |
| 2019-03 | CT#83 | CP-190025 | 148 | 1 | F | SMF Service Instance during AMF change | 15.3.0 |
| 2019-03 | CT#83 | CP-190025 | 149 | 1 | F | GMLC URI for Namf_Location EventNotify | 15.3.0 |
| 2019-03 | CT#83 | CP-190025 | 150 | 1 | F | Correction of keyAmfChangeInd | 15.3.0 |
| 2019-03 | CT#83 | CP-190025 | 151 | 1 | F | N2SmInformation in UeContextCreateData & UeContextCreatedData | 15.3.0 |
| 2019-03 | CT#83 | CP-190025 | 153 | | F | API version update | 15.3.0 |
| 2019-06 | CT#84 | CP-191036 | 154 | | F | ngapCause in UeContextCreatedData | 15.4.0 |
| 2019-06 | CT#84 | CP-191036 | 160 | | F | Correction N1 N2 Message Transfer when CM-IDLE | 15.4.0 |
| 2019-06 | CT#84 | CP-191036 | 161 | | F | Correction on CR0021 implementation | 15.4.0 |
| 2019-06 | CT#84 | CP-191036 | 162 | | F | Event Notify Failure Response | 15.4.0 |
| 2019-06 | CT#84 | CP-191036 | 164 | | F | UE Identities for Event Notification | 15.4.0 |
| 2019-06 | CT#84 | CP-191036 | 155 | 1 | F | Content Type | 15.4.0 |
| 2019-06 | CT#84 | CP-191036 | 163 | 1 | F | LPP Handling | 15.4.0 |
| 2019-06 | CT#84 | CP-191036 | 165 | 1 | F | AMF Event Alignment | 15.4.0 |
| 2019-06 | CT#84 | CP-191036 | 166 | 1 | F | Missing Loss Of Connectivity Event | 15.4.0 |
| 2019-06 | CT#84 | CP-191036 | 171 | 2 | F | Storage of OpenAPI specification files | 15.4.0 |
| 2019-06 | CT#84 | CP-191036 | 172 | 1 | F | Location header in redirect response | 15.4.0 |
| 2019-06 | CT#84 | CP-191036 | 173 | 1 | F | LMF Service Instance Id for N1N2MessageTransfer | 15.4.0 |
| 2019-06 | CT#84 | CP-191036 | 174 | | F | Remove Subscribed-Data-Report event type and SARI data type | 15.4.0 |
| 2019-06 | CT#84 | CP-191036 | 175 | 1 | F | Correction in PwsInformation Parameter | 15.4.0 |
| 2019-06 | CT#84 | CP-191036 | 177 | 1 | F | Copyright Note in OpenAPI Spec | 15.4.0 |
| 2019-06 | CT#84 | CP-191036 | 178 | 1 | F | Correction on EBI in PDU session context | 15.4.0 |
| 2019-06 | CT#84 | CP-191036 | 179 | 1 | F | Major API version | 15.4.0 |
| 2019-06 | CT#84 | CP-191036 | 181 | 1 | F | Status code of Namf_EventExposure Unsubscrive service operation | 15.4.0 |
| 2019-06 | CT#84 | CP-191036 | 187 | | F | TS 29.518 API version update | 15.4.0 |
| 2019-06 | CT#84 | CP-191046 | 182 | 2 | F | Corrections of the references to retrieve Callback URI from NRF for N1and N2 notifications | 16.0.0 |
| 2019-06 | CT#84 | CP-191049 | 159 | 2 | B | Updates to CreateUEContext for eNS Support | 16.0.0 |
| 2019-06 | CT#84 | CP-191054 | 168 | 3 | B | Update N2InformationNotification for I-SMF insertion, change and removal | 16.0.0 |
| 2019-06 | CT#84 | CP-191050 | 184 | 3 | B | Add NB-IoT specific UE Radio Access Capability in UE context | 16.0.0 |
| 2019-06 | CT#84 | CP-191050 | 185 | 1 | B | Update to the UEContextTransfer service for adding Gap timer | 16.0.0 |
| 2019-06 | CT#84 | CP-191048 | 186 | | B | TS 29.518 API version update | 16.0.0 |
| 2019-09 | CT#85 | CP-192110 | 0189 | 2 | A | Wrong Cardinality of lcsSupportedGADShapes in RequestPosInfo | 16.1.0 |
| 2019-09 | CT#85 | CP-192128 | 0190 | 1 | F | Correction for ngapMessageType | 16.1.0 |
| 2019-09 | CT#85 | CP-192128 | 0191 | 1 | F | NonUeN2InfoUnscribe for PWS | 16.1.0 |
| 2019-09 | CT#85 | CP-192188 | 0193 | 1 | B | Transfer 5G SRVCC Parameters between AMFs | 16.1.0 |
| 2019-09 | CT#85 | CP-192193 | 0194 | 1 | B | CreateUEContext – I-SMF and SM Context ID Information | 16.1.0 |
| 2019-09 | CT#85 | CP-192110 | 0197 | 1 | A | Use of ARP value for Priority Paging | 16.1.0 |
| 2019-09 | CT#85 | CP-192193 | 0198 | 1 | B | Correction of the smfChangeIndication | 16.1.0 |
| 2019-09 | CT#85 | CP-192110 | 0200 | | A | Signalling Old GUAMI to target AMF during the AMF planned removal procedure | 16.1.0 |
| 2019-09 | CT#85 | CP-192128 | 0201 | 1 | F | 5GS User State retrieval | 16.1.0 |
| 2019-09 | CT#85 | CP-192128 | 0202 | 1 | F | Forwarding UL N2 message to target AMF during AMF planned removal procedure | 16.1.0 |
| 2019-09 | CT#85 | CP-192128 | 0203 | 1 | F | MT SMS to UE in RRC INACTIVE state with NG-RAN paging failure | 16.1.0 |
| 2019-09 | CT#85 | CP-192128 | 0205 | | F | Corrections to Mapped Service Operations of Namf_Communication service | 16.1.0 |
| 2019-09 | CT#85 | CP-192110 | 0208 | 1 | A | Missing Location header | 16.1.0 |

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|---|---|---|---|---|---|---|---|
| 2019-09 | CT#85 | CP-192110 | 0210 | 1 | A | Missing status codes | 16.1.0 |
| 2019-09 | CT#85 | CP-192134 | 0211 | | B | Transfer Information of MA PDU Session between AMFs | 16.1.0 |
| 2019-09 | CT#85 | CP-192110 | 0214 | 3 | A | OpenAPI Correction on Location Header | 16.1.0 |
| 2019-09 | CT#85 | CP-192128 | 0215 | | F | Error response of the EBIAssignment | 16.1.0 |
| 2019-09 | CT#85 | CP-192135 | 0216 | | B | Namf_EventExposure service invoked by NWDAF | 16.1.0 |
| 2019-09 | CT#85 | CP-192193 | 0217 | | B | ETSUN_N1N2MessageTransfer Failure due to SM Context relocation needed | 16.1.0 |
| 2019-09 | CT#85 | CP-192132 | 0218 | 1 | F | Service Gap Time | 16.1.0 |
| 2019-09 | CT#85 | CP-192132 | 0221 | 2 | B | HLCom extended buffering in MT Service | 16.1.0 |
| 2019-09 | CT#85 | CP-192132 | 0223 | 2 | B | Small Data Rate Control Status | 16.1.0 |
| 2019-09 | CT#85 | CP-192123 | 0224 | 1 | F | Example of HTTP multipart message | 16.1.0 |
| 2019-09 | CT#85 | CP-192132 | 0225 | 1 | B | Extended Buffering Support in Communication Service | 16.1.0 |
| 2019-09 | CT#85 | CP-192120 | 0227 | | F | TS 29.518 API version update | 16.1.0 |
| 2019-10 | | | | | | Corrupted references fixed | 16.1.1 |
| 2019-12 | CT#86 | CP-193051 | 0229 | 1 | B | Target Access type in N1N2MessageTransfer Request for a MA PDU session | 16.2.0 |
| 2019-12 | CT#86 | CP-193036 | 0230 | | F | egiList and ncgiList in N2InformationTransferReqData not needed | 16.2.0 |
| 2019-12 | CT#86 | CP-193056 | 0231 | | B | Event exposure between AMF and SMF | 16.2.0 |
| 2019-12 | CT#86 | CP-193051 | 0233 | 1 | B | MA PDU session accepted indication | 16.2.0 |
| 2019-12 | CT#86 | CP-193031 | 0235 | 1 | A | Source AMF NGAP ID | 16.2.0 |
| 2019-12 | CT#86 | CP-193031 | 0239 | | A | N1N2MessageTransfer request during an on-going handover procedure | 16.2.0 |
| 2019-12 | CT#86 | CP-193036 | 0240 | | B | RIM Information Transfer procedure | 16.2.0 |
| 2019-12 | CT#86 | CP-193046 | 0241 | | B | User location report | 16.2.0 |
| 2019-12 | CT#86 | CP-193055 | 0244 | 4 | B | Update the service operation of AMF | 16.2.0 |
| 2019-12 | CT#86 | CP-193031 | 0246 | | A | Correction to ProvideLocInfo | 16.2.0 |
| 2019-12 | CT#86 | CP-193062 | 0248 | 3 | B | Transferring UE Radio Capability ID between AMFs | 16.2.0 |
| 2019-12 | CT#86 | CP-193031 | 0250 | 1 | A | Reference correction | 16.2.0 |
| 2019-12 | CT#86 | CP-193048 | 0251 | 1 | F | Reference correction | 16.2.0 |
| 2019-12 | CT#86 | CP-193049 | 0253 | 1 | F | Correction on MT Enable UE Reachability | 16.2.0 |
| 2019-12 | CT#86 | CP-193063 | 0254 | | F | Excluding security context in the UE context | 16.2.0 |
| 2019-12 | CT#86 | CP-193049 | 0255 | 1 | B | Adding Rate Control attributes to N1N2messageTransferReq data type | 16.2.0 |
| 2019-12 | CT#86 | CP-193049 | 0256 | 2 | B | Mobile Terminated Data Transfer for Control Plane CIoT 5GS Optimisation | 16.2.0 |
| 2019-12 | CT#86 | CP-193036 | 0257 | 1 | F | PDU Session Release for UE in RRC INACTIVE state with NG-RAN paging failure | 16.2.0 |
| 2019-12 | CT#86 | CP-193036 | 0260 | 2 | F | Add Corresponding OpenAPI descriptions in clause 5.1 | 16.2.0 |
| 2019-12 | CT#86 | CP-193164 | 0261 | 2 | B | Updating support for subscription-based access restriction | 16.2.0 |
| 2019-12 | CT#86 | CP-193166 | 0262 | 2 | B | AMF Location Service Operations for a Commercial and Deferred 5GC-MT-LR | 16.2.0 |
| 2019-12 | CT#86 | CP-193055 | 0263 | 1 | B | LMF identification for LMF change | 16.2.0 |
| 2019-12 | CT#86 | CP-193055 | 0264 | 1 | B | Location Service ProvidePositioningInfo | 16.2.0 |
| 2019-12 | CT#86 | CP-193122 | 0266 | 2 | B | NF/NF Service Set ID in UE Context Transfer | 16.2.0 |
| 2019-12 | CT#86 | CP-193031 | 0268 | 1 | A | Definition of hpcfId | 16.2.0 |
| 2019-12 | CT#86 | CP-193080 | 0270 | 3 | A | Secondary RAT Data Usage Report | 16.2.0 |
| 2019-12 | CT#86 | CP-193055 | 0273 | 1 | B | AMF forwarding Location services messages beween UE and LMF | 16.2.0 |
| 2019-12 | CT#86 | CP-193044 | 0275 | | F | TS 29.518 API version update | 16.2.0 |
| 2020-03 | CT#87 | CP-200017 | 0276 | 3 | F | SMF change indication during Inter-AMF registration | 16.3.0 |
| 2020-03 | CT#87 | CP-200020 | 0277 | 3 | F | DNN encoding in Namf_Communication API | 16.3.0 |
| 2020-03 | CT#87 | CP-200043 | 0279 | 2 | F | smsSupport attribute in UE context | 16.3.0 |
| 2020-03 | CT#87 | CP-200043 | 0280 | 2 | F | AMF event subscription without the "options" attribute | 16.3.0 |
| 2020-03 | CT#87 | CP-200039 | 0281 | 2 | D | Editorial corrections | 16.3.0 |
| 2020-03 | CT#87 | CP-200043 | 0282 | 1 | F | Correction of typos | 16.3.0 |
| 2020-03 | CT#87 | CP-200043 | 0283 | 2 | F | Class indication in subscription response | 16.3.0 |
| 2020-03 | CT#87 | CP-200043 | 0284 | 3 | F | Cause values for PWS errors detected by AMF | 16.3.0 |
| 2020-03 | CT#87 | CP-200039 | 0285 | 2 | F | Correction - formatting consistency | 16.3.0 |
| 2020-03 | CT#87 | CP-200020 | 0286 | 1 | B | 29518 CR optionality of ProblemDetails | 16.3.0 |
| 2020-03 | CT#87 | CP-200031 | 0287 | 1 | B | Additional Access Type in UE Context Transfer | 16.3.0 |
| 2020-03 | CT#87 | CP-200017 | 0288 | 1 | B | Granularity of the SMF change Indication | 16.3.0 |
| 2020-03 | CT#87 | CP-200179 | 0289 | 1 | B | V2X information in UE Context | 16.3.0 |
| 2020-03 | CT#87 | CP-200178 | 0290 | 1 | B | Availability after DDN Failure | 16.3.0 |
| 2020-03 | CT#87 | CP-200020 | 0294 | 1 | B | Ongoing registration or handover during paging | 16.3.0 |
| 2020-03 | CT#87 | CP-200033 | 0295 | 1 | B | 5G CIOT Attribute in UeContext | 16.3.0 |
| 2020-03 | CT#87 | CP-200030 | 0296 | 2 | B | Event Exposure invoked by NWDAF | 16.3.0 |
| 2020-03 | CT#87 | CP-200017 | 0297 | 1 | F | V-SMF insertion or removal | 16.3.0 |
| 2020-03 | CT#87 | CP-200033 | 0298 | | F | Feature definition for support of CIoT features | 16.3.0 |
| 2020-03 | CT#87 | CP-200033 | 0299 | | F | Mobile Terminated Data | 16.3.0 |
| 2020-03 | CT#87 | CP-200043 | 0300 | | F | UE_IN_NON_ALLOWED_AREA error in EnableUEReachability response | 16.3.0 |
| 2020-03 | CT#87 | CP-200035 | 0302 | 1 | B | SUPI pattern | 16.3.0 |
| 2020-03 | CT#87 | CP-200018 | 0303 | | B | LCS service authorization | 16.3.0 |
| 2020-03 | CT#87 | CP-200018 | 0305 | 3 | B | Cm state exposure | 16.3.0 |
| 2020-03 | CT#87 | CP-200052 | 0306 | | F | TS 29.518 API Rel16 API External doc update | 16.3.0 |

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|---|---|---|---|---|---|---|---|
| 2020-06 | CT#88e | CP-201054 | 0307 | | F | Storage of YAML files in ETSI Forge | 16.4.0 |
| 2020-06 | CT#88e | CP-201031 | 0308 | | F | V-SMF and I-SMF service instance Id | 16.4.0 |
| 2020-06 | CT#88e | CP-201054 | 0309 | 1 | F | N1N2Transfer Failure Notification for UEs in RRC Inactive state | 16.4.0 |
| 2020-06 | CT#88e | CP-201045 | 0310 | 1 | B | NPN extensions for Inter-AMF N2 Handover | 16.4.0 |
| 2020-06 | CT#88e | CP-201054 | 0311 | 1 | F | Supported Headers Tables for Response codes 2xx and 3xx | 16.4.0 |
| 2020-06 | CT#88e | CP-201054 | 0312 | 1 | F | Binary Data Types Table | 16.4.0 |
| 2020-06 | CT#88e | CP-201046 | 0313 | 1 | B | Maximum UP resources activation of 2 PDU sessions | 16.4.0 |
| 2020-06 | CT#88e | CP-201054 | 0314 | 1 | F | Add new Notifications Overview Tables | 16.4.0 |
| 2020-06 | CT#88e | CP-201054 | 0315 | | F | subscriptionId in AmfCreatedEventSubscription and AmfEventReport | 16.4.0 |
| 2020-06 | CT#88e | CP-201054 | 0316 | | F | Non-delivery of N1 message to UE due to Xn/N2 handover | 16.4.0 |
| 2020-06 | CT#88e | CP-201054 | 0318 | | F | Reference Corrections | 16.4.0 |
| 2020-06 | CT#88e | CP-201034 | 0319 | 1 | F | Optionality of ProblemDetails in TS29.518 cleanup | 16.4.0 |
| 2020-06 | CT#88e | CP-201034 | 0321 | | F | Default LocationFilter | 16.4.0 |
| 2020-06 | CT#88e | CP-201067 | 0322 | 2 | B | MDT Configuration | 16.4.0 |
| 2020-06 | CT#88e | CP-201043 | 0323 | 2 | B | Update the event subscription and notification on area of interest | 16.4.0 |
| 2020-06 | CT#88e | CP-201047 | 0324 | 2 | B | Authentication and Authorization status | 16.4.0 |
| 2020-06 | CT#88e | CP-201048 | 0325 | 1 | F | Stage 2 procedures for wireline access | 16.4.0 |
| 2020-06 | CT#88e | CP-201048 | 0326 | 1 | F | TWAP ID change reporting | 16.4.0 |
| 2020-06 | CT#88e | CP-201054 | 0328 | 2 | F | Periodic reporting | 16.4.0 |
| 2020-06 | CT#88e | CP-201054 | 0330 | 1 | F | Reasons for loss of connectivity | 16.4.0 |
| 2020-06 | CT#88e | CP-201023 | 0331 | 2 | F | UEContextTransfer - N3IWF/W-AGF/TNGF ID and RAN NGAP ID | 16.4.0 |
| 2020-06 | CT#88e | CP-201018 | 0339 | 2 | A | Binary IE Encoding | 16.4.0 |
| 2020-06 | CT#88e | CP-201054 | 0340 | 1 | F | Broadcast Empty Area List | 16.4.0 |
| 2020-06 | CT#88e | CP-201044 | 0341 | 1 | F | Clarification on EBI Allocation for MAPDU | 16.4.0 |
| 2020-06 | CT#88e | CP-201032 | 0342 | | F | Correct Reference on Location Procedures | 16.4.0 |
| 2020-06 | CT#88e | CP-201046 | 0343 | 4 | B | UE Maximum Availability Time | 16.4.0 |
| 2020-06 | CT#88e | CP-201023 | 0344 | 3 | A | Event of UE Reachability | 16.4.0 |
| 2020-06 | CT#88e | CP-201032 | 0345 | 1 | F | GUAMI in N1/N2 Message Notification | 16.4.0 |
| 2020-06 | CT#88e | CP-201032 | 0346 | 1 | F | LCS Correlation Id for NRPPa Transfer | 16.4.0 |
| 2020-06 | CT#88e | CP-201054 | 0347 | 1 | F | PWS Message Transfer Precedence | 16.4.0 |
| 2020-06 | CT#88e | CP-201054 | 0348 | 1 | F | Data type column in Resource URI variables Table | 16.4.0 |
| 2020-06 | CT#88e | CP-201054 | 0349 | 1 | F | Add custom operation Name | 16.4.0 |
| 2020-06 | CT#88e | CP-201046 | 0350 | 2 | B | Monitoring Event Information | 16.4.0 |
| 2020-06 | CT#88e | CP-201032 | 0351 | 2 | F | LMF indicating access type for transmission of LPP message | 16.4.0 |
| 2020-06 | CT#88e | CP-201032 | 0352 | 1 | F | UePrivacyRequirements for Location Request | 16.4.0 |
| 2020-06 | CT#88e | CP-201044 | 0354 | 1 | F | Condition of MA-PDU Session Context Transfer | 16.4.0 |
| 2020-06 | CT#88e | CP-201054 | 0355 | 1 | F | N2 PDU Session Modification for a UE in CM-IDLE state | 16.4.0 |
| 2020-06 | CT#88e | CP-201032 | 0356 | 1 | F | GMLC authorization in RequestPosInfo | 16.4.0 |
| 2020-06 | CT#88e | CP-201197 | 0357 | 1 | F | PC5 policy container from PCF | 16.4.0 |
| 2020-06 | CT#88e | CP-201054 | 0358 | 2 | F | Maximum number of reports | 16.4.0 |
| 2020-06 | CT#88e | CP-201054 | 0359 | | F | Correction for implementation error | 16.4.0 |
| 2020-06 | CT#88e | CP-201032 | 0362 | 1 | B | Indication of control plane CIoT 5GS optimization to an LMF | 16.4.0 |
| 2020-06 | CT#88e | CP-201043 | 0367 | 1 | F | Sampling ratio for AMF event exposure | 16.4.0 |
| 2020-06 | CT#88e | CP-201032 | 0368 | 1 | F | The result of location verification by UE | 16.4.0 |
| 2020-06 | CT#88e | CP-201043 | 0369 | 2 | F | AMF event exposure for any UE | 16.4.0 |
| 2020-06 | CT#88e | CP-201018 | 0371 | 1 | A | pwdErrorInfo should be pwsErrorInfo in openAPI | 16.4.0 |
| 2020-06 | CT#88e | CP-201073 | 0375 | | F | 29.518 Rel-16 API version and External doc update | 16.4.0 |
| 2020-09 | CT#89e | CP-202097 | 0376 | 2 | F | DAPS Handover information | 16.5.0 |
| 2020-09 | CT#89e | CP-202114 | 0378 | 3 | F | Clarification on hSmfId in PduSessionContext transferred to target AMF | 16.5.0 |
| 2020-09 | CT#89e | CP-202093 | 0379 | 2 | F | Clairification on Max Number of Reports | 16.5.0 |
| 2020-09 | CT#89e | CP-202093 | 0380 | 2 | F | Event Reort in Response to AMF Event Subscription Update | 16.5.0 |
| 2020-09 | CT#89e | CP-202109 | 0381 | 1 | F | SNSSAI during mobility procedure | 16.5.0 |
| 2020-09 | CT#89e | CP-202093 | 0382 | | F | Callback URI correction | 16.5.0 |
| 2020-09 | CT#89e | CP-202093 | 0383 | 1 | A | Definition of DRX | 16.5.0 |
| 2020-09 | CT#89e | CP-202093 | 0384 | 2 | A | Cardinality of AmfUpdateEventSubscriptionItem | 16.5.0 |
| 2020-09 | CT#89e | CP-202093 | 0385 | | F | Identifier of the NF service consumer sending an N1 message | 16.5.0 |
| 2020-09 | CT#89e | CP-202093 | 0386 | | F | Clarifications to EBI Assignment procedure | 16.5.0 |
| 2020-09 | CT#89e | CP-202043 | 0388 | | A | Correction of UE Context Transfer payload in case of UE initial registration | 16.5.0 |
| 2020-09 | CT#89e | CP-202043 | 0392 | 1 | A | Registration Status Update for PCF for UE Policy | 16.5.0 |
| 2020-09 | CT#89e | CP-202093 | 0394 | 1 | F | Additional PraId | 16.5.0 |
| 2020-09 | CT#89e | CP-202093 | 0395 | 1 | F | PCF Group Id | 16.5.0 |
| 2020-09 | CT#89e | CP-202040 | 0397 | 1 | A | Selected EPS NAS Security Algorithm_Rel16 | 16.5.0 |
| 2020-09 | CT#89e | CP-202112 | 0398 | 1 | F | Removal of EN on CP 5G CIoT Optimisation | 16.5.0 |
| 2020-09 | CT#89e | CP-202112 | 0399 | 1 | F | Correction of Notification or Verification only for UE Positioning | 16.5.0 |
| 2020-09 | CT#89e | CP-202108 | 0400 | 2 | F | Managing RACS ID for mobility across ePLMNs | 16.5.0 |
| 2020-09 | CT#89e | CP-202093 | 0401 | 1 | F | Correction of n2InfoNotifyUrl in figures | 16.5.0 |
| 2020-09 | CT#89e | CP-202112 | 0402 | 2 | F | Add Response Codes on operation provide-pos-info | 16.5.0 |
| 2020-09 | CT#89e | CP-202112 | 0403 | | F | Corrections on N2InformationNotification | 16.5.0 |
| 2020-09 | CT#89e | CP-202096 | 0407 | | F | 29.518 Rel-16 API version and External doc update | 16.5.0 |
| 2020-12 | CT#90e | CP-203050 | 0409 | 1 | F | Broadcast of Assistance Data by an LMF | 16.6.0 |
| 2020-12 | CT#90e | CP-203050 | 0410 | 1 | F | Serving Cell Id in N1MessageNotification | 16.6.0 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2020-12 | CT#90e | CP-203080 | 0411 | 3 | F | Supplement to UeContext | 16.6.0 |
| 2020-12 | CT#90e | CP-203030 | 0413 | | F | Clarification on usage of "locationAge" and "geoInfo" in ProvideLocInfo | 16.6.0 |
| 2020-12 | CT#90e | CP-203030 | 0414 | | F | Incorrect NOTE | 16.6.0 |
| 2020-12 | CT#90e | CP-203163 | 0415 | 1 | F | HTTP 3xx redirection | 16.6.0 |
| 2020-12 | CT#90e | CP-203048 | 0417 | 1 | F | IMS AS query for UE IP Reachability | 16.6.0 |
| 2020-12 | CT#90e | CP-203035 | 0418 | 1 | F | UE Reachability Status Change | 16.6.0 |
| 2020-12 | CT#90e | CP-203040 | 0420 | 2 | F | Transfer N2 SM Info Received from SMF to Target AMF | 16.6.0 |
| 2020-12 | CT#90e | CP-203048 | 0421 | 2 | F | Miscellaneous corrections | 16.6.0 |
| 2020-12 | CT#90e | CP-203045 | 0422 | 1 | F | Partial failure of event subscription | 16.6.0 |
| 2020-12 | CT#90e | CP-203054 | 0423 | | F | SBI Binding Level | 16.6.0 |
| 2020-12 | CT#90e | CP-203030 | 0425 | 2 | F | Current location of a UE | 16.6.0 |
| 2020-12 | CT#90e | CP-203030 | 0426 | 1 | F | CreateUEContext Failue | 16.6.0 |
| 2020-12 | CT#90e | CP-203041 | 0430 | 1 | F | Event Subscription Synchronization | 16.6.0 |
| 2020-12 | CT#90e | CP-203054 | 0431 | 1 | F | HPCF Set Id | 16.6.0 |
| 2020-12 | CT#90e | CP-203027 | 0433 | 1 | A | Initial Location | 16.6.0 |
| 2020-12 | CT#90e | CP-203030 | 0437 | 3 | F | Corrections for unused data types and OperationId in OpenAPI | 16.6.0 |
| 2020-12 | CT#90e | CP-203048 | 0438 | | F | User Location | 16.6.0 |
| 2020-12 | CT#90e | CP-203027 | 0439 | | A | Event subscription update | 16.6.0 |
| 2020-12 | CT#90e | CP-203036 | 0441 | | F | 29.518 Rel-16 API version and External doc update | 16.6.0 |
| 2020-12 | CT#90e | CP-203064 | 0412 | 1 | F | Essential corrections | 17.0.0 |
| 2020-12 | CT#90e | CP-203057 | 0424 | 1 | F | EBI and ARP mapping update | 17.0.0 |
| 2020-12 | CT#90e | CP-203057 | 0427 | | F | N51 interface between NEF and AMF | 17.0.0 |
| 2020-12 | CT#90e | CP-203055 | 0442 | | F | 29.518 Rel-17 API version and External doc update | 17.0.0 |
| 2021-03 | CT#91e | CP-210178 | 0443 | 2 | F | Subscription not found inconsistency | 17.1.0 |
| 2021-03 | CT#91e | CP-210021 | 0444 | 2 | B | NF discovery based on SUCI information | 17.1.0 |
| 2021-03 | CT#91e | CP-210177 | 0446 | 2 | A | Handover Reject during EPS to 5GS Handover with AMF Re-allocation | 17.1.0 |
| 2021-03 | CT#91e | CP-210157 | 0448 | 1 | A | Handover Cancel during EPS to 5GS Handover with AMF Re-allocation | 17.1.0 |
| 2021-03 | CT#91e | CP-210159 | 0450 | 1 | A | Encoding of Forward Relocation Request | 17.1.0 |
| 2021-03 | CT#91e | CP-210040 | 0452 | 1 | A | DNN and Selected DNN | 17.1.0 |
| 2021-03 | CT#91e | CP-210037 | 0454 | 1 | A | Binding information of AMF event subscriptions | 17.1.0 |
| 2021-03 | CT#91e | CP-210173 | 0456 | | A | Error Responses for Indirect Communication | 17.1.0 |
| 2021-03 | CT#91e | CP-210043 | 0458 | 1 | A | UE context transfer during Inter-PLMN mobility registration | 17.1.0 |
| 2021-03 | CT#91e | CP-210043 | 0460 | | A | User Location in ProvideLocInfo | 17.1.0 |
| 2021-03 | CT#91e | CP-210059 | 0462 | | A | EBI allocation for Emergency PDU sessions | 17.1.0 |
| 2021-03 | CT#91e | CP-210059 | 0464 | 1 | A | Implementation error | 17.1.0 |
| 2021-03 | CT#91e | CP-210049 | 0468 | 2 | A | Interworking S-NSSAI during EPS to 5GS handover with AMF Relocation | 17.1.0 |
| 2021-03 | CT#91e | CP-210041 | 0470 | 1 | A | Target Node in Location continuity for handover from NG-RAN | 17.1.0 |
| 2021-03 | CT#91e | CP-210030 | 0471 | 1 | F | Wrong Reference for Reachable of Regulatory | 17.1.0 |
| 2021-03 | CT#91e | CP-210043 | 0473 | 1 | A | Corrections on resource and notification URI | 17.1.0 |
| 2021-03 | CT#91e | CP-210043 | 0475 | | A | Storage of YAML files | 17.1.0 |
| 2021-03 | CT#91e | CP-210161 | 0477 | 1 | A | Add the missing MDT parameters for NR | 17.1.0 |
| 2021-03 | CT#91e | CP-210048 | 0479 | 1 | A | Corrections on Enhanced Coverage information | 17.1.0 |
| 2021-03 | CT#91e | CP-210048 | 0481 | 1 | A | UE Differentiation Information | 17.1.0 |
| 2021-03 | CT#91e | CP-210021 | 0482 | 2 | F | Clarification to Communication-Failure-Report | 17.1.0 |
| 2021-03 | CT#91e | CP-210155 | 0483 | 3 | B | PRA Information update | 17.1.0 |
| 2021-03 | CT#91e | CP-210046 | 0485 | | A | 4xx codes during event notification | 17.1.0 |
| 2021-03 | CT#91e | CP-210048 | 0487 | 1 | F | Correction on UE Reachability | 17.1.0 |
| 2021-03 | CT#91e | CP-210046 | 0489 | | A | Support of immediate report | 17.1.0 |
| 2021-03 | CT#91e | CP-210029 | 0490 | | F | 29.518 Rel-17 API version and External doc update | 17.1.0 |
| 2021-06 | CT#92e | CP-211076 | 0493 | | A | Indicating the Serving PLMN ID to the Target AMF during inter-AMF handover | 17.2.0 |
| 2021-06 | CT#92e | CP-211076 | 0495 | | A | PDU session contexts transfer during a UE initial registration | 17.2.0 |
| 2021-06 | CT#92e | CP-211063 | 0497 | 1 | A | LMF using AMF event exposure service | 17.2.0 |
| 2021-06 | CT#92e | CP-211065 | 0499 | 1 | F | PPI mapping | 17.2.0 |
| 2021-06 | CT#92e | CP-211083 | 0502 | 1 | A | Incomplete Implementation of CR | 17.2.0 |
| 2021-06 | CT#92e | CP-211047 | 0503 | 2 | F | RAN Node Level Location Accuracy | 17.2.0 |
| 2021-06 | CT#92e | CP-211054 | 0504 | 2 | F | UE Reachability with Not Allowed Areas | 17.2.0 |
| 2021-06 | CT#92e | CP-211023 | 0507 | 1 | F | hNRF from NSSF in home PLMN | 17.2.0 |
| 2021-06 | CT#92e | CP-211058 | 0508 | | F | OpenAPI Reference | 17.2.0 |
| 2021-06 | CT#92e | CP-211059 | 0511 | | F | NF type of consumer subscribing to AMF event | 17.2.0 |
| 2021-06 | CT#92e | CP-211039 | 0512 | 1 | B | Authorization and QoS data for ProSe services | 17.2.0 |
| 2021-06 | CT#92e | CP-211036 | 0513 | 2 | B | Event Exposure enhancement with Partitioning criteria at AMF | 17.2.0 |
| 2021-06 | CT#92e | CP-211067 | 0515 | 2 | A | Maximum Response Time in the EE subscription request | 17.2.0 |
| 2021-06 | CT#92e | CP-211028 | 0518 | | F | Data Type Description for Namf_MT Service API | 17.2.0 |
| 2021-06 | CT#92e | CP-211028 | 0519 | | F | Data Type Description for Namf_Location Service API | 17.2.0 |
| 2021-06 | CT#92e | CP-211030 | 0521 | 1 | B | New parameter Subscribed-UE-Slice-MBR added | 17.2.0 |
| 2021-06 | CT#92e | CP-211028 | 0522 | 1 | F | Terminating Domain Selection request during on-going Registration | 17.2.0 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2021-06 | CT#92e | CP-211028 | 0523 | 2 | F | Network Triggered Service Request for UE in CM-CONNECTED state outside of the validity area included in N1N2MessageTransfer Request | 17.2.0 |
| 2021-06 | CT#92e | CP-211065 | 0525 | | A | Network Provided Location Information for non-3GPP access | 17.2.0 |
| 2021-06 | CT#92e | CP-211076 | 0527 | 1 | A | Group subscription transfer during inter-AMF mobility | 17.2.0 |
| 2021-06 | CT#92e | CP-211077 | 0529 | 1 | A | IAB Authorization for Inter-AMF handover | 17.2.0 |
| 2021-06 | CT#92e | CP-211026 | 0530 | 1 | B | GMLC using AMF event exposure service | 17.2.0 |
| 2021-06 | CT#92e | CP-211032 | 0531 | 1 | B | N1N2MessageTransfer supporting MUSIM | 17.2.0 |
| 2021-06 | CT#92e | CP-211026 | 0533 | 1 | B | Add Local location | 17.2.0 |
| 2021-06 | CT#92e | CP-211076 | 0535 | 1 | A | Registration with AMF re-direction | 17.2.0 |
| 2021-06 | CT#92e | CP-211036 | 0536 | 1 | B | Analytics subscription information | 17.2.0 |
| 2021-06 | CT#92e | CP-211036 | 0537 | 1 | B | S-NSSAIs per TA mapping event | 17.2.0 |
| 2021-06 | CT#92e | CP-211059 | 0541 | | A | Redirect Response for Namf_Communication | 17.2.0 |
| 2021-06 | CT#92e | CP-211036 | 0545 | 1 | B | Support of Mute Reporting | 17.2.0 |
| 2021-06 | CT#92e | CP-211059 | 0546 | | A | Redirect Response for Namf_EventExposure | 17.2.0 |
| 2021-06 | CT#92e | CP-211059 | 0548 | | A | Redirect Response for Namf_MT | 17.2.0 |
| 2021-06 | CT#92e | CP-211059 | 0550 | 1 | A | Redirect Response for Namf_Location | 17.2.0 |
| 2021-06 | CT#92e | CP-211059 | 0555 | 1 | A | Missing 307 and 308 for Namf_Communication | 17.2.0 |
| 2021-06 | CT#92e | CP-211028 | 0556 | 1 | F | Data Type Description for Namf_Communication Service API | 17.2.0 |
| 2021-06 | CT#92e | CP-211028 | 0557 | 1 | F | Data Type Description for Namf_EventExposure Service API | 17.2.0 |
| 2021-06 | CT#92e | CP-211028 | 0558 | 1 | F | Corrections on cardinality issues | 17.2.0 |
| 2021-06 | CT#92e | CP-211062 | 0564 | | A | hSmfId in PduSessionContext transferred to target AMF | 17.2.0 |
| 2021-06 | CT#92e | CP-211050 | 0565 | | F | 29.518 Rel-17 API version and External doc update | 17.2.0 |
| 2021-09 | CT#93e | CP-212026 | 0568 | | C | Broadcast Empty Area List for Write-Replace-Warning Request NG-RAN | 17.3.0 |
| 2021-09 | CT#93e | CP-212026 | 0569 | 1 | F | Corrections to NGAP messages | 17.3.0 |
| 2021-09 | CT#93e | CP-212082 | 0572 | 1 | A | AM Policy Information | 17.3.0 |
| 2021-09 | CT#93e | CP-212051 | 0573 | 1 | F | The maxReports IE in AmfEvent or AmfEventMode | 17.3.0 |
| 2021-09 | CT#93e | CP-212051 | 0574 | | F | Resource archetype correction | 17.3.0 |
| 2021-09 | CT#93e | CP-212041 | 0575 | | B | NF Services consumed by DCCF | 17.3.0 |
| 2021-09 | CT#93e | CP-212051 | 0576 | | F | Binding Indication | 17.3.0 |
| 2021-09 | CT#93e | CP-212026 | 0578 | | F | Missing errors in AMF APIs | 17.3.0 |
| 2021-09 | CT#93e | CP-212034 | 0580 | 1 | B | Add UE Positioning Capabilities | 17.3.0 |
| 2021-09 | CT#93e | CP-212097 | 0582 | | A | 3xx description correction for SCP | 17.3.0 |
| 2021-09 | CT#93e | CP-212034 | 0585 | 1 | B | Multiple QoS Class | 17.3.0 |
| 2021-09 | CT#93e | CP-212037 | 0586 | | B | Dynamic management of group based event monitoring | 17.3.0 |
| 2021-09 | CT#93e | CP-212051 | 0591 | 1 | F | NfInstanceId of CBCF | 17.3.0 |
| 2021-09 | CT#93e | CP-212036 | 0592 | | F | Miscellaneous 5G ProSe related corrections and updates | 17.3.0 |
| 2021-09 | CT#93e | CP-212059 | 0593 | | F | 29.518 Rel-17 API version and External doc update | 17.3.0 |
| 2021-12 | CT#94e | CP-213097 | 0596 | | B | Introduction of new AMF services (MBSBroadcast and MBSCommunication) to clauses 4.1 and 5.1 | 17.4.0 |
| 2021-12 | CT#94e | CP-213097 | 0597 | | B | MBSBroadcast API - Overview of MBSBroadcast service | 17.4.0 |
| 2021-12 | CT#94e | CP-213097 | 0598 | 4 | B | MBSBroadcast API - ContextCreate service operation | 17.4.0 |
| 2021-12 | CT#94e | CP-213097 | 0599 | 2 | B | MBSBroadcast API - ContextUpdate service operation | 17.4.0 |
| 2021-12 | CT#94e | CP-213097 | 0600 | | B | MBSBroadcast API - ContextRelease service operation | 17.4.0 |
| 2021-12 | CT#94e | CP-213097 | 0601 | | B | MBSBroadcast API - Resources and methods overview | 17.4.0 |
| 2021-12 | CT#94e | CP-213097 | 0602 | | B | MBSBroadcast API – Resource Definition – Broadcast MBS session contexts collection | 17.4.0 |
| 2021-12 | CT#94e | CP-213097 | 0603 | | B | MBSBroadcast API – Resource Definition – Individual broadcast MBS session context | 17.4.0 |
| 2021-12 | CT#94e | CP-213097 | 0604 | 3 | B | MBSBroadcast API - Data Model & OpenAPI for ContextCreate and ContextStatusNotify service operations | 17.4.0 |
| 2021-12 | CT#94e | CP-213097 | 0605 | 3 | B | MBSBroadcast API - Data Model & OpenAPI for ContextUpdate service operation | 17.4.0 |
| 2021-12 | CT#94e | CP-213097 | 0606 | 1 | B | MBSBroadcast API - Data Model & OpenAPI for ContextRelease service operation | 17.4.0 |
| 2021-12 | CT#94e | CP-213097 | 0608 | 1 | B | Namf_MBSCommnuication service description | 17.4.0 |
| 2021-12 | CT#94e | CP-213097 | 0609 | 1 | B | Namf_MBSCommnuication resources and methods | 17.4.0 |
| 2021-12 | CT#94e | CP-213097 | 0610 | 2 | B | Namf_MBSCommnuication data type and openAPI | 17.4.0 |
| 2021-12 | CT#94e | CP-213097 | 0611 | 1 | B | Namf_MBSCommnuication error handling | 17.4.0 |
| 2021-12 | CT#94e | CP-213086 | 0612 | 2 | F | Transfer UE radio capability for paging between AMFs | 17.4.0 |
| 2021-12 | CT#94e | CP-213112 | 0613 | 1 | B | UUAA-MM status indication in UE Context | 17.4.0 |
| 2021-12 | CT#94e | CP-213112 | 0614 | 1 | B | N1N2MessageTransfer update | 17.4.0 |
| 2021-12 | CT#94e | CP-213112 | 0615 | 1 | B | Notification enhancement with additional filtering | 17.4.0 |
| 2021-12 | CT#94e | CP-213102 | 0616 | 1 | B | AMF event correction | 17.4.0 |
| 2021-12 | CT#94e | CP-213100 | 0617 | 2 | B | UEContext for SNPN | 17.4.0 |
| 2021-12 | CT#94e | CP-213097 | 0618 | 1 | B | EnableGroupReachability service procedure | 17.4.0 |
| 2021-12 | CT#94e | CP-213097 | 0619 | 1 | B | EnableGroupReachability resource and OpenAPI definition | 17.4.0 |
| 2021-12 | CT#94e | CP-213101 | 0622 | 1 | B | AM Policy Association modification Time synchronization enhancement | 17.4.0 |
| 2021-12 | CT#94e | CP-213108 | 0623 | 2 | B | Notification for SM Policy Association Events | 17.4.0 |
| 2021-12 | CT#94e | CP-213096 | 0624 | 1 | B | Higher Resolution Timestamp for Location Estimates | 17.4.0 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2021-12 | CT#94e | CP-213096 | 0625 | 1 | F | UE Positioning Capabilities Data Type | 17.4.0 |
| 2021-12 | CT#94e | CP-213085 | 0627 | 1 | F | Correction on MT service | 17.4.0 |
| 2021-12 | CT#94e | CP-213087 | 0628 | 3 | F | S-NSSAI in PDU session context | 17.4.0 |
| 2021-12 | CT#94e | CP-213138 | 0630 | 1 | A | Idle Status Indication | 17.4.0 |
| 2021-12 | CT#94e | CP-213096 | 0633 | | B | Add scheduled location time | 17.4.0 |
| 2021-12 | CT#94e | CP-213097 | 0641 | | B | MBSBroadcast API - ContextStatusNotify service operation | 17.4.0 |
| 2021-12 | CT#94e | CP-213097 | 0642 | | B | MBSBroadcast API – Resource Definition - ContextStatusNotify service operation | 17.4.0 |
| 2021-12 | CT#94e | CP-213117 | 0643 | | F | One time location report when the current location cannot be obtained | 17.4.0 |
| 2021-12 | CT#94e | CP-213087 | 0644 | 1 | B | Minimal Report Interval | 17.4.0 |
| 2021-12 | CT#94e | CP-213117 | 0645 | | F | Event Subscription Modification Pattern | 17.4.0 |
| 2021-12 | CT#94e | CP-213138 | 0647 | 1 | A | Immediate Reporting | 17.4.0 |
| 2021-12 | CT#94e | CP-213138 | 0649 | 1 | A | Resolve EN for Event Subscription Sync | 17.4.0 |
| 2021-12 | CT#94e | CP-213145 | 0651 | 1 | A | 5GS User State Correction | 17.4.0 |
| 2021-12 | CT#94e | CP-213148 | 0654 | 1 | A | Essential Correction on N1N2MessageSubscribe for UE Policy | 17.4.0 |
| 2021-12 | CT#94e | CP-213098 | 0655 | 1 | B | MT_EnableUEReachability supporting MUSIM | 17.4.0 |
| 2021-12 | CT#94e | CP-213098 | 0656 | | F | Update MUSIM references | 17.4.0 |
| 2021-12 | CT#94e | CP-213117 | 0657 | 1 | F | Correcting citation to N1 messages | 17.4.0 |
| 2021-12 | CT#94e | CP-213110 | 0658 | 1 | B | Addition of UEs for group based event subscription | 17.4.0 |
| 2021-12 | CT#94e | CP-213086 | 0659 | | F | 3gpp-Sbi-Consumer-Info in UE Context | 17.4.0 |
| 2021-12 | CT#94e | CP-213087 | 0660 | 1 | F | Corrections related to the description fields in the OpenAPI descriptions | 17.4.0 |
| 2021-12 | CT#94e | CP-213121 | 0661 | | F | 29.518 Rel-17 API version and External doc update | 17.4.0 |
| 2022-03 | CT#95 | CP-220023 | 0664 | | F | New application error for ProvideDomainSelectionInfo (T-ADS) for a deregistered UE | 17.5.0 |
| 2022-03 | CT#95 | CP-220072 | 0665 | | F | Correction to CreateUEContext service operation regarding 5G-SRVCC | 17.5.0 |
| 2022-03 | CT#95 | CP-220081 | 0666 | 1 | A | Release of old access resources during Intra-AMF HO between 3GPP and non-3GPP accesses | 17.5.0 |
| 2022-03 | CT#95 | CP-220084 | 0667 | 1 | A | Secondary RAT data usage reporting over N14 during Inter-AMF handover | 17.5.0 |
| 2022-03 | CT#95 | CP-220034 | 0668 | 1 | B | Schedule location time for AMF | 17.5.0 |
| 2022-03 | CT#95 | CP-220051 | 0669 | | B | RedCap indication in UE context | 17.5.0 |
| 2022-03 | CT#95 | CP-220035 | 0670 | | F | Miscellaneous corrections | 17.5.0 |
| 2022-03 | CT#95 | CP-220048 | 0671 | 1 | F | Correction on time synchronization error budget | 17.5.0 |
| 2022-03 | CT#95 | CP-220055 | 0672 | | F | Essential Clarification for SMF ID in N1 Container | 17.5.0 |
| 2022-03 | CT#95 | CP-220049 | 0673 | 2 | B | UE Access Behavior Trends Event | 17.5.0 |
| 2022-03 | CT#95 | CP-220049 | 0674 | 2 | B | UE Location Trends Event | 17.5.0 |
| 2022-03 | CT#95 | CP-220049 | 0675 | 1 | B | UE MM Transactions Report Event | 17.5.0 |
| 2022-03 | CT#95 | CP-220035 | 0676 | | B | Correct the IEs included in EnableGroupReachability request | 17.5.0 |
| 2022-03 | CT#95 | CP-220035 | 0677 | | B | Service operation definition of UEReachabilityInfoNotify | 17.5.0 |
| 2022-03 | CT#95 | CP-220035 | 0678 | 1 | B | Resource and data types definition of UEReachabilityInfoNotify | 17.5.0 |
| 2022-03 | CT#95 | CP-220024 | 0679 | 1 | F | Rerouted Registration Request message | 17.5.0 |
| 2022-03 | CT#95 | CP-220024 | 0681 | 1 | F | Corrections related to the description fields in the OpenAPI descriptions | 17.5.0 |
| 2022-03 | CT#95 | CP-220025 | 0682 | 1 | B | Inter-PLMN mobility of PDU sessions | 17.5.0 |
| 2022-03 | CT#95 | CP-220035 | 0683 | | F | Miscellaneous corrections to the Namf_MBSBroadcast API | 17.5.0 |
| 2022-03 | CT#95 | CP-220035 | 0684 | | B | N2 MBS Session Management Information in Namf_MBSBroadcast API | 17.5.0 |
| 2022-03 | CT#95 | CP-220035 | 0685 | 1 | B | N2 MBS Session Management Information in Namf_MBSCommunication API | 17.5.0 |
| 2022-03 | CT#95 | CP-220065 | 0693 | | B | Additional Subscribed Policy Request Triggers in 3GPP R17 | 17.5.0 |
| 2022-03 | CT#95 | CP-220084 | 0696 | | A | Essential Correction on Policy Trigger | 17.5.0 |
| 2022-03 | CT#95 | CP-220081 | 0698 | | A | Missing Subscribed Policy Request Triggers in 3GPP R16 | 17.5.0 |
| 2022-03 | CT#95 | CP-220025 | 0699 | 2 | B | SM Context Consistency Validation | 17.5.0 |
| 2022-03 | CT#95 | CP-220076 | 0701 | 1 | A | Add UE triggered policy provisioning pocedure in N1MessageNotify operation | 17.5.0 |
| 2022-03 | CT#95 | CP-220079 | 0705 | | A | V-SMF removal during Inter-AMF registration | 17.5.0 |
| 2022-03 | CT#95 | CP-220049 | 0706 | | F | Removal of Editor's Note | 17.5.0 |
| 2022-03 | CT#95 | CP-220025 | 0707 | 1 | F | 204 No Content in subscription modification response | 17.5.0 |
| 2022-03 | CT#95 | CP-220044 | 0709 | 1 | B | Update on AMF exposure events | 17.5.0 |
| 2022-03 | CT#95 | CP-220081 | 0711 | 1 | A | 200 OK in subscription modification response | 17.5.0 |
| 2022-03 | CT#95 | CP-220066 | 0712 | | F | 29.518 Rel-17 API version and External doc update | 17.5.0 |
| 2022-06 | CT#96 | CP-221043 | 0716 | | F | N1MessageNotify for ProSe | 17.6.0 |
| 2022-06 | CT#96 | CP-221023 | 0717 | 1 | F | NGAP MB-SMF related IE | 17.6.0 |
| 2022-06 | CT#96 | CP-221023 | 0718 | 1 | F | MBS Service Area Information for Location dependent MBS session | 17.6.0 |
| 2022-06 | CT#96 | CP-221023 | 0719 | | F | MBS Service Area in Namf_MBSCommunication N2MessageTransfer Request | 17.6.0 |
| 2022-06 | CT#96 | CP-221024 | 0720 | 2 | F | Location dependent MBS broadcast session | 17.6.0 |
| 2022-06 | CT#96 | CP-221043 | 0721 | | F | pc5QoSPara attribute name in ProseContext data type | 17.6.0 |
| 2022-06 | CT#96 | CP-221027 | 0722 | | F | Correction of typos in description fields | 17.6.0 |
| 2022-06 | CT#96 | CP-221023 | 0723 | | F | Paging strategy handling for multicast MBS session | 17.6.0 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2022-06 | CT#96 | CP-221039 | 0724 | | F | Disaster Roaming Registration | 17.6.0 |
| 2022-06 | CT#96 | CP-221027 | 0725 | | F | Query parameters not complying with 29.501 naming conventions | 17.6.0 |
| 2022-06 | CT#96 | CP-221045 | 0726 | 1 | F | Released PDU Sessions during Registration | 17.6.0 |
| 2022-06 | CT#96 | CP-221045 | 0727 | 1 | F | Mobility between HPLMN and VPLMN | 17.6.0 |
| 2022-06 | CT#96 | CP-221024 | 0729 | 3 | F | Adding MBS session ID to ContextCreateRspData | 17.6.0 |
| 2022-06 | CT#96 | CP-221024 | 0730 | 2 | F | Adding SNSSAI to ContextCreateReqData | 17.6.0 |
| 2022-06 | CT#96 | CP-221024 | 0733 | 1 | F | Corrctions on HTTP Response | 17.6.0 |
| 2022-06 | CT#96 | CP-221024 | 0734 | 1 | F | Broadcast MBS Session Release Require procedure | 17.6.0 |
| 2022-06 | CT#96 | CP-221024 | 0735 | | F | N2 MBS Info Change Indicator | 17.6.0 |
| 2022-06 | CT#96 | CP-221024 | 0736 | 1 | F | Signaling of NG-RAN Node ID from AMF to MB-SMF | 17.6.0 |
| 2022-06 | CT#96 | CP-221061 | 0738 | | A | Updp Subscription Callback Binding | 17.6.0 |
| 2022-06 | CT#96 | CP-221061 | 0740 | 2 | A | Essential Correction for PCF Bindings | 17.6.0 |
| 2022-06 | CT#96 | CP-221033 | 0741 | | F | PCF Provided UE Slice MBR in UE Context | 17.6.0 |
| 2022-06 | CT#96 | CP-221068 | 0743 | | A | SmfSelInfo in UE Context | 17.6.0 |
| 2022-06 | CT#96 | CP-221024 | 0745 | 1 | F | Remove non-existent ngapData in the ContextCreateReqData as required in openAPI | 17.6.0 |
| 2022-06 | CT#96 | CP-221024 | 0746 | 1 | B | Reporting NG-RAN failure to MB-SMF | 17.6.0 |
| 2022-06 | CT#96 | CP-221022 | 0747 | 1 | F | Event Reporting in RRC inactive state | 17.6.0 |
| 2022-06 | CT#96 | CP-221033 | 0748 | 1 | F | Subscribed-UE-Slice-MBR | 17.6.0 |
| 2022-06 | CT#96 | CP-221068 | 0750 | | A | pc5QoSPara attribute name in V2xContext data type | 17.6.0 |
| 2022-06 | CT#96 | CP-221051 | 0751 | | F | 29.518 Rel-17 API version and External doc update | 17.6.0 |
| 2022-09 | CT#97 | CP-222047 | 0754 | 1 | F | Current Location | 17.7.0 |
| 2022-09 | CT#97 | CP-222031 | 0755 | 1 | F | NGAP MB-SMF related IE | 17.7.0 |
| 2022-09 | CT#97 | CP-222026 | 0758 | 1 | F | Handling of N1N2MessageTransfer Failure when UE in Non-Allowed Area | 17.7.0 |
| 2022-09 | CT#97 | CP-222053 | 0760 | | F | Correct the description of the provide-loc-info and the cancel-pos-info | 17.7.0 |
| 2022-09 | CT#97 | CP-222063 | 0764 | 1 | A | MSC Server Identity in Namf_Location_EventNotify during SRVCC handover | 17.7.0 |
| 2022-09 | CT#97 | CP-222031 | 0765 | | F | Corrections for the N2MbsSmInfo data type | 17.7.0 |
| 2022-09 | CT#97 | CP-222031 | 0766 | | F | Correction for the Namf_MBSBroadcast API | 17.7.0 |
| 2022-09 | CT#97 | CP-222061 | 0768 | 1 | A | SMSF Set and Binding Info | 17.7.0 |
| 2022-09 | CT#97 | CP-222043 | 0769 | 1 | F | SMF Security and Capability in PDU Session Context | 17.7.0 |
| 2022-09 | CT#97 | CP-222054 | 0770 | 2 | F | Create UE Context with AMF Relocation | 17.7.0 |
| 2022-09 | CT#97 | CP-222065 | 0772 | 1 | A | Timestamp for Periodic Event Reporting during Mobility | 17.7.0 |
| 2022-09 | CT#97 | CP-222047 | 0774 | 1 | F | EE Subscription Notification Error Handling | 17.7.0 |
| 2022-09 | CT#97 | CP-222068 | 0776 | 1 | A | 409 Response on Xn HO and Intra-AMF N2 HO | 17.7.0 |
| 2022-09 | CT#97 | CP-222053 | 0777 | 1 | F | Essential Clarification for N2 Info Subscription | 17.7.0 |
| 2022-09 | CT#97 | CP-222068 | 0783 | | A | AMF relocation in EPS to 5GS handover | 17.7.0 |
| 2022-09 | CT#97 | CP-222035 | 0785 | 1 | F | Removal of Editor's Note | 17.7.0 |
| 2022-09 | CT#97 | CP-222036 | 0786 | 1 | F | Indication of Network Assisted Positioning method | 17.7.0 |
| 2022-09 | CT#97 | CP-222031 | 0787 | 3 | F | Clarification to OperationStatus | 17.7.0 |
| 2022-09 | CT#97 | CP-222031 | 0788 | 1 | F | Correction to ContextCreateReqData | 17.7.0 |
| 2022-09 | CT#97 | CP-222057 | 0789 | 1 | F | EPS interworking Info in UE Context | 17.7.0 |
| 2022-09 | CT#97 | CP-222058 | 0790 | | F | 29.518 Rel-17 API version and External doc update | 17.7.0 |
| 2022-09 | CT#97 | CP-222232 | 0793 | | F | Inserting missing clarification on Max Number of Reports | 17.7.0 |
| 2022-12 | CT#98 | CP-223057 | 0794 | | F | Feature bit alignment | 17.8.0 |
| 2022-12 | CT#98 | CP-223036 | 0797 | 1 | F | Add MBS Update procedure to the ContextCreate service operation | 17.8.0 |
| 2022-12 | CT#98 | CP-223060 | 0799 | 1 | F | Inter AMF mobility when UE is registered in both 3GPP and non-3GPP | 17.8.0 |
| 2022-12 | CT#98 | CP-223036 | 0801 | 1 | F | Missing clauses for Namf_MBSBroadcast Service | 17.8.0 |
| 2022-12 | CT#98 | CP-223036 | 0802 | 1 | F | Missing clauses for Namf_MBSCommunication Service | 17.8.0 |
| 2022-12 | CT#98 | CP-223099 | 0810 | 3 | A | Area of Interest Event Status from Old AMF | 17.8.0 |
| 2022-12 | CT#98 | CP-223069 | 0812 | 1 | A | Missing IMEISV in N1N2 Message Subscription | 17.8.0 |
| 2022-12 | CT#98 | CP-223051 | 0815 | 1 | F | PLMN with disaster condition | 17.8.0 |
| 2022-12 | CT#98 | CP-223062 | 0817 | 1 | F | Correction of boolean type IEs | 17.8.0 |
| 2022-12 | CT#98 | CP-223066 | 0822 | | F | 29.518 Rel-17 API version and External doc update | 17.8.0 |
| 2022-12 | CT#98 | CP-223030 | 0800 | 1 | B | Inter AMF mobility when UE is registered in SNPN with different access | 18.0.0 |
| 2022-12 | CT#98 | CP-223035 | 0803 | 1 | B | Support of RRC_INACTIVE with long eDRX | 18.0.0 |
| 2022-12 | CT#98 | CP-223036 | 0804 | | B | Multicast MBS session (de)activation or update after an AMF failure | 18.0.0 |
| 2022-12 | CT#98 | CP-223036 | 0805 | | F | MBS session update for Broadcast | 18.0.0 |
| 2022-12 | CT#98 | CP-223028 | 0806 | 1 | F | Missing mandatory status codes in OpenAPI | 18.0.0 |
| 2022-12 | CT#98 | CP-223027 | 0808 | 1 | B | Retry Timer for N1N2 Transfer Failure | 18.0.0 |
| 2022-12 | CT#98 | CP-223058 | 0813 | 3 | F | Event Subscription Termination Notification | 18.0.0 |
| 2022-12 | CT#98 | CP-223062 | 0816 | 1 | F | Value of ageOfLocationInformation | 18.0.0 |
| 2022-12 | CT#98 | CP-223033 | 0821 | | F | 29.518 Rel-18 API version and External doc update | 18.0.0 |
| 2023-03 | CT#99 | CP-230039 | 0825 | 1 | B | Support Unavailability Duration in Namf_EE API | 18.1.0 |
| 2023-03 | CT#99 | CP-230090 | 0830 | | A | Essential Corrections on Resource URI | 18.1.0 |
| 2023-03 | CT#99 | CP-230062 | 0835 | | D | Editorial correction to ContextRelease | 18.1.0 |
| 2023-03 | CT#99 | CP-230036 | 0836 | 1 | B | Protocol support for MBS Session in MOCN | 18.1.0 |
| 2023-03 | CT#99 | CP-230077 | 0838 | 1 | A | Update operation status and event for broadcast session release | 18.1.0 |
| 2023-03 | CT#99 | CP-230049 | 0839 | | F | Miscellaneous Corrections | 18.1.0 |

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|---|---|---|---|---|---|---|---|
| 2023-03 | CT#99 | CP-230030 | 0840 | 1 | B | Parameters used to consume Namf_MT_enableUeReachability service | 18.1.0 |
| 2023-03 | CT#99 | CP-230046 | 0841 | 1 | B | Manage Event Muting Impact on NFp | 18.1.0 |
| 2023-03 | CT#99 | CP-230029 | 0843 | | B | OAuth2 scopes in the Namf_MT API | 18.1.0 |
| 2023-03 | CT#99 | CP-230029 | 0844 | 1 | B | OAuth2 scopes in the Namf_Communication API | 18.1.0 |
| 2023-03 | CT#99 | CP-230032 | 0845 | 1 | B | Local LMF and GMLC selection | 18.1.0 |
| 2023-03 | CT#99 | CP-230077 | 0847 | | A | Incorrect enable-group-reachability operation resource URI in the OpenAPI | 18.1.0 |
| 2023-03 | CT#99 | CP-230088 | 0849 | | A | Add nextPeriodicReportTime IE to Namf_EventExposure OpenAPI | 18.1.0 |
| 2023-03 | CT#99 | CP-230032 | 0852 | 2 | B | Support of Event Report Allowed Area | 18.1.0 |
| 2023-03 | CT#99 | CP-230093 | 0855 | 1 | A | Essential Corrections on AMF Events | 18.1.0 |
| 2023-03 | CT#99 | CP-230029 | 0856 | 2 | B | PWS N2 Subscription Unavailability Indication | 18.1.0 |
| 2023-03 | CT#99 | CP-230094 | 0859 | 1 | A | Missed AM Policy Information in UE Context | 18.1.0 |
| 2023-03 | CT#99 | CP-230031 | 0860 | 1 | B | Multiple location report for MT-LR Immediate Location Request for the regulatory service | 18.1.0 |
| 2023-03 | CT#99 | CP-230032 | 0863 | 1 | B | UE Unaware Positioning | 18.1.0 |
| 2023-03 | CT#99 | CP-230032 | 0864 | 2 | B | Support of low power and high accuracy positioning | 18.1.0 |
| 2023-03 | CT#99 | CP-230032 | 0865 | 1 | B | Location service in PNI-NPN with signalling optimisation | 18.1.0 |
| 2023-03 | CT#99 | CP-230041 | 0867 | 1 | B | Satellite backhaul category change | 18.1.0 |
| 2023-03 | CT#99 | CP-230042 | 0868 | | B | Event exposure subscribed by the TSCTSF | 18.1.0 |
| 2023-03 | CT#99 | CP-230062 | 0869 | 1 | F | Correct the name of the retry after timer | 18.1.0 |
| 2023-03 | CT#99 | CP-230062 | 0870 | 1 | F | Correction on UE Context Release | 18.1.0 |
| 2023-03 | CT#99 | CP-230071 | 0872 | | F | 29.518 Rel-18 API version and External doc update | 18.1.0 |
| 2023-06 | CT#100 | CP-231028 | 0871 | 4 | F | Location header description | 18.2.0 |
| 2023-06 | CT#100 | CP-231043 | 0876 | 1 | B | Presence-In-AOI-Report event with adjustment of the AoI based on the UE's RA | 18.2.0 |
| 2023-06 | CT#100 | CP-231043 | 0877 | 1 | B | Presence-In-AOI-Report event for RAN timing synchronization status change | 18.2.0 |
| 2023-06 | CT#100 | CP-231025 | 0878 | | F | Correct the table of the HTTP status code for N1N2MessageTransfer | 18.2.0 |
| 2023-06 | CT#100 | CP-231031 | 0879 | 1 | B | Location service bi-directional continuity between EPS and 5GS | 18.2.0 |
| 2023-06 | CT#100 | CP-231031 | 0880 | 2 | B | Support of PRUs | 18.2.0 |
| 2023-06 | CT#100 | CP-231047 | 0885 | | F | Correction of Event muting mechanism | 18.2.0 |
| 2023-06 | CT#100 | CP-231064 | 0886 | | F | iwkSnssai in EPS to 5GS handover procedure with AMF relocation | 18.2.0 |
| 2023-06 | CT#100 | CP-231064 | 0887 | | F | Correction on service operations in Namf_Communication service | 18.2.0 |
| 2023-06 | CT#100 | CP-231081 | 0889 | 1 | A | Pattern of path in AmfUpdateEventOptionItem | 18.2.0 |
| 2023-06 | CT#100 | CP-231083 | 0891 | | A | Add group member UE(s) for a group subscription | 18.2.0 |
| 2023-06 | CT#100 | CP-231075 | 0893 | 1 | A | Missing finer periodicities than 1s and an infinite reporting amount | 18.2.0 |
| 2023-06 | CT#100 | CP-231031 | 0894 | 1 | B | PRU Indication in N1N2MessageTransfer | 18.2.0 |
| 2023-06 | CT#100 | CP-231028 | 0895 | 1 | B | Omit UE IDs for Number of UEs in Area Event | 18.2.0 |
| 2023-06 | CT#100 | CP-231064 | 0898 | 3 | B | Support of RACS feature | 18.2.0 |
| 2023-06 | CT#100 | CP-231043 | 0899 | 1 | B | Service Operations for Subscription and management of network timing synchronization status monitoring | 18.2.0 |
| 2023-06 | CT#100 | CP-231043 | 0900 | 1 | B | Data Type for Subscription and management of network timing synchronization status monitoring | 18.2.0 |
| 2023-06 | CT#100 | CP-231030 | 0901 | 1 | B | SMF as the NF consumer of Namf_MT_enableUeReachability service | 18.2.0 |
| 2023-06 | CT#100 | CP-231069 | 0906 | 1 | F | Essential Correction on 202 Location Header of N1N2MessageTransfer | 18.2.0 |
| 2023-06 | CT#100 | CP-231029 | 0908 | 1 | F | Event Synchronization with No EE Subscription Indication via UECM | 18.2.0 |
| 2023-06 | CT#100 | CP-231047 | 0910 | 1 | B | Variable reporting periodicity | 18.2.0 |
| 2023-06 | CT#100 | CP-231083 | 0912 | 1 | A | Correction the name of taiList attribute in TargetArea | 18.2.0 |
| 2023-06 | CT#100 | CP-231031 | 0914 | 2 | B | Add reporting indication | 18.2.0 |
| 2023-06 | CT#100 | CP-231042 | 0915 | 1 | B | Update on ProseContext | 18.2.0 |
| 2023-06 | CT#100 | CP-231069 | 0917 | | F | Correct the data type UeContextTransferReqData name | 18.2.0 |
| 2023-06 | CT#100 | CP-231069 | 0918 | | F | Correct the table in provide-pos-info response | 18.2.0 |
| 2023-06 | CT#100 | CP-231048 | 0919 | 2 | B | Partially allowed Network slice | 18.2.0 |
| 2023-06 | CT#100 | CP-231031 | 0920 | 2 | B | Requested LMF ID is not available | 18.2.0 |
| 2023-06 | CT#100 | CP-231090 | 0923 | 1 | F | Incorrect OpenAPI definition of aoiStateList in AmfEventSubscriptionAddInfo | 18.2.0 |
| 2023-06 | CT#100 | CP-231070 | 0925 | | F | 29.518 Rel-18 API version and External doc update | 18.2.0 |
| 2023-09 | CT#101 | CP-232033 | 0928 | 1 | F | Subscription authorization in Context Transfer | 18.3.0 |
| 2023-09 | CT#101 | CP-232043 | 0930 | 1 | B | Updating N1messageNotification during AMF re-allocation procedure to include partially Allowed NSSAI | 18.3.0 |
| 2023-09 | CT#101 | CP-232036 | 0933 | 1 | B | HR-SBO Allowed indication during intra-PLMN Handover | 18.3.0 |
| 2023-09 | CT#101 | CP-232033 | 0934 | 1 | F | Event subscription rejection with UE_NOT_SERVED_BY_AMF application error | 18.3.0 |
| 2023-09 | CT#101 | CP-232046 | 0935 | 2 | B | A2X information in UE Context | 18.3.0 |
| 2023-09 | CT#101 | CP-232057 | 0937 | 1 | B | Update the ProseContext data type | 18.3.0 |
| 2023-09 | CT#101 | CP-232035 | 0938 | 1 | F | Correction on Reporting Indication | 18.3.0 |
| 2023-09 | CT#101 | CP-232043 | 0939 | 1 | B | Slice Usage Control Information in UE Context | 18.3.0 |
| 2023-09 | CT#101 | CP-232063 | 0941 | | A | Missed HA GNSS Metrics Support over SBI | 18.3.0 |

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|---|---|---|---|---|---|---|---|
| 2023-09 | CT#101 | CP-232053 | 0942 | 4 | B | Support of Mobile Base Station Relay in AMF services | 18.3.0 |
| 2023-09 | CT#101 | CP-232058 | 0944 | 1 | F | Correction of NGAP IE type description in N2InfoContent | 18.3.0 |
| 2023-09 | CT#101 | CP-232058 | 0945 | | F | Correction of Unsubscribe procedure description | 18.3.0 |
| 2023-09 | CT#101 | CP-232058 | 0946 | | F | Correction of references to the tables in the description | 18.3.0 |
| 2023-09 | CT#101 | CP-232049 | 0949 | 2 | B | PC5 QoS parameters related to RSPP transport over PC5 | 18.3.0 |
| 2023-09 | CT#101 | CP-232035 | 0950 | 1 | B | Support on Indoor/Outdoor indication | 18.3.0 |
| 2023-09 | CT#101 | CP-232062 | 0953 | 1 | A | Add GNSS integrity requirement | 18.3.0 |
| 2023-09 | CT#101 | CP-232058 | 0954 | | F | Correction on the description of DateTime | 18.3.0 |
| 2023-09 | CT#101 | CP-232049 | 0956 | 2 | B | Update on AMF service for MT procedures for ranging_SL | 18.3.0 |
| 2023-09 | CT#101 | CP-232033 | 0957 | | F | Correcting the definition of the maxRespTime attribute | 18.3.0 |
| 2023-09 | CT#101 | CP-232035 | 0958 | 1 | B | Support on NLOS/LOS measurement indication | 18.3.0 |
| 2023-09 | CT#101 | CP-232060 | 0959 | | F | 29.518 Rel-18 API version and External doc update | 18.3.0 |
| 2023-12 | CT#102 | CP-233037 | 0936 | 2 | B | Periodic or triggered location events via user plane to an LCS Client or AF | 18.4.0 |
| 2023-12 | CT#102 | CP-233036 | 0955 | 3 | B | Updates to support Extended DRX for RRC-INACTIVE state with CN based MT communication handling | 18.4.0 |
| 2023-12 | CT#102 | CP-233028 | 0961 | 1 | B | Event subscription not authorized by the target AMF | 18.4.0 |
| 2023-12 | CT#102 | CP-233037 | 0963 | 2 | B | Add LCS-UP context | 18.4.0 |
| 2023-12 | CT#102 | CP-233044 | 0964 | | F | Incorrect numbering for ExpectedUeBehavior | 18.4.0 |
| 2023-12 | CT#102 | CP-233063 | 0966 | 2 | A | Correct the N2MbsSMinfo in the MBS broadcast service | 18.4.0 |
| 2023-12 | CT#102 | CP-233044 | 0967 | 1 | F | Editorial corrections and adding missing attributes in the procedure description | 18.4.0 |
| 2023-12 | CT#102 | CP-233041 | 0968 | 2 | B | Updates on the RAN Timing Synchronization Status Report related N2 information | 18.4.0 |
| 2023-12 | CT#102 | CP-233044 | 0971 | 1 | F | Missed Service Operations in Tables | 18.4.0 |
| 2023-12 | CT#102 | CP-233045 | 0972 | 2 | F | Clarification on muting mechanism in AMF event exposure service | 18.4.0 |
| 2023-12 | CT#102 | CP-233041 | 0976 | 1 | B | UE reconnection indication in UE context in AMF | 18.4.0 |
| 2023-12 | CT#102 | CP-233041 | 0977 | 1 | B | Update Non-UE N2 subscription and notification | 18.4.0 |
| 2023-12 | CT#102 | CP-233033 | 0978 | 1 | B | Support for resource sharing across multiple broadcast MBS Sessions during network sharing | 18.4.0 |
| 2023-12 | CT#102 | CP-233050 | 0980 | 2 | B | Update on AMF services for ranging_SL | 18.4.0 |
| 2023-12 | CT#102 | CP-233044 | 0981 | 1 | F | Clarification on Unavailability Period | 18.4.0 |
| 2023-12 | CT#102 | CP-233031 | 0983 | 1 | F | ProblemDetails RFC 7807 obsoleted by 9457 | 18.4.0 |
| 2023-12 | CT#102 | CP-233031 | 0984 | 1 | F | HTTP RFCs obsoleted by IETF RFC 9110, 9111 and 9113 | 18.4.0 |
| 2023-12 | CT#102 | CP-233037 | 0986 | 1 | F | Reporting Indication Definition Alignment to Stage 2 | 18.4.0 |
| 2023-12 | CT#102 | CP-233037 | 0987 | 1 | B | Multiple QoS for Deferred Location Service Continuation from 5GS to EPS | 18.4.0 |
| 2023-12 | CT#102 | CP-233031 | 0988 | 2 | B | TAI Range List in AreaOfValidity | 18.4.0 |
| 2023-12 | CT#102 | CP-233041 | 0991 | 1 | F | Network Timing Synchronization Status feature in Table 6.1.8-1 | 18.4.0 |
| 2023-12 | CT#102 | CP-233041 | 0992 | 1 | B | New event filters for Presence-In-AOI-Report event subscription targeting Any UE | 18.4.0 |
| 2023-12 | CT#102 | CP-233077 | 0993 | 2 | B | Broadcast Session Transport procedure support by the ContextStatusNotify service operation | 18.4.0 |
| 2023-12 | CT#102 | CP-233040 | 0995 | 1 | F | UE_NOT_RESPONDING Application Error for 504 Response Missing | 18.4.0 |
| 2023-12 | CT#102 | CP-233037 | 0996 | | F | Update the support of PRU related procedures | 18.4.0 |
| 2023-12 | CT#102 | CP-233056 | 1000 | 2 | F | Corrections to support Extended DRX in CM-IDLE state | 18.4.0 |
| 2023-12 | CT#102 | CP-233050 | 1003 | 1 | B | Update on delivery of N2 information for rangingsl | 18.4.0 |
| 2023-12 | CT#102 | CP-233056 | 1004 | 1 | F | Update on delivery of N2 information for 5G Prose | 18.4.0 |
| 2023-12 | CT#102 | CP-233060 | 1006 | | F | 29.518 Rel-18 API version and External doc update | 18.4.0 |
| 2024-03 | CT#103 | CP-240039 | 1009 | 1 | B | Subscription to TSS information Reporting using Namf_NonUeN2MessageTransfer | 18.5.0 |
| 2024-03 | CT#103 | CP-240039 | 1010 | | B | Event Filters for Subscriptions to Presence-In-AOI-Report events | 18.5.0 |
| 2024-03 | CT#103 | CP-240062 | 1013 | 1 | A | UeInfoList and UeInfo | 18.5.0 |
| 2024-03 | CT#103 | CP-240028 | 1014 | | F | Immediate MDT | 18.5.0 |
| 2024-03 | CT#103 | CP-240028 | 1015 | | F | MDT User Consent for PLMNs adjacent to the serving PLMN | 18.5.0 |
| 2024-03 | CT#103 | CP-240035 | 1016 | | B | Fill in the missing RAN3 references for 5MBS NGAP IEs | 18.5.0 |
| 2024-03 | CT#103 | CP-240028 | 1020 | 3 | B | Addition of voice support match indicator in UE context | 18.5.0 |
| 2024-03 | CT#103 | CP-240048 | 1023 | 1 | B | Distinction of MBSR and IAB node indicators | 18.5.0 |
| 2024-03 | CT#103 | CP-240045 | 1025 | 1 | B | Aligning distance and location terminology with stage 2 | 18.5.0 |
| 2024-03 | CT#103 | CP-240029 | 1028 | | B | EnableUEReachability supporting additional N2 indication | 18.5.0 |
| 2024-03 | CT#103 | CP-240029 | 1029 | 2 | C | EnableUEReachability to support reporting of DL data size | 18.5.0 |
| 2024-03 | CT#103 | CP-240053 | 1030 | 2 | F | N1N2MessageTransfer enhancements for eDRX | 18.5.0 |
| 2024-03 | CT#103 | CP-240042 | 1031 | 1 | B | Alternative S-NSSAIs for PDU Sessions in UE Context | 18.5.0 |
| 2024-03 | CT#103 | CP-240042 | 1032 | 1 | B | Network Slice Deregistration Inactive Remaining Time | 18.5.0 |
| 2024-03 | CT#103 | CP-240030 | 1034 | 1 | B | Support of user plane positioning connection management procedures | 18.5.0 |
| 2024-03 | CT#103 | CP-240045 | 1036 | | B | coordinate id in case of absolute locations | 18.5.0 |
| 2024-03 | CT#103 | CP-240045 | 1037 | 1 | F | Corrections on Application Layer ID | 18.5.0 |
| 2024-03 | CT#103 | CP-240028 | 1038 | 1 | F | Editorial Corrections | 18.5.0 |
| 2024-03 | CT#103 | CP-240042 | 1039 | 2 | F | Support of Network Slice Replacement and area restrictions at UE mobility | 18.5.0 |
| 2024-03 | CT#103 | CP-240066 | 1040 | 1 | A | Feature Negotiation for Analytics Subscriptions Support via N14 | 18.5.0 |
| 2024-03 | CT#103 | CP-240056 | 1041 | | F | 29.518 Rel-18 API version and External doc update | 18.5.0 |

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| 2024-06 | CT#104 | CP-241028 | 1043 | | F | Attribute Name Correction | 18.6.0 |
| 2024-06 | CT#104 | CP-241042 | 1044 | 1 | F | Correct the description related to partially allowed NSSAI | 18.6.0 |
| 2024-06 | CT#104 | CP-241042 | 1045 | 2 | F | Remove the EN related to partially allowed NSSAI | 18.6.0 |
| 2024-06 | CT#104 | CP-241042 | 1046 | 4 | B | Subscribe to event notification for Slice Service Area | 18.6.0 |
| 2024-06 | CT#104 | CP-241029 | 1048 | 1 | F | Correct SMF behaviour of sending a new Namf_MT_EnableUEReachability | 18.6.0 |
| 2024-06 | CT#104 | CP-241028 | 1052 | 1 | F | Callbacks | 18.6.0 |
| 2024-06 | CT#104 | CP-241057 | 1054 | 1 | A | MBS service area in the EnableGroupReachabilityReqData | 18.6.0 |
| 2024-06 | CT#104 | CP-241050 | 1056 | 1 | F | Clarify the semantics of the S-NSSAI and the NSI in the AmfEventArea datatype | 18.6.0 |
| 2024-06 | CT#104 | CP-241039 | 1057 | | B | Restoration of TSS monitoring upon an NG-RAN failure with or w/o restart | 18.6.0 |
| 2024-06 | CT#104 | CP-241056 | 1059 | 1 | A | Integrity Result | 18.6.0 |
| 2024-06 | CT#104 | CP-241028 | 1061 | 1 | F | To be Released PDU Session for Other Reasons during Handover | 18.6.0 |
| 2024-06 | CT#104 | CP-241066 | 1064 | 1 | A | Add GPSI to 5GC-MT-LR Procedure without UDM Query | 18.6.0 |
| 2024-06 | CT#104 | CP-241050 | 1066 | 1 | F | UE ID Usage for UE Related Resources | 18.6.0 |
| 2024-06 | CT#104 | CP-241045 | 1067 | 2 | F | Updates on feature description | 18.6.0 |
| 2024-06 | CT#104 | CP-241050 | 1070 | 2 | F | Resolving the case of conflicting Target identifiers in the event subscription | 18.6.0 |
| 2024-06 | CT#104 | CP-241053 | 1069 | 3 | A | Content of JSON Patch requests | 18.6.0 |
| 2024-06 | CT#104 | CP-241050 | 1072 | | F | Service name corrections | 18.6.0 |
| 2024-06 | CT#104 | CP-241050 | 1074 | | F | Correction of feature negotiation description | 18.6.0 |
| 2024-06 | CT#104 | CP-241049 | 1075 | 1 | B | Support for the XR Device with 2Rx | 18.6.0 |
| 2024-06 | CT#104 | CP-241029 | 1076 | 1 | B | Support for the eRedCap UEs | 18.6.0 |
| 2024-06 | CT#104 | CP-241028 | 1077 | 1 | B | Handling of unsupported event subscription | 18.6.0 |
| 2024-06 | CT#104 | CP-241045 | 1078 | | F | NGAP Reference to Ranging and Sidelink Positioning Service Information | 18.6.0 |
| 2024-06 | CT#104 | CP-241050 | 1083 | | F | Write-Replace Warning Indication & Stop Warning Indication | 18.6.0 |
| 2024-06 | CT#104 | CP-241050 | 1084 | | F | EBI allocation applying to PDU sessions using SSC mode 1 | 18.6.0 |
| 2024-06 | CT#104 | CP-241028 | 1085 | 1 | F | Handling of NW triggered SR during an on-going UE triggered SR race condition | 18.6.0 |
| 2024-06 | CT#104 | CP-241038 | 1086 | 1 | B | Remote UE Indication in Location Information | 18.6.0 |
| 2024-06 | CT#104 | CP-241030 | 1087 | 1 | F | Update the description of EventNotify service | 18.6.0 |
| 2024-06 | CT#104 | CP-241050 | 1088 | 1 | F | Updates on naming convention for enumeration | 18.6.0 |
| 2024-06 | CT#104 | CP-241045 | 1089 | 1 | F | Alignment with naming conventions | 18.6.0 |
| 2024-06 | CT#104 | CP-241045 | 1091 | 1 | F | Notification for privacy check on UE for RangingSl | 18.6.0 |
| 2024-06 | CT#104 | CP-241052 | 1092 | | F | 29.518 Rel-18 API version and External doc update | 18.6.0 |
| 2024-09 | CT#105 | CP-242053 | 1095 | 1 | F | subregTimer in IdleStatusIndication | 18.7.0 |
| 2024-09 | CT#105 | CP-242237 | 1102 | 3 | F | Network Slice Deregistration Inactive timer information | 18.7.0 |
| 2024-09 | CT#105 | CP-242053 | 1106 | 1 | F | Reachability-Report event | 18.7.0 |
| 2024-09 | CT#105 | CP-242053 | 1108 | | F | Correction to the RAN-ID-LIST feature in the MBSCommunication API | 18.7.0 |
| 2024-09 | CT#105 | CP-242053 | 1109 | 1 | F | Incorrect reference to a table note | 18.7.0 |
| 2024-09 | CT#105 | CP-242054 | 1116 | | F | 29.518 Rel-18 API version and External doc update | 18.7.0 |
| 2024-12 | CT#106 | CP-243022 | 1118 | 1 | F | PDU Session Priority | 18.8.0 |
| 2024-12 | CT#106 | CP-243022 | 1124 | 1 | F | Correction of unavailability period | 18.8.0 |
| 2024-12 | CT#106 | CP-243022 | 1131 | | F | Correction on LCS Correlation ID as UE Context ID | 18.8.0 |
| 2024-12 | CT#106 | CP-243051 | 1149 | 1 | F | Correction for the deferred 5GC-MT-LR procedure | 18.8.0 |
| 2024-12 | CT#106 | CP-243068 | 1158 | | F | API version and External doc update | 18.8.0 |
| 2025-03 | CT#107 | CP-250023 | 1174 | | F | Wrong cardinality in Namf_MBSBroadcast service | 18.9.0 |
| 2025-03 | CT#107 | CP-250136 | 1182 | | F | 29.518 Rel18 API version and External doc update | 18.9.0 |
| 2025-06 | CT#108 | CP-251052 | 1201 | | F | Add missing changes in the approved CR | 18.10.0 |
| 2025-06 | CT#108 | CP-251048 | 1204 | 1 | F | Solving UE shall have maximum one LCS-UPP | 18.10.0 |
| 2025-06 | CT#108 | CP-251042 | 1212 | 2 | A | Correction of UE access behavior trend and UE location trend events reports | 18.10.0 |
| 2025-06 | CT#108 | CP-251078 | 1221 | | F | API version and External doc update | 18.10.0 |
| 2025-09 | CT#109 | CP-252034 | 1230 | | F | IERSR for "UE Reachable for DL Traffic" and "Loss-of-Connectivity" events | 18.11.0 |
| 2025-12 | CT#110 | CP-253141 | 1266 | 2 | F | Correlation of LCS-UPP Connection during UPP-CM Procedure | 18.12.0 |
| 2026-03 | CT#111 | CP-260024 | 1286 | 2 | F | UE Reachability for Data | 18.13.0 |
| 2026-03 | CT#111 | CP-260025 | 1300 | | F | Correct placement of network slice timer information in UEContextTransfer | 18.13.0 |
| 2026-03 | CT#111 | CP-260039 | 1304 | | F | API version and External doc update | 18.13.0 |
| 2026-06 | CT#112 | CP-261044 | 1335 | 1 | F | Missed UE Policy Request Trigger for VPLMN Specific URSP | 18.14.0 |
| 2026-06 | CT#112 | CP-261087 | 1340 | | F | API version and External doc update | 18.14.0 |

# History

| Version | Date | Status |
|---|---|---|
| V18.5.0 | May 2024 | Publication |
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